

# Integration and Transform Frontiers for the Fabius–Rvachev System

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## Provenance

This volume preserves the absorbed theorem and proof bodies verbatim, with labels, citation keys, and asset paths mechanically prefixed per part. Later edits add clearly marked, status-only Lean crosswalk annotations; the historical source hashes below are unchanged and no source theorem statement has been altered. The absorbed drafts are deleted from the working tree; git history is the archive.

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- Fabius\_Integral\_Transforms\_Report/Fabius\_Integral\_Transforms\_Report.tex (second-wave arrival, folded in as Part XII on 2026-08-28; distinct from the same-named 2026-08-27 source of Part IV)  
SHA-256 09d547d89417f861efcaa34c799736497ce13e426bfc0d6121c249a73713fa9

### Current Lean status of the layer-cake core

The part-local status paragraphs and roadmaps below record the repository snapshots audited by the absorbed reports. A later formalization now supplies their common measure-theoretic first-order engine. The theorem `Fabius.interval_smul_setIntegral_subgraph` in `SubgraphFubini.lean` is a real- or complex-scalar, Banach-valued weighted Fubini identity for measurable subgraphs over arbitrary  $\sigma$ -finite measures. Its atom-preserving closed-supergraph dual is `Fabius.interval_smul_setIntegral_supergraph` in the same module. `SurvivalLayerCake.lean` specializes it to finite measures, first without a support hypothesis by clamping the stopping point and then, for measures carried by a compact interval, in partial and full stopped-primitive forms. The exact Rvachev specializations `Fabius.ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral_min` and `Fabius.ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral` are in `ProbabilityLaplaceMoments.lean`.

On the closed-supergraph side, `CDFLayerCake.lean` exports the support-free clamped identity `Fabius.intervalIntegral_cdf_smul_eq_integral_clamp`; the stronger partial terminal identity `Fabius.intervalIntegral_cdf_smul_eq_integral_max_of_ae_le`, which assumes only  $c \leq b$  and almost-everywhere upper support  $z \leq b$ ; its compact-support corollary `Fabius.intervalIntegral_cdf_smul_eq_integral_max_of_ae_mem_Icc`; and the full compact-support form `Fabius.intervalIntegral_cdf_smul_eq_integral_of_ae_mem_Icc`. All use closed lower rays and therefore retain atoms. The dyadic law's carried-support declarations `Fabius.ProbabilityRepresentation.ae_weightedSumDistribution_mem_Icc` and `Fabius.ProbabilityRepresentation.weightedSumDistribution_restrict_Icc` are owned by `ProbabilityRepresentation.lean`, not by the downstream moment module.

For inverse clocks, `InverseLayerCake.lean` first proves `Fabius.galoisConnection_Icc_restrict_of_lt_iff_lt`: interval preservation and the strict adjunction make the two compact-interval restrictions a Galois connection. Thus its generic explicit-primitive and pointwise- $C^1$  identities `Fabius.intervalIntegral_smul_intervalIntegral_of_lt_iff_lt` and `Fabius.intervalIntegral_smul_comp_of_lt_iff_lt` now derive the measurable clamped representative used by the subgraph proof rather than assuming either clock measurable. Their Fabius wrappers are `Fabius.intervalIntegral_smul_intervalIntegral_fabiusInv` and `Fabius.intervalIntegral_smul_comp_fabiusInv`; these four theorems allow real or complex weights and Banach-valued primitives.

The same module also proves the arbitrary-absolutely-continuous layer cake for a real-valued primitive and any real or complex  $L^1$  weight. The generic subtraction form is `Fabius.intervalIntegral_mul_comp_sub_of_lt_iff_lt_of_absolutelyContinuousOnInterval`, its  $\Psi(a) = 0$  corollary is `Fabius.intervalIntegral_mul_comp_of_lt_iff_lt_of_absolutelyContinuousOnInterval`, and the exact Fabius wrapper is `Fabius.intervalIntegral_mul_comp_fabiusInv_of_absolutelyContinuousOnInterval`. Absolute continuity supplies derivative integrability, so no separate integrability hypothesis is imposed on the complementary side. The module additionally exports `Fabius.intervalIntegral_mul_fabiusInv_eq`, the full weighted inverse-area identity, and the endpoint area theorems `Fabius.intervalIntegral_fabiusInv_eq_intervalIntegral_rvachevUp` and `Fabius.intervalIntegral_fabiusInv_eq_one_half`. The stronger proper-compact absolutely-continuous calculus is now in `InversePairIntegral.lean`. Its generic theorem `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` works on arbitrary ordered compact source and level intervals. It assumes interval preservation and only the strict-order adjunction  $C(x) < u \iff x < Q(u)$ ; in particular, it assumes measurability of neither clock. The proof derives monotonicity on both

closed intervals and constructs measurable clamped `IccExtend` versions internally. The exact Fabius specialization is `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv`, and taking both absolutely-continuous factors to be the identity gives `Fabius.intervalIntegral_fabiusReal_add_fabiusInv_eq_one`. For every proper compact cut  $y \in [a, b]$ , the generic restricted identity is `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt`; its exact Fabius wrapper is `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`. The identity-function specialization at an inverse endpoint is `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`. These are the other three public theorems of `InversePairIntegral.lean`, for six in total.

This inverse-layer-cake update does *not* claim reversed-endpoint identities, improper extensions through singular endpoints, higher inverse primitives, a fractional-order semigroup for merely interval-integrable inputs or noncomplete targets, Fabius-specific Riemann–Liouville formulas beyond the all-real-endpoint positive-order stopped-power and unit-endpoint moment identities, the causal support-truncated Rvachev primitive and semigroup, and the three positive-real single-step Fabius–Rvachev order shifts recorded in the next status box, shifted dyadic-lattice or endpoint-moment formulas, quantile-transmutation, or inverse-function fractional formulas, any Caputo or other fractional-derivative theorem, complex-order or complex-Mellin continuations, complex-valued arbitrary-absolutely-continuous primitives, reversed-endpoint layer cake, or named Fourier, Laplace, or Mellin consequences not literally exported by the declarations above. Those distinctions supersede only contrary historical status wording; the provenance hashes above remain hashes of the unmodified absorbed sources.

#### Current Lean status of the positive-real fractional Volterra core and semigroup

At compiled full-facade checkpoint 149332f9d, the later module `FractionalVolterra.lean` defines `Fabius.fractionalVolterra`, a total Gamma-normalized oriented interval integral for a real order, arbitrary real base and endpoint, and an input valued in any real normed space. The unconditional declarations `Fabius.fractionalVolterra_self`, `Fabius.fractionalVolterra_congr`, `Fabius.fractionalVolterra_smul`, `Fabius.fractionalVolterra_one`, and `Fabius.fractionalVolterra_nat_succ` give the base-point value, locality on the closed unoriented endpoint interval, real-scalar compatibility, the ordinary order-one primitive, and exact agreement with `normalizedVolterra` at every positive natural order. In the classical regime  $0 < \alpha$  and  $a \leq x$ , continuity on  $[a, x]$  gives kernel interval integrability by `Fabius.intervalIntegrable_fractionalVolterra_kernel`, and `Fabius.fractionalVolterra_add_input` gives additivity for two continuous inputs. If  $a \leq b$  and  $a \leq x$ , the operator-independent theorem `Fabius.intervalIntegral_eq_integral_min_of_eq_zero` truncates any interval-integrable kernel that vanishes on the open right tail at  $\min(x, b)$ . Under the same endpoint bounds,  $0 < \alpha$ , and continuity on  $[a, x]$ , its fractional specialization is `Fabius.fractionalVolterra_eq_intervalIntegral_min_of_eq_zero`. For  $0 < \alpha$ ,  $0 < \beta$ , and  $s < x$ , `Fabius.intervalIntegral_fractionalVolterra_betaKernel` proves the exact shifted beta-kernel value

$$\int_s^x (x-u)^{\alpha-1} (u-s)^{\beta-1} du = (x-s)^{\alpha+\beta-1} \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \quad (1)$$

The companion module `FractionalVolterraSemigroup.lean` proves by `Fabius.intervalIntegrable_fractionalVolterra_betaKernel` that the raw kernel is interval-integrable already for  $s \leq x$ . Under the strict inequality, `Fabius.intervalIntegral_fractionalVolterra_normalizedBetaKernel` gives

$$\int_s^x \frac{(x-u)^{\alpha-1}}{\Gamma(\alpha)} \frac{(u-s)^{\beta-1}}{\Gamma(\beta)} du = \frac{(x-s)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)}. \quad (2)$$

For a complete target,  $a \leq x$ , and an input continuous on  $[a, x]$ , `Fabius.fractionalVolterra_a_add` proves

$$I_{a+}^{\alpha+\beta} f(x) = I_{a+}^{\alpha} \left[ u \mapsto I_{a+}^{\beta} f(u) \right] (x). \quad (3)$$

The other three declarations in this six-theorem companion module evaluate shifted powers and constants in an arbitrary real Banach target. For  $v \in E$  and  $a < x$ ,

$$I_{a+}^{\alpha} \left[ t \mapsto \frac{(t-a)^{\beta-1}}{\Gamma(\beta)} v \right] (x) = \frac{(x-a)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} v, \quad \alpha, \beta > 0, \quad (4)$$

$$I_{a+}^{\alpha} [t \mapsto (t-a)^{\rho} v] (x) = \frac{\Gamma(\rho+1)(x-a)^{\alpha+\rho}}{\Gamma(\alpha+\rho+1)} v, \quad \alpha > 0, \rho > -1. \quad (5)$$

These are `Fabius.fractionalVolterra_normalized_rpow_smul` and `Fabius.fractionalVolterra_rpow_smul`. The declaration `Fabius.fractionalVolterra_const` also includes the degenerate endpoint:

$$I_{a+}^{\alpha} [t \mapsto v] (x) = \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} v \quad (\alpha > 0, a \leq x). \quad (6)$$

The four theorems in `FractionalVolterraCalculus.lean` add increasing-affine covariance for arbitrary real  $\alpha$ ,  $0 < c$ , and  $a \leq x$ , with no regularity, integrability, or completeness, and positive-order integration by parts across an ordinary derivative:

$$I_{a+}^{\alpha+1} g'(x) = I_{a+}^{\alpha} g(x) - \frac{(x-a)^{\alpha}}{\Gamma(\alpha+1)} g(a). \quad (7)$$

The declarations are `Fabius.fractionalVolterra_affine`, `Fabius.fractionalVolterra_comp_affine`, `Fabius.fractionalVolterra_add_one_deriv`, and `Fabius.fractionalVolterra_add_one_deriv_of_eq_zero`; the last removes the boundary term when  $g(a) = 0$ . The total definition `Fabius.rvachevFractionalPrimitive` in `FabiusFractionalVolterra.lean` sets  $U_{\beta}(x) = I_{(-1)+}^{\beta} \text{up}(x)$ . Its support, natural-order, and semigroup theorems are

$$U_{\beta}(x) = \frac{1}{\Gamma(\beta)} \int_{-1}^{\min(x,1)} (x-t)^{\beta-1} \text{up}(t) dt, \quad \beta > 0, -1 \leq x, \quad (8)$$

$$U_{n+1}(x) = J_{-1}^{n+1} \text{up}(x), \quad n \in \mathbb{N}, x \in \mathbb{R}, \quad (9)$$

$$U_{\alpha+\beta}(x) = I_{(-1)+}^{\alpha} U_{\beta}(x), \quad \alpha, \beta > 0, -1 \leq x. \quad (10)$$

They are `Fabius.rvachevFractionalPrimitive_eq_intervalIntegral_min`, `Fabius.rvachevFractionalPrimitive_nat_succ`, and `Fabius.rvachevFractionalPrimitive_add`. The remaining three theorems in the module are

$$I_{0+}^{\alpha+1} \mathcal{F}(x) = 2^{\alpha} I_{0+}^{\alpha} \mathcal{F}(x/2), \quad \alpha > 0, x \geq 0, \quad (11)$$

$$I_{0+}^{\alpha+1} F(x) = 2^{\alpha} I_{0+}^{\alpha} F(x/2), \quad \alpha > 0, 0 \leq x \leq 1, \quad (12)$$

$$I_{-1+}^{\alpha+1} \text{up}(x) = 2^{\alpha} I_{0+}^{\alpha} F((x+1)/2), \quad \alpha > 0, x \geq -1. \quad (13)$$

They are `Fabius.fractionalVolterra_add_one_extendedFabius_of_nonneg`, `Fabius.fractionalVolterra_add_one_fabiusReal`, and `Fabius.fractionalVolterra_add_one_rvachevUp`. Thus the four fractional operator modules expose two definitions and 26 theorems in total.

The four theorems in `FractionalCDFLayerCake.lean` combine this kernel with the closed-supergraph layer cake. They are the support-free clamped theorem `Fabius.fractionalVolte`



rra\_measureReal\_Iic\_eq\_integral\_clamp, the one-sided lower-supported positive-part theorem Fabius.fractionalVolterra\_measureReal\_Iic\_eq\_integral\_posPart, its direct probability-CDF wrapper Fabius.fractionalVolterra\_cdf\_eq\_integral\_posPart, and the compact endpoint-power theorem Fabius.fractionalVolterra\_measureReal\_Iic\_eq\_integral\_rpow\_of\_ae\_mem\_Icc. Writing  $\mu_X$  for the dyadic weighted-sum law,  $X$  for its coordinate series, and  $\mathbf{P}$  for the underlying product measure, the four declarations in FabiusFractionalIntegral.lean give, for every  $\alpha > 0$  and every real  $x$ ,

$$I_{0+}^{\alpha} F(x) = \frac{1}{\Gamma(\alpha + 1)} \int_{\mathbb{R}} (x - z)_{+}^{\alpha} d\mu_X(z) = \frac{1}{\Gamma(\alpha + 1)} \int_{[0,1]^{\mathbb{N}}} (x - X(\omega))_{+}^{\alpha} d\mathbf{P}(\omega), \quad (14)$$

including  $x \leq 0$ . At  $x = 1$ , reflection gives

$$I_{0+}^{\alpha} F(1) = \frac{1}{\Gamma(\alpha + 1)} \int_{\mathbb{R}} z^{\alpha} d\mu_X(z) = \frac{1}{\Gamma(\alpha + 1)} \int_{[0,1]^{\mathbb{N}}} X(\omega)^{\alpha} d\mathbf{P}(\omega). \quad (15)$$

The law/sample-space pairs are, respectively, Fabius.ProbabilityRepresentation.fractionalVolterra\_fabiusReal\_eq\_integral\_posPart, Fabius.ProbabilityRepresentation.fractionalVolterra\_fabiusReal\_eq\_uniformProduct\_integral\_posPart, Fabius.ProbabilityRepresentation.fractionalVolterra\_fabiusReal\_one\_eq\_integral\_rpow, and Fabius.ProbabilityRepresentation.fractionalVolterra\_fabiusReal\_one\_eq\_uniformProduct\_integral\_rpow.

This tranche proves no order-zero operator law, reversed-endpoint fractional interpretation, semigroup theorem for merely interval-integrable inputs or noncomplete targets, nonpositive-scale affine covariance, Caputo or other fractional derivative, complex-order theorem, iterated fractional shift or shifted dyadic lattice, endpoint-moment wrapper, transform/tail-series theorem, or quantile- or inverse-function fractional specialization. Formula (14) exactly discharges the positive-real integral line (1028), including its all-real endpoint range, and gives the positive-real slices of (96), (455), (1111), and (1329); the endpoint theorem gives their positive-real moment forms. Its positive-integer specialization also gives (31). Their complex-order continuations remain open, as does the fractional-derivative line (1029). The power and constant theorems make several boundary terms below reusable but do not by themselves package the Rvachev transform/tail, quantile-transmutation, weighted incomplete-beta, or inverse-function formulas. This status box supersedes only contrary historical formalization wording in the absorbed snapshots; the provenance hashes above remain unchanged.

### Current Lean status of analytic finite-series filters

AnalyticSeriesFilter.lean exports two definitions and seven theorems. The definitions are Fabius.finiteAnalyticSeriesFilter and Fabius.geometricAnalyticSeriesFilter. Under explicit unconditional summability of the weighted sampled series (so a zero-weight node imposes no convergence obligation), Fabius.summable\_finiteAnalyticSeriesFilter\_diagonal and Fabius.finiteAnalyticSeriesFilter\_eq\_tsum give diagonal summability and evaluation; Fabius.finiteAnalyticSeriesFilter\_eq\_head\_add\_tail\_of\_exact and Fabius.finiteAnalyticSeriesFilter\_eq\_constant\_add\_tail\_of\_exact\_zero give the exact head/tail decompositions; and Fabius.geometricAnalyticSeriesFilter\_eq\_constant\_add\_gaussian\_tsum, Fabius.geometricLagrangeAnalyticSeriesFilter\_eq\_constant\_add\_gaussian\_tsum, and Fabius.geometricLagrangeAnalyticSeriesFilter\_shifted give the denominator-free Gaussian residual tails. No conditional/boundary convergence, formal-to-analytic PowerSeries evaluation bridge, actual sinc-product instantiation, radius or uniform convergence, norm/sign/error bound, asymptotic



estimate, positivity, or Fabius-specific acceleration theorem is asserted.

### Current Lean status of finite complex Thue–Morse product bridges

`ThueMorseComplexProductBridge.lean` exports two definitions and eighteen theorems for the total complex finite-product layer. The normalized prefixes `Fabius.shiftedComplexSincPrefix` and `Fabius.complexLaplacePrefix` identify every finite dyadic Thue–Morse block with, respectively, its complex sinc and negative-Laplace products, including level zero and the removable origin. The denominator-free primary equalities have quotient corollaries away from zero, and `Fabius.complexLaplacePrefix_eq_exp_mul_shiftedComplexSincPrefix` gives the exact finite Fourier–Laplace rotation. No infinite-product convergence, analytic logarithm, tail estimate, affine lacunary cosine companion, or measure-specific transform theorem is asserted by this module.

### Current Lean status of the ordinary Cauchy–Stieltjes core

The reusable foundation is `MeasureCauchyTransform.lean`. Its three definitions `Fabius.measureCauchyDomain`, `Fabius.measureCauchyTransform`, and `Fabius.measureCauchyPower` package the oriented transform and every unnormalized kernel power of a real measure. The nine direct theorems are the integral and index-one forms, affine naturality for transforms and powers, openness, first derivative, holomorphy, and the transform/power recurrences of a uniform affine fixed-point law: `Fabius.measureCauchyTransform_apply`, `Fabius.measureCauchyPower_one`, `Fabius.measureCauchyPower_map_affine`, `Fabius.measureCauchyTransform_map_affine`, `Fabius.isOpen_measureCauchyDomain`, `Fabius.hasDerivAt_measureCauchyTransform`, `Fabius.analyticOn_measureCauchyTransform`, `Fabius.measureCauchyPower_succ_of_uniformAffineFixedPoint`, and `Fabius.hasDerivAt_measureCauchyTransform_of_uniformAffineFixedPoint`. The fixed-point interface assumes only a finite measure, support in an affine-invariant carrier, nonzero uniform and tail scales, and the displayed equality in law; the carrier need not be closed, compact, measurable, or nonempty, and the measure need not be a probability measure. Negative nonzero scales are allowed.

The adjacent generic bounded-support layer is `MeasureCauchyMomentLaurent.lean`, with exactly one public definition and fifteen public theorems. For a finite real measure and any scalar field  $\mathbb{K}$  with `RCLike` structure, the definition `Fabius.measureCauchyMoment` is

$$m_n(c) := \int_{\mathbb{R}} ((x : \mathbb{K}) - c)^n d\mu(x).$$

Given  $R \geq 0$ , the measure-theoretic relative-support hypothesis  $\|(x : \mathbb{K}) - c\| \leq R$  almost everywhere, and  $R < \|z - c\|$ , the exact generic identity is

$$\int_{\mathbb{R}} \frac{d\mu(x)}{z - x} = \sum_{k < N} \frac{m_k(c)}{(z - c)^{k+1}} + (z - c)^{-N} \int_{\mathbb{R}} \frac{((x : \mathbb{K}) - c)^N}{z - x} d\mu(x),$$

and its sharp mass-sensitive bound is

$$\left\| \int_{\mathbb{R}} \frac{d\mu(x)}{z - x} - \sum_{k < N} \frac{m_k(c)}{(z - c)^{k+1}} \right\| \leq \mu(\mathbb{R}) (\|z - c\| - R)^{-1} \left( \frac{R}{\|z - c\|} \right)^N.$$

No probability normalization is assumed: the zeroth moment is the total mass. The theorem inventory is exact. The field identity is `Fabius.inv_sub_eq_sum_range_add`; the five foundational results are `Fabius.measureCauchyMoment_zero`, `Fabius.measurable_inv_sub_rclike`, `Fabius.integrable_centered_pow_div_sub_of_ae_norm_sub_le`,

Fabius.integrable\_inv\_sub\_of\_ae\_norm\_sub\_le, and Fabius.norm\_measureCauchyMoment\_le. The five direct-integral Laurent results are Fabius.integral\_inv\_sub\_eq\_sum\_range\_measureCauchyMoment\_add, Fabius.norm\_integral\_inv\_sub\_sub\_sum\_range\_measureCauchyMoment\_le, Fabius.summable\_measureCauchyMoment\_laurent, Fabius.integral\_inv\_sub\_eq\_tsum\_measureCauchyMoment, and Fabius.hasSum\_measureCauchyMoment\_laurent. Finally, the four complex named-transform wrappers are Fabius.measureCauchyTransform\_eq\_sum\_range\_measureCauchyMoment\_add, Fabius.norm\_measureCauchyTransform\_sub\_sum\_range\_measureCauchyMoment\_le, Fabius.measureCauchyTransform\_eq\_tsum\_measureCauchyMoment, and Fabius.hasSum\_measureCauchyTransform\_measureCauchyMoment\_laurent.

GeometricUniformCauchy.lean specializes the preceding uniform-affine fixed-point interface to the law of  $X_q = (1 - q)U_0 + qX'_q$ . Its three definitions and six theorems are Fabius.geometricUniformStieltjesDomain, Fabius.geometricUniformStieltjesTransform, Fabius.geometricUniformStieltjesPower, Fabius.geometricUniformStieltjesTransform\_apply, Fabius.geometricUniformStieltjesPower\_one, Fabius.isOpen\_geometricUniformStieltjesDomain, Fabius.analyticOn\_geometricUniformStieltjesTransform, Fabius.hasDerivAt\_geometricUniformStieltjesTransform\_refinement, and Fabius.geometricUniformStieltjesPower\_succ. The definitions are total in real  $q$ , holomorphy assumes  $|q| < 1$ , and the divided DDE and adjacent-power recurrence also assume  $q \neq 0$ . Because the proof uses the exact series range as its invariant carrier, negative ratios are included; only the divided formulas exclude  $q = 0$ .

For a bounded Fabius solution  $F$ , the later module CauchyTransform.lean defines the report-sign transforms Fabius.rvachevCauchyTransform and Fabius.fabiusStieltjesTransform. In this status box write  $R_{\text{up}}(z) := \text{rvachevCauchyTransform } F \, z$  and  $S_{\text{unit}}(z) := \text{fabiusStieltjesTransform } z$ , independently of the part-local notation below. The declarations Fabius.rvachevCauchyTransform\_apply and Fabius.fabiusStieltjesTransform\_apply identify them with the measure integrals of  $(z - x)^{-1}$ , and Fabius.rvachevCauchyTransform\_eq\_integral\_rvachevUp gives the up-density form. The declarations Fabius.analyticOn\_rvachevCauchyTransform and Fabius.analyticOn\_fabiusStieltjesTransform prove holomorphy on the complements of the named complexified intervals  $[-1, 1]$  and  $[0, 1]$ , while Fabius.hasDerivAt\_rvachevCauchyTransform and Fabius.hasDerivAt\_fabiusStieltjesTransform identify the first derivatives with the negatives of the integrals of the squared Cauchy kernels. Finally, Fabius.fabiusStieltjesTransform\_eq\_two\_mul\_rvachevCauchyTransform and Fabius.rvachevCauchyTransform\_eq\_inv\_two\_mul\_fabiusStieltjesTransform prove

$$S_{\text{unit}}(z) = 2R_{\text{up}}(2z - 1), \quad R_{\text{up}}(w) = \frac{1}{2}S_{\text{unit}}\left(\frac{w + 1}{2}\right).$$

The equalities are total in Lean's Bochner-integral convention. Their classical analytic interpretation uses  $z \notin [0, 1]$  in the first formula and  $w \notin [-1, 1]$  in the second. The same module now also defines Fabius.rvachevCauchyPower and Fabius.fabiusStieltjesPower. The direct named bridge Fabius.fabiusStieltjesPower\_eq\_two\_pow\_mul\_rvachevCauchyPower proves  $S_n(z) = 2^n R_n(2z - 1)$  for every natural  $n$ . More importantly for the status of the absorbed reports, Fabius.hasDerivAt\_fabiusStieltjesTransform\_refinement directly exports

$$S'_{\text{unit}}(z) = 4(S_{\text{unit}}(2z) - S_{\text{unit}}(2z - 1))$$

on the unit slit domain: this is no longer merely an unnamed corollary through the affine bridge. The direct named theorems Fabius.fabiusStieltjesPower\_succ and Fabius.rvachevCauchyPower\_succ give both adjacent-order recurrences throughout their complex slit domains.

For completeness, the remaining public names in this exact module inventory are the two domain definitions `Fabius.rvachevCauchyDomain` and `Fabius.fabiusStieltjesDomain`, the index-one identities `Fabius.rvachevCauchyPower_one` and `Fabius.fabiusStieltjesPower_one`, and the openness declarations `Fabius.isOpen_rvachevCauchyDomain` and `Fabius.isOpen_fabiusStieltjesDomain`. Thus the module has six public definitions and seventeen public theorems; every one is crosswalked in this status box. This discharges the ordinary compact-support measure-transform content of the later formulas (639), (644), (781), (906), and (1046), but not every alternate quantile or inverse-integral presentation attached to those labels.

`CauchyCDF.lean` adds three atom-exact integration-by-parts declarations. For a finite measure,  $a \leq b$ , almost-everywhere support on  $[a, b]$ , and  $z$  outside the complexified interval, `Fabius.integral_inv_sub_eq_mass_smul_sub_intervalIntegral_measureReal_Iic` proves

$$\int \frac{d\mu(x)}{z-x} = \mu(\mathbb{R}) \frac{1}{z-b} - \int_a^b \frac{\mu((-\infty, t])}{(z-t)^2} dt.$$

The closed lower ray retains atoms. The probability normalization `Fabius.integral_inv_sub_eq_sub_intervalIntegral_cdf` exposes Mathlib's CDF, and `Fabius.fabiusStieltjesTransform_eq_inv_sub_one_sub_intervalIntegral_fabiusReal` gives, for every bounded Fabius solution and  $z \in \text{fabiusStieltjesDomain}$ ,

$$S_{\text{unit}}(z) = \frac{1}{z-1} - \int_0^1 \frac{F(t)}{(z-t)^2} dt.$$

Thus the Fabius-CDF formulas (490) and (1400) are discharged exactly. The survival companions are now exported by `CauchySurvival.lean`: `Fabius.integral\_inv\_sub\_eq\_mass\_smul\_add\_intervalIntegral\_measureReal\_Ioi` is the general survival integration by parts  $\int (z-x)^{-1} d\mu = \mu(\mathbb{R})(z-a)^{-1} + \int_a^b \mu((t, \infty))(z-t)^{-2} dt$  for any finite measure almost surely supported on  $[a, b]$ , and `Fabius.fabiusStieltjesTransform\_eq\_inv\_add\_intervalIntegral\_rvachevUp` is the Fabius instance  $S_{\text{unit}}(z) = z^{-1} + \int_0^1 \text{up}(t)(z-t)^{-2} dt$  — the transform through the up-density itself, with the closed kernel primitive `Fabius.intervalIntegral\_inv\_sub\_sq` shared by both. The full higher-power families are now exported by `CauchyHigherPowers.lean`: for every  $p \geq 1$ , `Fabius.integral\_pow\_inv\_sub\_eq\_mass\_smul\_sub\_intervalIntegral\_measureReal\_Iic` and `Fabius.integral\_pow\_inv\_sub\_eq\_mass\_smul\_add\_intervalIntegral\_measureReal\_Ioi` give the CDF and survival integration by parts at resolvent power  $p$ ,  $\int (z-x)^{-p} d\mu = \mu(\mathbb{R})(z-b)^{-p} - p \int_a^b \mu((-\infty, t])(z-t)^{-(p+1)} dt$  and its lower-endpoint survival companion, with the closed primitive `Fabius.intervalIntegral\_pow\_inv\_sub`; the first power recovers the formulas above. The orthogonal-polynomial arc now runs two steps further: `OrthogonalPolynomialJacobi.lean` computes the next exact Jacobi data of the up-law from the corpus's own moment recursion — the even moments  $c_2 = 19/675$ ,  $c_3 = 583/59535$  (`Fabius.moment_two`, `Fabius.moment_three`), the third Hankel determinant  $h_3 = 32/18225$  (`Fabius.hankelDet_three`), the Jacobi ratios  $a_2 = 32/2025$  and  $a_3 = 19808/7441875$  (`Fabius.hankelRatio_two`, `Fabius.hankelRatio_three`, the latter read off the norm-producing moment rather than a  $4 \times 4$  determinant),  $h_4$  (`Fabius.hankelDet_four`), and the next two monic orthogonal polynomials  $P_3 = x^3 - \frac{19}{75}x$  and  $P_4 = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}$  (`Fabius.upOrthoPolynomial_three`, `Fabius.upOrthoPolynomial_four`), with their pointwise and squared-norm corollaries. The *moment/Laurent layer* is now delivered on the symmetric side by `StieltjesMomentLaurent.lean`: for every real  $|z| > 1$  and every  $N$ ,  $\int (z-x)^{-1} d\mu_{\text{up}} = \sum_{k < N} m_k z^{-(k+1)} + z^{-N} \int x^N (z-x)^{-1} d\mu_{\text{up}}$  with the *exact* remainder (`Fabius.integral_inv_sub_eq_sum_upMoment_add`), the geometric tail bound ( $|z| -$

$1)^{-1}|z|^{-N}$  (Fabius.abs\_integral\_inv\_sub\_sub\_sum\_le), and the convergent series  $\int (z-x)^{-1} d\mu_{\text{up}} = \sum_{k \geq 0} m_k z^{-(k+1)}$  in tsum/HasSum form (Fabius.integral\_inv\_sub\_eq\_tsum\_upMoment, Fabius.hasSum\_upMoment\_laurent); the moments  $m_k = \text{Fabius.upMoment}$  are the Hankel data of the orthogonal-polynomial layer, so the same numbers drive the determinant polynomials, the three-term recurrence, and the asymptotic expansion of the transform. These named real up-measure results are the  $\mathbb{K} = \mathbb{R}$ ,  $c = 0$ ,  $R = 1$ , probability specialization of the generic MeasureCauchyMomentLaurent.lean layer, rather than a replacement of its arbitrary-center finite-measure interface. The compatibility theorems Fabius.ae\_norm\_sub\_zero\_le\_one\_rvachevMeasure and Fabius.measureCauchyMoment\_rvachevMeasure\_zero identify the generic almost-everywhere relative-ball condition and centered coefficients with the up-law's unit support and upMoment data in every RCLike field; the probability hypothesis reduces the generic total-mass factor to one. The *logarithmic fixed point* is delivered in real form by StieltjesLogFixedPoint.lean: the uniform digit integrates the Cauchy kernel to a logarithm,  $\int_{-1/2}^{1/2} (c-u)^{-1} du = \log \frac{c+1/2}{c-1/2}$  for  $c > \frac{1}{2}$  (Fabius.integral\_inv\_sub\_uniform\_half), and conditioning the transform on the refinement tail gives  $R_1(z) = \mathbb{E} \log \frac{2z-X+1}{2z-X-1}$  for real  $z > 1$  (Fabius.integral\_inv\_sub\_eq\_integral\_log) – the display of [theorem 12.1](#)'s fixed-point equation, proved through the convolution form of the refinement. The hierarchy itself follows by the same conditioning, with no differentiation under the integral: StieltjesResolventHierarchy.lean proves  $R_{n+1}(z) = \frac{2^n}{n} (R_n(2z-1) - R_n(2z+1))$  for every  $n \geq 1$  and real  $z > 1$  (Fabius.integral\_pow\_inv\_sub\_succ), from the exact digit-level kernel integral  $\int_{-1/2}^{1/2} (c-u)^{-(n+1)} du = ((c-\frac{1}{2})^{-n} - (c+\frac{1}{2})^{-n})/n$  (Fabius.integral\_pow\_inv\_sub\_uniform\_half) and the integrability of all resolvent powers off the support (Fabius.integrable\_pow\_inv\_sub\_rvachevMeasure). The *generalized-order layer* is delivered in real form by the same conditioning: StieltjesGeneralizedOrder.lean proves the fractional hierarchy  $R_\alpha(z) = \frac{2^{\alpha-1}}{\alpha-1} (R_{\alpha-1}(2z-1) - R_{\alpha-1}(2z+1))$  for every real  $\alpha > 1$  and  $z > 1$  (Fabius.integral\_rpow\_inv\_sub\_hierarchy), from the fractional digit-level kernel integral  $\int_{-1/2}^{1/2} (c-u)^{-\alpha} du = ((c-\frac{1}{2})^{1-\alpha} - (c+\frac{1}{2})^{1-\alpha})/(\alpha-1)$  (Fabius.integral\_rpow\_inv\_sub\_uniform\_half) and fractional-kernel integrability off the support; the integer hierarchy is the specialization  $\alpha = n+1$ . The *complex- $z$  layer* is now delivered by StieltjesCauchyTransform.lean and StieltjesHerglotz.lean: for every complex  $z$  with  $\|z\| > 1$  the expansion with exact remainder, the geometric tail bound, and the convergent Laurent series hold verbatim (Fabius.integral\_inv\_sub\_complex\_eq\_sum\_upMoment\_add, Fabius.norm\_integral\_inv\_sub\_complex\_sub\_sum\_le, Fabius.integral\_inv\_sub\_complex\_eq\_tsum\_upMoment), the kernel identity having been generalized to an arbitrary field (Fabius.inv\_sub\_eq\_sum\_range\_add); the named transform of the renormalization layer carries the same series (Fabius.rvachevCauchyTransform\_eq\_tsum\_upMoment), and the real program is its restriction (Fabius.integral\_inv\_sub\_complex\_ofReal). This is the  $\mathbb{K} = \mathbb{C}$ ,  $c = 0$ ,  $R = 1$  up-probability face of the generic direct-integral and named-transform theorems above; in particular, Fabius.measureCauchyMoment\_rvachevMeasure\_zero supplies the complex embedding of the same real Hankel moments, while Fabius.ae\_norm\_sub\_zero\_le\_one\_rvachevMeasure supplies the relative unit-ball hypothesis. Off the real axis the kernel is integrable with bound  $|\text{Im } z|^{-1}$  (Fabius.integrable\_inv\_sub\_complex\_of\_im\_ne\_zero), the transform reflects under conjugation,  $R(\bar{z}) = \overline{R(z)}$  (Fabius.rvachevCauchyTransform\_conj), and it maps the upper half-plane strictly into the lower with the quantitative Herglotz bound  $\text{Im } R(z) \leq -\text{Im } z / (\|z\| + 1)^2$  (Fabius.im\_rvachevCauchyTransform\_le, Fabius.im\_rvachevCauchyTransform\_neg, Fabius.im\_rvachevCauchyTransform\_pos) – minus a Herglotz–Nevanlinna function, the orientation data of the boundary theory. The entry identities of that theory are exact at every height: for every real  $x$  and every  $\varepsilon \neq 0$ ,  $\text{Im } R(x+i\varepsilon) = -\int \varepsilon / ((x-t)^2 + \varepsilon^2) d\mu_{\text{up}}(t)$  and  $\text{Re } R(x+i\varepsilon) = \int (x-t) / ((x-t)^2 + \varepsilon^2) d\mu_{\text{up}}(t)$



– the Poisson and conjugate-Poisson displays (`Fabius.im_rvachevCauchyTransform_eq`, `Fabius.re_rvachevCauchyTransform_eq`). `StieltjesInversion.lean` adds the kernel layer those displays run on: strict positivity, the sharp height bound  $\varepsilon/((x-t)^2 + \varepsilon^2) \leq \varepsilon^{-1}$  with its attainment at  $t = x$  (`Fabius.poisson_kernel_pos`, `Fabius.poisson_kernel_le_inv`, `Fabius.poisson_kernel_self`), the normalisation  $\int \pi^{-1}\varepsilon/((x-t)^2 + \varepsilon^2) dt = 1$  – inherited from Mathlib’s Cauchy density rather than re-proved – (`Fabius.integral_poisson_kernel_eq_one`), integrability against the up-measure, and the far-field localisation  $\int_{|t-x| \geq \delta} \leq \varepsilon/\delta^2 \rightarrow 0$  (`Fabius.setIntegral_poisson_far_le`, `Fabius.tendsto_setIntegral_poisson_far`) – the concentration half of an approximate identity. Consequently –  $\text{Im } R(x + i\varepsilon)/\pi$  is the up-measure integral of a probability density and lies in  $[0, 1/(\pi\varepsilon)]$  (`Fabius.inv_pi_mul_neg_im_rvachevCauchyTransform_mem_Icc`).

Both halves of the approximate identity are now formal. `PoissonApproximateIdentity.lean` supplies the Lebesgue-side tail *exactly*, not as an estimate: each half-line integrates to  $\pi/2 - \arctan(\delta/\varepsilon)$  (`Fabius.setIntegral_poisson_kernel_Ioi`, `Fabius.setIntegral_poisson_kernel_Iic`), so the two-sided tail is  $1 - \frac{2}{\pi} \arctan(\delta/\varepsilon)$  (`Fabius.setIntegral_poisson_density_compl_Ioo`) and the near mass  $\frac{2}{\pi} \arctan(\delta/\varepsilon)$ ; letting  $\varepsilon \rightarrow 0^+$  gives the concentration (`Fabius.tendsto_setIntegral_poisson_density_Ioo`, and the packaged `Fabius.poisson_kernel_approximate_identity`). Two theorems are carried purely to keep the reading honest: the tail is *strictly positive* at every  $\varepsilon > 0$  and  $\delta > 0$ , so 0 is a limit and never a value (`Fabius.setIntegral_poisson_density_compl_Ioo_pos`), and at  $\delta = \varepsilon$  the tail is exactly 1/2 (`Fabius.setIntegral_poisson_density_compl_Ioo_self`) – a calibration that any sign or factor-of-two slip would break.

`PoissonMassSwap.lean` supplies the other variable: for each fixed height the Poisson extension of the up-measure is itself a probability density in  $x$ ,  $\int_{\mathbb{R}} -\frac{1}{\pi} \text{Im } R(x + i\varepsilon) dx = 1$  (`Fabius.integral_inv_pi_mul_neg_im_rvachevCauchyTransform_eq_one`, `Fabius.integral_neg_im_rvachevCauchyTransform_eq_pi`), by a Tonelli swap over the nonnegative kernel.

With both inputs in place the inversion theorem itself is now formal. `StieltjesPerron.lean` proves

$$\lim_{\varepsilon \rightarrow 0^+} -\frac{1}{\pi} \int_{(a,b)} \text{Im } R(x + i\varepsilon) dx = \mu_{\text{up}}((a,b))$$

for every  $a < b$  (`Fabius.stieltjesPerron_inversion`, `Fabius.tendsto_setIntegral_inv_pi_mul_neg_im`). Notably there is *no* endpoint hypothesis: the classical requirement  $\mu(\{a\}) = \mu(\{b\}) = 0$  is discharged rather than assumed, because the up-measure is absolutely continuous (`Fabius.rvachevMeasure_singleton`), and the reading is backed by the guard `Fabius.rvachevMeasure_real_Icc_eq_Ioo` – closed and open intervals carry the same mass, so nothing was quietly dropped. The route is the expected one: the exchange (`Fabius.setIntegral_inv_pi_mul_neg_im_eq`) turns the  $x$ -integral into a  $\mu_{\text{up}}$ -integral of the interval mass  $P_\varepsilon(t) = \frac{1}{\pi} (\arctan \frac{b-t}{\varepsilon} - \arctan \frac{a-t}{\varepsilon})$ , whose pointwise limits are 1 inside, 0 outside and  $\frac{1}{2}$  at each endpoint (`Fabius.tendsto_poissonIntervalMass_of_mem_Ioo` and neighbours), and dominated convergence finishes. Here too the limits are guarded rather than described:  $P_\varepsilon$  is strictly positive and strictly below one everywhere (`Fabius.poissonIntervalMass_pos`, `Fabius.poissonIntervalMass_lt_one`), so neither limit is ever attained, and `Fabius.poissonIntervalMass_calibration` pins the  $\pi^{-1}$ .

`CauchyRenormalization.lean` adds five public theorems. The declarations `Fabius.mapsTo_rvachevCauchyDomain_two_mul_add_one` and `Fabius.mapsTo_rvachevCauchyDomain_two_mul_sub_one` prove that both dyadic branches preserve the slit domain. For every bounded Fabius solution and every  $z \in \text{rvachevCauchyDomain}$ , `Fabius.hasDerivAt_rvachevCauchyTransform_affineDifference` and `Fabius.deriv_rvachevCauchyTransform`

prove

$$R'_{\text{up}}(z) = 2(R_{\text{up}}(2z + 1) - R_{\text{up}}(2z - 1)).$$

This exactly discharges (785), (1050), and the ordinary first-order equation (1406). For every  $n \in \mathbb{N}$ , `Fabius.iteratedDeriv_rvachevCauchyTransform_eq_thueMorse_sum` proves

$$R_{\text{up}}^{(n)}(z) = 2^{\binom{n+1}{2}} \sum_{j=0}^{2^n-1} \varepsilon_j R_{\text{up}}(2^n z + 2^n - 1 - 2j).$$

The unit-interval Stieltjes derivative equation is now the direct named theorem `Fabius.hasDerivAt_fabiusStieltjesTransform_refinement`; the all-power affine bridge and both complex-slit adjacent-order recurrences are direct named theorems in the same module, as recorded above.

The real logarithmic fixed point for  $z > 1$ , the positive-integer real hierarchy, real-order lowering for  $\alpha > 1$ , the arbitrary-center finite-measure `RCLike` moment/Laurent expansion with its exact mass-sensitive remainder, its  $c = 0$ ,  $R = 1$  real and complex up-probability specializations, finite-height Herglotz–Poisson formulas, integrated interval Stieltjes–Perron inversion, and initial exact Jacobi data are formalized by the `MeasureCauchyMomentLaurent.lean`, `Stieltjes*.lean`, and `OrthogonalPolynomialJacobi.lean` layers described above. What remains open is the complex logarithmic continuation/logarithmic potential, named Laplace/Fourier transform forms, complex order, pointwise or nontangential Sokhotski–Plemelj and principal-value Hilbert-transform formulas, the full J-fraction/Padé package, and the separate explicit conversion of the all-order Thue–Morse orbit into a named higher-Cauchy-kernel integral identity. The generic CDF/survival integration-by-parts formulas at every positive power do not by themselves prove that last orbit formula. This status box supersedes only contrary historical formalization wording in the absorbed snapshots; their provenance hashes remain unchanged.

## Part I

# Antiderivatives of Monomially Weighted Fabius-Type Functions

## Abstract

Let  $F$  be the bounded Fabius distribution function, let  $up$  be Rvachev's compactly supported up-function, and let  $Q = F^{-1}$  on  $[0, 1]$ . This report constructs normalized antiderivatives of  $x^p F(x)$  and their higher-order Volterra analogues. For  $p \in \mathbb{N}_0$ , the answer is a finite closed expression in dyadically contracted Fabius values. More generally, for every complex exponent  $\alpha$ , every integer order  $n \geq 1$ , and  $0 < x \leq 1$ , we prove a locally uniformly and absolutely convergent Newton–Volterra expansion

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^\alpha F(t) dt = \sum_{r \geq 0} (-1)^r \binom{n+r-1}{r} \alpha^r x^{\alpha-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right).$$

Endpoint flatness makes this an entire function of  $\alpha$ , including all negative powers; nonnegative integral exponents are exactly the terminating cases. At  $x = 1$ , the coefficients become the Fabius moments and give a Newton interpolation formula, a two-parameter family of exact moment identities, and a logarithmic constant

$$\int_0^1 \frac{F(x)}{x} dx = -\mathbb{E} \log X = \sum_{n \geq 1} \frac{d_n}{n} = \log 2 + \frac{1}{2} \sum_{n \geq 1} \frac{c_n}{n}.$$

We derive global piecewise formulas, exponential generating functions, Bell–Bernoulli cumulant forms, formulas for  $F(1-x) = 1 - F(x)$ , complete piecewise antiderivatives of  $x^p up(x)$ , exact higher derivatives in Thue–Morse form, and Riemann–Liouville extensions of nonintegral order. For the inverse Fabius function we obtain closed change-of-variable formulas, including

$$\int_0^y Q(u) du = yQ(y) - F\left(\frac{Q(y)}{2}\right), \quad \mathcal{I}^\nu Q(y) = \frac{1}{\Gamma(\nu+1)} \int_0^{Q(y)} (y - F(t))^\nu dt.$$

Combining the inverse endpoint equivalent already developed in the repository with regular-variation arguments yields new asymptotics for weighted inverse primitives and for large quantile moments. The final sections separate proved statements from conjectural saddle refinements, record reproducible numerical checks, and propose further formalization and arithmetic directions. The integer core is now machine checked in Lean: this comprises the normalized Volterra operator, its identification with literal repeated integration, the finite polynomial and natural-monomial commutators, the global signed Fabius ladder, and the bounded formulas on the full range  $x \leq 1$ . The infinite complex-power series, Mellin consequences, remaining report-specific fractional formulas (including iterated, weighted, and complex-order formulas), global exterior pieces, Rvachev folds, and the improper-singular, higher-order, or fractional inverse-function formulas remain research-frontier material. The separate full-interval weighted inverse layer-cake identity and the complete and variable-upper proper-compact absolutely-continuous inverse-pair identities are covered by the current consolidated status note.

# 1 Scope, status, and corpus audit

## 1.1 The source corpus

The requested source tree was audited recursively at the repository snapshot identified in its own frontier inventory as commit 32d6d36c51d803289e6d6a0dc0c37753766eba47 (27 August 2026). That inventory records 67 distinct \*.tex paths under Analysis/FabiusFunction/docs. The audit included the live primary exposition, the mathematical glossary, the Lean walkthrough, the non-elementarity article, the consolidated non-formalized frontier volume, the source-paper transcriptions, and the frozen archive and draft families [1, 2, 3, 4, 5, 6].

Archived files contain many historical copies of material later consolidated into the live exposition or frontier volume. For nonduplication, the present work therefore used two passes:

1. a path-complete inventory and phrase audit over the recursive corpus;
2. theorem-level comparison against the live primary exposition and the consolidated frontier volume, with targeted checks of the inverse, repeated-integration, Newton/Richardson, moment, asymptotic, and non-elementarity source families.

At the audited snapshot, the corpus already contained the basic functional equations, exact dyadic values, moment recurrences, the sinc product, all-orders endpoint asymptotics, repeated-averaging limits, and extensive inverse-function work. It did *not* yet develop the normalized antiderivative ladder of this report, the finite monomial formula, the arbitrary-complex-power Newton–Volterra series, or the inverse-primitive reductions below. Literal searches for “antiderivative” and “indefinite integral” returned no such treatment in the live primary or consolidated frontier source. This establishes novelty *relative to the audited repository*; it is not a claim that no equivalent identity has ever appeared elsewhere. In particular, the base functional integral identity  $\int_0^x F(t) dt = F(x/2)$  is an immediate classical consequence of the defining equation. The subsequent Lean implementation described below changes the current formalization status, not this historical snapshot audit.

### Status convention used in this report

Every item labeled theorem, proposition, lemma, or corollary is either proved in place, derived immediately from a proved identity, or explicitly identified as an imported standard theorem; that label alone does not assert a Lean proof. In particular, the first formula in [theorem 13.4](#) is the classical Lagrange inverse-derivative formula; the report cites that result and proves only the coefficient conversion used here. “Repository input” means that an already established result is imported from the audited corpus. “Conjecture” means that the statement is supported by the preceding analysis but is not proved. Besides repository inputs—including derivative identities already present in the audited corpus and a post-snapshot Lean proof of the inverse endpoint equivalent—only the declarations in the crosswalk below are claimed to be machine checked by the present status update; all other new claims in this report remain non-formalized.

## 1.2 Lean formalization update

The reusable foundation is formalized in `FabiusFunction.NormalizedVolterra`. Its definition `normalizedVolterra` takes values in an arbitrary real normed vector space and uses oriented interval integrals. For continuous Banach-valued inputs, `iteratedPrimitive_eq_normalizedVolterra` identifies its Cauchy kernel with literal repeated application of the first primitive. Thus the formal result is not confined to the scalar Fabius case or to ordered endpoints.

The same module now has a generic affine and basepoint calculus with arbitrary real normed targets, focused-build verified at compiled checkpoint `e109088ed`. The division-free forward form `normalizedVolterra_affine` holds for every real scale  $c$ , including  $c = 0$  and negative  $c$ , while the inverse form `normalizedVolterra_comp_affine` assumes  $c \neq 0$ . `normalized`



`Volterra_basepoint_shift` assumes interval integrability only on the local pieces from  $a$  to  $b$  and from  $b$  to  $x$ , imposes no order on  $a, b, x$ , and includes  $n = 0$ . Its zero-tail specialization `normalizedVolterra_succ_eq_taylor_of_eq_zero` retains only integrability from  $a$  to  $b$ , is for positive order, and assumes vanishing on the open unoriented interval between  $b$  and  $x$ ; it requires neither endpoint order nor endpoint values.

| Mathematical content                                                             | Exact Lean declaration and module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normalized kernel and literal-iteration bridge                                   | <code>normalizedVolterra</code> , <code>iteratedPrimitive</code> , and <code>iteratedPrimitive_eq_normalizedVolterra</code> in <code>FabiusFunction.NormalizedVolterra</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Positive-real fractional kernel, elementary algebra, and positive-integer bridge | <code>fractionalVolterra</code> , <code>fractionalVolterra_self</code> , <code>fractionalVolterra_congr</code> , <code>fractionalVolterra_smul</code> , <code>intervalIntegrable_fractionalVolterra_kernel</code> , <code>intervalIntegral_eq_integral_min_of_eq_zero</code> , <code>fractionalVolterra_eq_intervalIntegral_min_of_eq_zero</code> , <code>fractionalVolterra_add_input</code> , <code>fractionalVolterra_one</code> , and <code>fractionalVolterra_nat_succ</code> , together with <code>intervalIntegral_fractionalVolterra_betaKernel</code> , in <code>FabiusFunction.FractionalVolterra</code> .                                                                                |
| Positive-real beta convolution and additive composition of orders                | <code>intervalIntegrable_fractionalVolterra_betaKernel</code> , <code>intervalIntegral_fractionalVolterra_normalizedBetaKernel</code> , and <code>fractionalVolterra_add</code> in <code>FabiusFunction.FractionalVolterraSemigroup</code> ; the last theorem assumes an ordered endpoint and a continuous Banach-valued input.                                                                                                                                                                                                                                                                                                                                                                     |
| Positive affine covariance and derivative order raising                          | <code>fractionalVolterra_affine</code> , <code>fractionalVolterra_comp_affine</code> , <code>fractionalVolterra_add_one_deriv</code> , and <code>fractionalVolterra_add_one_deriv_of_eq_zero</code> in <code>FabiusFunction.FractionalVolterraCalculus</code> ; affine covariance uses a positive scale and arbitrary real order, while order raising assumes a positive order, a continuous Banach-valued primitive, a right interior derivative, and interval integrability of that derivative.                                                                                                                                                                                                   |
| Positive-real causal Rvachev primitives and Fabius–Rvachev fractional shifts     | <code>rvachevFractionalPrimitive</code> , <code>rvachevFractionalPrimitive_eq_intervalIntegral_min</code> , <code>rvachevFractionalPrimitive_nat_succ</code> , <code>rvachevFractionalPrimitive_add</code> , <code>fractionalVolterra_add_one_extendedFabius_of_nonneg</code> , <code>fractionalVolterra_add_one_fabiusReal</code> , and <code>fractionalVolterra_add_one_rvachevUp</code> in <code>FabiusFunction.FabiusFractionalVolterra</code> ; the definition and natural-order bridge are total, the cutoff and semigroup assume positive order and $x \geq -1$ , and the final three shift theorems have endpoint domains, respectively, $x \geq 0$ , $0 \leq x \leq 1$ , and $x \geq -1$ . |
| Affine covariance for all real scales, and its nonzero-scale inverse             | <code>normalizedVolterra_affine</code> and <code>normalizedVolterra_comp_affine</code> in <code>FabiusFunction.NormalizedVolterra</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Mathematical content                                                        | Exact Lean declaration and module                                                                                                                                                          |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local basepoint shift and positive-order open-interval zero-tail Taylor jet | <code>normalizedVolterra_basepoint_shift</code> and <code>normalizedVolterra_succ_eq_taylor_of_eq_zero</code> in <code>FabiusFunction.NormalizedVolterra</code> .                          |
| Kernel raising and arbitrary finite polynomial weights                      | <code>normalizedVolterra_kernel_pow</code> , <code>normalizedVolterra_polynomial</code> , and <code>normalizedVolterra_monomial</code> in <code>FabiusFunction.NormalizedVolterra</code> . |
| Signed Fabius primitive ladder, all orders and all real endpoints           | <code>normalizedVolterra_extendedFabius</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .                                                                                      |
| Bounded Fabius primitive ladder on the proved initial half-line $x \leq 1$  | <code>normalizedVolterra_fabiusReal_of_le_one</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .                                                                                |
| Finite natural-monomial Fabius formulas                                     | <code>normalizedVolterra_pow_mul_extendedFabius</code> and <code>normalizedVolterra_pow_mul_fabiusReal_of_le_one</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .             |

The generic polynomial theorem permits every order  $n \in \mathbb{N}_0$ , every real base point and endpoint, every interval-integrable input with values in an arbitrary real normed vector space, and an arbitrary real polynomial. Its coefficients are Hasse derivatives and rising factorials, so the order-zero identity and all endpoint degeneracies require no side conditions. The Fabius specializations likewise include order zero. For the bounded function the proved dyadic formula holds on  $(-\infty, 1]$ , not merely on  $[0, 1]$ .

The affine and basepoint theorems above are general operator identities. They do not themselves calculate explicit Fabius or Rvachev-up coefficients, and they do not assemble any of the report's piecewise formulas.

The generic positive-real Riemann–Liouville kernel, its ordered-endpoint continuous-Banach-input semigroup, positive affine covariance, derivative order raising, and the three real-positive Fabius–Rvachev shifts in the new table rows are now formalized. No claim of this update extends to the infinite complex-power Newton–Volterra series or its normal convergence, the Mellin/Newton and negative-power consequences, a semigroup for merely interval-integrable inputs or noncomplete targets, or a complex-order, shifted, reflected, or weighted hierarchy semigroup beyond the causal Rvachev specialization. The all-real-endpoint positive-order Fabius stopped-power and unit-endpoint moment identities in `FabiusFractionalIntegral.lean` and the three single-step shifts are formalized; no claim extends beyond them to iterated or negative-branch shifts, fractional derivatives, hierarchy-weighted fractional monomials, dyadic-lattice or endpoint-up wrappers, the  $x > 1$  branches of the bounded global piecewise formula, folded/exterior Rvachev primitives other than the exported causal primitive, explicit exterior tail expansions or Fourier renormalization, or the improper-singular, higher-order, and fractional inverse-Fabius primitives or inverse asymptotics developed below. The separate full-interval weighted inverse layer-cake theorem and complete and variable-upper proper-compact absolutely-continuous inverse-pair theorems recorded in the consolidated status box have since crossed the frontier; the stronger improper-singular, higher-order, fractional, and asymptotic claims just listed have not.

### 1.3 What “closed form” means here

The repository proves strong non-elementarity statements for  $F$ ,  $\text{up}$ , and  $Q$  [5, 2]. Consequently an antiderivative of  $x^p F(x)$  cannot generally be elementary on any interior interval: if it were, its derivative divided by  $x^p$  on a subinterval not containing zero would make  $F$  elementary there. The meaningful native closed class is therefore the algebra generated by polynomials, rational constants, and finitely many dyadically scaled Fabius values. The finite formula in [theorem 5.1](#) is closed in

precisely that sense.

## 2 Normalization and probabilistic infrastructure

### 2.1 The three Fabius-type functions

We follow the repository's careful distinction between three objects. The bounded distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

with

$$F(1-x) = 1 - F(x), \quad x \in \mathbb{R}, \quad (16)$$

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (17)$$

The compactly supported Rvachev function is

$$\text{up}(x) = F(1 - |x|), \quad (18)$$

and the signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{s_2(m)} \text{up}(x - 2m - 1), \quad \mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (19)$$

Here  $s_2(m)$  is the binary digit sum. The bounded and signed functions coincide on  $(-\infty, 1]$ , but not globally [1, 11].

Let  $(U_j)_{j \geq 1}$  be independent uniform random variables on  $[0, 1]$ , and put

$$X = \sum_{j \geq 1} 2^{-j} U_j. \quad (20)$$

Then  $F$  is the distribution function of  $X$ , and  $X \stackrel{d}{=} 1 - X$ . The density of  $Y = 2X - 1$  is  $\text{up}$ . We use the moments

$$d_n = \mathbb{E}X^n, \quad c_n = \mathbb{E}(2X - 1)^{2n} = \int_{-1}^1 x^{2n} \text{up}(x) \, dx. \quad (21)$$

The repository establishes

$$d_0 = 1, \quad (n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (22)$$

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}), \quad (23)$$

$$c_n = \frac{2d_{2n+1}}{2n+1}. \quad (24)$$

The moment generating function

$$G(z) = \mathbb{E}e^{zX} = \sum_{n \geq 0} d_n \frac{z^n}{n!} = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j} \quad (25)$$

satisfies

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad G(-z) = e^{-z} G(z). \quad (26)$$

At imaginary arguments, the centered transform  $e^{-z} G(2z)$  becomes the familiar infinite sinc product for  $\text{up}$  [10, 9, 1].

## 2.2 Normalized integer and fractional primitives

For an integer  $n \geq 1$ , a real base point  $a$ , and an oriented endpoint  $x$ , define the normalized Volterra primitive

$$(\mathcal{J}_{n,a}f)(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt, \quad \mathcal{J}_{0,a}f = f. \quad (27)$$

The zero-based notation is  $\mathcal{J}_n = \mathcal{J}_{n,0}$ . For positive  $n$ , it is the unique solution of  $D^n u = f$  whose derivatives of orders  $0, \dots, n-1$  vanish at  $a$ . Every other  $n$ -fold antiderivative differs from it by a polynomial of degree at most  $n-1$ . For continuous Banach-valued inputs, the formal theorem `iteratedPrimitive_eq_normalizedVolterra` proves that (27) is exactly literal repeated integration.

For  $\operatorname{Re} \nu > 0$ , the Riemann–Liouville version is

$$(\mathcal{I}^\nu f)(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} f(t) dt. \quad (28)$$

For a real  $\nu > 0$  and  $x \geq 0$ , this is the zero-based specialization of `fractionalVolterra` in `FabiusFunction.FractionalVolterra`. That module also proves its elementary endpoint, locality, scalar, continuous-input additivity, order-one, positive-natural-order, and compact-right-support cutoff rules recorded in the consolidated status box. The companion `FabiusFunction.FractionalVolterraSemigroup` proves additive composition of two positive real orders for a continuous Banach-valued input on an ordered compact interval. `FabiusFunction.FractionalVolterraCalculus` additionally proves positive affine covariance and the Gamma-normalized one-step order-raising integration-by-parts law. Its specializations in `FabiusFunction.FabiusFractionalVolterra` define the causal primitive  $U_\beta = I_{(-1)+}^\beta$  up, identify its support-truncated integral, prove its positive-natural bridge and positive-order semigroup, and give  $I_{0+}^{\nu+1} F(x) = 2^\nu I_{0+}^\nu F(x/2)$  on  $[0, 1]$ , the signed version for  $x \geq 0$ , and the corresponding Up-to-Fabius bridge from base  $-1$ . These modules do not formalize the complex-order reading of this display, the shifted dyadic-lattice, endpoint-moment, transform/tail, or complex-power/weighted formulas below, or fractional derivative and inverse/quantile operators. The notation  $\mathcal{J}_n = \mathcal{I}^n$  will be used when the order is positive and integral. For a complex exponent  $\alpha$  and positive  $x$ , we take  $x^\alpha = \exp(\alpha \log x)$  with the real logarithm.

## 3 The dyadic antiderivative ladder

**Theorem 3.1** (Exact normalized primitive ladder). *For every  $n \in \mathbb{N}_0$  and every real  $x \leq 1$ ,*

$$(\mathcal{J}_n F)(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right). \quad (29)$$

*For the signed global extension, the same formula holds for every real  $x$ :*

$$(\mathcal{J}_n \mathcal{F})(x) = 2^{\binom{n}{2}} \mathcal{F}\left(\frac{x}{2^n}\right). \quad (30)$$

*The order-zero cases use the identity convention in (27).*

*Proof.* The case  $n = 0$  is immediate. For the signed extension set  $H_n(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n)$ . The global dilation equation gives

$$H'_n(x) = 2^{\binom{n}{2}-n} \mathcal{F}'(x/2^n) = 2^{\binom{n}{2}-n+1} \mathcal{F}(x/2^{n-1}) = H_{n-1}(x).$$

Iterating yields  $H_n^{(n)} = \mathcal{F}$ . Every derivative of  $H_n$  below order  $n$  is a constant multiple of a scaled value of  $\mathcal{F}$ , hence vanishes at zero because  $\mathcal{F}(0) = 0$ . Uniqueness with its zero initial jet proves (30). If  $x \leq 1$ , the oriented interval between zero and  $x$  lies in  $(-\infty, 1]$ , where  $F = \mathcal{F}$ ; moreover  $x/2^n \leq 1$ . Replacing  $\mathcal{F}$  by  $F$  on both sides therefore proves (29) on the stated full range.  $\square$

*Remark 3.2.* The range restriction  $x \leq 1$  in (29) is the full interval on which the bounded function is obtained directly from the signed extension. It is  $\mathcal{F}$ , not the bounded CDF, that obeys the clean dilation equation on all of  $\mathbb{R}$ . The proposed global bounded completion is instead piecewise polynomial; see [theorem 5.1](#) and its frontier-status note.

**Corollary 3.3** (Moment interpretation of the ladder). *For  $n \geq 1$  and  $x \geq 0$ ,*

$$(\mathcal{J}_n F)(x) = \frac{1}{n!} \mathbb{E}(x - X)_+^n. \quad (31)$$

*In particular,*

$$(\mathcal{J}_n F)(1) = \frac{d_n}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \quad (32)$$

*Proof.* Write  $F(t) = \mathbb{E}1_{\{X \leq t\}}$ , apply Tonelli’s theorem, and integrate  $(x - t)^{n-1}$  from  $t = X$  to  $t = x$ . At  $x = 1$ , symmetry gives  $\mathbb{E}(1 - X)^n = \mathbb{E}X^n = d_n$ .  $\square$

## 4 The Newton–Volterra expansion for arbitrary powers

### 4.1 A general commutator identity

The key algebraic mechanism does not depend on the Fabius function.

**Theorem 4.1** (Formalized polynomial Volterra commutator). *Let  $E$  be a real normed vector space, let  $a, x \in \mathbb{R}$ , let  $f : \mathbb{R} \rightarrow E$  be interval integrable between  $a$  and  $x$ , and let  $P \in \mathbb{R}[X]$ . For every  $n \in \mathbb{N}_0$ ,*

$$\mathcal{J}_{n,a}[P(t)f(t)](x) = \sum_{r=0}^{\deg_N P} (-1)^r n^{\bar{r}} P^{[r]}(x) \mathcal{J}_{n+r,a}f(x). \quad (33)$$

Here  $P^{[r]}$  is the  $r$ -th Hasse derivative,  $\deg_N$  is the natural degree, and  $n^{\bar{r}}$  is the rising factorial, with  $0^{\bar{0}} = 1$  and  $0^{\bar{r}} = 0$  for  $r > 0$ . Consequently (33) includes the order-zero identity without a separate exception.

*Proof.* Taylor’s finite polynomial identity about the upper endpoint is

$$P(t) = \sum_{r=0}^{\deg_N P} P^{[r]}(x)(t-x)^r = \sum_{r=0}^{\deg_N P} (-1)^r P^{[r]}(x)(x-t)^r.$$

Linearity reduces the claim to the kernel-raising identity

$$\mathcal{J}_{n,a}[(x-t)^r f(t)](x) = n^{\bar{r}} \mathcal{J}_{n+r,a}f(x).$$

For positive  $n$ , this follows by combining the powers of  $x - t$  and the factorial identity  $(n-1)!n^{\bar{r}} = (n+r-1)!$ . For  $n = 0$ , evaluation at  $t = x$  agrees with the stated zero rising-factorial convention. This proves all orders at once.  $\square$

This theorem is exactly `normalizedVolterra_polynomial`; its monomial specialization is `normalizedVolterra_monomial`, both in `FabiusFunction.NormalizedVolterra`. Taking  $P(t) = t^p$  gives  $P^{[r]}(x) = \binom{p}{r} x^{p-r}$ , so for  $n \geq 1$  it is equivalent to the finite part of the next theorem.

**Theorem 4.2** (Newton–Volterra commutator). *Let  $n \geq 1$ , let  $x > 0$ , and suppose the displayed integrals converge. Then for every complex  $\alpha$ ,*

$$\mathcal{J}_n[t^\alpha f(t)](x) = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} \alpha^r x^{\alpha-r} (\mathcal{J}_{n+r}f)(x), \quad (34)$$

whenever termwise integration is justified. Independently of this positivity condition, if  $\alpha = p \in \mathbb{N}_0$ , the series terminates at  $r = p$  and is an algebraic identity for every locally integrable  $f$  and every real oriented endpoint  $x$ :

$$\mathcal{J}_n[t^p f(t)](x) = \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} x^{p-r} (\mathcal{J}_{n+r} f)(x). \quad (35)$$

*Proof.* Inside the Volterra kernel, expand about the upper endpoint:

$$t^\alpha = x^\alpha \left(1 - \frac{x-t}{x}\right)^\alpha = \sum_{r \geq 0} (-1)^r \frac{\alpha^{\underline{r}}}{r!} x^{\alpha-r} (x-t)^r.$$

After integration, the factor multiplying  $\mathcal{J}_{n+r} f$  is

$$\frac{\Gamma(n+r)}{\Gamma(n)r!} = \binom{n+r-1}{r}.$$

For  $p \in \mathbb{N}_0$ , the binomial expansion is finite, so no convergence argument is needed.  $\square$

#### Formalization boundary for the commutator

The finite polynomial identity (33), including its natural-monomial specialization (35), is formalized in the greater normed-space-valued, arbitrary-base, all-order generality stated above. The infinite expansion (34) for nonintegral or complex  $\alpha$ , including the justification of termwise integration, is not yet formalized.

For  $p \geq 1$ , the first two operator recurrences encoded in (35) are

$$\mathcal{J}_n[t f(t)] = x \mathcal{J}_n f - n \mathcal{J}_{n+1} f, \quad (36)$$

$$\mathcal{J}_n[t^p f(t)] = x \mathcal{J}_n[t^{p-1} f(t)] - n \mathcal{J}_{n+1}[t^{p-1} f(t)]. \quad (37)$$

They provide a convenient triangular implementation without symbolic integration.

## 4.2 The complete Fabius formula

**Theorem 4.3** (Arbitrary-complex-power Fabius primitive). *Let  $n \geq 1$ ,  $0 < x \leq 1$ , and  $\alpha \in \mathbb{C}$ . Then the integral  $\mathcal{J}_n[t^\alpha F(t)](x)$  exists, is entire as a function of  $\alpha$ , and has the absolutely and locally uniformly convergent expansion*

$$\mathcal{J}_n[t^\alpha F(t)](x) = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} \alpha^{\underline{r}} 2^{\binom{n+r}{2}} x^{\alpha-r} F\left(\frac{x}{2^{n+r}}\right). \quad (38)$$

For  $\alpha = p \in \mathbb{N}_0$ , this terminates and becomes

$$\mathcal{J}_n[t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} 2^{\binom{n+r}{2}} x^{p-r} F\left(\frac{x}{2^{n+r}}\right). \quad (39)$$

The natural-power specialization has the stronger all-order form: for every  $n, p \in \mathbb{N}_0$  and every real  $x \leq 1$ ,

$$\mathcal{J}_n[t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{p}{r} n^{\bar{r}} x^{p-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (40)$$

For positive  $n$ , this is (39); the rising-factorial form also handles  $n = 0$  without an exceptional sum.

*Proof.* Substitution of (29) into (34) gives the formal expression. It remains to justify convergence for all  $\alpha$ .

Infinite flatness implies that for every integer  $M \geq 0$  there is a finite constant  $C_M$  such that  $F(t) \leq C_M t^M$  on  $[0, 1]$ . Hence, for every integer  $m \geq 1$ ,

$$0 \leq (\mathcal{J}_m F)(x) \leq \frac{C_M M!}{(m+M)!} x^{m+M}. \quad (41)$$

Indeed, this is the beta integral obtained by replacing  $F(t)$  with  $C_M t^M$ . The absolute value of the  $r$ -th term of (34), with  $m = n + r$ , is therefore at most

$$C_M M! x^{\operatorname{Re} \alpha + n + M} \binom{n+r-1}{r} \frac{|\alpha^r|}{(n+r+M)!}. \quad (42)$$

On a compact set of  $\alpha$ 's, the falling factorial is bounded by  $C r! r^B$  for some fixed  $B$ . Since  $\binom{n+r-1}{r} = O(r^{n-1})$ , choosing  $M > B + 1$  makes (42) summable uniformly on that compact set. This proves absolute local uniform convergence. The same flatness bound, with logarithmic powers inserted, permits differentiation under the defining integral and shows that it is entire in  $\alpha$ . The series and integral agree first in a half-plane where the ordinary binomial argument is immediate and then everywhere by the identity theorem; equivalently, one may use dominated convergence directly with (42).

For natural  $p$ , apply the finite monomial commutator (33) and then the bounded ladder (29). This argument has no convergence step and remains valid for every oriented endpoint  $x \leq 1$  and for order zero, proving (40).  $\square$

#### Formalization boundary for the Fabius formula

The finite identities (39) and (40) are formalized as `normalizedVolterra_pow_mul_fabiusReal_of_le_one`; the signed global version is `normalizedVolterra_pow_mul_extendedFabius`, both in `FabiusFunction.FabiusAntiderivatives`. The arbitrary-complex-power series (38), its entirety assertion, and its normal convergence are still non-formalized.

#### Main closed form for the ordinary antiderivative

For every real  $x \leq 1$  and  $p \in \mathbb{N}_0$ , the normalized ordinary primitive is

$$\int_0^x t^p F(t) dt = \sum_{r=0}^p (-1)^r \frac{p!}{(p-r)!} 2^{\binom{r+1}{2}} x^{p-r} F\left(\frac{x}{2^{r+1}}\right). \quad (43)$$

Thus only  $p + 1$  dyadically contracted Fabius evaluations are required.

A useful one-step recurrence for  $p \geq 1$  and  $x \leq 1$  follows by integrating by parts once and changing variables:

$$A_p(x) := \int_0^x t^p F(t) dt = x^p F(x/2) - p 2^p A_{p-1}(x/2), \quad A_0(x) = F(x/2). \quad (44)$$

Iteration of (44) is exactly (43).

### 4.3 Remainders and the negative-power series

For the ordinary primitive  $n = 1$ , repeated integration by parts gives an especially transparent exact remainder. For every integer  $N \geq 0$ ,

$$\int_0^x t^\alpha F(t) dt = \sum_{r=0}^N (-1)^r \alpha^r x^{\alpha-r} (\mathcal{J}_{r+1} F)(x) + R_N(\alpha, x), \quad (45)$$



$$R_N(\alpha, x) = (-1)^{N+1} \alpha^{N+1} \int_0^x t^{\alpha-N-1} (\mathcal{J}_{N+1} F)(t) dt. \quad (46)$$

If  $M > -\operatorname{Re} \alpha - 1$ , (41) yields

$$|R_N(\alpha, x)| \leq \frac{|\alpha^{N+1}| C_M M!}{(N+M+1)! (\operatorname{Re} \alpha + M + 1)} x^{\operatorname{Re} \alpha + M + 1}. \quad (47)$$

This estimate is useful even when direct evaluation of the extremely small Fabius terms would underflow.

For a negative integer exponent  $\alpha = -m$ ,  $m \geq 1$ , the signs cancel and every term is positive:

$$\int_0^x t^{-m} F(t) dt = \sum_{r=0}^{\infty} \frac{(m+r-1)!}{(m-1)!} 2^{\binom{r+1}{2}} x^{-m-r} F\left(\frac{x}{2^{r+1}}\right). \quad (48)$$

No principal value or regularization is involved: the flat zero of  $F$  makes  $x^{-m} F(x)$  integrable for every  $m$ . This behavior is sharply different from  $x^{-m} \operatorname{up}(x)$ , which is singular at the origin because  $\operatorname{up}(0) = 1$ .

**Corollary 4.4** (Endpoint localization). *Fix  $n \geq 1$  and  $\alpha \in \mathbb{C}$  with  $\operatorname{Re} \alpha \geq 0$ . As  $x \downarrow 0$ ,*

$$\mathcal{J}_n[t^\alpha F(t)](x) \sim x^\alpha (\mathcal{J}_n F)(x) = x^\alpha 2^{\binom{n}{2}} F(x/2^n). \quad (49)$$

*Proof.* The sharp endpoint theory in the audited corpus implies rapid variation: for every fixed  $0 < c < 1$ ,  $F(cx)/F(x) \rightarrow 0$  as  $x \downarrow 0$ . Normalize the nonnegative measure  $(x-t)^{n-1} F(t) dt$  on  $[0, x]$ . Its mass below  $(1-\delta)x$  is negligible relative to its total mass, because comparison with fixed dilates of  $F$  shows concentration in  $[(1-\delta)x, x]$ . On that interval  $(t/x)^\alpha \rightarrow 1$  uniformly as  $\delta \downarrow 0$ . Moreover,  $|(t/x)^\alpha| = (t/x)^{\operatorname{Re} \alpha} \leq 1$  on  $(0, x]$ , so the complementary part contributes at most twice its vanishing normalized mass. The quotient of the two sides of (49) is the expectation of  $(t/x)^\alpha$  under this normalized measure, hence tends to one.  $\square$

For  $\operatorname{Re} \alpha < 0$ , the same localization is expected and is consistent with the exact Newton–Volterra series, but the preceding concentration argument alone does not prove it:  $(t/x)^\alpha$  is then unbounded near zero. Extending (49) to that half-plane requires a separate uniform-integrability estimate from quantitative flatness (or from the sharp endpoint expansion), and no such extension is asserted here.

The first correction displayed by (38) is

$$\frac{\mathcal{J}_n[t^\alpha F(t)](x)}{x^\alpha \mathcal{J}_n F(x)} = 1 - n\alpha \frac{\mathcal{J}_{n+1} F(x)}{x \mathcal{J}_n F(x)} + \dots \quad (50)$$

The quotient on the right is governed by the repository’s Lambert phase and periodic saddle fluctuation; extracting its complete expansion is posed as a frontier problem in [section 16](#).

## 5 Global closed form for natural powers

For integer powers the bounded extension of  $F$  permits a complete all-real answer. Define

$$H_{n,p}(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^p F(t) dt, \quad (51)$$

with the usual oriented integral when  $x < 0$ .

**Formalization frontier**

The finite dyadic branch for every  $x \leq 1$ , including the zero branch on the negative half-line, follows from the formalized theorem `normalizedVolterra_pow_mul_fabiusReal_of_le_one`. The exterior moment-polynomial branch for  $x > 1$ , the resulting all-real piecewise assembly, and the matching-jet assertion at  $x = 1$  have not yet been formalized. Thus [theorem 5.1](#) remains in this frontier report.

**Theorem 5.1** (Global piecewise formula). *For  $n \geq 1$  and  $p \in \mathbb{N}_0$ ,*

$$H_{n,p}(x) = \begin{cases} 0, & x \leq 0, \\ \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} 2^{\binom{n+r}{2}} x^{p-r} F\left(\frac{x}{2^{n+r}}\right), & 0 \leq x \leq 1, \\ \frac{p!}{(p+n)!} x^{p+n} - \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} \frac{d_{p+j+1}}{p+j+1}, & x \geq 1. \end{cases} \quad (52)$$

*The pieces have matching derivatives through order  $n-1$  at zero and one.*

*Proof.* The first two cases follow from the support convention and [theorem 4.3](#). For  $x \geq 1$ , write

$$H_{n,p}(x) = \mathcal{J}_n[t^p](x) - \frac{1}{(n-1)!} \int_0^1 (x-t)^{n-1} t^p (1-F(t)) dt.$$

The first term is  $p!x^{p+n}/(p+n)!$ . Expand the finite polynomial kernel and use

$$\int_0^1 t^q (1-F(t)) dt = \frac{d_{q+1}}{q+1}, \quad (53)$$

which is the standard tail integral for  $\mathbb{E}X^{q+1}$ . Since the formula was derived from a single Volterra integral, matching jets are automatic.  $\square$

For the ordinary primitive, the last branch simplifies to

$$A_p(x) = \frac{x^{p+1} - d_{p+1}}{p+1}, \quad x \geq 1. \quad (54)$$

This makes clear that the normalized antiderivative is constant-polynomial outside the transition interval, exactly as repeated integration of a compact transition requires.

## 6 Definite integrals, Newton interpolation, and moment identities

### 6.1 The entire Mellin moment

Endpoint flatness makes

$$M(s) = \mathbb{E}X^s = \int_0^1 x^s dF(x) \quad (55)$$

an entire function of  $s \in \mathbb{C}$ . Integration by parts, initially in a right half-plane and then by analytic continuation, gives

$$\mathcal{A}(\alpha) := \int_0^1 x^\alpha F(x) dx = \frac{1 - M(\alpha+1)}{\alpha+1}. \quad (56)$$

The apparent singularity at  $\alpha = -1$  is removable.

Putting  $x = 1$  in [\(38\)](#) and using [\(32\)](#) gives the central Newton series.

**Theorem 6.1** (Moment Newton series). *For every  $\alpha \in \mathbb{C}$ ,*

$$\mathcal{A}(\alpha) = \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} \frac{d_{r+1}}{r+1}. \quad (57)$$

*More generally, for every integer  $n \geq 1$ ,*

$$\mathcal{J}_n[t^\alpha F(t)](1) = \frac{1}{(n-1)!} \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} \frac{d_{n+r}}{n+r}. \quad (58)$$

*Both series converge absolutely and locally uniformly in  $\alpha$ .*

The terminology is literal. If  $\Delta h(z) = h(z+1) - h(z)$ , then

$$\Delta^r \mathcal{A}(0) = \int_0^1 (x-1)^r F(x) dx = (-1)^r \frac{d_{r+1}}{r+1}. \quad (59)$$

Thus (57) is the Newton interpolation series of the entire function  $\mathcal{A}$  from its nonnegative integral values.

For  $p \in \mathbb{N}_0$ , the series terminates and combines with (56) to give

$$\int_0^1 x^p F(x) dx = \frac{1 - d_{p+1}}{p+1}. \quad (60)$$

Together with (23), this yields the exact finite dyadic identity

$$\sum_{r=0}^p (-1)^r \frac{p!}{(p-r)!} 2^{\binom{r+1}{2}} F(2^{-r-1}) = \frac{1 - d_{p+1}}{p+1}. \quad (61)$$

Every value in (61) is rational and may be replaced by the repository's explicit  $q$ -binomial or integer-sequence formulas.

## 6.2 A two-parameter finite-difference identity

Expanding  $(1-t)^{n-1}$  in the defining integral at  $x = 1$  produces an independent form. Equating it with (58) gives the following exact identity.

**Corollary 6.2** (Two-parameter moment identity). *For  $n \geq 1$  and  $p \in \mathbb{N}_0$ ,*

$$\sum_{r=0}^p (-1)^r \binom{p}{r} \frac{d_{n+r}}{n+r} = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{1 - d_{p+j+1}}{p+j+1}. \quad (62)$$

*More generally, for every  $\alpha \in \mathbb{C}$ , the finite left-hand side is replaced by its convergent Newton series and  $d_{p+j+1}$  by  $M(\alpha + j + 1)$ :*

$$\sum_{r \geq 0} (-1)^r \binom{\alpha}{r} \frac{d_{n+r}}{n+r} = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{1 - M(\alpha + j + 1)}{\alpha + j + 1}, \quad (63)$$

*where removable values are understood.*

### 6.3 Negative powers and logarithmic constants

For  $m \geq 1$ , (57) becomes the positive series

$$\int_0^1 x^{-m} F(x) dx = \sum_{n \geq 1} \frac{(m+n-2)!}{(m-1)! n!} d_n. \quad (64)$$

For  $m > 1$ , the Mellin form is

$$\int_0^1 x^{-m} F(x) dx = \frac{M(1-m) - 1}{m-1}. \quad (65)$$

At the removable case  $m = 1$ ,

$$\int_0^1 \frac{F(x)}{x} dx = -M'(0) = -\mathbb{E} \log X = \sum_{n \geq 1} \frac{d_n}{n}. \quad (66)$$

The centered moments give a second positive expansion. Since

$$M(s) = 2^{-s} \sum_{q \geq 0} \binom{s}{2q} c_q, \quad (67)$$

and  $\left. \frac{d}{ds} \binom{s}{2q} \right|_{s=0} = -1/(2q)$ , one obtains

$$\int_0^1 \frac{F(x)}{x} dx = \log 2 + \frac{1}{2} \sum_{q \geq 1} \frac{c_q}{q}. \quad (68)$$

Numerically, the accompanying code gives

$$\int_0^1 \frac{F(x)}{x} dx = 0.75824000040440865 \dots \quad (69)$$

## 7 Exponential master functions and the sinc product

The entire family of integer-power primitives is encoded by an incomplete Laplace transform. Applying [theorem 4.2](#) to  $e^{zt}$ , or simply expanding  $e^{-z(x-t)}$  in the Volterra kernel, gives the following.

**Theorem 7.1** (Exponential Volterra master). *For  $n \geq 1$ ,  $0 \leq x \leq 1$ , and  $z \in \mathbb{C}$ ,*

$$\mathcal{J}_n[e^{zt} F(t)](x) = e^{zx} \sum_{r=0}^{\infty} (-z)^r \binom{n+r-1}{r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (70)$$

The series is entire in  $z$ . At ordinary order  $n = 1$ ,

$$\int_0^x e^{zt} F(t) dt = e^{zx} \sum_{m \geq 1} (-z)^{m-1} 2^{\binom{m}{2}} F(x/2^m). \quad (71)$$

At  $x = 1$ ,

$$\int_0^1 e^{zt} F(t) dt = \frac{e^z - G(z)}{z} = e^z \frac{1 - G(-z)}{z}, \quad (72)$$

with the removable value  $1/2$  at  $z = 0$ .

Taking  $p$  derivatives with respect to  $z$  at zero recovers (39). Thus (70) is a compact generating object for all natural monomial weights and all integral antiderivative orders.

For the up-function, define the incomplete transform

$$U(z, x) = \int_{-1}^x e^{zt} \text{up}(t) dt. \quad (73)$$

On the left half, translation gives the particularly simple dyadic series

$$U(z, x) = e^{zx} \sum_{m \geq 1} (-z)^{m-1} 2^{\binom{m}{2}} F\left(\frac{x+1}{2^m}\right), \quad -1 \leq x \leq 0. \quad (74)$$

At the right endpoint,

$$\int_{-1}^1 e^{zt} \text{up}(t) dt = e^{-z} G(2z) = \frac{G(z) - G(-z)}{z}. \quad (75)$$

For  $z = 2\pi i\xi$ , this is precisely the infinite sinc product, up to the chosen Fourier normalization. Consequently all compact up-moments and their incomplete analogues arise by differentiating one dyadic Fabius series whose complete endpoint is the sinc product.

## 8 Bell and Bernoulli polynomial organization

Taking logarithms of (25) and using the classical Bernoulli expansion gives

$$\log G(z) = \frac{z}{2} + \sum_{q \geq 1} \frac{B_{2q}}{2q(2q)!(2^{2q} - 1)} z^{2q}. \quad (76)$$

Hence the cumulants of  $X$  are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2q} = \frac{B_{2q}}{2q(2^{2q} - 1)}, \quad \kappa_{2q+1} = 0 \quad (q \geq 1). \quad (77)$$

Writing  $Y_n$  for the complete exponential Bell polynomial,

$$d_n = Y_n(\kappa_1, \dots, \kappa_n). \quad (78)$$

Therefore every definite natural-power primitive has the explicit Bell–Bernoulli form

$$\int_0^1 x^p F(x) dx = \frac{1 - Y_{p+1}(\kappa_1, \dots, \kappa_{p+1})}{p+1}. \quad (79)$$

Substitution in (52) similarly expresses every exterior polynomial coefficient through Bernoulli numbers and Bell polynomials. This provides a direct bridge between the antiderivative problem and the Bell–Bernoulli infrastructure already emphasized in the repository.

## 9 Reflection transforms: $F(1 - x)$ and $1 - F(x)$

The global reflection identity (16) means that the two transformations requested in the problem are the same function:

$$F(1 - x) = 1 - F(x), \quad x \in \mathbb{R}. \quad (80)$$

For natural powers and every integer  $n \geq 1$ , linearity gives the complete normalized formula

$$\boxed{\mathcal{J}_n[x^p(1 - F(x))] = \frac{p!}{(p+n)!} x^{p+n} - H_{n,p}(x),} \quad (81)$$

where  $H_{n,p}$  is the global piecewise expression (52). The same formula therefore applies verbatim to  $x^p F(1-x)$ .

For arbitrary complex  $\alpha$  on  $x > 0$ , an ordinary antiderivative is

$$\int x^\alpha (1 - F(x)) dx = \begin{cases} \frac{x^{\alpha+1}}{\alpha+1} - \int_0^x t^\alpha F(t) dt + C, & \alpha \neq -1, \\ \log x - \int_0^x \frac{F(t)}{t} dt + C, & \alpha = -1, \end{cases} \quad (82)$$

where the Fabius term is evaluated by (38). Unlike the Fabius integral, a primitive normalized at zero exists only when  $\operatorname{Re} \alpha > -1$ , because  $1 - F(0) = 1$ . Formula (82) remains a valid local antiderivative on every positive interval avoiding zero for all  $\alpha$ .

At the complete endpoint, for  $\operatorname{Re} \alpha > -1$ ,

$$\int_0^1 x^\alpha (1 - F(x)) dx = \frac{M(\alpha+1)}{\alpha+1}. \quad (83)$$

Together, (56) and (83) split the elementary beta integral  $1/(\alpha+1)$  into the lower-CDF and upper-tail contributions.

## 10 Rvachev up-function: complete weighted primitives

### 10.1 First-order formulas and the singularity at zero

On  $0 \leq x \leq 1$ ,  $\operatorname{up}(x) = 1 - F(x)$ , so for  $\operatorname{Re} \alpha > -1$ ,

$$\int_0^x t^\alpha \operatorname{up}(t) dt = \frac{x^{\alpha+1}}{\alpha+1} - \sum_{r \geq 0} (-1)^r \alpha^r 2^{\binom{r+1}{2}} x^{\alpha-r} F\left(\frac{x}{2^{r+1}}\right). \quad (84)$$

The sum is the convergent Fabius series from (38). For  $\alpha = -1$ , replace the elementary term by  $\log x$ , obtaining a local primitive but not a convergent integral from zero.

This exposes an important structural contrast:

- $x^\alpha F(x)$  is integrable at zero for every  $\alpha \in \mathbb{C}$ , because  $F$  is infinitely flat;
- $x^\alpha \operatorname{up}(x)$  has the ordinary monomial threshold  $\operatorname{Re} \alpha > -1$ , because  $\operatorname{up}(0) = 1$ .

For a chosen upper-half-plane branch on the negative axis,  $(-u)^\alpha = e^{\pi i \alpha} u^\alpha$ , evenness gives, whenever the integrals converge,

$$\int_{-1}^x t^\alpha \operatorname{up}(t) dt = e^{\pi i \alpha} \left[ \int_0^1 u^\alpha \operatorname{up}(u) du - \int_0^{-x} u^\alpha \operatorname{up}(u) du \right], \quad -1 \leq x \leq 0. \quad (85)$$

For integer powers this becomes real and simplifies to finite Fabius sums.

### 10.2 All natural powers and all integer orders

Define the primitive based at the left support endpoint,

$$U_{n,p}(x) = \frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} t^p \operatorname{up}(t) dt, \quad n \geq 1, p \in \mathbb{N}_0. \quad (86)$$

Let  $H_{n,j}$  be as in (51), and define the full up-moments

$$m_k^{\operatorname{up}} = \int_{-1}^1 t^k \operatorname{up}(t) dt = \begin{cases} 0, & k \text{ odd}, \\ c_{k/2}, & k \text{ even}. \end{cases} \quad (87)$$

**Theorem 10.1** (Global weighted up-primitive). For  $n \geq 1$  and  $p \in \mathbb{N}_0$ ,

$$U_{n,p}(x) = \begin{cases} 0, & x \leq -1, \\ \sum_{j=0}^p \binom{p}{j} (-1)^{p-j} H_{n,j}(x+1), & -1 \leq x \leq 0, \\ \frac{(-1)^p}{(n-1)!} \sum_{j=0}^{n-1} \binom{n-1}{j} x^{n-1-j} \frac{d_{p+j+1}}{p+j+1} + \frac{p!}{(p+n)!} x^{p+n} - H_{n,p}(x), & 0 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} m_{p+j}^{\text{up}}, & x \geq 1. \end{cases} \quad (88)$$

Every branch is finite, and the interior branches reduce further to the scaled-Fabius formula (39).

*Proof.* For  $-1 \leq x \leq 0$ , set  $s = t + 1$ . Then  $\text{up}(t) = F(s)$ ,  $t^p = (s-1)^p$ , and the first finite sum follows. For  $0 \leq x \leq 1$ , split the integral at zero. The negative half is

$$\frac{(-1)^p}{(n-1)!} \int_0^1 (x+1-s)^{n-1} (1-s)^p F(s) ds.$$

Expand the polynomial in  $x$  and use

$$\int_0^1 (1-s)^q F(s) ds = \frac{d_{q+1}}{q+1},$$

which follows from [corollary 3.3](#). The positive half equals  $\mathcal{J}_n[t^p] - H_{n,p}$ . For  $x \geq 1$ , the full support has been crossed; expand  $(x-t)^{n-1}$  and insert the moments (87).  $\square$

The unweighted case is worth displaying separately. Put  $U_n = U_{n,0}$ . Then

$$U_n(x) = \begin{cases} 0, & x \leq -1, \\ 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), & -1 \leq x \leq 0, \\ \frac{1}{(n-1)!} \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{d_{j+1}}{j+1} x^{n-1-j} + \frac{x^n}{n!} - 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right), & 0 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{q=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2q} c_q x^{n-1-2q}, & x \geq 1. \end{cases} \quad (89)$$

For  $n = 1$ , this is the cumulative distribution function of  $Y = 2X - 1$ . For larger  $n$ , its right tail is the moment polynomial of the compact up density.

## 11 Exact higher derivatives and inverse differentiation

### 11.1 Thue–Morse derivative formulas

Let  $\varepsilon_k = (-1)^{s_2(k)}$ . Iterating Rvachev's functional-differential equation gives the standard exact derivative formula

$$\boxed{\text{up}^{(n)}(x) = 2^{\binom{n+1}{2}} \sum_{k=0}^{2^n-1} \varepsilon_k \text{up}(2^n x + 2^n - 1 - 2k).} \quad (90)$$



Using  $F'(x) = 2 \operatorname{up}(2x - 1)$ , one obtains, for  $m \geq 1$ ,

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} \sum_{k=0}^{2^{m-1}-1} \varepsilon_k \operatorname{up}(2^m x - 1 - 2k). \quad (91)$$

This version is valid globally for the bounded  $F$ . On the shrinking left cell  $0 < x < 2^{-m}$ , it collapses to

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} F(2^m x), \quad (92)$$

while the signed extension satisfies the same dilation formula on all of  $\mathbb{R}$ :

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (93)$$

Leibniz's rule therefore gives a closed formula for the requested derivatives. For  $p \in \mathbb{N}_0$ ,

$$D^n[x^p \mathcal{F}(x)] = \sum_{j=0}^{\min(n,p)} \binom{n}{j} \frac{p!}{(p-j)!} x^{p-j} 2^{\binom{n-j+1}{2}} \mathcal{F}(2^{n-j} x). \quad (94)$$

For arbitrary  $\alpha \in \mathbb{C}$  and  $x > 0$ , replace  $p!/(p-j)!$  by  $\alpha^j$  and sum through  $j = n$ . For the bounded function, use

$$D^n[x^\alpha F(x)] = \sum_{j=0}^n \binom{n}{j} \alpha^j x^{\alpha-j} F^{(n-j)}(x), \quad (95)$$

with (91). An identical Leibniz formula combined with (90) handles  $x^\alpha \operatorname{up}(x)$ .

## 11.2 The derivative/antiderivative dictionary

The formulas may be summarized as follows.

- Positive derivative order introduces a finite Thue–Morse superposition of dilated up-functions, or a single dilation for  $\mathcal{F}$ .
- Negative integral order on  $[0, 1]$  introduces a single contracted Fabius value, multiplied by  $2^{\binom{n}{2}}$ .
- Multiplication by a natural monomial turns either operation into a finite triangular sum; a nonintegral monomial changes the antiderivative sum from finite to absolutely convergent.

This is the exact operator closure that replaces an elementary integration table for the Fabius family.

## 12 Fractional antiderivative order

The integer ladder has a natural Riemann–Liouville continuation, although the single-dilation collapse is special to integral order.

**Proposition 12.1** (Probabilistic fractional primitive). *For  $\operatorname{Re} \nu > 0$  and  $x \geq 0$ ,*

$$(\mathcal{I}^\nu F)(x) = \frac{1}{\Gamma(\nu+1)} \mathbb{E}(x - X)_+^\nu. \quad (96)$$

At  $x = 1$ , symmetry gives

$$(\mathcal{I}^\nu F)(1) = \frac{M(\nu)}{\Gamma(\nu+1)}. \quad (97)$$

For  $x > 1$ ,

$$(\mathcal{I}^\nu F)(x) = \frac{x^\nu}{\Gamma(\nu+1)} \sum_{k \geq 0} (-1)^k \binom{\nu}{k} d_k x^{-k}, \quad (98)$$

for  $x = 1$ , endpoint flatness supplies the limiting interpretation.

*Proof.* Since  $\operatorname{Re} \nu > 0$ , the absolute value of the kernel is  $(x - t)^{\operatorname{Re} \nu - 1}$ , which is integrable on the finite triangular region arising after insertion of  $F(t) = \mathbb{E} \mathbf{1}_{\{X \leq t\}}$ . Fubini's theorem therefore gives (96); for positive real  $\nu$ , this reduces to the Tonelli argument in [corollary 3.3](#). At one, replace  $1 - X$  by the identically distributed  $X$ . For  $x > 1$ , expand  $(1 - X/x)^\nu$  and take expectations.  $\square$

**Theorem 12.2** (Fractional-order Newton–Volterra formula). *Let  $\operatorname{Re} \nu > 0$ ,  $0 < x \leq 1$ , and  $\alpha \in \mathbb{C}$ . Then*

$$\mathcal{I}^\nu[t^\alpha F(t)](x) = \sum_{r \geq 0} (-1)^r \frac{\nu^{\bar{r}}}{r!} \alpha^r x^{\alpha-r} (\mathcal{I}^{\nu+r} F)(x). \quad (99)$$

*The series converges absolutely and locally uniformly in  $(\alpha, \nu)$  on compact subsets of  $\mathbb{C} \times \{\operatorname{Re} \nu > 0\}$ . For  $p \in \mathbb{N}_0$ , it terminates:*

$$\mathcal{I}^\nu[t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{\nu+r-1}{r} \frac{p!}{(p-r)!} x^{p-r} (\mathcal{I}^{\nu+r} F)(x). \quad (100)$$

*At  $x = 1$ ,*

$$\mathcal{I}^\nu[t^\alpha F(t)](1) = \frac{1}{\Gamma(\nu)} \sum_{r \geq 0} (-1)^r \binom{\alpha}{r} \frac{M(\nu+r)}{\nu+r}. \quad (101)$$

*Proof.* Expand  $t^\alpha$  about  $x$  exactly as in [theorem 4.2](#). Raising the kernel power from  $\nu - 1$  to  $\nu + r - 1$  contributes  $\Gamma(\nu + r)/\Gamma(\nu) = \nu^{\bar{r}}$ . The flatness estimate used in [theorem 4.3](#), with gamma functions replacing factorials, proves normal convergence. Formula (101) follows from (97) and

$$\frac{\nu^{\bar{r}}}{r!} \alpha^r \frac{1}{\Gamma(\nu + r + 1)} = \frac{1}{\Gamma(\nu)} \binom{\alpha}{r} \frac{1}{\nu + r}.$$

$\square$

This fractional identity is an exact analytic continuation of the complete integer-order table. It also shows that the moment function  $M(s)$ , rather than a new special function, is the natural endpoint datum for fractional Fabius integration.

## 13 Antiderivatives of the inverse Fabius function

### 13.1 Exact first-order reductions

Let

$$Q(y) = F^{-1}(y), \quad 0 \leq y \leq 1. \quad (102)$$

The repository establishes strict monotonicity of  $F$ , smoothness of  $Q$  on  $(0, 1)$ , reflection

$$Q(1 - y) = 1 - Q(y), \quad (103)$$

and infinite endpoint slope [\[1, 2\]](#). Substitution  $u = F(t)$  yields an unexpectedly compact primitive.

**Theorem 13.1** (Closed primitive of the inverse). *For  $0 \leq y \leq 1$ ,*

$$\int_0^y Q(u) \, du = yQ(y) - F\left(\frac{Q(y)}{2}\right). \quad (104)$$

*In particular,*

$$\int_0^1 Q(u) \, du = \frac{1}{2}. \quad (105)$$

*Proof.* Put  $x = Q(y)$ . Then

$$\int_0^y Q(u) \, du = \int_0^x tF'(t) \, dt = xF(x) - \int_0^x F(t) \, dt.$$

Apply  $F(x) = y$  and the base ladder  $\int_0^x F(t) \, dt = F(x/2)$ . At  $y = 1$ , use  $Q(1) = 1$  and  $F(1/2) = 1/2$ .  $\square$

For a complex weight  $\alpha$ , the same substitution and one integration by parts give the following exact reduction to a power of  $F$ .

**Proposition 13.2** (Weighted inverse primitive). *Let  $0 < y \leq 1$  and  $x = Q(y)$ . If  $\alpha \neq -1$  and  $\operatorname{Re} \alpha \geq -1$ , then*

$$\int_0^y u^\alpha Q(u) \, du = \frac{y^{\alpha+1} Q(y)}{\alpha+1} - \frac{1}{\alpha+1} \int_0^{Q(y)} F(t)^{\alpha+1} \, dt. \quad (106)$$

At the exceptional exponent,

$$\int_0^y \frac{Q(u)}{u} \, du = Q(y) \log y - \int_0^{Q(y)} \log F(t) \, dt. \quad (107)$$

For real  $\alpha < -1$  and every  $y > 0$ , the improper integral diverges. At  $y = 0$ , the convergent cases above extend with value zero by taking the right-hand limit; this convention is needed because the displayed factors  $y^{\alpha+1}$  and  $\log y$  are not themselves defined at zero.

*Proof.* After  $u = F(t)$ , the left side is  $\int_0^x tF(t)^\alpha F'(t) \, dt$ . Use  $F^\alpha F' = (F^{\alpha+1})'/(\alpha+1)$ . On the boundary line  $\operatorname{Re} \alpha = -1$ , the endpoint factor still vanishes because of the multiplying  $t$ . At  $\alpha = -1$ , integrate  $t(\log F(t))'$ . The repository's endpoint asymptotic implies  $t \log F(t) \rightarrow 0$ . Finally,  $Q(u)$  decays more slowly than every positive power of  $u$ , so  $u^\alpha Q(u)$  cannot be integrable when  $\alpha < -1$ .  $\square$

At  $y = 1$  and  $p \in \mathbb{N}_0$ ,

$$\int_0^1 u^p Q(u) \, du = \frac{1}{p+1} \left( 1 - \int_0^1 F(t)^{p+1} \, dt \right). \quad (108)$$

If  $X_1, \dots, X_{p+1}$  are independent Fabius variables, then

$$\int_0^1 u^p Q(u) \, du = \frac{1}{p+1} \mathbb{E} \max(X_1, \dots, X_{p+1}). \quad (109)$$

Thus weighted quantile integrals are normalized expected maxima. At  $p = -1$ , formula (107) gives the convergent constant

$$\int_0^1 \frac{Q(u)}{u} \, du = - \int_0^1 \log F(t) \, dt. \quad (110)$$

### 13.2 Higher and fractional inverse primitives

The indicator representation

$$Q(u) = \int_0^1 \mathbf{1}_{\{F(t) \leq u\}} \, dt \quad (111)$$

turns all higher primitives into one-dimensional integrals of powers of  $y - F(t)$ .

**Theorem 13.3** (All orders for the inverse). *For  $\operatorname{Re} \nu > 0$  and  $0 \leq y \leq 1$ ,*

$$(\mathcal{I}^\nu Q)(y) = \frac{1}{\Gamma(\nu + 1)} \int_0^{Q(y)} (y - F(t))^\nu dt. \quad (112)$$

*For an integer  $n \geq 1$ , this is*

$$(\mathcal{J}_n Q)(y) = \frac{1}{n!} \int_0^{Q(y)} (y - F(t))^n dt. \quad (113)$$

*At the complete endpoint,*

$$(\mathcal{I}^\nu Q)(1) = \frac{1}{\Gamma(\nu + 1)} \int_0^1 F(t)^\nu dt, \quad (114)$$

*where reflection was used in the last step.*

*Proof.* Insert (111) into (28). Because  $\operatorname{Re} \nu > 0$ , the absolute value of the resulting kernel is integrable on the finite region  $0 \leq F(t) \leq u \leq y$ ; hence Fubini's theorem permits interchanging the integrals. For a fixed  $t \leq Q(y)$ , the variable  $u$  ranges from  $F(t)$  to  $y$ , and

$$\frac{1}{\Gamma(\nu)} \int_{F(t)}^y (y - u)^{\nu-1} du = \frac{(y - F(t))^\nu}{\Gamma(\nu + 1)}.$$

At  $y = 1$ , substitute  $t \mapsto 1 - t$  and use  $1 - F(t) = F(1 - t)$ . □

For a natural weight  $p$ , the same argument gives

$$\mathcal{I}^\nu [u^p Q(u)](y) = \frac{1}{\Gamma(\nu)} \int_0^{Q(y)} \int_{F(t)}^y (y - u)^{\nu-1} u^p du dt. \quad (115)$$

For  $y > 0$ , the inner integral is an incomplete beta function:

$$y^{p+\nu} [B(p + 1, \nu) - B_{F(t)/y}(p + 1, \nu)]. \quad (116)$$

When  $p$  is a nonnegative integer and  $\nu = n$  is a positive integer, it is a finite polynomial in  $y$  and  $F(t)$ .

### 13.3 Higher derivatives of the inverse

Let  $x = Q(y) \in (0, 1)$ . The classical Lagrange inverse-derivative formula [17] gives an exact all-orders identity that can be combined with the Thue–Morse derivative expression (91). We import that classical identity here rather than reprove Lagrange inversion.

**Theorem 13.4** (Lagrange formula for inverse derivatives). *For  $n \geq 1$ ,*

$$Q^{(n)}(y) = \frac{d^{n-1}}{dt^{n-1}} \left( \frac{t - x}{F(t) - F(x)} \right)^n \Big|_{t=x}. \quad (117)$$

*If*

$$a_m(x) = \frac{F^{(m+1)}(x)}{(m + 1)! F'(x)}, \quad (118)$$

*then equivalently*

$$Q^{(n)}(y) = \frac{(n - 1)!}{F'(x)^n} [z^{n-1}] \left( 1 + \sum_{m \geq 1} a_m(x) z^m \right)^{-n}. \quad (119)$$

Indeed, writing  $t = x + z$  gives

$$\frac{t - x}{F(t) - F(x)} = \frac{1}{F'(x)} \left( 1 + \sum_{m \geq 1} a_m(x) z^m \right)^{-1}.$$

Raising this identity to the  $n$ -th power and taking the  $(n - 1)$ -st derivative at  $z = 0$  multiplies the coefficient of  $z^{n-1}$  by  $(n - 1)!$ , proving the equivalence of (117) and (119).

The first cases are

$$Q' = \frac{1}{F'}, \quad (120)$$

$$Q'' = -\frac{F''}{(F')^3}, \quad (121)$$

$$Q''' = \frac{3(F'')^2 - F'F'''}{(F')^5}, \quad (122)$$

$$Q'''' = \frac{-15(F'')^3 + 10F'F''F''' - (F')^2F''''}{(F')^7}, \quad (123)$$

all evaluated at  $x = Q(y)$ . Inserting (91) makes each one a finite Thue–Morse expression in translated, dilated up-values. On a left dyadic cell, (92) further collapses every derivative of  $F$  to a single scaled Fabius value.

## 14 New inverse-primitive asymptotics

### 14.1 Repository input and slow variation

The consolidated frontier volume derives from the sharp Lambert expansion the endpoint equivalent

$$Q(y) \sim \frac{\sqrt{\log(1/y)}}{e\sqrt{\log 2}} \exp\left[-\sqrt{2 \log 2 \log(1/y)}\right] \quad (y \downarrow 0). \quad (124)$$

At the audited snapshot this statement was marked “Derived” rather than Lean-formalized in the source. The current Lean development now proves precisely this equivalence as `Fabius.fabiusInv_isEquivalent_fabiusInverseAsymptoticMain` in `FabiusInverseAsymptotic.lean`. Thus the results below are unconditional mathematical consequences of a now-formalized input, although their integral and order-statistic deductions have not themselves yet been translated into Lean [2].

Put

$$L = \log 2, \quad a = \sqrt{2L}, \quad K = \frac{1}{e\sqrt{L}}. \quad (125)$$

Then  $Q(y) \sim K\sqrt{T}e^{-a\sqrt{T}}$ , where  $T = \log(1/y)$ . For every fixed  $c > 0$ , replacing  $T$  by  $T - \log c$  changes this expression by a factor tending to one. Therefore

$$\frac{Q(cy)}{Q(y)} \longrightarrow 1, \quad y \downarrow 0, \quad (126)$$

so  $Q$  is slowly varying at zero.

## 14.2 Weighted primitives near zero

**Theorem 14.1** (Karamata law for inverse primitives). *Let  $\alpha > -1$  and  $\nu > 0$  be real. Then, as  $y \downarrow 0$ ,*

$$\mathcal{I}^\nu[u^\alpha Q(u)](y) \sim \frac{\Gamma(\alpha + 1)}{\Gamma(\alpha + \nu + 1)} y^{\alpha + \nu} Q(y). \quad (127)$$

*In particular,*

$$\int_0^y u^\alpha Q(u) du \sim \frac{y^{\alpha+1} Q(y)}{\alpha + 1}, \quad (128)$$

$$(\mathcal{I}^\nu Q)(y) \sim \frac{y^\nu Q(y)}{\Gamma(\nu + 1)}. \quad (129)$$

*Proof.* Scale  $u = ys$  in the defining integral:

$$\mathcal{I}^\nu[u^\alpha Q(u)](y) = \frac{y^{\alpha+\nu} Q(y)}{\Gamma(\nu)} \int_0^1 (1-s)^{\nu-1} s^\alpha \frac{Q(ys)}{Q(y)} ds.$$

Pointwise convergence to one follows from (126). Potter bounds for slowly varying functions provide an integrable dominating beta kernel. The limiting integral is  $B(\alpha + 1, \nu)/\Gamma(\nu) = \Gamma(\alpha + 1)/\Gamma(\alpha + \nu + 1)$ .  $\square$

The boundary exponent  $\alpha = -1$  is convergent but belongs to a different scale.

**Theorem 14.2** (Logarithmic inverse primitive). *As  $y \downarrow 0$ , with  $T = \log(1/y)$ ,*

$$\int_0^y \frac{Q(u)}{u} du \sim \frac{2\sqrt{T}}{a} Q(y) \quad (130)$$

$$\sim \frac{\sqrt{2}}{e \log 2} T \exp\left[-\sqrt{2 \log 2} T\right]. \quad (131)$$

*Proof.* With  $u = e^{-s}$ ,

$$\int_0^y \frac{Q(u)}{u} du = \int_T^\infty Q(e^{-s}) ds.$$

The input equivalent turns the integrand into  $K\sqrt{s}e^{-a\sqrt{s}}$ . Substituting  $v = \sqrt{s}$  gives

$$2K \int_{\sqrt{T}}^\infty v^2 e^{-av} dv = 2K e^{-a\sqrt{T}} \left( \frac{T}{a} + \frac{2\sqrt{T}}{a^2} + \frac{2}{a^3} \right).$$

The leading term is  $(2\sqrt{T}/a)Q(y)$ . A standard asymptotic sandwich transfers the equivalent from the model integrand to  $Q(e^{-s})$ .  $\square$

## 14.3 Large weighted quantile moments

For  $p \in \mathbb{N}_0$ , define

$$q_p = \int_0^1 u^p Q(u) du, \quad C_p = \int_0^1 (1-u)^p Q(u) du. \quad (132)$$

Reflection (103) gives the exact identity

$$q_p = \frac{1}{p+1} - C_p. \quad (133)$$

**Theorem 14.3** (Large quantile-moment correction). *As  $p \rightarrow \infty$  through the nonnegative integers,*

$$C_p \sim \frac{Q(1/p)}{p}, \quad (134)$$

*and therefore*

$$q_p = \frac{1}{p+1} - \frac{\sqrt{\log p}}{e\sqrt{\log 2} p} \exp\left[-\sqrt{2 \log 2 \log p}\right] (1 + o(1)). \quad (135)$$

*Proof.* Set  $u = t/p$ . Then

$$\frac{pC_p}{Q(1/p)} = \int_0^p \left(1 - \frac{t}{p}\right)^p \frac{Q(t/p)}{Q(1/p)} dt.$$

For fixed  $t > 0$ , the integrand tends to  $e^{-t}$  by slow variation. Potter bounds and the elementary exponential bound on the binomial kernel supply domination after splitting the integral into a compact interval and a tail. The limit is  $\int_0^\infty e^{-t} dt = 1$ . Substitute (124) at  $y = 1/p$  and use (133).  $\square$

Let  $M_n = \max(X_1, \dots, X_n)$  and  $m_n = \min(X_1, \dots, X_n)$ . From (109) and symmetry,

$$1 - \mathbb{E}M_n = \mathbb{E}m_n = \int_0^1 F(t)^n dt. \quad (136)$$

The theorem immediately gives a compact order-statistic equivalent.

**Corollary 14.4** (Extreme order statistics and high inverse order). *As  $n \rightarrow \infty$  through the positive integers,*

$$\mathbb{E}m_n = 1 - \mathbb{E}M_n \sim Q(1/n). \quad (137)$$

*Consequently,*

$$(\mathcal{J}_n Q)(1) = \frac{1}{n!} \int_0^1 F(t)^n dt \sim \frac{Q(1/n)}{n!}. \quad (138)$$

The correction  $Q(1/n)$  is stretched-logarithmic: it is smaller than every negative power of  $\log n$ , but larger than  $n^{-\varepsilon}$  for every fixed  $\varepsilon > 0$ . This scale is invisible to a conventional expansion in powers of  $1/n$ .

## 15 Numerical experiments and reproducibility

The archive accompanying this report contains `numerical_experiments.py`, a fully commented Python script using only the standard library and `mpmath`. It computes two independent moment systems:

$$(n+1)(2^n - 1)d_n = \sum_{k < n} \binom{n+1}{k} d_k, \quad (139)$$

$$(4^n - 1)c_n = \sum_{k < n} \binom{2n}{2k} \frac{c_k}{2(n-k) + 1}. \quad (140)$$

The second recurrence follows from  $H(2z) = (\sinh z/z)H(z)$  for the centered moment generating function and is independent of the first recurrence. Exact rational arithmetic verifies (60) for  $0 \leq p \leq 8$  and (62) for  $1 \leq n \leq 6$ ,  $0 \leq p \leq 8$ .

The script also checks the exponential master identity at  $z = 0.7 + 0.2i$ . The moment-series and independent infinite-product values of  $G(z)$  agree to approximately 24 decimal places, and the two sides of (72) agree to the same precision. Full machine-readable values are in `verification_output.txt` and `fractional_power_checks.csv`.



Table 2: Independent evaluations of  $\mathcal{A}(\alpha) = \int_0^1 x^\alpha F(x) dx$ . The first column uses the Newton series (57); the second uses (56) and the centered expansion (67). The shown discrepancies are truncation errors, not statistical errors.

| $\alpha$ | Newton–Fabius series           | Centered-moment formula        | absolute difference   |
|----------|--------------------------------|--------------------------------|-----------------------|
| 0.5      | 0.4209681733402671557991       | 0.4209681733402671557990       | $9.0 \times 10^{-23}$ |
| −0.5     | 0.6073863107431723970836       | 0.6073863107431723973782       | $2.9 \times 10^{-19}$ |
| −1       | 0.7582400004044086356798       | 0.7582400004044086568776       | $2.1 \times 10^{-17}$ |
| −2       | 1.3148861440057837837803       | 1.3148861440058189655548       | $3.5 \times 10^{-14}$ |
| 2.3      | 0.2591812728638398877191737152 | 0.2591812728638398877191737148 | $3.7 \times 10^{-28}$ |

### Numerical interpretation

The positive series for negative powers converge more slowly as the negative exponent grows. The displayed digits for  $\alpha = -2$  are therefore fewer than for positive or fractional exponents. The code reports both truncations and their discrepancy rather than presenting unsupported extra digits.

## 16 Conjectures and frontier directions

### 16.1 A periodic saddle expansion for weighted primitives

The exact Newton–Volterra series and the repository’s all-orders periodic Lambert expansion should combine into an all-orders local expansion of (49).

**Conjecture 16.1** (Weighted primitive saddle hierarchy). *For fixed  $n \geq 1$  and  $\alpha \in \mathbb{C}$ , let  $\lambda_n(x)$  be the exact Lambert phase associated with  $x/2^n$ . There exist smooth one-periodic functions  $P_{j,n,\alpha}$  such that, for every  $N \geq 1$ ,*

$$\frac{\mathcal{J}_n[t^\alpha F(t)](x)}{x^\alpha 2^{\binom{n}{2}} F(x/2^n)} = 1 + \sum_{j=1}^{N-1} \frac{P_{j,n,\alpha}(\lambda_n(x))}{\lambda_n(x)^j} + O(\lambda_n(x)^{-N}). \quad (141)$$

The first coefficient is obtained by inserting the phase expansion of  $\mathcal{J}_{n+1}F/(x\mathcal{J}_nF)$  into (50).

This conjecture is structurally plausible because each successive term of (38) shifts the exact phase by approximately one unit, and the periodic fluctuation is therefore sampled coherently rather than averaged away.

### 16.2 Second-order large-quantile correction

Ignoring the next periodic inverse correction for a moment, expansion of the explicit leading model in theorem 14.3 gives

$$\frac{Q(t/p)}{Q(1/p)} = 1 + \frac{a \log t}{2\sqrt{\log p}} + O\left(\frac{1 + |\log t|^2}{\log p}\right).$$

Since  $\int_0^\infty e^{-t} \log t dt = -\gamma$ , this suggests the following conditional refinement.

**Conjecture 16.2** (Universal first correction, modulo periodic inverse jets). *If the relative error in (124) is uniformly  $o((\log(1/y))^{-1/2})$  on fixed multiplicative windows, then*

$$\frac{pC_p}{Q(1/p)} = 1 - \frac{\gamma\sqrt{2\log 2}}{2\sqrt{\log p}} + o((\log p)^{-1/2}). \quad (142)$$

In the full problem, the repository’s periodic saddle jets may add a bounded oscillatory term at this same or the next scale; determining the precise interaction is open.

### 16.3 Large negative moments

The positive moment function  $M(s)$  is entire, so all negative moments  $N(r) = M(-r) = \mathbb{E}X^{-r}$  exist. Exact integration by parts gives

$$N(r) = 1 + r \int_0^1 x^{-r-1} F(x) dx. \quad (143)$$

Inserting the repository's exact-phase expansion and applying Laplace's method in the Lambert variable leads to the next conjecture.

**Conjecture 16.3** (Periodic large-negative-moment law). *As real  $r \rightarrow +\infty$ ,*

$$\log M(-r) = \frac{\log 2}{2} r^2 - r \log r + \left(1 + \frac{\log 2}{2}\right) r + \frac{1}{2} \log r + C + \Omega(r) + o(1), \quad (144)$$

where  $C$  is constant and  $\Omega$  is a bounded smooth one-periodic function determined by the saddle fluctuation  $\Psi$ . For integral  $r$ , the periodic phase is sampled at a fixed saddle class and may collapse into the constant term.

The quadratic coefficient and the terms through  $\frac{1}{2} \log r$  follow from the stationary phase near  $\lambda = r + 1/2$ ; the exact periodic saddle shift must still be controlled.

### 16.4 Holonomicity, arithmetic, and formalization problems

The following directions appear especially promising.

1. **Lean formalization.** The reusable first wave is complete: `FabiusFunction.NormalizedVolterra` supplies the real-normed-space operator, affine covariance for every real scale (with the inverse form restricted to nonzero scale), local basepoint shifts, open-tail Taylor jets, and the finite polynomial/monomial commutators; its literal-iteration/FTC bridge uses a Banach target; `FabiusFunction.FabiusAntiderivatives` supplies the signed and bounded integer ladders and their natural-monomial consequences. The new generic transport and jet identities do not themselves give the explicit Fabius/up coefficients or piecewise assembly. Independently, the positive-real fractional kernel, its continuous Banach-valued semigroup, positive affine covariance, the exact derivative order-raising boundary formula, the causal support-truncated Rvachev primitive and semigroup, and the one-step signed, bounded, and Up-to-Fabius shifts are now in `FractionalVolterra.lean`, `FractionalVolterraSemigroup.lean`, `FractionalVolterraCalculus.lean`, and `FabiusFractionalVolterra.lean`. The next finite targets are the exterior  $x > 1$  branch and matching jets in (52), followed independently by the inverse identity (104). The normal-convergence proof of (38), its Mellin consequences, and fractional order form a separate analytic wave.
2. **Dyadic arithmetic.** Substitute the repository's exact  $q$ -binomial evaluator into (39). Determine sharp 2-adic valuations and odd denominator factors of  $H_{n,p}(a/2^m)$ . The two-parameter identity (62) may create cancellations not visible term by term.
3. **Newton-series uniqueness.** Establish optimal growth classes in which (57) is the unique Newton interpolation of  $(1 - d_{p+1})/(p + 1)$ . The superpolynomial decay of  $d_n$  should yield stronger convergence than generic Carlson-type hypotheses.
4. **Non-holonomicity.** Test the conjecture that  $\mathcal{A}(\alpha)$  and  $M(\alpha)$  are not D-finite and that their integer samples are not P-recursive. The dyadic functional equation for  $G$ , the infinite Bernoulli cumulant sequence, and the periodic saddle fluctuation all point in this direction, but none alone is a proof.

5. **Fractional order and Mellin–Barnes methods.** Develop (99) as a two-complex-variable special function in  $(\alpha, \nu)$ . Its endpoint form (101) may admit Mellin–Barnes continuation and residue expansions tied to the sinc product.
6. **Inverse powers of the CDF.** Study  $\int_0^x F(t)^s dt$ , the sole new object in (106). For positive integral  $s$  it represents order statistics; near  $s = 0$  its one-sided derivatives involve  $\int (\log F)^k$ , whose growth should encode the quadratic-log endpoint.
7. **Fourier antiderivatives.** The complete transform (75) is an infinite sinc product, while the exterior branches of (88) are moment polynomials. A distributional Fourier theory that explicitly separates those polynomial tails from  $(2\pi i\xi)^{-n}\widehat{\text{up}}(\xi)$  should produce useful compact renormalized primitives.
8. **Order statistics and periodic corrections.** Push theorem 14.3 to all orders using the closed periodic inverse jets already present in the frontier volume. This would transfer the Lambert–W and periodic structure into a new probabilistic observable:  $\mathbb{E} \min(X_1, \dots, X_n)$ .

## 17 Summary of principal formulas

| Object                                         | Exact formula, leading result, and current Lean status                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generic affine covariance                      | <code>normalizedVolterra_affine</code> is a Lean-formalized forward identity for arbitrary real normed targets and every $c \in \mathbb{R}$ , including zero and negative scales; <code>normalizedVolterra_comp_affine</code> assumes $c \neq 0$ .                                                                                                                                                 |
| Generic positive-real fractional order raising | <code>fractionalVolterra_affine</code> and <code>fractionalVolterra_comp_affine</code> formalize positive affine covariance for arbitrary real order; for positive order, <code>fractionalVolterra_add_one_deriv</code> formalizes integration by parts with the exact left-endpoint boundary term.                                                                                                |
| One-step fractional Fabius–Rvachev shifts      | <code>fractionalVolterra_add_one_extendedFabius_of_nonneg</code> , <code>fractionalVolterra_add_one_fabiusReal</code> , and <code>fractionalVolterra_add_one_rvachevUp</code> are Lean formalized for real positive order on $x \geq 0$ , $[0, 1]$ , and $x \geq -1$ , respectively.                                                                                                               |
| Generic basepoint and zero-tail calculus       | <code>normalizedVolterra_basepoint_shift</code> is Lean formalized with local interval integrability on $a$ to $b$ and $b$ to $x$ , no endpoint order, and $n = 0$ included. The zero-tail theorem retains only integrability from $a$ to $b$ , is positive-order, and assumes vanishing on the open unoriented interval between $b$ and $x$ , with no endpoint order or endpoint-value condition. |
| Generic polynomial weight                      | Finite Hasse-derivative formula (33); Lean formalized for normed-space-valued inputs, arbitrary base and endpoint, and every $n \in \mathbb{N}_0$ .                                                                                                                                                                                                                                                |
| $n$ -fold primitive of $F$ for $x \leq 1$      | $\mathcal{J}_n F(x) = 2^{\binom{n}{2}} F(x/2^n)$ , Lean formalized for every $n \in \mathbb{N}_0$ . The signed analogue is formalized for every real $x$ .                                                                                                                                                                                                                                         |
| Natural monomial weight                        | Finite sum (40) is Lean formalized for $x \leq 1$ ; the $x > 1$ global completion in (52) remains a frontier.                                                                                                                                                                                                                                                                                      |
| Complex monomial weight                        | Absolutely convergent Newton–Volterra series (38), entire in $\alpha$ ; not yet Lean formalized.                                                                                                                                                                                                                                                                                                   |
| Complete endpoint                              | $\int_0^1 x^\alpha F(x) dx = (1 - M(\alpha + 1))/(\alpha + 1)$ and Newton series (57); Mellin/Newton frontier.                                                                                                                                                                                                                                                                                     |

| Object                      | Exact formula, leading result, and current Lean status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Negative integer weight     | Positive dyadic series (48); logarithmic constant (66)–(68); frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Reflection/complement       | $F(1-x) = 1-F(x)$ , and $\mathcal{J}_n[x^p(1-F)] = p!x^{p+n}/(p+n)! - H_{n,p}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Rvachev up-function         | Complete four-branch formula (88); exterior branches are moment polynomials; folded-primitive frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Higher derivatives          | Thue–Morse sums (90) and (91); single dilation for $\mathcal{F}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Fractional order            | The underlying positive-real ordered-endpoint operator and its positive-integer bridge are Lean formalized, as are raw beta-kernel integrability, the strict-interval raw and normalized beta values, the continuous-Banach-input semigroup, the positive-real Fabius probability formula (96), increasing-affine covariance, and the causal support-truncated Rvachev primitive and semigroup together with the three single-step Fabius–Rvachev order shifts. The weighted commutator (99), shifted dyadic-lattice and endpoint-moment wrappers, transform/tail expansions, and every complex-order assertion remain frontier. |
| Inverse first primitive     | $\int_0^y Q(u) du = yQ(y) - F(Q(y)/2)$ , as in (104); frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Inverse arbitrary order     | $\mathcal{I}^\nu Q(y) = \Gamma(\nu+1)^{-1} \int_0^{Q(y)} (y-F(t))^\nu dt$ ; frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Inverse endpoint asymptotic | Weighted Karamata law (127), logarithmic boundary (131), and large moment correction (135); frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## A Corpus comparison map

| Source family                            | Role in the nonduplication audit                                                                                                                                                                                                                                  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabius_Function_and_Rvachev_Up/          | Primary live exposition: bounded/signed conventions, shape, derivatives, moments, dyadic arithmetic, effective flatness, sharp Lambert asymptotics, and formalization status.                                                                                     |
| FabiusFunction_Mathematical_Glossary/    | Notation and cross-reference normalization; used to prevent collisions among $F$ , up, the signed extension, moments, and inverse conventions.                                                                                                                    |
| fabius_lean_walkthrough/                 | Map from human statements to Lean declarations; used to identify which imported facts are already formalized.                                                                                                                                                     |
| Non_Elementarity_of_the_Fabius_Function/ | Analytic-locus and non-elementarity results; establishes why the native finite Fabius formula, rather than an elementary primitive, is the appropriate notion of closed form.                                                                                     |
| Consolidated frontier volume             | Consolidated frontier volume: repeated averaging, inverse function, small-argument inversion, q-calculus, sinc-product decay, Lambert–W methods, periodic saddle jets, and other deferred claims.                                                                 |
| Frontier draft packages                  | Package-level checks of inverse, exponent-sequence/Newton, repeated-integration, Fourier-decay, Thue–Morse, and asymptotic reports. The Newton report concerns exponent sequences and q-Richardson reconstruction, not Newton series of monomial antiderivatives. |

| Source family | Role in the nonduplication audit                                                                                                                                                      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| papers/       | Source transcriptions and corrected papers, especially Arias de Reyna's compact support and arithmetic articles; used for historical formulas and moment normalizations.              |
| archive/      | Frozen early notes, primary-exposition revisions, q-calculus studies, and standalone reports. Historical descendants were treated as lineages rather than independent novelty claims. |

The decisive negative checks were not based only on titles. The live primary source and consolidated frontier source were searched for the literal phrases “antiderivative” and “indefinite integral”; neither contains a treatment of the present problem. The repeated-integration material concerns iterative averaging/convergence to up, not the Volterra primitives here. The inverse material treats regularity, dyadic inversion, non-elementarity, and endpoint asymptotics, but not (104), (112), or weighted quantile integrals. The Newton draft treats exponent sequences, sinc products, Prouhet cancellation, and q-Richardson reconstruction, not (57). These are snapshot-relative source-document comparisons. The later Lean modules listed in the status crosswalk now formalize the integer Volterra core together with generic affine transport, local basepoint shifts, and positive-order open-tail Taylor jets. Those generic declarations do not supply explicit Fabius/up coefficients or piecewise assembly; the complex-power, Mellin, folded, improper-singular or higher inverse, exterior-piecewise, and remaining report-specific fractional claims remain outside that formalized core. The complete and variable-upper proper-compact absolutely-continuous inverse-pair identities are now formalized separately in `InversePairIntegral.lean`.

## B Exact initial values and sample antiderivatives

The first moments computed from (22) are

$$\begin{aligned} d_0 &= 1, & d_1 &= \frac{1}{2}, & d_2 &= \frac{5}{18}, & d_3 &= \frac{1}{6}, \\ d_4 &= \frac{143}{1350}, & d_5 &= \frac{19}{270}, & d_6 &= \frac{1153}{23814}, & d_7 &= \frac{583}{17010}. \end{aligned}$$

The first centered moments are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}.$$

For every real  $x \leq 1$ , the first ordinary primitives are

$$\int_0^x F(t) \, dt = F(x/2), \tag{145}$$

$$\int_0^x tF(t) \, dt = xF(x/2) - 2F(x/4), \tag{146}$$

$$\int_0^x t^2 F(t) \, dt = x^2 F(x/2) - 4xF(x/4) + 16F(x/8), \tag{147}$$

$$\int_0^x t^3 F(t) \, dt = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16). \tag{148}$$

At  $x = 1$ , these equal respectively

$$\frac{1}{2}, \quad \frac{13}{36}, \quad \frac{5}{18}, \quad \frac{1207}{5400}. \tag{149}$$

For a first negative-power example,

$$\int_0^x \frac{F(t)}{t} dt = \sum_{r \geq 0} r! 2^{\binom{r+1}{2}} x^{-r-1} F\left(\frac{x}{2^{r+1}}\right). \quad (150)$$

For a fractional example,

$$\int_0^x t^{1/2} F(t) dt = \sum_{r \geq 0} (-1)^r 1/2^r 2^{\binom{r+1}{2}} x^{1/2-r} F\left(\frac{x}{2^{r+1}}\right). \quad (151)$$

## C Derivation of the centered-moment recurrence

Let

$$H(z) = \mathbb{E} e^{z(2X-1)} = \sum_{n \geq 0} c_n \frac{z^{2n}}{(2n)!}. \quad (152)$$

The distributional fixed point

$$2X - 1 \stackrel{d}{=} \frac{V + (2X' - 1)}{2}, \quad (153)$$

where  $V$  is uniform on  $[-1, 1]$  and  $X'$  is an independent copy of  $X$ , gives

$$H(2z) = \frac{\sinh z}{z} H(z). \quad (154)$$

Equating coefficients of  $z^{2n}$  yields

$$(4^n - 1)c_n = \sum_{k=0}^{n-1} \binom{2n}{2k} \frac{c_k}{2(n-k)+1}, \quad (155)$$

which is the independent recurrence used in the numerical checks.

## D Implementation notes

The supplied Python program has four layers.

1. Exact Fraction arithmetic generates the initial  $d_n$  and  $c_n$ , checks the terminating natural-power identity, and verifies the two-parameter identity with no rounding.
2. High-precision mpmath recurrences generate longer moment arrays.
3. The Newton series (57) and centered formula (67) are evaluated independently for positive, fractional, negative, and complex exponents.
4. The moment series for  $G$  is compared with the independent product (25), after which (72) is checked.

No Monte Carlo sampling is used. The default parameters in the packaged output use 85 decimal digits,  $d_0, \dots, d_{1500}$ , and  $c_0, \dots, c_{700}$ .

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## Part II

# Fabius Monomial Antiderivatives and Inverse-Quantile Integrals

### Abstract

Let  $F$  be the bounded Fabius distribution function, let  $\text{up}(x) = F(1 - |x|)$  be Rvachev's compactly supported even up-function, let  $\mathcal{F}$  be the signed global extension satisfying  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ , and let  $J = F^{-1}$  on  $[0, 1]$ . A recursive audit of the LaTeX documents under `Analysis/FabiusFunction/docs` finds extensive treatments of exact dyadic arithmetic, moments,  $q$ -binomial and Thue–Morse formulae, repeated-integration limits, Fourier sinc products, endpoint asymptotics, and inverse asymptotics, but no systematic calculus of indefinite monomial-weighted primitives.

The central observation is the exact zero-based primitive ladder

$$A_n(x) = 2^{\binom{n}{2}} F(x/2^n), \quad A'_n = A_{n-1}, \quad A_0 = F,$$

valid for  $x \leq 1$ . It immediately gives, for every  $p \in \mathbb{N}_0$ ,

$$\int x^p F(x) \, dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C.$$

For arbitrary  $\alpha \in \mathbb{C}$  the terminating sum becomes an absolutely and locally uniformly convergent series. Endpoint flatness is strong enough that this series is exact for every complex exponent, including all negative powers, rather than merely asymptotic. Consequently  $\mathcal{M}_F(s) = \int_0^1 x^{s-1} F(x) \, dx$  is entire and has an explicit dyadic Newton-type expansion. Differentiation in  $s$  produces logarithmic weights, with coefficients expressible through signed Stirling numbers or complete Bell polynomials.

Reflection gives primitives of  $F(1-x) = 1 - F(x)$ , and the same formulas provide piecewise primitives and all polynomial moments of  $\text{up}$ . For the inverse function a change of variables yields the exact nonlinear reduction

$$\int_0^y t^\beta J(t) \, dt = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int_0^{J(y)} F(x)^{\beta+1} \, dx,$$

with a separate logarithmic identity at  $\beta = -1$ . Positive integer powers of  $F$  in the remainder admit an explicit finite Thue–Morse spline limit. The repository's Lambert– $W$  inverse asymptotic then gives sharp endpoint laws for these inverse primitives, including a phase transition at  $\beta = -1$ . Exact rational checks and independent quasi-Monte Carlo experiments accompany the proofs. Several conjectural extensions and a concrete Lean formalization roadmap conclude the report.



**Current Lean formalization status.** The complete natural-order core is now machine-checked, and in greater generality than it is first presented in this report. The module `FabiusFunction.NormalizedVolterra` treats arbitrary real normed-space-valued inputs, arbitrary base points, oriented endpoints, and order zero. For continuous Banach-valued inputs, `iteratedPrimitive_eq_normalizedVolterra` identifies literal iteration with the Cauchy kernel; without a completeness assumption, `normalizedVolterra_polynomial` and `normalizedVolterra_monomial` prove the finite polynomial and natural-monomial commutators. Also for arbitrary real normed targets, the forward form `normalizedVolterra_affine` covers every real scale  $c$ , including  $c = 0$  and negative  $c$ , while its inverse form `normalizedVolterra_comp_affine` assumes  $c \neq 0$ . `normalizedVolterra_basepoint_shift` needs interval integrability only from  $a$  to  $b$  and from  $b$  to  $x$ , has no endpoint-order hypothesis, and includes  $n = 0$ . `normalizedVolterra_succ_eq_taylor_of_eq_zero` retains only integrability from  $a$  to  $b$ , is positive-order, and assumes vanishing on the open unoriented interval between  $b$  and  $x$ , with neither endpoint order nor endpoint values. The module `FabiusFunction.FabiusAntiderivatives` proves the signed global ladder and monomial formula in `normalizedVolterra_extendedFabius` and `normalizedVolterra_pow_mul_extendedFabius`. Its bounded counterparts `normalizedVolterra_fabiusReal_of_le_one` and `normalizedVolterra_pow_mul_fabiusReal_of_le_one` hold on the proved initial half-line  $x \leq 1$ , not merely on  $[0, 1]$ .

These four new declarations are generic transport and finite Taylor-jet identities. They do not themselves compute explicit Fabius/up coefficients or assemble piecewise formulas. They were focused-build verified at compiled checkpoint e109088ed.

The generic positive-real ordered-endpoint fractional kernel and its positive-integer bridge are now formalized in `FabiusFunction.FractionalVolterra`; `FabiusFunction.FractionalVolterraSemigroup` adds raw beta-kernel integrability, its normalized value, and additive composition of positive orders for continuous Banach-valued inputs on ordered compact intervals. `FabiusFunction.FractionalVolterraCalculus` adds positive affine covariance at arbitrary real order and ordinary-derivative order raising from an interval-integrable right interior derivative, while `FabiusFunction.FabiusFractionalVolterra` exports the total causal Rvachev fractional primitive, its support-truncated integral, natural-order bridge and positive-order semigroup, and the three positive-real single-step Fabius–Rvachev shifts. No Lean formalization is claimed here for the complex-power series or its convergence and remainder estimates, the logarithmic and Mellin consequences, a semigroup for merely interval-integrable inputs or noncomplete targets, a complex-order semigroup, shifted dyadic-lattice or endpoint-moment Rvachev wrappers, fractional derivatives, hierarchy-weighted fractional monomials, the negative-endpoint signed branch, or the explicit Rvachev tail series, piecewise coefficients, transform, or refinement formulas, the local, higher-order, or fractional inverse-function formulas and asymptotics, or the bounded function’s piecewise continuation for  $x > 1$ . Those portions remain research frontiers; the analytic arguments and numerical checks below do not change that status.

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# 1 Executive summary and nonduplication boundary

## 1.1 Corpus audit

The audit covered the living expositions, the consolidated frontier volume, the paper transcriptions, the glossary and Lean walkthrough, the currently visible frontier-draft families, and every frozen historical family listed by the repository archive manifest. The tree is actively edited, so the audit was performed at the document-family level: principal mathematical sources were read closely, draft reports were searched thematically and lexically, and auxiliary generated tables or frozen duplicates were checked through their parent documents and manifests. The principal interfaces are the current synthesis [10], the consolidated frontier volume [11], the inverse-frontier report [12], and the archive ledger [13].

The closest existing material is the repeated-integration dossier in the frontier volume. It studies how normalized repeated integrations of splines converge to up and gives exact convergence rates. Its object is a sequence of approximating functions. The present report instead fixes the limiting Fabius function and identifies its entire Volterra primitive tower. Searches for *antiderivative*, *indefinite integral*, and the corresponding monomial patterns returned no treatment of the formulas below in the current primary synthesis, consolidated frontier volume, or the new inverse, Newton, and spectral frontier reports.

---

All novelty labels in this report mean “not found in the recursively audited repository snapshot of 27 August 2026.” They are not claims that no equivalent identity has ever appeared elsewhere. Elementary consequences are included because they organize a useful calculus; the frontier claims concern the combined exact ladder, the all-complex-exponent convergence theorem, its Mellin and Bell consequences, and the inverse nonlinear reduction with its Thue–Morse realization.

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## 1.2 Contribution map

# 2 Established platform and notation

## 2.1 The bounded Fabius function

We use the repository’s normalization. The bounded Fabius function  $F : \mathbb{R} \rightarrow [0, 1]$  satisfies

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (156)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (157)$$

It is  $C^\infty$ , strictly increasing on  $[0, 1]$ , and flat at both endpoints. Let

$$X = \sum_{j \geq 1} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]. \quad (158)$$

Then  $F$  is the distribution function of  $X$ . In particular, for every  $z \in \mathbb{C}$ , all complex moments  $\mathbb{E}X^z$  exist after choosing the positive-real branch, because the law has infinite-order flatness at zero.

Write

$$d_n = \mathbb{E}X^n = n \int_0^1 t^{n-1} \text{up}(t) dt = n! 2^{\binom{n}{2}} F(2^{-n}), \quad n \in \mathbb{N}, \quad (159)$$

with  $d_0 = 1$ . The repository proves the triangular recurrence

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (160)$$

| Topic                      | Result                                                                                                                                       | Status                                                                                                                                                                                                                 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primitive ladder           | $n$ -fold zero-based primitive $2^{\binom{n}{2}}F(x/2^n)$ on $x \leq 1$ , its signed global form, and its Volterra kernel.                   | Lean-formalized, more generally                                                                                                                                                                                        |
| Polynomial weights         | Finite natural-monomial and arbitrary real-polynomial Volterra commutators.                                                                  | Lean-formalized, more generally                                                                                                                                                                                        |
| Complex powers             | Absolutely convergent exact series for every $\alpha \in \mathbb{C}$ , with an exact remainder and compact-uniform convergence.              | Frontier; not formalized                                                                                                                                                                                               |
| Mellin calculus            | Entire Mellin transform, complete monotonicity on the real axis, dyadic interpolation, and logarithmic derivatives.                          | Frontier; not formalized                                                                                                                                                                                               |
| Arithmetic bridge          | Complete moments as finite combinations of inverse-dyadic values, hence direct inputs for Bernoulli, Thue–Morse, and $q$ -binomial formulae. | Frontier here; arithmetic inputs formalized                                                                                                                                                                            |
| Rvachev transforms         | Piecewise antiderivatives, global normalized primitives, and fractional absolute moments.                                                    | Frontier; not formalized                                                                                                                                                                                               |
| Inverse Fabius             | Exact weighted reduction to $\int F^{\beta+1}$ , critical logarithmic formula, convergence threshold, and order-statistic interpretation.    | Frontier; not formalized                                                                                                                                                                                               |
| Nonlinear remainder        | Explicit finite Thue–Morse spline formula for $\int F^r$ , $r \in \mathbb{N}$ .                                                              | Frontier; not formalized                                                                                                                                                                                               |
| Endpoint inverse           | Weighted inverse-primitive equivalents, including the critical $\beta = -1$ law.                                                             | Frontier; informal consequence                                                                                                                                                                                         |
| Fractional primitive index | A continuous-index Volterra semigroup together with a $q$ -dilation transform and cocycle interpolating the integer ladder.                  | Generic positive-real ordered-continuous semigroup, positive affine covariance, derivative order raising, and the real one-step Fabius–Rvachev shifts formalized; iterated/complex $q$ -dilation structure conjectural |

Table 5: Main contributions and status.

and the Bernoulli recurrence

$$d_n = \frac{n2^n}{4(2^n - 1)}d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (161)$$

Thus any result expressed through the  $d_n$  is automatically connected to the corpus’s Bernoulli and exact dyadic layers.

## 2.2 Rvachev’s up-function and the signed extension

The even compactly supported bump is

$$\text{up}(x) = F(1 - |x|), \quad \text{supp up} = [-1, 1]. \quad (162)$$

Equivalently,  $\text{up}$  is the density of  $Y = 2X - 1$ . The signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{\text{wt}_2(m)} \text{up}(x - 2m - 1), \quad (163)$$

a locally finite sum satisfying

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (164)$$

It agrees with  $F$  on  $(-\infty, 1]$  but is not the bounded distribution function to the right of 1. This distinction is essential when extending primitive formulas globally.

### 2.3 Inverse function

Let

$$J = F^{-1} : [0, 1] \longrightarrow [0, 1]. \quad (165)$$

Then  $J(1 - y) = 1 - J(y)$  and  $J$  is smooth on  $(0, 1)$ , with infinite endpoint slope. The corpus establishes the leading lower-endpoint equivalent

$$J(y) \sim \frac{\sqrt{T}}{e\sqrt{\log 2}} \exp\left(-\sqrt{2(\log 2)T}\right), \quad T = \log \frac{1}{y} \longrightarrow \infty. \quad (166)$$

The sharper repository theory contains a periodic Gamma–zeta correction and Lambert– $W$  reversion; (166) is the interface needed for the first endpoint laws below.

## 3 The exact primitive ladder

### 3.1 A self-similar Volterra chain

For  $n \in \mathbb{N}_0$  define

$$A_0(x) = F(x), \quad A_n(x) = 2^{\binom{n}{2}} F(x/2^n) \quad (n \geq 1). \quad (167)$$

**Theorem 3.1** (Primitive ladder). *For  $x \leq 1$  and every  $n \geq 1$ ,*

$$A'_n(x) = A_{n-1}(x), \quad A_n(0) = 0. \quad (168)$$

*Consequently  $A_n$  is the  $n$ -fold zero-based primitive of  $F$ :*

$$\boxed{2^{\binom{n}{2}} F(x/2^n) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} F(t) dt.} \quad (169)$$

*For the signed extension  $\mathcal{F}$ , the same statements hold for every  $x \in \mathbb{R}$ .*

*Proof.* On  $(-\infty, 0)$  all the bounded candidates vanish identically, and flatness at zero supplies the same derivative identity there; the oriented Volterra integral also vanishes because  $F$  vanishes on  $(-\infty, 0]$ . For  $0 \leq x \leq 1$  and  $n \geq 1$ , one has  $0 \leq x/2^n \leq 1/2$ , so (157) gives

$$A'_n(x) = 2^{\binom{n}{2}-n} F'(x/2^n) = 2^{\binom{n}{2}-n+1} F(x/2^{n-1}) = 2^{\binom{n-1}{2}} F(x/2^{n-1}) = A_{n-1}(x).$$

All  $A_n(0)$  vanish. Iterating the fundamental theorem of calculus gives the Volterra kernel, with the negative-endpoint case interpreted as an oriented integral. For  $\mathcal{F}$ , equation (164) has no domain restriction, so the calculation is global. Equivalently, the bounded result on all of  $(-\infty, 1]$  follows from the global one and  $F = \mathcal{F}$  on that full initial half-line.  $\square$

**Remark 3.2** (Why the triangular exponent is inevitable). Let  $(\mathcal{J}f)(x) = \int_0^x f(t) dt$  and  $(Sf)(x) = f(x/2)$ . Then

$$\mathcal{J}S = 2S\mathcal{J}, \quad \mathcal{J}F = SF. \quad (170)$$

Therefore

$$\mathcal{J}^n F = 2^{0+1+\dots+(n-1)} S^n F = 2^{\binom{n}{2}} S^n F. \quad (171)$$

The coefficients are thus a  $q$ -Weyl phenomenon with  $q = 1/2$ , not an accidental by-product of repeated differentiation. This is the operator-level bridge to the half-base  $q$ -binomial calculus in the corpus.

### 3.2 Repeated integration by parts

**Theorem 3.3** (Finite weighted identity). *Let  $g \in C^N([0, x])$ , where  $0 \leq x \leq 1$ . Then*

$$\int_0^x g(t)F(t) dt = \sum_{k=0}^{N-1} (-1)^k g^{(k)}(x) A_{k+1}(x) + (-1)^N \int_0^x g^{(N)}(t) A_N(t) dt. \quad (172)$$

*The identity is global with  $F, A_n$  replaced by  $\mathcal{F}, 2^{\binom{n}{2}} \mathcal{F}(\cdot/2^n)$ .*

*Proof.* Use  $A'_{k+1} = A_k$  and integrate by parts successively. Every lower boundary term vanishes because  $A_k(0) = 0$ .  $\square$

## 4 Closed forms for nonnegative integer powers

### 4.1 The monomial formula

**Theorem 4.1** (Polynomial-weight antiderivative). *For every  $p \in \mathbb{N}_0$  and  $x \leq 1$ ,*

$$\int x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C. \quad (173)$$

*For the signed global extension  $\mathcal{F}$ , the same formula is valid for all  $x \in \mathbb{R}$ .*

*Proof.* For  $0 \leq x \leq 1$ , apply [theorem 3.3](#) with  $g(t) = t^p$  and  $N = p + 1$ . Since  $g^{(p+1)} = 0$ , the remainder vanishes. If  $x \leq 0$ , the oriented integral and every displayed scale term vanish because all their Fabius arguments are nonpositive. Adding an arbitrary constant converts the zero-based primitive to an indefinite antiderivative.  $\square$

The first cases are

$$\int F(x) dx = F(x/2) + C, \quad (174)$$

$$\int xF(x) dx = xF(x/2) - 2F(x/4) + C, \quad (175)$$

$$\int x^2 F(x) dx = x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \quad (176)$$

$$\int x^3 F(x) dx = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16) + C. \quad (177)$$

The cancellations are exact: differentiating the  $k$ th term produces one ordinary monomial derivative and one dilation derivative; the latter cancels the former contribution from the preceding scale.

**Corollary 4.2** (Arbitrary polynomial weight). *For a polynomial  $P$  of degree  $m$  and  $x \leq 1$ ,*

$$\int P(x)F(x) dx = \sum_{k=0}^m (-1)^k 2^{\binom{k+1}{2}} P^{(k)}(x) F(x/2^{k+1}) + C. \quad (178)$$

## 4.2 Continuation outside the unit interval

Formula (173) is a bounded- $F$  identity on  $(-\infty, 1]$ . For  $x > 1$ , define the normalized primitive

$$I_p(x) = \int_0^x t^p F(t) dt. \quad (179)$$

Then

$$I_p(x) = I_p(1) + \frac{x^{p+1} - 1}{p+1}, \quad x \geq 1, \quad (180)$$

while  $I_p(x) = 0$  for  $x \leq 0$ . Confusing this piecewise continuation with the global signed-extension formula is a normalization error:  $F$  is constant for  $x \geq 1$ , whereas  $\mathcal{F}$  continues in alternating compact blocks.

*Warning 4.3* (Formalization boundary beyond the natural finite core). The Lean theorems cover the bounded formulas through  $x \leq 1$  and the signed formulas globally. The elementary bounded continuation (180) for  $x > 1$  has not been packaged as a theorem in this antiderivative layer. Generic affine transport, basepoint shifts, and open-interval zero-tail Taylor jets are formalized, but they do not supply its explicit Fabius coefficients or its piecewise assembly. The positive-real single-step Fabius–Rvachev fractional shifts are now separately formalized, but not the later complex-power convergence theorem, Mellin continuation, iterated fractional-index lattice, Rvachev piecewise primitive formulas, or the local, higher-order, and fractional inverse-function reductions been formalized. They remain frontier mathematics even where this report gives an analytic proof.

## 4.3 Complete moments and exact arithmetic

Putting  $x = 1$  in (173) gives a finite inverse-dyadic formula.

**Corollary 4.4** (Moment bridge). For  $p \in \mathbb{N}_0$ ,

$$\int_0^1 x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} F(2^{-k-1}) \quad (181)$$

$$= \frac{1}{p+1} \sum_{j=1}^{p+1} (-1)^{j-1} \binom{p+1}{j} d_j \quad (182)$$

$$= \boxed{\frac{1 - d_{p+1}}{p+1}}. \quad (183)$$

*In particular, every integer-weight complete integral is rational.*

*Proof.* The first equality is evaluation of the primitive. Use  $2^{\binom{j}{2}} F(2^{-j}) = d_j/j!$  for the second. Finally, integration by parts against the probability measure  $dF$  gives

$$d_{p+1} = \int_0^1 x^{p+1} dF(x) = 1 - (p+1) \int_0^1 x^p F(x) dx.$$

□

Thus the antiderivative calculus is a direct consumer of every exact formula for  $F(2^{-n})$  in the repository. For example, with  $V_n = F(2^{-n})$ ,

$$V_0 = 1, \quad V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k, \quad (184)$$

so (181) is immediately executable in exact rational arithmetic. Alternatively one may insert the corpus's finite Thue–Morse formula or the translated  $q$ -binomial formula below. For  $m, n \in \mathbb{N}$  and arbitrary  $Q \in \mathbb{C}$ , its right-hand side is independent of  $Q$  (and equals the bounded value  $F(m/2^n)$  whenever  $m \leq 2^n$ ):

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}, \end{aligned} \quad (185)$$

or the Bernoulli recurrence (161). No new summation theorem is required: the primitive formula provides the missing finite linear functional that transports those arithmetic representations into integration.

The first values are

$$\begin{array}{c|cccccc} p & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline \int_0^1 x^p F(x) dx & \frac{1}{2} & \frac{13}{36} & \frac{5}{18} & \frac{1207}{5400} & \frac{251}{1350} & \frac{22661}{142884} \end{array} \quad (186)$$

#### 4.4 A scaled reflection identity

Because  $F(1-x) = 1 - F(x)$ , the same integrand has two different primitive representations. Equating them gives a family of functional identities.

**Proposition 4.5** (Integrated reflection). *Let  $I_j(x) = \int_0^x t^j F(t) dt$ . For  $p \in \mathbb{N}_0$  and  $0 \leq x \leq 1$ ,*

$$\sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1-x) = I_p(x) - \frac{x^{p+1}}{p+1} + \frac{d_{p+1}}{p+1}. \quad (187)$$

After inserting (173), this becomes a finite identity between values of  $F$  at the two dyadic scale towers  $x/2^r$  and  $(1-x)/2^r$ .

*Proof.* Substitution  $u = 1-x$  gives

$$\int x^p F(1-x) dx = - \sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1-x) + C.$$

The same integrand is  $x^p(1-F(x))$ , with primitive  $x^{p+1}/(p+1) - I_p(x)$ . Determine the constant at  $x = 1$  using (183).  $\square$

## 5 Fractional, negative, and complex powers

### 5.1 Flatness estimate

The all-complex-exponent result rests on more than formal repeated integration.

**Lemma 5.1** (Uniform Volterra flatness). *For every integer  $s \geq 0$  there is a constant  $C_s$  such that, for all  $n \geq 1$  and  $0 \leq x \leq 1$ ,*

$$0 \leq A_n(x) \leq C_s \frac{s!}{(s+n)!} x^{s+n} = C_s \frac{x^{s+n}}{(s+1)_n}. \quad (188)$$

*Proof.* Endpoint flatness gives  $F(t) \leq C_s t^s$  on  $[0, 1]$ . Insert this estimate into (169) and evaluate the beta integral:

$$A_n(x) \leq \frac{C_s}{(n-1)!} \int_0^x (x-t)^{n-1} t^s dt = C_s \frac{s!}{(s+n)!} x^{s+n}.$$

□

## 5.2 The exact complex-exponent series

For  $x > 0$  use the principal real logarithm and write  $x^\alpha = \exp(\alpha \log x)$ . Define

$$I_\alpha(x) = \int_0^x t^\alpha F(t) dt. \quad (189)$$

Unlike the corresponding integral with  $1 - F$ , this converges for every  $\alpha \in \mathbb{C}$ , because  $F$  is flat at zero.

**Theorem 5.2** (All-complex-exponent antiderivative). *For every  $\alpha \in \mathbb{C}$  and  $0 < x \leq 1$ ,*

$$I_\alpha(x) = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} 2^{\binom{k+1}{2}} x^{\alpha-k} F(x/2^{k+1}). \quad (190)$$

*The series is absolutely convergent and locally uniformly convergent in  $(\alpha, x) \in \mathbb{C} \times (0, 1]$ . It therefore defines an entire function of  $\alpha$ . For  $\alpha = p \in \mathbb{N}_0$  it terminates at  $k = p$  and reduces to (173).*

*For every  $N \geq 1$  the exact remainder after  $N$  terms is*

$$R_N(\alpha, x) = (-1)^N \alpha^{\underline{N}} \int_0^x t^{\alpha-N} A_N(t) dt. \quad (191)$$

*Proof.* Apply theorem 3.3 with  $g(t) = t^\alpha$ . This gives the first  $N$  terms and (191). Choose  $s$  so that  $\Re \alpha + s + 1 > 0$ . By lemma 5.1,

$$|R_N(\alpha, x)| \leq \frac{C_s |\alpha^{\underline{N}}|}{(s+1)_N} \frac{x^{\Re \alpha + s + 1}}{\Re \alpha + s + 1}. \quad (192)$$

For nonintegral  $\alpha$ ,  $\alpha^{\underline{N}} = (-1)^N \Gamma(N - \alpha) / \Gamma(-\alpha)$ , so the quotient in (192) is  $\mathcal{O}(N^{-\Re \alpha - s - 1})$ . For nonnegative integer  $\alpha$  it vanishes identically for large  $N$ . Uniform gamma-ratio estimates on compact  $\alpha$ -sets, with the polynomial definition of the falling factorial used across its removable singularities, give compact-uniform convergence. The same bound applied termwise proves absolute convergence. Entire dependence under the integral also follows directly from dominated convergence, using flatness of arbitrarily high order. □

**Corollary 5.3** (One-step scale recurrence). *For every  $\alpha \in \mathbb{C}$  and  $0 < x \leq 1$ ,*

$$I_\alpha(x) = x^\alpha F(x/2) - \alpha 2^\alpha I_{\alpha-1}(x/2). \quad (193)$$

*Here  $2^\alpha = \exp(\alpha \log 2)$ . Iterating (193) gives the finite expansion and exact remainder in theorem 5.2; letting the depth tend to infinity gives (190).*

*Proof.* Integrate once by parts with the primitive  $F(t/2)$ , and substitute  $t = 2u$  in the remaining integral. Convergence at the lower endpoint holds for every  $\alpha$  by flatness. □

**Remark 5.4** (A convergent integration-by-parts expansion). Ordinary repeated integration by parts often produces a divergent asymptotic series. Here the rapidly increasing dilation factor  $2^{\binom{k+1}{2}}$  is exactly defeated by the much smaller value  $F(x/2^{k+1})$ . The Volterra beta-factor  $(s+1)_{k+1}^{-1}$  exposes the mechanism. The result is a genuine convergent expansion for every complex exponent.



For  $x > 1$  the bounded-function continuation is

$$I_\alpha(x) = I_\alpha(1) + \begin{cases} \frac{x^{\alpha+1} - 1}{\alpha + 1}, & \alpha \neq -1, \\ \log x, & \alpha = -1, \end{cases} \quad (194)$$

where the first quotient has its usual removable value at  $\alpha = -1$ . For  $\mathcal{F}$  the series (190), with  $F$  replaced by  $\mathcal{F}$ , is valid for every  $x > 0$  because the dilation equation is global.

### 5.3 Negative integer powers

When  $\alpha = -m$ ,  $m \in \mathbb{N}$ , the two signs cancel because  $-m^{\bar{k}} = (-1)^k m^{\bar{k}}$ .

**Corollary 5.5** (Positive-term negative-power series). *For  $m \in \mathbb{N}$  and  $0 < x \leq 1$ ,*

$$\int_0^x \frac{F(t)}{t^m} dt = \sum_{k=0}^{\infty} m^{\bar{k}} 2^{\binom{k+1}{2}} x^{-m-k} F(x/2^{k+1}). \quad (195)$$

*All terms are nonnegative. In particular,*

$$\int_0^x \frac{F(t)}{t} dt = \sum_{k=0}^{\infty} k! 2^{\binom{k+1}{2}} x^{-k-1} F(x/2^{k+1}). \quad (196)$$

This positivity gives certified lower bounds from partial sums. Combining (192) with an explicit, formally verified flatness constant would turn it into a fully explicit rational enclosure at dyadic  $x$ .

### 5.4 Logarithmic powers, Stirling numbers, and Bell polynomials

Since  $I_\alpha(x)$  is entire,

$$\frac{\partial^r}{\partial \alpha^r} I_\alpha(x) = \int_0^x t^\alpha (\log t)^r F(t) dt. \quad (197)$$

Termwise differentiation in (190) gives

$$\begin{aligned} \int_0^x t^\alpha (\log t)^r F(t) dt &= \sum_{k \geq 0} (-1)^k 2^{\binom{k+1}{2}} F(x/2^{k+1}) x^{\alpha-k} \\ &\quad \times \sum_{\ell=0}^r \binom{r}{\ell} (\log x)^{r-\ell} \frac{d^\ell}{d\alpha^\ell} \alpha^{\bar{k}}. \end{aligned} \quad (198)$$

The derivative may be written without singularities through signed Stirling numbers  $s(k, j)$ :

$$\frac{d^\ell}{d\alpha^\ell} \alpha^{\bar{k}} = \sum_{j=\ell}^k s(k, j) j^{\bar{\ell}} \alpha^{j-\ell}. \quad (199)$$

Away from the removable roots  $0, 1, \dots, k-1$ , it also has the Bell form

$$\frac{d^\ell}{d\alpha^\ell} \alpha^{\bar{k}} = \alpha^{\bar{k}} \mathcal{B}_\ell(h_1, \dots, h_\ell), \quad (200)$$

where  $\mathcal{B}_\ell$  is the complete exponential Bell polynomial and

$$h_j = \psi^{(j-1)}(\alpha + 1) - \psi^{(j-1)}(\alpha - k + 1). \quad (201)$$

This supplies a direct antiderivative connection to the Bell-polynomial machinery used elsewhere in the corpus for saddle and inverse expansions.

At  $\alpha = 0$ ,  $\frac{d}{d\alpha} \alpha^{\bar{k}}|_{\alpha=0} = (-1)^{k-1} (k-1)!$  for  $k \geq 1$ , so

$$\int_0^x F(t) \log t dt = F(x/2) \log x - \sum_{k=1}^{\infty} (k-1)! 2^{\binom{k+1}{2}} x^{-k} F(x/2^{k+1}). \quad (202)$$

## 6 The Mellin transform as an entire dyadic interpolant

Define

$$\mathcal{M}_F(s) = \int_0^1 x^{s-1} F(x) dx. \quad (203)$$

**Theorem 6.1** (Entire Mellin series). *The function  $\mathcal{M}_F$  is entire and*

$$\mathcal{M}_F(s) = \sum_{k=0}^{\infty} (-1)^k s - 1^k 2^{\binom{k+1}{2}} F(2^{-k-1}). \quad (204)$$

For  $\Re s > 0$ ,

$$\mathcal{M}_F(s) = \frac{1 - \mathbb{E}X^s}{s}. \quad (205)$$

Since  $s \mapsto \mathbb{E}X^s$  is entire, this quotient continues to every  $s \in \mathbb{C}$ ; its singularity at  $s = 0$  is removable, with

$$\mathcal{M}_F(0) = -\mathbb{E} \log X. \quad (206)$$

*Proof.* Use [theorem 5.2](#) with  $\alpha = s - 1$ . For  $\Re s > 0$ , integration by parts against  $dF$  gives  $\mathbb{E}X^s = 1 - s\mathcal{M}_F(s)$ . Both sides are entire, so the identity continues to all  $s$ ; l'Hopital's rule gives (206).  $\square$

**Proposition 6.2** (Strict complete monotonicity and log-convexity). *For every real  $s$  and  $r \in \mathbb{N}_0$ ,*

$$(-1)^r \mathcal{M}_F^{(r)}(s) = \int_0^1 x^{s-1} (-\log x)^r F(x) dx > 0. \quad (207)$$

Consequently  $\mathcal{M}_F$  is strictly completely monotone on the entire real axis and strictly log-convex there.

*Proof.* Flatness justifies differentiation for every real  $s$ . Positivity is strict because the integrand is positive on a set of positive measure. Log-convexity follows from Cauchy–Schwarz applied to the measure  $x^{s-1} F(x) dx$ .  $\square$

The change of variables  $x = e^{-u}$  gives the bilateral-strength Laplace representation

$$\mathcal{M}_F(s) = \int_0^{\infty} e^{-su} F(e^{-u}) du, \quad (208)$$

which converges for every real  $s$  because  $F(e^{-u})$  decays faster than every ordinary exponential. Thus the inverse-dyadic values in (204) interpolate an entire, positive, log-convex Mellin object rather than merely its integer samples.

## 7 The transforms $1 - F(x)$ and $F(1 - x)$

The bounded normalization satisfies  $F(1 - x) = 1 - F(x)$  on all of  $\mathbb{R}$ . For  $x > 0$  and  $\alpha \neq -1$ , an indefinite primitive is

$$\int x^\alpha (1 - F(x)) dx = \frac{x^{\alpha+1}}{\alpha+1} - I_\alpha(x) + C. \quad (209)$$

At  $\alpha = -1$  it is

$$\int \frac{1 - F(x)}{x} dx = \log x - I_{-1}(x) + C. \quad (210)$$

The normalized integral from zero exists only for  $\Re \alpha > -1$ , because  $1 - F(0) = 1$ ; this contrasts with  $I_\alpha$ , which exists for every complex  $\alpha$ .

For  $p \in \mathbb{N}_0$ , substitution  $u = 1 - x$  gives the second finite form

$$\int x^p F(1 - x) dx = - \sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1 - x) + C. \quad (211)$$

Equations (209) and (211) are equal up to the constant computed in [proposition 4.5](#).

For example,

$$\int F(1 - x) dx = x - F(x/2) + C, \quad (212)$$

$$\int xF(1 - x) dx = \frac{x^2}{2} - xF(x/2) + 2F(x/4) + C. \quad (213)$$

## 8 Rvachev up-function primitives

### 8.1 Piecewise polynomial weights

On  $0 < x < 1$ ,  $\text{up}(x) = 1 - F(x)$ , so

$$\int x^p \text{up}(x) dx = \frac{x^{p+1}}{p+1} - I_p(x) + C. \quad (214)$$

On  $-1 < x < 0$ ,  $\text{up}(x) = F(1 + x)$  and, for integer  $p$ ,

$$\int x^p \text{up}(x) dx = \sum_{j=0}^p (-1)^{p-j} \binom{p}{j} I_j(1 + x) + C. \quad (215)$$

**Theorem 8.1** (Global normalized polynomial primitive of up). *Let*

$$H_p(x) = \int_{-\infty}^x t^p \text{up}(t) dt. \quad (216)$$

*Then*

$$H_p(x) = \begin{cases} 0, & x \leq -1, \\ \sum_{j=0}^p (-1)^{p-j} \binom{p}{j} I_j(1 + x), & -1 \leq x \leq 0, \\ \frac{(-1)^p d_{p+1}}{p+1} + \frac{x^{p+1}}{p+1} - I_p(x), & 0 \leq x \leq 1, \\ \frac{1 + (-1)^p}{p+1} d_{p+1}, & x \geq 1. \end{cases} \quad (217)$$

*Proof.* The first two branches follow from (215). Evenness gives

$$\int_{-1}^0 t^p \text{up}(t) dt = (-1)^p \int_0^1 t^p \text{up}(t) dt = \frac{(-1)^p d_{p+1}}{p+1}.$$

Add the positive-side primitive (214). The final branch is the complete moment.  $\square$

For  $p = 0$  this gives the particularly simple distribution function of the up-density:

$$H_0(x) = \begin{cases} 0, & x \leq -1, \\ F((1+x)/2), & -1 \leq x \leq 0, \\ \frac{1}{2} + x - F(x/2), & 0 \leq x \leq 1, \\ 1, & x \geq 1. \end{cases} \quad (218)$$

## 8.2 Fractional and negative powers

For  $0 < r \leq 1$  define the positive-side complement primitive

$$\Phi_\alpha(r) = \begin{cases} \frac{r^{\alpha+1}}{\alpha+1} - I_\alpha(r), & \alpha \neq -1, \\ \log r - I_{-1}(r), & \alpha = -1. \end{cases} \quad (219)$$

Then  $\Phi'_\alpha(r) = r^\alpha \text{up}(r)$  on  $(0, 1)$ . Hence for every complex  $\alpha$  it is a local antiderivative on the positive half of the support. If on  $-1 < x < 0$  one chooses the upper-half-plane branch  $x^\alpha = e^{i\pi\alpha}(-x)^\alpha$ , then

$$\int x^\alpha \text{up}(x) dx = \begin{cases} -e^{i\pi\alpha} \Phi_\alpha(-x) + C, & -1 < x < 0, \\ \Phi_\alpha(x) + C, & 0 < x < 1. \end{cases} \quad (220)$$

The lower-half-plane branch replaces  $e^{i\pi\alpha}$  by  $e^{-i\pi\alpha}$ . Since  $I_\alpha$  is given by (190), these are explicit convergent series on both open half-intervals, including every negative and fractional exponent.

A single primitive across the origin exists exactly when  $\Re\alpha > -1$  for the branch-free weight  $|x|^\alpha$ . Put

$$T_\alpha = \Phi_\alpha(1) = \frac{\mathbb{E}X^{\alpha+1}}{\alpha+1}. \quad (221)$$

Then the global normalized primitive is

$$\int_{-\infty}^x |t|^\alpha \text{up}(t) dt = \begin{cases} 0, & x \leq -1, \\ T_\alpha - \Phi_\alpha(-x), & -1 \leq x \leq 0, \\ T_\alpha + \Phi_\alpha(x), & 0 \leq x \leq 1, \\ 2T_\alpha, & x \geq 1. \end{cases} \quad (222)$$

In particular,

$$\int_{-1}^1 |x|^\alpha \text{up}(x) dx = \frac{2\mathbb{E}X^{\alpha+1}}{\alpha+1}, \quad \Re\alpha > -1. \quad (223)$$

The right side supplies a meromorphic continuation to all  $\alpha \in \mathbb{C}$ , with a simple pole of residue 2 at  $\alpha = -1$ . The pole records the genuine nonintegrability of  $|x|^{-1} \text{up}(x)$  at the central point, in contrast with the endpoint-flat integral  $I_{-1}$ .

For  $m \in \mathbb{N}_0$ ,

$$\int_{-1}^1 x^{2m} \text{up}(x) dx = \frac{2d_{2m+1}}{2m+1}, \quad \int_{-1}^1 x^{2m+1} \text{up}(x) dx = 0. \quad (224)$$

Thus the antiderivative formula recovers the moment bridge between the Fabius distribution and the infinite sinc product  $\widehat{\text{up}}(\xi) = \prod_{j \geq 1} \text{sinc}(2^{-j}\xi)$ : the Taylor coefficients of the product are exactly the complete even moments in (224).

## 9 The signed global extension

Since (164) holds on all of  $\mathbb{R}$ , the cleanest version of the calculus belongs to  $\mathcal{F}$ .

**Theorem 9.1** (Global polynomial and complex-power formulas). *For  $p \in \mathbb{N}_0$  and all  $x \in \mathbb{R}$ ,*

$$\int x^p \mathcal{F}(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} \mathcal{F}(x/2^{k+1}) + C. \quad (225)$$

For  $x > 0$  and  $\alpha \in \mathbb{C}$ ,

$$\int_0^x t^\alpha \mathcal{F}(t) dt = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} 2^{\binom{k+1}{2}} x^{\alpha-k} \mathcal{F}(x/2^{k+1}), \quad (226)$$

with absolute, locally uniform convergence.

The signed blocks in (163) therefore possess two complementary primitive descriptions: a finite scaled- $\mathcal{F}$  formula and a locally finite sum of shifted primitives of up. Equating them gives blockwise Thue–Morse identities that may be useful for formalizing the interaction between the global binary series and integration.

## 10 Integrals of the inverse Fabius function

### 10.1 The unweighted primitive

**Theorem 10.1** (Exact inverse primitive). For  $0 < y < 1$ ,

$$\int J(y) dy = yJ(y) - F(J(y)/2) + C. \quad (227)$$

The zero-based normalization is obtained with  $C = 0$ .

*Proof.* Set  $a = J(y)$ , so  $y = F(a)$ . Changing variables  $t = F(u)$  gives

$$\int_0^y J(t) dt = \int_0^a uF'(u) du = ay - \int_0^a F(u) du = yJ(y) - F(J(y)/2).$$

The last step is (174). □

At  $y = 1$ , (227) gives  $\int_0^1 J(y) dy = 1 - F(1/2) = 1/2$ , as also follows from inverse symmetry.

### 10.2 Arbitrary complex weights

For  $\Re\beta > -1$  define

$$K_\beta(y) = \int_0^y t^\beta J(t) dt. \quad (228)$$

**Theorem 10.2** (Weighted inverse reduction). Let  $0 < y \leq 1$  and put  $a = J(y)$ . If  $\beta \neq -1$  and the integrals are interpreted on any common local interval avoiding zero, then

$$\int y^\beta J(y) dy = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int^{J(y)} F(x)^{\beta+1} dx + C. \quad (229)$$

For  $\Re\beta > -1$ , the zero-based form is

$$K_\beta(y) = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int_0^{J(y)} F(x)^{\beta+1} dx. \quad (230)$$

At the critical exponent  $\beta = -1$ ,

$$K_{-1}(y) = J(y) \log y - \int_0^{J(y)} \log F(x) dx. \quad (231)$$

*Proof.* With  $x = J(y)$ ,

$$K_\beta(y) = \int_0^a x F(x)^\beta F'(x) dx.$$

For  $\beta \neq -1$ , integrate  $x d(F(x)^{\beta+1})/(\beta+1)$  by parts. At  $\beta = -1$ , integrate  $x d \log F(x)$ . The endpoint boundary  $x \log F(x) \rightarrow 0$  follows from the repository's small-argument asymptotic (and already from sufficiently strong flatness bounds).  $\square$

For  $p \in \mathbb{N}_0$  this becomes

$$K_p(y) = \frac{y^{p+1} J(y)}{p+1} - \frac{1}{p+1} \int_0^{J(y)} F(x)^{p+1} dx. \quad (232)$$

The reduction is exact, but nonlinear: unlike the forward monomial primitive, it introduces a power of  $F$ . This identifies the precise missing object needed for a closed inverse calculus.

### 10.3 Endpoint integrability threshold

**Proposition 10.3** (Sharp threshold). *The normalized integral  $K_\beta(y)$  is absolutely convergent at zero exactly when*

$$\Re \beta \geq -1. \quad (233)$$

For  $\Re \beta < -1$  it diverges, although the local antiderivative (229) remains valid away from zero.

*Proof.* Put  $t = e^{-u}$ . Up to a harmless phase,

$$\left| t^\beta J(t) dt \right| = e^{-(\Re \beta + 1)u} J(e^{-u}) du.$$

By (166),  $J(e^{-u})$  decays like  $\sqrt{u} e^{-\sqrt{2(\log 2)u}}$ , which is subexponential in  $u$  but integrable. The assertion follows according as the ordinary exponential factor decays, is absent, or grows.  $\square$

### 10.4 Complete inverse moments and order statistics

Let  $M_r = \max(X_1, \dots, X_r)$  for iid copies of  $X$ . Since  $\mathbb{P}(M_r \leq x) = F(x)^r$ ,

$$\int_0^1 F(x)^r dx = \mathbb{E}(1 - M_r). \quad (234)$$

Putting  $y = 1$  in (232) gives

$$\boxed{\int_0^1 y^p J(y) dy = \frac{\mathbb{E} M_{p+1}}{p+1}.} \quad (235)$$

Thus inverse moments are normalized expected maxima of Fabius random variables. This probabilistic meaning is complementary to the algebraic inverse-dyadic and Lambert–Bell analysis in the repository's inverse-frontier report.

At the critical exponent,

$$\int_0^1 \frac{J(y)}{y} dy = - \int_0^1 \log F(x) dx, \quad (236)$$

an entropy-like constant attached to the distribution function rather than to its density.

## 11 A finite Thue–Morse formula for the nonlinear inverse remainder

The integral of  $F^r$  in (232) is not reduced by the linear primitive ladder. It nevertheless has an exact finite-level representation. Let

$$X_N = \sum_{j=1}^N 2^{-j} U_j \quad (237)$$

and let  $F_N$  be its distribution function. Inclusion–exclusion for unequal box splines gives

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{m=0}^{2^N-1} (-1)^{\text{wt}_2(m)} (x - m/2^N)_+^N. \quad (238)$$

Moreover  $F_N \rightarrow F$  uniformly.

For an integer  $r \geq 1$ , a cutoff  $a \in [0, 1]$ , and a tuple  $\mathbf{m} = (m_1, \dots, m_r)$ , put

$$b(\mathbf{m}) = \max_i \frac{m_i}{2^N}, \quad \delta_i = b(\mathbf{m}) - \frac{m_i}{2^N}. \quad (239)$$

Define

$$Q_{N,r}(a; \mathbf{m}) = \mathbf{1}_{\{a > b(\mathbf{m})\}} \sum_{k_1, \dots, k_r=0}^N \frac{\prod_{i=1}^r \binom{N}{k_i} \delta_i^{N-k_i}}{1 + k_1 + \dots + k_r} \times (a - b(\mathbf{m}))^{1+k_1+\dots+k_r}. \quad (240)$$

**Theorem 11.1** (Finite Thue–Morse nonlinear integral). *For every  $N, r \in \mathbb{N}$  and  $a \in [0, 1]$ ,*

$$\int_0^a F_N(x)^r dx = \left( \frac{2^{N(N+1)/2}}{N!} \right)^r \sum_{m_1, \dots, m_r=0}^{2^N-1} (-1)^{\sum_i \text{wt}_2(m_i)} Q_{N,r}(a; \mathbf{m}). \quad (241)$$

$$\int_0^a F(x)^r dx = \lim_{N \rightarrow \infty} \int_0^a F_N(x)^r dx. \quad (242)$$

Consequently (232) has a completely explicit finite Thue–Morse spline limit for every  $p \in \mathbb{N}_0$ .

*Proof.* Expand the  $r$ th power of (238). For a fixed tuple, the product vanishes below  $b = b(\mathbf{m})$ . Above  $b$ , set  $z = x - b$  and write

$$\prod_{i=1}^r (x - m_i/2^N)^N = \prod_{i=1}^r (z + \delta_i)^N.$$

The binomial theorem followed by termwise integration in  $z$  gives (240). Uniform convergence  $F_N \rightarrow F$  and  $0 \leq F_N, F \leq 1$  give (242) by dominated convergence.  $\square$

The formula is exponentially expensive in both  $N$  and  $r$ . Its value is structural: it proves that the nonlinear remainder is still governed by the same finite Thue–Morse geometry, and it offers a starting point for orbit compression, Walsh transforms, and  $q$ -binomial acceleration.

## 12 Endpoint asymptotics for inverse primitives

Let

$$T = \log(1/y), \quad L = \log 2, \quad a = \sqrt{2L}, \quad C_0 = \frac{1}{e\sqrt{L}}. \quad (243)$$

Then (166) reads  $J(y) \sim C_0 \sqrt{T} e^{-a\sqrt{T}}$ .

**Theorem 12.1** (Subcritical weighted inverse primitives). *Let  $\rho = \beta + 1$  with  $\Re \rho > 0$ . As  $y \downarrow 0$ ,*

$$\boxed{K_\beta(y) \sim \frac{y^\rho J(y)}{\rho}}. \quad (244)$$

For real  $\rho > 0$ , the leading model admits the refined endpoint denominator

$$K_\beta(y) = \frac{y^\rho J(y)}{\rho + \frac{a}{2\sqrt{T}} - \frac{1}{2T}} (1 + \mathcal{O}(T^{-1}) + \text{periodic-transfer terms}), \quad (245)$$

where a fully rigorous coefficient at the displayed relative order requires inserting the corpus's periodic next term for  $J$ .

*Proof.* After  $t = e^{-u}$ ,

$$K_\beta(y) = \int_T^\infty e^{-\rho u} J(e^{-u}) du.$$

The factor  $J(e^{-u})$  is slowly varying on additive  $u$ -scales compared with  $e^{-\rho u}$ . Direct one-sided Laplace estimation using (166) gives (244). Differentiating the leading exponent  $\rho u + a\sqrt{u} - \frac{1}{2} \log u$  gives the denominator in (245); the repository's periodic correction must be retained before promoting the indicated structure to a complete second-term theorem.  $\square$

At the critical exponent the ordinary exponential disappears and the scale changes.

**Theorem 12.2** (Critical inverse primitive). *As  $y \downarrow 0$ ,*

$$K_{-1}(y) \sim \frac{\sqrt{2}}{e \log 2} T \exp\left(-\sqrt{2(\log 2)T}\right) \quad (246)$$

$$\sim \sqrt{\frac{2T}{\log 2}} J(y). \quad (247)$$

*Proof.* Now

$$K_{-1}(y) = \int_T^\infty J(e^{-u}) du \sim C_0 \int_T^\infty \sqrt{u} e^{-a\sqrt{u}} du.$$

Set  $v = \sqrt{u}$ . The integral becomes  $2C_0 \int_{\sqrt{T}}^\infty v^2 e^{-av} dv$ , whose leading term is  $(2C_0/a)T e^{-a\sqrt{T}}$ . Substitute the constants.  $\square$

For  $\Re \beta < -1$ , a primitive based at a fixed  $y_0 > 0$  diverges at zero with leading magnitude  $-y^{\beta+1} J(y)/(\beta+1)$ . Thus  $\beta = -1$  is a true transition: above it, the endpoint factor is  $1/(\beta+1)$ ; at it, an extra  $\sqrt{\log(1/y)}$  appears; below it, the same boundary term drives divergence.

## 13 Numerical experiments and exact verification

The accompanying script `experiments.py` is network-free and performs three logically separate checks.

1. It computes  $V_n = F(2^{-n})$  exactly from (184) using rational arithmetic.
2. It represents  $F(x/2^j)$  as a formal symbol  $F_j$  with derivative rule  $F_j' = 2^{1-j} F_{j-1}$  and verifies the cancellation in (173) exactly for  $0 \leq p \leq 12$ .
3. It compares the series (190) at  $x = 1$  with a scrambled Sobol simulation of the random series (158), using (205). This last item is only an independent numerical check.



The exact recurrence begins

$$V_0 = 1, \quad V_1 = \frac{1}{2}, \quad V_2 = \frac{5}{72}, \quad V_3 = \frac{1}{288}, \quad V_4 = \frac{143}{2073600}. \quad (248)$$

All formal derivative tests and the two moment evaluations (181)–(183) pass exactly.

| $\alpha$ | series, 240 terms  | Sobol mean        | scramble spread        |
|----------|--------------------|-------------------|------------------------|
| $-1$     | 0.7582400001750290 | 0.758239962896570 | $2.70 \times 10^{-7}$  |
| $-1/2$   | 0.6073863107353257 | 0.607386310704139 | $2.51 \times 10^{-8}$  |
| $1/2$    | 0.4209681733402818 | 0.420968175002922 | $4.38 \times 10^{-10}$ |
| $2.3$    | 0.2591812728638399 | 0.259181273425061 | $5.26 \times 10^{-11}$ |

Table 6: Independent check of  $I_\alpha(1)$ . Each Sobol mean uses four scrambles of  $2^{19}$  points and 28 random-series coordinates. The finite tail and QMC sampling error explain the displayed differences.

The slower apparent convergence at  $\alpha = -1$  is expected: the term quotient is only polynomial after the superexponential dyadic factors have canceled. The analytic proof allows an arbitrarily strong polynomial majorant by choosing a higher flatness order  $s$ , but the practical constant then worsens. This is a natural target for Richardson acceleration based on the exact tail shape.

## 14 Further deductions

### 14.1 An exponential-weight generating primitive

Taking  $g(x) = e^{zx}$  in the infinite version of theorem 3.3 gives, for  $0 < x \leq 1$ ,

$$\int_0^x e^{zt} F(t) dt = e^{zx} \sum_{k=0}^{\infty} (-z)^k 2^{\binom{k+1}{2}} F(x/2^{k+1}). \quad (249)$$

The series is entire in  $z$ . Expanding both sides recovers all polynomial primitives simultaneously. At  $x = 1$  it provides a direct dyadic expansion for the integrated moment-generating function of  $F$ .

### 14.2 A uniqueness principle for the dyadic coefficient row

Suppose a family of coefficients  $c_k$  satisfies

$$\int P(x) F(x) dx = \sum_{k \geq 0} c_k P^{(k)}(x) F(x/2^{k+1}) + C$$

for all polynomials  $P$ , with only finitely many nonzero terms for each  $P$ . Testing successively on  $1, x, x^2, \dots$  forces  $c_k = (-1)^k 2^{\binom{k+1}{2}}$ . Thus the triangular coefficient row in (178) is the unique local derivative functional built from the scale tower  $F(x/2^{k+1})$ .

### 14.3 A critical complement contrast

There is an instructive asymmetry at exponent  $-1$ :

$$\int_0^x \frac{F(t)}{t} dt < \infty, \quad \int_0^x \frac{1 - F(t)}{t} dt = \infty. \quad (250)$$

The first has the positive series (196); the second has only the local primitive (210). After folding, the same obstruction is the pole of the fractional up-moment at  $\alpha = -1$ .

## 15 Conjectures and frontier directions

**Conjecture 15.1** (Dyadic-tree algebra for nonlinear primitives). *For each integer  $r \geq 2$ , the functions*

$$Q_r(a) = \int_0^a F(x)^r dx$$

*admit an absolutely convergent expansion indexed by finite rooted dyadic trees. The vertex weights should be the same inverse-dyadic moments  $d_n$ , while tree symmetries should produce partial Bell polynomials and half-base Gaussian binomial coefficients. The finite spline formula (241) is expected to collapse to this tree expansion after Walsh–Thue–Morse orbit compression.*

**Conjecture 15.2** (Periodic Lambert transfer for inverse primitives). *Let the sharp inverse expansion be written in the repository’s phase variable as*

$$J(y) = J_0(y) \left( 1 + \frac{A_1(T, \theta)}{\sqrt{T}} + \frac{A_2(T, \theta)}{T} + \dots \right),$$

*where the  $A_j$  belong to the differential algebra generated by the periodic Gamma–zeta correction. Then, for every fixed  $\Re \beta > -1$ ,  $K_\beta(y)$  has an all-orders expansion in the same differential algebra, obtained by the formal resolvent*

$$\left( \beta + 1 + \frac{d}{dT} \right)^{-1}.$$

*At  $\beta = -1$  the resolvent becomes singular and raises the polynomial degree in  $\sqrt{T}$  by one, explaining (246).*

**Conjecture 15.3** (Sharp denominator law for polynomial primitives). *At a dyadic argument  $x = a/2^n$ , the reduced denominator of the finite primitive (173) has a predictable 2-adic valuation obtained by the minimum of the known valuations of  $F(a/2^{n+k+1})$ , except on explicitly classifiable cancellation congruence classes governed by Thue–Morse parity. The odd part should factor through the Mersenne products appearing in the half-base  $q$ -Pochhammer symbols.*

**Conjecture 15.4** (Stieltjes structure of the negative-power row). *For each  $m \in \mathbb{N}$  and fixed  $x \in (0, 1]$ , the positive terms in (195), after a natural factorial normalization, form a Stieltjes moment sequence. If true, diagonal Padé approximants would give alternating certified bounds and a substantially faster evaluation of  $I_{-m}(x)$ .*

**Conjecture 15.5** (Continuous primitive index and dilation cocycle). *Let  $A_\nu = \mathcal{J}^\nu F$  be the Riemann–Liouville fractional primitive, which already supplies the semigroup law  $A_{\nu+\mu} = \mathcal{J}^\nu A_\mu$ . There is a canonical transform  $\mathcal{T}_\nu$ , holomorphic in  $\nu$  for  $\Re \nu > 0$ , and a scalar cocycle  $c(\nu)$  such that*

$$A_\nu(x) = c(\nu) (\mathcal{T}_\nu F)(2^{-\nu} x), \quad c(n) = 2^{\binom{n}{2}}, \quad \mathcal{T}_n F = F \quad (n \in \mathbb{N}).$$

*In Mellin coordinates,  $c$  should be expressible through a half-base  $q$ -gamma or Barnes-type function, while  $\mathcal{T}_\nu$  should become the identity exactly at integer indices. The obstruction is that the integer formula uses the special vector  $F$  and does not arise from an operator identity on arbitrary functions. The positive-real semigroup premise for continuous Banach-valued inputs on ordered compact intervals is `fractionalVolterra_add`. The real one-step Fabius shift in `fractionalVolterra_add_one_fabiusReal` proves the  $n = 1$  order increment underlying this conjecture; the iterated packaging, holomorphic transform, scalar cocycle, and complex-order continuation asserted here remain conjectural.*

## 15.1 Promising research program

The following projects are concrete and separable.

1. **Lean continuation.** The natural-order primitive ladder and the finite polynomial/monomial commutators are complete, including their stronger real-normed-target Volterra forms, affine covariance, and local basepoint/zero-tail Taylor-jet calculus. The positive-real fractional core now also has positive affine covariance, derivative order raising, and one-step signed, bounded, and Up-to-Fabius shifts. These results do not supply iterated/complex-order packaging, fractional derivatives, weighted fractional formulas, endpoint lattices, or the report’s piecewise branches. The next analytic boundary is the complex series, which requires an effective flatness lemma, a falling-factorial derivative library, and a uniform remainder theorem.
2. **Certified fractional evaluation.** Combine the exact rational recurrence (184) with an explicit formal flatness estimate to produce interval bounds for (190) at rational  $\alpha$ .
3.  **$q$ -Richardson acceleration.** Determine the full asymptotic expansion of the positive tail in (195). The geometric scale suggests the  $q = 1/4$  Lagrange filters already developed in the frontier corpus. `AnalyticSeriesFilter.lean` now supplies their generic unconditional pointwise Gaussian-residual series identity under explicit nonzero-weight sample summability; proving that this particular tail satisfies those hypotheses, together with uniform bounds, signs, and an acceleration rate, remains part of the proposed work.
4. **Nonlinear Walsh compression.** Replace the  $2^{Nr}$  tuple sum in (241) by orbit sums under coordinate permutation, bitwise XOR, and dyadic reflection; search for Gaussian-binomial collapse.
5. **Inverse periodic transfer.** Insert the established periodic Lambert–Bell expansion of  $J$  into the endpoint integrals and prove [conjecture 15.2](#) with uniform phase derivatives.
6. **Fourier interpretation.** Express the even up-moments generated by (223) as fractional derivatives at the origin of the infinite sinc product, clarifying branch choices and possible Mellin–Barnes continuation.
7. **Arithmetic experiments.** Tabulate reduced denominators of (173) at general dyadic  $x$  and compare them with the formal 2-adic lower bounds and Mersenne factors in the current Lean corpus.

## 16 Lean crosswalk and remaining formalization frontier

The following are the exact current module and declaration names. The first module is independent of the Fabius function: it proves a reusable natural-order Volterra calculus for arbitrary real normed-space-valued inputs; the iteration/kernel bridge assumes completeness. This is strictly stronger than proving the report’s scalar identities one at a time.

| Module                                         | Declaration                                                                                                                                            | Machine-checked content                                                                                                |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>FabiusFunction.NormализedVolterra</code> | <code>volterraPrimitive</code> ,<br><code>iteratedPrimitive</code> ,<br><code>normalizedVolterra</code>                                                | Literal iteration and normalized Cauchy kernels at arbitrary base points and oriented endpoints, including order zero. |
| <code>FabiusFunction.NormализedVolterra</code> | <code>iteratedPrimitive_add</code> ,<br><code>iteratedPrimitive_succ_hasStrictDerivAt</code> , <code>iteratedPrimitive_eq_eq_normalizedVolterra</code> | Semigroup law, strict FTC ladder, and equality of literal iteration with the kernel formula for continuous inputs.     |

| Module                                     | Declaration                                                                                                                                         | Machine-checked content                                                                                                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_affine, normalizedVolterra_comp_affine                                                                                           | Forward affine covariance for arbitrary real normed targets and every real $c$ , including zero and negative scales; the inverse form assumes $c \neq 0$ .                                                                     |
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_basepoint_shift                                                                                                                  | Basepoint shift with interval integrability only from $a$ to $b$ and from $b$ to $x$ , no endpoint order, and every order including $n = 0$ .                                                                                  |
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_succ_eq_taylor_of_eq_zero                                                                                                        | Positive-order Taylor jet with integrability only from $a$ to $b$ and vanishing on the open unoriented interval between $b$ and $x$ ; neither endpoint order nor endpoint values are constrained.                              |
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_kernel_pow                                                                                                                       | All-order kernel raising with the exact rising-factorial coefficient.                                                                                                                                                          |
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_polynomial                                                                                                                       | Finite Hasse-derivative commutator for every real polynomial and every interval-integrable normed-space-valued input.                                                                                                          |
| FabiusFunction.NormalizedVolterra          | normalizedVolterra_monomial                                                                                                                         | Natural-monomial specialization, with every natural Volterra order and all base/endpoint cases included.                                                                                                                       |
| FabiusFunction.FractionalVolterra          | fractionalVolterra, fractionalVolterra_nat_succ                                                                                                     | Total Gamma-normalized real-order operator and its exact positive-natural-order bridge to normalizedVolterra.                                                                                                                  |
| FabiusFunction.FractionalVolterraSemigroup | fractionalVolterra_add                                                                                                                              | Addition of two positive real orders for a continuous Banach-valued input on an ordered compact interval, backed by the normalized beta-kernel theorem.                                                                        |
| FabiusFunction.FractionalVolterraCalculus  | fractionalVolterra_affine, fractionalVolterra_comp_affine                                                                                           | Positive affine covariance for arbitrary real order and its inverse-smul form; the scale is strictly positive and the endpoint is ordered.                                                                                     |
| FabiusFunction.FractionalVolterraCalculus  | fractionalVolterra_add_one_deriv, fractionalVolterra_add_one_deriv_of_eq_zero                                                                       | Positive-order integration by parts for a continuous Banach-valued primitive with a right interior derivative and interval-integrable derivative, including the exact left-boundary term and its zero-boundary specialization. |
| FabiusFunction.GlobalExtension             | fabiusReal_eq_extendedFabius_of_le_one                                                                                                              | Equality of the bounded function and signed extension throughout $(-\infty, 1]$ .                                                                                                                                              |
| FabiusFunction.FabiusAntiderivatives       | normalizedVolterra_extendedFabius                                                                                                                   | Complete signed primitive ladder for every natural order and every real endpoint.                                                                                                                                              |
| FabiusFunction.FabiusAntiderivatives       | normalizedVolterra_fabiusReal_of_le_one                                                                                                             | Complete bounded primitive ladder on the proved initial half-line $x \leq 1$ .                                                                                                                                                 |
| FabiusFunction.FabiusAntiderivatives       | normalizedVolterra_pow_mul_extendedFabius                                                                                                           | Finite natural-monomial formula for the signed extension, globally and at every natural Volterra order.                                                                                                                        |
| FabiusFunction.FabiusAntiderivatives       | normalizedVolterra_pow_mul_fabiusReal_of_le_one                                                                                                     | Finite natural-monomial formula for the bounded function when $x \leq 1$ .                                                                                                                                                     |
| FabiusFunction.FabiusFractionalVolterra    | rvachevFractionalPrimitive, rvachevFractionalPrimitive_eq_intervalIntegral_min, rvachevFractionalPrimitive_nat_succ, rvachevFractionalPrimitive_add | Total causal Rvachev fractional primitive, its positive-order support cutoff, its positive-natural bridge, and its positive-order semigroup on $x \geq -1$ .                                                                   |
| FabiusFunction.FabiusFractionalVolterra    | fractionalVolterra_add_one_extendedFabius_of_nonneg, fractionalVolterra_add_one_fabiusReal, fractionalVolterra_add_one_rvachevUp                    | Real-positive one-step shifts for the signed extension on $x \geq 0$ , the bounded function on $[0, 1]$ , and the Up-to-Fabius bridge based at $-1$ on $x \geq -1$ .                                                           |

Thus the former placeholder declarations for the integer ladder and finite polynomial weights are

retired. The generic affine and basepoint declarations do not calculate explicit Fabius/up coefficients or assemble report-specific piecewise branches. The remaining obligations are analytic, nonlinear, or explicit piecewise specializations:

- the complex-power series (190), including absolute and compact-local uniform convergence and a formal remainder bound;
- logarithmic weights and the entire Mellin theory derived from that series;
- fractional Fabius/Rvachev formulas beyond the formalized positive-real stopped-power, endpoint-moment, causal Rvachev semigroup, and three single-step order shifts, and any shifted, reflected, weighted, or complex-order semigroup interpolating the natural orders (the generic positive-real kernel and its ordered-continuous semigroup are formal);
- the iterated/lattice, endpoint-up-wrapper, hierarchy-weighted monomial, negative-endpoint, piecewise-coefficient, explicit exterior tail, refinement, and fractional-derivative Rvachev formulas beyond the causal primitive, including their Fourier sinc-product interpretations;
- every inverse-Fabius primitive, nonlinear-power reduction, Thue–Morse limit, and endpoint asymptotic in this report; and
- a named global piecewise primitive theorem for the bounded function to the right of 1 (distinct from the already formalized signed global ladder).

## 17 Conclusion

The Fabius differential equation contains an integration rule that is simpler than its differentiation rule: one integration merely halves the argument. Noncommutation of integration and halving accumulates the triangular factor  $2^{\binom{n}{2}}$ . Once this ladder is isolated, every polynomial-weighted antiderivative is finite, every complex-power antiderivative becomes a convergent series, and the Mellin transform becomes an entire dyadic interpolant. Reflection turns the same calculus into exact formulas for  $1 - F(x)$ ,  $F(1 - x)$ , and up; the signed extension makes it global.

The inverse function has a different but equally clean structure. Its weighted primitives reduce exactly to nonlinear powers of  $F$ , revealing the specific obstacle to an elementary inverse calculus. Those nonlinear powers still possess a finite Thue–Morse spline limit, and the Lambert– $W$  endpoint theory transfers to the primitives with a sharp critical transition at weight  $y^{-1}$ .

The main practical outcome is a compact toolbox: finite formulas for integer powers, convergent formulas for arbitrary complex powers, exact arithmetic at complete or dyadic endpoints, and a clear list of nonlinear and asymptotic problems remaining for future work.

## A Proof details for compact-uniform convergence

Let  $K \subset \mathbb{C}$  be compact and let  $\sigma_- = \min_{\alpha \in K} \Re \alpha$ . Choose an integer  $s$  with  $s + \sigma_- + 1 > 1$ . By lemma 5.1, the absolute value of the  $k$ th term of (190) is bounded by

$$C_s x^{\sigma_- + s + 1} \frac{|\alpha^k|}{(s + 1)_{k+1}}. \quad (251)$$

On a compact set, the Pochhammer polynomial admits a uniform estimate

$$|\alpha^k| \leq C_K \Gamma(k - \sigma_- + c_K) \quad (252)$$

for a fixed harmless  $c_K$ . Gamma-ratio asymptotics then give a summable majorant  $C_{K,s} k^{-s-\sigma_- - 2 + c_K}$  after increasing  $s$  if necessary. The same calculation with  $(s + 1)_N$  rather than  $(s + 1)_{N+1}$  proves

uniform vanishing of the remainder. Uniformity in  $x$  holds on every compact subinterval  $[\varepsilon, 1]$ ; at  $x = 0$  the primitive is identically zero but  $x^\alpha$  itself has no compact-uniform meaning across arbitrary complex  $\alpha$ .

## B Corpus inventory by document family

The following ledger records the families used to establish the nonduplication boundary. The public tree is live; names are those visible on 27 August 2026.

| Family                                           | Role in the audit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabius_Function_and_Rvachev_Up                   | Primary synthesis: definitions, probability, moments, exact inverse-dyadic and general dyadic values, Bernoulli recurrences, $q$ -binomial formulas, global binary series, Fourier products, asymptotics, and Lean crosswalk.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| semi-formalized-research-frontiers               | Consolidated integration, repeated-integration, dyadic web, inverse, small-argument, Fourier-decay, and gap-register dossiers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Glossary, non-elementarity, and Lean walkthrough | Normalization and notation checks; inverse and elementary-function boundaries; formalization status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Paper transcriptions                             | Fabius, Arias de Reyna, arithmetic, lacunary-product, and related source material used by the repository.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Current draft families                           | Fabius_Arithmetic_Rays_Frontier_Report, Fabius_Dyadic_Self_Sampling_Frontier_Package, Fabius_Half_Integer_Spectral_Frontier_Report, Fabius_Inverse_Frontier_Report_Source_and_PDF, Fabius_Newton_Rvachev_Frontier_Report, the two Fabius_Rvachev_Frontier_Report families, Fabius_Rvachev_New_Frontiers, Fabius_Rvachev_Thue_Morse_Frontier_Results, Spectral_Arithmetic_Pascal_Rvachev_Hierarchy, Thue_Morse_Formula_Atlas, the dyadic-inverse-Barnes, new-results, results-bundle, spectral-endpoint, and rvachev_up_fourier_decay families. These set the newest nonduplication boundary for inverse, Bell, Mellin, $q$ -Richardson, and sinc claims. |
| Archive manifest                                 | Frozen early notes; standalone non-elementarity, small-argument, and inverse dyadic reports; six primary-exposition generations; two supplements; two $q$ -calculus reports; two Lean reports; two glossary versions; and the frozen dyadic-asymptotic, alternative-representation, formula-to-asymptotic, $q$ -connection, Thue-Morse convergence, gap-register, repeated-integration, and formula-atlas reports.                                                                                                                                                                                                                                       |
| Auxiliary TeX fragments                          | Generated tables and display fragments in self-sampling and numerical bundles; checked through their parent reports and file inventories rather than treated as independent mathematical claims.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

The audit found integration in the senses of moments, probability, quadrature, and construction by repeated integration. It did not find the scale-ladder antiderivative formulas, the exact all-complex-exponent series, or the weighted inverse reductions of this report.

## C Reproducibility files

The accompanying archive contains:

- `Fabius_Monomial_Antiderivatives.tex`, this source;

- `Fabius_Monomial_Antiderivatives.pdf`, the rendered report;
- `experiments.py`, exact and numerical checks with detailed comments;
- `numerical_results.txt`, the full console report;
- `series_convergence.csv`, selected partial sums;
- `README.md` and `SHA256SUMS`.

Run

```
python experiments.py --output-dir .
```

to regenerate the numerical text and CSV files. The script requires Python 3, `mpmath`, `numpy`, and `scipy`; it makes no network requests.

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- [12] V. Reshetnikov, *Inverse Frontiers for the Fabius–Rvachev System*, draft report in ProveIt, 27 August 2026, [https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/Fabius\\_Inverse\\_Frontier\\_Report\\_Source\\_and\\_PDF](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF).

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## Part III

# Dyadic Primitive Ladders and Mellin–Newton Antiderivatives for the Fabius–Rvachev System

### Abstract

Let  $F$  be the bounded Fabius distribution function, let  $\mathcal{F}$  be the signed global Fabius extension, and let  $Q = F^{-1}$ . The existing PROVEIT corpus develops their differential equations, exact dyadic arithmetic, moment recurrences, half-base  $q$ -binomial formulae, infinite sinc products, lower-Lambert endpoint asymptotics, and Bell–Bernoulli coefficient calculus, but does not contain a systematic calculus for antiderivatives of monomially weighted Fabius functions. This report supplies such a calculus.

The key observation is an explicit primitive ladder. If

$$J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m),$$

then  $J_0 = \mathcal{F}$  and  $J'_{m+1} = J_m$ . Consequently every polynomial weight closes after finitely many integrations by parts. For  $p \in \mathbb{N}$ ,  $n \geq 1$ , and  $0 \leq x \leq 1$  we prove

$$I_0^n[x^p F(x)] = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}),$$

where  $I_0^n$  is the normalized  $n$ -fold primitive and  $n^{\bar{k}}$  is a rising factorial. The ordinary antiderivative is the case  $n = 1$ . A piecewise formula extends the result to the globally bounded  $F$ .

For an arbitrary complex exponent  $\alpha$ , the finite formula becomes a normally convergent Mellin–Newton series

$$I_0^n[x^\alpha F(x)] = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{\alpha-k} F(x/2^{n+k}).$$

It is entire in  $\alpha$ , terminates exactly at nonnegative integers, and becomes a positive-term series at negative integers. Complete integrals are governed by the entire moment interpolation  $D(s) = \mathbb{E}X^s$ :

$$\int_0^1 x^\alpha F(x) dx = \frac{1 - D(\alpha + 1)}{\alpha + 1} = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{d_{k+1}}{k + 1}.$$

This identifies dyadic Fabius samples with forward differences in the exponent, proves strict complete monotonicity and total positivity on the real exponent axis, and connects the antiderivative problem directly to the sinc-product cumulants and Bell–Bernoulli moments.

The same ladder yields shifted closed forms for  $x^p \text{up}(x)$  on  $[-1, 1]$  and formulas for the globally identical bounded-CDF transforms  $x^p F(1-x) = x^p (1 - F(x))$ , including their meromorphic exponent continuations. For the inverse function we obtain the compact quantile identity

$$\int_0^y Q(v)^\alpha dv = yQ(y)^\alpha - \alpha \int_0^{Q(y)} x^{\alpha-1} F(x) dx.$$

In particular,

$$\int Q(y) dy = yQ(y) - F(Q(y)/2) + C.$$

Higher inverse primitives lead naturally to powers  $F^j$ , or equivalently to order statistics of independent Fabius random variables.

Exact reciprocal-dyadic formulae, Thue–Morse and  $q$ -binomial substitutions, numerical experiments, a negative-power tail model driven by the proved log-periodic endpoint asymptotic, conjectures, a current Lean crosswalk, and a narrowed formalization plan complete the report. All statements are labeled as inherited, proved here, numerical, or conjectural. Priority outside the pinned corpus is not asserted.

# 1 Audit boundary, novelty discipline, and claim status

## 1.1 Pinned corpus and reading method

The source boundary is the exact `ProveIt` commit printed on the title page [12]. Its recursive documentation tree contains 76 indexed `.tex` paths. Every path was inventoried and searched; the living expositions, source-paper transcriptions, archived revisions, and frontier drafts were then grouped into mathematical lineages so that superseded copies were not mistaken for independent results. The families most relevant to the present problem—the primary Fabius–Rvachev exposition, repeated-integration studies, inverse-function reports, moment and  $q$ -calculus chapters, endpoint asymptotics, dyadic formulae, self-sampling/Appell material, and spectral/frontier dossiers—were read in depth. The full path inventory is recorded in [appendix C](#).

Targeted searches were made for *antiderivative*, *primitive*, *monomial*, products of  $F$  with powers, inverse-function areas, and candidate formula fragments such as  $F(x/2)$ . The archive does contain important repeated-integration work, but there the iteration is applied to boxes, splines, refinements, or convolution approximants in order to construct up; it is not a calculus of  $\int x^p F(x) dx$ . The inverse reports concern endpoint inversion and Lambert coordinates, not quantile antiderivatives. The Newton-basis frontier report concerns exponent sequences in infinite sinc products, not Newton interpolation in the power  $\alpha$  of the integrand.

**Novelty scope** The formula package below was not found in the pinned repository after path, phrase, and formula searches. This is a repository-relative novelty statement, not a claim of first publication in the worldwide literature. Classical ingredients such as repeated integration by parts, Newton series, Mellin transforms, order statistics, and total positivity are of course standard; the point is their unusually exact closure in the Fabius–Rvachev system.

## 1.2 Status convention

The report uses four statuses.

|                    |                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inherited</b>   | A theorem or definition already present in the pinned corpus, usually with a matching Lean declaration.                                                              |
| <b>Proved here</b> | A complete paper proof is supplied below from inherited facts. It is not claimed to be formalized unless a later source-status note names an exact Lean declaration. |
| <b>Numerical</b>   | A reproducible finite-precision or exact-rational experiment in the accompanying Python program.                                                                     |
| <b>Conjectural</b> | A statement suggested by the proved structure and experiments, with the missing analytic step identified.                                                            |

## 1.3 Current Lean crosswalk for the normalized Volterra layer

The novelty audit above remains pinned to the source snapshot on the title page. The following implementation-status note is later: the declarations listed here are present at the compiled Lean checkpoint `e109088ed`, verified by a focused build. Write

$$J_{n,a}f(x) := \begin{cases} f(x), & n = 0, \\ \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt, & n > 0. \end{cases} \quad (253)$$

In `FabiusFunction.NormalizedVolterra`, the exact crosswalk is as follows.

- `normalizedVolterra_affine` is the division-free forward identity

$$J_{n,ca+d}f(cx+d) = c^n J_{n,a}(t \mapsto f(ct+d))(x). \quad (254)$$

It holds for every real  $c$ , including  $c = 0$  and negative  $c$ , and includes  $n = 0$ . It has no integrability or completeness hypothesis and permits any real normed target.

- `normalizedVolterra_comp_affine` is the inverse-smul form

$$J_{n,a}(t \mapsto f(ct+d))(x) = (c^n)^{-1} J_{n,ca+d}f(cx+d), \quad c \neq 0. \quad (255)$$

The nonzero-scale hypothesis is essential to this divided form.

- `normalizedVolterra_basepoint_shift` states, assuming interval integrability of  $f$  from  $a$  to  $b$  and from  $b$  to  $x$ ,

$$J_{n,a}f(x) = J_{n,b}f(x) + \sum_{k=0}^{n-1} \frac{(x-b)^k}{k!} J_{n-k,a}f(b). \quad (256)$$

No ordering of  $a, b, x$  is imposed; at  $n = 0$  the sum is empty and the identity remains valid.

- `normalizedVolterra_succ_eq_taylor_of_eq_zero` states that if  $f$  is interval integrable from  $a$  to  $b$  and vanishes on the open unoriented interval between  $b$  and  $x$ , then

$$J_{n+1,a}f(x) = \sum_{k=0}^n \frac{(x-b)^k}{k!} J_{n+1-k,a}f(b). \quad (257)$$

No endpoint values or endpoint ordering are required, and the vanishing hypothesis is vacuous when  $b = x$ .

Thus the generic positive-order zero-tail assertion—the fact that a Volterra primitive is a finite Taylor polynomial after the input vanishes—is formally verified. This generic result does not by itself identify report-specific coefficients. Separately, `normalizedVolterra_pow_mul_extendedFabius` and `normalizedVolterra_pow_mul_fabiusReal_of_1e_one` already formalize the signed-global and bounded- $F$  natural-monomial coefficients (the latter on  $x \leq 1$ ). The survival, shifted-up, and exterior piecewise moment specializations displayed later retain the paper-level status assigned in this report.

## 1.4 Main new theorem package

The repository search found no counterpart of the following combined package:

1. the dyadic primitive ladder  $J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m)$ ;
2. the terminating polynomial formula for every monomial and every order of integration;
3. the normally convergent complex-exponent Mellin–Newton continuation;
4. the recovery of reciprocal-dyadic samples as forward differences in the exponent;
5. complete-moment generating functions and strict total positivity of the exponent kernel;
6. shifted polynomial primitives of up and a careful separation of  $F(1-x)$  from the globally compact bump;
7. the inverse-quantile primitive and its higher-order order-statistic hierarchy;
8. exact dyadic formulae for weighted primitives and inverse areas;
9. the negative-power tail model obtained by combining Newton terms with the proved log-periodic endpoint asymptotic.

## 2 Inherited Fabius–Rvachev data and notation

### 2.1 Three functions and one random variable

We keep the repository’s three-function convention. Historically, the distribution goes back to Jessen–Wintner and Fabius, while the compactly supported form was developed by the Rvachevs [1, 2, 3, 4]. The arithmetic and dyadic normalizations used here follow the modern treatments of Arias de Reyna and Haugland [5, 6, 7].

**Definition 2.1.** The *bounded Fabius function*  $F : \mathbb{R} \rightarrow [0, 1]$  is zero on  $(-\infty, 0]$ , one on  $[1, \infty)$ , and on the unit interval satisfies

$$F(1 - x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2). \quad (258)$$

The *Rvachev up-function* is

$$\text{up}(x) = F(1 - |x|), \quad \text{supp up} = [-1, 1]. \quad (259)$$

The *signed global extension*  $\mathcal{F}$  agrees with  $F$  on  $[0, 1]$ , vanishes on the negative half-line, and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (260)$$

On  $[-1, 1]$  one has  $\text{up}(x) = \mathcal{F}(x + 1)$ .

The bounded function is the distribution function of

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]. \quad (261)$$

The law is symmetric:  $X \stackrel{d}{=} 1 - X$ . Its moments are

$$d_s := D(s) := \mathbb{E}X^s \quad (s \in \mathbb{C}), \quad d_n = D(n) \quad (n \in \mathbb{N}). \quad (262)$$

The notation  $d_s$  is used only informally for noninteger  $s$ ;  $D$  is the analytic object. The corpus proves for integer  $n \geq 1$

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt, \quad (263)$$

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (264)$$

$$F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (265)$$

Infinite flatness at both endpoints implies that  $D(s) = \mathbb{E}X^s$  extends to an entire function: for every compact  $K \subset \mathbb{C}$ , the integrand  $X^s$  is uniformly integrable for  $s \in K$ . It also implies

$$d_n = \mathcal{O}_A(n^{-A}) \quad \text{for every } A > 0. \quad (266)$$

This last, deliberately weak form is enough for all Newton-series convergence arguments.

## 2.2 Integral operators and factorials

For  $n \geq 1$  define the normalized  $n$ -fold primitive from the origin by

$$(I_0^n f)(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) \, dt, \quad I_0^0 f = f. \quad (267)$$

Thus  $(I_0^n f)^{(n)} = f$ , and all derivatives of orders  $0, \dots, n-1$  vanish at zero. For a base point  $a$  we similarly write

$$(I_a^n f)(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) \, dt. \quad (268)$$

We use

$$z^{\underline{k}} = z(z-1) \cdots (z-k+1), \quad z^{\overline{k}} = z(z+1) \cdots (z+k-1), \quad (269)$$

with either product equal to 1 at  $k=0$ . Hence  $\binom{z}{k} = z^{\underline{k}}/k!$  and  $n^{\overline{k}} = (n+k-1)!/(n-1)!$  for  $n \in \mathbb{N}_{>0}$ .

## 3 The dyadic primitive ladder

### 3.1 Exact closure under integration

**Theorem 3.1** (Dyadic primitive ladder). *For  $m \in \mathbb{N}$  put*

$$J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m). \quad (270)$$

*Then*

$$J_0 = \mathcal{F}, \quad J'_{m+1} = J_m, \quad J_m^{(m)} = \mathcal{F}. \quad (271)$$

*For  $0 \leq x \leq 1$ ,*

$$(I_0^m F)(x) = J_m(x) = 2^{\binom{m}{2}} F(x/2^m). \quad (272)$$

*Proof.* Differentiate (270) and use (260):

$$J'_{m+1}(x) = 2^{\binom{m+1}{2}} 2^{-(m+1)} 2\mathcal{F}(x/2^m) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m) = J_m(x).$$

Every  $J_m$  and every derivative below order  $m$  vanishes at zero because  $\mathcal{F}$  is flat there. Hence  $J_m$  is exactly the normalized primitive (267). On the unit interval  $\mathcal{F} = F$  and all arguments  $x/2^m$  remain in  $[0, 1]$ .  $\square$

**Interpretation** The usual obstruction to a finite antiderivative formula is that integrating a special function creates a new special function. Here integration creates no new object: it only dilates the argument by  $2^{-1}$  and multiplies by a triangular power of two. Polynomial weights can therefore be removed by finitely many integrations by parts.

### 3.2 A useful moment bound for the ladder

**Lemma 3.2.** *For  $m \geq 1$  and  $0 \leq x \leq 1$ ,*

$$0 \leq J_m(x) \leq x^m J_m(1) = x^m \frac{d_m}{m!}. \quad (273)$$

*Proof.* The Cauchy representation and monotonicity of  $F$  give

$$\begin{aligned} J_m(x) &= \frac{x^m}{(m-1)!} \int_0^1 (1-u)^{m-1} F(xu) \, du \\ &\leq \frac{x^m}{(m-1)!} \int_0^1 (1-u)^{m-1} F(u) \, du = x^m J_m(1). \end{aligned}$$

If  $X$  has distribution function  $F$ , Tonelli and symmetry yield

$$J_m(1) = \frac{1}{m!} \mathbb{E}(1-X)^m = \frac{d_m}{m!}.$$

□

This estimate is the convergence engine for the complex-exponent theory. Notice the exact cancellation that will later occur:

$$\frac{n^{\bar{k}}}{(n+k)!} = \frac{1}{(n-1)!(n+k)}.$$
 (274)

## 4 Polynomial weights: terminating closed forms

### 4.1 A polynomial theorem

**Theorem 4.1** (Polynomially weighted  $n$ -fold primitive). *Let  $P$  be a polynomial of degree  $d$  and  $n \geq 1$ . On every interval on which the signed extension is used consistently,*

$$I_0^n[P\mathcal{F}](x) = \sum_{k=0}^d (-1)^k \frac{n^{\bar{k}}}{k!} P^{(k)}(x) 2^{\binom{n+k}{2}} \mathcal{F}(x/2^{n+k}).$$
 (275)

For  $0 \leq x \leq 1$ ,  $\mathcal{F}$  may be replaced by the bounded  $F$  throughout.

*Proof.* Write  $J_m = I_0^m \mathcal{F}$ . Repeated integration by parts gives

$$I_0^n[P\mathcal{F}] = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} P^{(k)} J_{n+k}.$$

The coefficient can be proved by induction on  $k$ : each new integration transfers a derivative to  $P$  and multiplies the Cauchy-kernel order by  $(n+k)/(k+1)$ . Since  $P^{(k)} = 0$  for  $k > d$ , the expansion terminates. Substitution of (270) proves the formula. One can also verify it directly by differentiating  $n$  times: adjacent terms cancel telescopically, leaving  $P\mathcal{F}$ . □

### 4.2 Monomial formula and examples

**Corollary 4.2** (Closed form for  $p \in \mathbb{N}$ ). *For  $p \in \mathbb{N}$ ,  $n \geq 1$ , and  $0 \leq x \leq 1$ ,*

$$I_0^n[x^p F(x)] = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}).$$
 (276)

*In particular, an ordinary antiderivative is*

$$\int x^p F(x) \, dx = \sum_{k=0}^p (-1)^k p^{\bar{k}} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C.$$
 (277)

The first cases are strikingly compact:

$$\int F(x) \, dx = F(x/2) + C, \quad (278)$$

$$\int xF(x) \, dx = xF(x/2) - 2F(x/4) + C, \quad (279)$$

$$\int x^2 F(x) \, dx = x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \quad (280)$$

$$\int x^3 F(x) \, dx = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16) + C. \quad (281)$$

The dyadic factors  $1, 2, 8, 64, \dots$  are  $2^{\binom{k+1}{2}}$ , not a geometric sequence; the triangular exponent is forced by iterating the dilation derivative (the displayed coefficients also contain the falling-factorial factors).

*Remark 4.3* (General  $n$ th antiderivative). Formula (276) is the unique  $n$ th antiderivative whose first  $n$  Taylor coefficients at zero vanish. The general  $n$ th antiderivative is obtained by adding an arbitrary polynomial of degree at most  $n - 1$ .

### 4.3 A recurrence in the exponent

Let

$$A_\alpha(x) = \int_0^x t^\alpha F(t) \, dt. \quad (282)$$

For nonnegative integer  $\alpha$  this is already covered by (277), but the recurrence below remains meaningful for every complex exponent.

**Proposition 4.4** (Dilation recurrence). *For  $0 < x \leq 1$  and  $\alpha \in \mathbb{C}$ ,*

$$A_\alpha(x) = x^\alpha F(x/2) - \alpha 2^\alpha A_{\alpha-1}(x/2). \quad (283)$$

*For  $0 < x \leq 1/2$  one also has*

$$(\alpha + 1)A_\alpha(x) = x^{\alpha+1} F(x) - 2^{-\alpha-1} A_{\alpha+1}(2x). \quad (284)$$

*Proof.* For the first formula integrate by parts against the primitive  $F(t/2)$  and substitute  $t = 2u$  in the remaining integral. For the second, integrate  $A_\alpha$  by parts in the opposite direction and use  $F'(t) = 2F(2t)$  on the left half of the unit interval. Infinite flatness makes the endpoint terms at zero vanish for every complex  $\alpha$ .  $\square$

### 4.4 The globally bounded $F$ : exact piecewise formula

The signed extension is the natural object for the pure ladder, but applications often want the bounded CDF. The normalized primitive then has a simple piecewise completion.

**Theorem 4.5** (Global bounded-CDF formula). *Let  $p \in \mathbb{N}$  and  $n \geq 1$ . For  $x \leq 0$ ,*

$$I_0^n[t^p F(t)](x) = 0. \quad (285)$$

*For  $0 \leq x \leq 1$  use (276). For  $x \geq 1$ ,*

$$I_0^n[t^p F(t)](x) = \frac{p!}{(p+n)!} x^{p+n} - \sum_{\ell=0}^{n-1} \frac{(-1)^\ell x^{n-1-\ell}}{\ell!(n-1-\ell)!} \frac{d_{p+\ell+1}}{p+\ell+1}. \quad (286)$$

*Proof.* For  $x \geq 1$ , write  $F = 1 - (1 - F)$ . The primitive of  $t^p$  is the first term. The survival function  $1 - F$  is supported on  $[0, 1]$ , so expand  $(x - t)^{n-1}$  and use

$$\int_0^1 t^m (1 - F(t)) \, dt = \frac{\mathbb{E}X^{m+1}}{m+1} = \frac{d_{m+1}}{m+1}.$$

The left half-line is immediate from the bounded extension  $F = 0$  there.  $\square$

Thus outside  $[0, 1]$  the answer is a polynomial whose coefficients are exact rational moments. The apparently non-elementary content is confined to the active unit interval.

## 5 Fractional, negative, and complex powers

### 5.1 The Mellin–Newton expansion

Fix the real logarithm for  $x > 0$ , so  $x^\alpha = \exp(\alpha \log x)$ .

**Theorem 5.1** (Complex-exponent  $n$ -fold primitive). *For every  $n \geq 1$ ,  $0 < x \leq 1$ , and  $\alpha \in \mathbb{C}$ ,*

$$I_0^n[t^\alpha F(t)](x) = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{\alpha-k} F(x/2^{n+k}). \quad (287)$$

*The series converges absolutely and locally uniformly in  $\alpha$  and uniformly for  $x$  in each compact subinterval of  $(0, 1]$ . It therefore defines an entire function of  $\alpha$ . For  $\alpha = p \in \mathbb{N}$  it terminates and becomes (276).*

*Proof.* In the Cauchy kernel (267), write

$$t^\alpha = x^\alpha \left(1 - \frac{x-t}{x}\right)^\alpha = x^\alpha \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{(x-t)^k}{x^k}.$$

On  $[\epsilon, x]$  the binomial series is uniformly convergent. Its  $k$ th integrated term is

$$(-1)^k \binom{\alpha}{k} x^{\alpha-k} n^{\bar{k}} J_{n+k}(x),$$

which is the term in (287). To pass to zero and prove normal convergence, apply lemma 3.2 and (274):

$$\left| \binom{\alpha}{k} n^{\bar{k}} x^{\alpha-k} J_{n+k}(x) \right| \leq \frac{x^{\Re \alpha + n}}{(n-1)!} \left| \binom{\alpha}{k} \right| \frac{d_{n+k}}{n+k}. \quad (288)$$

On a compact set of exponents,  $|\binom{\alpha}{k}|$  has at most polynomial growth in  $k$ , whereas (266) beats every polynomial. The Weierstrass test proves all stated forms of convergence and justifies termwise integration.  $\square$

For  $n = 1$  this reads

$$A_\alpha(x) = \sum_{k=0}^{\infty} (-1)^k \alpha^k 2^{\binom{k+1}{2}} x^{\alpha-k} F(x/2^{k+1}). \quad (289)$$

The infinite series is not merely an asymptotic integration-by-parts expansion: endpoint flatness makes it globally convergent for every complex exponent.



## 5.2 Negative powers and positivity

For  $m \in \mathbb{N}_{>0}$ ,  $(-1)^k - m^{\bar{k}} = m^{\bar{k}}$ , so

$$A_{-m}(x) = \sum_{k=0}^{\infty} m^{\bar{k}} 2^{\binom{k+1}{2}} x^{-m-k} F(x/2^{k+1}). \quad (290)$$

Every term is positive. Therefore every truncation is a rigorous lower bound, and the remainder has no cancellation. This is unusually favorable for certified computation. For a general negative real exponent  $\alpha = -a$ ,  $a > 0$ ,

$$A_{-a}(x) = x^{-a} \sum_{k \geq 0} \binom{a+k-1}{k} k! 2^{\binom{k+1}{2}} x^{-k} F(x/2^{k+1}). \quad (291)$$

At  $x = 1$ , the factorial and dyadic factors collapse to moments, as developed in [appendix 6](#).

*Remark 5.2* (Why all negative powers are integrable). Although  $x^\alpha$  itself is nonintegrable at zero when  $\Re \alpha \leq -1$ , the Fabius factor is flatter than every power. For each  $N$  there is  $C_N$  with  $F(x) \leq C_N x^N$  near zero, so  $x^\alpha F(x)$  is integrable for every fixed complex  $\alpha$ . The entire dependence on  $\alpha$  is a direct analytic expression of infinite flatness.

## 5.3 Exponent interpolation and exact forward differences

Define

$$B_{n,x}(\alpha) = x^{-\alpha} I_0^n [t^\alpha F(t)](x). \quad (292)$$

Then (287) is a Newton series in  $\alpha$ .

**Theorem 5.3** (Exponent–dyadic duality). For  $k \in \mathbb{N}$ ,

$$\Delta^k B_{n,x}(0) = (-1)^k n^{\bar{k}} 2^{\binom{n+k}{2}} x^{-k} F(x/2^{n+k}), \quad (293)$$

where  $\Delta h(\alpha) = h(\alpha + 1) - h(\alpha)$ . Equivalently,

$$F(x/2^{n+k}) = \frac{(-1)^k x^k}{n^{\bar{k}} 2^{\binom{n+k}{2}}} \Delta^k B_{n,x}(0). \quad (294)$$

At  $n = x = 1$ ,

$$d_{k+1} = (-1)^k (k+1) \Delta^k M(0), \quad M(\alpha) := A_\alpha(1). \quad (295)$$

*Proof.* The Newton basis satisfies  $\Delta^j \binom{\alpha}{k} |_{\alpha=0} = \delta_{jk}$ . Apply this to (287) after multiplication by  $x^{-\alpha}$ . The last identity uses (265).  $\square$

A new reconstruction principle Knowing the complete integral  $\alpha \mapsto \int_0^1 x^\alpha F(x) dx$  at the non-negative integers determines every reciprocal-dyadic sample by finite differences. Conversely, the reciprocal-dyadic ladder is precisely the Newton coefficient sequence of the exponent interpolation. Thus dyadic argument scale and discrete exponent scale are dual coordinates.

## 6 Complete integrals, moments, and special-function structure

### 6.1 Entire Mellin interpolation

**Theorem 6.1** (Complete weighted integral). *The function*

$$M(\alpha) = \int_0^1 x^\alpha F(x) \, dx \quad (296)$$

*is entire and satisfies*

$$M(\alpha) = \frac{1 - D(\alpha + 1)}{\alpha + 1} = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} \frac{d_{k+1}}{k+1}. \quad (297)$$

*The quotient has a removable singularity at  $\alpha = -1$ , with*

$$M(-1) = -D'(0) = -\mathbb{E} \log X = \sum_{k \geq 0} \frac{d_{k+1}}{k+1}. \quad (298)$$

*For  $p \in \mathbb{N}$ ,*

$$\int_0^1 x^p F(x) \, dx = \frac{1 - d_{p+1}}{p+1}. \quad (299)$$

*For  $m \in \mathbb{N}_{>0}$ ,*

$$M(-m) = \sum_{k \geq 0} \binom{m+k-1}{k} \frac{d_{k+1}}{k+1}. \quad (300)$$

*Proof.* For  $\Re \alpha > -1$ , Stieltjes integration by parts gives

$$D(\alpha + 1) = \int_0^1 x^{\alpha+1} \, dF(x) = 1 - (\alpha + 1)M(\alpha).$$

Both sides extend entirely because of endpoint flatness, proving the first equality and the removable value. The second equality is (287) at  $n = x = 1$ , followed by  $2^{\binom{k+1}{2}} F(2^{-k-1}) = d_{k+1}/(k+1)!$  and  $\alpha^{\underline{k}} = k! \binom{\alpha}{k}$ . The integer and negative-integer cases follow.  $\square$

The complete  $n$ -fold primitive has an equally clean series.

**Corollary 6.2.** *For  $n \geq 1$  and  $\alpha \in \mathbb{C}$ ,*

$$I_0^n[t^\alpha F(t)](1) = \frac{1}{(n-1)!} \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{d_{n+k}}{n+k}. \quad (301)$$

### 6.2 Ordinary and exponential generating functions

Let

$$G(z) = \mathbb{E} e^{zX} = \sum_{n \geq 0} d_n \frac{z^n}{n!}. \quad (302)$$

The corpus proves  $G(2z) = ((e^z - 1)/z)G(z)$ . The complete polynomial integrals have the exponential generating function

$$\sum_{p \geq 0} M(p) \frac{z^p}{p!} = \int_0^1 e^{zx} F(x) \, dx = \frac{e^z - G(z)}{z}, \quad (303)$$

with the removable value  $1/2$  at zero. Their ordinary generating function is

$$\sum_{p \geq 0} M(p) z^{p+1} = -\log(1-z) - \sum_{n \geq 1} \frac{d_n z^n}{n} = \mathbb{E} \log \frac{1-zX}{1-z}, \quad |z| < 1. \quad (304)$$

For the positive half of the up-function,

$$\sum_{p \geq 0} z^{p+1} \int_0^1 x^p \operatorname{up}(x) \, dx = \mathbb{E}[-\log(1-zX)]. \quad (305)$$

These identities turn moment recurrences into recurrences for whole families of weighted antiderivatives.

### 6.3 The sinc product, Bernoulli cumulants, and Bell moments

Put  $Y = 2X - 1$ . Its density is up on  $[-1, 1]$ , and its moment generating function is the hyperbolic continuation of the infinite sinc product:

$$C(z) = \mathbb{E} e^{zY} = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (306)$$

The even cumulants are

$$\kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (307)$$

With the corpus notation  $c_r = \mathbb{E} Y^{2r} = \int_{-1}^1 x^{2r} \operatorname{up}(x) \, dx$ , the complete exponential Bell polynomials give

$$c_r = \mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r}). \quad (308)$$

Consequently the entire moment interpolation has the absolutely convergent Bell–Bernoulli form

$$D(s) = 2^{-s} \sum_{r=0}^{\infty} \binom{s}{2r} c_r = 2^{-s} \sum_{r=0}^{\infty} \binom{s}{2r} \mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r}). \quad (309)$$

Combining (309) with (297) expresses every fractional or negative complete Fabius integral in Bernoulli numbers and Bell polynomials. This is not necessarily the fastest numerical representation, but it makes the connection with the corpus's all-orders coefficient calculus explicit.

### 6.4 Complete monotonicity and strict total positivity

**Theorem 6.3** (Exponent kernel). *For real  $\alpha$  and every  $m \in \mathbb{N}$ ,*

$$(-1)^m M^{(m)}(\alpha) = \int_0^1 x^\alpha (-\log x)^m F(x) \, dx > 0. \quad (310)$$

*Thus  $M$  is strictly completely monotone on the whole real axis and strictly log-convex. More strongly, if  $a_1 < \dots < a_r$  and  $b_1 < \dots < b_r$  are real, then*

$$\det[M(a_i + b_j)]_{i,j=1}^r > 0. \quad (311)$$

*In particular, every Hankel determinant  $\det[M(\alpha + i + j)]_{i,j=0}^{r-1}$  is positive.*

*Proof.* Differentiation under the integral is justified by flatness, and the integrand in (310) is positive on a set of positive measure. Log-convexity follows from Cauchy–Schwarz. For total positivity, apply Andreief's identity to the positive measure  $F(x) dx$ :

$$\det[M(a_i + b_j)] = \frac{1}{r!} \int_{(0,1)^r} \det[x_j^{a_i}] \det[x_j^{b_i}] \prod_{j=1}^r F(x_j) dx_j.$$

On the chamber  $0 < x_1 < \cdots < x_r < 1$ , both generalized Vandermonde determinants have the same strict sign; their product is positive. The measure of that chamber is positive.  $\square$

*Remark 6.4.* The theorem supplies a large family of nonlinear inequalities among the exact values  $M(p) = (1 - d_{p+1})/(p + 1)$ . For example,

$$M(a + b)M(c + d) > M(a + d)M(c + b)$$

whenever  $a < c$  and  $b < d$ . Such inequalities are invisible in the triangular moment recurrence but automatic in the exponent-integral representation.

## 7 Reflection, survival, and Rvachev's up-function

### 7.1 The two reflected bounded-CDF transforms, and the genuinely different bump

For the bounded CDF with its constant extensions, symmetry is global:

$$F(1 - x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (312)$$

Indeed, the two sides agree by the usual symmetry on  $[0, 1]$ , are both 1 for  $x \leq 0$ , and are both 0 for  $x \geq 1$ . Thus the two transformations explicitly requested in the problem have the same antiderivative theory for the bounded  $F$ .

The compact bump is different on the negative half-line:

$$\text{up}(x) = F(1 - |x|), \quad \text{up}(x) = F(1 - x) = 1 - F(x) \quad (x \geq 0). \quad (313)$$

It vanishes outside  $[-1, 1]$ , whereas the reflected/survival transform equals 1 on  $(-\infty, 0]$ . If  $F$  is replaced by the signed global extension  $\mathcal{F}$ , then  $\mathcal{F}(1 - x)$  and  $1 - \mathcal{F}(x)$  are genuinely different outside the unit interval. Domain labels are therefore still essential.

### 7.2 The survival transform $1 - F(x)$

**Theorem 7.1** (Weighted survival primitives). *For  $p \in \mathbb{N}$ ,  $n \geq 1$ , and  $0 \leq x \leq 1$ ,*

$$I_0^n[t^p F(1 - t)](x) = I_0^n[t^p(1 - F(t))](x) = \frac{p!}{(p + n)!} x^{p+n} - I_0^n[t^p F(t)](x), \quad (314)$$

where the last term is the finite sum (276). For complex  $\alpha$  with  $\Re \alpha > -1$ ,

$$I_0^n[t^\alpha F(1 - t)](x) = I_0^n[t^\alpha(1 - F(t))](x) = \frac{\Gamma(\alpha + 1)}{\Gamma(\alpha + n + 1)} x^{\alpha+n} - I_0^n[t^\alpha F(t)](x). \quad (315)$$

The right side meromorphically continues to all  $\alpha$ , with possible simple poles at  $-1, -2, \dots, -n$ .

For  $x \geq 1$ , the survival primitive becomes a polynomial of degree  $n - 1$ :

$$I_0^n[t^p(1 - F(t))](x) = \sum_{\ell=0}^{n-1} \frac{(-1)^\ell x^{n-1-\ell}}{\ell!(n-1-\ell)!} \frac{d_{p+\ell+1}}{p+\ell+1}. \quad (316)$$

For integer  $p$  and  $x \leq 0$ , since  $1 - F(t) = 1$  along the integration path,

$$I_0^n[t^p(1 - F(t))](x) = \frac{p!}{(p+n)!} x^{p+n}. \quad (317)$$

At  $x = 1$  the positive half of up has the Mellin transform

$$U(\alpha) := \int_0^1 x^\alpha \text{up}(x) \, dx = \frac{D(\alpha+1)}{\alpha+1}. \quad (318)$$

Its meromorphic continuation has exactly one pole, at  $\alpha = -1$ , with residue 1. The finite part is

$$\text{FP}_{\alpha=-1} U(\alpha) = D'(0) = \int_0^1 \frac{\text{up}(x) - 1}{x} \, dx. \quad (319)$$

This one-pole simplification is a consequence of the infinite flatness of  $\text{up}(x) - 1$  at zero.

### 7.3 A shifted primitive ladder for the compact bump

**Theorem 7.2** (Shifted Rvachev ladder). *For  $m \geq 1$  and  $-1 \leq x \leq 1$ , define*

$$K_m(x) = 2^{\binom{m}{2}} F((x+1)/2^m). \quad (320)$$

*Then  $K'_1 = \text{up}$ ,  $K'_{m+1} = K_m$ , and  $K_m$  is the normalized  $m$ -fold primitive from  $-1$ . Consequently, for  $p \in \mathbb{N}$  and  $n \geq 1$ ,*

$$I_{-1}^n[t^p \text{up}(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F((x+1)/2^{n+k}). \quad (321)$$

*Proof.* On the support,  $\text{up}(x) = \mathcal{F}(x+1)$ . Shift the global ladder of [theorem 3.1](#); for  $m \geq 1$  the argument  $(x+1)/2^m$  lies in  $[0, 1]$ , so the bounded  $F$  is sufficient. The polynomial formula follows exactly as in [theorem 4.1](#), now with base point  $-1$ .  $\square$

For fractional powers on the negative half of the support, it is natural to use  $(-x)^\alpha$ . If  $-1 < x < 0$ , substitution  $u = x+1$  and a uniformly convergent binomial series give

$$\int_{-1}^x (-t)^\alpha \text{up}(t) \, dt = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} A_k(1+x). \quad (322)$$

At  $x = 0$  the formula continues for  $\Re \alpha > -1$  and then meromorphically. The even Mellin transform of the full bump is

$$\int_{-1}^1 |x|^\alpha \text{up}(x) \, dx = \frac{2D(\alpha+1)}{\alpha+1}. \quad (323)$$

## 8 Higher derivatives and mixed orders

### 8.1 Derivatives of monomially weighted Fabius functions

The corpus gives

$$F^{(j)}(x) = 2^{\binom{j+1}{2}} \mathcal{F}(2^j x) \quad (0 < x < 1). \quad (324)$$

Leibniz's rule immediately gives the derivative companion to the primitive formula.

**Theorem 8.1** (Closed derivative formula). *For  $n \in \mathbb{N}$ ,  $\alpha \in \mathbb{C}$ , and  $0 < x < 1$ ,*

$$\frac{d^n}{dx^n} (x^\alpha F(x)) = \sum_{j=0}^n \binom{n}{j} \alpha^{n-j} x^{\alpha-n+j} 2^{\binom{j+1}{2}} \mathcal{F}(2^j x). \quad (325)$$

*On  $-1 < x < 1$ , replace the final factor by  $\mathcal{F}(2^j(x+1))$  to obtain the corresponding formula for  $x^\alpha \operatorname{up}(x)$  on a chosen branch of  $x^\alpha$ .*

Thus positive and negative operator orders are both finite on polynomial weights: positive orders use expanding dyadic arguments, while antiderivatives use contracting arguments. The signed extension is the common carrier that makes both directions globally meaningful.

## 8.2 A bilateral operator notation

For integer  $r$  define  $\mathcal{D}^r$  by ordinary differentiation when  $r \geq 0$  and by the normalized primitive  $I_0^{-r}$  when  $r < 0$ . For a monomial weight  $x^p$ , both signs of  $r$  produce finite sums. A useful research question is whether there is a natural fractional operator  $\mathcal{D}^z$  satisfying a dyadic semigroup law and interpolating both (287) and (325). Classical Riemann–Liouville operators interpolate in the order of integration, whereas the present Newton series interpolates in the exponent of the monomial. The two complex parameters should be studied jointly.

# 9 Integrals and derivatives of the inverse Fabius function

## 9.1 The quantile area identity

Let  $Q = F^{-1} : [0, 1] \rightarrow [0, 1]$ . Strict monotonicity on the open unit interval and endpoint continuity are inherited from the corpus.

**Theorem 9.1** (Inverse-power primitive). *For  $0 < y \leq 1$ , put  $q = Q(y)$ . Then for every  $\alpha \in \mathbb{C}$ ,*

$$\mathcal{Q}_\alpha(y) := \int_0^y Q(v)^\alpha dv = yq^\alpha - \alpha A_{\alpha-1}(q). \quad (326)$$

*Both sides are entire in  $\alpha$ . At  $y = 0$  the common continuous value is 0; this endpoint is understood by continuation rather than by the indeterminate symbol  $0^\alpha$ . At  $y = 1$ ,*

$$\int_0^1 Q(v)^\alpha dv = D(\alpha). \quad (327)$$

*Proof.* Substitute  $v = F(x)$  in the Lebesgue–Stieltjes sense:

$$\int_0^y Q(v)^\alpha dv = \int_0^q x^\alpha dF(x).$$

Integration by parts gives  $q^\alpha F(q) - \alpha \int_0^q x^{\alpha-1} F(x) dx$ . The endpoint term at zero vanishes for every complex exponent because  $F$  is infinitely flat. Since  $F(q) = y$ , this is (326). The complete identity is the quantile representation of moments.  $\square$

**Corollary 9.2** (An elementary-looking inverse antiderivative). *An ordinary antiderivative of  $Q$  is*

$$\int Q(y) dy = yQ(y) - F\left(\frac{Q(y)}{2}\right) + C. \quad (328)$$

More generally, for  $p \in \mathbb{N}_{>0}$ ,

$$\int_0^y Q(v)^p \, dv = yq^p - p \sum_{k=0}^{p-1} (-1)^k p - 1^k 2^{\binom{k+1}{2}} q^{p-1-k} F(q/2^{k+1}). \quad (329)$$

Differentiating (326) at  $\alpha = 0$  gives

$$\int_0^y \log Q(v) \, dv = y \log q - A_{-1}(q). \quad (330)$$

Formula (328) is the exact area-under-an-inverse identity specialized to the Fabius primitive ladder. Its compactness is a useful diagnostic: differentiating the right side, the two  $Q'$  terms cancel because  $\frac{1}{2}F'(Q/2) = F(Q) = y$ .

For negative complete inverse powers,

$$D(-m) = 1 + mM(-m-1) = 1 + m \sum_{k \geq 0} \binom{m+k}{k} \frac{d_{k+1}}{k+1}. \quad (331)$$

Again the series has positive terms.

## 9.2 Weighted inverse integrals and the nonlinear hierarchy

**Proposition 9.3** (Beta-weighted quantiles). For  $r \in \mathbb{N}$ ,  $0 < y \leq 1$ ,  $q = Q(y)$ , and  $\alpha \in \mathbb{C}$ ,

$$\int_0^y v^r Q(v)^\alpha \, dv = \frac{y^{r+1} q^\alpha}{r+1} - \frac{\alpha}{r+1} \int_0^q x^{\alpha-1} F(x)^{r+1} \, dx. \quad (332)$$

*Proof.* After  $v = F(x)$ , write  $F(x)^r \, dF(x) = d(F(x)^{r+1})/(r+1)$  and integrate by parts.  $\square$

This introduces

$$H_{\beta,j}(q) = \int_0^q x^\beta F(x)^j \, dx, \quad (333)$$

a nonlinear hierarchy absent from the linear primitive ladder. Expanding the Cauchy kernel and using (332) yields:

**Theorem 9.4** (Higher inverse primitive). For  $n \geq 1$ ,

$$I_0^n[Q^\alpha](y) = \frac{y^n q^\alpha}{n!} - \frac{\alpha}{n!} \sum_{j=1}^n (-1)^{j-1} \binom{n}{j} y^{n-j} H_{\alpha-1,j}(q). \quad (334)$$

The linear Fabius ladder therefore solves the first inverse primitive completely, but higher inverse primitives expose powers of  $F$ . This is not a failure of the method: it identifies the exact new objects required at the next level.

## 9.3 Order-statistic interpretation

Let  $X_1, \dots, X_n$  be independent copies of  $X$ .

**Theorem 9.5** (Order statistics). For real parts in the domain of the displayed expectation,

$$I_0^n[Q^\alpha](1) = \frac{1}{n!} \mathbb{E} [\min(X_1, \dots, X_n)^\alpha]. \quad (335)$$

For  $\Re \alpha > 0$  this is

$$I_0^n[Q^\alpha](1) = \frac{\alpha}{n!} \int_0^1 t^{\alpha-1} (1 - F(t))^n dt = \frac{\alpha}{n!} \int_0^1 t^{\alpha-1} \text{up}(t)^n dt. \quad (336)$$

In particular,

$$I_0^n Q(1) = \frac{1}{n!} \int_0^1 \text{up}(t)^n dt. \quad (337)$$

More generally, for  $r, s \in \mathbb{N}$ ,

$$\int_0^1 y^r (1-y)^s Q(y)^\alpha dy = B(r+1, s+1) \mathbb{E} \left[ X_{(r+1:r+s+1)}^\alpha \right]. \quad (338)$$

*Proof.* The minimum of  $n$  independent uniform variables has density  $n(1-y)^{n-1}$ ; applying the quantile map  $Q$  gives the minimum of  $n$  independent Fabius variables. The survival formula for a positive power gives (336). The Beta density yields (338).  $\square$

This probabilistic interpretation suggests studying the nonlinear integrals of  $F^j$  through order statistics rather than through direct functional equations.

## 9.4 Derivatives of the inverse

For  $0 < y < 1$  and  $q = Q(y)$ ,

$$Q^{(n)}(y) = \left[ \left( \frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \right]_{x=q}. \quad (339)$$

Faà di Bruno's formula then gives

$$\frac{d^n}{dy^n} Q(y)^\alpha = \sum_{k=1}^n \alpha^k Q(y)^{\alpha-k} B_{n,k}(Q', Q'', \dots, Q^{(n-k+1)}), \quad (340)$$

where  $B_{n,k}$  are partial Bell polynomials. Combining (339) with the Fabius derivative ladder expresses every derivative in terms of  $q$ ,  $\mathcal{F}(2^j q)$ , and rational operations. Endpoint blow-up is encoded by the reciprocal factors  $1/F'(q)$  and is naturally analyzed in the lower-Lambert coordinate already developed in the corpus.

# 10 Exact dyadic arithmetic, Thue–Morse sums, and $q$ -binomials

## 10.1 Reciprocal-dyadic weighted primitives

**Theorem 10.1** (Exact reciprocal-dyadic formula). For  $p, r \in \mathbb{N}$  and  $n \geq 1$ ,

$$I_0^n[x^p F(x)](2^{-r}) = 2^{-r(n+p)-\binom{r}{2}} \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} \frac{d_{r+n+k}}{(r+n+k)!}. \quad (341)$$

In particular the value is rational.

*Proof.* Insert  $x = 2^{-r}$  into (276), then use (265) at index  $r+n+k$ . The power of two in each summand is

$$2^{-r(p-k)} 2^{\binom{n+k}{2} - \binom{r+n+k}{2}} = 2^{-r(n+p)-\binom{r}{2}},$$

independent of  $k$ . This cancellation is the arithmetic signature of the primitive ladder.  $\square$



## 10.2 Exact inverse areas at reciprocal-dyadic ordinates

If  $q = 2^{-r}$  and  $y = F(q)$ , then (329) simplifies similarly.

**Theorem 10.2** (Dyadic quantile area). *For  $p \in \mathbb{N}_{>0}$  and  $r \in \mathbb{N}$ ,*

$$\int_0^{F(2^{-r})} Q(v)^p \, dv = 2^{-rp} \binom{r}{2} \left[ \frac{d_r}{r!} - p \sum_{k=0}^{p-1} (-1)^k p - 1 \frac{d_{r+k+1}}{(r+k+1)!} \right]. \quad (342)$$

At  $r = 2$ ,  $q = 1/4$ ,  $y = F(q) = 5/72$ , the first six exact values are

| $p$ | $\int_0^{5/72} Q(v)^p \, dv$ |
|-----|------------------------------|
| 1   | 1/72                         |
| 2   | 1493/518400                  |
| 3   | 53/86400                     |
| 4   | 48781/365783040              |
| 5   | 10777/365783040              |
| 6   | 196806103/29847896064000     |

These were evaluated independently by exact rational arithmetic and checked by numerical quantile quadrature.

## 10.3 Arbitrary dyadic arguments and Thue–Morse signs

The corpus proves, for  $0 \leq a \leq 2^N$ ,

$$F(a/2^N) = \frac{2^{-(\binom{N+1}{2})}}{N!} \sum_{h=0}^{a-1} (-1)^{s_2(h)} \sum_{j=0}^{\lfloor N/2 \rfloor} \binom{N}{2j} c_j (2a - 2h - 1)^{N-2j}. \quad (343)$$

Putting  $x = a/2^r$  in (276) and applying (343) with  $N = r + n + k$  gives an entirely finite closed form for every weighted primitive at every dyadic argument. It involves only rational centered moments  $c_j$ , binomial coefficients, powers of two, and the signed Thue–Morse sequence  $(-1)^{s_2(h)}$ . We leave it in this compositional form because expanding the three finite sums obscures the ladder structure and offers no computational advantage.

Write

$$(a; q)_N = \prod_{\ell=0}^{N-1} (1 - aq^\ell), \quad \left[ \begin{matrix} N \\ j \end{matrix} \right]_q = \frac{(q; q)_N}{(q; q)_j (q; q)_{N-j}}, \quad (344)$$

with the usual empty-product convention [11]. The alternative translated half-base  $q$ -binomial identity in the corpus is, for  $m, N \in \mathbb{N}$ ,

$$\mathcal{F}(m/2^N) = \frac{1}{2^{N^2} (1/2; 1/2)_N} \sum_{j=0}^N \frac{\left[ \begin{matrix} N \\ j \end{matrix} \right]_{1/2}}{4^{\binom{j}{2}} (N+j)!} \sum_{h=0}^{m2^j-1} (-1)^{s_2(h)} (h - m2^j + R)^{N+j}, \quad (345)$$

independent of the translation parameter  $R \in \mathbb{C}$ . Substitution of (345) into each ladder term supplies a  $q$ -binomial version of the antiderivative formula. This explains how  $q$ -Pochhammer symbols enter the integral problem: not through integration itself, which is elementary in the ladder basis, but through exact evaluation of the contracted Fabius samples.

## 11 Numerical experiments and reproducibility

### 11.1 Independent representations

The accompanying program `fabius_integrals_experiments.py` uses two independent representations.

1. A dyadic grid with  $2^{18} + 1 = 262145$  nodes is iterated under the integral/reflection fixed point for  $F$ . This checks pointwise finite primitives, shifted up-primitives, and inverse-quantile areas.
2. The exact rational recurrence (264) and a separate floating version through  $d_{900}$  check complete Newton sums, exact dyadic values, and negative-power tails without differentiating a flat function numerically.

The principal error maxima in the default run were

| Test family                                           | maximum absolute discrepancy |
|-------------------------------------------------------|------------------------------|
| Finite $F$ primitive versus direct quadrature         | $1.85 \times 10^{-10}$       |
| Shifted up primitive versus direct quadrature         | $2.39 \times 10^{-7}$        |
| Exact dyadic inverse areas versus quantile quadrature | $4.04 \times 10^{-9}$        |
| Grid value $F(1/4)$ versus $5/72$                     | $3.23 \times 10^{-12}$       |

The larger up-error is caused by direct quadrature across a compact-support endpoint and by interpolation, not by instability of the closed form.

### 11.2 Fractional and negative complete integrals

With 900 moment terms the computed values include

| $\alpha$   | $M(\alpha) = \int_0^1 x^\alpha F(x) \, dx$ |
|------------|--------------------------------------------|
| -2         | 1.3148861440026451                         |
| -1         | 0.7582400004044057                         |
| -1/2       | 0.6073863107431697                         |
| 1/2        | 0.4209681733402678                         |
| 3/2        | 0.3146395582080128                         |
| $\sqrt{2}$ | 0.3218439940718418                         |

Grid quadrature agrees to roughly  $10^{-9}$  or better in these tests despite the negative powers.

### 11.3 Exact checks

For  $p = 0, \dots, 12$ , exact rational arithmetic verifies term by term that

$$\sum_{k=0}^p (-1)^k \binom{p}{k} \frac{d_{k+1}}{k+1} = \frac{1 - d_{p+1}}{p+1}. \quad (346)$$

The CSV files retain numerator/denominator strings, not floating reconstructions. The inverse checks at  $q = 1/4$  likewise compute (342) with `fractions.Fraction` before comparison with quadrature.

## 12 Asymptotic tails for negative powers

### 12.1 Moment asymptotics inherited from the endpoint theory

Let  $L = \log 2$ . Combining the corpus's effective dyadic endpoint theorem with (265) and Stirling's formula yields the moment-scale structure

$$\log d_n = -\frac{(\log n)^2}{2L} + \frac{1}{2} \log n + C_M + \Psi(\log_2 n) + \mathcal{O}\left(\frac{(\log n)^2}{n}\right), \quad (347)$$

where  $\Psi$  is the same smooth nonconstant one-periodic correction whose Fourier coefficients are explicit Gamma-zeta values in the corpus, a standard Mellin-harmonic-sum phenomenon [9], and  $C_M = C_F + \frac{1}{2} \log(2\pi)$ . In the sharp source theorem the periodic argument is  $\theta_n = \log_2 \lambda_n$ , equivalently  $\Psi(\lambda_n)$ , where  $\lambda_n - \log_2 \lambda_n = n$ . Replacing  $\theta_n$  by  $\log_2 n$  changes the value by  $\mathcal{O}((\log n)/n)$ , already inside the displayed remainder. The exact theorem also retains the lower-Lambert saddle term  $-\log^2 n/(2L^2 n)$  rather than absorbing it; the branch conventions are those standard for Lambert  $W$  [8]. Formula (347) is used here only to motivate a tail scale; none of the exact integral identities depends on it.

### 12.2 A first hazard approximation

For  $a > 0$ , the negative-power complete series has positive terms

$$t_k(a) = \binom{a+k-1}{k} \frac{d_{k+1}}{k+1}, \quad M(-a) = \sum_{k \geq 0} t_k(a). \quad (348)$$

The term envelope itself has logarithmic derivative

$$-\frac{d}{dx} \log t_x(a) \approx \frac{1}{x} \left( \frac{\log x}{L} - (a - \tfrac{3}{2}) - \frac{\Psi'(\log_2 x)}{L} \right). \quad (349)$$

For the tail, however, the correct endpoint-Laplace variable is  $u = \log x$ :  $\int_N^\infty t_x(a) dx = \int_{\log N}^\infty e^u t_{e^u}(a) du$ . The Jacobian contributes one additional unit to the power-law exponent. The effective endpoint denominator is therefore

$$H_a(N) = \frac{\log N}{L} - (a - \tfrac{1}{2}) - \frac{\Psi'(\log_2 N)}{L}, \quad (350)$$

which suggests

$$\sum_{k \geq N} t_k(a) \sim \frac{N t_N(a)}{H_a(N)} = \frac{N t_N(a)}{\frac{\log N}{L} - (a - \tfrac{1}{2}) - \Psi'(\log_2 N)/L}. \quad (351)$$

Ignoring the periodic derivative gives the simple estimator

$$\hat{R}_N(a) = \frac{N t_N(a)}{\log N/L - (a - \tfrac{1}{2})}. \quad (352)$$

The experiment compares this with a 900-term reference tail. Representative ratios  $\hat{R}_N/R_N$  are

| $a$ | $N = 50$ | $N = 100$ | $N = 200$ | $N = 400$ |
|-----|----------|-----------|-----------|-----------|
| 1/2 | 0.950    | 0.974     | 0.990     | 1.000     |
| 1   | 0.959    | 0.980     | 0.994     | 1.002     |
| 2   | 0.987    | 0.996     | 1.003     | 1.010     |
| 3   | 1.037    | 1.022     | 1.019     | 1.021     |

At the smallest tails, double precision and the finite 900-term reference begin to matter; the agreement should therefore be read as structural evidence, not certification.

**Conjecture 12.1** (Log-periodic negative-power tail). *For every compact interval  $0 < a_0 \leq a \leq a_1 < \infty$ , the remainder of (348) satisfies (351) uniformly as  $N \rightarrow \infty$ , with a relative error  $o(1)$ . A second-order expansion is obtained by the usual endpoint Laplace correction from the first two derivatives of the hazard.*

**Missing proof step.** One needs a sufficiently uniform differentiable version of the moment asymptotic along the integer ray, including control of the periodic derivative and second differences. The repository already formalizes the continuous negative-Laplace logarithm and its first periodic derivative estimate. The remaining bridge is a discrete transfer theorem from that saddle object to  $d_n$  and then to the weighted tail.

### 12.3 Acceleration

The Newton series converges faster than every algebraic rate but slower than a fixed geometric rate in the term index. Standard Richardson extrapolation on an arithmetic grid is therefore mismatched. The natural acceleration should use either

- the hazard variable  $u_N = \log N/L - (a - 1/2)$ , or
- a geometric sequence  $N_j \approx N_0 2^j$ , phase-locked so that  $\log_2 N_j$  has constant fractional part and the periodic factor cancels.

**Conjecture 12.2** (Phase-locked  $q$ -Richardson acceleration). *For fixed  $a > 0$  and a fixed dyadic phase, there are explicitly computable linear combinations of the partial sums at  $N, 2N, 4N, \dots, 2^r N$  that cancel the first  $r$  terms of the hazard expansion while preserving positivity-based enclosures after pairing successive transforms.*

This would connect the present exponent-Newton series with the corpus's existing  $q$ -Richardson methods, but the asymptotic variable is now the truncation index rather than a truncated sinc-product scale.

## 13 Further frontier directions

### 13.1 The nonlinear power hierarchy

The most immediate new object is  $H_{\beta,j}$  from (333). Splitting at  $1/2$ , using  $F'(x) = 2F(2x)$ , and reflecting  $F(1-x) = 1 - F(x)$  should produce coupled dyadic functional equations among  $H_{\beta,j}$  and binomial combinations of lower powers. The following questions appear tractable.

*Open problem 13.1.* Derive a triangular functional system for  $H_{\beta,j}(x) = \int_0^x t^\beta F(t)^j dt$  under  $x \mapsto 2x$  and  $x \mapsto 1-x$ , and determine whether integer  $\beta, j$  give exact rational values at dyadic  $x$ .

*Open problem 13.2.* Express the complete values  $H_{\beta,j}(1)$  as moments of Fabius order statistics, then seek recurrences analogous to (264). The case  $j = 2$  is the first genuinely nonlinear test.

### 13.2 Two-parameter fractional calculus

The report interpolates the monomial exponent  $\alpha$  while keeping the integration order  $n$  integral. One may instead define the Riemann–Liouville family

$$I_0^\nu[x^\alpha F(x)] = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} t^\alpha F(t) dt, \quad \Re \nu > 0. \quad (353)$$

Expanding as before gives

$$I_0^\nu[x^\alpha F(x)] = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{\Gamma(\nu + k)}{\Gamma(\nu)} x^{\alpha-k} J_{\nu+k}(x), \quad (354)$$

provided a meaningful analytic interpolation  $J_z$  of the discrete primitive ladder is chosen. The integer data  $J_m(x) = 2^{\binom{m}{2}} F(x/2^m)$  suggests a  $q$ -difference or Mellin construction, but there is no canonical interpolation yet.

**Conjecture 13.3** (Canonical interpolated ladder). *There exists a distinguished jointly holomorphic family  $J_z(x)$  on a right half-plane, characterized by positivity on the real axis, suitable growth bounds,  $J_m$  at integers, and a functional relation interpolating  $J'_{m+1} = J_m$ . With this choice, (354) gives a jointly meromorphic two-parameter calculus in  $(\alpha, \nu)$ .*

### 13.3 Complex zeros and total positivity

The entire functions  $D$  and  $M$  satisfy

$$D(s+1) = 1 - (s+1)M(s). \quad (355)$$

The strict total positivity of  $M(a+b)$  constrains its real zeros (there are none) and its Padé approximants. A natural complex-analytic program is to locate zeros of  $D$  and  $M$ , look for zero-free half-planes, and test whether the Stieltjes/Laplace structure yields a Laguerre–Pólya or Pick-type property after a suitable change of variables. Complete monotonicity alone does not force such a classification, so numerical claims should be made cautiously.

### 13.4 Characterization by polynomial closure

*Open problem 13.4* (Rigidity). Classify smooth functions  $f$  for which there are constants  $a_m, b_m$  such that every polynomially weighted primitive  $\int P(x)f(x) dx$  is a finite linear combination of polynomials times  $f(a_mx + b_m)$ . Does nontrivial closure under all monomials force a dilation differential equation  $f'(x) = cf(\lambda x)$ , up to affine changes and elementary summands?

The Fabius system should be a model case of a broader “finite primitive module” over the polynomial ring.

### 13.5 Certified computation

Positive negative-power terms make interval certification straightforward if exact rational moments are available. The main difficulty is a rigorous tail enclosure. Three routes are promising:

1. derive monotone upper and lower hazard bounds from the formal negative-Laplace estimates;
2. bound  $d_n$  directly by optimized beta integrals using effective endpoint inequalities;
3. perform ball-arithmetic evaluation of a moderate prefix and attach a rigorously controlled log-periodic saddle tail.

Such an algorithm would return certified enclosures for  $M(-a)$ ,  $D(-a)$ , and inverse negative moments, not merely high-precision decimal approximations.

## 14 Remaining Lean formalization roadmap

At compiled checkpoint e109088ed, the generic affine, base-point-change, and zero-tail Taylor-jet calculus is already covered by the four declarations in [equations \(254\) to \(257\)](#). In particular, the generic polynomial shape after leaving the support is no longer a roadmap item, and the signed-global and bounded- $F$  natural-monomial coefficients are already formalized separately. Identifying the remaining survival, folded-up, and exterior piecewise coefficients is a distinct specialization. The remaining work can therefore be narrowed to:

1. add narrowly packaged specializations for the report’s survival, shifted up, and exterior piecewise coefficient formulas, without reproving the generic Volterra algebra or the already formalized signed/bounded  $F$  cases;
2. establish [lemma 3.2](#) using the probability measure already present in the development;
3. prove normal convergence of [\(287\)](#) on compact exponent sets;
4. define the entire functions  $D$  and  $M$ , prove [\(297\)](#), and formalize the forward-difference extraction [\(293\)](#);
5. define the inverse substitution as a pushforward/quantile integral and prove [\(326\)](#);
6. formalize the positive-half Mellin transform [\(318\)](#) and its one-pole continuation;
7. add order-statistic formulations after the nonlinear hierarchy has a stable API.

The principal analytically substantial step is local normal convergence in  $\alpha$ . It can be reduced to the already formalized flatness estimates plus a standard compact bound on  $\binom{\alpha}{k}$ . The generic integer polynomial and zero-tail algebra is already in Lean; the remaining report-specific coefficient identifications and the inverse identity require no new deep analysis.

## 15 Conclusion

The antiderivative problem is governed by one exact mechanism. The signed differential law  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$  turns into a contracting primitive ladder with triangular powers of two. Polynomial multiplication is then a nilpotent perturbation: repeated integration by parts terminates. Replacing an integer exponent by a complex exponent removes nilpotence but produces a normally convergent Newton series whose coefficients are precisely the dyadic ladder samples.

This viewpoint unifies several apparently separate topics. Complete weighted integrals are entire Mellin transforms of the Fabius distribution; their Newton coefficients are rational moments; those moments arise from the infinite sinc product through Bernoulli cumulants and Bell polynomials; exact dyadic evaluation imports Thue–Morse and half-base  $q$ -binomial formulae; and negative-power tails inherit the lower-Lambert log-periodic endpoint structure. The inverse function introduces no new object at first order: its primitive is the ordinary area-under-an-inverse formula closed by  $F(Q/2)$ . New nonlinear objects first appear at higher inverse order, where powers of  $F$  encode order statistics.

The most concrete next results are a rigorous log-periodic tail theorem, a dyadic functional system for  $H_{\beta,j}$ , and Lean formalization of the still-open exponent–Newton duality and report-specific coefficient crosswalk beyond the generic Volterra layer. These are narrow enough to attack directly and broad enough to connect the exact integral calculus back to the main arithmetic and asymptotic architecture of the repository.

## A Technical convergence details

### A.1 Rapid decay of the moments

For completeness, here is the elementary estimate used in [theorem 5.1](#). For every  $A > 0$ , endpoint flatness of  $up$  at 1 gives  $up(t) \leq C_A(1-t)^A$  on  $[0, 1]$ . Then

$$d_n = n \int_0^1 t^{n-1} up(t) dt \quad (356)$$

$$\leq C_A n \int_0^1 t^{n-1} (1-t)^A dt = C_A n B(n, A+1) = \mathcal{O}_A(n^{-A}). \quad (357)$$

No sharp endpoint asymptotic is needed.

### A.2 Compact bounds for generalized binomial coefficients

For a compact  $K \subset \mathbb{C}$ , choose  $R$  with  $|\alpha| \leq R$  on  $K$ . The recurrence  $\binom{\alpha}{k+1} = \binom{\alpha}{k}(\alpha-k)/(k+1)$  and a finite initial adjustment imply

$$\sup_{\alpha \in K} \left| \binom{\alpha}{k} \right| \leq C_K (1+k)^{R+1}. \quad (358)$$

Together with (266) and (288), this proves local uniform absolute convergence. Sharper asymptotics  $\binom{\alpha}{k} \sim (-1)^k k^{-\alpha-1} / \Gamma(-\alpha)$  away from nonnegative integers are useful for tail analysis but unnecessary for existence.

### A.3 Termwise exponent differentiation

Normal convergence permits differentiation with respect to  $\alpha$ . For example,

$$\int_0^x t^\alpha (\log t)^m F(t) dt = \frac{\partial^m}{\partial \alpha^m} A_\alpha(x), \quad (359)$$

with the derivative obtained termwise from (289). At  $\alpha = 0$  this gives convergent series for antiderivatives of  $(\log x)^m F(x)$ ; at  $\alpha = -r$  it gives  $x^{-r}(\log x)^m F(x)$ . Thus logarithmic weights are included at essentially no extra cost.

## B Selected exact formula table

| Object                 | Exact representation                                                                                |
|------------------------|-----------------------------------------------------------------------------------------------------|
| $I_0^n[x^p F]$         | $\sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k})$              |
| $I_0^n[x^\alpha F]$    | Same sum over $k \geq 0$ with $\binom{p}{k}$ replaced by $\binom{\alpha}{k}$ ; normally convergent. |
| $M(\alpha)$            | $(1 - D(\alpha + 1))/(\alpha + 1) = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} d_{k+1}/(k+1)$         |
| $I_0^n[x^\alpha F](1)$ | $\frac{1}{(n-1)!} \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} d_{n+k}/(n+k)$                           |
| $I_0^n[x^p(1-F)]$      | $p! x^{p+n}/(p+n)! - I_0^n[x^p F]$ on $[0, 1]$                                                      |

| Object                     | Exact representation                                                                       |
|----------------------------|--------------------------------------------------------------------------------------------|
| $I_{-1}^n[x^p \text{ up}]$ | $\sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F((x+1)/2^{n+k})$ |
| $\int_0^y Q(v)^\alpha dv$  | $yQ(y)^\alpha - \alpha A_{\alpha-1}(Q(y))$                                                 |
| $\int Q(y) dy$             | $yQ(y) - F(Q(y)/2) + C$                                                                    |
| $I_0^n[Q^\alpha](1)$       | $\frac{1}{n!} \mathbb{E}[\min(X_1, \dots, X_n)^\alpha]$                                    |

## C Recursive corpus inventory

The following 76 .tex paths were returned by the pinned recursive documentation inventory. Historical copies are retained because novelty checking must include superseded formulae, not only the current exposition.

| No. | Path relative to Analysis/FabiusFunction/docs/                                                  |
|-----|-------------------------------------------------------------------------------------------------|
| 1   | papers/ubiq15/ubiq15-corrected.tex                                                              |
| 2   | papers/arXiv-2211.13570v1/document.tex                                                          |
| 3   | papers/arXiv-1702.05442v1/09-Function.tex                                                       |
| 4   | papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex                                                 |
| 5   | papers/arXiv-1502.06738v2/Lacunary_products.tex                                                 |
| 6   | fabius_lean_walkthrough/fabius_lean_walkthrough.tex                                             |
| 7   | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex                                            |
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| 18  | archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex              |
| 19  | Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex             |
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| 24  | archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex   |
| 25  | archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex  |
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| 41  | semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/spectral_check.tex                   |
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| 43  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex                 |
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| 53  | semi-formalized-research-frontiers/drafts/Fabius_Derivative_Norm_Spectrum_bundle/Fabius_Derivative_Norm_Spectrum.tex        |
| 54  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex  |
| 55  | semi-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex  |
| 56  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex  |
| 57  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex  |
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| 59  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex                                                                         |
| 60  | semi-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex                                                                        |
| 61  | semi-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/fabius_frontier_dyadic_inverse_barnes.tex                                                                      |
| 62  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex                                                                      |
| 63  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex                                                                      |
| 64  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex                                                                      |
| 65  | archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex                                                                                 |
| 66  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex                                                                   |
| 67  | semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex                                                                |
| 68  | semi-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex                                                                |
| 69  | semi-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex                                                               |
| 70  | semi-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex                                                               |
| 71  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex                                                            |
| 72  | archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex                                                          |
| 73  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex                                                    |
| 74  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex                                          |
| 75  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex                                          |
| 76  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-a-n-infinite-product-of-sinc-functions.tex |

## D Reproduction commands and generated files

From the extracted archive directory:

```
python fabius_integrals_experiments.py --output-dir .
pdflatex -interaction=nonstopmode -halt-on-error
Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error
Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error
Integration_and_Transform_Frontiers.tex
```

The program generates:

- grid\_inverse\_power\_checks.csv;
- fabius\_finite\_primitive\_checks.csv;
- rvachev\_shifted\_primitive\_checks.csv;
- complete\_integer\_exact\_checks.csv;

- `fractional_complete_series_checks.csv`;
- `negative_power_tail_checks.csv`;
- `inverse_quantile_exact_checks.csv`;
- `newton_series_convergence.png`;
- `run_summary.txt`.

The script contains comments explaining every representation and every nontrivial scaling factor. NumPy and SciPy are used only for grid/quadrature checks; exact rational identities use Python's standard-library `Fraction` class.

## References

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## Part IV

# Integral Calculus and Transform Dualities for the Fabius–Rvachev System

## Abstract

This report develops an integral calculus for the Fabius function  $F$ , its inverse  $Q = F^{-1}$ , and Rvachev’s compactly supported function  $\text{up}$ . The starting point is the probability model

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

where the  $U_k$  are independent uniform random variables on  $[0, 1]$ ,  $F$  is the distribution function of  $Y$ , and  $\text{up}$  is the density of  $Z$ . This model turns integration into a combination of quantile transport, moment calculus, and infinite-product transforms.

The principal results are: a two-function inverse-pair integration identity containing all weighted area formulas for  $F$  and  $Q$ ; a bivariate Laplace–Fourier duality; an entire complex moment function  $\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv$  with two rapidly convergent Newton-type series; exact series for negative and fractional powers; a stable recurrence for negative moments and a positive normalized limit  $C_*$ , giving

$$\mathfrak{M}(-r) \sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)};$$

Riemann–Liouville primitives of arbitrary order for powers of  $F$  and  $Q$ ; beta-weighted formulas, Bernoulli-polynomial integrals, order-statistic representations, Cauchy transforms, and Gini/autocorrelation integrals expressed through the infinite sinc product. Numerical experiments are supplied in a reproducible Python script.

The weighted-law, CDF, and affine fixed-point portion of the post-snapshot Lean audit, checked at compiled source checkpoint d312c0603, now provides the generic continuous-linear naturality and conditional operator/scalar affine-recursion substrate for weighted uniform laws, together with the named geometric weights  $(1 - q)q^n$ , their probability law, self-similarity, reflection, absolute continuity, null singletons, exact support, and half-base bridges. The new real-CDF layer also proves continuity, reflection, and the midpoint value for  $|q| < 1$ , exact CDF tails for  $0 \leq q < 1$ , and, for  $0 < q < 1$ , both conditioning integrals, an explicit Radon–Nikodym density with compact support, and  $C^\infty$  regularity. At  $q = 1/2$ , the CDF and density are identified on all of  $\mathbb{R}$  with the bounded Fabius function and the affine Rvachev up-function. The generic affine-independent-copy layer now also proves that, on every complete second-countable real inner-product target, a measurable digit map and a probability digit law determine at most one probability fixed point whenever  $|q| < 1$ . Its geometric specialization characterizes the named law for the full range  $|q| < 1$ , including  $q = 0$  and negative  $q$ , without support, density, or moment hypotheses on a competing law. This does not formalize the report’s improper-singular inverse-pair extensions, fractional-power, complex-Mellin, rapid-decay/Fourier-inversion, explicit Bernoulli-cumulant or Bell-moment formulas and asymptotics, shape, or higher-derivative-wrapper consequences. A later mainline tranche, rechecked in the current combined full-facade build, does prove the exact finite residual product and pointwise convergence of its phase-bearing prefixes for every  $|q| < 1$ ; finite characteristic-function, MGF, and CGF tail factorizations were already available in the geometric dictionary. The existing `geometricSincProduct` packages the same sinc factors, after frequency rescaling, as a `tprod`; what remains separate is a named characteristic-function bridge to it (or explicit `HasProd` theorem) and compact-uniform convergence of the characteristic prefixes. A still later module, `InversePairIntegral.lean`, does formalize the complete and variable-upper proper-compact absolutely-continuous inverse-pair identities.

Here “new” means absent from the recursively audited LaTeX corpus of the referenced ProveIt directory at the time of writing. It is not a claim of worldwide priority. Each statement is labelled in context as proved, computational, or conjectural.

# 1 Corpus audit, scope, and novelty boundary

## 1.1 What was reviewed

The repository was traversed recursively through the documentation root, including the main monographs, glossary, Lean walkthrough, non-elementarity article, papers, canonical research-frontier volume, current frontier drafts, and frozen archive. The archive map was used to identify historical duplicates and their canonical successors; living source files and current drafts were checked directly at theorem and section level, with targeted full-text searches for integrals, inverse-function area formulas, Mellin transforms, fractional integrals, order statistics, Stieltjes transforms, and negative moments. The repository’s own normalized audit records sixty-seven distinct LaTeX paths at one baseline snapshot, while the current tree contains later additions as well.

The following table states the boundary used in this report.

| Imported baseline                                                                                                         | How it is used here                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Probability interpretation of $F$ , compact support and refinement equations for up, and the infinite sinc product        | Normalization and the common probabilistic language for all integrals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Generic weighted-uniform laws and support-free quantile transport                                                         | The formal source now contains arbitrary Banach-valued distribution splitting and reflection, exact interval mass for non-negative real weights, continuous-linear and scalar naturality, conditional operator/scalar affine recursions, the named geometric-uniform probability law with exact support and null singletons, its continuous CDF and explicit smooth compactly supported density in the positive-ratio range, a support-free atom-compatible quantile pushforward, its compact specialization, and the clamped Fabius inverse theorem. |
| Exact dyadic values, rational moment recurrences, Bernoulli and $q$ -binomial structure                                   | Inputs to the new Mellin/Newton continuation; these arithmetic results are not reproved in full.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lambert- $W$ endpoint and inverse asymptotics, periodic phase corrections                                                 | Used only for comparison and future asymptotic directions. The negative-moment theorem below is instead obtained from an exact moment recurrence.                                                                                                                                                                                                                                                                                                                                                                                                     |
| Dyadic self-sampling quadrature, finite-prefix inverse acceleration, Rvachev-Appell and inverse-moment polynomial systems | Explicitly excluded from novelty claims; only their standard Appell consequence for repeated primitives outside the support is recalled.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Earlier integration-frontier material on elementary weighted inverse identities                                           | Treated as precursor material. The present contribution is the full two-function calculus, complex-parameter continuation, fractional-power duality, and transform consequences.                                                                                                                                                                                                                                                                                                                                                                      |
| Normalized integer-order Volterra calculus and signed-global Fabius primitives                                            | Now formalized independently in <code>NormalizedVolterra.lean</code> and <code>FabiusAntiderivatives.lean</code> : arbitrary-base oriented kernels, literal iteration, strict FTC, the order semigroup, Taylor reconstruction, polynomial weights, and exact dyadic Fabius primitive ladders.                                                                                                                                                                                                                                                         |

**Formalization update: what has crossed the frontier**

At repository snapshot 0770442945d65bac5530c4f214c6c52c4cb9fbdc, Lean proves the complete *integer-order* substrate used by this report, in greater generality than the Fabius-only statements: the normalized Volterra operator is Banach-valued, based at an arbitrary real point, and uses oriented endpoints. Literal iteration equals the Cauchy kernel for continuous inputs; strict FTC, additive composition of orders, Taylor reconstruction, kernel raising, and finite polynomial commutators are all formal. For every  $n, p \in \mathbb{N}$ , the signed extension has exact global formulas for  $J_0^n \mathcal{F}$  and  $J_0^n(t^p \mathcal{F}(t))$ ; the bounded  $F$  has the same formulas for  $x \leq 1$ .

The integer Volterra results do *not* by themselves formalize inverse-pair substitutions, noninteger or complex powers, negative-moment continuation, order statistics, or transform identities. A separate `FractionalVolterra.lean` module now supplies the generic positive-real ordered-endpoint kernel, elementary algebra, positive-integer compatibility, and compact-right-support cutoff, together with the strict-interval shifted beta-kernel value. Its companion `FractionalVolterraSemigroup.lean` proves raw beta-kernel integrability, the normalized beta value, and additive composition of two positive orders for a continuous Banach-valued input on an ordered compact interval. `FractionalVolterraCalculus.lean` proves increasing-affine covariance and positive-order integration by parts across an interval-integrable right interior derivative with its exact left-endpoint boundary term, and `FabiusFractionalVolterra.lean` defines the total causal Rvachev fractional primitive, proves its support-truncated integral, positive-natural bridge, and positive-order semigroup on  $x \geq -1$ , and proves the signed, bounded, and folded single-step Fabius–Rvachev shifts on, respectively,  $x \geq 0$ ,  $[0, 1]$ , and  $x \geq -1$  from base  $-1$ . These modules prove no semigroup for merely interval-integrable inputs or noncomplete targets, fractional derivative, complex order, or shifted dyadic-lattice, endpoint-moment, transform/tail-series, negative-endpoint signed, explicit piecewise-coefficient, refinement, inverse, or quantile specialization. Support-free measure-level quantile transport, together with its compact specialization, is now formal separately, as recorded below. The full-interval explicit-primitive and pointwise- $C^1$  inverse layer-cake forms have also since been formalized, and now derive clock measurability from the compact-interval Galois connection. The real-primitive, real- or complex-weighted arbitrary- $L^1$ /absolutely-continuous subtraction theorem, its normalized corollary, and its Fabius wrapper are formal as well. Moreover, `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` in `InversePairIntegral.lean` proves the complete absolutely-continuous identity on arbitrary ordered compact source and level intervals. It requires no measurability hypothesis on either clock: interval preservation and the strict-order adjunction force monotonicity, after which the proof uses measurable clamped `IccExtend` versions. Its proper compact variable-upper companion is `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt`, with exact Fabius wrapper `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`. Thus both proper-compact formulas in [theorem 3.1](#) have crossed the frontier; its improper-singular extension has not. The identity specialization of the variable-upper wrapper is `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`.

**Results not found in the audited corpus**

The following package appears to be new relative to the repository corpus: the universal  $(A, B)$  inverse-pair identity in [theorem 3.1](#); the entire Mellin function and its Newton/central series in [theorems 4.2](#) and [4.4](#); the mixed reflected Mellin function; the negative-moment recurrence, periodic limiting profile, and constant  $C_*$  in [theorems 5.1](#) and [5.2](#); the fractional duality of [theorem 7.1](#); beta and resolvent transforms; and the sinc-product Gini formula in [theorem 10.2](#). This is a snapshot-relative corpus claim: both proper-compact absolutely-continuous formulas of [theorem 3.1](#) are now proved Lean declarations, while the improper-singular extension remains

part of this frontier package.

## 1.2 Conventions

All powers of positive real variables use the real logarithm:  $x^s = \exp(s \log x)$ . The totalized Fabius distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1).$$

Its inverse  $Q : (0, 1) \rightarrow (0, 1)$  is the ordinary increasing inverse; endpoint values are understood by continuity. Angular frequency is used for Fourier transforms:  $\hat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$ .

## 2 Probability model and transform baseline

### 2.1 The dyadic uniform sum

Let  $U_1, U_2, \dots$  be independent random variables uniformly distributed on  $[0, 1]$ , and put

$$Y = \sum_{k=1}^{\infty} \frac{U_k}{2^k}, \quad Z = 2Y - 1 = \sum_{k=1}^{\infty} \frac{2U_k - 1}{2^k}. \quad (360)$$

Then  $Y \in [0, 1]$ ,  $Z \in [-1, 1]$ , and the laws are symmetric:

$$Y \stackrel{d}{=} 1 - Y, \quad Z \stackrel{d}{=} -Z. \quad (361)$$

The Fabius function is the distribution function of  $Y$ ,

$$F(x) = \mathbb{P}(Y \leq x), \quad (362)$$

and  $\text{up}$  is the density of  $Z$ . Thus

$$F'(x) = 2 \text{up}(2x - 1), \quad \text{up}(x) = \begin{cases} F(x + 1), & -1 \leq x \leq 0, \\ F(1 - x) = 1 - F(x), & 0 \leq x \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (363)$$

In particular  $F(x) + F(1 - x) = 1$  on  $[0, 1]$ , and the quantile obeys

$$Q(1 - v) = 1 - Q(v). \quad (364)$$

The self-decomposition

$$Y \stackrel{d}{=} \frac{U + Y'}{2} \quad (365)$$

holds with  $U \sim \text{Unif}[0, 1]$  independent of a copy  $Y'$  of  $Y$ . It will generate both the positive- and negative-moment recurrences.

### Formal arbitrary-weight law and geometric CDF underlying the dyadic split

The weighted-law, CDF, and affine fixed-point API recorded in this box was checked and frozen at compiled full-facade checkpoint d312c0603. The sinc-prefix declarations below are later mainline additions rechecked in the current combined full-facade build; the report's original novelty snapshot remains the one stated in its provenance metadata.

The current Lean source proves the distributional mechanism before specializing the weights. For

every norm-summable sequence in a real Banach space with its Borel structure, the weighted uniform-coordinate pushforward is a probability law, splits as an independent uniform head plus the shifted-tail law, and is invariant under reflection through half the total vector weight. Nonnegative real weights give mass one to their natural interval and mass zero to its complement. The law definition itself remains total even without measurability. More generally, the weighted head–tail joint-law theorem accepts an arbitrary independently chosen summable tail-weight sequence  $v$ . The shifted tail  $v_n = w_{n+1}$  appears only when that theorem is specialized to split one parent law  $\mu_w$ .

The declarations for this probability, splitting, reflection, and carried-set layer are `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_comp`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution`, `Fabius.ProbabilityRepresentation.isProbabilityMeasure_weightedUniformDistribution`, `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tailWeightedUniformSeries`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_split`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_reflection`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_Icc`, and `Fabius.ProbabilityRepresentation.weightedUniformDistribution_compl_Icc`. The exact dyadic-to-generic law bridge is `Fabius.ProbabilityRepresentation.weightedSumDistribution_eq_weightedUniformDistribution`.

The module now also proves the generic measure-theoretic engine behind affine self-similarity. If measurable  $g$  and  $\Phi$  satisfy  $f(\omega) = \Phi(\omega_0, g(\tau\omega))$  almost everywhere, then

$$f_*\mathbf{P} = \Phi_*(\lambda_{[0,1]} \otimes g_*\mathbf{P}),$$

with no separate measurability assumption on  $f$ ; a pointwise version is a corollary. For a norm-summable sequence in a real Banach space and a continuous linear map  $L$ ,  $L(S_w) = S_{Lw}$ , and the corresponding Borel laws satisfy  $L_*\mu_w = \mu_{Lw}$ ; the target normed space need not be complete. Pointwise scalar naturality needs neither summability nor completeness, whereas its law-level version retains source completeness, Borel structure, and summability. If  $w_{n+1} = A(w_n)$  for a continuous linear endomorphism  $A$ , then

$$A_*\mu_w = \mu_{w+}, \quad S_w(\omega) = \omega_0 w_0 + A(S_w(\tau\omega)), \quad \mu_w = ((u, z) \mapsto ww_0 + Az)_*(\lambda_{[0,1]} \otimes \mu_w).$$

The scalar case  $A = c \text{ id}$  is valid for arbitrary real  $c$ , subject only to the displayed norm-summability hypothesis on the weights at the series-splitting and law levels.

The eleven declarations in this naturality tranche are `Fabius.ProbabilityRepresentation.uniformProduct_map_of_ae_head_tail_recursion`, `Fabius.ProbabilityRepresentation.uniformProduct_map_of_head_tail_recursion`, `Fabius.ProbabilityRepresentation.weightedUniformSeries_map`, `Fabius.ProbabilityRepresentation.weightedUniformSeries_smul`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_map`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_smul`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_map_eq_shift`, `Fabius.ProbabilityRepresentation.weightedUniformSeries_split_of_shift_eq_map`, `Fabius.ProbabilityRepresentation.weightedUniformDistribution_split_of_shift_eq_map`, `Fabius.ProbabilityRepresentation.weightedUniformSeries_geometric_split`, and `Fabius.ProbabilityRepresentation.weightedUniformDistribution_geometric_split`. The last two generic names express only a conditional scalar weight shift; the named specialization is a separate layer.

The seven compatibility declarations are `Fabius.ProbabilityRepresentation.weightedUniformSeries_smul_weights`, `Fabius.ProbabilityRepresentation.uniformProd`



uct\_map\_head\_tail\_function, Fabius.ProbabilityRepresentation.weightedUniformDistribution\_isProbabilityMeasure, Fabius.ProbabilityRepresentation.weightedUniformDistribution\_smul\_weights, Fabius.ProbabilityRepresentation.uniformProduct\_map\_head\_tail\_weightedUniformSeries, Fabius.ProbabilityRepresentation.weightedUniformDistribution\_unitInterval, and Fabius.ProbabilityRepresentation.weightedUniformDistribution\_compl\_unitInterval. The law-level scalar compatibility identity is oriented as  $\mu_{cw} = (x \mapsto cx)_* \mu_w$ .

The separate module `AffineIndependentCopy.lean` supplies one definition and eight theorems for the contractive measure-level mechanism. In the namespace `ProbabilityTheory`, they are `affineIndependentCopyLaw`, `affineIndependentCopyLaw_isProbabilityMeasure`, `affineIndependentCopyLaw_eq_map_prod`, `charFun_affineIndependentCopyLaw`, `charFun_eq_mul_charFun_of_affineIndependentCopy_fixedPoint`, `charFun_iterate_of_affineIndependentCopy_fixedPoint`, `eq_of_charFun_affine_recurrence`, `affineIndependentCopyLaw_fixedPoint_unique`, and `affineIndependentCopy_map_fixedPoint_unique`. The operator sends a candidate law  $\mu$  to the convolution of the pushed-forward digit law with the scalar image  $(q \cdot)_* \mu$ ; the API identifies this convolution with the direct product-map law, proves probability preservation, factors its characteristic function, and iterates the fixed-point recurrence through the residual argument  $q^n t$ . On a complete second-countable real inner-product space with its Borel structure, a measurable digit map over an arbitrary probability digit space and two probability candidate laws, the hypothesis  $|q| < 1$  makes the fixed point unique. This includes  $q = 0$  and every negative contraction and uses no support, absolute-continuity, density, or moment hypothesis.

For every real  $q$ , the named weights are  $w_n(q) = (1 - q)q^n$ , with exact zero and successor formulas. Under  $|q| < 1$  they are norm-summable with total mass one; their series is continuous and measurable, and the probability law satisfies the exact head-tail product identity, affine self-similarity, reflection, absolute continuity, null singletons, and equality of topological support with the exact series range. Under  $0 \leq q < 1$ , the weights are nonnegative, the series lies in  $[0, 1]$ , the law gives mass one to that interval and zero to its complement, and its topological support is exactly  $[0, 1]$ . The twenty-three declarations in `GeometricUniformLaw.lean` are `Fabius.ProbabilityRepresentation.geometricUniformWeight`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_zero`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_succ`, `Fabius.ProbabilityRepresentation.hasSum_geometricUniformWeight`, `Fabius.ProbabilityRepresentation.summable_norm_geometricUniformWeight`, `Fabius.ProbabilityRepresentation.geometricUniformWeight_nonneg`, `Fabius.ProbabilityRepresentation.geometricUniformSeries`, `Fabius.ProbabilityRepresentation.continuous_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.measurable_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_split`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_reflect`, `Fabius.ProbabilityRepresentation.geometricUniformSeries_mem_Icc`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_isProbabilityMeasure`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_absolutelyContinuous`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_nullSingletonClass`, `Fabius.ProbabilityRepresentation.uniformProduct_map_head_tail_geometricUniformSeries`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_selfSimilar`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_reflection`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_Icc`, `Fabius.ProbabilityRepresentation.geometricUniformDistribution_compl_Icc`, `Fabius.ProbabilityRepresentation.geometricUniformD`

istribution\_support\_eq\_range, and `Fabius.ProbabilityRepresentation.geometricUniformDistribution_support_eq_1cc`. The exact half-base series and law bridges are `Fabius.ProbabilityRepresentation.weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `Fabius.ProbabilityRepresentation.weightedSumDistribution_eq_geometricUniformDistribution_one_half`. The further module `GeometricUniformUniqueness.lean` exports the single corollary `Fabius.ProbabilityRepresentation.eq_geometricUniformDistribution_of_selfSimilar`. For every real  $q$  with  $|q| < 1$ , it says that a probability measure  $\mu$  satisfying

$$\mu = ((u, x) \mapsto (1 - q)u + qx)_* (\lambda_{[0,1]} \otimes \mu)$$

is exactly `Fabius.ProbabilityRepresentation.geometricUniformDistribution(q)`. The statement includes  $q = 0$  and negative  $q$  and imposes no carried-support, density, or moment condition on  $\mu$ .

The finite four-face theorem `Fabius.geometric_tail_dictionary_geometricUniform` already packages measure, characteristic-function, MGF, and CGF tail factorizations. The six public declarations in `GeometricSincFactorization.lean` are `Fabius.charFun_geometricUniformDigit`, `Fabius.charFun_geometricUniformDistribution_prefix`, `Fabius.charFun_geometricUniformDistribution_prefix_sinc`, `Fabius.tendsto_prefix_sinc_charFun`, `Fabius.charFun_weightedSumDistribution_prefix_sinc`, and `Fabius.tendsto_prefix_sinc_charFun_weightedSumDistribution`. The digit formula is unconditional in real  $q$ . For every  $|q| < 1$ ,  $m \in \mathbb{N}$ , and  $t \in \mathbb{R}$ , writing  $\phi_q(t) = \text{charFun}(\mu_q)(t)$ , the phase-collapsed finite residual formula is

$$\phi_q(t) = e^{i(1-q^m)t/2} \left( \prod_{k=0}^{m-1} \text{sinc} \left( \frac{(1-q)q^k t}{2} \right) \right) \phi_q(q^m t),$$

and the phase-bearing prefix without the residual converges pointwise to  $\phi_q(t)$ . This scope includes  $q = 0$  and negative  $q$ , and the final two declarations are the exact  $q = 1/2$  weighted-sum wrappers.

The three generic declarations in `ContinuousCDF.lean` are `Fabius.continuous_cdf_of_nullSingleton`, `Fabius.cdf_reflection_sub`, and `Fabius.measure_eq_withDensity_of_cdf_hasDerivAt`. They turn null singletons into CDF continuity, turn affine reflection of a real probability law into its all-real CDF reflection, and identify a law with Lebesgue measure having density of  $\text{Real} \circ f$  from an everywhere pointwise CDF derivative. The last result assumes neither prior absolute continuity nor continuity of  $f$ .

The twenty-seven declarations in `GeometricUniformCDF.lean` are `Fabius.ProbabilityRepresentation.geometricUniformCDF`, `Fabius.ProbabilityRepresentation.monotone_geometricUniformCDF`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_nonneg`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_le_one`, `Fabius.ProbabilityRepresentation.measurable_geometricUniformCDF`, `Fabius.ProbabilityRepresentation.continuous_geometricUniformCDF`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_reflection`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_one_half`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_zero_of_nonpos`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_one_of_one_le`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_eq_integral`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_eq_intervalIntegral`, `Fabius.ProbabilityRepresentation.geometricUniformDensity`, `Fabius.ProbabilityRepresentation.geometricUniformCDF_hasDerivAt`, `Fabius.ProbabilityRepresentation.deriv_geometricUniformCDF`, `Fabius.ProbabilityRepresenten`

tation.continuous\_geometricUniformDensity, Fabius.ProbabilityRepresentation.geometricUniformDensity\_nonneg, Fabius.ProbabilityRepresentation.geometricUniformDensity\_zero\_of\_nonpos, Fabius.ProbabilityRepresentation.geometricUniformDensity\_zero\_of\_one\_le, Fabius.ProbabilityRepresentation.support\_geometricUniformDensity\_subset\_Ioo, Fabius.ProbabilityRepresentation.support\_geometricUniformDensity\_subset\_Icc, Fabius.ProbabilityRepresentation.tsupport\_geometricUniformDensity\_subset\_Icc, Fabius.ProbabilityRepresentation.geometricUniformDensity\_hasCompactSupport, Fabius.ProbabilityRepresentation.geometricUniformDensity\_reflection, Fabius.ProbabilityRepresentation.geometricUniformDistribution\_eq\_withDensity, Fabius.ProbabilityRepresentation.contDiff\_geometricUniformCDF, and Fabius.ProbabilityRepresentation.contDiff\_geometricUniformDensity. The CDF definition and its monotonicity, bounds, and measurability are total in real  $q$ . Continuity, reflection, and the midpoint identity hold for  $|q| < 1$ ; its exterior values are formal for  $0 \leq q < 1$ . Under  $0 < q < 1$ , the two conditioning identities hold on all of  $\mathbb{R}$ , the explicit formula is an everywhere pointwise derivative and the exact Radon–Nikodym density, and the density is nonnegative, symmetric, compactly supported, and  $C^\infty$ . The ordinary support theorem is containment in  $(0, 1)$  and the topological-support theorem is containment in  $[0, 1]$ ; these should not be confused with the law’s exact-support theorem above.

Finally, ProbabilityRepresentation.lean gives the all-real half-base bridges Fabius.ProbabilityRepresentation.weightedSumCDF\_eq\_geometricUniformCDF\_one\_half, Fabius.ProbabilityRepresentation.geometricUniformCDF\_one\_half\_eq\_fabiusReal, and Fabius.ProbabilityRepresentation.geometricUniformDensity\_one\_half\_eq\_rvachevUp. Thus the dyadic CDF and density identities used at the start of this report are now formal.

The definition Fabius.ProbabilityRepresentation.geometricUniformDensity is totalized, but its classical density and Radon–Nikodym interpretation are not formal here for  $q \leq 0$ . The pointwise phase-bearing characteristic-function prefix limit is formal; geometricSincProduct already packages the same rescaled sinc factors as a tprod. Still outside the formal layer are a named characteristic-function bridge to that product (or explicit HasProd theorem), compact-uniform convergence of the characteristic prefixes, the rapid-decay/Fourier-inversion route, closed higher-derivative refinements and endpoint-flatness wrappers, shape results, further transform formulas, and explicit Bernoulli-cumulant/Bell-moment formulas and asymptotics. The affine measure fixed point is unique by Fabius.ProbabilityRepresentation.eq\_geometricUniformDistribution\_of\_selfSimilar for every  $|q| < 1$ . The oriented transform DDE and adjacent Cauchy-power recurrence are also formal for every nonzero  $|q| < 1$ , including negative ratios, in GeometricUniformCauchy.lean; these are distinct from the characteristic-function product and real-space density-derivative hierarchies.

## 2.2 Moment-generating and Fourier products

Define

$$P(t) = \mathbb{E}e^{tY}, \quad \mathcal{U}(t) = \mathbb{E}e^{tZ}. \quad (366)$$

Independence in [equation \(360\)](#) gives the locally uniformly convergent products

$$P(t) = \prod_{k=1}^{\infty} \frac{e^{t/2^k} - 1}{t/2^k}, \quad (367)$$

$$\mathcal{U}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}, \quad (368)$$

$$P(t) = e^{t/2} \mathcal{U}(t/2). \quad (369)$$

The characteristic function of the up law is therefore

$$\phi(t) = \widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \frac{\sin(t/2^k)}{t/2^k}. \quad (370)$$

The scaling equation

$$P(2t) = \frac{e^t - 1}{t} P(t) \quad (371)$$

is the exponential generating-function form of the moment recurrence below.

### 2.3 Rational moments, cumulants, and Bell polynomials

Write

$$m_n = \mathbb{E}Y^n = \int_0^1 Q(v)^n dv, \quad \mu_n = \mathbb{E}Z^n = \int_{-1}^1 x^n \text{up}(x) dx. \quad (372)$$

Taking the  $n$ -th moment in [equation \(365\)](#) gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (373)$$

Thus every  $m_n$  is rational. The first values are

$$1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \dots \quad (374)$$

The logarithm of [equation \(368\)](#) and  $\log(\sinh z/z) = \sum_{r \geq 1} 2^{2r-1} B_{2r} z^{2r} / (r(2r)!)$  give the cumulants

$$\kappa_1(Y) = \frac{1}{2}, \quad \kappa_{2r}(Y) = \frac{B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0 \quad (r \geq 1), \quad (375)$$

$$\kappa_{2r}(Z) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Z) = 0. \quad (376)$$

Consequently  $m_n$  and  $\mu_n$  are complete exponential Bell polynomials in these cumulants. A useful bridge between the one-sided and centered moments is

$$\mu_{2j} = 2 \int_0^1 x^{2j} \text{up}(x) dx = \frac{2m_{2j+1}}{2j+1}, \quad \mu_{2j+1} = 0. \quad (377)$$

For example

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (378)$$

## 3 A universal integration calculus for $F$ and $Q$

The classical area identity for a function and its inverse is only the first member of a much larger family. The following theorem is the main algebraic engine of the report.

### Exact boundary of the formal quantile theorems

The foundational measure-transport theorem is support-free. Let  $\mu$  be any probability measure on  $\mathbb{R}$ , and let  $Q$  be almost-everywhere measurable for uniform measure on  $[0, 1]$ . If, for every  $x \in \mathbb{R}$ ,

$$Q(y) \leq x \iff y \leq \text{cdf}_\mu(x) \quad \text{for almost every } y \in [0, 1],$$

then  $Q_*\lambda_{[0,1]} = \mu$ . There is no support assumption on  $\mu$  and no requirement that  $Q$  take its source into  $[0, 1]$ . A compact specialization replaces the almost-everywhere sublevel hypothesis by a pointwise order characterization on the unit square, together with a.e. support and interval preservation. Both theorems permit atoms.

The Fabius specialization uses the globally clamped inverse but restricts the uniform source to  $[0, 1]$ ; every strongly measurable Banach-valued statistic has the corresponding integral identity, with no integrability assumption because both Bochner integrals are totalized.

The exact declarations are `Fabius.map_quantile_eq`, `Fabius.map_inverseCDF_volume_restrict_Icc`, `Fabius.map_fabiusInv_restrict_Icc_eq_weightedSumDistribution`, and `Fabius.integral_comp_fabiusInv_restrict_Icc_eq_weightedSumDistribution` in `QuantileTransport.lean`. Those pushforward results alone do not prove the two-function identity below. Separately, `Fabius.intervalIntegral_smul_intervalIntegral_fabiusInv` and `Fabius.intervalIntegral_smul_comp_fabiusInv` in `InverseLayerCake.lean` now prove its full-unit-interval layer-cake core for an explicit interval-integrable primitive and for a pointwise- $C^1$  primitive, respectively. Their generic forms no longer assume clock measurability: the helper `Fabius.galoisConnection_Icc_restrict_of_lt_iff_lt` makes the interval restrictions a Galois connection and supplies the monotonicity used to construct a measurable clamped inverse. For a real-valued absolutely continuous primitive and a real or complex  $L^1$  weight, the generic subtraction and normalized forms are `Fabius.intervalIntegral_mul_comp_sub_of_lt_iff_lt_of_absolutelyContinuousOnInterval` and `Fabius.intervalIntegral_mul_comp_of_lt_iff_lt_of_absolutelyContinuousOnInterval`; their exact Fabius wrapper is `Fabius.intervalIntegral_mul_comp_fabiusInv_of_absolutelyContinuousOnInterval`. No separate complementary-integrability hypothesis is required.

The complete proper-compact absolutely-continuous theorem is now `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` in `InversePairIntegral.lean`. It is more general than the unit-square statement: the source and level intervals are arbitrary ordered compact intervals, and the clocks need only preserve those intervals and satisfy  $C(x) < u \iff x < Q(u)$ . Neither clock is assumed measurable. Lean derives their monotonicity on the closed intervals and uses measurable clamped `IccExtend` versions internally. Its exact Fabius wrapper is `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv`, so [equation \(380\)](#) is machine checked exactly. The identity-factor corollary `Fabius.intervalIntegral_fabiusReal_add_fabiusInv_eq_one` is the corresponding area sum. The proper compact variable-upper theorem `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt` has exact Fabius wrapper `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`, so [equation \(379\)](#) is machine checked exactly as well. Taking both factors to be the identity at an inverse endpoint gives `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`. The theorem below as a whole remains a frontier claim only because it also asserts improper extensions through singular endpoints.

**Theorem 3.1** (two-function inverse-pair identity). *Let  $A, B$  be absolutely continuous on  $[0, 1]$ . For*

every  $z \in [0, 1]$ ,

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'(x)B(F(x)) \, dx + \int_0^{F(z)} A(Q(v))B'(v) \, dv. \quad (379)$$

At  $z = 1$ ,

$$A(1)B(1) - A(0)B(0) = \int_0^1 A'(x)B(F(x)) \, dx + \int_0^1 A(Q(v))B'(v) \, dv. \quad (380)$$

The formulas remain valid under the usual improper-integral hypotheses whenever an endpoint is singular.

*Proof.* Differentiate the product  $A(x)B(F(x))$ , integrate from 0 to  $z$ , and in the term containing  $F'(x)$  use the increasing substitution  $v = F(x)$ , so that  $x = Q(v)$ .  $\square$

*Remark 3.2.* No special functional equation of the Fabius function is needed in [theorem 3.1](#); it is a quantile identity. What is special here is that the moments and transforms of this particular quantile can be calculated by dyadic products and rational recurrences.

### 3.1 Power duality and logarithmic derivatives

Taking  $A(x) = x^p$ ,  $B(v) = v^q$  gives the following Mellin-type duality.

**Corollary 3.3** (power duality). *Initially for  $\Re p, \Re q > 0$ ,*

$$p \int_0^z x^{p-1} F(x)^q \, dx + q \int_0^{F(z)} Q(v)^p v^{q-1} \, dv = z^p F(z)^q. \quad (381)$$

*In particular,*

$$p \int_0^1 x^{p-1} F(x)^q \, dx + q \int_0^1 v^{q-1} Q(v)^p \, dv = 1. \quad (382)$$

*Both sides admit meromorphic or entire continuation in parameters whenever the endpoint flatness cancels the apparent singularities.*

For all logarithmic derivatives at once, define

$$J_{r,s}(p, q) = \int_0^1 x^{p-1} (\log x)^r F(x)^q (\log F(x))^s \, dx, \quad (383)$$

$$K_{r,s}(p, q) = \int_0^1 v^{q-1} (\log v)^s Q(v)^p (\log Q(v))^r \, dv. \quad (384)$$

Differentiating [equation \(382\)](#) gives, for  $r + s > 0$ ,

$$pJ_{r,s} + rJ_{r-1,s} + qK_{r,s} + sK_{r,s-1} = 0, \quad (385)$$

with  $pJ_{0,0} + qK_{0,0} = 1$ . Equation [equation \(385\)](#) is a compact general formula for every mixed derivative of the two Mellin kernels.



### 3.2 A bivariate Laplace–Fourier identity

The exponential choice  $A(x) = e^{sx}$ ,  $B(v) = e^{tv}$  produces an especially simple transform relation.

**Theorem 3.4** (bivariate inverse-pair transform). *For all  $s, t \in \mathbb{C}$  and  $z \in [0, 1]$ ,*

$$s \int_0^z e^{sx+tF(x)} dx + t \int_0^{F(z)} e^{tv+sQ(v)} dv = e^{sz+tF(z)} - 1. \quad (386)$$

Hence

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1. \quad (387)$$

The identities are entire in  $(s, t)$ ; the cases  $s = 0$  or  $t = 0$  are understood by continuity.

Every mixed power integral involving  $x$  and  $F(x)$ , together with its quantile dual, is obtained by coefficient extraction from [equation \(387\)](#). Purely imaginary  $s, t$  give a bivariate Fourier identity.

### 3.3 Weighted antiderivatives

Taking  $B(v) = v$  and a primitive  $A$  of a prescribed weight yields the useful master formula

$$\int_0^x A'(t)F(t) dt = A(x)F(x) - A(0)F(0) - \int_0^{F(x)} A(Q(v)) dv. \quad (388)$$

For monomial weights this becomes the promised indefinite integral.

**Corollary 3.5** (monomial–Fabius antiderivative). *For  $p \neq -1$ , initially where the integrals converge and then by continuation,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1}F(x) - \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (389)$$

The value at  $p = -1$  is removable:

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log Q(v) dv. \quad (390)$$

On  $[0, 1]$ , where  $\text{up} = 1 - F$ ,

$$\int_0^x t^p \text{up}(t) dt = \frac{x^{p+1}(1 - F(x)) + \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (391)$$

More generally, from [equation \(381\)](#),

$$\int_0^x t^{p-1} F(t)^q dt = \frac{x^p F(x)^q}{p} - \frac{q}{p} \int_0^{F(x)} Q(v)^p v^{q-1} dv. \quad (392)$$

Thus products of  $F$  and the positive half of  $\text{up}$  reduce to beta-weighted quantile integrals after expanding  $(1 - F)^n$ .

## 4 The entire complex moment function

**Definition 4.1.** For  $s \in \mathbb{C}$ , set

$$\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv. \quad (393)$$

For nonnegative integers,  $\mathfrak{M}(n) = m_n$ .

### 4.1 Entire continuation

The endpoint flatness of  $F$  is much stronger than what is needed for ordinary positive moments: for every  $N > 0$ ,

$$F(x) = O_N(x^N) \quad (x \downarrow 0). \quad (394)$$

**Theorem 4.2** (entire quantile Mellin function). *The function  $\mathfrak{M}$  in [equation \(393\)](#) is entire. For every  $r \geq 0$ ,*

$$\mathfrak{M}^{(r)}(s) = \int_0^1 Q(v)^s (\log Q(v))^r dv = \mathbb{E}[Y^s (\log Y)^r]. \quad (395)$$

*In particular all negative moments and all logarithmic moments exist.*

*Proof.* Let  $K \subset \mathbb{C}$  be compact and choose  $A > 0$  with  $\Re s \geq -A$  on  $K$ . For every fixed  $r$ , the integrand is bounded in modulus by  $Y^{-A} |\log Y|^r$ . Integration by parts gives

$$\mathbb{E}Y^{-A} = 1 + A \int_0^1 x^{-A-1} F(x) dx < \infty,$$

using [equation \(394\)](#) with an exponent larger than  $A$ ; inserting powers of  $|\log x|$  does not alter convergence. Dominated differentiation on compact sets proves holomorphy and [equation \(395\)](#).  $\square$

A first transform triangle follows immediately:

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad (396)$$

$$\int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}, \quad (397)$$

$$\int_0^1 Q(v)^{s-1} dv = \mathfrak{M}(s-1). \quad (398)$$

The first quotient has a removable singularity at  $s = 0$  and is entire; the second is the meromorphic continuation of the classical Mellin transform of  $\text{up}|_{[0,1]}$  and has exactly one pole, at  $s = 0$ , with residue 1. Evenness gives

$$\int_{-1}^1 |x|^{s-1} \text{up}(x) dx = \frac{2\mathfrak{M}(s)}{s}. \quad (399)$$

### 4.2 Superpolynomial moment decay

**Lemma 4.3.** *For every  $N > 0$ ,*

$$m_k = O_N(k^{-N}) \quad (k \rightarrow \infty). \quad (400)$$

*Proof.* For  $k > 0$ , the tail formula and symmetry give

$$m_k = k \int_0^1 x^{k-1} (1 - F(x)) dx = k \int_0^1 x^{k-1} F(1-x) dx.$$

By [equation \(394\)](#),  $F(1-x) \leq C_N(1-x)^N$ , hence

$$m_k \leq C_N k B(k, N+1) = O(k^{-N}).$$

$\square$



### 4.3 Two globally convergent Newton series

**Theorem 4.4** (Newton and central-even series). *Both of the following series converge locally uniformly for all  $s \in \mathbb{C}$ :*

$$\mathfrak{M}(s) = \sum_{k=0}^{\infty} (-1)^k \binom{s}{k} m_k, \quad (401)$$

$$\mathfrak{M}(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j} = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{2m_{2j+1}}{2j+1}. \quad (402)$$

*They therefore represent the entire function  $\mathfrak{M}$  on the whole plane.*

*Proof.* For  $0 < Y < 1$ , expand  $Y^s = (1 - (1 - Y))^s$ . Taking expectations and using  $1 - Y \stackrel{d}{=} Y$  gives equation (401). On every compact set in  $s$ , the binomial coefficients have at most polynomial growth in  $k$ ; lemma 4.3 therefore justifies absolute, locally uniform convergence and interchange of sum and expectation.

For equation (402), write  $Y = (1 + Z)/2$ , expand  $(1 + Z)^s$ , and use the evenness of the law of  $Z$ . The relation equation (377) and lemma 4.3 again give locally uniform convergence.  $\square$

Several useful specializations are immediate. For an integer  $r \geq 1$ ,

$$\mathfrak{M}(-r) = \sum_{k=0}^{\infty} \binom{r+k-1}{k} m_k. \quad (403)$$

Every term is rational and positive. In particular,

$$\mathfrak{M}(-1) = \sum_{k=0}^{\infty} m_k. \quad (404)$$

Differentiation of equation (401) at the origin yields logarithmic moments in terms of signed Stirling numbers of the first kind  $s(k, r)$ :

$$\mathfrak{M}^{(r)}(0) = r! \sum_{k=r}^{\infty} \frac{(-1)^k s(k, r)}{k!} m_k. \quad (405)$$

For  $r = 1$ ,

$$-\mathbb{E} \log Y = \sum_{k=1}^{\infty} \frac{m_k}{k} = \int_0^1 \frac{F(x)}{x} dx. \quad (406)$$

Numerically,

$$-\mathbb{E} \log Y = 0.758240000404408656 \dots \quad (407)$$

### 4.4 A mixed reflected Mellin function

The inverse symmetry equation (364) suggests the two-variable transform

$$\mathfrak{M}_2(a, b) = \int_0^1 Q(v)^a Q(1-v)^b dv = \mathbb{E}[Y^a (1-Y)^b]. \quad (408)$$

**Proposition 4.5.** *The function  $\mathfrak{M}_2$  is entire on  $\mathbb{C}^2$ , symmetric in  $a, b$ , and has the locally uniformly convergent expansion*

$$\mathfrak{M}_2(a, b) = \sum_{k=0}^{\infty} (-1)^k \binom{b}{k} \mathfrak{M}(a+k). \quad (409)$$

It also has the central expansion

$$\mathfrak{M}_2(a, b) = 2^{-a-b} \sum_{j,k \geq 0} (-1)^k \binom{a}{j} \binom{b}{k} \mu_{j+k}. \quad (410)$$

If either parameter is a nonnegative integer, the corresponding binomial expansion terminates.

*Proof.* Endpoint flatness at both 0 and 1 gives local domination for arbitrary negative real parts, proving entire dependence on two parameters. Expand  $(1 - Y)^b$  to obtain [equation \(409\)](#); superpolynomial decay of shifted moments gives local uniformity. The central formula follows from  $Y = (1 + Z)/2$ ,  $1 - Y = (1 - Z)/2$ .  $\square$

For example,

$$\int_0^1 Q(v)Q(1-v) dv = \frac{2}{9}, \quad \int_0^1 Q(v)^2 Q(1-v)^2 dv = \frac{34}{675}, \quad (411)$$

and

$$\int_0^1 \frac{dv}{Q(v)(1-Q(v))} = 2\mathfrak{M}(-1). \quad (412)$$

## 5 Negative moments: an exact recurrence and a new asymptotic constant

The Newton series [equation \(403\)](#) is effective for moderate negative orders. A second construction is much more stable for large orders and reveals a new normalization law.

### 5.1 The recurrence

For  $s > 0$ , define

$$A(s) = \mathfrak{M}(-s) = \mathbb{E}Y^{-s}, \quad D(s) = \mathbb{E}(1 + Y)^{-s}. \quad (413)$$

Conditioning in [equation \(365\)](#) and integrating the uniform variable gives the following exact relation.

**Theorem 5.1** (negative-moment recurrence). *For every real  $s > 0$ ,*

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)). \quad (414)$$

Equivalently, for  $s > 1$ ,

$$A(s) = \frac{2^s}{s-1} (A(s-1) - D(s-1)). \quad (415)$$

Moreover  $D(s)$  has the rapidly convergent centered series

$$D(s) = \left(\frac{2}{3}\right)^s \sum_{j=0}^{\infty} \frac{(s)_{2j}}{(2j)! 3^{2j}} \mu_{2j}, \quad (416)$$

where  $(s)_n = s(s+1) \cdots (s+n-1)$ .

*Proof.* By [equation \(365\)](#), for  $s > 0$ ,

$$A(s+1) = 2^{s+1} \mathbb{E} \int_0^1 (U + Y')^{-s-1} dU.$$

For  $y > 0$ ,

$$\int_0^1 (u+y)^{-s-1} du = \frac{y^{-s} - (1+y)^{-s}}{s},$$

which gives equation (414). For equation (416), write  $1+Y = (3+Z)/2$ , expand  $(1+Z/3)^{-s}$ , and use the vanishing of odd moments. The factor  $3^{-2j}$  makes the series geometrically stable in addition to the superpolynomial decay of  $\mu_{2j}$ .  $\square$

The first values generated by this recurrence are

$$\begin{aligned} A(1) &= 2.31488614400581905\dots, \\ A(2) &= 6.55899183845822858\dots, \\ A(3) &= 24.3890791927221220\dots, \\ A(4) &= 128.368907270620532\dots, \\ A(5) &= 1025.15473735960333\dots \end{aligned} \quad (417)$$

## 5.2 A normalized limit and a periodic phase profile

Define

$$H(s) = \frac{\Gamma(s)A(s)}{2^{s(s+1)/2}}, \quad \rho(s) = \frac{D(s)}{A(s)}. \quad (418)$$

Then equation (414) becomes

$$H(s+1) = H(s)(1 - \rho(s)). \quad (419)$$

Both factors are positive. Since  $\mathbb{P}(Y \leq 1/2) = 1/2$ ,

$$A(s) \geq 2^{s-1}, \quad D(s) \leq 1, \quad 0 < \rho(s) \leq 2^{1-s}. \quad (420)$$

This elementary estimate is enough to prove convergence.

**Theorem 5.2** (periodic limiting profile). *For  $0 < \theta \leq 1$ , the product*

$$C(\theta) = H(\theta) \prod_{j=0}^{\infty} (1 - \rho(\theta + j)) \quad (421)$$

*converges to a positive number. It extends by  $C(\theta+1) = C(\theta)$  to a positive, continuous, one-periodic function. As  $n \rightarrow \infty$ ,*

$$A(n+\theta) = C(\theta) \frac{2^{(n+\theta)(n+\theta+1)/2}}{\Gamma(n+\theta)} (1 + O_{\theta}(2^{-n})). \quad (422)$$

For integer  $r \rightarrow \infty$ ,

$$\boxed{\mathfrak{M}(-r) = C_* \frac{2^{r(r+1)/2}}{\Gamma(r)} (1 + O(2^{-r}))}, \quad C_* = C(1) > 0. \quad (423)$$

*Proof.* Iteration of equation (419) gives

$$H(n+\theta) = H(\theta) \prod_{j=0}^{n-1} (1 - \rho(\theta + j)).$$

The sum of the tail bounds in equation (420) is geometric. Hence the infinite product converges to a positive limit, and its omitted tail differs from 1 by  $O_{\theta}(2^{-n})$ . This proves equation (422). Shifting the initial point by one and using equation (419) proves exact periodicity. Uniform convergence on compact subintervals gives continuity; exact periodicity then removes the apparent endpoint at  $\theta = 0$ . The integer formula is the case  $\theta = 1$ .  $\square$

For integers the normalized sequence

$$C_r = \frac{(r-1)! \mathfrak{M}(-r)}{2^{r(r+1)/2}} \quad (424)$$

is strictly decreasing and satisfies

$$C_{r+1} = C_r \left( 1 - \frac{D(r)}{A(r)} \right), \quad C_r \downarrow C_*. \quad (425)$$

The product representation is

$$C_* = \frac{\mathfrak{M}(-1)}{2} \prod_{j=1}^{\infty} \left( 1 - \frac{D(j)}{A(j)} \right). \quad (426)$$

The computation described in [appendix 12](#) gives

$$C_* = 0.750721980407870959 \dots \quad (427)$$

The rapid stabilization is visible in

| $r$   | 1          | 2          | 4          | 6          | 8          | 12           |
|-------|------------|------------|------------|------------|------------|--------------|
| $C_r$ | 1.15744307 | 0.81987398 | 0.75216157 | 0.75072879 | 0.75072199 | 0.7507219804 |

### 5.3 Logarithmic asymptotics and entire-function growth

Stirling's expansion in [equation \(423\)](#) gives a complete algebraic asymptotic series:

$$\begin{aligned} \log \mathfrak{M}(-r) &= \frac{\log 2}{2} r^2 - r \log r + \left( 1 + \frac{\log 2}{2} \right) r + \frac{1}{2} \log r + \log \frac{C_*}{\sqrt{2\pi}} \\ &\quad - \frac{1}{12r} + \frac{1}{360r^3} - \frac{1}{1260r^5} + \dots, \end{aligned} \quad (428)$$

where

$$\log \frac{C_*}{\sqrt{2\pi}} = -1.20565842815389633 \dots \quad (429)$$

This refines the growth one would obtain by a direct saddle-point transfer from the Lambert- $W$  endpoint expansion.

There is also a clean entire-function consequence. If  $M_{\mathfrak{M}}(R) = \max_{|s|=R} |\mathfrak{M}(s)|$ , then

$$|\mathfrak{M}(s)| \leq \mathbb{E} Y^{\Re s} \leq \mathbb{E} Y^{-R} = \mathfrak{M}(-R), \quad |s| = R, \quad (430)$$

while equality occurs at  $s = -R$ . Thus

$$M_{\mathfrak{M}}(R) = \mathfrak{M}(-R). \quad (431)$$

**Corollary 5.3.** *The entire function  $\mathfrak{M}$  has order 2 and type  $(\log 2)/2$ .*

### 5.4 Singular Fabius integrals

Putting  $s = p + 1$  in [equation \(396\)](#) shows that

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p+1)}{p+1} \quad (432)$$

is an entire function of  $p$ , despite the apparent pole at  $p = -1$ . Therefore all negative powers are integrable against  $F$ . For integers  $k \geq 2$ ,

$$\int_0^1 \frac{F(x)}{x^k} dx = \frac{\mathfrak{M}(1-k) - 1}{k-1}. \quad (433)$$

The first examples are

$$\begin{aligned} \int_0^1 \frac{F(x)}{x} dx &= 0.758240000404408656\dots, \\ \int_0^1 \frac{F(x)}{x^2} dx &= 1.31488614400581905\dots, \\ \int_0^1 \frac{F(x)}{x^3} dx &= 2.77949591922911429\dots, \\ \int_0^1 \frac{F(x)}{x^4} dx &= 7.79635973090737400\dots \end{aligned} \quad (434)$$

By [equation \(423\)](#),

$$\int_0^1 \frac{F(x)}{x^k} dx \sim C_* \frac{2^{k(k-1)/2}}{\Gamma(k)}. \quad (435)$$

This is an asymptotic law for a natural family of increasingly singular antiderivatives.

## 6 Definite weighted integrals

### 6.1 Polynomial and analytic weights

Let  $g$  be integrable and let  $G(x) = \int_0^x g(t) dt$ . Taking  $A = G$ ,  $B(v) = v$  in [theorem 3.1](#) gives

$$\int_0^1 g(x)F(x) dx = G(1) - \mathbb{E}G(Y), \quad (436)$$

$$\int_0^1 g(x)(1 - F(x)) dx = \mathbb{E}G(Y). \quad (437)$$

For a polynomial  $g$ , every quantity on the right is a rational combination of the moments  $m_n$ . For an entire  $g$ , the same formulas give convergent series whenever the Taylor expansion can be integrated termwise.

The monomial cases are

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p+1)}{p+1}, \quad (438)$$

$$\int_0^1 x^p \text{up}(x) dx = \frac{\mathfrak{M}(p+1)}{p+1} \quad (\text{up} = 1 - F \text{ on } [0, 1]). \quad (439)$$

For  $p \in \mathbb{N}$  these are rational. For example,

| $p$                            | 0   | 1     | 2    | 3         |
|--------------------------------|-----|-------|------|-----------|
| $\int_0^1 x^p F(x) dx$         | 1/2 | 13/36 | 5/18 | 1207/5400 |
| $\int_0^1 x^p \text{up}(x) dx$ | 1/2 | 5/36  | 1/18 | 143/5400  |

For symmetric weights  $g(x) = g(1-x)$ , symmetry alone gives

$$\int_0^1 g(x)F(x) dx = \frac{1}{2} \int_0^1 g(x) dx. \quad (440)$$

## 6.2 Bernoulli-polynomial weights

Let

$$\beta_n = \mathbb{E}B_n(Y). \quad (441)$$

Their exponential generating function follows from [equation \(367\)](#):

$$\sum_{n=0}^{\infty} \beta_n \frac{t^n}{n!} = \frac{tP(t)}{e^t - 1}. \quad (442)$$

Symmetry implies  $\beta_{2j+1} = 0$ . Since  $B'_{n+1} = (n+1)B_n$ , [equation \(436\)](#) gives, for  $n \geq 1$ ,

$$\int_0^1 B_n(x)F(x) dx = -\frac{\beta_{n+1} - B_{n+1}}{n+1}. \quad (443)$$

The first nontrivial expected Bernoulli values are

$$\beta_2 = -\frac{1}{18}, \quad \beta_4 = \frac{23}{1350}, \quad \beta_6 = -\frac{1543}{119070}, \quad \beta_8 = \frac{1196113}{65063250}. \quad (444)$$

This supplies a direct integral bridge between the Bernoulli-polynomial structure and the Fabius distribution.

## 6.3 Beta weights

For  $\Re a, \Re b > 0$ , define

$$I_F(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1}F(x) dx. \quad (445)$$

**Theorem 6.1** (beta-weighted Fabius integrals). *For  $\Re a, \Re b > 0$ ,*

$$I_F(a, b) = \mathbb{E}B_Y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} \mathfrak{M}(b+n), \quad (446)$$

$$\int_0^1 x^{a-1}(1-x)^{b-1}(1-F(x)) dx = \mathbb{E}B_Y(a, b) = \sum_{n=0}^{\infty} \frac{(1-b)_n}{n!(a+n)} \mathfrak{M}(a+n). \quad (447)$$

Here  $B_y$  is the incomplete beta function. If  $a$  in [equation \(446\)](#), or  $b$  in [equation \(447\)](#), is a positive integer, the corresponding series terminates. If both parameters are positive integers, the integrals are rational.

*Proof.* Use  $F(x) = \mathbb{E}\mathbf{1}_{\{Y \leq x\}}$  and Fubini:

$$I_F(a, b) = \mathbb{E} \int_Y^1 x^{a-1}(1-x)^{b-1} dx = \mathbb{E}B_{1-Y}(b, a) = \mathbb{E}B_Y(b, a),$$

where the last equality uses  $1-Y \stackrel{d}{=} Y$ . The hypergeometric expansion

$$B_y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} y^{b+n}$$

gives [equation \(446\)](#). The endpoint flatness and [lemma 4.3](#) justify convergence beyond the ordinary boundary restrictions of the pointwise hypergeometric series. The second identity is analogous.  $\square$

For example, symmetry or direct termination gives

$$\int_0^1 x(1-x)F(x) dx = \frac{1}{12}. \quad (448)$$

Differentiating [equation \(446\)](#) in  $a$  and  $b$  produces all beta-logarithmic integrals with powers of  $\log x$  and  $\log(1-x)$ .

## 6.4 A dyadic half-interval identity

The lower-half differential equation  $F'(x) = 2F(2x)$  for  $0 \leq x \leq 1/2$  leads to exact quantile truncations.

**Proposition 6.2.** *For every complex  $p$ , with removable values interpreted by continuation,*

$$\int_0^{1/2} Q(v)^p dv = 2^{-p} \frac{1 - \Re(p+1)}{p+1}, \quad (449)$$

$$\int_0^{1/2} x^p F(x) dx = \frac{2^{-p-1}}{p+1} \left[ \frac{1}{2} - \frac{1 - \Re(p+2)}{p+2} \right]. \quad (450)$$

In particular,

$$\int_0^{1/2} F(x) dx = \frac{5}{72}. \quad (451)$$

*Proof.* Change variables  $v = F(x)$  in the first integral and use the lower-half equation:

$$\int_0^{1/2} Q(v)^p dv = \int_0^{1/2} x^p F'(x) dx = 2^{-p} \int_0^1 y^p F(y) dy.$$

This is [equation \(449\)](#). Substitute it into the incomplete formula [equation \(389\)](#) at  $x = 1/2$  to obtain [equation \(450\)](#).  $\square$

## 7 Fractional and integer antiderivatives

Repeated integration has a particularly transparent probabilistic meaning. The same calculus extends without change from integer order to the Riemann–Liouville operator. For  $\Re \alpha > 0$ , write

$$(I_0^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (452)$$

### Integer versus fractional proof status

For  $n \in \mathbb{N}$ , the operator  $I_0^n$  here is the formal normalized Volterra operator  $J_0^n$ . In particular, Lean proves

$$I_0^n F(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right) \quad (x \leq 1),$$

including  $n = 0$ , and proves the analogous global identity for the signed extension. It also proves every natural-monomial weight as a finite dyadic sum. The displayed noninteger- $\alpha$  truncated-moment representation below and its inverse companion remain outside that formal result; the separate one-step positive-real-order Fabius and up shifts are now formalized.

### 7.1 Fractional inverse-pair duality

**Theorem 7.1** (fractional power duality). *Let  $\Re \alpha > 0$ . Initially for  $\Re q > 0$ ,*

$$(I_0^\alpha F^q)(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv. \quad (453)$$

*Initially for  $\Re p > 0$ ,*

$$(I_0^\alpha Q^p)(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx. \quad (454)$$

*Both formulas continue in  $p$  or  $q$  wherever the endpoint terms admit continuation.*

*Proof.* Since  $d(x - t)^\alpha = -\alpha(x - t)^{\alpha-1} dt$ , integration by parts gives

$$I_0^\alpha F^q(x) = \frac{1}{\Gamma(\alpha + 1)} \int_0^x (x - t)^\alpha d(F(t)^q).$$

The boundary terms vanish when  $\Re q > 0$ . Using  $d(F^q) = qF^{q-1} dF$  and substituting  $v = F(t)$  proves [equation \(453\)](#). The proof of [equation \(454\)](#) is identical after interchanging the inverse pair.  $\square$

The case  $q = 1$  has a direct truncated-moment interpretation.

**Corollary 7.2.** For  $\Re \alpha > 0$ ,

$$(I_0^\alpha F)(x) = \frac{\mathbb{E}[(x - Y)_+^\alpha]}{\Gamma(\alpha + 1)}. \quad (455)$$

In particular,

$$(I_0^\alpha F)(1) = \frac{\mathfrak{M}(\alpha)}{\Gamma(\alpha + 1)}. \quad (456)$$

*Proof.* In [equation \(453\)](#), the integral over  $v$  is the expectation of  $(x - Q(V))_+^\alpha$  for a uniform  $V$ . Since  $Q(V) \stackrel{d}{=} Y$ , this is [equation \(455\)](#). At  $x = 1$ , use  $1 - Y \stackrel{d}{=} Y$ .  $\square$

Formula [equation \(456\)](#) says that the entire complex moment function is exactly the endpoint value of the fractional antiderivative, multiplied by  $\Gamma(\alpha + 1)$ . Thus the Newton series in [equation \(401\)](#) is at the same time a global series for fractional repeated integration.

## 7.2 Repeated primitives of Rvachev's up function

For  $\Re \alpha > 0$ , define the left-anchored fractional integral

$$(I_{-1}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \int_{-1}^x (x - t)^{\alpha-1} \text{up}(t) dt. \quad (457)$$

Because  $\text{up}$  is the density of  $Z$ ,

$$(I_{-1}^\alpha \text{up})(x) = \frac{\mathbb{E}[(x - Z)_+^{\alpha-1}]}{\Gamma(\alpha)}. \quad (458)$$

At the right endpoint,

$$(I_{-1}^\alpha \text{up})(1) = \frac{2^{\alpha-1} \mathfrak{M}(\alpha - 1)}{\Gamma(\alpha)}. \quad (459)$$

For an integer  $n \geq 1$ , [equation \(458\)](#) becomes

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}[(x - Z)_+^{n-1}]. \quad (460)$$

## 7.3 Appell polynomial tails

For  $x \geq 1$ , the positive-part signs in [equations \(455\)](#) and [\(460\)](#) disappear. Hence

$$(I_0^n F)(x) = \frac{1}{n!} \mathbb{E}(x - Y)^n, \quad x \geq 1, \quad (461)$$

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x - Z)^{n-1}, \quad x \geq 1. \quad (462)$$

The polynomial sequences

$$P_n(x) = \mathbb{E}(x - Y)^n, \quad R_n(x) = \mathbb{E}(x - Z)^n \quad (463)$$



are Appell sequences:

$$P'_n(x) = nP_{n-1}(x), \quad R'_n(x) = nR_{n-1}(x). \quad (464)$$

Their exponential generating functions are

$$\sum_{n \geq 0} P_n(x) \frac{t^n}{n!} = e^{tx} P(-t), \quad (465)$$

$$\sum_{n \geq 0} R_n(x) \frac{t^n}{n!} = e^{tx} \mathcal{U}(-t). \quad (466)$$

These polynomial tails and their inverse-moment projection role are part of the existing Rvachev–Appell research in the repository; here they arise as the exterior part of the fractional-truncated-moment formula.

#### 7.4 Ordinary $n$ -fold antiderivatives as incomplete moments

For an integer  $n \geq 1$ , the anchored  $n$ -fold antiderivative of  $F$  is

$$I_0^n F(x) = \frac{1}{n!} \int_0^{F(x)} (x - Q(v))^n dv. \quad (467)$$

The same expression can be expanded finitely:

$$I_0^n F(x) = \frac{1}{n!} \sum_{j=0}^n (-1)^j \binom{n}{j} x^{n-j} \int_0^{F(x)} Q(v)^j dv. \quad (468)$$

Thus every repeated primitive is determined by incomplete quantile moments. At a dyadic  $x$ , these can in turn be reduced by the exact dyadic machinery already developed in the repository.

For inverse powers, [equation \(454\)](#) gives the complementary expression

$$I_0^n (Q^p)(v) = \frac{p}{n!} \int_0^{Q(v)} x^{p-1} (v - F(x))^n dx. \quad (469)$$

The special case  $p = 1$  is a fractional extension of the area identity:

$$I_0^n Q(v) = \frac{1}{n!} \int_0^{Q(v)} (v - F(x))^n dx. \quad (470)$$

For  $n = 1$ , this is

$$\int_0^v Q(w) dw = \int_0^{Q(v)} (v - F(x)) dx, \quad (471)$$

the usual area formula in a form adapted to the Fabius quantile.

## 8 Derivatives of arbitrary order

### 8.1 Derivatives of the Fabius function

On the lower dyadic branch, repeated differentiation of  $F'(x) = 2F(2x)$  gives

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 < x < 2^{-n}. \quad (472)$$

The full interval is obtained by the known dyadic/Thue–Morse reflection formulas. We do not duplicate those formulas here; the role of [equation \(472\)](#) is to show why every inverse derivative becomes extremely singular near the endpoints.

## 8.2 Inverse derivatives and Bell polynomials

Let  $f_j = F^{(j)}(x)$ . Define rational differential expressions recursively by

$$\mathcal{R}_1(x) = \frac{1}{F'(x)}, \quad \mathcal{R}_{n+1}(x) = \frac{1}{F'(x)} \frac{d}{dx} \mathcal{R}_n(x). \quad (473)$$

**Proposition 8.1** (all inverse derivatives). *For  $0 < v < 1$ ,*

$$Q^{(n)}(v) = \mathcal{R}_n(Q(v)). \quad (474)$$

*Equivalently, for  $n \geq 2$ ,*

$$Q^{(n)}(v) = -\frac{1}{F'(Q(v))} \sum_{k=2}^n F^{(k)}(Q(v)) B_{n,k}(Q'(v), \dots, Q^{(n-k+1)}(v)), \quad (475)$$

where  $B_{n,k}$  is the partial exponential Bell polynomial.

*Proof.* The operator relation follows by applying  $d/dv = (F'(x))^{-1} d/dx$  at  $x = Q(v)$ . The Bell recurrence follows by differentiating  $F(Q(v)) = v$   $n$  times and isolating the term containing  $Q^{(n)}$ .  $\square$

The first expressions are

$$\begin{aligned} Q' &= \frac{1}{f_1}, \\ Q'' &= -\frac{f_2}{f_1^3}, \\ Q''' &= \frac{3f_2^2 - f_1 f_3}{f_1^5}, \\ Q'''' &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{f_1^7}, \end{aligned} \quad (476)$$

with every  $f_j$  evaluated at  $x = Q(v)$ . Combining these formulas with the Lambert- $W$  inverse endpoint expansion from the repository gives asymptotics of any fixed derivative order.

## 8.3 Derivatives of integral transforms

Since  $\mathfrak{M}$  is entire,

$$\mathfrak{M}^{(n)}(s) = \int_0^1 Q(v)^s (\log Q(v))^n dv. \quad (477)$$

The bilateral up transform satisfies, away from its complex zeros,

$$\mathcal{U}^{(n)}(t) = \mathcal{U}(t) B_n(L_1(t), \dots, L_n(t)), \quad (478)$$

where  $B_n$  is the complete exponential Bell polynomial and

$$L_j(t) = \frac{d^j}{dt^j} \log \mathcal{U}(t) = \sum_{k=1}^{\infty} 2^{-kj} \ell^{(j)}(t/2^k), \quad \ell(z) = \log \frac{\sinh z}{z}. \quad (479)$$

At  $t = 0$ , these are precisely the cumulants in [equation \(376\)](#); the continuation of [equation \(478\)](#) through zeros is obtained by cancelling the apparent logarithmic singularities.

## 9 Laplace, Fourier, Mellin, and resolvent transforms

### 9.1 One-sided Laplace transforms

Integration by parts against the distribution measure  $dF$  gives, for all  $t \in \mathbb{C}$  with removable values at  $t = 0$ ,

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad (480)$$

$$\int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}. \quad (481)$$

Consequently

$$\int_0^1 x^n e^{tx} F(x) dx = \frac{d^n}{dt^n} \left( \frac{e^t - P(t)}{t} \right), \quad (482)$$

$$\int_0^1 x^n e^{tx} \text{up}(x) dx = \frac{d^n}{dt^n} \left( \frac{P(t) - 1}{t} \right). \quad (483)$$

Purely imaginary  $t$  yields the corresponding sine and cosine transforms. On the full line,

$$\int_{-1}^1 e^{tx} \text{up}(x) dx = \mathcal{U}(t), \quad \int_{-1}^1 e^{i\omega x} \text{up}(x) dx = \phi(\omega). \quad (484)$$

### 9.2 Stieltjes transforms and all resolvent powers

For  $z \in \mathbb{C} \setminus [0, 1]$ , define

$$G_Y(z) = \mathbb{E} \frac{1}{z - Y} = \int_0^1 \frac{dv}{z - Q(v)}. \quad (485)$$

For  $|z| > 1$ ,

$$G_Y(z) = \sum_{n=0}^{\infty} \frac{m_n}{z^{n+1}}. \quad (486)$$

Similarly, for  $z \notin [-1, 1]$ ,

$$G_Z(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (487)$$

Applying [equation \(436\)](#) to a resolvent primitive gives all higher powers. For every integer  $n \geq 1$ ,

$$\int_0^1 \frac{F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[ \frac{1}{(z - 1)^n} - \mathbb{E} \frac{1}{(z - Y)^n} \right], \quad (488)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[ \mathbb{E} \frac{1}{(z - Y)^n} - \frac{1}{z^n} \right]. \quad (489)$$

For  $n = 1$ ,

$$\int_0^1 \frac{F(x)}{(z - x)^2} dx = \frac{1}{z - 1} - G_Y(z), \quad (490)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^2} dx = G_Y(z) - \frac{1}{z}. \quad (491)$$

Using the logarithmic primitive, with a branch analytic off  $[0, 1]$ ,

$$\int_0^1 \frac{F(x)}{z-x} dx = \mathbb{E} \log(z-Y) - \log(z-1), \quad (492)$$

$$\int_0^1 \frac{1-F(x)}{z-x} dx = \log z - \mathbb{E} \log(z-Y). \quad (493)$$

These formulas furnish analytic continuations and boundary-value formulas for Hilbert transforms by approaching the cut from above and below.

### 9.3 Entire-kernel calculus

If  $h(x) = \sum_{n \geq 0} a_n x^n$  is entire and the series is dominated on the compact support, then

$$\int_{-1}^1 h(x) \text{up}(x) dx = \sum_{n=0}^{\infty} a_n \mu_n, \quad (494)$$

$$\int_0^1 h(x) dF(x) = \sum_{n=0}^{\infty} a_n m_n. \quad (495)$$

For instance,

$$\int_{-1}^1 J_0(tx) \text{up}(x) dx = \sum_{n=0}^{\infty} \frac{(-1)^n (t^2/4)^n}{(n!)^2} \mu_{2n}. \quad (496)$$

Together with [equation \(436\)](#), this gives convergent moment series for Bessel, Airy, hypergeometric, and polylogarithmic kernels without requiring a separate quadrature of the flat endpoint function.

## 10 Products, order statistics, and correlation integrals

Integrals containing products of  $F$  and up are generally not rational, but they have exact probabilistic meanings that lead to useful transform formulas and asymptotics.

### 10.1 Products of distribution and survival powers

Let  $Y_1, \dots, Y_{m+n}$  be independent copies of  $Y$ , and define

$$A_m = \max(Y_1, \dots, Y_m), \quad B_n = \min(Y_{m+1}, \dots, Y_{m+n}). \quad (497)$$

The two blocks are independent and

$$\mathbb{P}(A_m \leq x < B_n) = F(x)^m (1 - F(x))^n. \quad (498)$$

**Theorem 10.1** (weighted order-statistic gap). *For integers  $m, n \geq 1$  and  $\Re p > -1$ ,*

$$\int_0^1 x^p F(x)^m (1 - F(x))^n dx = \frac{1}{p+1} \mathbb{E} [(B_n^{p+1} - A_m^{p+1}) \mathbf{1}_{\{A_m < B_n\}}]. \quad (499)$$

*In particular,*

$$\int_0^1 F(x)^m (1 - F(x))^n dx = \mathbb{E}(B_n - A_m)_+. \quad (500)$$

*Proof.* Insert the indicator of the event in [equation \(498\)](#), integrate, and apply Fubini. For each realization, the allowed interval is  $[A_m, B_n)$  if  $A_m < B_n$ , and empty otherwise. Its  $x^p$ -integral is the right side of [equation \(499\)](#).  $\square$

Since  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$ , this theorem covers all integer products  $F^m \text{up}^n$  on the positive half of the support. Two useful special cases are

$$\int_0^1 F(x)^m dx = \mathbb{E}(1 - A_m) = \mathbb{E} \min(Y_1, \dots, Y_m), \quad (501)$$

and

$$\int_0^1 F(x)^2 dx = \frac{1}{2} - \int_0^1 F(x)(1 - F(x)) dx. \quad (502)$$

When the density factor is present, substitution gives a closed beta value for arbitrary complex exponents in their convergence domain:

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a + 1, b + 1). \quad (503)$$

Using  $F'(x) = 2 \text{up}(2x - 1)$ ,

$$2 \int_0^1 F(x)^a (1 - F(x))^b \text{up}(2x - 1) dx = B(a + 1, b + 1). \quad (504)$$

## 10.2 Quantiles and ordinary order statistics

If  $Y_{r:n}$  is the  $r$ -th order statistic of  $n$  independent Fabius variables, then  $Y_{r:n} = Q(V_{r:n})$  in distribution, where  $V_{r:n}$  has beta density. Consequently

$$\mathbb{E}Y_{r:n}^p = \frac{1}{B(r, n - r + 1)} \int_0^1 v^{r-1} (1 - v)^{n-r} Q(v)^p dv. \quad (505)$$

Symmetry gives

$$\mathbb{E}Y_{r:n}^p = \mathbb{E}(1 - Y_{n+1-r:n})^p, \quad \mathbb{E}Y_{r:n} + \mathbb{E}Y_{n+1-r:n} = 1. \quad (506)$$

Expanding the beta polynomial in [equation \(505\)](#) reduces every integer  $n, r$  to a finite combination of the mixed quantile integrals appearing in [equation \(382\)](#).

## 10.3 The Gini integral and the sinc product

For two independent copies  $Y, Y'$ , the standard distribution identity is

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y - Y'|. \quad (507)$$

The difference is  $(Z - Z')/2$ , and its characteristic function is  $\phi(t/2)^2$ . The absolute-moment formula

$$\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty \frac{1 - \Re \varphi_X(t)}{t^2} dt \quad (508)$$

therefore yields an integral containing only the infinite sinc product.

**Theorem 10.2** (Gini–sinc identity). *With  $\phi$  as in [equation \(370\)](#),*

$$\boxed{\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2\pi} \int_0^\infty \frac{1 - \phi(t)^2}{t^2} dt.} \quad (509)$$

Numerically,

$$\int_0^1 F(x)(1 - F(x)) dx = 0.0955632320160066 \dots, \quad (510)$$

so

$$\int_0^1 F(x)^2 dx = 0.4044367679839934 \dots \quad (511)$$

This is a direct link between a nonlinear real-space integral of  $F$  and the square of the Rvachev infinite product. It does not appear to simplify to the rational moment constants.

## 10.4 Autocorrelation and $L^2$ energy

Define

$$\mathcal{C}(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x+a) dx. \quad (512)$$

Since  $\text{up}$  is even, the Wiener–Khinchin formula gives

$$\mathcal{C}(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} \phi(t)^2 dt. \quad (513)$$

At  $a = 0$ , Parseval’s identity gives

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = \frac{1}{\pi} \int_0^\infty \phi(t)^2 dt = 0.808873535967971 \dots \quad (514)$$

More generally, derivatives of  $\mathcal{C}$  at the origin are convergent spectral moments whenever the corresponding weighted sinc-product integral converges; the fast decay results developed elsewhere in the repository provide quantitative truncation bounds.

## 11 Additional integral-transform identities

### 11.1 A transform relation for inverse resolvents

The general identity [theorem 3.1](#) can be applied in the  $v$ -variable as well. Taking  $A(x) = x$  and  $B(v) = \log(z/(z-v))$  gives

$$\int_0^1 \frac{Q(v)}{z-v} dv + \int_0^1 \log \frac{z}{z-F(x)} dx = \log \frac{z}{z-1}. \quad (515)$$

With  $B(v) = (z-v)^{-1}$ ,

$$\int_0^1 \frac{Q(v)}{(z-v)^2} dv + \int_0^1 \frac{dx}{z-F(x)} = \frac{1}{z-1}. \quad (516)$$

These are nonlinear resolvent dualities: a transform of the quantile is paired with a composition transform of the distribution function.

### 11.2 Frullani representations for logarithmic constants

For  $0 < Y \leq 1$ ,

$$-\log Y = \int_0^\infty \frac{e^{-tY} - e^{-t}}{t} dt. \quad (517)$$

Averaging and using  $P(-t)$  gives

$$\int_0^1 \frac{F(x)}{x} dx = \int_0^\infty \frac{P(-t) - e^{-t}}{t} dt, \quad (518)$$

where

$$P(-t) = \prod_{k=1}^\infty \frac{1 - e^{-t/2^k}}{t/2^k}. \quad (519)$$

For  $a > 0$ , the negative moment itself has the Mellin–Laplace representation

$$\mathfrak{M}(-a) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} P(-t) dt. \quad (520)$$

This is an independent numerical route to [equation \(403\)](#) and connects the negative-moment constant  $C_*$  to the dyadic product on the positive Laplace axis.

### 11.3 A smoothing difference equation for $\mathfrak{M}$

Let

$$N(s) = \mathbb{E}(1 + Y)^s = \sum_{k=0}^{\infty} \binom{s}{k} m_k. \quad (521)$$

The series is entire because of [lemma 4.3](#). Integrating the uniform variable in [equation \(365\)](#) with an arbitrary complex exponent gives the entire difference relation

$$(s + 1)2^s \mathfrak{M}(s) = N(s + 1) - \mathfrak{M}(s + 1), \quad (522)$$

with the value at  $s = -1$  interpreted by continuity. At nonnegative integer  $s$ , coefficient comparison recovers [equation \(373\)](#); at negative  $s$ , it is equivalent to [equation \(414\)](#). Thus the positive rational moments and the superexponentially growing negative moments are two sides of one entire difference equation.

## 12 Numerical experiments and reproducibility

The accompanying script `experiments.py` computes all numerical values in this report without direct sampling of  $Y$ . It uses four independent structures:

1. exact rational arithmetic for low moments via [equation \(373\)](#);
2. a positive, extended-precision moment recurrence based on a normalized  $\text{Binomial}(n, 1/2)$  probability vector;
3. the Newton series [equation \(401\)](#), centered series [equation \(416\)](#), and stable negative recurrence [equation \(414\)](#);
4. adaptive quadrature of the sinc products in [equations \(509\)](#) and [\(514\)](#), with zeros supplied as break points and explicit algebraic tail bounds from the first several sinc factors.

The script used 12000 positive moments. Representative values are

| Quantity                              | Numerical value                                |
|---------------------------------------|------------------------------------------------|
| $\mathfrak{M}(1/2)$                   | 0.696 306 844 628 413 801                      |
| $\mathfrak{M}(-1/2)$                  | 1.489 254 605 665 521 67                       |
| $\mathfrak{M}(-1)$                    | 2.314 886 144 005 819 05                       |
| $-\mathbb{E} \log Y$                  | $7.582\,400\,004\,044\,086\,57 \times 10^{-1}$ |
| $C_*$                                 | 0.750 721 980 407 870 959                      |
| $\int_0^1 F(1 - F)$                   | 0.095 563 232 016 006 6                        |
| $\int_{\mathbb{R}} \text{up}(x)^2 dx$ | 0.808 873 535 967 971                          |

The fractional values provide useful checks that are not forced by rational arithmetic:

$$\begin{aligned} \mathfrak{M}(1/2) &= 0.696306844628413801\dots, \\ \mathfrak{M}(-1/2) &= 1.48925460566552167\dots, \\ \mathfrak{M}(3/2) &= 0.368547739989599266\dots, \\ \mathfrak{M}(5/2) &= 0.213401104479966542\dots \end{aligned} \quad (523)$$

The following samples of the periodic phase profile in [equation \(421\)](#) are strikingly close:

| $\theta$ | $C(\theta)$            |
|----------|------------------------|
| 0.10     | 0.75072198040787095903 |
| 0.25     | 0.75072198040787095922 |
| 0.50     | 0.75072198040787096020 |
| 0.75     | 0.75072198040787096019 |
| 1.00     | 0.75072198040787095923 |

The spread is at the level of  $10^{-18}$ , close to the effective precision of the long-double moment input; it motivates but does not prove [conjecture 13.1](#). The JSON file shipped with the report records all displayed data, the truncation size, and the Fourier cutoff.

### 12.1 Numerical error notes

The moment recurrence is cancellation-free. Truncating the Newton series after  $K$  terms is controlled by the superpolynomial tail  $m_k = O_N(k^{-N})$ ; comparing  $K = 6000, 8000, 10000, 12000$  stabilizes the displayed fractional values. For  $C_*$ , the recurrence itself supplies an a posteriori bound: after index  $r$ , the remaining relative product error is at most a constant multiple of  $\sum_{j \geq r} 2^{1-j} = O(2^{-r})$ .

For the Gini integral with cutoff  $T$ ,

$$\int_T^\infty \frac{1 - \phi(t)^2}{t^2} dt = \frac{1}{T} - \int_T^\infty \frac{\phi(t)^2}{t^2} dt.$$

Bounding the first three squared sinc factors gives an omitted correction  $O(T^{-7})$ . For the  $L^2$  integral, five factors give an  $O(T^{-9})$  tail. The script uses  $T = 500$ , making these errors smaller than the digits printed.

## 13 Conjectures and frontier directions

### 13.1 Near-rigidity of the negative-moment phase

**Conjecture 13.1** (phase rigidity). *The continuous one-periodic limiting profile  $C(\theta)$  in [equation \(421\)](#) is constant:*

$$C(\theta) \equiv C_*. \quad (524)$$

A weaker and perhaps more realistic alternative is that  $C$  is nonconstant but its first Fourier harmonics are suppressed by the same spectral mechanism responsible for the extremely small periodic oscillations in the endpoint asymptotics studied elsewhere in the repository. The exact recurrence reduces this question to the Fourier analysis of the periodic product [equation \(421\)](#). A rigorous high-precision test should compute  $A(\theta)$  independently by [equation \(520\)](#), rather than by the same moment data used in the recurrence.

### 13.2 Completely regular growth of the entire Mellin function

The maximum modulus is known exactly through [equation \(431\)](#), but directional cancellation remains open.

**Conjecture 13.2** (indicator function). *The entire function  $\mathfrak{M}$  has completely regular growth of order 2 with indicator*

$$h(\vartheta) = \frac{\log 2}{2} \max\{0, -\cos \vartheta\}^2. \quad (525)$$

The upper bound follows immediately from  $|\mathfrak{M}(re^{i\vartheta})| \leq \mathfrak{M}(r \max(0, -\cos \vartheta))$ . The difficult part is a matching lower bound away from the negative real axis, where oscillation in  $e^{i\Im(s) \log Y}$  may create zeros and exceptional directions.



### 13.3 Arithmetic nature of the new constants

The quantities

$$C_*, \quad \mathfrak{M}(-r), \quad -\mathbb{E} \log Y, \quad \int_0^1 F(1-F), \quad \int_{\mathbb{R}} \text{up}^2 \quad (526)$$

are defined by positive rational series or infinite sinc products, but no reduction to standard constants is visible.

**Conjecture 13.3.** *The constants  $C_*$ ,  $\mathfrak{M}(-r)$  for every positive integer  $r$ , and  $-\mathbb{E} \log Y$  are irrational; at least one of the stronger properties of transcendence or algebraic independence should hold among them.*

This is deliberately stated as a research target, not as numerical evidence for a specific transcendence theorem. Possible tools include Mahler-type functional equations for  $P(t)$ , arithmetic estimates for the rational Newton partial sums, and Padé approximants controlled by the moment recurrence.

### 13.4 Complete monotonicity of normalized errors

The sequence  $C_r$  is decreasing by [equation \(425\)](#). Numerics suggest much more.

**Conjecture 13.4.** *The error sequence  $C_r - C_*$  is completely monotone:*

$$(-1)^k \Delta^k (C_r - C_*) \geq 0 \quad (r \geq 1, k \geq 0). \quad (527)$$

A proof might represent  $C_r - C_*$  as a Hausdorff moment sequence derived from the positive tail product  $\prod_{j \geq r} (1 - \rho(j))$ .

### 13.5 Order-statistic endpoint regimes

When the sample size  $n$  grows, the minimum  $Y_{1:n}$  probes precisely the Lambert- $W$  endpoint regime. Since

$$\mathbb{P}(Y_{1:n} > x) = (1 - F(x))^n,$$

its moments are

$$\mathbb{E} Y_{1:n}^p = p \int_0^1 x^{p-1} (1 - F(x))^n dx. \quad (528)$$

A promising direction is a uniform saddle-point expansion in  $p$  and  $n$ , with the periodic endpoint phase transported into order-statistic oscillations. The corresponding maximum follows by symmetry. This would connect the present integral calculus directly to the repository's inverse-Lambert and half-integer spectral programs.

### 13.6 General base and $q$ -deformations

Replace the dyadic weights by  $b^{-k}$ , normalized so the support remains  $[0, 1]$ . The distributional smoothing, inverse-pair calculus, and fractional identities survive unchanged, while the product  $P(t)$ , Bernoulli cumulants, and negative-moment normalization acquire base  $b$ . The expected leading law is

$$\mathbb{E} Y_b^{-r} \asymp C_b \frac{b^{r(r+1)/2}}{\Gamma(r)}. \quad (529)$$

Determining  $C_b$ , its phase profile, and its relationship with  $q = 1/b$  binomial coefficients is a natural extension.

### 13.7 Formalization roadmap

The most accessible Lean targets are:

1. package the remaining improper-singular form of [theorem 3.1](#); its complete and variable-upper proper-compact absolutely-continuous forms are already formal in greater interval-theoretic generality;
2. the power identity [equation \(382\)](#) and its polynomial corollaries;
3. the positive and negative moment recurrences;
4. positivity, monotonicity, and convergence of  $C_r$ ;
5. the order-statistic gap identity;
6. finite truncations and certified tail bounds for the Newton and sinc-product series.

The entire-function and fractional-integral portions require more analytic infrastructure, but can be decomposed into dominated-convergence, holomorphic-parameter integration, and beta-integral lemmas.

## 14 Summary of principal formulas

For convenient reference, the central formulas are collected here.

### Distribution and transforms

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

$$P(t) = \prod_{k \geq 1} \frac{e^{t/2^k} - 1}{t/2^k}, \quad \phi(t) = \prod_{k \geq 1} \frac{\sin(t/2^k)}{t/2^k},$$

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad \int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}.$$

### Inverse-pair integration

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'B \circ F + \int_0^{F(z)} A \circ Q B',$$

$$p \int_0^1 x^{p-1} F(x)^q dx + q \int_0^1 v^{q-1} Q(v)^p dv = 1,$$

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1.$$

### Entire moment calculus

$$\mathfrak{M}(s) = \int_0^1 Q(v)^s dv,$$

$$\mathfrak{M}(s) = \sum_{k \geq 0} (-1)^k \binom{s}{k} m_k = 2^{-s} \sum_{j \geq 0} \binom{s}{2j} \mu_{2j},$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad \int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}.$$

### Negative moments and fractional primitives

$$\begin{aligned}
 A(s+1) &= \frac{2^{s+1}}{s} (A(s) - D(s)), \\
 \mathfrak{M}(-r) &\sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)}, \\
 I_0^\alpha F^q(x) &= \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv, \\
 I_0^\alpha Q^p(v) &= \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx.
 \end{aligned}$$

## 15 Conclusion

The probability/quantile viewpoint converts the integral theory of the Fabius–Rvachev system into a coherent calculus rather than a list of isolated evaluations. The inverse-pair identity organizes indefinite weighted integrals; the entire moment function organizes fractional and negative powers; Riemann–Liouville operators organize repeated antiderivatives; and the infinite sinc product organizes correlation integrals. The negative-moment recurrence is especially useful: it extracts a new constant and an analytic periodic phase directly from the dyadic smoothing law, with exponentially fast normalization.

The next decisive steps are to determine whether the phase profile is exactly constant, to connect its Fourier modes with the spectral corrections already present in endpoint asymptotics, and to formalize the remaining improper-singular inverse-pair scope, complex-order and iterated/lattice fractional identities, fractional derivatives, remaining hierarchy-weighted and endpoint-wrapper fractional identities, and moment recurrences. The reusable integer-order Volterra layer and its exact Fabius specialization are already formal; these remaining directions are sufficiently concrete to support both high-precision experiments and Lean proofs.

## A Exact low-order data

The exact positive moments used by the script are

| $n$ | $m_n = \mathbb{E}Y^n$ | $\mu_n = \mathbb{E}Z^n$ when shown |
|-----|-----------------------|------------------------------------|
| 0   | 1                     | 1                                  |
| 1   | 1/2                   | 0                                  |
| 2   | 5/18                  | 1/9                                |
| 3   | 1/6                   | 0                                  |
| 4   | 143/1350              | 19/675                             |
| 5   | 19/270                | 0                                  |
| 6   | 1153/23814            | 583/59535                          |
| 7   | 583/17010             | 0                                  |
| 8   | 1616353/65063250      | 132809/32531625                    |
| 9   | 132809/7229250        | 0                                  |
| 10  | 134926369/9762090030  | 46840699/24405225075               |

The centered entries can be generated more economically from  $\mu_{2j} = 2m_{2j+1}/(2j+1)$  than by expanding  $(2Y-1)^{2j}$ , which avoids severe cancellation at high order.

## B Repository audit categories

The recursive review used the following categories.

| Category                           | Role in the audit                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core expositions                   | Definitions, probability law, smoothness, dyadic derivatives, symmetry, Fourier product, moment arithmetic.                                                                 |
| Glossary and Lean walk-through     | Normalization checks and correspondence with formalized objects.                                                                                                            |
| Canonical research-frontier volume | Primary non-duplication comparison for asymptotics, Lambert- $W$ , Bell/Bernoulli, $q$ -binomial, interpolation, and extrapolation themes.                                  |
| Current drafts                     | Checks against dyadic self-sampling, inverse-function expansions, Rvachev–Appell systems, half-integer spectral effects, Fourier decay, and certification algorithms.       |
| Papers                             | Historical and arithmetic results, exact dyadic values, Fourier coefficients, and endpoint asymptotics.                                                                     |
| Frozen archive                     | Historical variants and absorbed integration material, checked through the archive map and canonical successors to prevent counting old snapshots as independent frontiers. |

## C Reproducibility instructions

From the directory containing this report, run

```
python experiments.py --moments 12000 --cutoff 500
```

The script writes `numerical_results.json` and `numerical_results.tex`. The latter is included directly in the PDF table. Reducing `--moments` to 2000 gives a quick check; values involving the phase profile should use at least 10000 moments. The code requires NumPy, SciPy, and mpmath.

## References

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**Part V****Integral and Transform Frontiers for the Fabius and Rvachev Functions****Abstract**

This report develops a systematic integral calculus for the Fabius function  $F$ , Rvachev's compactly supported function  $u$ , and the inverse Fabius function  $G = F^{-1}$ . It begins with a recursive audit of the current TeX corpus in the ProveIt repository, including the main monograph, the consolidated non-formalized frontier document, imported papers, current draft packages, and historical archives. Existing derivative, moment, Fourier-product, Lambert- $W$ , Bell-polynomial, Bernoulli-polynomial, and inverse-endpoint results are treated as baselines rather than reannounced as new.

The principal new results are: a finite dyadic closed form for  $\int x^p F(x) dx$  when  $p \in \mathbb{N}_0$ ; an absolutely and locally uniformly convergent dyadic series for arbitrary complex powers, including negative and fractional exponents; two-parameter recurrences for fractional Volterra integrals; complementary Mellin transforms controlled by an entire complex moment function; an entire Newton interpolation of complex moments from rational integer moments; exact local primitives and all-order derivative formulas for the inverse Fabius function; a sharp  $L^q$  threshold for  $G'$ ; refined inverse-primitive endpoint asymptotics carrying the repository's log-periodic Lambert phase; Stieltjes, logarithmic-potential, autocorrelation, and order-statistic transforms; and a rational Jacobi continued fraction generated by the Fabius moments. Numerical experiments based on finite Thue-Morse splines independently check the main formulas. Unproved conjectures and a Lean formalization roadmap are included.

# 1 Scope, corpus audit, and novelty protocol

## 1.1 Repository review

The live directory

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>

was recursively audited on 27 August 2026. All discoverable TeX paths were inventoried. Current synthesis documents and research drafts were examined through their raw TeX, imported papers were checked for the definitions and results relevant to integration, and historical copies were compared through their content indexes so that superseded duplicates were not counted as independent evidence.

The main baselines were:

- `Fabius_Function_and_Rvachev_Up.tex`, the principal monograph;
- `semi-formalized-research-frontiers.tex`, the consolidated frontier synthesis;
- the Lean walkthrough, glossary, and non-elementarity report;
- the imported TeX sources of Arias de Reyna, Haugland, Aistleitner–Hofer–Larcher, Tóth, and related papers;
- the archive index and its historical report families.

The current draft inbox contained the following additional packages:

- |                                                              |                                                                |
|--------------------------------------------------------------|----------------------------------------------------------------|
| • <code>Fabius_Arithmetic_Rays_Frontier_Report</code>        | • <code>Fabius_Rvachev_Thue_Morse_Frontier_Results</code>      |
| • <code>Fabius_Dyadic_Self_Sampling_Frontier_Package</code>  | • <code>Spectral_Arithmetic_Pascal_Rvachev_Hierarchy</code>    |
| • <code>Fabius_Half_Integer_Spectral_Frontier_Report</code>  | • <code>Thue_Morse_Formula_Atlas</code>                        |
| • <code>Fabius_Inverse_Frontier_Report_Source_and_PDF</code> | • <code>fabius_frontier_dyadic_inverse_barne_s_report</code>   |
| • <code>Fabius_Newton_Rvachev_Frontier_Report</code>         | • <code>fabius_frontier_new_results</code>                     |
| • <code>Fabius_Rvachev_Frontier_Report-2</code>              | • <code>fabius_frontier_results</code>                         |
| • <code>Fabius_Rvachev_Frontier_Report</code>                | • <code>fabius_frontier_results_bundle</code>                  |
| • <code>Fabius_Rvachev_New_Frontiers</code>                  | • <code>fabius_frontier_spectral_endpoint_report_bundle</code> |
|                                                              | • <code>rvachev_up_fourier_decay</code>                        |

The primitive identity  $F(x/2) = \int_0^x F(t) dt$  is implicit in the audited spectral hierarchy and therefore is used here as a baseline. Likewise, the repository already contains extensive work on integer moments, dyadic values, Fourier sinc products, endpoint asymptotics, inverse asymptotics, and Bell-polynomial reversion. The novelty claims below are deliberately narrower: “new” means not located explicitly in this audited corpus or in the targeted external literature search, not an absolute historical-priority claim.

## 1.2 Main questions addressed

The report studies:

- normalized and indefinite antiderivatives of  $F$ ,  $\text{up}$ ,  $1 - F$ ,  $F(1 - x)$ , and  $G$ ;

- polynomial, negative, fractional, complex, exponential, logarithmic, and resolvent weights;
- integer and fractional repeated integration, and all-order differentiation of the inverse;
- Mellin, Laplace, Fourier, Stieltjes/Cauchy, logarithmic-potential, and correlation transforms;
- products and compositions such as  $F^a(1 - F)^b$ ,  $\text{up}(2G - 1)^r$ , and shifted products of  $\text{up}$ ;
- exact arithmetic consequences, numerical verification, conjectures, and formalization targets.

## 2 Definitions and probabilistic normalization

### 2.1 The bounded Fabius function and Rvachev's function

Let  $F : \mathbb{R} \rightarrow [0, 1]$  be the bounded Fabius distribution function, with

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (530)$$

and, on  $0 \leq x \leq 1/2$ ,

$$F'(x) = 2F(2x). \quad (531)$$

It obeys

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1. \quad (532)$$

Rvachev's function is

$$\text{up}(u) = \begin{cases} F(1 - |u|), & |u| \leq 1, \\ 0, & |u| > 1. \end{cases} \quad (533)$$

It is even, nonnegative,  $C^\infty$ , supported on  $[-1, 1]$ , and normalized by  $\int_{-1}^1 \text{up}(u) \, du = 1$ . The two functions satisfy

$$F'(x) = 2 \text{up}(2x - 1), \quad 0 < x < 1, \quad (534)$$

and, on  $[0, 1]$ ,

$$\text{up}(x) = 1 - F(x) = F(1 - x). \quad (535)$$

The repository also uses a signed global extension  $\mathcal{F}$  that agrees with  $F$  on  $[0, 1]$  and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad x \in \mathbb{R}. \quad (536)$$

It is indispensable when derivatives move the argument outside  $[0, 1]$ .

### 2.2 Random-series model

Let  $U_1, U_2, \dots$  be independent uniform random variables on  $[0, 1]$  and set

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}. \quad (537)$$

Then  $F$  is the distribution function of  $X$ . If  $V_j = 2U_j - 1$  and

$$Y = 2X - 1 = \sum_{j=1}^{\infty} \frac{V_j}{2^j}, \quad (538)$$

then  $Y$  has density  $\text{up}$ . In particular,

$$\text{up}(t) = \mathbb{P}(X > t), \quad 0 \leq t \leq 1. \quad (539)$$

This survival-function interpretation is the quickest route to many definite integrals.

Let

$$D(z) = \mathbb{E}X^z, \quad d_n = D(n). \quad (540)$$

Because the density is infinitely flat at 0, all negative moments exist and  $D$  extends to an entire function; this is proved in [proposition 7.1](#).



### 2.3 Endpoint flatness and the inverse

Every derivative of  $F$  vanishes at 0, every derivative of  $1 - F$  vanishes at 1, and every derivative of  $\text{up} - 1$  vanishes at 0. Thus, for each  $M \in \mathbb{N}$ ,

$$F(x) = O(x^M), \quad F'(x) = O(x^M) \quad (x \downarrow 0). \quad (541)$$

These bounds justify complex powers, repeated integration by parts, and normal convergence arguments that would fail for an ordinary density with algebraic endpoint behavior.

The restriction of  $F$  to  $[0, 1]$  is strictly increasing. Its inverse

$$G : (0, 1) \rightarrow (0, 1), \quad G = F^{-1}, \quad (542)$$

obeys

$$G(1 - y) = 1 - G(y). \quad (543)$$

## 3 Three master integration principles

**Theorem 3.1** (CDF–survival integration). *Let  $H$  be absolutely continuous on  $[0, 1]$  and suppose the following integrals exist. Then*

$$\int_0^1 H'(t) \text{up}(t) \, dt = \mathbb{E}[H(X) - H(0)], \quad (544)$$

$$\int_0^1 H'(t) F(t) \, dt = H(1) - \mathbb{E}H(X). \quad (545)$$

*Proof.* For (544), use (539) and Fubini:

$$\int_0^1 H'(t) \mathbb{P}(X > t) \, dt = \mathbb{E} \int_0^X H'(t) \, dt.$$

Equation (545) follows either similarly or by adding the two formulas and using  $F + \text{up} = 1$ .  $\square$

**Theorem 3.2** (Quantile push-forward). *For every Borel function  $\Phi$  for which either side is absolutely integrable,*

$$\int_0^1 \Phi(G(y)) \, dy = \int_0^1 \Phi(x) \, dF(x) = \mathbb{E}\Phi(X). \quad (546)$$

*Proof.* This is the probability-integral transform, or directly the substitution  $y = F(x)$ .  $\square$

**Theorem 3.3** (Universal beta family). *For  $\Re a > -1$  and  $\Re b > -1$ ,*

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) \, dx = B(a + 1, b + 1). \quad (547)$$

*Equivalently,*

$$\int_0^1 F(x)^a \text{up}(x)^b \text{up}(2x - 1) \, dx = \frac{1}{2} B(a + 1, b + 1). \quad (548)$$

*Proof.* Put  $y = F(x)$  and use (534).  $\square$

The point of [theorems 3.1](#) and [3.2](#) is not merely probabilistic elegance. They convert any tractable ordinary antiderivative  $H$  into an exact Fabius or Rvachev integral. Monomials give moments, exponentials give Laplace products, logarithms give entropy and logarithmic potentials, and resolvents give Stieltjes transforms.

## 4 Integer-order primitives and derivative hierarchies

### 4.1 Normalized primitives of $F$

From (531),

$$\frac{d}{dx}F(x/2) = F(x), \quad 0 \leq x \leq 1. \quad (549)$$

Thus  $F(x/2)$  is the primitive normalized to vanish at  $x = 0$ .

For  $n \in \mathbb{N}$ , write

$$I_0^n f(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt. \quad (550)$$

**Theorem 4.1** (Exact  $n$ th primitive of  $F$ ). For  $n \in \mathbb{N}$  and  $0 \leq x \leq 1$ ,

$$\boxed{I_0^n F(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right)}. \quad (551)$$

Every other  $n$ th antiderivative differs from the right side by a polynomial of degree at most  $n-1$ .

*Proof.* The case  $n = 1$  is (549). If the formula holds at order  $n$ , then

$$I_0^{n+1} F(x) = 2^{\binom{n}{2}} \int_0^x F(t/2^n) dt = 2^{\binom{n}{2}+n} F(x/2^{n+1}),$$

which is the required triangular exponent.  $\square$

For the bounded extension and  $x \geq 1$ , the  $n$ th derivative is 1, so matching derivatives at  $x = 1$  gives

$$I_0^n F(x) = \frac{(x-1)^n}{n!} + \sum_{k=0}^{n-1} \frac{2^{\binom{n-k}{2}} F(2^{-(n-k)})}{k!} (x-1)^k. \quad (552)$$

For the signed global extension, a single formula is valid on all of  $\mathbb{R}$ :

$$\frac{d^n}{dx^n} \left[ 2^{\binom{n}{2}} \mathcal{F}(x/2^n) \right] = \mathcal{F}(x). \quad (553)$$

### 4.2 Derivatives and primitives of up

The global CDF of the centered density is

$$\int_{-\infty}^x \text{up}(t) dt = F\left(\frac{x+1}{2}\right). \quad (554)$$

Consequently:

**Theorem 4.2** (Left  $n$ th primitive of up). For  $n \in \mathbb{N}$  and  $x \leq 1$ ,

$$\boxed{\frac{1}{(n-1)!} \int_{-\infty}^x (x-t)^{n-1} \text{up}(t) dt = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right)}. \quad (555)$$

For  $x \geq 1$ , it becomes the moment polynomial

$$\sum_{k=0}^{n-1} \frac{(-1)^k \mu_k}{k!(n-1-k)!} x^{n-1-k}, \quad \mu_k = \int_{-1}^1 t^k \text{up}(t) dt. \quad (556)$$

*Proof.* Equation (554) is the case  $n = 1$ . Induct as in [theorem 4.1](#), now mapping the lower endpoint  $-1$  to 0. For  $x \geq 1$ , expand  $(x-t)^{n-1}$  over the compact support.  $\square$

On  $0 \leq x \leq 1$  one also has the convenient centered primitive

$$\int_0^x \text{up}(t) dt = x - F(x/2) = F\left(\frac{x+1}{2}\right) - \frac{1}{2}. \quad (557)$$

The forward derivative hierarchy is

$$\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x), \quad (558)$$

and, for  $-1 \leq x \leq 1$ ,

$$\text{up}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n(x+1)). \quad (559)$$

The antiderivative hierarchy (551) is therefore the exact inverse triangular scaling of the derivative hierarchy.

## 5 Dyadic antiderivatives with polynomial, fractional, and complex weights

This section contains the main antiderivative results of the report. The elementary dilation primitive in equation (549) becomes substantially more useful after one allows a monomial weight and then analytically continues the exponent.

### 5.1 A finite closed form for nonnegative integer powers

For  $0 < x \leq 1$  and  $\nu \in \mathbb{C}$  put

$$A_\nu(x) = \int_0^x t^\nu F(t) dt, \quad t^\nu = e^{\nu \log t}. \quad (560)$$

Endpoint flatness makes this integral locally uniformly convergent for every  $\nu \in \mathbb{C}$ ; hence  $A_\nu(x)$  is entire as a function of  $\nu$ .

**Lemma 5.1** (One-step dyadic reduction). *For  $0 < x \leq 1$  and every  $\nu \in \mathbb{C}$ ,*

$$A_\nu(x) = x^\nu F(x/2) - \nu 2^\nu A_{\nu-1}(x/2). \quad (561)$$

*Proof.* On  $[0, 1]$ ,  $\frac{d}{dt} F(t/2) = F(t)$ . Integration by parts gives

$$A_\nu(x) = [t^\nu F(t/2)]_0^x - \nu \int_0^x t^{\nu-1} F(t/2) dt.$$

The lower boundary vanishes for all complex  $\nu$  because  $F$  is flat at 0. In the remaining integral put  $t = 2u$ . □

**Theorem 5.2** (Closed monomial antiderivative). *For every  $p \in \mathbb{N}_0$  and  $0 \leq x \leq 1$ ,*

$$\boxed{\int_0^x t^p F(t) dt = \sum_{j=0}^p (-1)^j \frac{p!}{(p-j)!} 2^{\binom{j+1}{2}} x^{p-j} F\left(\frac{x}{2^{j+1}}\right)}. \quad (562)$$

*In particular,*

$$\int_0^x t F(t) dt = x F(x/2) - 2 F(x/4), \quad (563)$$

$$\int_0^x t^2 F(t) dt = x^2 F(x/2) - 4x F(x/4) + 16 F(x/8), \quad (564)$$

$$\int_0^x t^3 F(t) dt = x^3 F(x/2) - 6x^2 F(x/4) + 48x F(x/8) - 384 F(x/16). \quad (565)$$

For  $x \geq 1$ , the bounded extension is obtained by adding the elementary tail

$$A_\nu(x) = A_\nu(1) + \begin{cases} \frac{x^{\nu+1} - 1}{\nu + 1}, & \nu \neq -1, \\ \log x, & \nu = -1, \end{cases} \quad (566)$$

where the first quotient has a removable singularity at  $\nu = -1$ .

*Proof.* Iterate (561). At the  $j$ th step the falling factorial contributes  $p!/(p-j)!$ . The powers of 2 telescope to

$$2^{jp-j(j-1)/2} 2^{-j(p-j)} = 2^{j(j+1)/2}.$$

The process terminates because  $p^{p+1} = 0$ . Formula (566) follows from  $F = 1$  on  $[1, \infty)$ .  $\square$

*Remark 5.3.* At  $x = 1$ , integration by parts against the probability measure  $dF$  gives

$$\int_0^1 t^p F(t) dt = \frac{1 - d_{p+1}}{p+1}, \quad \int_0^1 t^p \text{up}(t) dt = \frac{d_{p+1}}{p+1}. \quad (567)$$

Since the integer moments  $d_n$  are rational, all these definite integrals are rational.

## 5.2 A convergent dyadic series for arbitrary complex powers

The recurrence does not terminate for a nonintegral exponent, but endpoint flatness converts it into a genuinely convergent series rather than merely a formal asymptotic expansion.

**Theorem 5.4** (Complex-weight dyadic antiderivative series). *Fix  $0 < x \leq 1$ . For every  $\nu \in \mathbb{C}$ ,*

$$A_\nu(x) = \sum_{j=0}^{\infty} (-1)^j \nu^{\underline{j}} 2^{\binom{j+1}{2}} x^{\nu-j} F\left(\frac{x}{2^{j+1}}\right). \quad (568)$$

*The series converges absolutely and locally uniformly in  $\nu$ . After  $N$  terms the exact remainder is*

$$R_N = (-1)^N \nu^{\underline{N}} 2^{N\nu - \binom{N}{2}} A_{\nu-N}(x/2^N). \quad (569)$$

*Consequently the sum is entire in  $\nu$  and may be differentiated termwise on compact subsets of the  $\nu$ -plane.*

*Proof.* Iterating (561)  $N$  times gives the displayed partial sum and (569). It remains to prove  $R_N \rightarrow 0$ . Flatness means that for every  $M \in \mathbb{N}$  there is  $C_M$  such that  $0 \leq F(v) \leq C_M v^M$  on  $[0, 1]$ . From theorem 4.1,

$$\begin{aligned} 2^{\binom{N}{2}} F(u/2^N) &= \frac{1}{(N-1)!} \int_0^u (u-v)^{N-1} F(v) dv \\ &\leq C_M \frac{M!}{(N+M)!} u^{N+M}. \end{aligned} \quad (570)$$

After changing variables in (569),

$$R_N = (-1)^N \nu^{\underline{N}} \int_0^x u^{\nu-N} [2^{\binom{N}{2}} F(u/2^N)] du.$$

For  $\nu$  in a fixed compact set, choose  $M$  so that  $\Re \nu + M > -1$  uniformly. Bound (570) then gives

$$|R_N| \leq C_{M,K,x} \frac{|\nu^{\underline{N}}|}{(N+M)!}.$$

The quotient tends to zero faster than a chosen inverse power of  $N$ , locally uniformly in  $\nu$ ; at nonnegative integral  $\nu$  it eventually vanishes identically. This also proves normal convergence.  $\square$

**Corollary 5.5** (Negative powers). For  $m \in \mathbb{N}$  and  $0 < x \leq 1$ ,

$$\int_0^x \frac{F(t)}{t^m} dt = \sum_{j=0}^{\infty} m^{\bar{j}} 2^{\binom{j+1}{2}} x^{-m-j} F\left(\frac{x}{2^{j+1}}\right). \quad (571)$$

Thus every negative-power antiderivative is finite despite its apparently singular algebraic weight.

*Proof.* Insert  $\nu = -m$  in (568) and use  $-m^{\bar{j}} = (-1)^j m^{\bar{j}}$ .  $\square$

**Corollary 5.6** (Weights against up). For  $0 < x \leq 1$  and initially  $\Re \nu > -1$ ,

$$\int_0^x t^{\nu} \text{up}(t) dt = \frac{x^{\nu+1}}{\nu+1} - A_{\nu}(x). \quad (572)$$

The right-hand side is the meromorphic continuation in  $\nu$ , with a single simple pole at  $\nu = -1$  of residue 1. The regularized integral

$$\int_0^x t^{\nu} (\text{up}(t) - 1) dt = -A_{\nu}(x) \quad (573)$$

continues to an entire function of  $\nu$ .

**Frontier contribution.** Equations (568) and (571), including normal convergence for arbitrary complex exponent, were not located explicitly in the audited corpus. They supply the requested series form for fractional and negative powers without expanding the nowhere-analytic function  $F$  in an ordinary Taylor series.

### 5.3 Repeated weighted antiderivatives

**Corollary 5.7** (Integer-order Volterra primitives with a complex monomial). For  $n \in \mathbb{N}$ ,  $0 < x \leq 1$ , and  $\nu \in \mathbb{C}$ ,

$$I_0^n [t^{\nu} F(t)](x) = \frac{1}{(n-1)!} \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} x^{n-1-r} A_{\nu+r}(x). \quad (574)$$

For  $\nu = p \in \mathbb{N}_0$ , substituting (562) gives a finite double sum of dyadically scaled values of  $F$ ; for general  $\nu$ , substituting (568) gives a normally convergent series.

*Proof.* Expand  $(x-t)^{n-1}$  in the defining Volterra integral and collect  $A_{\nu+r}(x)$ .  $\square$

## 6 Fractional Volterra integration and a two-parameter dyadic recurrence

**Definition 6.1.** For  $\Re \alpha > 0$  and  $0 < x \leq 1$ , the left Riemann–Liouville integral is

$$(I_0^{\alpha} f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (575)$$

**Theorem 6.2** (Fractional dilation recurrence). For  $\Re \alpha > 1$  and  $0 < x \leq 1$ ,

$$I_0^{\alpha} F(x) = 2^{\alpha-1} I_0^{\alpha-1} F(x/2). \quad (576)$$

More generally, if  $m \in \mathbb{N}$  and  $m < \Re \alpha$ ,

$$I_0^{\alpha} F(x) = 2^{m\alpha-m(m+1)/2} I_0^{\alpha-m} F(x/2^m). \quad (577)$$

At  $\alpha = n \in \mathbb{N}$ , taking  $m = n-1$  recovers (551).

*Proof.* Substitute  $t = 2u$  and use  $F(2u) = F'(u)/2$ :

$$I_0^\alpha F(x) = \frac{2^{\alpha-1}}{\Gamma(\alpha)} \int_0^{x/2} (x/2 - u)^{\alpha-1} F'(u) du.$$

The boundary terms vanish, and one integration by parts yields (576). Iteration gives (577).  $\square$

After reindexing the report's order as  $\alpha = \beta + 1$ , the real slice  $\beta > 0$  of (576) is exactly `Fabius.fractionalVolterra_add_one_fabiusReal`. The declaration does not assert the complex-order scope or export the iterated formula (577).

For a weighted kernel define

$$J_{\alpha,\nu}(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} t^\nu F(t) dt. \quad (578)$$

Again, endpoint flatness makes  $J_{\alpha,\nu}$  entire in  $\nu$  for each fixed  $\alpha$  with  $\Re \alpha > 0$ .

**Theorem 6.3** (Two-parameter fractional recurrence). *For  $0 < x \leq 1$ ,  $\Re \alpha > 1$ , and  $\nu \in \mathbb{C}$ ,*

$$J_{\alpha,\nu}(x) = 2^{\alpha+\nu-1} [J_{\alpha-1,\nu}(x/2) - \nu J_{\alpha,\nu-1}(x/2)]. \quad (579)$$

*The relation reduces the pair  $(\alpha, \nu)$  along a two-dimensional dyadic lattice. When  $\nu$  is a nonnegative integer, all branches that lower  $\nu$  terminate; when  $\alpha$  is an integer, the remaining quantities reduce to corollary 5.7.*

*Proof.* After  $t = 2u$  and  $F(2u) = F'(u)/2$ ,

$$J_{\alpha,\nu}(x) = \frac{2^{\alpha+\nu-1}}{\Gamma(\alpha)} \int_0^{x/2} (x/2 - u)^{\alpha-1} u^\nu F'(u) du.$$

Integrating by parts differentiates the product of the power and Volterra kernel. Its two terms are precisely  $J_{\alpha-1,\nu}$  and  $-\nu J_{\alpha,\nu-1}$ .  $\square$

**Remark 6.4** (Fractional derivatives). For  $m - 1 < \Re \alpha < m$ , the Riemann–Liouville derivative  $D_0^\alpha F = D^m I_0^{m-\alpha} F$  can be reduced by (576) before differentiation. Because every derivative of  $F$  at 0 vanishes, Caputo and Riemann–Liouville derivatives coincide on  $F$  for positive nonintegral orders. A useful formalization target is to prove this statement in a function space where the dyadic dilation operator and the fractional Volterra semigroup are both bounded.

## 7 Mellin transforms, logarithmic moments, and Newton interpolation

### 7.1 The entire complex moment function

**Proposition 7.1** (Entire moment interpolation). *The function*

$$D(z) = \int_0^1 x^z dF(x) = \int_0^1 x^z F'(x) dx$$

*is entire on  $\mathbb{C}$ . Its derivatives are*

$$D^{(m)}(z) = \int_0^1 x^z (\log x)^m dF(x). \quad (580)$$

*The integer moments satisfy  $d_0 = 1$  and*

$$(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{d_k}{n - k + 1}, \quad n \geq 1. \quad (581)$$

*In particular all  $d_n$  are rational.*

*Proof.* The density  $F'$  is flat at 0. On each compact set of  $z$ -values, a sufficiently high power bound for  $F'$  dominates  $x^{\Re z} |\log x|^m$  near zero, so differentiation under the integral is valid to every order. For (581), use the distributional fixed point

$$X \stackrel{d}{=} \frac{U + X'}{2},$$

where  $U$  is uniform on  $[0, 1]$  and  $X'$  is an independent copy of  $X$ . Expanding  $(U + X')^n$  and isolating the  $k = n$  term yields the recurrence.  $\square$

The first few moments are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \dots \quad (582)$$

## 7.2 Mellin transforms of $F$ and $\text{up}$

**Theorem 7.2** (Complementary Mellin transforms). *For  $z \in \mathbb{C}$ ,*

$$\mathcal{M}_F(z) := \int_0^1 x^{z-1} F(x) \, dx = \frac{1 - D(z)}{z}, \quad (583)$$

where the quotient has a removable singularity at  $z = 0$  and is entire. Initially for  $\Re z > 0$ ,

$$\mathcal{M}_{\text{up},+}(z) := \int_0^1 x^{z-1} \text{up}(x) \, dx = \frac{D(z)}{z}. \quad (584)$$

The latter identity is its meromorphic continuation to  $\mathbb{C}$ ; it has only one pole, a simple pole at  $z = 0$  with residue 1.

*Proof.* Stieltjes integration by parts gives

$$D(z) = 1 - z \int_0^1 x^{z-1} F(x) \, dx$$

where first justified in a right half-plane and then continued by proposition 7.1. Equation (584) follows from  $\text{up} = 1 - F$  on  $[0, 1]$ .  $\square$

**Corollary 7.3** (Logarithmic integral hierarchy). *For  $m \in \mathbb{N}_0$ ,*

$$\int_0^1 \frac{(\log x)^m}{x} F(x) \, dx = -\frac{D^{(m+1)}(0)}{m+1}. \quad (585)$$

Equivalently,

$$\int_0^1 \frac{(\log x)^m}{x} (\text{up}(x) - 1) \, dx = \frac{D^{(m+1)}(0)}{m+1}. \quad (586)$$

In particular,

$$\int_0^1 \frac{F(x)}{x} \, dx = -\mathbb{E} \log X. \quad (587)$$

*Proof.* Differentiate (583) at  $z = 0$  and use the Taylor expansion of  $D$ .  $\square$

### 7.3 A Newton series reconstructed from rational moments

At  $x = 1$ , the dyadic points in (568) can be eliminated by the repository identity

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}). \quad (588)$$

This produces an interpolation formula that reconstructs every complex moment from the rational moments at nonnegative integers.

**Corollary 7.4** (Newton–Mellin interpolation). *For every  $\nu \in \mathbb{C}$ ,*

$$\frac{1 - D(\nu + 1)}{\nu + 1} = \sum_{j=0}^{\infty} (-1)^j \nu^{\underline{j}} \frac{d_{j+1}}{(j+1)!}, \quad (589)$$

with the left side interpreted removably at  $\nu = -1$ . Equivalently, for every  $z \in \mathbb{C}$ ,

$$D(z) = 1 - z \sum_{j=0}^{\infty} (-1)^j z - 1^{\underline{j}} \frac{d_{j+1}}{(j+1)!}. \quad (590)$$

The series converges locally uniformly on  $\mathbb{C}$ .

*Proof.* Set  $x = 1$  in (568), use (588), and combine with (583). Normal convergence was proved in theorem 5.4. It can also be seen directly: endpoint flatness at 1 makes  $d_n = O_M(n^{-M})$  for every  $M$ , whereas  $z - 1^{\underline{j}}/j!$  has at most polynomial growth on compact  $z$ -sets.  $\square$

**Corollary 7.5** (A rapidly convergent logarithmic constant).

$$-\mathbb{E} \log X = \int_0^1 \frac{F(x)}{x} dx = \sum_{n=1}^{\infty} \frac{d_n}{n} = 0.758239999 \dots \quad (591)$$

The decimal is numerical, while the series identity is exact.

*Proof.* Put  $z = 0$  in (589); then  $(-1)^j - 1^{\underline{j}} = j!$ . The probabilistic identity  $\sum_{n \geq 1} d_n/n = \mathbb{E}[-\log(1 - X)] = \mathbb{E}[-\log X]$  provides an independent check using symmetry.  $\square$

**Frontier contribution.** Formula (590) is an entire Newton interpolation of a non-elementary complex moment function from rational data. It creates a direct bridge between the dyadic Fabius values, the moment arithmetic, and complex Mellin analysis.

### 7.4 A complex fixed-point equation

**Proposition 7.6** (Moment fixed point). *For all  $z \in \mathbb{C}$ , by removable continuation at  $z = -1$ ,*

$$D(z) = \frac{2^{-z}}{z+1} \mathbb{E}[(1+X)^{z+1} - X^{z+1}]. \quad (592)$$

At  $z = -1$  this reads

$$D(-1) = 2\mathbb{E} \log \frac{1+X}{X}. \quad (593)$$

*Proof.* Condition on  $X'$  in  $X \stackrel{d}{=} (U + X')/2$  and integrate  $(u + X')^z$  over  $0 \leq u \leq 1$ .  $\square$

One may expand  $(1+X)^{z+1}$  in a generalized binomial series. The rapid decay of  $d_n$  then gives, initially in a standard convergence region and subsequently by continuation,

$$D(z) = \frac{2^{-z}}{z+1} \left[ \sum_{n=0}^{\infty} \binom{z+1}{n} d_n - D(z+1) \right]. \quad (594)$$

This identity is a promising starting point for analytic difference equations in the Mellin variable.



### 7.5 Mellin–Laplace bridges

Let  $B(s) = \mathbb{E}e^{-sX}$ , developed in the next section. For  $\Re z < 0$ ,

$$D(z) = \frac{1}{\Gamma(-z)} \int_0^\infty s^{-z-1} B(s) \, ds. \quad (595)$$

More generally, if  $m < \Re z < m + 1$  with  $m \in \mathbb{N}_0$ , then

$$D(z) = \frac{1}{\Gamma(-z)} \int_0^\infty s^{-z-1} \left[ B(s) - \sum_{k=0}^m \frac{(-s)^k d_k}{k!} \right] \, ds. \quad (596)$$

The subtractions regularize the origin, while  $m < \Re z$  controls the polynomial tail at infinity. These formulas convert asymptotics of the infinite product  $B$  into complex-moment information and are suitable for saddle-point analysis of negative moments.

## 8 Laplace, Fourier, and Bell–Bernoulli transforms

### 8.1 The one-sided Laplace product

**Theorem 8.1** (Laplace product and three exact transforms). *For  $s \in \mathbb{C}$ ,*

$$B(s) = \mathbb{E}e^{-sX} = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad (597)$$

*with each zero-over-zero factor understood by continuity. The product is entire and satisfies*

$$B(s) = \frac{1 - e^{-s/2}}{s/2} B(s/2). \quad (598)$$

*Moreover,*

$$\int_0^1 e^{-sx} \operatorname{up}(x) \, dx = \frac{1 - B(s)}{s}, \quad (599)$$

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{B(s) - e^{-s}}{s}, \quad (600)$$

$$\int_0^1 e^{-sG(y)} \, dy = B(s). \quad (601)$$

*All quotients are interpreted by removable continuation at  $s = 0$ .*

*Proof.* Independence in (537) gives (597). The first two integral formulas follow from the master identities in theorem 3.1 with an exponential antiderivative. The last is the quantile substitution (546).  $\square$

For every  $p \in \mathbb{N}_0$ , differentiation supplies weighted transforms without any new quadrature:

$$\int_0^1 x^p e^{-sx} \operatorname{up}(x) \, dx = (-1)^p \frac{d^p}{ds^p} \frac{1 - B(s)}{s}, \quad (602)$$

$$\int_0^1 x^p e^{-sx} F(x) \, dx = (-1)^p \frac{d^p}{ds^p} \frac{B(s) - e^{-s}}{s}, \quad (603)$$

$$\int_0^1 G(y)^p e^{-sG(y)} \, dy = (-1)^p B^{(p)}(s). \quad (604)$$

For a complex power, Mellin inversion or the series of theorem 5.4 provides the corresponding continuation.

## 8.2 Bernoulli cumulants and Bell-polynomial derivatives

Let  $B_{2m}$  denote the Bernoulli numbers with  $B_2 = 1/6$ . Near the origin,

$$\log B(s) = -\frac{s}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!(2^{2m}-1)} s^{2m}, \quad |s| < 4\pi. \quad (605)$$

Thus the cumulants of  $X$  are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2m} = \frac{B_{2m}}{2m(2^{2m}-1)}, \quad \kappa_{2m+1} = 0 \quad (m \geq 1). \quad (606)$$

If  $\mathcal{B}_n$  is the complete exponential Bell polynomial, then

$$d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (607)$$

At a general transform parameter,

$$B^{(n)}(s) = B(s) \mathcal{B}_n((\log B)'(s), \dots, (\log B)^{(n)}(s)). \quad (608)$$

Equations (604) and (608) therefore express every exponentially damped inverse moment through Bernoulli data and Bell polynomials.

## 8.3 Bilateral transform of up and the sinc product

Since  $Y = 2X - 1$  has density up,

$$\int_{-1}^1 e^{-su} \text{up}(u) \, du = e^s B(2s) = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}. \quad (609)$$

With the angular-frequency convention

$$\phi_Y(\omega) = \int_{-1}^1 e^{i\omega u} \text{up}(u) \, du,$$

one obtains

$$\phi_Y(\omega) = \prod_{j=1}^{\infty} \text{sinc}(\omega/2^j), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (610)$$

In the repository convention  $\widehat{f}(\xi) = \int f(x) e^{-2\pi i \xi x} \, dx$ , this becomes

$$\widehat{\text{up}}(\xi) = \prod_{n=0}^{\infty} \text{sinc}(\pi \xi / 2^n). \quad (611)$$

Polynomially weighted Fourier transforms follow from

$$\int_{-1}^1 u^n e^{i\omega u} \text{up}(u) \, du = i^{-n} \phi_Y^{(n)}(\omega). \quad (612)$$

Away from the zeros of the product,  $\phi_Y^{(n)}$  can again be organized by Bell polynomials applied to the logarithmic derivatives of the sinc factors. At a zero, the entire derivative in (612), rather than a singular logarithmic representation, is the correct object.

## 9 Integral calculus of the inverse Fabius function

### 9.1 Global inverse moments and local primitives

The quantile push-forward immediately turns every inverse-Fabius moment into a moment of  $X$ .

**Theorem 9.1** (Entire inverse-power transform). *For every  $z \in \mathbb{C}$ ,*

$$\int_0^1 G(y)^z dy = D(z). \quad (613)$$

*The integral converges for all complex  $z$ , including arbitrarily large negative real part.*

*Proof.* Equation (613) is (546) with  $\Phi(x) = x^z$ . Convergence follows from [proposition 7.1](#); equivalently,  $G(y)$  tends to zero more slowly than every positive power of  $y$ , so negative powers of  $G$  remain integrable at  $y = 0$ .  $\square$

**Theorem 9.2** (Exact local inverse-power primitive). *Let  $0 \leq y \leq 1$ , put  $x = G(y)$ , and let  $p \in \mathbb{C}$ . Then*

$$\boxed{\int_0^y G(t)^p dt = yx^p - pA_{p-1}(x).} \quad (614)$$

*For  $p \in \mathbb{N}$ , [theorem 5.2](#) makes the right side a finite sum over  $F(x/2^j)$ ,  $1 \leq j \leq p$ . For arbitrary complex  $p$ , [theorem 5.4](#) gives a normally convergent dyadic series.*

*Proof.* Use  $t = F(u)$ , so  $u = G(t)$  and  $dt = F'(u) du$ :

$$\int_0^y G(t)^p dt = \int_0^x u^p F'(u) du = yx^p - p \int_0^x u^{p-1} F(u) du.$$

The lower boundary vanishes for every complex  $p$  by endpoint flatness.  $\square$

For example,

$$\int_0^y G(t) dt = yG(y) - F(G(y)/2), \quad (615)$$

$$\int_0^y G(t)^2 dt = yG(y)^2 - 2G(y)F(G(y)/2) + 4F(G(y)/4), \quad (616)$$

$$\int_0^y G(t)^3 dt = yG(y)^3 - 3G(y)^2 F(G(y)/2) + 12G(y)F(G(y)/4) - 48F(G(y)/8). \quad (617)$$

At  $y = 1$ , these identities reduce to  $\int_0^1 G^p = d_p$ .

**Corollary 9.3** (A weighted CDF–quantile duality). *For  $p \in \mathbb{C}$  and  $\Re q > 0$ ,*

$$\int_0^1 G(y)^p y^{q-1} dy = \frac{1}{q} - \frac{p}{q} \int_0^1 x^{p-1} F(x)^q dx. \quad (618)$$

*The endpoint at  $x = 0$  causes no restriction on  $p$  because  $F(x)^q$  is flat whenever  $\Re q > 0$ .*

*Proof.* Put  $y = F(x)$  and integrate  $x^p d(F(x)^q)/q$  by parts.  $\square$

A second universal identity, useful for inverse derivatives, is

$$\int_0^1 \Phi(G(y))G'(y) dy = \int_0^1 \Phi(x) dx \quad (619)$$

whenever either side is absolutely integrable.

## 9.2 All derivatives of the inverse

Set  $f = F'$ . Define differential polynomials  $Q_n[f]$  recursively by

$$Q_1 = 1, \quad Q_{n+1} = fQ'_n - (2n-1)f'Q_n. \quad (620)$$

**Theorem 9.4** (All-order inverse derivative formula). *If  $x = G(y)$ , then for every  $n \geq 1$ ,*

$$G^{(n)}(y) = \frac{Q_n[f](x)}{f(x)^{2n-1}}. \quad (621)$$

The first cases are

$$\begin{aligned} G'(y) &= \frac{1}{F'(x)}, \\ G''(y) &= -\frac{F''(x)}{F'(x)^3}, \\ G'''(y) &= \frac{3F''(x)^2 - F'(x)F'''(x)}{F'(x)^5}, \end{aligned} \quad (622)$$

$$G^{(4)}(y) = \frac{-15F''(x)^3 + 10F'(x)F''(x)F'''(x) - (F'(x))^2F^{(4)}(x)}{F'(x)^7}. \quad (623)$$

Using the signed extension,

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad 0 < x < 1, \quad (624)$$

so every inverse derivative is a rational expression in finitely many dyadically dilated values of  $\mathcal{F}$ .

*Proof.* The differentiation operator in the inverse coordinate is  $\frac{d}{dy} = f(x)^{-1} \frac{d}{dx}$ . Assuming (621) at order  $n$  and differentiating gives

$$\frac{1}{f} \frac{d}{dx} \left( Q_n f^{-(2n-1)} \right) = \frac{fQ'_n - (2n-1)f'Q_n}{f^{2n+1}},$$

which is exactly (620).  $\square$

The recurrence is a compact alternative to a full Faà di Bruno expansion. The  $Q_n$  are closely related to the inverse-function Bell polynomials; (620) is often more efficient computationally because it preserves the natural denominator  $f^{2n-1}$ .

## 9.3 Sharp integrability of the first inverse derivative

**Theorem 9.5** ( $L^q$  threshold for  $G'$ ). *For  $q > 0$ ,*

$$\int_0^1 |G'(y)|^q dy = \int_0^1 F'(x)^{1-q} dx. \quad (625)$$

Consequently,

$$G' \in L^q(0, 1) \iff 0 < q \leq 1. \quad (626)$$

In particular  $G \in W^{1,1}(0, 1)$  but  $G \notin W^{1,q}(0, 1)$  for any  $q > 1$ .

*Proof.* Substitute  $y = F(x)$  in the left side. If  $0 < q \leq 1$ , the exponent  $1 - q$  is nonnegative and  $F'$  is bounded. If  $q > 1$ , flatness gives  $F'(x) \leq C_N x^N$  near 0 for every  $N$ . Choose  $N$  with  $N(q-1) \geq 1$ ; then  $F'(x)^{1-q}$  dominates a nonintegrable multiple of  $x^{-N(q-1)}$ .  $\square$

The complementary density–quantile identity is

$$\int_0^1 G'(y)^{-q} dy = \int_0^1 F'(x)^{q+1} dx, \quad (627)$$

whenever finite. Two notable specializations are

$$\begin{aligned} \int_0^1 \log G'(y) dy &= - \int_0^1 F'(x) \log F'(x) dx \\ &= - \int_{-1}^1 \text{up}(u) \log(2 \text{up}(u)) du, \end{aligned} \quad (628)$$

$$\int_0^1 \text{up}(2G(y) - 1)^r dy = \int_{-1}^1 \text{up}(u)^{r+1} du, \quad r > -1. \quad (629)$$

Formula (628) identifies the mean logarithmic slope of the quantile with the differential entropy of the Fabius density.

## 10 Endpoint asymptotics of inverse primitives

The audited inverse-function reports derive a Lambert– $W$  endpoint law with a small periodic fluctuation. Write

$$T = \log(1/y), \quad L = \log 2, \quad \rho = \sqrt{\frac{2T}{L}}, \quad \beta = \frac{1}{L} + \frac{1}{2}. \quad (630)$$

Their leading inversion gives

$$G(y) \sim \frac{\sqrt{T}}{e\sqrt{L}} \exp\left(-\sqrt{2LT}\right), \quad y \downarrow 0. \quad (631)$$

In particular  $G$  is slowly varying at zero:  $G(cy)/G(y) \rightarrow 1$  for each fixed  $c > 0$ .

Let  $\Psi$  denote the bounded period-1 phase function in the repository’s refined inverse asymptotic and define

$$a(\rho) = \frac{\Psi'(\rho + \beta) - 1}{L}. \quad (632)$$

The logarithmic elasticity is

$$\epsilon(y) := \frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{a(\rho)}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right). \quad (633)$$

**Theorem 10.1** (Refined inverse-power primitive). *Assume the derivative-controlled form of the all-orders inverse phase expansion underlying (633). For each fixed  $p \in \mathbb{C}$ , as  $y \downarrow 0$ ,*

$$A_{p-1}(G(y)) = yG(y)^p \left[ \frac{1}{\rho} + \frac{a(\rho) - p}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right], \quad (634)$$

$$\int_0^y G(t)^p dt = yG(y)^p \left[ 1 - \frac{p}{\rho} + \frac{p(p - a(\rho))}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right]. \quad (635)$$

In particular,

$$F(G(y)/2) = yG(y) \left[ \frac{1}{\rho} + \frac{a(\rho) - 1}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right]. \quad (636)$$

The period-1 ripple survives integration, but first appears one order below the leading correction.

*Proof.* Putting  $u = G(t)$  in  $A_{p-1}(G(y))$  gives the exact identity

$$A_{p-1}(G(y)) = \int_0^y G(t)^p \epsilon(t) dt. \quad (637)$$

Let  $y = e^{-T}$  and  $H(T) = G(e^{-T})^p \epsilon(e^{-T})$ . Then

$$A_{p-1}(G(y)) = \int_T^\infty e^{-u} H(u) du = e^{-T} (H(T) + H'(T) + O(H''(T))).$$

The derivative-controlled version of (633) gives

$$H'(T) = G(e^{-T})^p \left[ -\frac{p}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right], \quad H''(T) = O\left(G(e^{-T})^p \frac{\log \rho}{\rho^3}\right).$$

Substitution proves (634). Formula (635) follows from (614), and (636) is the case  $p = 1$  because  $A_0(x) = F(x/2)$ .  $\square$

Without the second-order phase information, Karamata's theorem already gives the robust leading form

$$\int_0^y t^{q-1} G(t)^p dt \sim \frac{y^q G(y)^p}{q}, \quad q > 0, \quad (638)$$

for every fixed real  $p$ . Equation (635) resolves the first two corrections in the important case  $q = 1$ .

## 11 Stieltjes, Cauchy, and logarithmic transforms

For  $w \in \mathbb{C} \setminus [0, 1]$  define the Cauchy–Stieltjes transform

$$S(w) = \mathbb{E} \frac{1}{w - X} = \int_0^1 \frac{dF(x)}{w - x}. \quad (639)$$

**Theorem 11.1** (Four equivalent Stieltjes representations). *For  $w \notin [0, 1]$ ,*

$$S(w) = \int_0^1 \frac{dy}{w - G(y)}, \quad (640)$$

$$= \frac{1}{w} + \int_0^1 \frac{\text{up}(t)}{(w - t)^2} dt, \quad (641)$$

$$= \frac{1}{w - 1} - \int_0^1 \frac{F(t)}{(w - t)^2} dt. \quad (642)$$

For  $|w| > 1$ ,

$$S(w) = \sum_{n=0}^{\infty} \frac{d_n}{w^{n+1}}. \quad (643)$$

The transform of Rvachev's centered density is

$$\int_{-1}^1 \frac{\text{up}(u)}{w - u} du = \frac{1}{2} S\left(\frac{w+1}{2}\right), \quad w \notin [-1, 1]. \quad (644)$$

*Proof.* Equation (640) is the quantile push-forward. Apply theorem 3.1 to  $H(t) = (w - t)^{-1}$  to obtain (641); Stieltjes integration by parts gives (642). Geometric expansion gives (643). Finally use  $u = 2x - 1$  and  $dF(x) = 2 \text{up}(2x - 1) dx$ .  $\square$

A primitive in the spectral variable produces logarithmic kernels.

**Corollary 11.2** (Logarithmic potentials). *Choose a branch of the logarithm on a simply connected domain avoiding  $[0, 1]$ . Then*

$$\int_0^1 \frac{F(x)}{w-x} dx = \mathbb{E} \log(w-X) - \log(w-1), \quad (645)$$

$$\int_0^1 \frac{\text{up}(x)}{w-x} dx = \log w - \mathbb{E} \log(w-X), \quad (646)$$

$$\int_0^1 \log(w-G(y)) dy = \mathbb{E} \log(w-X). \quad (647)$$

*Differentiating any one of these equations recovers the corresponding Stieltjes transform.*

*Proof.* Integrate by parts using  $\frac{d}{dx} \log(w-x) = -(w-x)^{-1}$  and apply the quantile push-forward to the last formula.  $\square$

Boundary values of  $S$  encode the Fabius density through the Sokhotski–Plemelj formula. Thus the Cauchy transform packages moment arithmetic outside the interval and reconstructs  $F'$  on the cut. Its continued-fraction expansion, determined by the Hankel determinants of  $(d_n)$ , is developed below.

## 12 Products, order statistics, and autocorrelation

### 12.1 Products of $F$ and $1-F$

For positive integers  $m, n$ , let  $A_m = \max(X_1, \dots, X_m)$  and  $B_n = \min(X'_1, \dots, X'_n)$ , with the two samples independent. Then

$$\int_0^1 F(x)^m (1-F(x))^n dx = \mathbb{E}(B_n - A_m)_+. \quad (648)$$

Indeed, the integrand is  $\mathbb{P}(A_m \leq x < B_n)$ , and integration measures the random interval length. Two useful special cases are

$$\int_0^1 F(x)^m dx = 1 - \mathbb{E}A_m, \quad \int_0^1 \text{up}(x)^m dx = \mathbb{E}B_m. \quad (649)$$

Symmetry gives equality of the two integrals.

The first mixed product admits several especially transparent forms.

**Theorem 12.1** (A Fabius Gini integral). *Let  $X'$  be an independent copy of  $X$ . Then*

$$C_F := \int_0^1 F(x)(1-F(x)) dx = \frac{1}{2} \mathbb{E}|X - X'|, \quad (650)$$

$$= \frac{1}{\pi} \int_0^\infty \frac{1 - |\phi_X(t)|^2}{t^2} dt, \quad (651)$$

where

$$\phi_X(t) = e^{it/2} \prod_{j=1}^\infty \text{sinc}(t/2^{j+1}). \quad (652)$$

Equivalently,

$$C_F = \frac{1}{2\pi} \int_0^\infty \frac{1 - \phi_Y(t)^2}{t^2} dt. \quad (653)$$

Furthermore,

$$\int_{-1}^1 \text{up}(u)^2 \, du = 1 - 2C_F, \quad \int_0^1 F(x)^2 \, dx = \frac{1}{2} - C_F. \quad (654)$$

Numerically,

$$C_F \approx 0.0955632. \quad (655)$$

*Proof.* The standard one-dimensional Gini identity gives  $\mathbb{E}|X - X'| = 2 \int F(1 - F)$ . The Fourier representation follows from  $|z| = \frac{2}{\pi} \int_0^\infty (1 - \cos tz) t^{-2} \, dt$  and the characteristic function of  $X - X'$ . Since  $X = (Y + 1)/2$ , (653) follows by scaling. Finally, on  $[0, 1]$  one has  $\text{up} = 1 - F$ , and symmetry gives  $\int F^2 = \int (1 - F)^2$ . Expanding  $F(1 - F)$  proves (654).  $\square$

## 12.2 The autocorrelation transform

Define

$$R(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x - a) \, dx. \quad (656)$$

Because  $\text{up}$  is even,  $R = \text{up} * \text{up}$ ; it is an even  $C^\infty$  probability density supported on  $[-2, 2]$ . In angular-frequency convention,

$$R(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\omega a} \prod_{j=1}^{\infty} \text{sinc}^2(\omega/2^j) \, d\omega. \quad (657)$$

If  $\mu_n = \int u^n \text{up}(u) \, du$ , then

$$\int_{-2}^2 a^n R(a) \, da = \sum_{k=0}^n \binom{n}{k} \mu_k \mu_{n-k}. \quad (658)$$

Only even indices survive. More generally, correlations of derivatives are obtained by multiplying the integrand in (657) by powers of  $i\omega$ . These identities translate questions about products of shifted Rvachev functions into the arithmetic of a squared lacunary sinc product.

## 13 Numerical experiments

### 13.1 Finite Thue–Morse spline model

The accompanying program uses the finite random sum

$$Y_n = \sum_{j=1}^n \frac{V_j}{2^j} \quad (659)$$

and its compactly supported spline density. Let  $\tau_j = (-1)^{s_2(j)}$ ,  $A_n = 1 - 2^{-n}$ , and  $(r)_+ = \max(r, 0)$ . Then

$$p_n(y) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \tau_j \left( y + A_n - \frac{j}{2^{n-1}} \right)_+^{n-1}, \quad (660)$$

$$P_n(y) = \frac{2^{\binom{n}{2}}}{n!} \sum_{j=0}^{2^n-1} \tau_j \left( y + A_n - \frac{j}{2^{n-1}} \right)_+^n. \quad (661)$$

The finite Fabius CDF is  $F_n(x) = P_n(2x - 1)$ . Long-double arithmetic and symmetry are used to suppress cancellation. Exact rational moments are generated independently from (581).



### 13.2 Results and reproducibility

The first table gives the largest residual among several test points for  $\int_0^x F_n(t) dt \approx F_n(x/2)$ , together with the finite-rank approximation to  $C_F$ . The second tests (562) at  $x = 0.83$  and rank 10. The third shows convergence of (590) at  $z = 0.5 + 0.7i$ .

| Spline rank | max. primitive residual | $\int_0^1 F_n(1 - F_n)$ |
|-------------|-------------------------|-------------------------|
| 6           | $1.873e - 05$           | 0.095552260300          |
| 8           | $1.170e - 06$           | 0.095562546323          |
| 10          | $7.313e - 08$           | 0.095563189160          |

| $p$ | quadrature     | dyadic formula | difference     |
|-----|----------------|----------------|----------------|
| 0   | 0.330437473102 | 0.330437457904 | $1.520e - 08$  |
| 1   | 0.205934882125 | 0.205935065265 | $-1.831e - 07$ |
| 2   | 0.135359629291 | 0.135359392297 | $2.370e - 07$  |
| 3   | 0.092437619662 | 0.092438027029 | $-4.074e - 07$ |
| 5   | 0.046620893512 | 0.046622072851 | $-1.179e - 06$ |

| Newton terms | approximation error | approximation                 |
|--------------|---------------------|-------------------------------|
| 5            | $8.605e - 03$       | $0.601640735 - 0.321223181 i$ |
| 10           | $1.041e - 03$       | $0.598727644 - 0.314110902 i$ |
| 20           | $6.790e - 05$       | $0.598823615 - 0.313135185 i$ |
| 40           | $2.260e - 06$       | $0.598859558 - 0.313079898 i$ |
| 60           | $2.590e - 07$       | $0.598861332 - 0.313078830 i$ |
| 100          | $1.245e - 07$       | $0.598861546 - 0.313078775 i$ |

The beta integral (547) was also evaluated directly for several nonintegral exponent pairs; residuals were between  $4 \times 10^{-16}$  and  $6 \times 10^{-14}$ . A rank-10, 400-node quantile quadrature checked (601) with errors

$$5.7 \times 10^{-9}, \quad 3.6 \times 10^{-8}, \quad 7.3 \times 10^{-8} \quad (662)$$

for  $s = 0.7, 2.5, 5$ , respectively. The partial sums  $\sum_{n \leq 200} d_n/n$  give 0.758239999524, agreeing with (591).

The finite spline is not forced to obey the infinite-rank dyadic primitive identity exactly; its residual falls by about a factor of 16 whenever the rank increases by 2, consistent with convergence to the limiting law. By contrast, the beta substitution is exact for every smooth finite-rank CDF, so its residual is essentially floating-point roundoff.

The archive contains `numerical_experiments.py`. Running

```
python numerical_experiments.py
```

regenerates `numerical_results.txt` and `numerical_results.tex`. The code documents every normalization, the finite spline formula, quadrature, bisection inversion, and rational-moment recurrence.

## 14 An arithmetic continued fraction generated by the moments

The Stieltjes transform admits one more exact structure that appears not to have been exploited in the audited corpus. Let  $P_n$  be the monic polynomials orthogonal with respect to  $dF$  on  $[0, 1]$ , normalized by  $P_{-1} = 0, P_0 = 1$ .

### Post-snapshot Lean status of the finite Gram–Stieltjes algebra

The post-snapshot development begins with `FiniteMomentGram.lean` and `GramStieltjes.lean`, and the current `FiniteMomentJacobi.lean` completes their generic finite recurrence layer. They take an arbitrary scalar sequence `moment : Nat → R`; they do not identify that sequence with the Fabius moments  $d_n$  or with the moments of any measure. This measure-free generality complements, rather than replaces, the up-measure-specific pipeline described later in this section. In `FiniteMomentGram.lean`, `Fabius.momentFunctional` is the polynomial linear functional determined by its values on powers of  $X$ . The declarations `Fabius.momentFunctional_of_1`, `linearMap`, `Fabius.momentFunctional_injective`, and `Fabius.momentFunctional_map` give recovery, uniqueness, and scalar base change. The symmetric bilinear form `Fabius.momentPairing`, the matrix `Fabius.momentHankelMatrix`, its determinant `Fabius.momentHankelDet`, and the degree-bounded restriction `Fabius.finiteMomentPairing` are likewise generic. The exact matrix theorem `Fabius.finiteMomentPairing_toMatrix` identifies the restricted pairing in the monomial basis with the Hankel matrix. Over a commutative integral domain, `Fabius.finiteMomentPairing_nondegenerate_iff` identifies nondegeneracy with nonvanishing of the corresponding Hankel determinant.

In `GramStieltjes.lean`, `Fabius.gramStieltjesNumerator` is the fraction-free polynomial obtained from the last column of the adjugate of the order- $(n+1)$  Hankel matrix. The declarations `Fabius.momentPairing_gramStieltjesNumerator_X_pow_eq_zero`, `Fabius.momentPairing_gramStieltjesNumerator_X_pow_eq_det`, `Fabius.gramStieltjesNumerator_coeff_top`, and `Fabius.momentPairing_gramStieltjesNumerator_self` prove, respectively, its lower-power orthogonality, its pairing with  $X^n$ , its degree- $n$  coefficient, and its self-pairing in terms of consecutive Hankel determinants. Over a field, after assuming the order- $n$  determinant is nonzero, `Fabius.gramStieltjesPolynomial_isMonicOfDegree`, `Fabius.momentPairing_gramStieltjesPolynomial_eq_zero`, and `Fabius.eq_gramStieltjesPolynomial_of_isMonicOfDegree_of_orthogonal` give the monic degree- $n$  normalization, orthogonality to every lower-degree polynomial, and uniqueness. The exact self-pairing quotient is `Fabius.momentPairing_gramStieltjesPolynomial_self`; its nonzero corollary assumes both consecutive determinants are nonzero.

`FiniteMomentJacobi.lean` defines the norm quotient `Fabius.gramStieltjesNorm`, the diagonal coefficient `Fabius.gramStieltjesJacobiDiagonal`, and the subdiagonal coefficient `Fabius.gramStieltjesJacobiSubdiagonal`. The theorem `Fabius.gramStieltjesJacobiSubdiagonal_eq_det_ratio` proves

$$\frac{h_{n+1}}{h_n} = \frac{\Delta_{n+2}\Delta_n}{\Delta_{n+1}^2}.$$

The theorem `Fabius.gramStieltjesPolynomial_three_term_zero` proves  $XP_0 = P_1 + \alpha_0 P_0$  from the nonzero first minor, and `Fabius.gramStieltjesPolynomial_three_term` proves every higher monic three-term recurrence from exactly the three nonzero minors of orders  $n, n+1, n+2$ . These theorems are finite field algebra and assume neither a measure nor positivity. The separate up-measure pipeline in `MomentHankelMatrix.lean`, `MomentHankelValues.lean`, and the `OrthogonalPolynomial*.lean` modules now supplies strict positivity of the Hankel matrices and ratios, identifies the exact moments, constructs the normalized polynomials, proves parity and the symmetric three-term recurrence, and evaluates the first displayed Jacobi data. The generic modules above remain useful because their determinant and orthogonality algebra assumes neither a measure nor positivity. The comparison module `OrthogonalPolynomialGramBridge.lean` proves that the generic functional for `upMoment` is integration against the canonical up measure and identifies both Hankel objects, both fraction-free determinant polynomials, and both monic normalizations. Its Jacobi wrappers identify the generic norm with the up-specific Hankel ratio, prove the generic diagonal is zero, and identify the generic

subdiagonal with the quotient of consecutive ratios. Root location and simplicity, Gaussian quadrature, finite or infinite  $J$ -fraction identification, and continued-fraction convergence remain outside the formal corpus.

**Proposition 14.1** (Rational Jacobi parameters). *There are positive rational numbers  $b_n$  such that*

$$P_{n+1}(x) = \left(x - \frac{1}{2}\right) P_n(x) - b_n P_{n-1}(x), \quad n \geq 0, \quad (663)$$

where the term containing  $b_0$  is absent. In particular,

$$b_1 = \frac{1}{36}, \quad b_2 = \frac{8}{225}. \quad (664)$$

Consequently the Cauchy transform has the Stieltjes  $J$ -fraction

$$S(w) = \frac{1}{w - \frac{1}{2} - \frac{b_1}{w - \frac{1}{2} - \frac{b_2}{w - \frac{1}{2} - \ddots}}}. \quad (665)$$

Every finite convergent is a rational function with rational coefficients.

*Proof.* Gram–Schmidt applied to  $1, x, x^2, \dots$  uses only the rational moments  $d_n$ , so every coefficient of  $P_n$  and every squared norm  $h_n = \int P_n^2 dF$  is rational. Positivity of the measure gives  $h_n > 0$  and  $b_n = h_n/h_{n-1} > 0$ . Symmetry under  $x \mapsto 1-x$  implies  $P_n(1-x) = (-1)^n P_n(x)$ , hence  $\int x P_n(x)^2 dF(x) = h_n/2$  and every diagonal Jacobi coefficient equals  $1/2$ . The standard resolvent expansion of the Jacobi matrix gives (665). Finally  $b_1 = \text{Var}(X) = 1/36$ ; using the fourth centered moment  $19/10800$  gives  $b_2 = 8/225$ .  $\square$

This rational Jacobi matrix is a new arithmetic object attached to the Fabius law. Determining valuations, denominator growth, and the rate at which  $b_n$  approaches the universal interval value is a natural continuation of the repository's dyadic arithmetic program.

## 15 Conjectures and frontier directions

### 15.1 Higher inverse derivatives

Let

$$R_n(y) = \frac{y^n G^{(n)}(y)}{G(y)}, \quad \delta = y \frac{d}{dy}. \quad (666)$$

Direct differentiation gives the exact recurrence

$$R_1 = \epsilon, \quad R_{n+1} = (\epsilon - n)R_n + \delta R_n. \quad (667)$$

The first-order elasticity and the all-orders Lambert phase strongly suggest the following.

**Conjecture 15.1** (All-order inverse endpoint law). *For every fixed  $n \geq 1$ ,*

$$G^{(n)}(y) \sim (-1)^{n-1} (n-1)! \frac{G(y)}{y^n \rho(y)}, \quad y \downarrow 0. \quad (668)$$

Consequently, for  $n \geq 1$  and  $q > 0$ ,

$$G^{(n)} \in L^q(0, 1) \iff 0 < q \leq \frac{1}{n}. \quad (669)$$

The case  $n = 1$  is [theorem 9.5](#). At the borderline  $q = 1/n$ , the residual factor  $G(y)^{1/n}/\rho(y)^{1/n}$  should make the logarithmic-coordinate integral convergent.

A proof should follow by establishing derivative-stable error estimates for the all-orders inverse phase, then inducting in (667). This is more delicate than formal differentiation because an undifferentiated asymptotic remainder does not control its derivatives.

### 15.2 Negative moments and a dual endpoint saddle

The entire function  $D$  grows rapidly on the negative real axis. The dyadic endpoint formula, Stirling's approximation, and the forward saddle suggest the coarse scale

**Conjecture 15.2** (Negative-moment saddle). *As  $r \rightarrow +\infty$ ,*

$$\log D(-r) = \frac{\log 2}{2} r^2 - r \log r + O(r). \quad (670)$$

*After subtracting the complete smooth expansion in  $r$ , a bounded period-1 function of the corrected phase  $r - \log_2 r + O(1)$  should remain.*

This problem is a Mellin-dual counterpart of the small- $x$  and inverse-Fabius endpoint analysis. Equation (595) offers a second route through the large- $s$  saddle of  $B(s)$ .

### 15.3 Autocorrelation arithmetic

The squared sinc product in (657) has a refinement mask obtained by squaring the original mask. It is reasonable to expect a dyadic arithmetic theory for  $R(a)$  analogous to that of  $\text{up}(a)$ , but with triangular rather than uniform digit laws.

**Conjecture 15.3** (Dyadic autocorrelation arithmetic). *For every dyadic rational  $a \in [-2, 2]$ , the value  $R(a)$  belongs to the field generated by the Fabius dyadic values and, in fact, is rational after the natural power-of-two normalization determined by the spline rank. Its odd-prime denominators should be governed by products of factors  $2^{2m} - 1$  with multiplicities reflecting the squared mask.*

Even a weaker exact recurrence for  $R(k/2^m)$  would be useful: it would turn shifted products of  $\text{up}$  into arithmetic data and would provide certified values of  $\|\text{up}\|_2^2 = R(0)$ .

### 15.4 Fractional operator spectrum

Equation (579) suggests studying the dilation–Volterra operator algebra generated by

$$(Df)(x) = f(x/2), \quad (V_\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt.$$

The Fabius function is a cyclic vector for a noncommutative relation between these operators. Promising questions include:

- meromorphic continuation of  $J_{\alpha,\nu}$  jointly in  $(\alpha, \nu)$  after canonical endpoint subtractions;
- spectral decompositions of the dilation generator in Mellin coordinates;
- base- $b$  analogues, replacing the dyadic triangular exponents by  $b$ -adic quadratic forms;
- a fractional analogue of the signed derivative hierarchy (536).

### 15.5 Mellin zeros, growth, and acceleration

The Newton series (590) can be evaluated entirely from rational moments. Its convergence is superalgebraic but not generally geometric. The endpoint phase suggests splitting  $d_n$  into a smooth saddle part and a tiny periodic residual, then applying Euler or Nørlund acceleration to the Newton basis. This may reveal:

- the order and indicator function of the entire function  $D(z)$ ;
- zero patterns of  $D$ ,  $1 - D$ , and the regularized Mellin transforms;
- functional relations between those zeros and the zeros of the Laplace product  $B$ ;
- certified high-precision negative moments, inaccessible to direct quadrature.

### 15.6 Orthogonal-polynomial arithmetic

The rational coefficients  $b_n$  in (663) deserve a separate arithmetic study. General orthogonal-polynomial theory predicts interval-universal limiting behavior because the density is positive on  $(0, 1)$ , but the superflat endpoints may leave a slow, log-periodic signature in the convergence rate. Concrete problems are:

- determine 2-adic and odd-prime valuations of  $b_n$ ;
- derive a nonlinear recurrence directly from the moment recurrence (581);
- identify the first log-periodic term in the large- $n$  Jacobi asymptotics;
- connect Padé approximants of  $S$  to exact dyadic evaluations of  $F$ .

## 16 Novelty ledger and formalization roadmap

| Result                                     | Status in audit                           | Main proof mechanism                                             |
|--------------------------------------------|-------------------------------------------|------------------------------------------------------------------|
| $\int_0^x F = F(x/2)$ and integer iterates | baseline, partly implicit                 | Fabius differential dilation and normalization                   |
| Finite formula (562)                       | not located explicitly                    | repeated integration by parts with shrinking argument            |
| Complex series (568)                       | new relative to audit                     | exact recurrence plus endpoint-flat factorial bound              |
| Newton–Mellin formula (590)                | new relative to audit                     | dyadic endpoint/moment identity at $x = 1$                       |
| Fractional recurrence (579)                | new relative to audit                     | Riemann–Liouville kernel and one integration by parts            |
| Local inverse primitive (614)              | not located explicitly                    | quantile substitution followed by Stieltjes integration by parts |
| All inverse derivatives (621)              | standard inverse calculus, new packaging  | differential-polynomial recurrence preserving $F'^{2n-1}$        |
| Sharp $L^q$ threshold (626)                | not located explicitly                    | change of variables and infinite endpoint flatness               |
| Refined inverse primitive (635)            | new consequence of repository asymptotics | elasticity identity and Watson expansion                         |
| Stieltjes/logarithmic transforms           | new systematic synthesis                  | master CDF–survival and quantile identities                      |

| Result                        | Status in audit        | Main proof mechanism                                             |
|-------------------------------|------------------------|------------------------------------------------------------------|
| Gini/autocorrelation formulas | new synthesis          | order statistics and Fourier representation of absolute distance |
| Rational Jacobi parameters    | not located explicitly | rational moments, symmetry, and Gram–Schmidt                     |

The following sequence is realistic for Lean formalization:

1. formalize  $A_p$  for  $p \in \mathbb{N}_0$ , prove (561), and derive the finite sum by induction;
2. formalize the  $n$ th Volterra primitive (574) using finite binomial expansion;
3. define  $D(z)$  first on real exponents, prove the moment recurrence and Mellin identities, then add complex analyticity;
4. prove the finite Newton identities at integer  $z$  before introducing normal convergence of (590);
5. formalize the inverse local primitive and the  $Q_n$  recurrence, which require only one-dimensional calculus;
6. treat  $L^q$  divergence using the existing endpoint-flat lemmas;
7. postpone the refined periodic asymptotics until derivative-controlled error bounds are available in the formal library.

## 17 Conclusion

The Fabius–Rvachev system is unusually well suited to integration: differentiation doubles the argument, while normalized integration halves it. Once this elementary observation is combined with endpoint flatness, it yields more than the familiar primitive  $F(x/2)$ . Polynomial weights produce finite dyadic sums; arbitrary complex weights produce normally convergent dyadic series; the endpoint values turn those series into an entire Newton interpolation of complex moments; and quantile substitution transfers the same structure to the inverse Fabius function.

The transform picture is similarly unified. Mellin transforms are controlled by the entire moment function  $D$ , Laplace transforms by the product  $B$ , Fourier transforms by the lacunary sinc product, and Stieltjes transforms by a rational-moment Jacobi matrix. Products of  $F$  become order-statistic gaps, while shifted products of up become autocorrelations of the squared sinc product. The resulting formulas are exact, computationally testable, and closely connected to the repository’s existing themes of Thue–Morse cancellation,  $q$ -binomial arithmetic, Bernoulli/Bell expansions, and Lambert– $W$  endpoint inversion.

The strongest immediate research opportunities are to formalize the complex-weight series and Newton interpolation, prove derivative-stable all-orders inverse asymptotics, develop the arithmetic of the rational Jacobi parameters and autocorrelation values, and use the Mellin–Laplace bridge to understand negative complex moments.

## A Selected exact formulas at a glance

For convenience, the central formulas are collected here:

$$I_0^n F(x) = 2^{\binom{n}{2}} F(x/2^n),$$

$$\int_0^x t^p F(t) dt = \sum_{j=0}^p (-1)^j \frac{p!}{(p-j)!} 2^{\binom{j+1}{2}} x^{p-j} F(x/2^{j+1}),$$

$$\begin{aligned}
A_\nu(x) &= \sum_{j \geq 0} (-1)^j \nu^j 2^{\binom{j+1}{2}} x^{\nu-j} F(x/2^{j+1}), \\
\mathcal{M}_F(z) &= \frac{1 - D(z)}{z}, \quad \mathcal{M}_{\text{up},+}(z) = \frac{D(z)}{z}, \\
D(z) &= 1 - z \sum_{j \geq 0} (-1)^j z - 1^j \frac{d_{j+1}}{(j+1)!}, \\
B(s) &= \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \\
\int_0^y G(t)^p dt &= yG(y)^p - pA_{p-1}(G(y)), \\
G^{(n)}(y) &= \frac{Q_n[F'](G(y))}{F'(G(y))^{2n-1}}, \\
S(w) &= \int_0^1 \frac{dy}{w - G(y)} = \sum_{n \geq 0} \frac{d_n}{w^{n+1}}.
\end{aligned}$$

## B Bibliographic and repository notes

The repository was inspected through its live GitHub directory pages and raw TeX views on 27 August 2026. Because the draft and archive trees contain superseded copies, an archived duplicate was used to identify earlier claims but was not counted as an independent source. The accompanying report is self-contained and does not require the repository to compile.

## References

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## Part VI

# Integral and Transform Frontiers for the Fabius–Rvachev System

### Abstract

The Fabius function, its inverse, Rvachev’s compactly supported up-function, and the signed global self-differential extension form an unusually rigid calculus: differentiation is conjugate to dyadic dilation. This report develops the corresponding *integration* theory. The first headline result is a two-sided integer operator orbit

$$D^r \mathcal{F}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x), \quad r \in \mathbb{Z},$$

where negative powers are the distinguished zero-initial antiderivatives. It yields closed forms for all integer-order primitives of  $F$ ,  $1 - F$ , and  $\text{up}$ , and finite formulas for polynomially weighted primitives. Repeated integration by parts then gives convergent series for arbitrary complex monomial powers, including negative and fractional powers.

A second group of results concerns the inverse  $I = F^{-1}$ . In particular,

$$\int_0^y I(v) \, dv = yI(y) - F\left(\frac{I(y)}{2}\right),$$

and integer powers  $I^m$  have finite closed primitives. An entire Mellin interpolant

$$D(z) = \int_0^1 x^z F'(x) \, dx = 2^{-z} \sum_{j \geq 0} \binom{z}{2j} c_j$$

simultaneously represents all complex moments of the inverse:  $D(z) = \int_0^1 I(y)^z \, dy$ . This produces explicit rational-moment series for negative powers and formulas for logarithmic inverse moments.

The strongest distributional result is an exact pushforward theorem. If  $Q_m = 2^{m(m+1)/2}$ , then for every  $m \geq 1$ ,

$$\frac{|F^{(m)}(U)|}{Q_m} \stackrel{d}{=} F(U), \quad U \sim \text{Unif}[0, 1].$$

Consequently all  $L^p$ , Orlicz, Lorentz, and rearrangement-invariant norms of every normalized derivative are known at once. It also gives the sharp integrability law  $I' \in L^p(0, 1)$  exactly for  $0 < p \leq 1$ , together with an exact norm identity.

The report further develops Laplace, Fourier, Mellin, Cauchy–Stieltjes, correlation, Riemann–Liouville, and Volterra-resolvent transforms. The log-periodic factor already present in the repository is used to prove a new obstruction: the integer self-differential/dilation identity cannot extend exactly to a noninteger standard fractional derivative. Numerical experiments based on Fourier inversion verify the new formulas to high precision. Novelty claims are relative to the recursively audited repository corpus, not assertions of worldwide priority.

# 1 Scope, corpus audit, and status of the results

## 1.1 Repository corpus and non-duplication boundary

The recursive audit covered the LaTeX material under

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The tree contains a current expository monograph, a merged research-frontier volume, separate inverse-function and Newton–Rvachev reports, Thue–Morse and spectral reports, Fourier-decay studies, archived predecessor drafts, a Lean walkthrough, a glossary, and transcriptions of primary papers. The live Newton–Rvachev report records a 67-source snapshot, while the archive explicitly preserves superseded and duplicated versions. Therefore the audit treated a current document and its archived ancestors as one mathematical source rather than as independent evidence of novelty.

The corpus already develops, in depth:

- the functional-differential equations, signed block extension, derivative formulae, partition of unity, and probabilistic random-series model;
- exact values at dyadic rationals,  $q$ -Pochhammer and  $q$ -binomial formulae, Thue–Morse cancellation, and moment recurrences;
- the infinite sinc/cosine Fourier products and sharp Fourier decay;
- endpoint asymptotics, Lambert– $W$  saddle coordinates, periodic corrections, and Richardson-type acceleration;
- local inverse germs, inverse endpoint asymptotics, and formal series machinery;
- repeated integration as an approximation process converging to up.

What was not found as a developed subject is a systematic calculus of indefinite antiderivatives, polynomial/fractional weighted primitives, exact inverse-function primitives, Mellin interpolation of inverse moments, derivative equimeasurability, or the fractional-dilation obstruction proved here. Some ingredients used below are of course known in the repository—especially the derivative formula, block signs, moments, Fourier product, and periodic Laplace factor. The novelty lies in the new integral consequences and their synthesis.

*Remark 1.1* (Priority discipline). “New” below means not duplicated in the inspected repository. The report has not performed an exhaustive search of every historical source in every language, so it does not claim global first publication. Each statement is marked as proved, conjectural, or computationally supported.

## 1.2 Proof-status summary

| Result family                                       | Status | Main input                                                    |
|-----------------------------------------------------|--------|---------------------------------------------------------------|
| Two-sided integer derivative/primitive orbit        | proved | $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ , flatness at the origin |
| Polynomial and complex-power weighted primitives    | proved | finite/infinite integration by parts with flatness estimates  |
| Closed primitive and power primitives of $F^{-1}$   | proved | change of variables $y = F(x)$ and the primitive ladder       |
| Fractional causal primitives and Volterra resolvent | proved | Riemann–Liouville convolution and compact support             |

| Result family                                      | Status        | Main input                                                |
|----------------------------------------------------|---------------|-----------------------------------------------------------|
| Laplace/Fourier/Mellin/Cauchy transform identities | proved        | probability model, integration by parts, moment expansion |
| Derivative pushforward/equimeasurability theorem   | proved        | Thue–Morse block decomposition and exact sign balance     |
| Sharp $L^p$ threshold for $(F^{-1})'$              | proved        | pushforward theorem and endpoint flatness                 |
| No exact noninteger fractional-dilation collapse   | proved        | periodic Laplace factor and nonzero first Fourier mode    |
| Higher inverse-derivative $L^p$ threshold          | conjectural   | Lambert– $W$ endpoint asymptotics and low-order formulas  |
| Numerical constants and checks                     | computational | FFT inversion of the infinite sinc product                |

### 1.3 Current Lean crosswalk for the normalized Volterra layer

The proof labels above record the status of the report’s mathematical arguments. A later implementation checkpoint now verifies the reusable operator layer: the declarations below are present at compiled Lean commit `e109088ed`, verified by a focused build. Put

$$J_{n,a}f(x) := \begin{cases} f(x), & n = 0, \\ \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) \, dt, & n > 0. \end{cases} \quad (671)$$

The exact declarations in `FabiusFunction.NormalizedVolterra` are:

- `normalizedVolterra_affine`, the division-free identity

$$J_{n,ca+d}f(cx+d) = c^n J_{n,a}(t \mapsto f(ct+d))(x), \quad (672)$$

for every real  $c$ , including  $c = 0$  and negative scales, and including  $n = 0$ . It requires neither integrability nor completeness and works in any real normed target;

- `normalizedVolterra_comp_affine`, the inverse-smul form

$$J_{n,a}(t \mapsto f(ct+d))(x) = (c^n)^{-1} J_{n,ca+d}f(cx+d), \quad c \neq 0, \quad (673)$$

whose nonzero-scale hypothesis is required by the division;

- `normalizedVolterra_basepoint_shift`, which, under interval integrability from  $a$  to  $b$  and from  $b$  to  $x$ , proves

$$J_{n,a}f(x) = J_{n,b}f(x) + \sum_{k=0}^{n-1} \frac{(x-b)^k}{k!} J_{n-k,a}f(b). \quad (674)$$

There is no ordering hypothesis on  $a, b, x$ , and  $n = 0$  is included;

- `normalizedVolterra_succ_eq_taylor_of_eq_zero`, which assumes integrability only from  $a$  to  $b$  and vanishing of  $f$  on the open unoriented interval between  $b$  and  $x$ , and proves

$$J_{n+1,a}f(x) = \sum_{k=0}^n \frac{(x-b)^k}{k!} J_{n+1-k,a}f(b). \quad (675)$$

It imposes neither endpoint values nor endpoint order; when  $b = x$ , the vanishing condition is vacuous.

Accordingly, the generic assertion that a positive-order Volterra primitive becomes a finite Taylor polynomial after the input vanishes is Lean formalized. Separately, `normalizedVolterra_pow_mul_extendedFabius` and `normalizedVolterra_pow_mul_fabiusReal_of_le_one` formalize the signed-global and bounded- $F$  natural-monomial coefficients, the latter on  $x \leq 1$ . The generic operator theorem alone does not identify the displayed  $1 - F$ , shifted-up, or exterior piecewise moment coefficients; those report-specific evaluations retain their existing paper-proof status.

## 2 Normalization and structural identities

### 2.1 The three related functions

Let  $F : [0, 1] \rightarrow [0, 1]$  be the standard Fabius distribution function, normalized by

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x).$$

Its density can be written on the whole unit interval as

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 \leq x \leq 1. \quad (676)$$

The even, compactly supported Rvachev function is

$$\operatorname{up}(t) = \begin{cases} F(t + 1), & -1 \leq t \leq 0, \\ F(1 - t), & 0 \leq t \leq 1, \\ 0, & |t| \geq 1. \end{cases} \quad (677)$$

It is a probability density:  $\int_{-1}^1 \operatorname{up}(t) dt = 1$ .

The global signed self-differential extension is denoted by  $\mathcal{F}$ . It agrees with  $F$  on  $[0, 1]$ , vanishes on  $(-\infty, 0]$ , and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (678)$$

Writing  $\varepsilon_k = (-1)^{s_2(k)}$ , where  $s_2(k)$  is the binary digit sum, its block form is

$$\mathcal{F}(x) = \varepsilon_k \operatorname{up}(x - 2k - 1), \quad 2k \leq x \leq 2k + 2. \quad (679)$$

The first  $2^r$  Thue–Morse signs have equally many plus and minus signs whenever  $r \geq 1$ .

### 2.2 Derivatives, moments, and transforms already available

Iteration of (678) gives

$$\mathcal{F}^{(m)}(x) = Q_m \mathcal{F}(2^m x), \quad Q_m := 2^{m(m+1)/2}. \quad (680)$$

All derivatives of  $F$  vanish at both endpoints. Define the even moments

$$c_j := \int_{-1}^1 t^{2j} \operatorname{up}(t) dt. \quad (681)$$

They are rational and begin

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \dots$$

With the Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ ,

$$\widehat{\operatorname{up}}(\xi) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi \xi}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi \xi}{2^m}\right)^m. \quad (682)$$

The bilateral moment-generating transform is

$$M(s) := \int_{-1}^1 e^{st} \text{up}(t) dt = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}. \quad (683)$$

These established identities are the raw material for the integration theory below.

### 3 The exact antiderivative ladder

#### 3.1 A two-sided integer differential orbit

For  $n \geq 1$ , put

$$A_n := 2^{n(n-1)/2}. \quad (684)$$

Notice that  $A_n = Q_n/2^n$ .

**Theorem 3.1** (Distinguished  $n$ -fold primitive of the global extension). *For  $x \geq 0$  and  $n \geq 1$ ,*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} \mathcal{F}(t) dt = A_n \mathcal{F}\left(\frac{x}{2^n}\right). \quad (685)$$

*Equivalently, with zero-initial negative powers of differentiation,*

$$\boxed{D^r \mathcal{F}(x) = 2^{r(r+1)/2} \mathcal{F}(2^r x), \quad r \in \mathbb{Z}.} \quad (686)$$

*Proof.* For positive  $r$ , this is (680). For  $r = -n$ , differentiate  $A_n \mathcal{F}(x/2^n)$  exactly  $n$  times. The chain rule and (680) give the coefficient

$$A_n 2^{-n^2} Q_n = 2^{n(n-1)/2 - n^2 + n(n+1)/2} = 1,$$

so the  $n$ -th derivative is  $\mathcal{F}(x)$ . Every derivative of the proposed primitive of order below  $n$  vanishes at  $x = 0$ , because  $\mathcal{F}$  is flat there. These are precisely the initial conditions characterizing the Volterra integral on the left.  $\square$

**Corollary 3.2** (All local primitives of  $F$ ). *On  $[0, 1]$ , every  $n$ -fold indefinite primitive of  $F$  is*

$$D^{-n} F(x) = A_n F\left(\frac{x}{2^n}\right) + P_{n-1}(x), \quad (687)$$

*where  $P_{n-1}$  is an arbitrary polynomial of degree at most  $n-1$ . The zero-initial primitive is the first term alone.*

The first cases are exceptionally simple:

$$\int_0^x F(t) dt = F(x/2), \quad (688)$$

$$\int_0^x (x-t) F(t) dt = 2F(x/4), \quad (689)$$

$$\frac{1}{2} \int_0^x (x-t)^2 F(t) dt = 8F(x/8). \quad (690)$$

Thus the antiderivatives are not merely expressible by quadrature: they remain in the same self-differential orbit.

**Corollary 3.3** (Primitives of up,  $1 - F$ , and affine copies). *For  $-1 \leq x \leq 1$ ,*

$$\frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} \text{up}(t) dt = A_n F\left(\frac{x+1}{2^n}\right). \quad (691)$$

*For  $0 \leq x \leq 1$ ,*

$$D^{-n}(1 - F(x)) = (-1)^n A_n F\left(\frac{1-x}{2^n}\right) + P_{n-1}(x). \quad (692)$$

*More generally, for  $a \neq 0$ , on every interval where the relevant global branch is fixed,*

$$D^{-n}F(ax+b) = a^{-n} A_n \mathcal{F}\left(\frac{ax+b}{2^n}\right) + P_{n-1}(x). \quad (693)$$

*The condition  $a \neq 0$  is required by this inverse-scaled formula. It is distinct from the division-free forward covariance (672), which remains valid when the scale is zero.*

*Proof.* For (691), differentiating the right side  $n$  times gives  $\mathcal{F}(x+1) = \text{up}(x)$  on  $[-1, 1]$ . Formula (692) follows from  $1 - F(x) = F(1-x)$ . The affine formula is the chain rule for  $a \neq 0$ .  $\square$

### 3.2 Global causal primitives of up

The scaled formula (691) is local because  $\mathcal{F}(x+1)$  ceases to equal  $\text{up}(x)$  outside its central block. The global causal primitive becomes a moment polynomial after the support is passed.

**Proposition 3.4** (Complete piecewise form). *Let*

$$U_n(x) := \frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} \text{up}(t) dt,$$

*with the upper limit understood as  $\min(x, 1)$  once  $x > 1$ . Then*

$$U_n(x) = \begin{cases} 0, & x \leq -1, \\ A_n F((x+1)/2^n), & -1 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2j} c_j x^{n-1-2j}, & x \geq 1. \end{cases} \quad (694)$$

*Proof.* Only the last branch needs comment. Expand  $(x-t)^{n-1}$ ; odd moments vanish by evenness of up, and the even moments are  $c_j$ .  $\square$

**Remark 3.5** (Formal status of the zero tail). At compiled checkpoint e109088ed, the declaration

```
normalizedVolterra_succ_eq_taylor_of_eq_zero
```

formally supplies the generic finite Taylor polynomial once the integrand vanishes beyond the support. It does not by itself prove the explicit  $c_j$ -coefficient identification in (694), nor the analogous displayed  $F$ - or up-specific coefficients; those specializations remain at the report's paper-proof status.

### 3.3 Definite dyadic integrals become exact arithmetic

A direct endpoint consequence is useful for the arithmetic side of the repository.

**Corollary 3.6** (Polynomial integrals over dyadic intervals). *If  $P \in \mathbb{Q}[x]$  and  $a, b \in [0, 1]$  are dyadic rationals, then  $\int_a^b P(x)F(x) dx$  is rational. The same holds for  $P(x)\text{up}(x)$  on dyadic subintervals of  $[-1, 1]$ .*

*Proof.* The finite endpoint formula in [theorem 4.1](#) below involves only rational coefficients and values of  $F$  at dyadic arguments. Those values are rational by the established arithmetic theory.  $\square$

This creates a direct bridge from antiderivatives to the repository's  $q$ -binomial formulae: every such integral can be converted into a finite expression in exactly computable dyadic values and then into the preferred  $q$ -arithmetic form.

## 4 Polynomial, analytic, fractional, and negative weights

### 4.1 A finite operator formula for every polynomial weight

Define the zero-initial primitive ladder

$$\mathcal{A}_j(x) := A_j F(x/2^j), \quad j \geq 1, \quad (695)$$

so  $\mathcal{A}'_j = \mathcal{A}_{j-1}$ , with  $\mathcal{A}_0 = F$ .

**Theorem 4.1** (Polynomially weighted  $n$ -fold primitives). *Let  $P$  be a polynomial of degree  $d$ , and let  $n \geq 1$ . Then*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} P(t) F(t) dt = \sum_{k=0}^d (-1)^k \binom{n+k-1}{k} P^{(k)}(x) \mathcal{A}_{n+k}(x). \quad (696)$$

Consequently the general indefinite  $n$ -fold primitive is the right side plus an arbitrary polynomial of degree at most  $n-1$ .

*Proof.* Differentiate the right side. The contributions containing  $P^{(k)} \mathcal{A}_{n+k-1}$  cancel by Pascal's identity. Repeating this  $n$  times leaves  $P(x)F(x)$ . Every derivative of order below  $n$  vanishes at the origin because all members of the  $\mathcal{A}$ -ladder are flat there. This identifies the Volterra solution uniquely.  $\square$

For a monomial  $P(x) = x^p$ ,  $p \in \mathbb{N}$ , the formula is

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^p F(t) dt = \sum_{k=0}^p (-1)^k \binom{n+k-1}{k} \frac{p!}{(p-k)!} x^{p-k} \mathcal{A}_{n+k} F(x/2^{n+k}). \quad (697)$$

In particular,

$$\int_0^1 x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{k(k+1)/2} F(2^{-k-1}). \quad (698)$$

This is a finite rational closed form. The first examples are

$$\int_0^1 F(x) dx = \frac{1}{2}, \quad \int_0^1 xF(x) dx = \frac{13}{36}, \quad \int_0^1 x^2 F(x) dx = \frac{5}{18}. \quad (699)$$

Since  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$ ,

$$\int_0^1 x^p \text{up}(x) dx = \frac{1}{p+1} - \int_0^1 x^p F(x) dx. \quad (700)$$

Formula (700) is compatible with the half-moment sequences already studied in the repository; the new contribution here is the finite antiderivative route.

## 4.2 Convergent series for arbitrary complex powers

For  $\alpha \in \mathbb{C}$ , write  $\alpha^{\underline{k}} = \alpha(\alpha - 1) \cdots (\alpha - k + 1)$  and use the real logarithm to define  $x^\alpha = e^{\alpha \log x}$  for  $x > 0$ .

**Theorem 4.2** (Complex-power weighted primitive). *For every  $\alpha \in \mathbb{C}$ , every integer  $n \geq 1$ , and every  $x > 0$ ,*

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^\alpha F(t) dt = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} \alpha^{\underline{k}} x^{\alpha-k} \mathcal{A}_{n+k}(x). \quad (701)$$

The integral and series are locally uniform entire functions of  $\alpha$ . If  $\alpha \in \mathbb{N}$ , the series terminates and reduces to (697). Negative and fractional powers require no endpoint regularization because  $F$  is flat at zero.

*Proof.* Repeated integration by parts gives the first  $N$  terms and a remainder containing  $\alpha^{\underline{N+1}}$  times a higher primitive. Flatness implies that for every integer  $r \geq 0$  there is  $C_r$  such that  $0 \leq F(t) \leq C_r t^r$  on  $[0, x]$ . Consequently

$$0 \leq \mathcal{A}_m(x) \leq C_r \frac{r! x^{m+r}}{(m+r)!}. \quad (702)$$

The factorial in the denominator dominates the falling factorial and the polynomial factor  $\binom{n+k-1}{k}$ ; choosing  $r$  large gives absolute, locally uniform convergence and sends the remainder to zero. The same bound permits differentiation with respect to  $\alpha$ .  $\square$

The first-order case is especially transparent:

$$\int_0^x t^\alpha F(t) dt = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} x^{\alpha-k} 2^{k(k+1)/2} F(x/2^{k+1}). \quad (703)$$

It is a convergent self-similar transseries whose scales descend geometrically while its coefficients grow factorially; endpoint flatness makes the net terms summable.

## 4.3 Analytic weights and exponential resolvents

The same argument applies to a sufficiently controlled analytic weight  $g$ :

$$\mathcal{I}^n(gF)(x) = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} g^{(k)}(x) \mathcal{A}_{n+k}(x), \quad (704)$$

provided the derivative growth is dominated by the factorial decay in (702). For  $g(x) = e^{\lambda x}$ , this gives the entire series

$$\mathcal{I}^n(e^{\lambda x} F(x)) = e^{\lambda x} \sum_{k=0}^{\infty} (-\lambda)^k \binom{n+k-1}{k} \mathcal{A}_{n+k} F(x/2^{n+k}). \quad (705)$$

This is a local resolvent expansion entirely in scaled Fabius values.

## 5 Closed primitives involving the inverse Fabius function

Let

$$I := F^{-1} : (0, 1) \rightarrow (0, 1). \quad (706)$$

It obeys  $I(1-y) = 1 - I(y)$  and

$$I'(y) = \frac{1}{F'(I(y))}. \quad (707)$$



### 5.1 A closed first primitive

**Theorem 5.1** (Exact area primitive of the inverse). For  $0 \leq y \leq 1$ ,

$$\int_0^y I(v) dv = yI(y) - F\left(\frac{I(y)}{2}\right). \quad (708)$$

Thus an indefinite primitive is  $yI(y) - F(I(y)/2) + C$ .

*Proof.* Put  $q = I(y)$ , change variables  $v = F(x)$ , and integrate by parts:

$$\int_0^y I(v) dv = \int_0^q xF'(x) dx = qy - \int_0^q F(x) dx.$$

The final integral is  $F(q/2)$  by (688).  $\square$

At  $y = 1/2$ ,  $I(y) = 1/2$ , and  $F(1/4) = 5/72$ , so

$$\int_0^{1/2} I(v) dv = \frac{1}{4} - \frac{5}{72} = \frac{13}{72}. \quad (709)$$

At  $y = 1$ , the formula gives  $\int_0^1 I = 1/2$ , as required by symmetry.

### 5.2 Powers of the inverse

**Theorem 5.2** (Primitive of an arbitrary complex inverse power). For  $\alpha \in \mathbb{C}$ ,  $0 < y < 1$ , and  $q = I(y)$ ,

$$\int_0^y I(v)^\alpha dv = yq^\alpha - \alpha \int_0^q x^{\alpha-1} F(x) dx \quad (710)$$

$$= yq^\alpha - \alpha \sum_{k=0}^{\infty} (-1)^k \alpha - 1^k q^{\alpha-1-k} 2^{k(k+1)/2} F(q/2^{k+1}). \quad (711)$$

The series terminates when  $\alpha$  is a nonnegative integer.

*Proof.* After  $v = F(x)$ , integrate  $x^\alpha F'(x)$  by parts. The boundary term at zero vanishes for every complex  $\alpha$  because  $F$  is flatter than every power. Then apply theorem 4.2 with exponent  $\alpha - 1$ .  $\square$

For  $m \in \mathbb{N}$ , this is the finite closed form

$$\int_0^y I(v)^m dv = yq^m - m \sum_{k=0}^{m-1} (-1)^k \frac{(m-1)!}{(m-1-k)!} q^{m-1-k} 2^{k(k+1)/2} F(q/2^{k+1}). \quad (712)$$

### 5.3 All repeated primitives: a layer-cake formula

**Theorem 5.3** ( $n$ -fold primitive of an inverse power). For  $n \geq 1$ ,  $\operatorname{Re} \alpha > 0$ , and  $q = I(y)$ ,

$$\frac{1}{(n-1)!} \int_0^y (y-v)^{n-1} I(v)^\alpha dv = \frac{\alpha}{n!} \int_0^q x^{\alpha-1} (y-F(x))^n dx. \quad (713)$$

In particular,

$$\frac{1}{(n-1)!} \int_0^y (y-v)^{n-1} I(v) dv = \frac{1}{n!} \int_0^{I(y)} (y-F(x))^n dx. \quad (714)$$

*Proof.* Use  $I(v)^\alpha = \int_0^{I(v)} \alpha x^{\alpha-1} dx$ . For real  $\alpha > 0$ , Tonelli's theorem applies directly; for complex  $\operatorname{Re} \alpha > 0$ , absolute convergence gives Fubini's theorem (or equivalently analytic continuation from the positive real axis). The condition  $x < I(v)$  is equivalent to  $F(x) < v$ . The inner integral is  $(n-1)!^{-1} \int_{F(x)}^y (y-v)^{n-1} dv = (y-F(x))^n/n!$ .  $\square$

Expansion of the right side gives a finite reduction to integrals of  $x^{\alpha-1} F(x)^k$ . It also provides a stable numerical representation near the singular endpoints, where direct differentiation of  $I$  is ill-conditioned.

## 5.4 A general weighted area identity

For every integrable weight  $w$ , Fubini gives

$$\int_0^1 w(y) I(y) dy = \int_0^1 \left( \int_{F(x)}^1 w(y) dy \right) dx. \quad (715)$$

This identity is formalized exactly for every interval-integrable real weight as `Fabius.intervalIntegral_mul_fabiusInv_eq` in `InverseLayerCake.lean`. The theorem is a full- $[0, 1]$  identity; it does not by itself package the repeated or fractional inverse primitives elsewhere in this volume. The complementary complete and variable-upper two-function absolutely-continuous identities on proper compact intervals are now formalized, without measurability assumptions on either clock, as `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` and `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt` in `InversePairIntegral.lean`; their exact Fabius wrappers are `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv` and `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`, and the partial inverse-area specialization is `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`. With  $w(y) = y^n$ ,

$$\int_0^1 y^n I(y) dy = \frac{1 - \int_0^1 F(x)^{n+1} dx}{n+1}. \quad (716)$$

This converts mixed quantile moments into nonlinear Fabius moments without numerical inversion.

## 6 Causal fractional integrals and the Volterra resolvent

### 6.1 Riemann–Liouville primitives of $\text{up}$

For  $\beta > 0$ , define

$$U_\beta(x) := \frac{1}{\Gamma(\beta)} \int_{-1}^{\min(x,1)} (x-t)^{\beta-1} \text{up}(t) dt, \quad x > -1. \quad (717)$$

The total Lean definition is `Fabius.rvachevFractionalPrimitive`, and `Fabius.rvachevFractionalPrimitive_eq_intervalIntegral_min` proves this display for every  $\beta > 0$  and  $-1 \leq x$ , including the endpoint omitted by the report’s historical  $x > -1$  convention. For positive integer  $\beta = n$ , `Fabius.rvachevFractionalPrimitive_nat_succ` identifies it with the corresponding `normalizedVolterra` primitive; the separate explicit piecewise coefficient formula in [proposition 3.4](#) is not claimed by that bridge.

**Proposition 6.1** (Transform and tail expansion). *Let  $V_\beta(y) = U_\beta(y-1)$ ,  $y > 0$ . For  $\text{Re } s > 0$ ,*

$$\mathcal{L}[V_\beta](s) = s^{-\beta} e^{-s} M(s). \quad (718)$$

For  $x > 1$ ,

$$U_\beta(x) = \frac{x^{\beta-1}}{\Gamma(\beta)} \sum_{j=0}^{\infty} \binom{\beta-1}{2j} \frac{c_j}{x^{2j}}. \quad (719)$$

*The series also converges at  $x = 1$ , because  $\text{up}$  is flat at both support endpoints. When  $\beta$  is a positive integer it terminates, recovering the moment polynomial in [\(694\)](#).*

*Proof.* After shifting the support to  $[0, 2]$ , the transform of the density is  $e^{-s} M(-s) = e^{-s} M(s)$ . Riemann–Liouville integration multiplies the Laplace transform by  $s^{-\beta}$ . For the tail, expand  $(x-t)^{\beta-1}$  in powers of  $t/x$ ; odd moments vanish. Endpoint flatness gives termwise convergence at the boundary case  $x = 1$ .  $\square$

The semigroup law

$$U_{\alpha+\beta} = \mathcal{I}^\alpha U_\beta \quad (720)$$

follows from the beta integral. This gives a natural continuous interpolation between the exact integer primitives, although [appendix 13](#) shows that it cannot collapse to one simply scaled Fabius function at noninteger order. The generic positive-real ordered-endpoint continuous-Banach-input identity underlying this calculation is Lean formalized as `Fabius.fractionalVolterra_add`, and the exact support-truncated Rvachev specialization is exported as `Fabius.rvachevFractionalPrimitive_add`. The transform (718), the tail series (719), and their boundary convergence remain report-local.

## 6.2 Volterra resolvent

For  $\lambda \in \mathbb{C}$ , define

$$R_\lambda(x) := \int_{-1}^{\min(x,1)} e^{\lambda(x-t)} \text{up}(t) \, dt. \quad (721)$$

**Theorem 6.2** (Resolvent series). *The function  $R_\lambda$  is entire in  $\lambda$  and*

$$R_\lambda(x) = \sum_{n=1}^{\infty} \lambda^{n-1} U_n(x). \quad (722)$$

On the support interval,

$$R_\lambda(x) = \sum_{n=1}^{\infty} \lambda^{n-1} A_n F((x+1)/2^n), \quad -1 \leq x \leq 1. \quad (723)$$

It is the unique solution of

$$R'_\lambda(x) = \text{up}(x) + \lambda R_\lambda(x), \quad R_\lambda(-1) = 0. \quad (724)$$

For  $x \geq 1$ ,

$$R_\lambda(x) = e^{\lambda x} M(\lambda). \quad (725)$$

At  $x = 1$ , this yields the scale-value/product identity

$$e^\lambda M(\lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} 2^{n(n-1)/2} F(2^{1-n}). \quad (726)$$

It is an entire generating function for inverse-dyadic Fabius values at a different normalization from the usual moment generating series.

## 7 Laplace and Fourier transforms on the unit interval

### 7.1 Closed transforms of $F$ and the positive half of $\text{up}$

**Theorem 7.1** (One-sided exponential transforms). *For  $s \in \mathbb{C} \setminus \{0\}$ ,*

$$\int_0^1 e^{sx} F(x) \, dx = \frac{e^s - e^{s/2} M(s/2)}{s}, \quad (727)$$

$$\int_0^1 e^{sx} \text{up}(x) \, dx = \frac{e^{s/2} M(s/2) - 1}{s}. \quad (728)$$

The removable values at  $s = 0$  are both  $1/2$ .

*Proof.* Integration by parts gives

$$s \int_0^1 e^{sx} F(x) dx = e^s - \int_0^1 e^{sx} F'(x) dx.$$

With  $t = 2x - 1$ , (676) turns the last integral into  $e^{s/2} M(s/2)$ . The second formula follows from  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$ .  $\square$

Setting  $s = -2\pi i\xi$  gives a one-sided Fourier transform not explicit in the repository's full-line up product:

$$\int_0^1 F(x) e^{-2\pi i\xi x} dx = \frac{e^{-2\pi i\xi} - e^{-\pi i\xi} \widehat{\text{up}}(\xi/2)}{-2\pi i\xi}. \quad (729)$$

The transform of the positive half of up is obtained by replacing the numerator with  $e^{-\pi i\xi} \widehat{\text{up}}(\xi/2) - 1$ .

## 7.2 A Bromwich-compatible primitive representation

Use the conventional one-sided Laplace transform  $\mathcal{L}_-[f](s) = \int_0^\infty e^{-st} f(t) dt$ . Equation (727), with  $s \mapsto -s$ , gives

$$\mathcal{L}_-[F\mathbf{1}_{[0,1]}](s) = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}. \quad (730)$$

For  $0 < x < 1$  and any admissible Bromwich line  $\text{Re } s = c > 0$ , the causal primitive therefore has the representation

$$\mathcal{A}_n(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{sx} s^{-n} \mathcal{L}_-[F\mathbf{1}_{[0,1]}](s) ds. \quad (731)$$

The exact scaled formula  $\mathcal{A}_n(x) = A_n F(x/2^n)$  is thus a nontrivial collapse of a Bromwich integral on the unit interval. At fractional order, the analogous global collapse fails by the periodic obstruction in [appendix 13](#).

# 8 An entire Mellin interpolant and inverse moments

## 8.1 Definition and binomial moment expansion

Define

$$D(z) := \int_0^1 x^z F'(x) dx. \quad (732)$$

Because  $F'$  is flat at zero, the integral converges locally uniformly for every  $z \in \mathbb{C}$ ; hence  $D$  is entire.

**Theorem 8.1** (Entire binomial-moment interpolation). *For all  $z \in \mathbb{C}$ ,*

$$D(z) = \int_{-1}^1 \left( \frac{1+t}{2} \right)^z \text{up}(t) dt = 2^{-z} \sum_{j=0}^{\infty} \binom{z}{2j} c_j. \quad (733)$$

*The series converges locally uniformly on  $\mathbb{C}$ . At nonnegative integers,  $D(n)$  is the usual Fabius-distribution moment and the series terminates.*

*Proof.* The first equality follows from  $t = 2x - 1$  in (676). Expand  $(1+t)^z$  binomially. Odd terms integrate to zero. Although the binomial series has boundary radius one, the moments  $c_j$  decay faster than every power because up is flat at  $|t| = 1$ ; this gives local uniform convergence after integration.  $\square$

For a positive integer  $m$ ,

$$D(-m) = 2^m \sum_{j=0}^{\infty} \binom{m+2j-1}{2j} c_j. \quad (734)$$

Thus every negative inverse moment is represented by a rapidly convergent positive series of rational terms.

## 8.2 Mellin transforms of $F$ and up

Let

$$H(z) := \int_0^1 x^{z-1} F(x) dx. \quad (735)$$

Flatness again makes  $H$  entire. Integration by parts gives

$$\boxed{H(z) = \frac{1 - D(z)}{z}}, \quad (736)$$

where  $z = 0$  is removable. For  $m \geq 1$ ,

$$\int_0^1 x^{-m-1} F(x) dx = \frac{D(-m) - 1}{m}. \quad (737)$$

These genuinely singular-looking integrals are finite because of endpoint flatness.

On the positive half of up, initially for  $\operatorname{Re} z > 0$ ,

$$\int_0^1 x^{z-1} \operatorname{up}(x) dx = \frac{D(z)}{z}. \quad (738)$$

The right side is its meromorphic continuation to  $\mathbb{C}$ , with a single simple pole at  $z = 0$  and residue one. All prospective poles at negative integers are erased by the flatness of up at  $x = 1$  and the fact that its Taylor series at the origin is the constant one.

## 8.3 Every complex moment of the inverse

**Theorem 8.2** (Inverse-moment identity). *For every  $z \in \mathbb{C}$ ,*

$$\boxed{\int_0^1 I(y)^z dy = D(z) = 2^{-z} \sum_{j=0}^{\infty} \binom{z}{2j} c_j}. \quad (739)$$

*In particular, all positive, fractional, negative, and imaginary moments of the inverse are values of one entire function.*

*Proof.* Use  $y = F(x)$ , so  $I(y) = x$  and  $dy = F'(x) dx$ . Convergence for every complex  $z$  follows either from endpoint asymptotics or directly from equality with the entire function  $D$ .  $\square$

**Corollary 8.3** (Two-variable inverse exponential–Mellin transform). *The function*

$$\mathcal{D}(z, s) := \int_0^1 I(y)^z e^{sI(y)} dy \quad (740)$$

*is entire on  $\mathbb{C}^2$ , and*

$$\boxed{\mathcal{D}(z, s) = \int_0^1 x^z e^{sx} F'(x) dx = \sum_{\ell=0}^{\infty} \frac{s^\ell}{\ell!} D(z + \ell)}. \quad (741)$$

In particular, the entire exponential transform of the inverse has the closed product

$$\int_0^1 e^{sI(y)} dy = e^{s/2} M(s/2) = e^{s/2} \prod_{j=1}^{\infty} \frac{\sinh(s/2^{j+1})}{s/2^{j+1}}. \quad (742)$$

Thus its characteristic function is obtained by setting  $s = -2\pi i\xi$ , and every inverse moment is recovered by differentiation at  $s = 0$ .

*Proof.* The substitution  $y = F(x)$  gives the first integral in (741); expansion of  $e^{sx}$  gives the locally uniform series. Setting  $z = 0$  and then  $t = 2x - 1$  yields  $e^{s/2} \int_{-1}^1 e^{st/2} \text{up}(t) dt = e^{s/2} M(s/2)$ .  $\square$

This transform is strictly log-convex on the real axis. More strongly, for every  $\sigma \in \mathbb{R}$  and integer  $k \geq 0$ ,

$$(-1)^k D^{(k)}(\sigma) = \int_0^1 (-\log x)^k x^\sigma F'(x) dx > 0. \quad (743)$$

Thus  $D$  is completely monotone on every right half-line, despite extending entirely across the complex plane.

#### 8.4 Logarithmic moments and harmonic-number series

Differentiating (739) at zero gives

$$\int_0^1 (\log I(y))^r dy = D^{(r)}(0). \quad (744)$$

The first two are

$$\int_0^1 \log I(y) dy = -\log 2 - \frac{1}{2} \sum_{j=1}^{\infty} \frac{c_j}{j}, \quad (745)$$

$$\int_0^1 (\log I(y))^2 dy = (\log 2)^2 + (\log 2) \sum_{j=1}^{\infty} \frac{c_j}{j} + \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j} c_j, \quad (746)$$

where  $H_n$  is the harmonic number. Numerically,

$$\int_0^1 \log I(y) dy = -0.758240000404409\dots, \quad \int_0^1 (\log I(y))^2 dy = 0.719845639798448\dots \quad (747)$$

Higher derivatives can be organized by signed Stirling numbers or complete Bell polynomials applied to  $2^{-z}$  and the binomial series.

### 9 Exact equimeasurability of every derivative

This section contains the strongest new definite-integral theorem in the report.

#### 9.1 Pushforward law

**Theorem 9.1** (Derivative pushforward theorem). *Let  $m \geq 1$ ,  $Q_m = 2^{m(m+1)/2}$ , and let  $\Phi$  be any bounded Borel function. Then*

$$\int_0^1 \Phi\left(\frac{|F^{(m)}(x)|}{Q_m}\right) dx = \int_0^1 \Phi(F(x)) dx. \quad (748)$$

For the signed derivatives,

$$\int_0^1 \Phi\left(\frac{F'(x)}{2}\right) dx = \int_0^1 \Phi(F(x)) dx, \quad (749)$$

$$\int_0^1 \Phi\left(\frac{F^{(m)}(x)}{Q_m}\right) dx = \frac{1}{2} \int_0^1 (\Phi(F(x)) + \Phi(-F(x))) dx, \quad m \geq 2. \quad (750)$$

*Proof.* By (680), the normalized derivative is  $\mathcal{F}(2^m x)$ . As  $x$  traverses  $[0, 1]$ , the argument traverses  $[0, 2^m]$ , which consists of  $2^{m-1}$  blocks of length two. On block  $k$ , (679) gives  $\varepsilon_k \text{up}(t)$ ,  $-1 \leq t \leq 1$ , with Jacobian  $2^{-m}$ . The unsigned block integral is independent of  $k$ , and

$$\frac{1}{2} \int_{-1}^1 \Phi(\text{up}(t)) dt = \int_0^1 \Phi(F(x)) dx$$

by evenness and  $\text{up}(t) = F(1-t)$  for  $0 \leq t \leq 1$ . For  $m = 1$  there is one positive block. For  $m \geq 2$ , the first  $2^{m-1}$  Thue–Morse signs are exactly balanced, giving the signed mixture.  $\square$

In probabilistic notation, if  $U \sim \text{Unif}[0, 1]$ ,

$$\frac{|F^{(m)}(U)|}{Q_m} \stackrel{d}{=} F(U), \quad (751)$$

and for  $m \geq 2$  the sign is an independent fair Rademacher variable at the level of the pushforward distribution.

## 9.2 Distribution function and rearrangements

Since  $F$  is increasing,

$$\lambda \left\{ x \in [0, 1] : \frac{|F^{(m)}(x)|}{Q_m} \leq a \right\} = I(a), \quad 0 \leq a \leq 1. \quad (752)$$

Thus the quantile of the normalized derivative magnitude is exactly  $F$  itself, and its cumulative distribution is the inverse Fabius function. Equivalently, the decreasing rearrangement is

$$\left( \frac{|F^{(m)}|}{Q_m} \right)^*(t) = F(1-t). \quad (753)$$

This identifies every rearrangement-invariant norm, not only power norms.

**Corollary 9.2** (All  $L^p$  norms and moments). *For every real  $p > 0$  and  $m \geq 1$ ,*

$$\int_0^1 |F^{(m)}(x)|^p dx = Q_m^p \int_0^1 F(x)^p dx, \quad \|F^{(m)}\|_p = Q_m \|F\|_p. \quad (754)$$

*In particular,*

$$\int_0^1 |F^{(m)}(x)| dx = \frac{Q_m}{2}. \quad (755)$$

*For  $m \geq 2$ , all odd signed moments vanish, while*

$$\int_0^1 F^{(m)}(x)^{2r} dx = Q_m^{2r} \int_0^1 F(x)^{2r} dx. \quad (756)$$

More generally, (748) gives exact Orlicz modulars  $\int \Psi(|F^{(m)}|/Q_m) = \int \Psi(F)$  and the same equality in Lorentz and every other rearrangement-invariant function space. The oscillatory complexity of high derivatives is therefore measure-theoretically invisible after the single factor  $Q_m$  is removed.

## 10 Inverse derivatives and sharp integrability

### 10.1 An exact $L^p$ transform

**Theorem 10.1** (Sharp inverse-derivative identity). *For every real  $p \leq 1$ ,*

$$\int_0^1 I'(y)^p dy = 2^{1-p} \int_0^1 F(x)^{1-p} dx. \quad (757)$$

*For every  $p > 1$ , the integral diverges. Hence*

$$I' \in L^p(0, 1) \iff 0 < p \leq 1. \quad (758)$$

*In particular  $I \in W^{1,1}(0, 1)$ , but  $I \notin W^{1,p}(0, 1)$  for any  $p > 1$ .*

*Proof.* Use  $y = F(x)$  and (707):

$$\int_0^1 I'(y)^p dy = \int_0^1 F'(x)^{1-p} dx.$$

Apply [theorem 9.1](#) with  $m = 1$  and power  $1 - p \geq 0$ , obtaining  $2^{1-p} \int F^{1-p}$ . If  $p > 1$ , endpoint flatness gives, for every  $N$ ,  $F'(x) \leq C_N x^N$  near zero. Therefore  $F'(x)^{1-p} \geq C_N^{1-p} x^{-N(p-1)}$ ; choosing  $N(p-1) \geq 1$  forces divergence.  $\square$

Writing  $p = -q$  gives, for  $q \geq -1$ ,

$$\int_0^1 I'(y)^{-q} dy = 2^{q+1} \int_0^1 F(x)^{q+1} dx. \quad (759)$$

At  $q = 1$ ,

$$\int_0^1 \frac{dy}{I'(y)} = 4 \int_0^1 F(x)^2 dx \quad (760)$$

$$= 2 \int_{-1}^1 \text{up}(t)^2 dt \quad (761)$$

$$= 2 \int_{\mathbb{R}} |\widehat{\text{up}}(\xi)|^2 d\xi \quad (762)$$

$$= 2 \int_{\mathbb{R}} \prod_{n=0}^{\infty} \text{sinc}^2\left(\frac{\pi\xi}{2^n}\right) d\xi. \quad (763)$$

Numerically this constant is

$$1.617747071935941\dots \quad (764)$$

### 10.2 Entropy-type identities

The power identity

$$\int_0^1 F'(x)^r dx = 2^r \int_0^1 F(x)^r dx, \quad r \geq 0, \quad (765)$$

can be differentiated with respect to  $r$ . At  $r = 1$ ,

$$\int_0^1 F'(x) \log F'(x) dx = \log 2 + 2 \int_0^1 F(x) \log F(x) dx. \quad (766)$$

Higher derivatives in  $r$  relate all logarithmic power moments of the density to those of the distribution function.



### 10.3 All higher inverse derivatives

Define rational differential expressions recursively by

$$R_1(x) = \frac{1}{F'(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{F'(x)}. \quad (767)$$

Then

$$I^{(n)}(y) = R_n(I(y)). \quad (768)$$

The first cases are

$$I''(y) = -\frac{F''(q)}{F'(q)^3}, \quad (769)$$

$$I'''(y) = \frac{3F''(q)^2 - F'''(q)F'(q)}{F'(q)^5}, \quad (770)$$

with  $q = I(y)$ . Substituting (680) expresses every numerator in terms of finitely many values  $\mathcal{F}(2^j q)$ . For example,

$$I''(y) = -\frac{\mathcal{F}(4q)}{\mathcal{F}(2q)^3}. \quad (771)$$

This exact multiscale form is well suited to endpoint asymptotics and to Lean formalization.

**Conjecture 10.2** (Higher inverse-derivative threshold). *For every integer  $n \geq 1$ ,*

$$I^{(n)} \in L^p(0, 1) \iff 0 < p \leq \frac{1}{n}. \quad (772)$$

*The endpoint  $p = 1/n$  is expected to converge because the Lambert- $W$  correction contributes a stretched-exponential factor beyond the borderline  $y^{-1}$  scale.*

The conjecture is a theorem for  $n = 1$  by [theorem 10.1](#). The recursion (767) and the repository's sharp endpoint asymptotics strongly suggest  $I^{(n)}(y) \asymp y^{-n} I(y) / \sqrt{\log(1/y)}$  up to explicit slowly varying and periodic factors, which gives the stated threshold. A rigorous proof requires uniform control of cancellations among the Bell-polynomial terms.

## 11 Nonlinear beta chains and Bell-polynomial integrals

### 11.1 Universal beta integrals

The substitution  $y = F(x)$  immediately yields a large exact family.

**Theorem 11.1** (Beta-chain identity). *If  $\operatorname{Re} a > -1$  and  $\operatorname{Re} b > -1$ , then*

$$\boxed{\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a + 1, b + 1).} \quad (773)$$

*If  $H(t) = F((t + 1)/2) = \frac{1}{2} + \int_0^t \operatorname{up}(u) du$ , then equivalently*

$$\int_{-1}^1 H(t)^a (1 - H(t))^b \operatorname{up}(t) dt = B(a + 1, b + 1). \quad (774)$$

This is useful whenever a complicated combination contains exactly one density factor. For example,

$$\int_0^1 F(x)^n F'(x) dx = \frac{1}{n+1}, \quad \int_0^1 F(x)^n (1-F(x))^m F'(x) dx = \frac{n!m!}{(n+m+1)!}. \quad (775)$$

If  $B_n(x)$  is the Bernoulli polynomial, then its zero mean on  $[0, 1]$  gives the explicit Bernoulli family

$$\int_0^1 B_n(F(x)) F'(x) dx = \int_{-1}^1 B_n(H(t)) \text{up}(t) dt = 0, \quad n \geq 1. \quad (776)$$

This places the Bernoulli-polynomial identities in the same quantile chain as the beta integrals rather than treating them as isolated substitutions.

## 11.2 Bell-polynomial total derivatives

Let  $B_{n,k}$  denote the partial exponential Bell polynomial, and let  $\Phi \in C^n([0, 1])$ . Faà di Bruno's formula gives

$$\frac{d^n}{dx^n} \Phi(F(x)) = \sum_{k=1}^n \Phi^{(k)}(F(x)) B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)). \quad (777)$$

Since every positive derivative of  $F$  vanishes at both endpoints, integration gives the exact nonlinear family

$$\boxed{\sum_{k=1}^n \int_0^1 \Phi^{(k)}(F(x)) B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)) dx = 0, \quad n \geq 2.} \quad (778)$$

For  $n = 1$ , the value is  $\Phi(1) - \Phi(0)$ . The first nontrivial examples are

$$\int_0^1 (\Phi''(F) F'^2 + \Phi'(F) F'') dx = 0, \quad (779)$$

$$\int_0^1 (\Phi'''(F) F'^3 + 3\Phi''(F) F' F'' + \Phi'(F) F''') dx = 0. \quad (780)$$

After substituting (680), these become exact multiscale integral relations among the scales  $2x, 4x, 8x, \dots$  of  $\mathcal{F}$ . Bell and Bernoulli polynomial weights can be inserted for  $\Phi$ , creating a systematic family tied directly to the repository's existing polynomial machinery.

## 12 Cauchy–Stieltjes transforms and correlations

### 12.1 Cauchy transform of up

Define, for  $z \in \mathbb{C} \setminus [-1, 1]$ ,

$$C(z) := \int_{-1}^1 \frac{\text{up}(t)}{z-t} dt. \quad (781)$$

For  $|z| > 1$ ,

$$C(z) = \sum_{j=0}^{\infty} \frac{c_j}{z^{2j+1}}. \quad (782)$$

Let  $X$  have density  $\text{up}$ . The random-series fixed point is

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (783)$$

where  $U$  is uniform on  $[-1, 1]$  and independent of  $X' \stackrel{d}{=} X$ . Conditioning on  $X'$  gives the logarithmic transform equation

$$C(z) = \mathbb{E} \log \left( \frac{2z - X + 1}{2z - X - 1} \right), \quad (784)$$

with the branch normalized at infinity. Differentiation yields

$$C'(z) = 2(C(2z + 1) - C(2z - 1)). \quad (785)$$

Thus the Cauchy transform inherits a functional-differential equation dual to that of up.

**Corollary 12.1** (Stieltjes transform of the inverse). *For  $z \in \mathbb{C} \setminus [0, 1]$ ,*

$$S_I(z) := \int_0^1 \frac{dy}{z - I(y)} = 2C(2z - 1). \quad (786)$$

Consequently, for  $|z| > 1$ ,

$$S_I(z) = \sum_{n=0}^{\infty} \frac{D(n)}{z^{n+1}}, \quad (787)$$

and the transform satisfies

$$S'_I(z) = 4(S_I(2z) - S_I(2z - 1)). \quad (788)$$

*Proof.* Substitute  $y = F(x)$  and then  $t = 2x - 1$ . Formula (787) is the geometric expansion, while (788) follows from (785) and the affine relation in (786).  $\square$

## 12.2 Autocorrelation and a sinc-square integral

Let

$$\mathcal{R}(a) := \int_{\mathbb{R}} \text{up}(x) \text{up}(x + a) dx. \quad (789)$$

Then  $\mathcal{R}$  is even, supported on  $[-2, 2]$ , and

$$\widehat{\mathcal{R}}(\xi) = |\widehat{\text{up}}(\xi)|^2. \quad (790)$$

At the two integer shifts,

$$\mathcal{R}(0) = 2 \int_0^1 F(x)^2 dx, \quad (791)$$

$$\mathcal{R}(1) = \int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} - \int_0^1 F(x)^2 dx. \quad (792)$$

Hence

$$\mathcal{R}(0) + 2\mathcal{R}(1) = 1. \quad (793)$$

The Parseval term in (762) is  $2\mathcal{R}(0)$ ; together with (793), it may also be written as  $2 - 4\mathcal{R}(1)$ .

## 13 The exact obstruction to fractional self-differential scaling

The integer orbit (686) tempts one to write  $D^\alpha \mathcal{F}(x) \stackrel{?}{=} 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x)$  for noninteger  $\alpha$ . The temptation is extremely strong numerically because the relevant discrepancy is only a few parts in  $10^6$ . It is nevertheless false for standard fractional differentiation.

### 13.1 Laplace transform of the global signed extension

For  $s > 0$ , define

$$L(s) := \int_0^\infty e^{-sx} \mathcal{F}(x) dx. \quad (794)$$

The block decomposition gives

$$L(s) = e^{-s} M(s) \prod_{j=1}^{\infty} (1 - e^{-2^j s}). \quad (795)$$

Taking Laplace transforms in  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$  gives the exact dilation equation

$$L(s/2) = sL(s). \quad (796)$$

Put  $s = 2^t$ . The general positive solution of (796) has the form

$$L(2^t) = 2^{-t(t+1)/2} P(t), \quad P(t+1) = P(t). \quad (797)$$

The repository's Mellin analysis of the corresponding dyadic periodization computes the nonzero Fourier modes of  $\log P$ ; in the normalization used there they are proportional to

$$-\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log 2}, \quad \chi_k = \frac{2\pi i k}{\log 2}, \quad k \neq 0. \quad (798)$$

In particular, the first mode is nonzero, so  $P$  has minimal period one.

### 13.2 Fractional defect theorem

For  $\alpha \in \mathbb{R}$ , define the Laplace fractional operator  $\mathbf{D}^\alpha$  on  $\mathcal{F}$  by

$$\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s) = s^\alpha L(s), \quad (799)$$

using the principal positive power for  $s > 0$ . For positive  $\alpha$  this agrees with the Riemann–Liouville derivative because all initial data vanish; for negative  $\alpha$  it is the causal fractional integral.

**Theorem 13.1** (No noninteger exact dilation collapse). *Let  $\alpha \in \mathbb{R}$ , and define*

$$(T_\alpha \mathcal{F})(x) := 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x). \quad (800)$$

*Then*

$$\frac{\mathcal{L}[T_\alpha \mathcal{F}](s)}{\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s)} = \frac{P(\log_2 s - \alpha)}{P(\log_2 s)}. \quad (801)$$

*Consequently*

$$T_\alpha \mathcal{F} = \mathbf{D}^\alpha \mathcal{F} \iff \alpha \in \mathbb{Z}. \quad (802)$$

*Proof.* Scaling in the Laplace transform gives

$$\mathcal{L}[T_\alpha \mathcal{F}](s) = 2^{\alpha(\alpha-1)/2} L(s/2^\alpha).$$

Substitute (797); all quadratic and power factors cancel, leaving (801). Equality for all  $s$  is equivalent to  $P(t - \alpha) = P(t)$  for all  $t$ . Because  $P$  has minimal period one, this occurs exactly when  $\alpha \in \mathbb{Z}$ .  $\square$

The defect is exactly log-periodic in  $s$ , not an asymptotic error. For  $\alpha = 1/2$ , numerical evaluation gives oscillations of roughly  $-4.251 \times 10^{-6}$  to  $4.251 \times 10^{-6}$ . This explains why a naive numerical experiment can falsely suggest a perfect fractional extension.

## 14 Numerical experiments

### 14.1 Method

The accompanying script reconstructs  $\text{up}$  from (682) on a period-four FFT grid with  $2^{18}$  points. Because the exact support is  $[-1, 1]$ , the periodic copies do not overlap. It then forms  $F(x) = \text{up}(x - 1)$  on  $[0, 1]$ , evaluates moments by Simpson quadrature, and uses monotone interpolation only for the inverse. The code is fully commented and included in the archive.

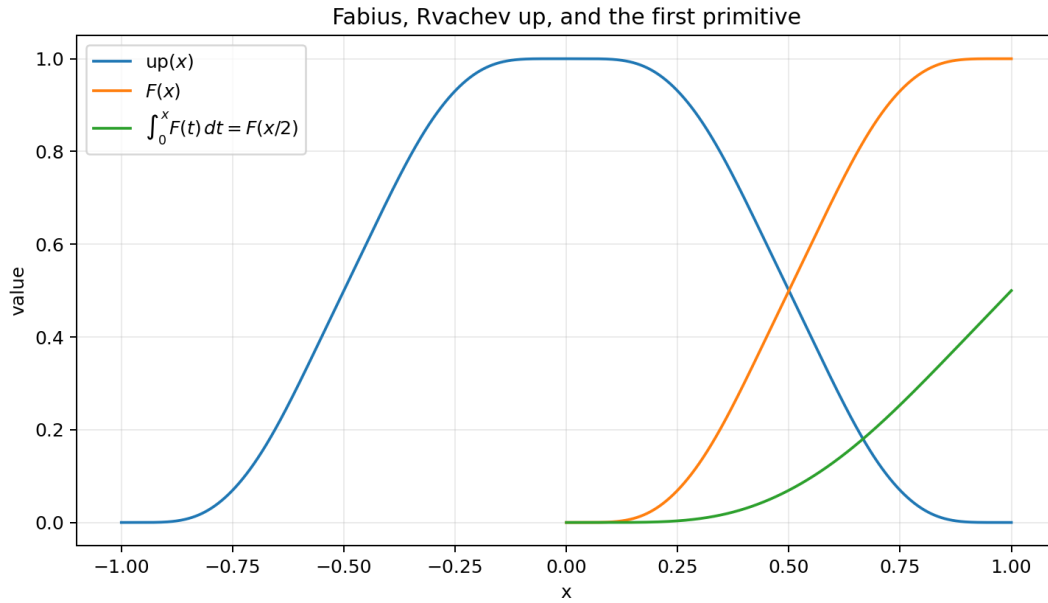


Figure 1: The Fabius function, Rvachev’s up-function, and the exact first primitive  $\int_0^x F(t) dt = F(x/2)$ .

### 14.2 Representative checks

| Identity                                 | Test range               | observed discrepancy        |
|------------------------------------------|--------------------------|-----------------------------|
| Integer primitive ladder for $F$         | orders 1 through 5       | below $4 \times 10^{-11}$   |
| Integer primitive ladder for $\text{up}$ | orders 1 through 5       | below $8 \times 10^{-11}$   |
| Complex-power series, $\alpha = -1.5$    | 100 terms at $x = 0.73$  | $1.6 \times 10^{-7}$        |
| Complex-power series, $\alpha = 0.5$     | 100 terms at $x = 0.73$  | $1.3 \times 10^{-10}$       |
| Laplace products                         | four real/complex points | below $3 \times 10^{-15}$   |
| Mellin series, $z = 1 + 2i$              | 321 moments              | $2.4 \times 10^{-16}$       |
| Derivative pushforward moments           | orders 1 through 6       | below $2.1 \times 10^{-12}$ |
| Inverse primitive                        | five $y$ -values         | about $3.7 \times 10^{-8}$  |
| Inverse exponential/Stieltjes transforms | four complex tests       | below $3.5 \times 10^{-8}$  |

Table 16: Numerical verification. The larger inverse-primitive discrepancy comes from linear inversion of a finite grid, not from the closed formula.

The computed nonlinear constants include

$$\int_0^1 F(x)^2 dx = 0.404436767983985 \dots, \quad (803)$$

$$\int_0^1 F(x)^3 dx = 0.356655151975978 \dots, \quad (804)$$

$$\int_{-1}^1 \text{up}(x)^2 dx = 0.808873535967971 \dots \quad (805)$$

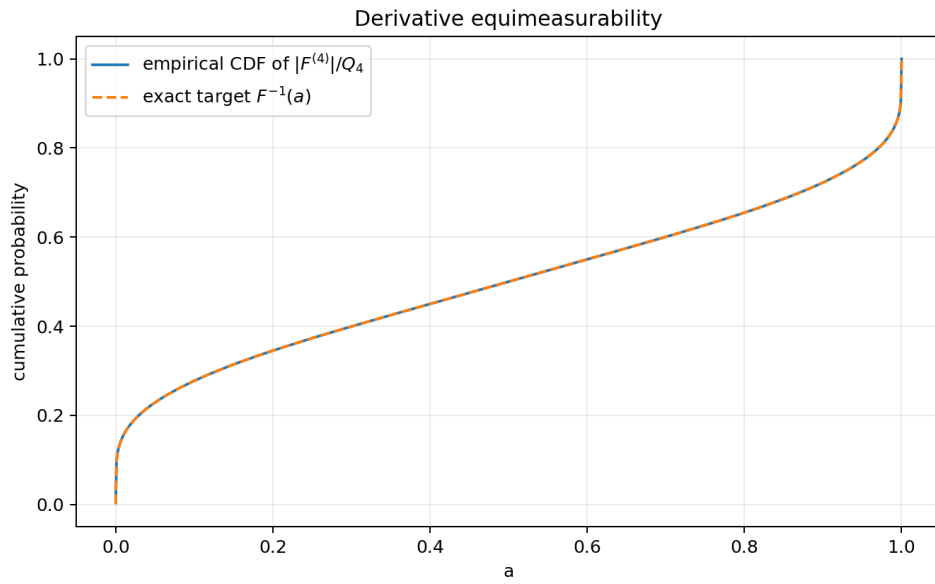


Figure 2: The empirical CDF of  $|F^{(4)}|/Q_4$  lies on the exact target  $I = F^{-1}$ , illustrating [theorem 9.1](#).

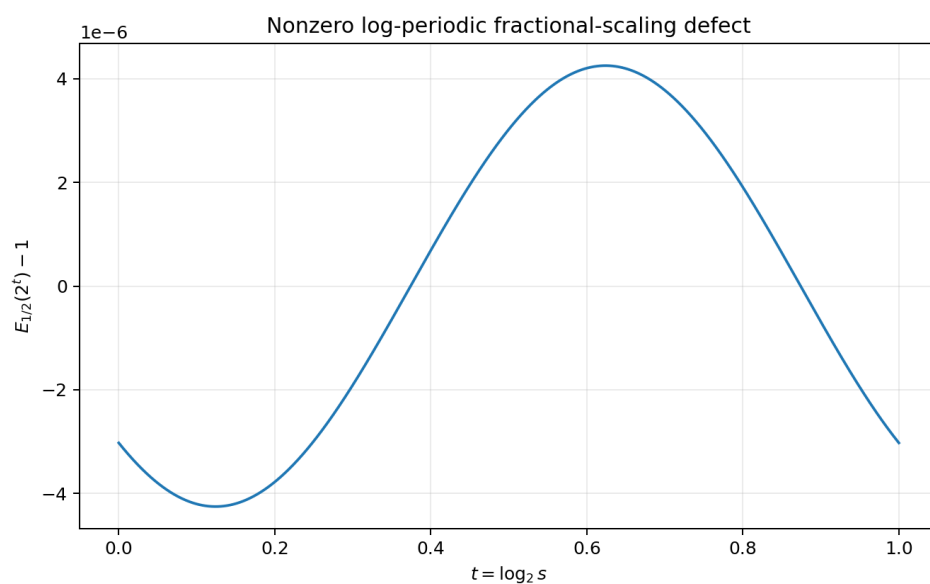


Figure 3: One logarithmic period of the nonzero defect  $P(t - 1/2)/P(t) - 1$ . Its parts-per-million size makes the false fractional scaling law numerically seductive.

## 15 Further conjectures and research directions

### 15.1 Higher inverse derivatives

The sharp threshold in [conjecture 10.2](#) is the most immediate analytic problem. A proof should combine:

1. the exact recursion ([767](#));
2. the all-orders Lambert- $W$  expansion for  $F$  and  $F'$  near zero;
3. a noncancellation theorem for the leading inverse Bell polynomial;
4. symmetry to transfer the result to  $y \uparrow 1$ .

A successful proof would determine the precise Lorentz endpoint space and may reveal a log-periodic refinement at  $p = 1/n$ .

### 15.2 Growth and zeros of the entire inverse-moment function

**Conjecture 15.1** (Order-two, log-periodically modulated growth). *The entire function  $D(z) = \int_0^1 I(y)^z dy$  has order two. Along the negative real axis,*

$$\log D(-r) = \frac{\log 2}{2} r^2 + O(r \log r), \quad (806)$$

*with a nonconstant periodic correction in the natural dyadic saddle phase. Its complex zero counting function should inherit the same order and a dyadic modulation.*

This should be approachable by applying the repository's endpoint saddle machinery to the Mellin integral defining  $D(-r)$ . The exact series ([733](#)) gives an independent Newton/binomial route and a stringent numerical cross-check.

### 15.3 Arithmetic nature of singular integrals

The constants

$$D(-m), \quad H(-m) = \frac{D(-m) - 1}{m}, \quad D^{(r)}(0) \quad (807)$$

are explicit convergent series of rational Rvachev moments, but their arithmetic nature is unknown. It is reasonable to ask whether any are rational, algebraic, or provably transcendental. Their dyadic product origin suggests connections with Mahler functions and special values of  $q$ -products, but the entire interpolation does not appear to be a standard Mahler function in  $z$ .

### 15.4 Fractional primitive linear independence

**Conjecture 15.2** (No finite scaled-Fabius representation at noninteger order). *For  $\beta \notin \mathbb{N}$ , the causal fractional primitive  $U_\beta$  in ([717](#)) cannot be represented on a nonempty interval as a finite linear combination of affine-scaled copies of  $F$ ,  $1 - F$ , or up with algebraic coefficient functions.*

[Theorem 13.1](#) proves the simplest one-term global version. A finite combination would imply a finite exponential-polynomial relation among shifted periodic factors in Laplace space; nonvanishing of all Fourier modes is a promising route to linear independence.



## 15.5 Dyadic autocorrelations and $q$ -arithmetic

The values  $\mathcal{R}(k/2^n)$  mix two shifted Fabius pieces. A useful program is to:

- derive a finite-scale recursion for dyadic autocorrelations from the random fixed point (783);
- express the result through sinc-square products and Thue–Morse blocks;
- determine whether any nontrivial dyadic shifts have rational or controlled 2-adic values;
- connect the recursion to the exact  $q$ -binomial arrays already formalized.

The integer relation  $\mathcal{R}(0) + 2\mathcal{R}(1) = 1$  is the first boundary datum.

## 15.6 Formalization roadmap

At compiled checkpoint e109088ed, the generic affine, base-point-change, and zero-tail polynomial calculus in equations (672) to (675) is no longer a roadmap item, and the signed-global and bounded- $F$  natural-monomial coefficients are formalized separately. The remaining Lean-ready work is more narrowly:

1. add narrowly packaged specializations for the report's  $1 - F$ , shifted-up, and exterior piecewise moment-coefficient formulas, without reproving the generic Volterra layer or the already formalized signed/bounded  $F$  cases;
2. the inverse primitive, by change of variables and integration by parts;
3. the pushforward theorem, by a finite block partition and Thue–Morse balance;
4. the inverse  $L^p$  identity and divergence threshold;
5. the beta-chain and Bell-polynomial integral identities.

The Mellin-entire extension and fractional obstruction require more analytic infrastructure but have short conceptual proofs once local uniform convergence and the periodic Laplace factor are available.

## 16 Conclusion

The self-differential identity does considerably more than generate derivatives. With the correct zero-initial normalization it creates an exact two-sided integer calculus, and this calculus survives multiplication by polynomials in a finite form and by complex powers in a convergent infinite form. The inverse function is equally rigid: its first primitive is closed in terms of one scaled Fabius value, all integer powers have finite primitives, and every complex inverse moment lies on one entire binomial-moment interpolant.

The derivative pushforward theorem reveals a hidden conservation law: after removal of the triangular factor  $Q_m$ , every derivative magnitude has exactly the same value distribution. This collapses infinitely many norm computations to one base integral and gives the sharp Sobolev failure of the inverse. Finally, the tiny nonconstant log-periodic factor explains both the near-validity and the exact failure of naive fractional self-similarity.

These results open a coherent frontier joining antiderivatives, quantiles, integral transforms, dyadic arithmetic, Thue–Morse signs, Bell polynomials, and Lambert- $W$  asymptotics. The remaining formal targets are the report-specific and analytic statements not covered by the compiled generic Volterra layer.

## A Corpus map used in the audit

The recursive repository material was grouped as follows.

| Corpus component                         | Role in the audit                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabius_Function_and_Rvachev_Up           | Current foundational exposition: definitions, derivative orbit, moments, probability, Fourier product, exact arithmetic, and asymptotics.                               |
| non-formalized-research-frontiers        | Canonical merged frontier synthesis: dyadic formula web, asymptotic completion, Thue–Morse rates, periodic Laplace factors, and research-status boundaries.             |
| Inverse frontier report                  | Inverse germs, reversion, endpoint inversion, and Lambert– $W$ corrections; checked to separate the new inverse integral identities from existing local inverse series. |
| Newton–Rvachev frontier report           | Generalized scale sequences, Newton products, Mellin endpoint poles, and a recursive 67-source ledger.                                                                  |
| Fourier-decay draft family               | Sharp decay of the infinite sinc product and periodic corrections; used to avoid restating spectral asymptotics as new integral results.                                |
| Thue–Morse and dyadic arithmetic reports | Exact sign blocks, $q$ -binomial identities, interpolation, and rational dyadic values; used in the block proof and the rational-integral corollary.                    |
| Repeated-integration reports and archive | Approximation of up by iterative integration; distinguished from the exact antiderivatives of the limiting function developed here.                                     |
| Papers and transcriptions                | Fabius, Rvachev, Arias de Reyna, Haugland, and related primary/background sources.                                                                                      |
| Lean walkthrough and glossary            | Notation, formal dependencies, and theorem naming.                                                                                                                      |

Searches for *antiderivative*, *equimeasurable*, inverse  $L^p$  laws, and the specific Mellin identity (739) returned no developed counterpart in the current canonical volumes or live inverse report. Existing moment and transform formulae have been credited as inputs wherever they are reused.

## B Selected formulas for direct use

For convenience, the main computational identities are collected here.

$$\begin{aligned}
\mathcal{I}^n F(x) &= 2^{n(n-1)/2} F(x/2^n), \\
\mathcal{I}^n \text{up}(x) &= 2^{n(n-1)/2} F((x+1)/2^n), \quad -1 \leq x \leq 1, \\
\mathcal{I}^n(PF)(x) &= \sum_{k=0}^{\deg P} (-1)^k \binom{n+k-1}{k} P^{(k)}(x) 2^{(n+k)(n+k-1)/2} F(x/2^{n+k}), \\
\int_0^y I(v) \, dv &= yI(y) - F(I(y)/2), \\
\int_0^1 I(y)^z \, dy &= 2^{-z} \sum_{j \geq 0} \binom{z}{2j} c_j, \\
\int_0^1 e^{sI(y)} \, dy &= e^{s/2} M(s/2), \\
\int_0^1 \frac{dy}{z - I(y)} &= 2C(2z - 1),
\end{aligned}$$

$$\begin{aligned}\int_0^1 \Phi(|F^{(m)}|/Q_m) \, dx &= \int_0^1 \Phi(F) \, dx, \\ \int_0^1 I'(y)^p \, dy &= 2^{1-p} \int_0^1 F(x)^{1-p} \, dx \quad (p \leq 1), \\ \int_0^1 e^{sx} F(x) \, dx &= \frac{e^s - e^{s/2} M(s/2)}{s}, \\ \mathcal{L}[U_\beta(\cdot - 1)](s) &= s^{-\beta} e^{-s} M(s).\end{aligned}$$

## C Reproducibility files

The archive accompanying this report contains:

- `fabius_integral_frontiers.tex` — this source;
- `fabius_integral_frontiers.pdf` — the rendered report;
- `numerical_experiments.py` — commented FFT and quadrature code;
- `numerical_results.txt` — machine-generated numerical output;
- three PNG figures used in the report.

The script fixes no random seed because it uses deterministic quadrature and Fourier inversion rather than Monte Carlo sampling.

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## Part VII

# Integral Transforms and Fractional Calculus for the Fabius and Rvachev Functions

## Abstract

Let  $F$  be the bounded Fabius distribution function, let  $\mathcal{F}$  be Rvachev's compactly supported up-function, let  $\mathcal{F}$  be the signed Thue–Morse continuation, and let  $G = F^{-1}$  on  $[0, 1]$ . This report audits the recursive LaTeX documentation tree of the `ProveIt` Fabius project and develops an integral and fractional-calculus layer that is not systematically present in the audited corpus. The novelty claims are deliberately relative to that corpus, not claims of exhaustive worldwide priority.

The ordinary dilation equation yields complete  $n$ -fold primitive and derivative ladders. For example,

$$I_{0+}^n F(x) = 2^{\binom{n}{2}} F(x/2^n), \quad 0 \leq x \leq 1,$$

and the same identity is global for  $\mathcal{F}$ . Polynomial and analytic weights are handled by finite or convergent operator expansions. A nonlinear change-of-variables theorem converts fractional integrals of  $h(F)$  into integrals involving the inverse  $G$ , and conversely. It includes the explicit primitive

$$\int G(y) dy = yG(y) - F\left(\frac{G(y)}{2}\right) + C$$

and exact formulas for fractional integrals and derivatives of  $G$  that avoid differentiating its singular endpoint asymptotics.

The probability model  $Y = \sum_{j \geq 1} 2^{-j} V_j$ ,  $V_j \sim \text{Unif}[0, 1]$ , and  $Z = 2Y - 1$  organizes the definite integrals. Its complex moment  $\mu(s) = \mathbb{E}(Y^s)$  gives Mellin transforms, repeated-primitive endpoint values, logarithmic moments, and the exact fractional dyadic lattice

$$(I_{0+}^\beta F)(2^{-n}) = 2^{-n\beta - \binom{n}{2}} \frac{\mu(n + \beta)}{\Gamma(n + \beta + 1)}.$$

The integer moments follow from Bernoulli cumulants and complete Bell polynomials. Laplace, Fourier, Mellin, Stieltjes, Cauchy, autocorrelation, and fractional Sobolev transforms are developed in a common normalization.

The main frontier theorem introduces the continuous dyadic Appell family

$$(\mathfrak{A}_q \mathcal{F})(x) = 2^{q(q-1)/2} \mathcal{F}(2^{-q}x), \quad q \in \mathbb{R}.$$

It interpolates all integer primitives and derivatives but is not the Riemann–Liouville semigroup at noninteger order. Its exact Laplace-space defect is

$$\frac{\mathcal{L}\{\mathfrak{A}_q \mathcal{F}\}(s)}{s^{-q} \mathcal{L}\{\mathcal{F}\}(s)} = \exp(P(\log_2 s + q) - P(\log_2 s)),$$

where  $P$  is the same one-periodic Gamma–zeta correction that occurs in the audited Lambert– $W$  endpoint theory. Since every nonzero Fourier coefficient of its centered part is nonzero, the defect vanishes identically exactly for  $q \in \mathbb{Z}$ . Thus integer calculus is the rigid zero-defect sublattice of a natural continuous dyadic interpolation.

Finally, the endpoint quadratic law implies a Gevrey-2, zero-radius Taylor barrier for the Mellin transform of the inverse, and moment expansions give convergent exterior series for the Riesz fractional Laplacian and potential of  $\mathcal{F}$ . All numerical evidence is generated by the commented Python program distributed with the report.

**Proof-status convention.** Statements marked **proved here** are proved in this report from definitions and standard analysis. Statements marked **derived from audited inputs** combine such arguments with a theorem already present in the audited repository, chiefly the exact periodic

negative-Laplace decomposition or the endpoint quadratic law. Decimal values and plots are **numerically checked**. Conjectures are marked **conjectural/open**. Since the audit snapshot, Lean has formalized the natural-order normalized Volterra kernel, the signed-global and bounded Fabius primitive ladders, all natural-monomial weights, and the support-free quantile pushforward with its Banach-valued substitution corollary. Later modules also formalize the positive-real stopped-power formulas, increasing-affine covariance, ordinary-derivative order raising, and the three single-step Fabius–Rvachev shifts on their explicit endpoint domains. They do not imply the nonlinear inverse transmutation, complex-order or Mellin continuation, fractional derivatives, iterated/lattice shifts, an endpoint-up wrapper, or named piecewise up-tail results developed here; those remain frontier claims.

# 1 Corpus audit, scope, and novelty discipline

## 1.1 Recursive source boundary

The source boundary was the public recursive tree under `ProveIt/Analysis/FabiusFunction/docs` as accessed on 27 August 2026. The tree contains living synthesis documents, a consolidated non-formalized-frontier volume, active frontier drafts, historical snapshots, and corrected or transcribed source papers. A contemporaneous recursive inventory in the tree records 67 distinct `.tex` paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (twelve of them Fourier-decay dossiers), and seven transcribed or corrected papers. Historical copies were read as lineages: repetition across a frozen draft and its successor does not count as independent prior art.

The audit treated as established repository material, and therefore did not claim as new:

- the separation of the bounded  $F$ , the compactly supported up, and the signed global continuation  $\mathcal{F}$ ;
- exact dyadic values, moment formulas, Thue–Morse finite sums, half-base  $q$ -Pochhammer and Gaussian-binomial identities;
- repeated-integration splines and numerical integration experiments already occurring in the broad frontier notebooks;
- the infinite sinc product, its integer zeros, spectral zeta/determinant material, half-integer identities, Fourier-decay dossiers, and Pascal-weighted generalizations;
- the exact negative-Laplace decomposition, its nonconstant log-periodic Gamma–zeta term, the lower Lambert- $W_{-1}$  saddle, all-order Bell-polynomial expansions, and inverse endpoint asymptotics;
- Lagrange and Richardson extrapolation systems already developed in the repository.

Targeted searches were then made for `fractional calculus`, `Riemann–Liouville`, `Caputo`, general fractional antiderivatives, inverse fractional transmutation, and formulas equivalent to the continuous-order dyadic defect below. No systematic counterpart was found in the living/current lineage. The present report is therefore designed to be complementary: old inputs are recalled briefly, while proofs and frontier claims concentrate on integral calculus.

## 1.2 Headline additions relative to the audited tree

The following package appears to be new relative to the audited tree:

1. a complete ordinary and fractional weighted-antiderivative calculus based on the kernels  $I_{0+}^\alpha F$  and  $I_{-1+}^\alpha$  up;
2. nonlinear Riemann–Liouville transmutation identities between  $F$  and  $G = F^{-1}$ ;
3. exact fractional dyadic values in terms of the analytic moment  $\mu(n + \beta)$ ;
4. a continuous dyadic Appell interpolation of all integer derivatives and primitives;
5. the exact Gamma–zeta multiplier measuring its failure to equal the Riemann–Liouville semi-group, together with integer-order rigidity;
6. exterior moment series for the Riesz fractional Laplacian and Riesz potential of up;
7. a Gevrey-2, zero-radius Taylor theorem for the inverse Mellin transform at its natural boundary;
8. an inverse-derivative  $L^r$  threshold and several mixed  $F$ – $G$  integral bridges.

The word “new” below always means new relative to the audited repository and apparently nonstandard. A worldwide priority assertion would require a separate bibliographic search far beyond the source-comparison purpose of this report.

## 2 Four equivalent languages and normalization

### 2.1 The three functions

**Definition 2.1** (Bounded Fabius function). Following the classical and project normalizations [1, 3, 12], the bounded Fabius function is the  $C^\infty$  map  $F : \mathbb{R} \rightarrow [0, 1]$  satisfying

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (808)$$

$$F(1 - x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (809)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (810)$$

Its inverse on  $[0, 1]$  is denoted by

$$G = F^{-1} : [0, 1] \longrightarrow [0, 1].$$

**Definition 2.2** (Rvachev up-function). The compactly supported even bump is

$$\text{up}(x) = F(1 - |x|) = \begin{cases} F(x + 1), & x \leq 0, \\ F(1 - x), & x > 0. \end{cases} \quad (811)$$

Thus  $\text{supp up} = [-1, 1]$ ,  $\text{up}(0) = 1$ , and

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1. \quad (812)$$

**Definition 2.3** (Signed Thue–Morse continuation). Let  $\varepsilon_n = (-1)^{s_2(n)}$ , where  $s_2(n)$  is the binary digit sum. Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (813)$$

The sum is locally finite,  $\mathcal{F} = F$  on  $(-\infty, 1]$ , and the audited global equation is

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad x \in \mathbb{R}. \quad (814)$$

The bounded and signed objects must not be interchanged to the right of 1. The global object is ideal for pure dilation calculus; the bounded object is ideal for probability, Mellin transforms, and inverse-function questions.

### 2.2 Probability model and moments

Let  $V_1, V_2, \dots$  be independent uniform random variables on  $[0, 1]$  and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad Z = 2Y - 1. \quad (815)$$

Then  $F$  is the CDF of  $Y$ , while  $\text{up}$  is the density of  $Z$ ; this random-series model is classical in the subject [4, 5, 13]. Consequently

$$F'(x) = 2 \text{up}(2x - 1), \quad \int_{-1}^1 \text{up}(x) dx = 1. \quad (816)$$

We use

$$\mu(s) = \mathbb{E}(Y^s) = \int_0^1 x^s F'(x) dx, \quad m_n = \mathbb{E}(Z^n) = \int_{-1}^1 x^n \text{up}(x) dx. \quad (817)$$

Because  $F'$  is flat at both endpoints,  $\mu(s)$  is entire in  $s$ . The symmetry  $Y \stackrel{d}{=} 1 - Y$  and  $Z \stackrel{d}{=} -Z$  will repeatedly remove odd terms.

### 2.3 Integral conventions

We use the standard left-sided Riemann–Liouville and Caputo conventions [7, 8, 10]. For  $a \in \mathbb{R}$  and  $\operatorname{Re} \alpha > 0$ , the left Riemann–Liouville integral is

$$(I_{a+}^{\alpha} f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t) dt. \quad (818)$$

We abbreviate

$$\mathcal{J}_{\alpha}(x) = (I_{0+}^{\alpha} F)(x), \quad \mathcal{K}_{\alpha}(x) = (I_{-1+}^{\alpha} \operatorname{up})(x). \quad (819)$$

The order-zero convention is  $I^0 f = f$ . For  $n-1 < \alpha < n$ , the left Riemann–Liouville and Caputo derivatives are

$$D_{a+}^{\alpha} f = \frac{d^n}{dx^n} I_{a+}^{n-\alpha} f, \quad (820)$$

$${}^C D_{a+}^{\alpha} f = I_{a+}^{n-\alpha} f^{(n)}. \quad (821)$$

If  $f^{(k)}(a) = 0$  for  $0 \leq k < n$ , they coincide. This applies to  $F$  at 0 and to  $\operatorname{up}$  at  $-1$  for every finite order.

Our Fourier convention is

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-itx} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t) e^{itx} dt. \quad (822)$$

Here  $\operatorname{sinc} z = \sin z/z$  with  $\operatorname{sinc} 0 = 1$ .

## 3 Ordinary primitives and the full integer-order ladder

### 3.1 The first primitive and its totalization

**Theorem 3.1** (First primitive of the bounded function). *For every real  $x$ ,*

$$\int_0^x F(t) dt = \begin{cases} 0, & x \leq 0, \\ F(x/2), & 0 \leq x \leq 1, \\ x - \frac{1}{2}, & x \geq 1. \end{cases} \quad (823)$$

The middle formula is [proved here](#).

*Proof.* For  $0 \leq x \leq 1$ , differentiation and [equation \(810\)](#) give

$$\frac{d}{dx} F(x/2) = \frac{1}{2} F'(x/2) = F(x),$$

and both sides vanish at  $x = 0$ . Symmetry gives  $\int_0^1 F = 1/2$ . Since  $F = 1$  to the right of 1, the final branch is  $1/2 + x - 1$ .  $\square$

**Corollary 3.2** (Complement and reflected primitives). *For  $0 \leq x \leq 1$ ,*

$$\int_0^x (1 - F(t)) dt = x - F(x/2), \quad (824)$$

$$\int_0^x F(1-t) dt = x - F(x/2), \quad (825)$$

$$\int_0^x \operatorname{up}(t) dt = x - F(x/2). \quad (826)$$



### 3.2 All integer primitives

**Theorem 3.3** (Dyadic primitive hierarchy). *For every integer  $n \geq 0$ ,*

$$(I_{0+}^n F)(x) = 2^{\binom{n}{2}} F(x/2^n), \quad 0 \leq x \leq 1. \quad (827)$$

*For the signed continuation, the same formula holds globally:*

$$(I_{0+}^n \mathcal{F})(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n), \quad x \in \mathbb{R}, \quad (828)$$

*where the anchored iterated integral is understood for  $x < 0$  with its usual orientation. Both statements are [proved here](#).*

*Proof.* The case  $n = 0$  is tautological and  $n = 1$  is [theorem 3.1](#). If  $H_n(x) = 2^{\binom{n}{2}} F(x/2^n)$ , then, while  $0 \leq x \leq 1$ ,

$$H'_n(x) = 2^{\binom{n}{2}-n} F'(x/2^n) = 2^{\binom{n}{2}-n+1} F(x/2^{n-1}) = H_{n-1}(x).$$

The endpoint value is  $H_n(0) = 0$ . For  $\mathcal{F}$ , the same calculation uses the global dilation equation [equation \(814\)](#) without a domain restriction.  $\square$

**Corollary 3.4** (Endpoint primitive values and dyadic values). *For  $n \geq 1$ ,*

$$(I_{0+}^n F)(1) = \frac{\mu(n)}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \quad (829)$$

*Thus*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{\mu(n)}{n!}. \quad (830)$$

*Proof.* The kernel formula gives

$$(I_{0+}^n F)(1) = \frac{1}{(n-1)!} \int_0^1 (1-t)^{n-1} F(t) dt.$$

By symmetry and integration by parts this equals  $\mu(n)/n!$ . The second equality is [equation \(827\)](#) at  $x = 1$ .  $\square$

### 3.3 Integer derivatives

**Theorem 3.5** (Derivative dilation ladder). *For every integer  $n \geq 0$ ,*

$$\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x), \quad x \in \mathbb{R}. \quad (831)$$

*Consequently, whenever  $0 \leq x \leq 2^{-n}$ ,*

$$F^{(n)}(x) = 2^{\binom{n+1}{2}} F(2^n x). \quad (832)$$

*This is an audited input, restated to pair it with [theorem 3.3](#).*

**Corollary 3.6** (Local derivative moments). *For  $n \geq 0$  and  $\operatorname{Re} p > -1$ ,*

$$\int_0^{2^{-n}} x^p F^{(n)}(x) dx = 2^{\binom{n}{2}-np} \frac{1 - \mu(p+1)}{p+1}. \quad (833)$$

*The right side extends holomorphically in  $p$  after removing the apparent singularity.*

*Proof.* Use [equation \(832\)](#), put  $u = 2^n x$ , and use the Mellin identity proved in [theorem 6.1](#) below.  $\square$

### 3.4 The up-function primitive hierarchy

**Theorem 3.7** (CDF and repeated primitives of up). *For every  $x \in \mathbb{R}$ ,*

$$\int_{-\infty}^x \text{up}(t) dt = F\left(\frac{x+1}{2}\right). \quad (834)$$

*For each integer  $n \geq 1$ ,*

$$(I_{-1+}^n \text{up})(x) = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), \quad -1 \leq x \leq 2^{n-1} - 1. \quad (835)$$

*In particular, it holds throughout the support interval  $[-1, 1]$ . These identities are [proved here](#).*

*Proof.* Differentiate  $F((x+1)/2)$  and use the unified identity  $F'(u) = 2 \text{up}(2u-1)$ . The function vanishes at  $x = -1$ . Repeated differentiation of the right side in [equation \(835\)](#) gives the preceding order whenever  $(x+1)/2^n \leq 1/2$ ; the base case is [equation \(834\)](#).  $\square$

**Proposition 3.8** (Polynomial continuation beyond the support). *For  $n \geq 1$  and  $x \geq 1$ ,*

$$(I_{-1+}^n \text{up})(x) = \frac{1}{(n-1)!} \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2j} m_{2j} x^{n-1-2j}. \quad (836)$$

*Thus every repeated primitive of the compact bump becomes an explicit moment polynomial once the integration point has passed the support. The statement is [proved here](#).*

*Proof.* For  $x \geq 1$  the kernel representation is

$$(I_{-1+}^n \text{up})(x) = \frac{1}{(n-1)!} \int_{-1}^1 (x-t)^{n-1} \text{up}(t) dt.$$

Expand the polynomial and use  $m_{2j+1} = 0$ .  $\square$

## 4 Weighted antiderivatives and analytic weights

### 4.1 Monomial weights: a finite closed form

**Theorem 4.1** (Weighted Riemann–Liouville expansion). *Let  $p \in \mathbb{N}_0$  and  $\text{Re } \alpha > 0$ . Then*

$$I_{0+}^\alpha [t^p F(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}(x), \quad (837)$$

*where  $(\alpha)_k = \Gamma(\alpha+k)/\Gamma(\alpha)$ . This is [proved here](#).*

*Proof.* Inside the defining integral, expand

$$t^p = (x - (x-t))^p = \sum_{k=0}^p (-1)^k \binom{p}{k} x^{p-k} (x-t)^k.$$

The  $k$ th term changes the kernel order from  $\alpha$  to  $\alpha+k$  and contributes  $\Gamma(\alpha+k)/\Gamma(\alpha) = (\alpha)_k$ .  $\square$

**Corollary 4.2** (Closed  $n$ th antiderivative of  $x^p F(x)$ ). *For integers  $n \geq 1$ ,  $p \geq 0$ , and  $0 \leq x \leq 1$ ,*

$$I_{0+}^n [t^p F(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (n)_k 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}). \quad (838)$$

For  $n = 1$ , an ordinary indefinite antiderivative is

$$\int x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C \quad (839)$$

throughout  $0 \leq x \leq 1$ . Replacing  $F$  by  $\mathcal{F}$  makes both formulas global.

**Example 4.3.** The first cases are

$$\begin{aligned} \int F(x) dx &= F(x/2) + C, \\ \int xF(x) dx &= xF(x/2) - 2F(x/4) + C, \\ \int x^2 F(x) dx &= x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \\ \int x^3 F(x) dx &= x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16) + C. \end{aligned}$$

The rapidly growing coefficients are offset by the extraordinary flatness of the dyadically contracted  $F$  terms.

## 4.2 Complex powers and dyadic-shell series

For noninteger powers, a finite elementary combination is not expected, but two exact representations remain available.

**Proposition 4.4** (Binomial fractional-weight series). *Let  $\operatorname{Re} \alpha > 0$ ,  $\operatorname{Re} p > -1$ , and  $x > 0$ . Then*

$$I_{0+}^{\alpha}[t^p F(t)](x) = \sum_{k=0}^{\infty} (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}(x). \quad (840)$$

*The series converges absolutely. It terminates when  $p \in \mathbb{N}_0$ .*

*Proof.* Use the binomial series  $t^p = x^p(1 - (x - t)/x)^p$  and dominate with  $0 \leq \mathcal{J}_{\alpha+k}(x) \leq x^{\alpha+k}/\Gamma(\alpha + k + 1)$ . The resulting scalar terms are  $\mathcal{O}(k^{-\operatorname{Re} p - 2})$ .  $\square$

**Proposition 4.5** (Dyadic-shell series for arbitrary powers). *For fixed  $x \in (0, 1]$  and every  $p \in \mathbb{C}$ ,*

$$\int_0^x t^p F(t) dt = x^{p+1} \sum_{n=0}^{\infty} 2^{-n(p+1)} \int_{1/2}^1 u^p F(x2^{-n}u) du. \quad (841)$$

*The series is absolutely and locally uniformly convergent in  $p$ ; hence the weighted primitive is entire in the exponent. This is [proved here](#).*

*Proof.* Partition  $(0, x]$  into the shells  $[x2^{-n-1}, x2^{-n}]$  and substitute  $t = x2^{-n}u$ . Endpoint flatness of  $F$  dominates every factor  $2^{n|\operatorname{Re} p|}$  on compact  $p$ -sets.  $\square$

## 4.3 Truncated moment–quantile formula

**Theorem 4.6** (Power-weighted primitive through the inverse). *For  $p \neq -1$  and  $0 < x \leq 1$ ,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} F(x) - \int_0^{F(x)} G(u)^{p+1} du}{p+1}. \quad (842)$$

At  $p = -1$  the removable logarithmic form is

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log G(u) du. \quad (843)$$

These formulas are **proved here** and extend holomorphically in  $p$ .

*Proof.* Stieltjes integration by parts gives

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} F(x) - \int_0^x t^{p+1} dF(t)}{p+1}.$$

The monotone substitution  $u = F(t)$  turns the Stieltjes integral into  $\int_0^{F(x)} G(u)^{p+1} du$ . The logarithmic identity follows by using  $\log t$  as the antiderivative of  $1/t$  and then the same substitution.  $\square$

#### 4.4 Analytic weights

**Theorem 4.7** (Local analytic-weight expansion). *Suppose  $g$  is analytic in a disk centered at  $x > 0$  that contains  $[0, x]$ , and  $\operatorname{Re} \alpha > 0$ . Then*

$$I_{0+}^\alpha [g(t)F(t)](x) = \sum_{k=0}^{\infty} \frac{(-1)^k (\alpha)_k}{k!} g^{(k)}(x) \mathcal{J}_{\alpha+k}(x). \quad (844)$$

The convergence is absolute under any Cauchy estimate with radius strictly larger than  $x$ .

*Proof.* Insert the Taylor series  $g(t) = \sum_{k \geq 0} g^{(k)}(x)(t-x)^k/k!$  into [equation \(818\)](#) and integrate termwise.  $\square$

**Corollary 4.8** (Exponential and trigonometric weights). *For  $\lambda \in \mathbb{C}$ ,*

$$I_{0+}^\alpha [e^{\lambda t} F(t)](x) = e^{\lambda x} \sum_{k=0}^{\infty} \frac{(-\lambda)^k (\alpha)_k}{k!} \mathcal{J}_{\alpha+k}(x). \quad (845)$$

Taking real or imaginary parts gives the corresponding cosine and sine formulas.

#### 4.5 Weights for up

The same finite algebra works after shifting the lower endpoint.

**Proposition 4.9** (Shifted weighted up-integrals). *For  $p \in \mathbb{N}_0$ ,  $\operatorname{Re} \alpha > 0$ , and  $x \geq -1$ ,*

$$I_{-1+}^\alpha [(t+1)^p \operatorname{up}(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k (x+1)^{p-k} \mathcal{K}_{\alpha+k}(x). \quad (846)$$

Ordinary powers  $t^p$  follow by expanding  $t^p = ((t+1) - 1)^p$ .

## 5 Nonlinear transmutation between a function and its inverse

### 5.1 A fractional change-of-variables theorem

**Theorem 5.1** (Fabius–quantile transmutation). *Let  $\operatorname{Re} \alpha > 0$ ,  $0 \leq x \leq 1$ , and  $h \in C^1([0, 1])$ . Then*

$$I_{0+}^\alpha [h(F)](x) = \frac{h(0)x^\alpha}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^{F(x)} h'(u)(x-G(u))^\alpha du. \quad (847)$$

Dually, for  $0 \leq y \leq 1$ ,

$$I_{0+}^{\alpha}[h(G)](y) = \frac{h(0)y^{\alpha}}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^{G(y)} h'(x)(y - F(x))^{\alpha} dx. \quad (848)$$

Both formulas are **proved here**.

*Proof.* For a differentiable  $f$ ,

$$I_{0+}^{\alpha}f(x) = \frac{f(0)x^{\alpha}}{\Gamma(\alpha+1)} + \frac{1}{\Gamma(\alpha+1)} \int_0^x (x-t)^{\alpha} f'(t) dt.$$

Apply this to  $f = h \circ F$  and substitute  $u = F(t)$ , then to  $f = h \circ G$  and substitute  $x = G(t)$ , equivalently  $t = F(x)$ .  $\square$

**Corollary 5.2** (Fractional integrals of the inverse and its powers). For  $\operatorname{Re} \alpha > 0$ ,

$$I_{0+}^{\alpha}G(y) = \frac{1}{\Gamma(\alpha+1)} \int_0^{G(y)} (y - F(x))^{\alpha} dx, \quad (849)$$

$$I_{0+}^{\alpha}[G^q](y) = \frac{q}{\Gamma(\alpha+1)} \int_0^{G(y)} x^{q-1}(y - F(x))^{\alpha} dx, \quad \operatorname{Re} q > 0. \quad (850)$$

At the right endpoint,

$$I_{0+}^{\alpha}G(1) = \frac{1}{\Gamma(\alpha+1)} \int_0^1 F(x)^{\alpha} dx. \quad (851)$$

*Proof.* Set  $h(x) = x$  or  $x^q$  in [equation \(848\)](#). At  $y = 1$ , use symmetry to replace  $(1 - F(x))^{\alpha}$  by  $F(x)^{\alpha}$  under the integral.  $\square$

## 5.2 Ordinary inverse antiderivatives

**Theorem 5.3** (Elementary primitive of  $G$ ). For  $0 \leq y \leq 1$ ,

$$\boxed{\int G(y) dy = yG(y) - F\left(\frac{G(y)}{2}\right) + C.} \quad (852)$$

In particular,

$$\int_0^1 G(y) dy = \frac{1}{2}. \quad (853)$$

The formula is **proved here**.

*Proof.* Take  $\alpha = 1$  in [equation \(849\)](#):

$$\int_0^y G(u) du = yG(y) - \int_0^{G(y)} F(x) dx.$$

The remaining integral is  $F(G(y)/2)$  by [theorem 3.1](#). At  $y = 1$ , use  $G(1) = 1$  and  $F(1/2) = 1/2$ .  $\square$

**Theorem 5.4** (Power-weighted inverse primitive). For  $p \neq -1$ ,

$$\int y^p G(y) dy = \frac{y^{p+1}G(y) - \int_0^{G(y)} F(x)^{p+1} dx}{p+1} + C. \quad (854)$$

At  $p = -1$ ,

$$\int \frac{G(y)}{y} dy = G(y) \log y - \int_0^{G(y)} \log F(x) dx + C. \quad (855)$$

Both identities are **proved here**.

*Proof.* Integrate by parts and use  $G'(y) dy = dx$  under  $y = F(x)$ .  $\square$

### 5.3 Mixed moments of $F$ and $G$

**Proposition 5.5** (Master mixed-moment bridge). *Whenever the integrals converge,*

$$\int_0^1 y^a G(y)^b dy = \int_0^1 F(x)^a x^b F'(x) dx. \quad (856)$$

*In particular,*

$$\int_0^1 G(y)^b dy = \mu(b), \quad (857)$$

*and, for  $\operatorname{Re} a > -1$  and  $\operatorname{Re} b > 0$ ,*

$$\int_0^1 y^a G(y)^b dy = \frac{1}{a+1} \left( 1 - b \int_0^1 x^{b-1} F(x)^{a+1} dx \right). \quad (858)$$

*These formulas are **proved here**.*

*Proof.* Substitute  $y = F(x)$  in the left side. Formula [equation \(857\)](#) is the case  $a = 0$ . For [equation \(858\)](#), write  $F^a F' = (a+1)^{-1} (F^{a+1})'$  and integrate by parts.  $\square$

### 5.4 Ordinary and fractional derivatives of $G$

**Proposition 5.6** (Recursive inverse derivatives). *On  $(0, 1)$  define*

$$R_1(x) = \frac{1}{F'(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{F'(x)}. \quad (859)$$

*Then*

$$G^{(n)}(y) = R_n(G(y)). \quad (860)$$

*For example,*

$$G'(y) = \frac{1}{F'(G(y))}, \quad (861)$$

$$G''(y) = -\frac{F''(G(y))}{F'(G(y))^3}, \quad (862)$$

$$G'''(y) = \frac{3F''(G(y))^2 - F'(G(y))F'''(G(y))}{F'(G(y))^5}. \quad (863)$$

*Proof.* Differentiate  $F(G(y)) = y$  and iterate the chain rule. The recurrence follows because  $\frac{d}{dy} R_n(G(y)) = R'_n(G(y))/F'(G(y))$ .  $\square$

**Theorem 5.7** (Fractional derivative of the inverse). *For  $0 < \alpha < 1$  and  $0 < y < 1$ , the Riemann–Liouville and Caputo derivatives agree and*

$$D_{0+}^\alpha G(y) = \frac{1}{\Gamma(1-\alpha)} \int_0^{G(y)} (y - F(x))^{-\alpha} dx. \quad (864)$$

*The value is finite for every interior  $y$  and diverges at both endpoints  $y \downarrow 0$  and  $y \uparrow 1$ . The identity is **proved here**; the two endpoint statements are **derived from audited inputs** (using the audited endpoint quadratic law).*

*Proof.* Because  $G(0) = 0$  and  $G$  is absolutely continuous,  $D_{0+}^\alpha G = I_{0+}^{1-\alpha} G'$ . Substitute  $u = F(x)$ , for which  $G'(u) du = dx$ . Interior finiteness follows from the local behavior of a monotone  $C^1$  change of variables away from the endpoints. At  $y = 1$ , the integral contains  $(1 - F(x))^{-\alpha}$  near  $x = 1$ ; by symmetry and the endpoint quadratic law this is not integrable. At the opposite endpoint,

$$D_{0+}^\alpha G(y) \geq \frac{y^{-\alpha} G(y)}{\Gamma(1-\alpha)}.$$

The endpoint quadratic law implies  $F(x) = o(x^N)$  for every  $N > 0$ , hence  $G(y)/y^\alpha \rightarrow \infty$  as  $y \downarrow 0$ .  $\square$

### 5.5 Sharp $L^r$ threshold for the inverse slope

**Theorem 5.8** (Inverse-slope integrability). *For  $r > 0$ ,*

$$\int_0^1 G'(y)^r dy = \int_0^1 F'(x)^{1-r} dx = 2^{-r} \int_{-1}^1 \text{up}(t)^{1-r} dt. \quad (865)$$

*This integral is finite if and only if  $0 < r \leq 1$ . The identity is **proved here**; the “only if” direction is **derived from audited inputs** (using the audited endpoint quadratic law).*

*Proof.* Use  $y = F(x)$  and  $G'(F(x)) = 1/F'(x)$ , then  $F'(x) = 2 \text{up}(2x - 1)$ . If  $r \leq 1$ , the exponent  $1 - r$  is nonnegative, so boundedness and compact support suffice. If  $r > 1$ , near  $x = 0$  the endpoint law

$$-\log F(x) \sim \frac{(\log(1/x))^2}{2 \log 2}$$

implies that  $F'(x)^{1-r}$  grows faster than  $\exp(c(\log(1/x))^2)$  for some  $c > 0$ , and its integral diverges after  $x = e^{-u}$ .  $\square$

**Corollary 5.9** (Negative powers and entropy). *For  $q > -1$ ,*

$$\int_0^1 G'(y)^{-q} dy = 2^q \int_{-1}^1 \text{up}(t)^{q+1} dt. \quad (866)$$

Moreover,

$$\int_0^1 \log G'(y) dy = - \int_0^1 F'(x) \log F'(x) dx. \quad (867)$$

*Thus the mean logarithmic slope of the quantile is the differential entropy of the Fabius density.*

## 6 Definite integrals, moments, and Bell–Bernoulli structure

### 6.1 Mellin transforms of $F$ and $\text{up}$

**Theorem 6.1** (Mellin–moment bridge). *For  $\text{Re } s > 0$ ,*

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mu(s)}{s}, \quad (868)$$

$$\int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mu(s)}{s}. \quad (869)$$

*The first expression extends to an entire function of  $s$ . For  $\text{Re } q > -1$ ,*

$$\mathbb{E}|Z|^q = \frac{2\mu(q+1)}{q+1}. \quad (870)$$

*All three identities are **proved here**.*

*Proof.* Integrate [equation \(868\)](#) by parts against  $x^s/s$ . On  $[0, 1]$ ,  $\text{up}(x) = 1 - F(x)$ , which gives [equation \(869\)](#). Evenness gives the final formula.  $\square$

**Corollary 6.2** (Power-weighted definite integrals). *For  $\text{Re } p > -1$ ,*

$$\int_0^1 x^p F(x) dx = \frac{1 - \mu(p+1)}{p+1}, \quad (871)$$

$$\int_0^1 x^p \text{up}(x) dx = \frac{\mu(p+1)}{p+1}. \quad (872)$$

*At integer  $p$ , the values are rational.*

**Corollary 6.3** (Logarithmic Fabius moments). *For  $n \in \mathbb{N}_0$ ,*

$$\int_0^1 \frac{(\log x)^n}{x} F(x) dx = -\frac{\mu^{(n+1)}(0)}{n+1}. \quad (873)$$

*Proof.* Differentiate the entire continuation of  $(1 - \mu(s))/s$  at  $s = 0$ .  $\square$

## 6.2 Bernoulli cumulants and complete Bell moments

The moment generating function is

$$\mathcal{A}(t) = \mathbb{E}e^{tY} = \prod_{j=1}^{\infty} \frac{e^{t/2^j} - 1}{t/2^j}. \quad (874)$$

Its logarithm immediately exposes Bernoulli numbers.

**Proposition 6.4** (Cumulants). *Let  $\kappa_n$  be the cumulants of  $Y$ . Then*

$$\kappa_1 = \frac{1}{2}, \quad \kappa_n = \frac{B_n}{n(2^n - 1)} \quad (n \geq 2), \quad (875)$$

where  $B_n$  are Bernoulli numbers with  $B_2 = 1/6$ . Consequently

$$\mu_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (876)$$

where  $\mathcal{B}_n$  is the complete exponential Bell polynomial. The cumulants of  $Z = 2Y - 1$  are  $2^n \kappa_n$  for  $n \geq 2$ , and its odd moments vanish.

*Proof.* For  $V \sim \text{Unif}[0, 1]$ ,

$$\log \mathbb{E}e^{tV} = \frac{t}{2} + \sum_{n \geq 2} \frac{B_n}{n} \frac{t^n}{n!}.$$

Scale by  $2^{-j}$ , sum over  $j$ , and use  $\sum_{j \geq 1} 2^{-jn} = 1/(2^n - 1)$ . The standard moment–cumulant relation gives [equation \(876\)](#).  $\square$

Table 18: Exact moments generated from Bernoulli cumulants.

| $n$ | $\mu_n = \mathbb{E}(Y^n)$      | $m_n = \mathbb{E}(Z^n)$        |
|-----|--------------------------------|--------------------------------|
| 0   | 1                              | 1                              |
| 1   | $\frac{1}{2}$                  | 0                              |
| 2   | $\frac{5}{18}$                 | $\frac{1}{9}$                  |
| 3   | $\frac{1}{6}$                  | 0                              |
| 4   | $\frac{143}{1350}$             | $\frac{19}{675}$               |
| 5   | $\frac{19}{270}$               | 0                              |
| 6   | $\frac{1153}{23814}$           | $\frac{583}{59535}$            |
| 7   | $\frac{583}{17010}$            | 0                              |
| 8   | $\frac{1616353}{65063250}$     | $\frac{132809}{32531625}$      |
| 9   | $\frac{132809}{7229250}$       | 0                              |
| 10  | $\frac{134926369}{9762090030}$ | $\frac{46840699}{24405225075}$ |



### 6.3 Bernoulli-polynomial integrals

**Proposition 6.5** (Bernoulli-polynomial bridge). *Let  $B_n(x)$  be the Bernoulli polynomial. For  $n \geq 0$ ,*

$$\int_0^1 B_n(x) F(x) dx = \frac{B_{n+1}(1) - \mathbb{E}B_{n+1}(Y)}{n+1}. \quad (877)$$

*The expectation is an explicit rational combination of the moments in [equation \(876\)](#). Its exponential generating function is*

$$\sum_{n=0}^{\infty} \mathbb{E}B_n(Y) \frac{t^n}{n!} = \frac{t}{e^t - 1} \mathcal{A}(t). \quad (878)$$

*Symmetry implies  $\mathbb{E}B_n(Y) = 0$  for every odd  $n$ .*

*Proof.* Use  $B'_{n+1} = (n+1)B_n$  and integrate by parts against  $F$ . The generating function is the classical generating function of Bernoulli polynomials averaged at  $Y$ . Finally,  $B_n(1-x) = (-1)^n B_n(x)$  and  $Y \stackrel{d}{=} 1-Y$ .  $\square$

### 6.4 Moments of ordinary derivatives

**Theorem 6.6** (Derivative moments). *For  $n \geq 1$ , the Mellin transform of  $F^{(n)}$  is the entire identity*

$$\int_0^1 x^p F^{(n)}(x) dx = (-1)^{n-1} p^{\underline{n-1}} \mu(p-n+1), \quad p \in \mathbb{C}. \quad (879)$$

*For integer  $p < n-1$ , the falling factorial vanishes. For integers  $p, n \geq 0$ ,*

$$\int_{-1}^1 x^p \text{up}^{(n)}(x) dx = \begin{cases} (-1)^n p^{\underline{n}} m_{p-n}, & p \geq n, \\ 0, & p < n. \end{cases} \quad (880)$$

*These are [proved here](#).*

*Proof.* Integrate by parts until only  $F'$  remains; every higher derivative vanishes at both endpoints. The last integral is  $\mu(p-n+1)$ . Flatness makes both sides entire in  $p$ . The compactly supported up formula follows from  $n$  integrations by parts.  $\square$

### 6.5 Beta integrals and products

**Proposition 6.7** (Probability-integral-transform beta family). *For every integrable  $\varphi$ ,*

$$\int_0^1 \varphi(F(x)) F'(x) dx = \int_0^1 \varphi(u) du. \quad (881)$$

*Therefore, for  $\text{Re } a, \text{Re } b > -1$ ,*

$$\int_0^1 F(x)^a (1-F(x))^b F'(x) dx = B(a+1, b+1). \quad (882)$$

*On the left half,*

$$\int_0^{1/2} F(x)^a \text{up}(x)^b F(2x) dx = \frac{1}{2} B_{1/2}(a+1, b+1). \quad (883)$$

*If  $H(z) = F((z+1)/2)$  is the CDF of up, then*

$$\int_{-1}^1 \text{up}(z) H(z)^a (1-H(z))^b dz = B(a+1, b+1). \quad (884)$$

*Proof.* Use  $u = F(x)$  in [equation \(881\)](#); then use  $F' = 2F(2x)$  on the left half. The last formula is the same probability-integral transform for  $H$ .  $\square$

**Proposition 6.8** (Powers of the CDF and order statistics). *For  $m > 0$  and  $0 \leq x \leq 1$ ,*

$$\int_0^x F(t)^m dt = xF(x)^m - m \int_0^{F(x)} u^{m-1} G(u) du. \quad (885)$$

*If  $m \in \mathbb{N}$  and  $Y_1, \dots, Y_m$  are independent copies of  $Y$ , then*

$$\int_0^1 F(x)^m dx = 1 - \mathbb{E} \max(Y_1, \dots, Y_m) = \int_0^1 (1 - F(x))^m dx. \quad (886)$$

*Proof.* Integrate  $F^m$  by parts and substitute  $u = F(t)$ . The CDF of the maximum is  $F^m$ ; symmetry gives the last equality.  $\square$

## 6.6 A mixed product and a sinc-product integral

**Theorem 6.9** (Gini–Fourier identity). *Let*

$$\Phi(t) = \widehat{\text{up}}(t) = \prod_{j=1}^{\infty} \text{sinc}(t/2^j). \quad (887)$$

*Then*

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y_1 - Y_2| \quad (888)$$

$$= \frac{1}{4} \mathbb{E}|Z_1 - Z_2| \quad (889)$$

$$= \frac{1}{2\pi} \int_0^{\infty} \frac{1 - \Phi(t)^2}{t^2} dt. \quad (890)$$

*Numerically,*

$$\int_0^1 F(x)(1 - F(x)) dx = 0.095563232016 \dots \quad (891)$$

*The identities are [proved here](#); the decimal is [numerically checked](#).*

*Proof.* For every continuous CDF  $C$ ,  $\mathbb{E}|X_1 - X_2| = 2 \int C(1 - C)$ . The affine relation  $Z = 2Y - 1$  gives the second line. Finally,  $\mathbb{E}|W| = (2/\pi) \int_0^{\infty} (1 - \text{Re } \varphi_W(t)) t^{-2} dt$ , while the characteristic function of  $Z_1 - Z_2$  is  $\Phi(t)^2$ .  $\square$

## 7 Laplace, Fourier, Mellin, Cauchy, and correlation transforms

### 7.1 Laplace transforms of $F$ and $\text{up}$

Define the negative moment transform

$$\phi(s) = \mathbb{E} e^{-sY} = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad \text{Re } s \geq 0. \quad (892)$$

**Theorem 7.1** (Laplace dictionary). *For  $\text{Re } s > 0$ ,*

$$\int_0^{\infty} e^{-sx} F(x) dx = \frac{\phi(s)}{s}, \quad (893)$$

$$\int_0^1 e^{-sx} F(x) dx = \frac{\phi(s) - e^{-s}}{s}, \quad (894)$$

$$\int_0^1 e^{-sx} \text{up}(x) dx = \frac{1 - \phi(s)}{s}. \quad (895)$$

The bilateral moment and Fourier transforms of  $\text{up}$  are

$$\int_{-1}^1 e^{sx} \text{up}(x) dx = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}, \quad (896)$$

$$\widehat{\text{up}}(t) = \prod_{j=1}^{\infty} \text{sinc}(t/2^j). \quad (897)$$

*Proof.* Integration by parts against the probability measure  $dF$  gives  $\mathcal{L}F = \phi/s$ . Subtract the tail  $\int_1^{\infty} e^{-sx} dx$  for the unit interval and use  $\text{up} = 1 - F$  on  $[0, 1]$ . Independence in [equation \(815\)](#) gives the products for  $Z$ .  $\square$

**Corollary 7.2** (Transforms of fractional primitives and derivatives). For  $\text{Re } \alpha > 0$ ,

$$\mathcal{L}\{I_{0+}^{\alpha} F\}(s) = s^{-\alpha-1} \phi(s). \quad (898)$$

For noninteger  $\alpha > 0$ , because all initial data of  $F$  vanish at 0,

$$\mathcal{L}\{D_{0+}^{\alpha} F\}(s) = s^{\alpha-1} \phi(s). \quad (899)$$

Thus Bromwich inversion gives

$$I_{0+}^{\alpha} F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} s^{-\alpha-1} \phi(s) e^{sx} ds, \quad (900)$$

$$D_{0+}^{\alpha} F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} s^{\alpha-1} \phi(s) e^{sx} ds. \quad (901)$$

## 7.2 Laplace and Mellin transforms of the inverse

**Theorem 7.3** (Inverse-transform dictionary). For  $\text{Re } s > 0$ ,

$$\int_0^1 e^{-sy} G(y) dy = \frac{\int_0^1 e^{-sF(x)} dx - e^{-s}}{s}, \quad (902)$$

$$\int_0^1 e^{-sy} G(y)^q dy = \frac{q}{s} \int_0^1 x^{q-1} (e^{-sF(x)} - e^{-s}) dx, \quad \text{Re } q > 0. \quad (903)$$

For  $\text{Re } s > 0$ ,

$$\int_0^1 y^{s-1} G(y) dy = \frac{1 - \int_0^1 F(x)^s dx}{s}. \quad (904)$$

All formulas are [proved here](#).

*Proof.* Integrate by parts, then use  $G'(y) dy = dx$  under  $y = F(x)$ . For the power formula, write the boundary term  $e^{-s}$  as  $qe^{-s} \int_0^1 x^{q-1} dx$ .  $\square$

### 7.3 Stieltjes and Cauchy transforms

**Proposition 7.4** (Stieltjes transform of the Fabius law). *For  $z \in \mathbb{C} \setminus [0, 1]$ ,*

$$S_Y(z) = \int_0^1 \frac{dF(x)}{z-x} = \int_0^1 \frac{dy}{z-G(y)} = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (905)$$

*If  $C_{\text{up}}(w) = \int_{-1}^1 \text{up}(t)/(w-t) dt$ , then*

$$S_Y(z) = 2C_{\text{up}}(2z-1). \quad (906)$$

**Proposition 7.5** (Ordinary Cauchy transforms). *For  $z \in \mathbb{C} \setminus [0, 1]$ , with a consistent logarithm branch,*

$$\int_0^1 \frac{F(x)}{z-x} dx = \mathbb{E} \log \left( \frac{z-Y}{z-1} \right) = \sum_{n=1}^{\infty} \frac{1-\mu_n}{nz^n}, \quad |z| > 1, \quad (907)$$

$$\int_0^1 \frac{G(y)}{z-y} dy = \int_0^1 \log \left( \frac{z-F(x)}{z-1} \right) dx. \quad (908)$$

*Proof.* For [equation \(907\)](#), integrate by parts with antiderivative  $-\log(z-x)$ , then expand the logarithms at infinity. The inverse formula is identical, followed by  $y = F(x)$ .  $\square$

### 7.4 Autocorrelation, convolution, and Sobolev energies

**Proposition 7.6** (Autocorrelation). *The autocorrelation*

$$C(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x+a) dx \quad (909)$$

*has the Fourier representation*

$$C(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ita} \Phi(t)^2 dt. \quad (910)$$

*More generally, every  $m$ -fold convolution of  $\text{up}$  has transform  $\Phi^m$  and is the density of a sum of  $m$  independent copies of  $Z$ .*

**Definition 7.7** (Fractional Sobolev energy). *For  $\alpha \geq 0$ , define*

$$\mathcal{E}_{\alpha} = \int_{\mathbb{R}} \text{up}(x) (-\Delta)^{\alpha/2} \text{up}(x) dx = \frac{1}{\pi} \int_0^{\infty} t^{\alpha} \Phi(t)^2 dt. \quad (911)$$

*The rapid decay of  $\Phi$  makes this finite for every  $\alpha \geq 0$ .*

**Proposition 7.8** (Energy refinement identity and log-convexity). *The product relation  $\Phi(2t) = \text{sinc}(t)\Phi(t)$  implies*

$$\mathcal{E}_{\alpha} = \frac{2^{\alpha+1}}{\pi} \int_0^{\infty} t^{\alpha} \text{sinc}(t)^2 \Phi(t)^2 dt. \quad (912)$$

*The map  $\alpha \mapsto \mathcal{E}_{\alpha}$  is log-convex on  $[0, \infty)$ .*

*Proof.* Substitute  $u = 2t$  in [equation \(911\)](#). Log-convexity is Holder's inequality applied to the positive measure  $\Phi(t)^2 dt/\pi$  and the functions  $t^{\alpha}$ .  $\square$

The accompanying computation gives

|                        |                |                |               |               |                |
|------------------------|----------------|----------------|---------------|---------------|----------------|
| $\alpha$               | 0              | 1/2            | 1             | 2             | 4              |
| $\mathcal{E}_{\alpha}$ | 0.808873535968 | 0.939432032833 | 1.28502241154 | 3.23549414387 | 51.7679063017. |

(913)

These decimals are **numerically checked**.

## 8 Riemann–Liouville and Caputo calculus on dyadic scales

### 8.1 The exact order-shift equation

**Theorem 8.1** (Fractional order shift). *For  $\operatorname{Re} \alpha > 0$  and  $0 \leq x \leq 1$ ,*

$$\boxed{\mathcal{J}_{\alpha+1}(x) = 2^\alpha \mathcal{J}_\alpha(x/2).} \quad (914)$$

*For  $\mathcal{F}$ , the same identity holds for every real  $x$ . This is [proved here](#).*

For real  $\alpha > 0$ , the bounded statement is exactly `Fabius.fractionalVolterra_add_one_fabiusReal`; `Fabius.fractionalVolterra_add_one_extendedFabius_of_nonneg` gives the signed statement for  $x \geq 0$ . The report’s complex-order scope and signed  $x < 0$  extension are not asserted, and the iterated corollary below is not separately exported.

*Proof.* Since  $F(t) = \frac{d}{dt}F(t/2)$  for  $0 \leq t \leq 1$ , integration by parts gives

$$\mathcal{J}_{\alpha+1}(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} F(t/2) dt.$$

Set  $t = 2u$ . The global version uses  $\mathcal{F}(t) = \frac{d}{dt}\mathcal{F}(t/2)$  for all  $t$ . □

**Remark 8.2** (Exact Lean boundary of the order shift). For real  $\alpha > 0$ , the bounded identity in (914) is exactly `Fabius.fractionalVolterra_add_one_fabiusReal`. The declaration `Fabius.fractionalVolterra_add_one_extendedFabius_of_nonneg` proves the signed version for  $x \geq 0$ . Thus the real positive-order portion is machine checked; the report’s complex-order reading and its signed claim at negative endpoints are not supplied by those declarations. Iteration to (915) and the endpoint-moment lattice below are likewise separate frontier obligations.

**Corollary 8.3** (Integer shift of a fractional base order). *For  $n \in \mathbb{N}_0$ ,  $\operatorname{Re} \beta > 0$ , and  $0 \leq x \leq 1$ ,*

$$\mathcal{J}_{n+\beta}(x) = 2^{n\beta+\binom{n}{2}} \mathcal{J}_\beta(x/2^n). \quad (915)$$

### 8.2 Endpoint values and the fractional dyadic lattice

**Theorem 8.4** (Fractional endpoint moment). *For  $\operatorname{Re} \alpha > 0$ ,*

$$\mathcal{J}_\alpha(1) = \frac{\mu(\alpha)}{\Gamma(\alpha+1)}. \quad (916)$$

*Consequently, for  $n \in \mathbb{N}_0$  and  $\operatorname{Re} \beta > 0$ ,*

$$\boxed{\mathcal{J}_\beta(2^{-n}) = 2^{-n\beta-\binom{n}{2}} \frac{\mu(n+\beta)}{\Gamma(n+\beta+1)}.} \quad (917)$$

*These identities are [proved here](#).*

*Proof.* Symmetry transforms the defining integral into

$$\mathcal{J}_\alpha(1) = \frac{1}{\Gamma(\alpha)} \int_0^1 u^{\alpha-1} (1-F(u)) du.$$

The tail-moment formula  $\mu(\alpha) = \alpha \int_0^1 u^{\alpha-1} (1-F(u)) du$  proves the first claim. Put  $x = 1$  in [equation \(915\)](#) and solve for  $\mathcal{J}_\beta(2^{-n})$ . □

**Remark 8.5.** At  $\beta = 0$ , interpreted by continuity, [equation \(917\)](#) becomes the classical integer-moment dyadic formula [equation \(830\)](#). The new point is that the lattice extends continuously and analytically in the order.

### 8.3 Fractional calculus for up

**Theorem 8.6** (Up–Fabius fractional bridge). *For  $\operatorname{Re} \alpha > 0$  and  $x \geq -1$ ,*

$$\mathcal{K}_{\alpha+1}(x) = 2^\alpha \mathcal{J}_\alpha\left(\frac{x+1}{2}\right). \quad (918)$$

*At the right support endpoint,*

$$\mathcal{K}_\alpha(1) = 2^{\alpha-1} \frac{\mu(\alpha-1)}{\Gamma(\alpha)}, \quad \alpha > 0, \quad (919)$$

*where the moment is understood by its entire continuation. These are [proved here](#).*

For real  $\alpha > 0$ , the first display (918) is exactly `Fabius.fractionalVolterra_add_one_rv` `achevUp`. That declaration does not export the complex-order continuation, the endpoint display, or the shifted lattice below.

*Proof.* The function  $H(x) = F((x+1)/2)$  is the CDF of up, hence  $H' = \text{up}$ . Integrate the  $\mathcal{K}_{\alpha+1}$  kernel by parts and substitute  $u = (t+1)/2$ . At  $x = 1$ , alternatively write

$$\mathcal{K}_\alpha(1) = \Gamma(\alpha)^{-1} \mathbb{E}(1-Z)^{\alpha-1} = 2^{\alpha-1} \Gamma(\alpha)^{-1} \mathbb{E}(1-Y)^{\alpha-1}$$

and use symmetry. □

**Corollary 8.7** (Shifted fractional dyadic lattice). *For  $n \in \mathbb{N}_0$  and  $\operatorname{Re} \beta > 0$ ,*

$$\mathcal{K}_{\beta+1}(-1 + 2^{1-n}) = 2^{\beta-n\beta-\binom{n}{2}} \frac{\mu(n+\beta)}{\Gamma(n+\beta+1)}. \quad (920)$$

### 8.4 Fractional derivatives of $F$

**Theorem 8.8** (Fractional derivative on the first dyadic cell). *For  $0 < \beta < 1$  and  $0 \leq x \leq 1/2$ ,*

$$D_{0+}^\beta F(x) = {}^C D_{0+}^\beta F(x) = 2^\beta \mathcal{J}_{1-\beta}(2x). \quad (921)$$

*At dyadic points,*

$$D_{0+}^\beta F(2^{-n-1}) = 2^{\beta-n(1-\beta)-\binom{n}{2}} \frac{\mu(n+1-\beta)}{\Gamma(n+2-\beta)}. \quad (922)$$

*The formulas are [proved here](#).*

*Proof.* Endpoint flatness gives  $D^\beta F = I^{1-\beta} F'$ . On  $[0, 1/2]$ , insert  $F'(t) = 2F(2t)$  and put  $u = 2t$ . Then use [equation \(917\)](#). □

**Corollary 8.9** (General mixed integer/fractional derivative). *Let  $q = m + \beta$  with  $m \in \mathbb{N}_0$  and  $0 < \beta < 1$ . For  $0 \leq x \leq 2^{-m-1}$ ,*

$$D_{0+}^{m+\beta} F(x) = 2^{\binom{m+1}{2} + m\beta} D_{0+}^\beta F(2^m x). \quad (923)$$

*Combining with [equation \(922\)](#) gives an explicit moment formula on every compatible dyadic lattice.*

## 8.5 Complements and right-sided fractional integrals

**Proposition 8.10** (Fractional complements). *For  $0 \leq x \leq 1$  and  $\operatorname{Re} \alpha > 0$ ,*

$$I_{0+}^{\alpha}[1 - F](x) = \frac{x^{\alpha}}{\Gamma(\alpha + 1)} - \mathcal{J}_{\alpha}(x), \quad (924)$$

$$I_{1-}^{\alpha}F(x) = \frac{(1 - x)^{\alpha}}{\Gamma(\alpha + 1)} - \mathcal{J}_{\alpha}(1 - x), \quad (925)$$

$$I_{1-}^{\alpha}[1 - F](x) = \mathcal{J}_{\alpha}(1 - x). \quad (926)$$

Here  $I_{1-}^{\alpha}f(x) = \Gamma(\alpha)^{-1} \int_x^1 (t - x)^{\alpha-1} f(t) dt$ .

*Proof.* The first formula is linearity. In the right-sided integrals put  $u = 1 - t$  and use  $F(1 - u) = 1 - F(u)$ .  $\square$

## 9 A continuous dyadic Appell family and its exact fractional defect

This section contains the central frontier result. Integer integration and differentiation fit a natural continuous dilation interpolation. The interpolation is almost, but not exactly, Riemann–Liouville calculus. The discrepancy is the repository’s Gamma–zeta periodic term.

### 9.1 The continuous-order dyadic family

**Definition 9.1** (Dyadic Appell family). For  $q \in \mathbb{R}$  and any function  $f$  on  $\mathbb{R}$ , define

$$(\mathfrak{A}_q f)(x) = 2^{q(q-1)/2} f(2^{-q}x). \quad (927)$$

We apply this operator principally to  $f = \mathcal{F}$ .

**Theorem 9.2** (Interpolation, derivative ladder, and cocycle). *For real  $q, \alpha, \beta$ :*

$$\mathfrak{A}_n \mathcal{F} = I_{0+}^n \mathcal{F} \quad (n \in \mathbb{N}_0), \quad (928)$$

$$\mathfrak{A}_{-n} \mathcal{F} = \mathcal{F}^{(n)} \quad (n \in \mathbb{N}_0), \quad (929)$$

$$\frac{d}{dx} \mathfrak{A}_q \mathcal{F} = \mathfrak{A}_{q-1} \mathcal{F}, \quad (930)$$

$$\mathfrak{A}_{\alpha} \mathfrak{A}_{\beta} = 2^{-\alpha\beta} \mathfrak{A}_{\alpha+\beta}. \quad (931)$$

All statements are **proved here**.

*Proof.* The integer identities are [equations \(828\) and \(831\)](#). Differentiate [equation \(927\)](#) and use  $\mathcal{F}' = 2\mathcal{F}(2\cdot)$ . Direct composition of the two dilations gives the scalar cocycle  $2^{-\alpha\beta}$ .  $\square$

**Remark 9.3.** The cocycle is not the Riemann–Liouville semigroup law. Thus  $q \mapsto \mathfrak{A}_q$  is an Appell interpolation of a single function’s integer ladder, not a fractional-integration semigroup on an arbitrary function space. The next theorem quantifies the difference exactly.

### 9.2 Laplace transform of the signed continuation

Let

$$\mathcal{L}(s) = \int_0^{\infty} e^{-sx} \mathcal{F}(x) dx, \quad s > 0. \quad (932)$$

**Theorem 9.4** (Thue–Morse Laplace completion). *For  $s > 0$ ,*

$$\mathcal{L}(s) = e^{-s} \left( \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j} \right) \left( \prod_{j=0}^{\infty} (1 - e^{-2^{j+1}s}) \right) \quad (933)$$

$$= \phi(2s) \prod_{j=0}^{\infty} (1 - e^{-2^{j+1}s}). \quad (934)$$

*It satisfies the exact functional equation*

$$\mathcal{L}(s/2) = s\mathcal{L}(s). \quad (935)$$

*The statements are **proved here**.*

*Proof.* On the block  $[2n, 2n+2]$ ,  $\mathcal{F}(x) = \varepsilon_n \operatorname{up}(x - 2n - 1)$ . Summing the block Laplace transforms gives

$$\mathcal{L}(s) = e^{-s} M_Z(-s) \sum_{n \geq 0} \varepsilon_n e^{-2ns}.$$

Evenness makes  $M_Z(-s) = M_Z(s)$ , and the Thue–Morse generating product is  $\sum \varepsilon_n q^n = \prod_{j \geq 0} (1 - q^{2^j})$ . Since  $e^{-s} M_Z(s) = \phi(2s)$ , both products follow. Alternatively, Laplace-transforming  $\mathcal{F}' = 2\mathcal{F}(2x)$  gives  $s\mathcal{L}(s) = \mathcal{L}(s/2)$  because  $\mathcal{F}(0) = 0$ .  $\square$

### 9.3 Exact completion of the periodic endpoint term

Put  $L = \log 2$ . The audited repository proves, for the same  $\phi$ , an exact Lambert– $W$ /Gamma–zeta endpoint decomposition [12, 13, 11]:

$$\log \phi(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + P(\log_2 s) + E(s), \quad (936)$$

where  $P$  is smooth and one-periodic and

$$E(s) = -\sum_{j=0}^{\infty} \log(1 - e^{-2^j s}). \quad (937)$$

The repository also proves that the centered function  $\Psi = P - \int_0^1 P$  has Fourier coefficients

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad k \neq 0, \quad (938)$$

and every one of them is nonzero.

**Theorem 9.5** (Exact bilateral completion). *For every  $s > 0$ ,*

$$\boxed{\log \mathcal{L}(s) = -\frac{(\log s)^2}{2L} - \frac{1}{2} \log s + P(\log_2 s).} \quad (939)$$

*This is **derived from audited inputs** (using the audited decomposition).*

*Proof.* Apply equation (936) to  $2s$ . Periodicity gives  $P(\log_2(2s)) = P(\log_2 s)$ , and the quadratic terms simplify to  $-(\log s)^2/(2L) - (\log s)/2$ . By equation (937),  $E(2s)$  is the negative logarithm of the second product in equation (934), so it cancels exactly.  $\square$



### 9.4 The exact Riemann–Liouville defect

The ideal order- $q$  Riemann–Liouville action on a flat function has Laplace multiplier  $s^{-q}$ . The actual transform of the dyadic interpolation is

$$\mathcal{L}\{\mathfrak{A}_q\mathcal{F}\}(s) = 2^{q(q+1)/2} \mathcal{L}(2^q s). \quad (940)$$

**Theorem 9.6** (Gamma–zeta fractional-dilation defect). *For  $q \in \mathbb{R}$  and  $s > 0$ , define*

$$R_q(s) = \frac{\mathcal{L}\{\mathfrak{A}_q\mathcal{F}\}(s)}{s^{-q} \mathcal{L}(s)}. \quad (941)$$

Then

$$R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)) = \exp(\Psi(\log_2 s + q) - \Psi(\log_2 s)). \quad (942)$$

Moreover,

$$R_{q+1}(s) = R_q(s), \quad (943)$$

$$R_n(s) = 1 \quad (n \in \mathbb{Z}), \quad (944)$$

$$R_q(s) \equiv 1 \text{ for all } s > 0 \iff q \in \mathbb{Z}. \quad (945)$$

The theorem is **derived from audited inputs** (using [theorem 9.5](#) and the audited nonvanishing of the Gamma–zeta Fourier coefficients).

*Proof.* Use [equations \(939\) and \(940\)](#); the quadratic and linear logarithmic terms cancel exactly, leaving the difference of  $P$ . Periodicity of  $P$  gives [equations \(943\) and \(944\)](#). If  $R_q \equiv 1$ , then  $\Psi(u + q) = \Psi(u)$  for every  $u$ . The Fourier mode  $k = 1$  is nonzero by [equation \(938\)](#), so  $e^{2\pi i q} = 1$  and  $q \in \mathbb{Z}$ . The converse was already shown.  $\square$

**Corollary 9.7** (Uniform defect bounds). *Let  $\text{osc } P = \max P - \min P$ . Then*

$$e^{-\text{osc } P} \leq R_q(s) \leq e^{\text{osc } P} \quad (q \in \mathbb{R}, s > 0). \quad (946)$$

The numerical Gamma–zeta series gives

$$\text{osc } P = 4.2507727262 \dots \times 10^{-6}, \quad (947)$$

so the natural dyadic interpolation is extraordinarily close to fractional calculus, but rigorously never identical at a noninteger order.

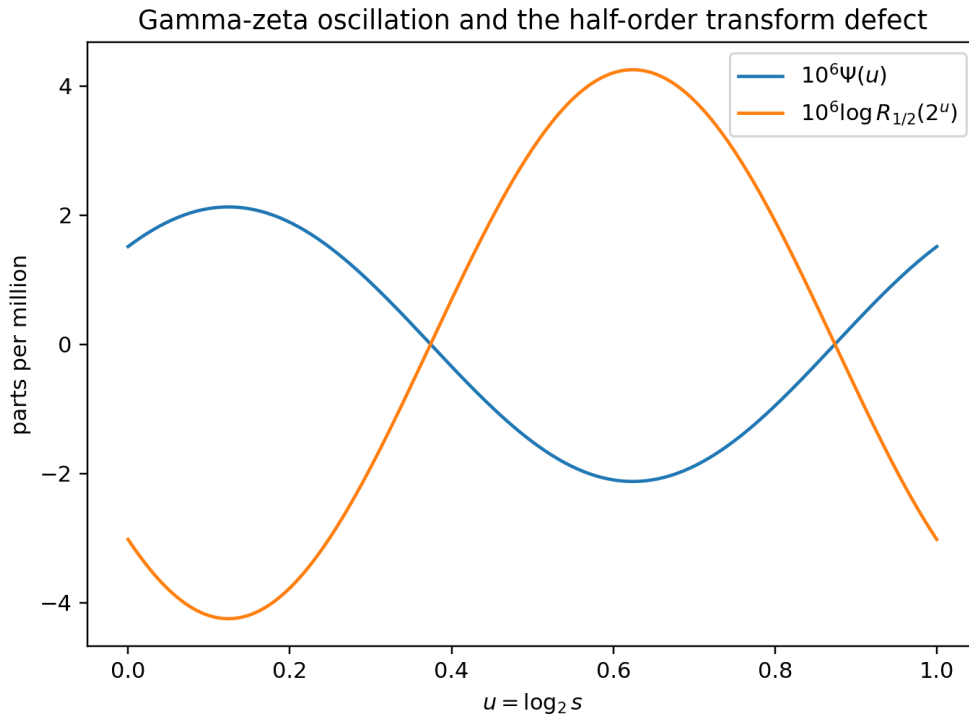


Figure 4: The centered Gamma–zeta oscillation  $\Psi(u)$  and the exact half-order defect  $\log R_{1/2}(2^u) = \Psi(u + 1/2) - \Psi(u)$ . The vertical scale is in parts per million.

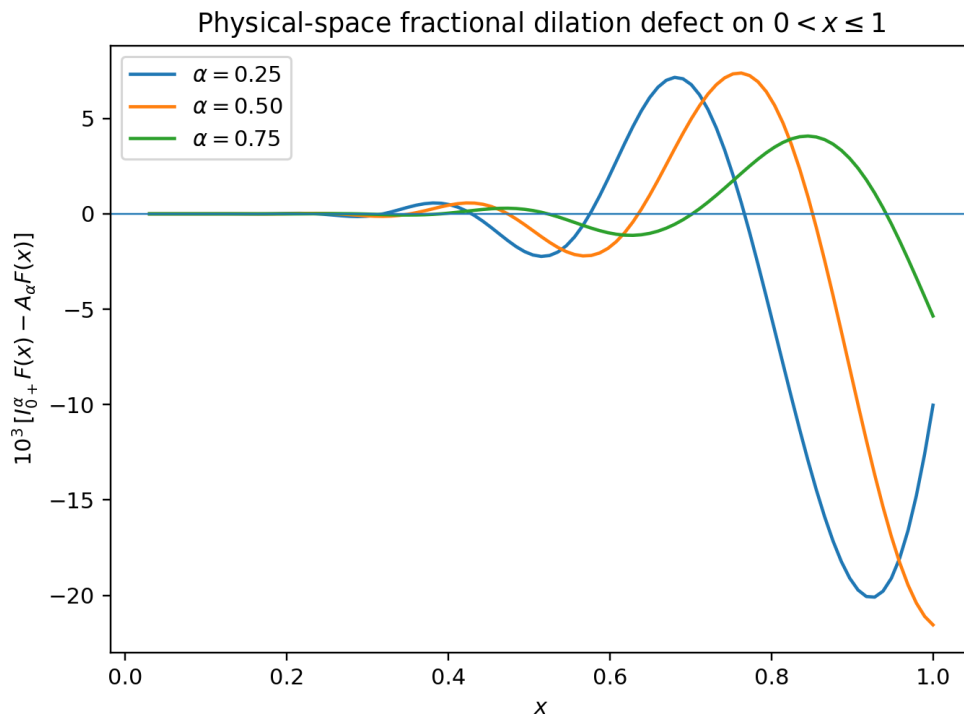


Figure 5: Physical-space comparison of a Riemann–Liouville fractional integral of  $F$  with the continuous dyadic interpolation. The visible difference is rescaled; the underlying identity is exact only at integer order.

## 9.5 Interpretation

The defect theorem gives a precise answer to a natural but previously ambiguous question: what should “half an antiderivative” mean for the Fabius dilation equation? There are two canonical answers.

- Riemann–Liouville calculus preserves the additive order semigroup and has multiplier  $s^{-q}$ .
- Dyadic Appell interpolation preserves the exact Fabius dilation shape and the derivative ladder  $\partial_x \mathfrak{A}_q = \mathfrak{A}_{q-1}$ .

They agree on the integer lattice. Away from it, their ratio is not an uncontrolled error but a smooth log-periodic multiplier with explicit Gamma–zeta Fourier coefficients. The same periodic object already controls the sharp Lambert- $W$  endpoint asymptotics; it therefore acts as an obstruction simultaneously in physical endpoint asymptotics and in continuous-order operator interpolation.

## 10 Whole-line fractional operators acting on up

The left-sided operators remember a base point. Translation-invariant fractional calculus is better expressed through the Riesz fractional Laplacian and potential.

### 10.1 Fractional Laplacian outside the support

For  $0 < \alpha < 2$ , use the standard singular-integral normalization [9]:

$$(-\Delta)^{\alpha/2} f(x) = C_{1,\alpha} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(y)}{|x - y|^{1+\alpha}} dy, \quad C_{1,\alpha} = \frac{2^\alpha \Gamma((1 + \alpha)/2)}{\sqrt{\pi} |\Gamma(-\alpha/2)|}. \quad (948)$$

**Theorem 10.1** (Exact exterior moment series). *For  $0 < \alpha < 2$  and  $|x| > 1$ ,*

$$\boxed{(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \sum_{j=0}^{\infty} \frac{(1 + \alpha)_{2j}}{(2j)!} \frac{m_{2j}}{|x|^{1+\alpha+2j}}.} \quad (949)$$

*The series converges absolutely and locally uniformly on  $|x| > 1$ . This is [proved here](#).*

*Proof.* Outside the support,  $\text{up}(x) = 0$ , so

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \int_{-1}^1 \frac{\text{up}(y)}{|x - y|^{1+\alpha}} dy.$$

For  $x > 1$ , expand  $(x - y)^{-1-\alpha} = x^{-1-\alpha} \sum_{k \geq 0} (1 + \alpha)_k (y/x)^k / k!$ . Odd moments vanish. The case  $x < -1$  is identical by evenness.  $\square$

The first terms are

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} |x|^{-1-\alpha} \left( 1 + \frac{(1 + \alpha)(2 + \alpha)}{18x^2} + \frac{19(1 + \alpha)_4}{16200x^4} + \dots \right). \quad (950)$$

At  $x = 2$ , six even moments reproduce direct quadrature to roughly  $10^{-7}$  for the tested orders  $\alpha = 0.5, 0.7, 1.3$ ; this is [numerically checked](#).

## 10.2 Riesz potentials

For  $0 < \alpha < 1$ , define the one-dimensional Riesz potential with Fourier multiplier  $|t|^{-\alpha}$  by

$$I_{\alpha}^R f(x) = c_{1,\alpha} \int_{\mathbb{R}} |x - y|^{\alpha-1} f(y) dy, \quad c_{1,\alpha} = \frac{\Gamma((1-\alpha)/2)}{2^{\alpha} \sqrt{\pi} \Gamma(\alpha/2)}. \quad (951)$$

**Theorem 10.2** (Exterior Riesz-potential series). *For  $0 < \alpha < 1$  and  $|x| > 1$ ,*

$$I_{\alpha}^R \text{up}(x) = c_{1,\alpha} |x|^{\alpha-1} \sum_{j=0}^{\infty} \frac{(1-\alpha)_{2j}}{(2j)!} \frac{m_{2j}}{|x|^{2j}}. \quad (952)$$

*The series is absolutely and locally uniformly convergent.*

*Proof.* Expand  $|x - y|^{\alpha-1} = |x|^{\alpha-1} (1 - \text{sgn}(x)y/|x|)^{-(1-\alpha)}$  and use evenness.  $\square$

## 10.3 Transform-side fractional derivatives

By Fourier multiplication,

$$(-\Delta)^{\alpha/2} \text{up}(t) = |t|^{\alpha} \prod_{j=1}^{\infty} \text{sinc}(t/2^j). \quad (953)$$

This supplies a direct integral transform for all translation-invariant fractional derivatives of  $\text{up}$ . It also makes the energies in [equation \(911\)](#) the diagonal quadratic form of the operator.

# 11 The inverse Mellin transform has a Gevrey-2 boundary

This section uses only the audited endpoint quadratic law

$$-\log F(e^{-u}) \sim \frac{u^2}{2 \log 2}, \quad u \rightarrow \infty. \quad (954)$$

It yields a new regularity classification for the inverse Mellin transform.

Define

$$A(s) = \int_0^1 F(x)^s dx, \quad \text{Re } s > 0, \quad (955)$$

and

$$Q(x) = -\log F(x), \quad a_n = \int_0^1 Q(x)^n dx. \quad (956)$$

By [equation \(904\)](#),

$$\mathcal{M}_G(s) := \int_0^1 y^{s-1} G(y) dy = \frac{1 - A(s)}{s}. \quad (957)$$

**Theorem 11.1** (Sharp logarithmic-moment growth). *The moments of  $Q = -\log F$  satisfy*

$$\lim_{n \rightarrow \infty} \left( \frac{a_n}{(2n)!} \right)^{1/n} = \frac{1}{2 \log 2}. \quad (958)$$

*This is **derived from audited inputs** (using the audited endpoint quadratic law).*

*Proof.* Put  $x = e^{-u}$ . Then

$$a_n = \int_0^\infty Q(e^{-u})^n e^{-u} du.$$

Let  $c = (2 \log 2)^{-1}$ . For every  $\varepsilon > 0$ , there is  $U$  such that  $(c - \varepsilon)u^2 \leq Q(e^{-u}) \leq (c + \varepsilon)u^2$  for  $u \geq U$ . The upper bound splits the integral at  $U$  and compares the tail with  $(c + \varepsilon)^n \Gamma(2n + 1)$ . The compact part is exponentially negligible after division by  $(2n)!$ . The lower bound compares with the incomplete gamma tail  $(c - \varepsilon)^n \int_U^\infty u^{2n} e^{-u} du$ , whose ratio to  $(2n)!$  tends to 1. Take  $n$ th roots and then  $\varepsilon \downarrow 0$ .  $\square$

**Theorem 11.2** (Gevrey-2 but nonanalytic inverse Mellin boundary). *The functions  $A(s)$  and  $\mathcal{M}_G(s)$  have derivatives of every order from the right at  $s = 0$ , with*

$$A^{(n)}(0+) = (-1)^n a_n, \quad (959)$$

$$\mathcal{M}_G^{(n)}(0+) = -\frac{A^{(n+1)}(0+)}{n+1}. \quad (960)$$

*Their formal Taylor series at 0 have radius of convergence zero. More precisely, the derivative growth is of Gevrey order 2 and of no smaller Gevrey order in the root-growth sense. Consequently neither transform has a holomorphic continuation through  $s = 0$  that agrees with its defining integral on  $\operatorname{Re} s > 0$ . This is **derived from audited inputs**.*

*Proof.* Every  $a_n$  is finite by endpoint flatness, so differentiation under the integral from the right gives [equation \(959\)](#). Divide the Taylor identity  $1 - A(s)$  by  $s$  to obtain [equation \(960\)](#). By [equation \(958\)](#),  $a_n$  is asymptotic in root scale to  $c^n (2n)!$ , and  $(2n)!$  is of order  $4^n (n!)^2$  up to subexponential factors. Hence the derivatives have Gevrey-2 growth. The Taylor coefficients  $a_n/n!$  have unbounded  $n$ th root, so the radius is zero. A holomorphic continuation would have a positive Taylor radius, a contradiction.  $\square$

*Remark 11.3.* This theorem explains a sharp asymmetry. The direct moment  $\mu(s) = \mathbb{E}Y^s$  is entire because it integrates against the flat density  $F'$ . The inverse Mellin transform probes powers of  $F$  itself; at  $s = 0$  it sees the full quadratic logarithmic endpoint profile and becomes only Gevrey-2. The Lambert- $W$  endpoint geometry is therefore visible not only pointwise in  $G$  but also as a transform-domain analyticity barrier.

## 12 Numerical experiments and reproducibility

The accompanying program `numerical_experiments.py` is a verification layer, not a source of proof. It uses three independent representations:

1. exact rational cumulants and moments generated from Bernoulli numbers;
2. a finite weighted Irwin–Hall spline for  $\sum_{j=1}^N 2^{-j} V_j$ , with exact Thue–Morse inclusion–exclusion signs and the unresolved tail replaced by its mean;
3. infinite Laplace/Fourier products and the Gamma–zeta Fourier series of  $P$ .

At spline level  $N = 11$ , the tests give:

- ordinary primitive errors below  $2.0 \times 10^{-8}$  at the sampled points;
- fractional shift errors below  $1.3 \times 10^{-8}$ ;
- fractional dyadic-lattice errors below  $1.3 \times 10^{-8}$ ;
- direct versus transmuted inverse fractional integrals agreeing to about  $2.2 \times 10^{-10}$ ;
- the mixed product value in [equation \(891\)](#);

- exterior fractional-Laplacian moment series agreeing with direct quadrature to  $10^{-7}$ – $10^{-6}$  using six terms;
- the exact integer-order defect  $R_n = 1$  and the order periodicity  $R_{q+1} = R_q$  to the working precision.

Representative half-order values are

$$\begin{aligned} R_{1/2}(0.1) &= 1.000004008430907\dots, \\ R_{1/2}(1) &= 0.999996976545554\dots, \\ R_{1/2}(10) &= 0.999998632323868\dots \end{aligned} \quad (961)$$

The complete machine-readable output is included as `numerical_results.txt`.

### 12.1 Numerical cautions

The finite-spline formula is an alternating sum with severe cancellation if evaluated too far on the right. The code therefore computes only on the stable left half and imposes the exact reflection  $F(1-x) = 1 - F(x)$ . Fractional kernels are desingularized by substitutions before adaptive quadrature. The Gamma–zeta defect is evaluated at high precision because its amplitude is only a few parts in  $10^6$ .

No decimal in this report is used to assert equality. Each exact identity has an independent symbolic proof in the preceding sections.

## 13 Conjectures and research directions

### 13.1 Higher inverse derivatives

The first inverse derivative has the sharp threshold in [theorem 5.8](#). The Lambert endpoint heuristic suggests a complete hierarchy.

**Conjecture 13.1** (Higher inverse-derivative threshold). *For every integer  $n \geq 1$  and  $r > 0$ ,*

$$G^{(n)} \in L^r(0, 1) \iff 0 < r \leq \frac{1}{n}. \quad (962)$$

*conjectural/open*

A formal endpoint differentiation of  $G(y) \approx \exp[-\sqrt{2 \log 2 \log(1/y)}]$  predicts  $G^{(n)}(y) \approx y^{-n} G(y)$  times powers of  $\log(1/y)$ . After  $y = e^{-v}$ , the  $L^r$  integrand has leading exponential  $\exp((nr-1)v - r\sqrt{2Lv})$ , giving exactly the stated threshold, including convergence on the boundary  $nr = 1$ .

### 13.2 Fractional endpoint transseries

**Conjecture 13.2** (Shifted Lambert–Gamma–zeta transseries). *For fixed noninteger  $\alpha$ , the functions  $I_{0+}^\alpha F(x)$  and  $D_{0+}^\alpha F(x)$  admit full small- $x$  transseries in the same lower-Lambert coordinate as  $F$ , with the periodic Gamma–zeta phase shifted explicitly by  $\alpha$ . The leading logarithmic quadratic term is unchanged; the algebraic prefactor and periodic phase acquire computable  $\alpha$ -dependent corrections.*

*conjectural/open*

The Bromwich representations [equations \(900\)](#) and [\(901\)](#) make this a concrete saddle problem. Multiplication by  $s^{\mp\alpha}$  changes the logarithmic saddle equation by an additive  $\alpha \log s$ , while the periodic  $P(\log_2 s)$  survives.

### 13.3 Physical-space sign structure of the dyadic defect

Define for  $q > 0$

$$\Delta_q(x) = I_{0+}^q \mathcal{F}(x) - \mathfrak{A}_q \mathcal{F}(x). \quad (963)$$

Its Laplace transform is known exactly from [theorem 9.6](#), but its zeros and sign changes in physical space are not.

**Conjecture 13.3** (Finite sign complexity on a dyadic cell). *For every  $q \in (0, 1)$ ,  $\Delta_q$  has finitely many sign changes on  $[0, 1]$ , and their number is uniformly bounded in  $q$ . The symmetry  $q \leftrightarrow 1 - q$  visible in the dominant Fourier mode of the multiplier should induce an approximate pairing of zero locations.* **conjectural/open**

A rigorous attack could combine complete monotonicity of Laplace kernels, variation-diminishing estimates, and a certified interval evaluation of the Gamma–zeta series.

### 13.4 A Jackson or $q$ -fractional calculus at base $1/2$

The repository’s Gaussian-binomial and half-base  $q$ -Pochhammer formulas suggest that the dyadic Appell family may be only one coordinate of a richer  $q = 1/2$  fractional calculus. Promising goals are:

- construct a Jackson-type fractional integral whose integer values reproduce the exact repeated Fabius primitives;
- identify its semigroup defect as a  $q$ -cocycle related to [equation \(931\)](#);
- express its values at dyadic rationals by analytic continuations of Gaussian-binomial coefficients;
- compare its Mellin symbol with the Gamma–zeta multiplier  $R_q$ .

This direction may provide the most direct bridge from fractional calculus to the repository’s  $q$ -binomial arithmetic.

### 13.5 Arithmetic and closed forms for Sobolev energies

The energies  $\mathcal{E}_{2m} = \|\text{up}^{(m)}\|_2^2$  and their fractional analogues are natural periods of the infinite sinc product. It is unknown whether any nontrivial member has a closed form in familiar constants.

**Conjecture 13.4** (No elementary gamma–zeta reduction for generic energy). *Except for identities forced by scaling or Parseval, the values  $\mathcal{E}_\alpha$  at generic rational  $\alpha \geq 0$  are not finite combinations of algebraic numbers, powers of  $\pi$ , and ordinary zeta values. They should instead define a new family of dyadic sinc periods.* **conjectural/open**

Useful next steps are high-precision computation, integer-relation searches with strict complexity penalties, Mellin–Barnes representations, and comparison with the spectral zeta already developed for the zero divisor of  $\Phi$ .

### 13.6 Nonlinear fractional Volterra equations for $G$

Equation [equation \(849\)](#) can be read as a nonlinear Volterra equation because  $G$  occurs both as the integration limit and as the inverse hidden in  $F$ . Differentiating in  $y$  gives a family of Abel-type equations. Questions include uniqueness in weighted Holder classes, stable numerical inversion, and whether the order variable  $\alpha$  can be used as a regularization parameter for computing  $G$  near its singular endpoints.

### 13.7 Complex order and monodromy

The operator  $\mathfrak{A}_q$  is entire in complex  $q$  after a branch for  $2^q$  is fixed. The exact defect becomes

$$R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)).$$

For complex  $q$ , the Fourier series of  $P$  grows in one imaginary direction and decays in the other. Determining the maximal strip of analytic continuation, the zero/pole structure of  $R_q$ , and the monodromy of complex-order fractional inversions is a concrete frontier problem.

### 13.8 Formalization program

A practical Lean development can proceed in increasing order of analytic difficulty:

- ✓ use the now-formalized integer primitive ladder and weighted polynomial formulas in `NormalizedVolterra.lean` and `FabiusAntiderivatives.lean`;
- ✓ use the now-formalized positive-real Riemann–Liouville kernel API; its increasing-affine and ordinary-derivative calculus and the three single-step Fabius–Rvachev order shifts are also now formalized;
- 1. formalize the remaining endpoint-up wrapper and fractional dyadic lattice;
- ✓ use the now-formalized support-free quantile pushforward and compact Fabius substitution from `QuantileTransport.lean`; the stronger nonlinear  $F$ – $G$  transmutation theorem remains to be proved;
- 2. define  $\mathfrak{A}_q$  and prove the cocycle and derivative ladder;
- 3. combine the existing exact periodic decomposition with the Thue–Morse Laplace product to formalize the defect and rigidity theorem;
- 4. only then address Riesz kernels and the Gevrey boundary theorem.

The unfinished early stages require mostly real integration, Gamma identities, and existing Fabius lemmas. The sixth stage reuses the strongest current Gamma–zeta formalization and would make the report’s flagship theorem machine checked.

## 14 Summary of exact formulas

For convenience, the principal identities are collected here.

| Object                        | Exact formula                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------|
| First primitive               | $\int_0^x F(t) dt = F(x/2), 0 \leq x \leq 1.$                                                   |
| $n$ th primitive              | $I_{0+}^n F(x) = 2^{\binom{n}{2}} F(x/2^n).$                                                    |
| $n$ th derivative             | $\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x).$                                 |
| Up CDF                        | $\int_{-\infty}^x \text{up}(t) dt = F((x+1)/2).$                                                |
| Weighted fractional primitive | $I^\alpha[t^p F] = \sum_{k=0}^p (-1)^k \binom{p}{k} (\alpha)_k x^{p-k} \mathcal{J}_{\alpha+k}.$ |
| Inverse primitive             | $\int G(y) dy = yG(y) - F(G(y)/2) + C.$                                                         |



| Object                        | Exact formula                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Inverse fractional integral   | $I^\alpha G(y) = \Gamma(\alpha + 1)^{-1} \int_0^{G(y)} (y - F(x))^\alpha dx.$                                             |
| Inverse fractional derivative | $D^\alpha G(y) = \Gamma(1 - \alpha)^{-1} \int_0^{G(y)} (y - F(x))^{-\alpha} dx, 0 < \alpha < 1.$                          |
| Mellin transform              | $\int_0^1 x^{s-1} F(x) dx = (1 - \mu(s))/s.$                                                                              |
| Fractional endpoint           | $\mathcal{J}_\alpha(1) = \mu(\alpha)/\Gamma(\alpha + 1).$                                                                 |
| Fractional dyadic lattice     | $\mathcal{J}_\beta(2^{-n}) = 2^{-n\beta - \binom{n}{2}} \mu(n + \beta)/\Gamma(n + \beta + 1).$                            |
| Laplace transform             | $\mathcal{L}F(s) = \phi(s)/s.$                                                                                            |
| Fourier transform of up       | $\widehat{\text{up}}(t) = \prod_{j \geq 1} \text{sinc}(t/2^j).$                                                           |
| Dyadic Appell family          | $\mathfrak{A}_q \mathcal{F}(x) = 2^{q(q-1)/2} \mathcal{F}(2^{-q}x).$                                                      |
| Exact fractional defect       | $R_q(s) = \exp(P(\log_2 s + q) - P(\log_2 s)).$                                                                           |
| Riesz tail                    | $(-\Delta)^{\alpha/2} \text{up}(x) = -C_{1,\alpha} \sum_{j \geq 0} (1 + \alpha)_{2j} m_{2j} / ((2j)!  x ^{1+\alpha+2j}).$ |
| Inverse Mellin boundary       | $\lim_{n \rightarrow \infty} (a_n / (2n)!)^{1/n} = 1/(2 \log 2).$                                                         |

## A Detailed proof supplements

### A.1 Entire continuation in power exponents

A recurring statement above is that an apparently restricted Mellin integral is entire. The mechanism is endpoint flatness. If  $h \in C^\infty([0, 1])$  and every derivative of  $h$  vanishes at 0, then for each  $N$  there is  $C_N$  with  $|h(x)| \leq C_N x^N$  on  $[0, 1]$ . Hence

$$\int_0^1 x^s h(x) dx$$

converges locally uniformly for  $\text{Re } s > -N - 1$ . Since  $N$  is arbitrary, the Mellin transform is entire. This applies to  $F$ ,  $F'$ , and every  $F^{(n)}$  near the left endpoint. It also justifies differentiation under the integral and the analytic continuation used in [equations \(841\) and \(879\)](#).

### A.2 Moment recurrence

The distributional fixed point

$$Y \stackrel{d}{=} \frac{V + Y'}{2}, \quad (964)$$

with  $V \sim \text{Unif}[0, 1]$  and  $Y' \stackrel{d}{=} Y$  independent gives, for integer  $n \geq 1$ ,

$$(2^n - 1)\mu_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{\mu_k}{n - k + 1}. \quad (965)$$

This is a convenient exact alternative to the Bell-polynomial construction. The two methods provide independent symbolic checks of the moment table.

### A.3 Why the inverse Mellin integral stops at $\operatorname{Re} s = 0$

For  $\operatorname{Re} s < 0$ ,  $F(x)^s = \exp(|\operatorname{Re} s|Q(x))$ . With  $x = e^{-u}$  and  $Q(e^{-u}) \sim cu^2$ , the integrand behaves like  $\exp(|\operatorname{Re} s|cu^2 - u)$ , which is not integrable. Thus  $A(s)$  has a genuine natural boundary for its defining integral at  $\operatorname{Re} s = 0$ . The transform  $\mathcal{M}_G(s)$  itself remains finite at  $s = 0$ , but its derivatives exhibit the same Gevrey-2 growth and it cannot cross the boundary analytically.

### A.4 A compact derivation of the Gamma-zeta defect

Writing  $u = \log s$ ,  $v = u/L$ , [equation \(939\)](#) is

$$\log \mathcal{L}(s) = -\frac{u^2}{2L} - \frac{u}{2} + P(v).$$

The transform of  $\mathfrak{A}_q \mathcal{F}$  is  $2^{q(q+1)/2} \mathcal{L}(2^q s)$ . Therefore

$$\begin{aligned} \log R_q(s) &= \frac{q(q+1)}{2}L + qu + \log \mathcal{L}(e^{u+qL}) - \log \mathcal{L}(e^u) \\ &= P(v+q) - P(v). \end{aligned}$$

Every polynomial term cancels. This cancellation is why the tiny periodic correction is not a secondary asymptotic detail here: it is the entire noninteger defect.

## B Numerical files and execution

The archive contains:

- `numerical_experiments.py`: the fully commented Python program;
- `numerical_results.txt`: exact moments and numerical checks;
- `moment_table.tex`: the generated rational moment table;
- `fractional_dilation_defect.png`: physical-space comparison;
- `gamma_zeta_fractional_defect.png`: transform multiplier plot.

Run from the report directory with

```
python numerical_experiments.py
```

The program requires Python 3 with `numpy`, `scipy`, `sympy`, `mpmath`, and `matplotlib`. All output paths are relative to the script.

## C Audit notes and source map

The recursive audit was organized by mathematical lineage rather than by filename count. The principal living inputs were:

- the formal-primary exposition `Fabius_Function_and_Rvachev_Up.tex`;
- the consolidated `semi-formalized-research-frontiers.tex`;
- the mathematical glossary and Lean walkthrough;
- active drafts on arithmetic/dyadic sampling, inverse asymptotics, Newton–Rvachev hierarchies, half-integer spectral identities, and sinc-product Fourier decay;

- corrected/transcribed primary papers by Fabius, Rvachev, and Arias de Reyna as retained in the repository.

Searches made at the absorbed report's snapshot used both terminology and formula fragments. At that snapshot no current-lineage match was found for the Riemann–Liouville order shift [equation \(914\)](#), the quantile transmutation [equations \(847\)](#) and [\(848\)](#), the exact defect [equation \(942\)](#), or the Gevrey limit [equation \(958\)](#). The audit does not exclude an independent result elsewhere in the mathematical literature.

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## Part VIII

# Integral, Transform, and Fractional-Calculus Frontiers for the Fabius–Rvachev System

## Abstract

The Fabius–Rvachev documentation corpus already contains a mature theory of the Thue–Morse sequence, the bounded Fabius function, Rvachev’s compactly supported up-function, exact dyadic values, repeated integration, infinite sinc products,  $q$ -binomial acceleration, and lower-Lambert endpoint asymptotics. This report performs a non-duplication audit of that recursive LaTeX corpus and develops an integral calculus that joins several structures not previously assembled there.

The main exact device is probabilistic. If  $X = \sum_{k \geq 1} 2^{-k} U_k$ , where the  $U_k$  are independent uniform variables on  $[0, 1]$ , then the bounded Fabius function  $F$  is the distribution function of  $X$ , while on  $[0, 1]$ ,  $\text{up}(x) = 1 - F(x) = \mathbb{P}(X > x)$ . This converts weighted integrals of  $F$  and up into expectations of primitives and truncated powers. It yields closed forms for monomial weights, exponential kernels, Mellin transforms, all integer repeated antiderivatives, and fractional Riemann–Liouville integrals. For  $p > -1$  and  $\alpha > 0$ , the fractional antiderivatives of  $x^p F(x)$  and  $x^p \text{up}(x)$  are expressed by incomplete beta kernels; at the endpoint they become convergent series in generalized moments  $m(s) = \mathbb{E}X^s$ . Reflection of the Fabius law gives an entire Newton series  $m(s) = \sum_{n \geq 0} (-1)^n \binom{s}{n} m_n$ , thereby covering fractional and negative powers by rational integer moments.

For the inverse Fabius function  $Q = F^{-1}$ , a layer-cake formula gives its ordinary and fractional antiderivatives, Laplace/Fourier transforms, mixed Mellin identities, and all quantile moments. A particularly direct bridge is

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1) (I_{0+}^\alpha Q)(1).$$

For the full up-law, the Cauchy transform  $S(z) = \int_{-1}^1 \text{up}(t)/(z - t) dt$  satisfies the new exact functional-differential equation  $S'(z) = 2[S(2z + 1) - S(2z - 1)]$ . Combining this with the rational moment sequence resolves a continued-fraction direction explicitly posed in an active repository draft:  $S$  has a symmetric Jacobi fraction with positive rational coefficients, whose first values are computed exactly and whose limit is  $1/4$ . Finally, nonlinear power integrals are identified with order statistics and shown to obey  $\int_0^1 \text{up}(x)^p dx \sim Q(1/p)$ .

All theorems designated as proved are supplied with arguments. Numerical work is used only as an independent check and is reproduced by the commented Python program included with the report. Novelty claims are relative to the audited repository, not assertions of exhaustive worldwide priority.

**Status discipline.** **Background / corpus result.** marks facts already present in the repository or standard literature. **Proved in this report.** marks deductions proved here and not located in the audited corpus in the displayed form. **Conditional statement.** requires an explicit regularity hypothesis. **Conjectural.** is an open proposal. Numerical agreement never substitutes for proof.

**Current Lean boundary.** The natural-order normalized Volterra kernel, its signed-global and bounded Fabius ladders, and every natural-monomial specialization are formalized in `NormalizedVolterra.lean` and `FabiusAntiderivatives.lean`. The positive-real fractional Volterra core, continuous-input semigroup, affine and ordinary-derivative calculus, stopped-power formulas, the total causal Rvachev primitive with its support cutoff, natural-order bridge, and semigroup, and three single-step Fabius–Rvachev shifts are now formalized in the fractional modules recorded in the consolidated status box. The support-free quantile pushforward and its Banach-valued substitution corollary are formalized in `QuantileTransport.lean`. The

survival and inverse layer-cake kernels are formalized in `SurvivalLayerCake.lean` and `InverseLayerCake.lean`. Finally, `InversePairIntegral.lean` proves the complete and variable-upper proper-compact absolutely-continuous inverse-pair formulas on arbitrary ordered source and level intervals, deriving monotonicity and measurable clamped extensions from the strict-order adjunction rather than assuming either clock measurable. The variable-upper generic and Fabius declarations are `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt` and `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`; the partial inverse-area specialization is `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`. Beyond the causal primitive declarations and the single positive-real up order shift based at  $-1$ , named iterated, explicit exterior-tail, endpoint-up, refinement, reflected, and piecewise support-tail formulas, higher inverse primitives, complex-power or Mellin continuation, fractional derivatives, iterated/lattice and endpoint-up wrappers, and wrappers not literally exported by these modules remain frontier material; reversed endpoints and improper singular endpoints are not covered by the proper-compact tranche.

# 1 Corpus audit and the non-duplication boundary

## 1.1 Coverage protocol

The directory `Analysis/FabiusFunction/docs` is a layered research environment rather than a single paper. At the pinned 27 August 2026 snapshot, an existing audit inventory records all 67 distinct `*.tex` paths. The archive README explains that historical files are frozen provenance objects and that byte-identical current duplicates are pruned. The non-formalized frontier directory is explicitly a quarantine boundary: its claims are not silently promoted into the formalized primary exposition [4, 3, 7].

The present audit used four passes.

1. The 67-path inventory supplied the recursive coverage scaffold.
2. Living primary manuscripts, the consolidated frontier volume, the Thue–Morse atlas, inverse-function reports, repeated-integration reports, and every active top-level frontier report were reopened on the current branch.
3. Frozen archive entries were compared by mathematical lineage, so that a superseded copy did not count as an independent novelty witness, while a distinctive historical argument remained part of the boundary.
4. Candidate phrases and neighboring concepts were searched thematically: Riemann–Liouville and Caputo calculus, fractional primitives, inverse layer-cake formulae, Cauchy and Hilbert transforms, Stieltjes/Jacobi fractions, order statistics, power integrals, and generalized negative moments.

This procedure located substantial adjacent work. In particular, the corpus already contains rational up-moments, Hankel determinants, monic orthogonal polynomials, Gaussian and Lobatto quadrature, and repeated-integration hierarchies. One active report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as a future direction [5]. Therefore the continued fraction below is presented as a resolution of a recorded gap, not as a topic absent from the research program.

## 1.2 What is imported and what is added

| Theme                      | Existing corpus foundation                                                                                            | Addition in this report                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Probability dictionary     | $F$ as a smooth distribution function; up as the density of a symmetric random series; exact Laplace/Fourier products | Universal survival-kernel calculus for arbitrary weights, including partial antiderivatives and truncated moments |
| Repeated integration       | Integer repeated-integral formulae and exact derivative towers                                                        | One incomplete-beta theorem interpolating all integer and fractional orders for $x^p F$ and $x^p \text{up}$       |
| Moments and Bernoulli data | Rational integer moments from Bernoulli cumulants and Bell recurrences                                                | Entire Newton interpolation $m(s)$ , Mellin continuation, negative powers, and logarithmic integrals              |
| Inverse Fabius             | Dyadic inverse germs, Richardson acceleration, endpoint Lambert–Gamma–zeta asymptotics, inverse elasticity            | Layer-cake transforms, quantile antiderivatives, mixed Mellin identities, and fractional $Q$ -calculus            |
| Orthogonal polynomials     | Hankel norms and positive Gaussian quadrature; J-fraction posed as an open problem                                    | Cauchy renormalization equation, explicit symmetric J-fraction, exact rational coefficients, and the $1/4$ limit  |
| Nonlinear integrals        | Endpoint/inverse asymptotics sufficient to prove slow variation of $Q$                                                | Order-statistic formulae and $\int \text{up}^p \sim Q(1/p)$ , with a second-order frontier conjecture             |
| Fractional derivatives     | No Riemann–Liouville/Caputo treatment located                                                                         | Exact fractional refinement equations for $F$ , $\text{up}$ , and $Q$ , plus a full-line Riesz refinement law     |

*Remark 1.1* (Priority boundary). The formulae below often use classical probability, Mellin, fractional-calculus, and orthogonal-polynomial tools. The novelty assertion is the corpus-relative combination and the displayed Fabius–Rvachev specializations. A worldwide priority claim would require a dedicated multilingual search using all historical normalizations of the Fabius and up-functions.

## 2 Normalizations and the probabilistic platform

### 2.1 The bounded Fabius law and the up-law

Let  $U_1, U_2, \dots$  be independent and uniformly distributed on  $[0, 1]$ , and put

$$X = \sum_{k=1}^{\infty} \frac{U_k}{2^k}. \quad (966)$$

Then  $0 < X < 1$  almost surely. Let

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1, \quad (967)$$

and extend it by  $F(x) = 0$  for  $x < 0$ ,  $F(x) = 1$  for  $x > 1$ . This is the bounded Fabius function in the normalization used throughout the repository. It is a strictly increasing  $C^\infty$  bijection of  $[0, 1]$ , flat at both endpoints, and satisfies

$$F(1-x) = 1 - F(x), \quad F'(x) = 2 \text{up}(2x-1). \quad (968)$$

Let  $V_k = 2U_k - 1$ , uniform on  $[-1, 1]$ , and define

$$Y = 2X - 1 = \sum_{k=1}^{\infty} \frac{V_k}{2^k}. \quad (969)$$

The density of  $Y$  is Rvachev's even up-function up, supported on  $[-1, 1]$ , normalized by  $\int \text{up} = 1$  and  $\text{up}(0) = 1$ . On the positive half of its support,

$$\boxed{\text{up}(x) = F(1 - x) = 1 - F(x) = \mathbb{P}(X > x), \quad 0 \leq x \leq 1.} \quad (970)$$

Thus the right half of the up-bump is exactly the survival function of the Fabius probability law.

The fixed-point decompositions

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad Y \stackrel{d}{=} \frac{V + Y'}{2} \quad (971)$$

use independent copies  $X', Y'$ . They imply

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)) \quad (972)$$

and, on  $0 < x < 1/2$ ,  $F'(x) = 2F(2x)$ . These are established corpus identities [1, 10].

## 2.2 Entire exponential products and rational cumulants

Define

$$A(z) = \mathbb{E}e^{zX} = \prod_{k=1}^{\infty} \frac{e^{z/2^k} - 1}{z/2^k}, \quad (973)$$

$$\Phi(z) = \mathbb{E}e^{zY} = e^{-z}A(2z) = \prod_{k=1}^{\infty} \frac{\sinh(z/2^k)}{z/2^k}. \quad (974)$$

Both are entire. With Bernoulli numbers in the convention  $B_1 = -1/2$ , the cumulants of  $X$  are

$$\kappa_1(X) = \frac{1}{2}, \quad \boxed{\kappa_n(X) = \frac{B_n}{n(2^n - 1)}, \quad n \geq 2,} \quad (975)$$

and those of  $Y$  are

$$\kappa_1(Y) = 0, \quad \boxed{\kappa_n(Y) = \frac{2^n B_n}{n(2^n - 1)}, \quad n \geq 2.} \quad (976)$$

Consequently all integer moments are rational and are generated by complete Bell polynomials:

$$m_n := \mathbb{E}X^n = \mathcal{B}_n(\kappa_1(X), \dots, \kappa_n(X)), \quad \mu_n := \mathbb{E}Y^n = \mathcal{B}_n(\kappa_1(Y), \dots, \kappa_n(Y)). \quad (977)$$

The first values are

$$(m_0, m_1, m_2, m_3, m_4, m_5, m_6) = \left(1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}\right), \quad (978)$$

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8) = \left(1, \frac{1}{9}, \frac{19}{675}, \frac{583}{59535}, \frac{132809}{32531625}\right). \quad (979)$$

## 3 The universal survival-kernel calculus

### 3.1 Arbitrary kernels and partial antiderivatives

**Theorem 3.1** (Survival-kernel identity). *Let  $k \in L^1(0, 1)$ , and put  $K(x) = \int_0^x k(t) dt$ . Then*

$$\boxed{\int_0^1 k(t) \text{up}(t) dt = \mathbb{E}K(X).} \quad (980)$$



More generally, for  $0 \leq x \leq 1$ ,

$$\boxed{\int_0^x k(t) \text{up}(t) dt = \mathbb{E}K(\min\{X, x\})}. \quad (981)$$

Both identities remain valid for complex-valued kernels whenever the displayed integrals are absolutely convergent. **Proved in this report.**

#### Exact Lean crosswalk

The partial and full identities are machine checked, in the stronger form of an interval-integrable kernel with values in an arbitrary real Banach space, as `Fabius.ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral_min` and `Fabius.ProbabilityRepresentation.intervalIntegral_rvachevUp_smul_eq_integral` in `ProbabilityLaplaceMoments.lean`. Taking the Banach space to be  $\mathbb{C}$  gives the complex-kernel case. Their reusable finite-measure versions, including a support-free clamped theorem, are `Fabius.intervalIntegral_survival_smul_eq_integral_clamp`, `Fabius.intervalIntegral_survival_smul_eq_integral_min_of_ae_mem_Icc`, and `Fabius.intervalIntegral_survival_smul_eq_integral_of_ae_mem_Icc` in `SurvivalLayerCake.lean`.

*Proof.* By (970),  $\text{up}(t) = \mathbb{E}\mathbf{1}_{\{t < X\}}$ . Tonelli–Fubini gives

$$\int_0^x k(t) \text{up}(t) dt = \mathbb{E} \int_0^x k(t) \mathbf{1}_{\{t < X\}} dt = \mathbb{E} \int_0^{\min(X, x)} k(t) dt.$$

Taking  $x = 1$  proves (980).  $\square$

**Editorial Lean status (28 August 2026).** The full- and partial-endpoint identities (980) and (981) are formalized in a strictly stronger Banach-valued family. The full-endpoint theorem is `Fabius.ProbabilityRepresentation.interval_Icc_intervalIntegral_eq_intervalIntegral_smul_rvachevUp`; its arbitrary compact-interval finite-measure engine is `Fabius.interval_Icc_intervalIntegral_eq_intervalIntegral_smul_survival`. The clipped finite-measure and Fabius declarations are `Fabius.interval_Icc_intervalIntegral_min_eq_intervalIntegral_smul_survival` and `Fabius.ProbabilityRepresentation.interval_Icc_intervalIntegral_min_eq_intervalIntegral_smul_rvachevUp`. Only Bochner integrability is assumed, so the family covers real and complex kernels. The full- and partial-endpoint scalar wrappers `Fabius.ProbabilityRepresentation.intervalIntegral_mul_rvachevUp_eq_integral_Icc_intervalIntegral` and `Fabius.ProbabilityRepresentation.intervalIntegral_mul_rvachevUp_eq_integral_Icc_intervalIntegral_min` have exactly the orientations of the two displayed report identities.

**Corollary 3.2** (Monomial antiderivatives). For  $\Re p > -1$  and  $0 \leq x \leq 1$ ,

$$\int_0^x t^p \text{up}(t) dt = \frac{\mathbb{E} \min(X, x)^{p+1}}{p+1}, \quad (982)$$

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} - \mathbb{E} \min(X, x)^{p+1}}{p+1}. \quad (983)$$

At  $x = 1$ , writing  $m(s) = \mathbb{E}X^s$ ,

$$\boxed{\int_0^1 t^p \text{up}(t) dt = \frac{m(p+1)}{p+1}}, \quad (984)$$

$$\boxed{\int_0^1 t^p F(t) \, dt = \frac{1 - m(p+1)}{p+1}.} \quad (985)$$

**Proved in this report.**

For nonnegative integers, this gives for example

|                                           | $p = 0$ | $p = 1$ | $p = 2$ |
|-------------------------------------------|---------|---------|---------|
| $\int_0^1 x^p \operatorname{up}(x) \, dx$ | 1/2     | 5/36    | 1/18    |
| $\int_0^1 x^p F(x) \, dx$                 | 1/2     | 13/36   | 5/18    |

*Remark 3.3* (Negative powers). The survival integral in (984) has the sharp local restriction  $\Re p > -1$ , because  $\operatorname{up}(0) = 1$ . In contrast,  $F$  is flat at zero, so  $\int_0^1 x^p F(x) \, dx$  converges for every  $p \in \mathbb{C}$ ; the right side of (985) supplies its entire continuation after removing the apparent singularity at  $p = -1$ . In particular,

$$\int_0^1 \frac{F(x)}{x} \, dx = -m'(0) = \mathbb{E}[-\log X]. \quad (986)$$

### 3.2 Exponential kernels and ordinary transforms

Taking  $k(t) = e^{zt}$  in theorem 3.1 yields the entire identities

$$\boxed{\int_0^1 \operatorname{up}(t) e^{zt} \, dt = \frac{A(z) - 1}{z},} \quad (987)$$

$$\boxed{\int_0^1 F(t) e^{zt} \, dt = \frac{e^z - A(z)}{z},} \quad (988)$$

with removable values at  $z = 0$ . Differentiation under the integral gives all monomial-exponential combinations:

$$\int_0^1 t^n \operatorname{up}(t) e^{zt} \, dt = \frac{d^n}{dz^n} \left( \frac{A(z) - 1}{z} \right), \quad (989)$$

$$\int_0^1 t^n F(t) e^{zt} \, dt = \frac{d^n}{dz^n} \left( \frac{e^z - A(z)}{z} \right). \quad (990)$$

At  $z = -2\pi i\xi$ , these are the finite-interval Fourier transforms of  $F$  and the positive half of  $\operatorname{up}$ .

### 3.3 Derivative-weighted beta integrals

**Theorem 3.4** (Probability-integral-transform family). *Let  $a, b \in \mathbb{C}$  with  $\Re a, \Re b > -1$ . Then*

$$\boxed{\int_0^1 F(x)^a (1 - F(x))^b F'(x) \, dx = B(a+1, b+1).} \quad (991)$$

More generally, for any integrable  $\varphi$ ,

$$\int_0^1 \varphi(F(x)) F'(x) \, dx = \int_0^1 \varphi(u) \, du. \quad (992)$$

Equivalently, since  $F'(x) = 2 \operatorname{up}(2x - 1)$ , this is an exact family of integrals mixing the  $\operatorname{up}$ -function with powers of the Fabius function. **Proved in this report.**

*Proof.* Use the substitution  $u = F(x)$ , legitimate because  $F$  is a smooth strictly increasing bijection of  $[0, 1]$ .  $\square$

The quantile-weighted extension is

$$\int_0^1 x^r F(x)^a (1 - F(x))^b F'(x) dx = \int_0^1 Q(u)^r u^a (1 - u)^b du, \quad (993)$$

where  $Q = F^{-1}$ . Thus every beta-weighted moment of the inverse is a forward Fabius integral.

## 4 Generalized moments and the Mellin transform

### 4.1 An entire Newton series from reflection

The flat endpoints imply more than the existence of all negative moments.

**Lemma 4.1** (Super-polynomial decay of integer moments). *For every  $N > 0$ ,*

$$m_n = \mathbb{E}X^n = \mathcal{O}_N(n^{-N}) \quad (n \rightarrow \infty). \quad (994)$$

*Proved in this report.*

*Proof.* By reflection,  $\mathbb{P}(X > 1 - t) = F(t)$ . Endpoint flatness gives  $F(t) = \mathcal{O}_M(t^M)$  for every  $M$ . The layer-cake formula

$$m_n = n \int_0^1 x^{n-1} \mathbb{P}(X > x) dx$$

followed by  $x = 1 - t$ , a split at fixed small  $t$ , and  $(1 - t)^{n-1} \leq e^{-(n-1)t}$  gives  $m_n = \mathcal{O}_M(n^{-M})$ .  $\square$

**Theorem 4.2** (Entire Newton interpolation of Fabius moments). *For every  $s \in \mathbb{C}$ ,*

$$m(s) := \mathbb{E}X^s = \sum_{n=0}^{\infty} (-1)^n \binom{s}{n} m_n. \quad (995)$$

*The series converges absolutely and locally uniformly in  $s$ , hence defines an entire function. Its coefficients  $m_n$  are rational Bell polynomials in the Bernoulli–Mersenne cumulants (975). Proved in this report.*

*Proof.* The reflection law gives  $1 - X \stackrel{d}{=} X$ . For every fixed  $0 < X < 1$ ,

$$X^s = (1 - (1 - X))^s = \sum_{n \geq 0} (-1)^n \binom{s}{n} (1 - X)^n.$$

On compact subsets of the  $s$ -plane, the generalized binomial coefficients grow at most polynomially in  $n$ , while lemma 4.1 makes  $m_n$  smaller than every inverse power. Thus expectation and summation may be interchanged, and local uniform convergence follows.  $\square$

Important specializations are

$$m(-r) = \sum_{n=0}^{\infty} \binom{n+r-1}{n} m_n, \quad r \in \mathbb{N}, \quad (996)$$

$$-m'(0) = \sum_{n=1}^{\infty} \frac{m_n}{n}, \quad (997)$$

so (986) has a convergent series of rational terms. Numerically,

$$\int_0^1 \frac{F(x)}{x} dx = 0.7582399961\dots, \quad \int_0^1 \frac{F(x)}{x^2} dx = m(-1) - 1 = 1.314885\dots \quad (998)$$

The decimals are checks, not claimed closed constants.

## 4.2 A three-function Mellin dictionary

**Theorem 4.3** (Mellin transform triad). *For  $\Re s > 0$ ,*

$$\mathcal{M}[\text{up}|_{[0,1]}](s) = \int_0^1 x^{s-1} \text{up}(x) \, dx = \frac{m(s)}{s}, \quad (999)$$

$$\mathcal{M}[F](s) = \int_0^1 x^{s-1} F(x) \, dx = \frac{1 - m(s)}{s}, \quad (1000)$$

$$\mathcal{M}[F'](s) = \int_0^1 x^{s-1} F'(x) \, dx = m(s-1). \quad (1001)$$

*The first extends meromorphically to  $\mathbb{C}$  with one simple pole at  $s = 0$ , of residue one. The second and third extend entire. **Proved in this report.***

*Proof.* The first identity is (984) with  $p = s - 1$ . The second follows from  $F = 1 - \text{up}$ , and the third is the definition of expectation under the density  $F'$ . Entire continuation uses theorem 4.2; the zero of  $1 - m(s)$  at  $s = 0$  removes the apparent pole in (1000).  $\square$

The Mellin triad isolates a striking analytic asymmetry. The positive half of the up-function equals one at the origin and therefore contributes the single Mellin pole. The Fabius function and its density are flat there, and their Mellin transforms are entire.

## 5 Inverse Fabius integral calculus

Let

$$Q = F^{-1} : [0, 1] \rightarrow [0, 1]. \quad (1002)$$

It satisfies  $Q(1 - u) = 1 - Q(u)$ .

### 5.1 Area identity and ordinary antiderivatives

**Theorem 5.1** (Inverse-area antiderivative). *For  $0 \leq u \leq 1$ ,*

$$\boxed{\int_0^u Q(v) \, dv = uQ(u) - \int_0^{Q(u)} F(x) \, dx.} \quad (1003)$$

*In particular,  $\int_0^1 Q(u) \, du = 1/2$ . **Proved in this report.***

The terminal  $u = 1$  conclusion is exactly `Fabius.intervalIntegral_fabiusInv_eq_one_half`; the companion theorem `Fabius.intervalIntegral_fabiusInv_eq_intervalIntegral_rvachevUp` identifies the same inverse area with the unit-interval area under `up` and follows the full weighted identity `Fabius.intervalIntegral_mul_fabiusInv_eq`. The one-half theorem is proved directly from inverse reflection. The proper-compact two-function theorem `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv` in `InversePairIntegral.lean` also specializes, with both factors equal to the identity, to the exact area-sum corollary `Fabius.intervalIntegral_fabiusReal_add_fabiusInv_eq_one`. The variable- $u$  identity (1003) is exactly `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`; it is the identity-function specialization of `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`, whose generic proper-compact parent is `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt`.

*Proof.* The rectangle  $[0, Q(u)] \times [0, u]$  is partitioned, up to the graph, into the area below  $F$  and the area to the left of  $Q$ . Equivalently, substitute  $v = F(x)$  and integrate by parts.  $\square$

## 5.2 A general layer-cake transform

**Theorem 5.2** (Forward-inverse layer cake). *Let  $\mathbb{K} = \mathbb{R}$  or  $\mathbb{C}$ , let  $\phi \in L^1((0, 1); \mathbb{K})$ , and let  $\Psi : \mathbb{R} \rightarrow \mathbb{R}$  be absolutely continuous on  $[0, 1]$ , with  $\Psi(0) = 0$ . Then*

$$\int_0^1 \phi(u) \Psi(Q(u)) \, du = \int_0^1 \Psi'(x) \left( \int_{F(x)}^1 \phi(u) \, du \right) \, dx. \quad (1004)$$

*Proved in this report.*

### Exact Lean scope of the inverse layer cake

The displayed formula is exactly `Fabius.intervalIntegral_mul_comp_fabiusInv_of_absolutelyContinuousOnInterval` in `InverseLayerCake.lean`. Its generic compact-rectangle parent is the stronger subtraction identity

$$\int_c^d \phi(u) (\Psi(Q(u)) - \Psi(a)) \, du = \int_a^b \Psi'(x) \left( \int_{C(x)}^d \phi(u) \, du \right) \, dx, \quad (1005)$$

namely `Fabius.intervalIntegral_mul_comp_sub_of_lt_iff_lt_of_absolutelyContinuousOnInterval`; `Fabius.intervalIntegral_mul_comp_of_lt_iff_lt_of_absolutelyContinuousOnInterval` is its  $\Psi(a) = 0$  corollary. Both accept real or complex  $L^1$  weights and a real-valued absolutely continuous primitive. Absolute continuity supplies derivative integrability, so no separate integrability assumption is needed on the right. Ordered source and level intervals may degenerate.

The order helper `Fabius.galoisConnection_Icc_restrict_of_lt_iff_lt` proves that the compact-interval clock restrictions form a Galois connection. Consequently the generic explicit-primitive and pointwise- $C^1$  theorems now derive the measurable clamped inverse used by the subgraph argument and assume no clock measurability. Their Fabius forms retain the older, wider codomain interfaces: `Fabius.intervalIntegral_smul_intervalIntegral_fabiusInv` takes an interval-integrable Banach-valued kernel and writes  $\Psi$  explicitly as its primitive; `Fabius.intervalIntegral_smul_comp_fabiusInv` gives the displayed composition when  $\Psi(0) = 0$ , its derivative is continuous on  $[0, 1]$ , and the derivative identity holds pointwise there. Both allow real or complex scalar weights and Banach-valued primitives. The arbitrary-AC theorem does not extend that primitive codomain: it is real-valued. The separate `InversePairIntegral.lean` tranche exports the proper compact variable-upper identity just crosswalked above. Neither tranche exports reversed-endpoint, improper-singular, higher- or fractional-inverse, or named Fourier, Laplace, or Mellin corollaries.

*Proof.* Write  $\Psi(Q(u)) = \int_0^1 \Psi'(x) \mathbf{1}_{\{x < Q(u)\}} \, dx$ . Since  $x < Q(u)$  is equivalent to  $F(x) < u$ , Fubini gives the result.  $\square$

For real  $p > 0$ , taking  $\Psi(x) = x^p$  gives

$$\int_0^1 \phi(u) Q(u)^p \, du = p \int_0^1 x^{p-1} \left( \int_{F(x)}^1 \phi(u) \, du \right) \, dx. \quad (1006)$$

The report extends the same display to complex  $p$  with  $\Re p > 0$ . That complex-valued-primitive continuation is not part of the arbitrary-AC Lean wrapper. Taking  $\phi = 1$  proves the exact quantile-moment identity

$$\int_0^1 Q(u)^p \, du = m(p), \quad \Re p > 0, \quad (1007)$$

with analytic continuation supplied by [theorem 4.2](#) where desired.

### 5.3 Laplace, Fourier, and mixed Mellin transforms of the inverse

For  $s \neq 0$ , [theorem 5.2](#) with  $\Psi(x) = x$  gives

$$\int_0^1 e^{-su} Q(u) \, du = \frac{1}{s} \int_0^1 (e^{-sF(x)} - e^{-s}) \, dx. \quad (1008)$$

Similarly, for  $t \neq 0$ ,

$$\int_0^1 e^{itu} Q(u) \, du = \frac{1}{it} \int_0^1 (e^{it} - e^{itF(x)}) \, dx. \quad (1009)$$

These identities replace a transform of a difficult inverse by nonlinear but forward evaluations of  $F$ .

A two-parameter Mellin identity follows by taking  $\phi(u) = u^{s-1}$ .

**Theorem 5.3** (Mixed forward-inverse Mellin identity). *For  $\Re s > 0$ ,  $\Re p > 0$ ,*

$$\int_0^1 u^{s-1} Q(u)^p \, du = \frac{1}{s} - \frac{p}{s} \int_0^1 x^{p-1} F(x)^s \, dx. \quad (1010)$$

*Proved in this report.*

This makes nonlinear powers  $F^s$  and inverse moments two faces of one transform. For integer  $s$ , the right side has the order-statistic interpretation developed in [appendix 10](#).

### 5.4 All ordinary derivatives of the inverse

Put  $f = F'$ , and, after differentiating with respect to the spatial variable  $x$ , define

$$R_1(x) = \frac{1}{f(x)}, \quad R_{n+1}(x) = \frac{R'_n(x)}{f(x)}. \quad (1011)$$

Then

$$Q^{(n)}(u) = R_n(Q(u)), \quad 0 < u < 1. \quad (1012)$$

This is simply the operator formula

$$Q^{(n)}(u) = \left( \frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \Big|_{x=Q(u)}. \quad (1013)$$

The first cases are

$$Q' = f^{-1}, \quad Q'' = -\frac{f'}{f^3}, \quad (1014)$$

$$Q''' = \frac{3(f')^2 - f f''}{f^5}, \quad Q'''' = \frac{-15(f')^3 + 10 f f' f'' - f^2 f'''}{f^7}, \quad (1015)$$

all evaluated at  $x = Q(u)$ . The recurrence is preferable to a single large Faà di Bruno polynomial when high orders are required.

## 6 Integer-order derivative and antiderivative hierarchies

### 6.1 The Thue–Morse derivative tower

Let  $s_2(j)$  denote the binary digit sum. Iterating (972) gives the established exact formula

$$\text{up}^{(n)}(x) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} (-1)^{s_2(j)} \text{up}(2^n x + 2^n - 1 - 2j). \quad (1016)$$

On the left edge of the Fabius interval,

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 \leq x \leq 2^{-n}. \quad (1017)$$

These are imported corpus results; their role here is to connect the integral formulae to arbitrary derivative order.

For any nonnegative integer  $m$ , integration by parts and endpoint flatness give

$$\int_{-1}^1 x^m \text{up}^{(n)}(x) dx = \begin{cases} (-1)^n \frac{m!}{(m-n)!} \mu_{m-n}, & m \geq n, \\ 0, & m < n, \end{cases} \quad (1018)$$

and, for  $n \geq 1$ ,

$$\int_0^1 x^p F^{(n)}(x) dx = (-1)^{n-1} p^{n-1} m(p-n+1), \quad (1019)$$

initially where the integration-by-parts calculation is classical and then by analytic continuation. Exponential weights similarly yield

$$\int_{-1}^1 e^{zx} \text{up}^{(n)}(x) dx = (-z)^n \Phi(z), \quad (1020)$$

$$\int_0^1 e^{zx} F^{(n)}(x) dx = (-z)^{n-1} A(z), \quad n \geq 1. \quad (1021)$$

### 6.2 All repeated antiderivatives as truncated moments

Throughout the truncated-moment formulas, powers of a positive part mean

$$u_+^s := \mathbf{1}_{\{u>0\}} u^s.$$

Thus the zeroth truncated power is the strict-positivity indicator, while negative exponents are evaluated only where  $u > 0$ . Replacing  $>$  by  $\geq$  does not change any expectation below because the relevant laws are continuous.

For integer  $n \geq 1$ ,

$$(I_{0+}^n F)(x) = \frac{1}{n!} \mathbb{E}(x - X)_+^n, \quad (1022)$$

$$(I_{-1+}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x - Y)_+^{n-1}, \quad (1023)$$

$$(I_{0+}^n Q)(u) = \frac{1}{n!} \int_0^{Q(u)} (u - F(x))^n dx. \quad (1024)$$

The first and third are special cases of fractional theorems below. For  $x > 1$ , the up antiderivative becomes the finite moment polynomial

$$(I_{-1+}^n \text{ up})(x) = \frac{1}{(n-1)!} \sum_{r=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2r} \mu_{2r} x^{n-1-2r}. \quad (1025)$$

Thus the rational up-moments are precisely the coefficients of all exterior ordinary antiderivatives.

## 7 Fractional calculus

### 7.1 Conventions

For  $\alpha > 0$ , the left Riemann–Liouville integral is

$$(I_{a+}^\alpha g)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} g(t) dt. \quad (1026)$$

For  $a \leq x$ , the real-normed-space-valued version of this positive-real kernel is `FabiusFunction.FractionalVolterra`, with its semigroup in `FabiusFunction.FractionalVolterraSemigroup` and the exact API recorded in the consolidated status box. `FabiusFunction.FractionalVolterraCalculus` additionally proves increasing-affine covariance and ordinary-derivative order raising, and `FabiusFunction.FabiusFractionalVolterra` defines the total causal Rvachev fractional primitive, proves its support-truncated integral, natural-order bridge and positive-order semigroup, and proves the three positive-real single-step Fabius–Rvachev shifts. These operator modules prove no reversed-endpoint fractional interpretation, semigroup for merely interval-integrable inputs or noncomplete targets, fractional derivative, complex order, shifted dyadic-lattice, endpoint-moment, transform/tail-series, or inverse/quantile identity. Separately, `FabiusFunction.FabiusFractionalIntegral` proves the all-real-endpoint positive-order Fabius stopped-power identity (14) and both unit-endpoint moment forms (15). It does not supply additional Rvachev transform/tail series, negative-endpoint signed branches, full piecewise/refinement hierarchy, quantile-transmutation, inverse-function, complex-order, or fractional-derivative formulas. The raw beta kernel is interval-integrable for two positive orders and a non-strict endpoint inequality, while its raw and normalized evaluations assume a strict endpoint inequality. For  $n-1 < \alpha < n$ ,

$$D_{a+}^\alpha g = \frac{d^n}{dx^n} I_{a+}^{n-\alpha} g, \quad {}^C D_{a+}^\alpha g = I_{a+}^{n-\alpha} g^{(n)}. \quad (1027)$$

These conventions follow the standard monograph of Samko, Kilbas, and Marichev [12]. Because  $F$  and up are flat at their relevant left endpoints, the usual boundary correction terms between Riemann–Liouville and Caputo derivatives vanish whenever both sides are classically defined.

### 7.2 Fractional primitives and derivatives of the Fabius function

**Theorem 7.1** (Fractional truncated-power formula for  $F$ ). *For  $\alpha > 0$  and real  $x$ , with  $F$  extended as a distribution function,*

$$(I_{0+}^\alpha F)(x) = \frac{1}{\Gamma(\alpha+1)} \mathbb{E}(x-X)_+^\alpha. \quad (1028)$$

For  $0 < \alpha < 1$  and  $0 < x < 1$ ,

$$(D_{0+}^\alpha F)(x) = ({}^C D_{0+}^\alpha F)(x) = \frac{1}{\Gamma(1-\alpha)} \mathbb{E}(x-X)_+^{-\alpha}. \quad (1029)$$

*Proved in this report.*



*Proof.* Insert  $F(t) = \mathbb{E}\mathbf{1}_{\{X \leq t\}}$  into (1026) and interchange expectation and integration. The inner integral from  $X$  to  $x$  equals  $(x - X)^\alpha / \alpha$ . For  $0 < \alpha < 1$ , differentiate the formula with  $\alpha$  replaced by  $1 - \alpha$ . The local singularity is integrable because the density of  $X$  is continuous. Equality with the Caputo derivative uses  $F(0) = 0$ .  $\square$

The random fixed point (971) gives a fractional refinement law which interpolates the ordinary repeated-integration hierarchy.

**Theorem 7.2** (Fractional Fabius refinement). *Let  $J_\alpha(x) = (I_{0+}^\alpha F)(x)$ , with  $J_\alpha(x) = 0$  for  $x < 0$ . Then for every  $\alpha > 0$ ,*

$$J_\alpha(x) = 2^{-\alpha} (J_{\alpha+1}(2x) - J_{\alpha+1}(2x - 1)). \quad (1030)$$

On  $0 < x < 1/2$ , the corresponding derivative scaling is

$$D_{0+}^{\alpha+1} F(x) = 2^{\alpha+1} (D_{0+}^\alpha F)(2x), \quad (1031)$$

whenever interpreted classically, and otherwise in the distributional sense. *Proved in this report.*

*Proof.* From  $X \stackrel{d}{=} (U + X')/2$ ,

$$\mathbb{E}(x - X)_+^\alpha = 2^{-\alpha} \mathbb{E}(2x - U - X')_+^\alpha.$$

Averaging first over  $U \in [0, 1]$  gives the difference of two  $(\alpha + 1)$ -st truncated powers divided by  $\alpha + 1$ , proving (1030). The derivative law follows from  $F'(x) = 2F(2x)$ , the scaling rule for left fractional derivatives, and endpoint flatness.  $\square$

### 7.3 Weighted fractional antiderivatives: the incomplete-beta master theorem

For  $a, b > 0$ , write

$$B_z(a, b) = \int_0^z t^{a-1} (1 - t)^{b-1} dt. \quad (1032)$$

**Theorem 7.3** (Fractional monomial–Fabius/up integrals). *Let  $\alpha > 0$ ,  $\Re p > -1$ , and  $0 < x \leq 1$ . Then*

$$I_{0+}^\alpha [t^p \text{up}(t)](x) = \frac{x^{p+\alpha}}{\Gamma(\alpha)} \mathbb{E}[B_{\min(X/x, 1)}(p+1, \alpha)], \quad (1033)$$

$$I_{0+}^\alpha [t^p F(t)](x) = \frac{x^{p+\alpha}}{\Gamma(\alpha)} \mathbb{E}[\mathbf{1}_{\{X < x\}} B_{1-X/x}(\alpha, p+1)]. \quad (1034)$$

At  $x = 1$ ,

$$I_{0+}^\alpha [t^p \text{up}(t)](1) = \frac{1}{\Gamma(\alpha)} \sum_{j=0}^{\infty} \frac{(1-\alpha)_j}{j! (p+j+1)} m(p+j+1), \quad (1035)$$

$$I_{0+}^\alpha [t^p F(t)](1) = \frac{\Gamma(p+1)}{\Gamma(p+\alpha+1)} - I_{0+}^\alpha [t^p \text{up}(t)](1). \quad (1036)$$

The series converges absolutely for  $\alpha > 0$ . If  $\alpha = n \in \mathbb{N}$ , it terminates:

$$I_{0+}^n [t^p \text{up}(t)](1) = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{m(p+j+1)}{p+j+1}. \quad (1037)$$

*Proved in this report.*

*Proof.* For the up-function, insert  $\text{up}(t) = \mathbb{E}1_{\{t < X\}}$  into the Riemann–Liouville integral. The inner range is  $0 < t < \min(X, x)$ ; scaling  $t = xr$  gives (1033). For  $F$ , the inner range is  $X < t < x$ , producing the complementary incomplete beta integral in (1034).

At  $x = 1$ , expand  $(1 - t)^{\alpha-1} = \sum_{j \geq 0} (1 - \alpha)_j t^j / j!$  in  $B_X(p + 1, \alpha)$ , then take expectations. The coefficients are  $\mathcal{O}(j^{-\alpha})$ , the extra denominator is  $j^{-1}$ , and  $|m(p + j + 1)| \leq 1$  in the initial domain, giving absolute convergence. For integer  $\alpha$ , the Pochhammer symbol truncates.  $\square$

**Remark 7.4** (Why this answers several integral questions at once). Setting  $\alpha = 1$  recovers the ordinary antiderivatives (982)–(983). Integer  $\alpha > 1$  gives every repeated antiderivative as a finite combination of generalized moments. Noninteger  $\alpha$  gives a convergent moment series. Negative and fractional  $p$  are allowed down to the sharp barrier  $\Re p = -1$  for the up-integral; the Fabius integral itself continues beyond that barrier because of endpoint flatness.

## 7.4 Fractional primitives of the up-function

Define, for  $\alpha > 0$ ,

$$H_\alpha(x) = (I_{-1+}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \mathbb{E}(x - Y)_+^{\alpha-1}. \quad (1038)$$

**Theorem 7.5** (Fractional up-refinement). *For all  $\alpha > 0$  and real  $x$ ,*

$$H_\alpha(x) = 2^{-\alpha} (H_{\alpha+1}(2x + 1) - H_{\alpha+1}(2x - 1)). \quad (1039)$$

For  $x > 1$ ,

$$H_\alpha(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)} \sum_{r=0}^{\infty} \binom{\alpha-1}{2r} \mu_{2r} x^{-2r}. \quad (1040)$$

The exterior series converges absolutely; for integer  $\alpha$  it reduces to (1025). *Proved in this report.*

*Proof.* Use  $Y \stackrel{d}{=} (V + Y')/2$  and average first over uniform  $V \in [-1, 1]$ . The integral of  $(2x - Y' - V)_+^{\alpha-1}$  over  $V$  produces the difference of two  $\alpha$ -th powers and the factor  $2^{-\alpha}$ . For  $x > 1$ , expand  $(1 - Y/x)^{\alpha-1}$ ; symmetry removes all odd moments.  $\square$

For general derivative order  $n - 1 < \alpha < n$ , the one-sided Riemann–Liouville derivative is exactly

$$D_{-1+}^\alpha \text{up}(x) = \frac{d^n}{dx^n} H_{n-\alpha}(x). \quad (1041)$$

This formula is computationally useful because theorem 7.5 reduces every  $H_\beta$  to scaled copies one order higher.

## 7.5 Fractional calculus of the inverse Fabius function

**Theorem 7.6** (Fractional inverse layer cake). *For  $\alpha > 0$  and  $0 \leq u \leq 1$ ,*

$$(I_{0+}^\alpha Q)(u) = \frac{1}{\Gamma(\alpha + 1)} \int_0^{Q(u)} (u - F(x))^\alpha dx. \quad (1042)$$

At the endpoint,

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1) (I_{0+}^\alpha Q)(1). \quad (1043)$$

For  $0 < \alpha < 1$ ,

$$(D_{0+}^\alpha Q)(u) = ({}^C D_{0+}^\alpha Q)(u) = \frac{1}{\Gamma(1 - \alpha)} \int_0^{Q(u)} (u - F(x))^{-\alpha} dx. \quad (1044)$$

*Proved in this report.*

*Proof.* Apply [theorem 5.2](#) with  $\phi(v) = (u - v)_+^{\alpha-1}/\Gamma(\alpha)$  and  $\Psi(x) = x$ . At  $u = 1$ , use  $1 - F(x) = \text{up}(x)$ . For the derivative, differentiate the formula for  $I_{0+}^{1-\alpha}Q$ . The moving-boundary term vanishes because its positive power is zero at  $x = Q(u)$ , and the remaining singularity is integrable for  $\alpha < 1$ . Since  $Q(0) = 0$ , the Riemann–Liouville and Caputo derivatives agree.  $\square$

## 7.6 A full-line Riesz refinement law

Let  $D_R^\alpha$  be the full-line Riesz derivative with Fourier multiplier  $|2\pi\xi|^\alpha$ . Applying it to [\(972\)](#), and using the exact scaling law, gives

$$\frac{d}{dx}(D_R^\alpha \text{up})(x) = 2^{\alpha+1} [(D_R^\alpha \text{up})(2x+1) - (D_R^\alpha \text{up})(2x-1)]. \quad (1045)$$

The identity holds in tempered distributions for every  $\alpha \geq 0$ , and classically wherever the Riesz derivative is represented by an ordinary function. It is the fractional-order analogue of the differential refinement equation with no ambiguity from a moving one-sided terminal.

# 8 Cauchy, Stieltjes, and Hilbert transforms

## 8.1 The Cauchy transform and its representations

Define

$$S(z) = \int_{-1}^1 \frac{\text{up}(t)}{z-t} dt, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (1046)$$

Since  $\text{up}$  is even and has moments  $\mu_{2r}$ ,

$$S(z) = \sum_{r=0}^{\infty} \frac{\mu_{2r}}{z^{2r+1}}, \quad |z| > 1. \quad (1047)$$

For  $\Re z > 1$ , the exponential resolvent gives

$$S(z) = \int_0^\infty e^{-zs} \Phi(s) ds. \quad (1048)$$

The random fixed point  $Y = (V + Y')/2$  also gives the nonlinear logarithmic identity

$$S(z) = \int_{-1}^1 \text{up}(y) \log\left(\frac{2z-y+1}{2z-y-1}\right) dy, \quad (1049)$$

with a branch chosen consistently off  $[-1, 1]$ .

## 8.2 An exact functional-differential equation

**Theorem 8.1** (Cauchy renormalization equation). *The Cauchy transform of the up-law satisfies*

$$S'(z) = 2(S(2z+1) - S(2z-1)), \quad z \notin [-1, 1]. \quad (1050)$$

*Proved in this report.*

*Proof.* Differentiate under the integral and integrate by parts, using the flat endpoint values  $\text{up}(\pm 1) = 0$ :

$$S'(z) = - \int_{-1}^1 \frac{\text{up}(t)}{(z-t)^2} dt = \int_{-1}^1 \frac{\text{up}'(t)}{z-t} dt.$$

Insert [\(972\)](#). In the first term use  $y = 2t + 1$ ; in the second use  $y = 2t - 1$ . The two transformed integrals are exactly  $S(2z+1)$  and  $S(2z-1)$ .  $\square$

Expanding (1050) at infinity reproduces the rational moment recurrence. Conversely, the equation plus  $S(z) \sim 1/z$  packages the whole moment sequence into one analytic renormalization law.

For the Fabius density, define

$$T(z) = \int_0^1 \frac{F'(x)}{z-x} dx = 2S(2z-1). \quad (1051)$$

Then

$$T'(z) = 4(T(2z) - T(2z-1)). \quad (1052)$$

The positive-half up resolvent has a particularly simple expectation form:

$$\boxed{\int_0^1 \frac{\text{up}(x)}{z-x} dx = \mathbb{E} \log \frac{z}{z-X}.} \quad (1053)$$

### 8.3 Boundary values and the Hilbert transform

For  $-1 < x < 1$ , the Sokhotski–Plemelj boundary values are

$$S(x+i0) = \text{p. v.} \int_{-1}^1 \frac{\text{up}(t)}{x-t} dt - i\pi \text{up}(x). \quad (1054)$$

Thus the real part is  $\pi$  times the Hilbert transform under the convention

$$(\mathcal{H} \text{up})(x) = \frac{1}{\pi} \text{p. v.} \int_{-1}^1 \frac{\text{up}(t)}{x-t} dt. \quad (1055)$$

Taking nontangential limits in (1050) yields coupled refinement equations for  $\mathcal{H} \text{up}$  and  $\text{up}$ , with the expected changes when  $2x \pm 1$  enters or leaves the support. This gives a systematic route to boundary-value computation without direct principal-value quadrature.

## 9 The Jacobi continued fraction of the up-law

### 9.1 From rational moments to a J-fraction

Let  $P_n$  be the monic orthogonal polynomial of degree  $n$  for the probability measure  $\text{up}(x) dx$ . Existing corpus work proves positivity of the Hankel determinants and develops these polynomials for Gaussian quadrature [5]. Evenness forces the diagonal recurrence parameters to vanish, so

$$P_{n+1}(x) = xP_n(x) - a_n P_{n-1}(x), \quad P_0 = 1, \quad P_1 = x, \quad (1056)$$

where

$$a_n = \frac{h_n}{h_{n-1}} = \frac{\Delta_{n+1}\Delta_{n-1}}{\Delta_n^2} > 0. \quad (1057)$$

Here  $h_n = \int P_n^2 \text{up}$  and  $\Delta_n = \det(\mu_{i+j})_{0 \leq i,j < n}$ . Rational moments imply  $a_n \in \mathbb{Q}_{>0}$ .

**Theorem 9.1** (Symmetric Jacobi fraction). *The Cauchy transform  $S$  has the locally uniform continued fraction*

$$\boxed{S(z) = \frac{1}{z - \frac{a_1}{z - \frac{a_2}{z - \frac{a_3}{z - \ddots}}}}, \quad z \in \mathbb{C} \setminus [-1, 1].} \quad (1058)$$

All coefficients are positive rationals. The first exact values are displayed in [table 20](#). Moreover,

$$\lim_{n \rightarrow \infty} a_n = \frac{1}{4}. \quad (1059)$$

**Proved in this report.**

*Proof.* The Stieltjes–Jacobi correspondence for a compactly supported positive measure gives (1058); symmetry gives zero diagonal parameters, and (1057) gives positivity and rationality. The up-density is strictly positive almost everywhere on  $(-1, 1)$ . Rakhmanov’s theorem for orthogonal polynomials on  $[-1, 1]$  therefore gives orthonormal off-diagonal coefficients tending to  $1/2$ ; squaring converts them to the monic coefficients  $a_n \rightarrow 1/4$  [14].  $\square$

Table 20: First Jacobi coefficients of the monic up-polynomials, computed by exact rational Gram–Schmidt.

| $n$ | $a_n$ (exact)                         | decimal        |
|-----|---------------------------------------|----------------|
| 1   | $\frac{1}{9}$                         | 0.1111111111   |
| 2   | $\frac{32}{225}$                      | 0.1422222222   |
| 3   | $\frac{619}{3675}$                    | 0.168435374150 |
| 4   | $\frac{835232}{4640643}$              | 0.179981955087 |
| 5   | $\frac{1789043898625}{9272274723657}$ | 0.192945523288 |

The first polynomials are

$$P_0(x) = 1, \quad P_1(x) = x, \quad (1060)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad P_3(x) = x^3 - \frac{19}{75}x, \quad (1061)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (1062)$$

The convergents of (1058) are diagonal Padé approximants to the moment expansion (1047). Hence the functional equation (1050) can be used to test or accelerate rational approximants across dyadic affine scales.

**Conjecture 9.2** (Monotone Nevai approach). *The monic Jacobi coefficients of the up-law satisfy*

$$0 < a_1 < a_2 < a_3 < \cdots < \frac{1}{4}. \quad (1063)$$

**Conjectural.**

The first twelve coefficients support the conjecture, but endpoint flatness can produce subtle high-order recurrence effects. A proof may require total positivity of suitable Hankel-ratio arrays, a comparison theorem for even weights, or a direct use of (1050) in Jacobi coordinates.

## 10 Nonlinear powers and order statistics

### 10.1 Exact order-statistic identities

Let  $X_1, X_2, \dots$  be independent copies of  $X$ , and let  $Q = F^{-1}$ .

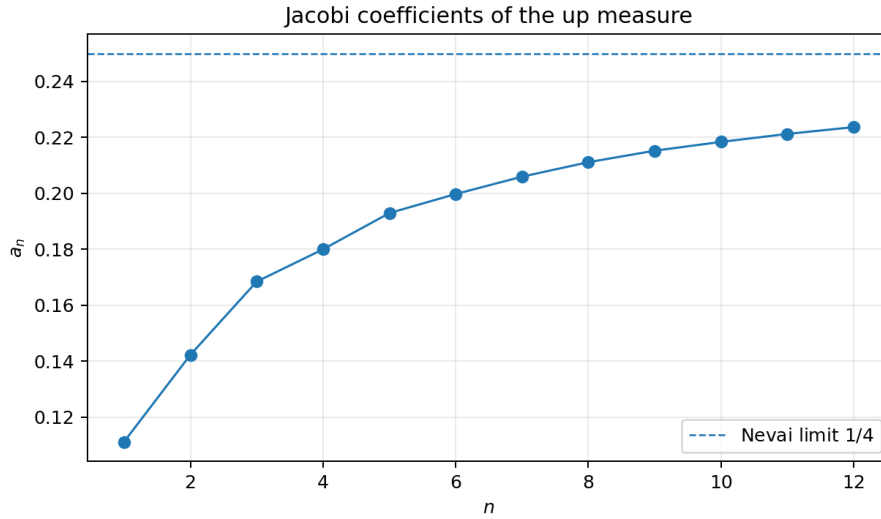


Figure 6: The first twelve exact Jacobi coefficients (shown in decimal form) and the rigorous limiting value  $1/4$ . Monotonicity is suggested but not proved.

**Theorem 10.1** (Power integrals as minima). *For every integer  $p \geq 1$ ,*

$$\int_0^1 \text{up}(x)^p dx = \int_0^1 F(x)^p dx = \mathbb{E} \min(X_1, \dots, X_p). \quad (1064)$$

*Equivalently,*

$$\int_0^1 \text{up}(x)^p dx = p \int_0^1 Q(v)(1-v)^{p-1} dv. \quad (1065)$$

*Proved in this report.*

*Proof.* The survival function of the minimum is  $\text{up}(x)^p$ , so its expectation is the first integral. Reflection gives  $\int F^p = \int \text{up}^p$ . If  $B_p = \min(U_1, \dots, U_p)$  for uniform variables, then  $B_p$  has density  $p(1-v)^{p-1}$ , and  $\min(X_1, \dots, X_p) \stackrel{d}{=} Q(B_p)$ .  $\square$

**Theorem 10.2** (Mixed powers as a random gap). *Let  $m, n \geq 1$ , let  $A = \max(X_1, \dots, X_m)$ , and let  $B = \min(X_{m+1}, \dots, X_{m+n})$ . Then*

$$\int_0^1 F(x)^m \text{up}(x)^n dx = \mathbb{E}(B - A)_+. \quad (1066)$$

*Proved in this report.*

*Proof.* For fixed  $x$ ,  $F(x)^m \text{up}(x)^n = \mathbb{P}(A \leq x < B)$ . Integrate the indicator over  $x \in [0, 1]$  and use Fubini.  $\square$

Fractional exponents are linked to the inverse without introducing a noninteger number of samples: (1043) gives

$$\int_0^1 \text{up}(x)^\alpha dx = \Gamma(\alpha + 1)(I_{0+}^\alpha Q)(1), \quad \alpha > 0. \quad (1067)$$

## 10.2 Large-power asymptotics from inverse slow variation

The inverse-frontier corpus proves the endpoint elasticity [6]

$$\frac{uQ'(u)}{Q(u)} \rightarrow 0 \quad (u \downarrow 0), \quad (1068)$$

with a sharper lower-Lambert and periodic expansion. It follows that  $Q$  is slowly varying at zero:

$$\frac{Q(cu)}{Q(u)} \rightarrow 1 \quad (u \downarrow 0), \quad c > 0. \quad (1069)$$

**Theorem 10.3** (Power-integral scale). *As  $p \rightarrow \infty$  through positive integers,*

$$\int_0^1 up(x)^p dx \sim Q(1/p). \quad (1070)$$

*Proved in this report.*

*Proof.* From (1065), substitute  $v = t/p$ :

$$\frac{\int_0^1 up(x)^p dx}{Q(1/p)} = \int_0^p \frac{Q(t/p)}{Q(1/p)} \left(1 - \frac{t}{p}\right)^{p-1} dt.$$

For fixed  $t > 0$ , slow variation gives convergence of the ratio to one, and the beta kernel converges to  $e^{-t}$ . Potter bounds for slowly varying functions provide an integrable majorant after splitting at  $t = 1$ ; the far tail is exponentially negligible. Dominated convergence gives  $\int_0^\infty e^{-t} dt = 1$ .  $\square$

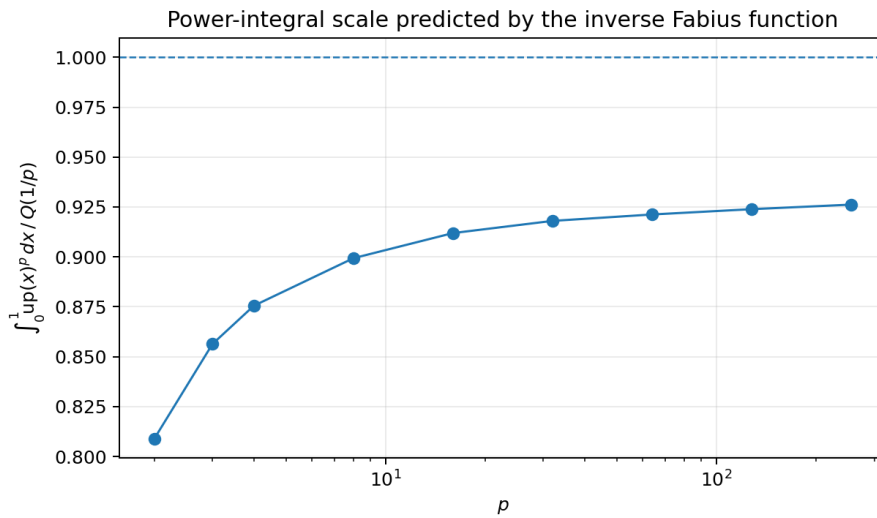


Figure 7: Numerical approach of  $\int_0^1 up(x)^p dx / Q(1/p)$  toward the rigorous limit one. The slow approach reflects the lower-Lambert endpoint scale.

Define the small elasticity

$$h_p = \frac{p^{-1}Q'(p^{-1})}{Q(p^{-1})}. \quad (1071)$$

The endpoint theory gives  $h_p \rightarrow 0$ , at a logarithmically slow rate.

**Proposition 10.4** (Conditional second-order law). *Let  $B_p$  have the Beta(1,  $p$ ) law. Assume the local self-neglecting property*

$$\frac{tuQ'(tu)/Q(tu)}{uQ'(u)/Q(u)} \rightarrow 1 \quad (1072)$$

*uniformly for  $t$  in compact subsets of  $(0, \infty)$ , and assume that the family*

$$Z_p = \frac{Q(B_p)/Q(1/p) - 1}{h_p} \quad (1073)$$

*is uniformly integrable. Then*

$$\int_0^1 \text{up}(x)^p \, dx = Q(1/p)(1 - \gamma h_p + o(h_p)), \quad (1074)$$

*where  $\gamma$  is Euler's constant. More precisely,*

$$Z_p \implies \log E, \quad E \sim \text{Exp}(1). \quad (1075)$$

*A second-order Potter bound dominating  $Z_p$  is one sufficient replacement for the stated uniform-integrability assumption. **Conditional statement.***

*Proof under the stated hypotheses.* Since  $pB_p \Rightarrow E$ , integrate the logarithmic derivative of  $Q$  between  $1/p$  and  $t/p$ . The self-neglecting condition gives

$$\log \frac{Q(t/p)}{Q(1/p)} = h_p \log t + o(h_p)$$

locally uniformly. Exponentiation yields the distributional limit. The stated uniform integrability permits passage to expectations, and  $\mathbb{E} \log E = -\gamma$  gives (1074).  $\square$

**Conjecture 10.5** (Second-order endpoint regularity). *The self-neglecting condition (1072) and the required uniform integrability follow from the repository's all-orders inverse lower-Lambert expansion, uniformly on every compact multiplicative  $t$ -range. Consequently (1074) holds without an additional hypothesis. **Conjectural.***

## 11 Correlation and transform families

Although autocorrelation identities are standard Fourier consequences, they provide useful definite integrals involving two copies of the up-function. Define

$$C(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x+a) \, dx = (\text{up} * \text{up})(a). \quad (1076)$$

With the Fourier convention  $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} \, dx$ ,

$$\widehat{C}(\xi) = |\widehat{\text{up}}(\xi)|^2 = \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi \xi}{2^j}\right). \quad (1077)$$

Thus

$$C(a) = \int_{\mathbb{R}} \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi \xi}{2^j}\right) e^{2\pi i a \xi} \, d\xi. \quad (1078)$$

More generally,

$$\int_{\mathbb{R}} \text{up}^{(m)}(x) \text{up}^{(n)}(x+a) \, dx = \int_{\mathbb{R}} (-2\pi i \xi)^m (2\pi i \xi)^n |\widehat{\text{up}}(\xi)|^2 e^{2\pi i a \xi} \, d\xi. \quad (1079)$$



At  $a = 0$ ,

$$\| \text{up}^{(m)} \|_2^2 = (2\pi)^{2m} \int_{\mathbb{R}} \xi^{2m} \prod_{j=0}^{\infty} \text{sinc}^2\left(\frac{\pi\xi}{2^j}\right) d\xi. \quad (1080)$$

These formulae turn Sobolev energies and shifted derivative overlaps into integral transforms of the squared infinite sinc product. The corpus's Fourier decay theory supplies convergence at every derivative order.

## 12 Numerical verification and reproducibility

### 12.1 Exact and floating-point layers

The companion program `numerical_experiments.py` separates proof from experiment.

1. Bernoulli cumulants, moments, Gram polynomials, and Jacobi coefficients are computed with Python's exact `Fraction` arithmetic.
2. A uniform grid on  $[-1, 1]$  is iterated under the positive fixed-point operator

$$(Tg)(x) = \int_{2x-1}^{2x+1} g(t) dt.$$

The resulting density is used only to cross-check analytic identities.

3. Adaptive or grid quadrature checks the Cauchy equation, fractional refinement, inverse fractional layer cake, order-statistic formulae, and generalized-moment Newton series.

With 65,537 grid points and 25 refinement iterations, the main diagnostics were

| Check                                          | observed residual / error                               |
|------------------------------------------------|---------------------------------------------------------|
| Mass of reconstructed up                       | $2.2 \times 10^{-16}$ from one                          |
| Variance versus exact $1/9$                    | $5.2 \times 10^{-11}$                                   |
| Maximum symmetry defect                        | $1.5 \times 10^{-14}$                                   |
| Cauchy equation at three complex points        | relative $2.9 \times 10^{-11}$ to $5.6 \times 10^{-11}$ |
| Fractional up-refinement                       | $1.3 \times 10^{-12}$ to $7.7 \times 10^{-7}$           |
| Inverse fractional identity                    | $9.4 \times 10^{-11}$ to $1.3 \times 10^{-8}$           |
| Newton series for $m(-1/2), m(1/2), m(3/2)$    | $3.3 \times 10^{-8}$ to $8.0 \times 10^{-12}$           |
| Logarithmic identity $\int F(x)/x dx = -m'(0)$ | $4.4 \times 10^{-9}$                                    |

The full exact fractions through  $a_{12}$ , every test point, precision setting, and residual are recorded in `experiments_summary.txt`. The script is deterministic and uses no Monte Carlo simulation.

### 12.2 Reproduction command

From the extracted archive directory:

```
python numerical_experiments.py --output-dir reproduced
pdflatex -interaction=nonstopmode -halt-on-error Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error Integration_and_Transform_Frontiers.tex
```

The generated table `jacobi_coefficients.tex` and all three figures are included in the PDF build.

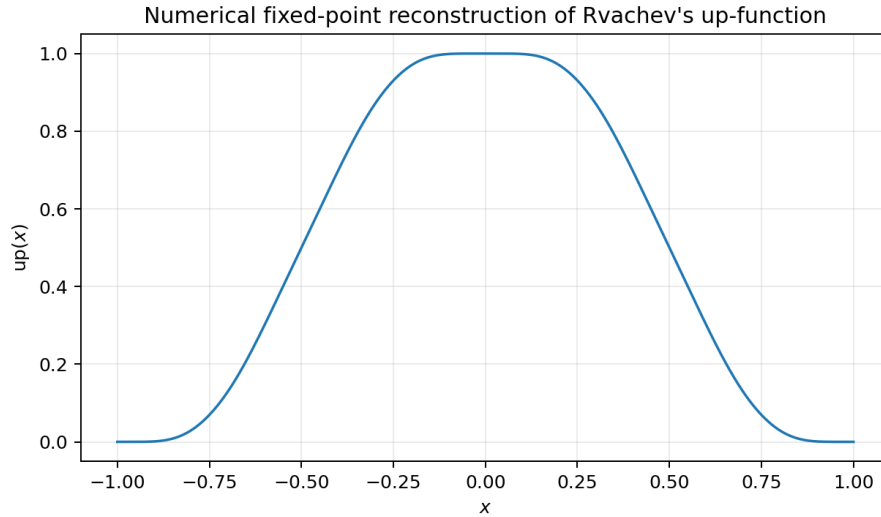


Figure 8: Positive fixed-point reconstruction used for numerical cross-checks. The exact theory, not this discretization, is the source of all theorems.

## 13 Conjectures and future research directions

### 13.1 Cauchy renormalization as a spectral equation

Equation (1050) deserves treatment as a boundary-value renormalization problem in its own right.

*Open problem 13.1* (Uniqueness from analytic data). Characterize the weakest growth and Herglotz conditions under which

$$S'(z) = 2[S(2z + 1) - S(2z - 1)], \quad S(z) \sim z^{-1},$$

uniquely determines the up-law. A positive answer would provide a transform- space definition of up independent of the infinite convolution.

*Open problem 13.2* (Dyadic Padé acceleration). Combine the J-fraction convergents with the affine values  $S(2z \pm 1)$  in (1050) to build certified rational approximants on  $\mathbb{C} \setminus [-1, 1]$ . Determine whether the dyadic equation improves the usual Markov convergence bounds for Stieltjes continued fractions.

### 13.2 Jacobi arithmetic and endpoint flatness

The exact rational coefficients  $a_n$  form a new arithmetic sequence attached to the up-law. Beyond conjecture 9.2, natural questions are:

1. determine sharp denominator-clearing products in terms of Mersenne factors  $2^k - 1$  or  $4^k - 1$ ;
2. find the first correction to  $a_n = 1/4 + o(1)$ , and decide whether endpoint flatness produces a logarithmic, stretched-exponential, or oscillatory rate;
3. translate the Cauchy functional equation into a closed hierarchy for Jacobi parameters or Hankel ratios;
4. construct a Lean proof of the first exact coefficients from the already formalized rational moment recurrence.

### 13.3 Fractional endpoint asymptotics

The repository's endpoint expansion controls  $F(x)$  and  $Q(u)$  through a lower-Lambert variable and periodic Gamma–zeta terms. The expectation kernels in (1028), (1034), and (1042) suggest a larger transseries.

**Conjecture 13.3** (Fractional lower-Lambert transseries). *For fixed noninteger  $\alpha > 0$ , the endpoint asymptotic expansions of  $I_{0+}^\alpha F$ ,  $D_{0+}^\alpha F$ , and  $I_{0+}^\alpha Q$  are organized by the same exact lower-Lambert phase as  $F$  and  $Q$ , with coefficients obtained by beta/gamma shifts of the existing periodic saddle jets.*  
**Conjectural.**

A practical route is to insert the known endpoint density expansion into the truncated-power expectations, split at the saddle scale, and carry the beta kernel through the existing uniform remainder estimates.

### 13.4 Nonlinear power integrals

The first-order equivalence (1070) is rigorous. The next frontier is to prove the self-neglecting property required in proposition 10.4 and then propagate the complete inverse transseries through the beta kernel. This should produce

$$\int_0^1 \text{up}(x)^p dx = Q(1/p) (1 - \gamma h_p + c_2 h_p^2 + \text{periodic corrections} + \dots), \quad (1081)$$

where  $c_2$  should involve moments of  $\log E$ , hence  $\pi^2/6$  and  $\gamma^2$ , while the periodic terms inherit the Gamma–zeta phase of the inverse Fabius expansion.

### 13.5 Fractional formalization roadmap

#### Kernel layer

The survival-kernel obligation in theorem 3.1 is discharged, including its partial form and Banach-valued kernels. The inverse obligation in theorem 5.2 is discharged for an explicit interval-integrable primitive, for the pointwise- $C^1$  composition form, and for the report's exact real-primitive, real- or complex-weighted arbitrary- $L^1$ /absolutely-continuous signature. The strict adjunction supplies a Galois connection, so all three generic interfaces now derive clock measurability internally. The complete proper-compact two-function identity is additionally discharged for arbitrary real-valued absolutely continuous factors by `InversePairIntegral.lean`, with clock measurability derived internally. Its variable-upper proper-compact companion and exact Fabius specialization are discharged there as well. Complex-valued arbitrary-AC primitives, reversed endpoints, improper singular endpoints, higher and fractional inverse primitives, and the report's named transform consequences remain outside this kernel tranche.

*Formalization obligation 13.4* (Special-function layer — discharged). Introduce incomplete beta integrals and prove theorem 7.3 first for positive real parameters. Complex continuation can remain outside the initial formalization.

*Post-snapshot status.* This obligation is discharged by `IncompleteBetaMaster.lean`: `Fabius.incompleteBeta` is the incomplete beta integral  $B_z(a, b) = \int_0^z u^{a-1}(1-u)^{b-1} du$ , and both displays of theorem 7.3 are formal at positive real parameters: `Fabius.fractionalVolterra_rpow_mul_rvachevUp` is (1033) and `Fabius.fractionalVolterra_rpow_mul_fabiusReal` is (1034), for  $\alpha > 0$ ,  $p \geq 0$ , and  $0 < x \leq 1$ , with the clamp  $\min(X/x, 1)$  implementing the stopping (the  $F$ -display's indicator  $\mathbf{1}_{\{X < x\}}$  appears as  $B_0 = 0$  at clamped samples). The proofs run through the survival and lower-CDF layer cakes and one scaling substitution  $t = xu$  (`Fabius.integral_`

`rpow_mul_rpow_eq_incompleteBeta` and its reflection); the exponent range  $p \in (-1, 0)$  and complex continuation remain outside, as the obligation allows.

*Formalization obligation 13.5* (Transform layer — partially discharged). The compact-interval transforms and powers, their first derivatives, the named unit and centered DDEs, the all-power affine bridge, both complex-slit adjacent-order recurrences, the atom-exact CDF and survival formulas at every positive kernel power, and (1050) are direct named results of `MeasureCauchyTransform.lean`, `GeometricUniformCauchy.lean`, `CauchyTransform.lean`, `CauchyCDF.lean`, `CauchySurvival.lean`, `CauchyHigherPowers.lean`, and `CauchyRenormalization.lean`. The real logarithmic fixed point, real integer and real-order hierarchies, real/complex exterior Laurent series, finite-height Herglotz–Poisson identities, integrated interval Stieltjes–Perron inversion, and initial Jacobi data are also discharged by direct named theorems in the merged Stieltjes and orthogonal-polynomial modules.

The Laurent obligation is now discharged at greater generality by the one-definition, fifteen-theorem `MeasureCauchyMomentLaurent.lean` API: arbitrary centers and `RCLike` coefficients, almost-everywhere relative-ball support, finite total mass, the exact finite remainder and its sharp mass factor, and the full `Summable/tsum/HasSum` series are all direct named results. The older real and complex up-measure statements remain the named  $c = 0, R = 1$  probability cases; `Fabius.a_e_norm_sub_zero_le_one_rvachevMeasure` and `Fabius.measureCauchyMoment_rvachevMeasure_zero` are the exact support and coefficient bridges to that specialization.

The remaining obligations are complex logarithmic continuation, Laplace/Fourier transform forms, complex order, pointwise/nontangential Plemelj and principal-value Hilbert theory, the complete J-fraction/Padé package, and the separate explicit higher-kernel Thue–Morse orbit formula.

*Formalization obligation 13.6* (Orthogonal-polynomial layer). Reuse the corpus’s Hankel-positivity and Gaussian-quadrature infrastructure to construct  $a_n = h_n/h_{n-1}$ , the finite J-fraction convergents, and exact certificates for the first coefficients. Rakhmanov’s asymptotic theorem may be imported as a separately documented analytic dependency.

*Post-snapshot status.* The construction’s first stage is now formal: `MomentHankelMatrix.lean` proves the Hankel moment matrices of the up-measure positive definite (`Fabius.momentHankel_posDef`: the quadratic form is the squared-polynomial moment `Fabius.dotProduct_momentHankel_mulVec`, a nonzero coefficient vector gives a nonzero polynomial with finitely many roots, and the up-measure charges  $(-1, 1)$  with full mass while annihilating finite sets), so every Hankel determinant  $h_n$  is positive (`Fabius.hankelDet_pos`) and the ratios  $a_n = h_{n+1}/h_n$  are well-defined and positive (`Fabius.hankelRatio_pos`). The exact-certificate stage has begun: `MomentHankelValues.lean` identifies the measure moments with the exact rational moment recursion (`Fabius.upMoment_even`; odd moments vanish) and evaluates the first data in closed form —  $h_1 = 1, h_2 = \frac{1}{9}, a_0 = 1$ , and  $a_1 = \frac{1}{9}$ , the variance (`Fabius.hankelDet_two`, `Fabius.hankelRatio_one`). The orthogonal polynomials themselves are now constructed: `OrthogonalPolynomialConstruction.lean` realizes  $h_n P_n(x)$  as the bordered Hankel determinant (`Fabius.hankelOrthoValue`) and proves the orthogonality relations  $\int P_n(x) x^j d\mu_{\text{up}} = 0$  for  $j < n$  (`Fabius.integral_hankelOrthoValue_mul_pow`) by the classical duplicated-row argument, formalized through the last-row cofactor expansion. The same module now packages the construction as a genuine element of  $\mathbb{R}[x]$ : `Fabius.hankelOrthoPolynomial` is the cofactor expansion as a `Polynomial`, its evaluation recovers the bordered determinant (`Fabius.hankelOrthoPolynomial_eval`), its coefficients are the signed last-row minors (`Fabius.hankelOrthoPolynomial_coeff`), and its top coefficient is the Hankel determinant itself,  $[x^n](h_n P_n) = h_n$  (`Fabius.hankelOrthoPolynomial_coeff_top`: deleting the monomial row and the last column leaves exactly the plain Hankel block, with sign  $(-1)^{n+n} = 1$ ) — so with positivity the polynomial is genuinely of degree  $n$  and monic after division by  $h_n$  (`Fabius.upOrthoPolynomial`, with `Fabius.upOrthoPolynomial_monice` and exact degree). The layer now runs considerably deeper. *Values*: the integral of  $h_n P_n$  against its own leading monomial is the *next* Hankel determinant — the  $n$ -shifted moment row completes the  $(n+1)$ -th Hankel

block (Fabius.integral\_hankelOrthoValue\_mul\_pow\_self) – giving the **norm identity**  $\int P_n^2 d\mu_{\text{up}} = a_n = h_{n+1}/h_n$  (Fabius.integral\_upOrthoPolynomial\_sq), with the concrete anchors  $P_0 = 1$ ,  $P_1 = x$ , and  $\int x^2 d\mu_{\text{up}} = a_1 = \frac{1}{9}$  (Fabius.upOrthoPolynomial\_one, Fabius.integral\_sq\_eq\_hankelRatio\_one). *Structure*: orthogonality holds against every polynomial of degree  $< n$  (Fabius.integral\_hankelOrthoValue\_mul\_eval), a nonzero polynomial has positive squared mass (Fabius.integral\_sq\_eval\_pos), and the monic orthogonal polynomial is *unique* (Fabius.eq\_upOrthoPolynomial\_of\_monic\_of\_orthogonal). *Symmetry*: the up-measure is negation-invariant (Fabius.integral\_comp\_neg\_rvachevMeasure), so by uniqueness  $P_n(-x) = (-1)^n P_n(x)$  (Fabius.upOrthoPolynomial\_comp\_neg\_X), the odd pairings  $\int P_n(x) x^j d\mu_{\text{up}}$  with  $n + j$  odd all vanish, and the Jacobi diagonal is identically zero,  $\int x P_n(x)^2 d\mu_{\text{up}} = 0$  (Fabius.integral\_mul\_sq\_upOrthoPolynomial\_eq\_zero). *Recurrence*: on these inputs the **three-term recurrence** is itself formal,

$$x P_{n+1} = P_{n+2} + \frac{a_{n+1}}{a_n} P_n, \quad x P_0 = P_1$$

(Fabius.upOrthoPolynomial\_three\_term, Fabius.upOrthoPolynomial\_three\_term\_zero, with a pointwise form): the defect  $x P_{n+1} - P_{n+2} - (a_{n+1}/a_n) P_n$  has degree  $\leq n+1$  (the monic leading terms cancel) and is orthogonal to  $x^j$  for all  $j < n+2$  – below  $n$  by orthogonality, at  $j = n$  by the exact cancellation of the two Hankel-ratio moments, at  $j = n+1$  by parity – so it has zero squared mass and vanishes (Fabius.eq\_zero\_of\_natDegree\_lt\_of\_orthogonal). The recurrence coefficient is the ratio of consecutive squared norms, and the  $n = 0$  instance extracts the concrete  $P_2 = x^2 - \frac{1}{9}$  (Fabius.upOrthoPolynomial\_two). The J-fraction layer thus has its three-term engine; the second-kind polynomials, the continued fraction of the Stieltjes transform, and the deeper certificates remain open.

## 14 Conclusion

The Fabius–Rvachev system admits a broader integral calculus than its original differential and Fourier definitions suggest. Three probability identities supply the organizing principle:

$$\text{up}(x) = \mathbb{P}(X > x), \quad F(x) = \mathbb{P}(X \leq x), \quad Q(u) = \inf\{x : F(x) \geq u\}. \quad (1082)$$

From them follow:

1. arbitrary-kernel and monomial antiderivatives as truncated moments;
2. an entire generalized-moment function reconstructed from rational moments by a Newton series;
3. a Mellin triad distinguishing the single pole of  $\text{up}$  from the entire transforms of  $F$  and  $F'$ ;
4. exact layer-cake transforms and fractional derivatives of the inverse;
5. incomplete-beta formulae interpolating ordinary, repeated, negative-power, and fractional antiderivatives;
6. a Cauchy functional-differential equation and the explicit Jacobi fraction requested by an existing frontier report;
7. order-statistic interpretations and a sharp inverse-Fabius scale for nonlinear power integrals.

The most promising next step is to combine the new kernel formulae with the repository’s all-orders lower-Lambert endpoint theory. That synthesis should turn several conditional statements here—especially the second-order power integral and fractional endpoint transseries—into explicit theorems with periodic Gamma–zeta coefficients.

## A Exact formula sheet

| Object                       | Exact formula                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------|
| Fabius moment                | $m(s) = \mathbb{E}X^s = \sum_{n \geq 0} (-1)^n \binom{s}{n} m_n$                                |
| Mellin of positive up-half   | $\int_0^1 x^{s-1} \text{up}(x) \, dx = m(s)/s$                                                  |
| Mellin of $F$                | $\int_0^1 x^{s-1} F(x) \, dx = (1 - m(s))/s$                                                    |
| Monomial up integral         | $\int_0^1 x^p \text{up}(x) \, dx = m(p+1)/(p+1), \Re p > -1$                                    |
| Monomial Fabius integral     | $\int_0^1 x^p F(x) \, dx = (1 - m(p+1))/(p+1), \text{entire in } p$                             |
| Laplace/exponential up-half  | $\int_0^1 \text{up}(x) e^{zx} \, dx = (A(z) - 1)/z$                                             |
| Laplace/exponential Fabius   | $\int_0^1 F(x) e^{zx} \, dx = (e^z - A(z))/z$                                                   |
| Quantile moments             | $\int_0^1 Q(u)^p \, du = m(p)$                                                                  |
| Inverse primitive            | $\int_0^u Q(v) \, dv = uQ(u) - \int_0^{Q(u)} F(x) \, dx$                                        |
| Fractional $F$ primitive     | $I_{0+}^\alpha F(x) = \mathbb{E}(x - X)_+^\alpha / \Gamma(\alpha + 1)$                          |
| Fractional inverse primitive | $I_{0+}^\alpha Q(u) = \Gamma(\alpha + 1)^{-1} \int_0^{Q(u)} (u - F(x))^\alpha \, dx$            |
| Power/fractional bridge      | $\int_0^1 \text{up}(x)^\alpha \, dx = \Gamma(\alpha + 1) I_{0+}^\alpha Q(1)$                    |
| Cauchy equation              | $S'(z) = 2[S(2z + 1) - S(2z - 1)]$                                                              |
| Jacobi fraction              | $S(z) = 1/(z - a_1/(z - a_2/(z - \cdots)))$ , $a_n \in \mathbb{Q}_{>0}$ , $a_n \rightarrow 1/4$ |
| Power asymptotic             | $\int_0^1 \text{up}(x)^p \, dx \sim Q(1/p)$                                                     |

## B Corpus families inspected

The recursive audit was organized by mathematical lineage. The principal families were:

1. the living primary exposition *Fabius Function and Rvachev Up*;
2. the mathematical glossary, non-elementarity article, and Lean walkthrough;
3. the consolidated non-formalized frontier volume and provenance/gap register;
4. repeated-integration and up-from-integration manuscripts;
5. dyadic formula,  $q$ -connection, asymptotic bridge, and small-argument manuscripts;
6. inverse-function, inverse-saddle, and lower-Lambert reports;
7. the Thue–Morse formula atlas and its archived drafts;
8. active arithmetic-ray, self-sampling, half-integer spectral, exponent- sequence/Newton, spectral-Pascal, finite-block, sinc-Richardson, and Fourier-decay reports;
9. imported TeX sources for the classical Fabius, Rvachev, arithmetic, and compact-support literature;
10. frozen archive variants retained by the repository’s provenance map.

The exact 67-path list at the pinned commit is recorded in the inventory appendix of the exponent-sequence/Newton report [7]. Current descendants and newly active draft folders were rechecked separately so that the audit did not stop at the pinned snapshot.

## C Reproducibility manifest

The consolidated volume and its retained companion assets comprise:

| Path                                    | Purpose                                               |
|-----------------------------------------|-------------------------------------------------------|
| Integration_and_Transform_Frontiers.tex | Complete LaTeX manuscript                             |
| Integration_and_Transform_Frontiers.pdf | Compiled and visually inspected report                |
| numerical_experiments.py                | Commented exact/numerical verification program        |
| experiments_summary.txt                 | Full deterministic diagnostics and exact coefficients |
| jacobi_coefficients.tex                 | Generated exact table used in the manuscript          |
| up_reconstruction.png                   | Fixed-point density diagnostic                        |
| jacobi_coefficients.png                 | Exact-coefficient trend figure                        |
| power_quantile_ratio.png                | Power-integral asymptotic diagnostic                  |
| requirements.txt                        | Python package versions used                          |
| README.md                               | Build and reproduction instructions                   |
| SHA256SUMS                              | Checksums for packaged artifacts                      |

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## Part IX

# Fractional Integral Calculus and Complex-Order Transform Hierarchies for the Fabius–Rvachev System

### Abstract

Let  $F$  be the bounded Fabius distribution function, let  $Q = F^{-1}$  on  $(0, 1)$ , and let  $\text{up}$  denote Rvachev’s even compactly supported density. The current ProveIt corpus already contains a substantial ordinary and Riemann–Liouville integration theory: exact integer primitive ladders, finite formulas for monomial weights, entire Newton–Mellin continuations, inverse-area identities, moment recurrences, sinc-product transforms, and a Lean formalization of the normalized integer Volterra ladder. This report uses those results as a baseline and develops a genuinely fractional and complex-order layer that was not found as a systematic subject in the recursively audited corpus.

The first new construction is an exact fractional calculus in the *Fabius clock*. Conjugating ordinary Riemann–Liouville and Caputo operators by the change of variable  $y = F(x)$  gives

$$(I_F^\alpha h)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt, \quad D_F = \frac{1}{F'} \frac{d}{dx}.$$

It has an exact semigroup, diagonalizes powers of  $F$ , and admits Mittag–Leffler eigenfunctions. The dual  $Q$ -clock calculus has the same properties. In ordinary time, a transmutation theorem gives, for  $0 < \beta < 1$ ,

$${}^C D_{0+}^\beta [H(Q(y))] = \frac{1}{\Gamma(1-\beta)} \int_0^{Q(y)} (y - F(x))^{-\beta} H'(x) dx.$$

For  $H(x) = x^r$  this yields a closed integral formula for the Caputo derivative of every positive power of the inverse Fabius function and the endpoint law

$${}^C D_{0+}^\beta Q(y)^r \sim \frac{Q(y)^r y^{-\beta}}{\Gamma(1-\beta)} \quad (y \downarrow 0).$$

A second exact bridge connects standard fractional derivatives of  $F$  to fractional primitives of  $\text{up}$ :

$$D_{0+}^\beta F(x) = 2^\beta (I_{-1+}^{1-\beta} \text{up})(2x - 1), \quad 0 < \beta < 1.$$

On the whole line, the Weyl family  $W_\sigma = \mathcal{F}^{-1}[(i\xi)^\sigma \widehat{\text{up}}(\xi)]$  satisfies the complex-order refinement hierarchy

$$W_\sigma(x) = 2^\sigma (W_{\sigma-1}(2x + 1) - W_{\sigma-1}(2x - 1)).$$

This continuously interpolates Rvachev’s differential–difference equation.

The transform side is organized by the entire complex-order generalized Stieltjes function

$$S_\lambda(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z - t)^\lambda} dt.$$

It obeys a dyadic order-lowering equation, an exact derivative hierarchy, and an even-moment series. It also controls nonlocal operators. For  $0 < \beta < 2$  and  $|x| > 1$ ,

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} S_{1+\beta}(|x|),$$

so the exterior fractional Laplacian has a full rational-moment expansion. A parallel formula is proved for Riesz potentials. Numerical Fourier experiments suggest that  $(-\Delta)^{\beta/2} \text{up}$  has exactly one positive zero for every  $0 < \beta < 2$ ; this is stated explicitly as a conjecture.

The report also develops complex-order resolvents containing  $F$  and  $Q$ , fractional autocorrelation energies of the infinite sinc product, a catalog of definite and indefinite integral identities, reproducible numerical checks, and a Lean formalization roadmap. “New” throughout means new relative to the pinned repository corpus after a targeted collision audit; it is not a claim of universal publication priority.

# 1 Scope, recursive corpus audit, and proof status

## 1.1 Audit endpoint and moving-tree caveat

The requested documentation root was inspected recursively at the commit shown on the title page:

<https://github.com/VladimirReshetnikov/ProveIt/tree/3adc011108796355f1a874a5036f28098408d370/Analysis/FabiusFunction/docs>.

At the audit endpoint, the indexed recursive query returned 89 TeX paths. The branch was actively changing during the investigation: the final pinned merge added both a new integral-transform report and a Lean implementation of the normalized integer Volterra foundation. For that reason, every novelty comparison below is tied to the immutable commit 3adc011108796355f1a874a5036f28098408d370, not merely to the branch name `main`.

The audit was path-complete and content-deduplicated. Current expositions, frontier reports, source-paper transcriptions, the glossary, the Lean walkthrough, and mathematically distinctive archival versions were examined. Byte-identical or superseded historical copies were treated as provenance, not as independent mathematical sources. The principal families were:

- the live Fabius–Rvachev exposition, glossary, non-elementarity study, and Lean walkthrough;
- exact dyadic,  $q$ -binomial, Thue–Morse, Bell/Bernoulli, and Richardson-extrapolation reports;
- the inverse-Fabius and Lambert– $W$  endpoint reports;
- repeated-integration, monomial-antiderivative, integral-transform, and derivative-norm reports;
- the complete Fourier-decay campaign for the infinite sinc product;
- archived predecessor documents and seven transcribed primary papers.

## 1.2 What the repository already contains

The following material is treated as *corpus baseline* rather than relabelled as a contribution of this report [9, 10, 11]:

- the probability model, functional–differential equations, signed Thue–Morse extension, partition of unity, and all integer derivatives;
- exact dyadic values, rational moment recurrences,  $q$ -Pochhammer and Gaussian-binomial formulas;
- sharp Fourier decay, zero multiplicities, lobe signs, and spectral determinant interpretations of the sinc product;
- Lambert– $W$  endpoint asymptotics and their small periodic correction;
- inverse-function germs, Bell-polynomial inverse derivatives, and inverse endpoint asymptotics;
- normalized integer Volterra primitives, monomially weighted primitive formulas, and their recent Lean formalization;
- Riemann–Liouville fractional primitives, inverse-area formulas, Mellin/Newton moment continuations, beta weights, order statistics, integer resolvents, and several correlation integrals;
- integer derivative equimeasurability and all rearrangement-invariant norm identities.

Targeted searches were performed for *Caputo*, *fractional Laplacian*, *Riesz potential*, *Weyl derivative*, *fractional calculus with respect to  $F$* , and the exact formula shapes proved below. The only Caputo occurrence located was a short remark noting that Caputo and Riemann–Liouville derivatives coincide for  $F$  because of endpoint flatness. No developed theory of the other structures was found.

### 1.3 Status ledger

| Result family                                                             | Status          | Main input                                                         |
|---------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------|
| Integer derivative and primitive ladders; monomial Newton–Volterra series | corpus baseline | signed self-differential equation and endpoint flatness            |
| Inverse-area identities and standard Riemann–Liouville primitives         | corpus baseline | quantile substitution and Volterra integration                     |
| Fabius-clock and inverse-clock fractional calculi                         | proved here     | exact conjugacy by $y = F(x)$ and $x = Q(y)$                       |
| Caputo/Riemann–Liouville transmutation for $H \circ Q$                    | proved here     | inverse-chain operator $(F')^{-1}D$                                |
| Endpoint laws for fractional integrals and Caputo derivatives of $Q^r$    | proved here     | rapid variation and convexity of $F$ near zero                     |
| Standard fractional bridge between $F$ and up                             | proved here     | $F^{(n)}(x) = 2^n \text{up}^{(n-1)}(2x - 1)$                       |
| Whole-line Weyl complex-order refinement hierarchy                        | proved here     | Fourier multiplier calculus and the sinc refinement product        |
| Generalized Stieltjes order hierarchy                                     | proved here     | random self-similarity $Z \stackrel{d}{=} (V + Z')/2$              |
| Complex-order resolvents involving $F$ , $Q$ , and $1 - F$                | proved here     | integration by parts after quantile transport                      |
| Exterior Riesz fractional-Laplacian and potential expansions              | proved here     | compact support and the generalized Stieltjes series               |
| One-node law for $(-\Delta)^{\beta/2}$ up                                 | conjectural     | nonperiodic singular-integral profiles for seven fractional orders |
| Fractional endpoint phase correction and fractional-energy rigidity       | conjectural     | Lambert phase and integer spectral moment interpolation            |

#### Priority and regularity discipline

A theorem labelled “proved here” is supplied with a proof in this document, but has not necessarily been formalized in Lean. Novelty is relative to the pinned repository corpus. Standard constructions from fractional calculus are cited rather than claimed as new; the new content is their exact specialization and interaction with the Fabius–Rvachev structure. Whenever a Caputo derivative requires endpoint regularity that the inverse  $Q$  may not possess at higher order, the hypothesis is stated explicitly.

## 2 Normalization and structural input

### 2.1 The bounded function, inverse, and compact density

Following the probabilistic construction of Fabius and the compact-support formulation of Rvachev/Arias de Reyna [1, 2, 3], let  $U_1, U_2, \dots$  be independent uniform random variables on  $[0, 1]$  and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad Z = 2Y - 1 = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad (1083)$$

where  $V_j = 2U_j - 1$  is uniform on  $[-1, 1]$ . The bounded Fabius function is the CDF of  $Y$ :

$$F(x) = \mathbb{P}(Y \leq x), \quad F(x) = 0 \ (x \leq 0), \quad F(x) = 1 \ (x \geq 1). \quad (1084)$$

The variable  $Z$  has the even density up supported on  $[-1, 1]$ . Thus

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 < x < 1, \quad (1085)$$

$$\operatorname{up}(t) = F(1 - |t|), \quad |t| \leq 1, \quad (1086)$$

$$F(1 - x) = 1 - F(x), \quad Q(1 - y) = 1 - Q(y). \quad (1087)$$

On  $[0, 1]$ ,  $\operatorname{up}(x) = 1 - F(x) = F(1 - x)$ . The inverse  $Q = F^{-1}$  is continuous on  $[0, 1]$ , smooth on  $(0, 1)$ , and satisfies

$$Q'(y) = \frac{1}{F'(Q(y))} = \frac{1}{2 \operatorname{up}(2Q(y) - 1)}. \quad (1088)$$

The left-half self-differential equation is

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}, \quad (1089)$$

and Rvachev's global differential-difference equation is

$$\operatorname{up}'(x) = 2(\operatorname{up}(2x + 1) - \operatorname{up}(2x - 1)). \quad (1090)$$

Every derivative of  $F$  vanishes at 0 and 1, and every derivative of  $\operatorname{up}$  vanishes at  $\pm 1$ .

## 2.2 Fourier products and moments

We use the angular-frequency convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\xi) e^{i\xi x} d\xi. \quad (1091)$$

Then

$$\phi(\xi) := \widehat{\operatorname{up}}(\xi) = \prod_{j=1}^{\infty} \operatorname{sinc}\left(\frac{\xi}{2^j}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}, \quad (1092)$$

and

$$\phi(\xi) = \operatorname{sinc}(\xi/2) \phi(\xi/2). \quad (1093)$$

The one-sided moment generating function is

$$P(s) = \mathbb{E}e^{sY} = \prod_{j=1}^{\infty} \frac{e^{s/2^j} - 1}{s/2^j}, \quad P(2s) = \frac{e^s - 1}{s} P(s). \quad (1094)$$

Write

$$m_n = \mathbb{E}Y^n, \quad c_k = \mathbb{E}Z^{2k} = \int_{-1}^1 t^{2k} \operatorname{up}(t) dt. \quad (1095)$$

All these moments are rational. The first centered values are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}. \quad (1096)$$

The recurrence

$$(2^n - 1)m_n = \sum_{j=0}^{n-1} \binom{n}{j} \frac{m_j}{n - j + 1} \quad (1097)$$

will be used for numerical coefficients.

### 2.3 Endpoint rapid variation

The repository's Lambert– $W$  analysis implies, in particular, that  $F$  is rapidly varying at 0:

$$\frac{F(tx)}{F(x)} \longrightarrow 0 \quad (x \downarrow 0) \quad \text{for every fixed } 0 \leq t < 1. \quad (1098)$$

For  $x \leq 1/2$ ,  $F$  is convex because differentiating [equation \(1089\)](#) gives  $F''(x) = 4F'(2x) \geq 0$ . Consequently

$$F(tx) \leq tF(x), \quad 0 \leq t \leq 1, \quad 0 \leq x \leq \frac{1}{2}. \quad (1099)$$

The combination of [equations \(1098\)](#) and [\(1099\)](#) is the key to the fractional inverse endpoint limits in [appendix 6](#).

## 3 Existing ordinary and Volterra calculus: a compact atlas

This section fixes notation and records the strongest formulas already present in the current corpus. They are indispensable inputs, but they are not novelty claims of this report.

### 3.1 Integer derivative and primitive orbits

Let  $\mathcal{F}$  be the signed global extension satisfying  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$  and agreeing with  $F$  on  $(-\infty, 1]$ . For  $n \in \mathbb{N}_0$ ,

$$D^n \mathcal{F}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x). \quad (1100)$$

For the normalized zero-based Volterra operator

$$(J^n f)(x) = \begin{cases} f(x), & n = 0, \\ \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt, & n \geq 1, \end{cases} \quad (1101)$$

its exact inverse orbit is

$$J^n \mathcal{F}(x) = 2^{\binom{n}{2}} \mathcal{F}\left(\frac{x}{2^n}\right). \quad (1102)$$

The same formula holds for bounded  $F$  whenever the entire integration segment lies in the agreement range; in particular, for  $0 \leq x \leq 1$ . Every other  $n$ -fold antiderivative differs from [equation \(1102\)](#) by a polynomial of degree at most  $n-1$ .

### 3.2 Monomially weighted primitives

For  $x > 0$ ,  $n \geq 1$ , and complex  $a$ , endpoint flatness makes the following Newton–Volterra series locally normally convergent in  $(a, x)$ :

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^a F(t) dt = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} a^r x^{a-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (1103)$$

For  $a \in \mathbb{N}_0$  it terminates and is a finite native closed form. Negative and fractional powers are covered by the same convergent series. On  $[0, 1]$ , weights multiplying  $1 - F = \text{up}$  follow by subtracting  $J^n(t^a F(t))$  from the elementary Volterra integral of  $t^a$ .

### 3.3 Inverse primitives and ordinary derivatives

For  $q = Q(y)$  and  $\Re p > 0$ ,

$$\int_0^y Q(v)^p dv = yq^p - p \int_0^q x^{p-1} F(x) dx, \quad (1104)$$

$$\int_0^y Q(v) dv = yQ(y) - F\left(\frac{Q(y)}{2}\right). \quad (1105)$$

The  $n$ -fold inverse primitive has the incomplete-moment form

$$I_{0+}^n Q^p(y) = \frac{p}{n!} \int_0^{Q(y)} x^{p-1} (y - F(x))^n dx. \quad (1106)$$

For ordinary derivatives, introduce

$$L_F = \frac{1}{F'(x)} \frac{d}{dx}. \quad (1107)$$

Then the exact inverse-chain identity is

$$\frac{d^n}{dy^n} H(Q(y)) = (L_F^n H)(Q(y)). \quad (1108)$$

In particular,  $Q^{(n)} = (L_F^n \text{id}) \circ Q$ . Equivalently, differentiating  $F(Q(y)) = y$  gives the Bell-polynomial recurrence

$$Q^{(n)}(y) = -\frac{1}{F'(Q(y))} \sum_{k=2}^n F^{(k)}(Q(y)) B_{n,k}(Q'(y), \dots, Q^{(n-k+1)}(y)). \quad (1109)$$

### 3.4 Standard Riemann–Liouville primitives

For  $\Re \alpha > 0$ , using the standard Riemann–Liouville convention [5, 6],

$$(I_{0+}^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (1110)$$

The corpus already proves the probabilistic formulas

$$I_{0+}^\alpha F(x) = \frac{\mathbb{E}[(x-Y)_+^\alpha]}{\Gamma(\alpha+1)}, \quad (1111)$$

$$I_{-1+}^\alpha \text{up}(x) = \frac{\mathbb{E}[(x-Z)_+^{\alpha-1}]}{\Gamma(\alpha)}, \quad (1112)$$

$$I_{0+}^\alpha Q^p(y) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(y)} x^{p-1} (y - F(x))^\alpha dx. \quad (1113)$$

For  $\Re \alpha > 1$ ,

$$I_{0+}^\alpha F(x) = 2^{\alpha-1} I_{0+}^{\alpha-1} F(x/2), \quad (1114)$$

which interpolates the integer primitive ladder but does not collapse a standard noninteger derivative to a single dilated Fabius value.

## 4 Exact fractional calculus in the Fabius and inverse clocks

Standard fractional calculus measures memory in the Euclidean difference  $x - t$ . The Fabius function supplies a second natural clock: probability mass  $F(x) - F(t)$ . Fractional calculus with respect to a function is classical in general form [7]; the exact Fabius specialization below appears to be absent from the audited corpus.

### 4.1 Conjugacy definition

Define the composition operator

$$(\mathcal{C}_F h)(y) = h(Q(y)), \quad (\mathcal{C}_F^{-1} g)(x) = g(F(x)). \quad (1115)$$

**Definition 4.1** (Fabius-clock integral and derivative). For  $\Re \alpha > 0$ ,

$$I_F^\alpha = \mathcal{C}_F^{-1} I_{0+}^\alpha \mathcal{C}_F. \quad (1116)$$

The first-order Fabius-clock derivative is

$$D_F = \mathcal{C}_F^{-1} D \mathcal{C}_F = \frac{1}{F'} \frac{d}{dx}. \quad (1117)$$

For  $n - 1 < \Re \beta < n$ , transport the usual Riemann–Liouville and Caputo operators [4, 6] by defining

$${}^{\text{RL}}D_F^\beta = D_F^n I_F^{n-\beta}, \quad (1118)$$

$${}^{\text{C}}D_F^\beta = I_F^{n-\beta} D_F^n. \quad (1119)$$

**Theorem 4.2** (kernel, semigroup, and exact conjugacy). For suitable  $h$ ,

$$(I_F^\alpha h)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt. \quad (1120)$$

Moreover,

$$I_F^\alpha I_F^\gamma = I_F^{\alpha+\gamma}, \quad {}^{\text{C}}D_F^\beta = \mathcal{C}_F^{-1} {}^{\text{C}}D_{0+}^\beta \mathcal{C}_F, \quad {}^{\text{RL}}D_F^\beta = \mathcal{C}_F^{-1} D_{0+}^\beta \mathcal{C}_F. \quad (1121)$$

*Proof.* Evaluate equation (1116) at  $y = F(x)$  and substitute  $v = F(t)$ :

$$\frac{1}{\Gamma(\alpha)} \int_0^{F(x)} (F(x) - v)^{\alpha-1} h(Q(v)) dv = \frac{1}{\Gamma(\alpha)} \int_0^x (F(x) - F(t))^{\alpha-1} h(t) F'(t) dt.$$

Every operator identity is then transported from ordinary fractional calculus by conjugacy. In particular, the semigroup is literal rather than formal.  $\square$

### 4.2 Power laws and Mittag–Leffler eigenfunctions

**Theorem 4.3** (diagonal action on powers). For  $\Re \lambda > -1$  and  $\Re \alpha > 0$ ,

$$I_F^\alpha F(x)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda + \alpha + 1)} F(x)^{\lambda+\alpha}. \quad (1122)$$

For the Riemann–Liouville derivative, wherever both sides are defined,

$${}^{\text{RL}}D_F^\beta F(x)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda - \beta + 1)} F(x)^{\lambda-\beta}. \quad (1123)$$

For  $n - 1 < \beta < n$ , the same formula holds for the Caputo derivative when  $\Re \lambda > n - 1$ ; the usual low-degree polynomial exceptions apply.

*Proof.* Under  $\mathcal{C}_F$ , the function  $F(x)^\lambda$  becomes  $y^\lambda$ . The assertions are exactly the beta-integral power laws for ordinary Riemann–Liouville and Caputo operators.  $\square$

**Corollary 4.4** (fractional eigenproblem). For  $0 < \beta \leq 1$  and  $a \in \mathbb{C}$ ,

$${}^{\text{C}}D_F^\beta E_\beta(a F(x)^\beta) = a E_\beta(a F(x)^\beta), \quad E_\beta(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\beta k + 1)}. \quad (1124)$$

Thus the ordinary exponential  $e^{a F(x)}$  is recovered at  $\beta = 1$ .

### 4.3 The dual inverse clock

The inverse function supplies the dual clock on the probability interval. Define

$$(I_Q^\alpha k)(y) = \frac{1}{\Gamma(\alpha)} \int_0^y (Q(y) - Q(v))^{\alpha-1} k(v) Q'(v) dv, \quad D_Q = \frac{1}{Q'} \frac{d}{dy}. \quad (1125)$$

This is conjugate to ordinary fractional calculus by  $x = Q(y)$ . Hence

$$I_Q^\alpha Q(y)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda + \alpha + 1)} Q(y)^{\lambda+\alpha}, \quad (1126)$$

$${}^{\text{RL}}D_Q^\beta Q(y)^\lambda = \frac{\Gamma(\lambda + 1)}{\Gamma(\lambda - \beta + 1)} Q(y)^{\lambda-\beta}. \quad (1127)$$

The pair  $(I_F, D_F)$  and  $(I_Q, D_Q)$  is therefore exactly dual under the mutually inverse coordinate maps  $F$  and  $Q$ .

*Remark 4.5* (What this does and does not say). The Fabius-clock derivative is not the standard fractional derivative of  $F$  as a function of Euclidean  $x$ . It is a nonlinear-coordinate operator in which the intricate Fabius geometry is moved into the kernel. Its advantage is exact algebra; the standard Caputo and Weyl derivatives studied below retain the original Euclidean clock and are analytically more delicate.

## 5 Caputo and Riemann–Liouville transmutation through the inverse

The ordinary inverse-chain rule [equation \(1108\)](#) has a nonlocal fractional counterpart. It converts a singular derivative of  $H \circ Q$  in the  $y$  variable into an integral over the much better behaved  $x$  variable.

### 5.1 Left fractional integrals and derivatives

**Theorem 5.1** (general inverse-composition transmutation). *Let  $q = Q(y)$  and  $\Re \alpha > 0$ . If  $H$  is integrable against the displayed kernels, then*

$$I_{0+}^\alpha [H \circ Q](y) = \frac{1}{\Gamma(\alpha)} \int_0^q (y - F(x))^{\alpha-1} H(x) F'(x) dx, \quad (1128)$$

$$= \frac{H(0)y^\alpha}{\Gamma(\alpha + 1)} + \frac{1}{\Gamma(\alpha + 1)} \int_0^q (y - F(x))^\alpha H'(x) dx, \quad (1129)$$

where the second identity assumes absolute continuity of  $H$ . For  $0 < \beta < 1$ ,

$${}^{\text{C}}D_{0+}^\beta [H \circ Q](y) = \frac{1}{\Gamma(1 - \beta)} \int_0^q (y - F(x))^{-\beta} H'(x) dx, \quad (1130)$$

and

$$D_{0+}^\beta [H \circ Q](y) = \frac{H(0)y^{-\beta}}{\Gamma(1 - \beta)} + {}^{\text{C}}D_{0+}^\beta [H \circ Q](y). \quad (1131)$$

*Proof.* Substitute  $v = F(x)$  in the definition of  $I_{0+}^\alpha [H(Q(v))]$  to obtain [equation \(1128\)](#). Since

$$d(y - F(x))^\alpha = -\alpha(y - F(x))^{\alpha-1} F'(x) dx,$$

integration by parts gives [equation \(1129\)](#). For the Caputo derivative,

$$\frac{d}{dv} H(Q(v)) = H'(Q(v)) Q'(v).$$

After  $v = F(x)$ , the factors  $Q'(F(x))F'(x)$  cancel exactly, leaving [equation \(1130\)](#). The final identity is the standard Riemann–Liouville/Caputo endpoint correction.  $\square$



**Corollary 5.2** (powers of the inverse Fabius function). *For  $r > 0$  and  $0 < \beta < 1$ ,*

$${}^C D_{0+}^\beta Q(y)^r = \frac{r}{\Gamma(1-\beta)} \int_0^{Q(y)} x^{r-1} (y - F(x))^{-\beta} dx. \quad (1132)$$

*Because  $Q(0) = 0$ , the Riemann–Liouville and Caputo derivatives coincide for  $Q^r$  in this range whenever the integral is finite.*

## 5.2 All positive noninteger orders

Let  $n - 1 < \beta < n$ . From [equation \(1108\)](#),

$$\frac{d^n}{dy^n} [H(Q(y))] = (L_F^n H)(Q(y)). \quad (1133)$$

Therefore:

**Proposition 5.3** (higher-order Caputo transmutation). *Suppose  $H \circ Q$  belongs to the classical Caputo domain on  $[0, y]$  and  $(L_F^n H)(x)F'(x)$  is integrable against the displayed kernel. Then*

$${}^C D_{0+}^\beta [H \circ Q](y) = \frac{1}{\Gamma(n-\beta)} \int_0^{Q(y)} (y - F(x))^{n-\beta-1} (L_F^n H)(x) F'(x) dx. \quad (1134)$$

*The Riemann–Liouville derivative is obtained by adding the usual finite endpoint jet*

$$\sum_{k=0}^{n-1} \frac{(H \circ Q)^{(k)}(0+)}{\Gamma(k-\beta+1)} y^{k-\beta}, \quad (1135)$$

*when these endpoint values exist.*

### Higher-order inverse derivatives

The inverse Fabius function has singular derivatives at 0 and 1. Consequently, a formula obtained by formally inserting  $H(x) = x$  into [equation \(1134\)](#) need not define a classical Caputo derivative for every  $\beta > 1$ . The theorem is deliberately stated in terms of the actual Caputo domain. Distributional or weighted-space extensions are promising, but require a separate endpoint-domain analysis.

## 5.3 Right-sided transmutation

For  $0 < \beta < 1$  and  $q = Q(y)$ , the right-sided counterparts are

$$I_{1-}^\alpha [H \circ Q](y) = \frac{H(1)(1-y)^\alpha}{\Gamma(\alpha+1)} - \frac{1}{\Gamma(\alpha+1)} \int_q^1 (F(x) - y)^\alpha H'(x) dx, \quad (1136)$$

$${}^C D_{1-}^\beta [H \circ Q](y) = -\frac{1}{\Gamma(1-\beta)} \int_q^1 (F(x) - y)^{-\beta} H'(x) dx, \quad (1137)$$

$$D_{1-}^\beta [H \circ Q](y) = \frac{H(1)(1-y)^{-\beta}}{\Gamma(1-\beta)} + {}^C D_{1-}^\beta [H \circ Q](y). \quad (1138)$$

They follow by the same substitution and integration by parts. Symmetry  $Q(1-y) = 1 - Q(y)$  converts left endpoint statements into right endpoint statements without recomputing the Fabius asymptotics.

## 6 Fractional endpoint laws for inverse powers

The integral formulas above permit endpoint asymptotics without differentiating the Lambert– $W$  inverse expansion term by term.

**Theorem 6.1** (fractional primitive and Caputo endpoint laws). *Let  $r > 0$  and  $\alpha > 0$ . Then*

$$I_{0+}^{\alpha} Q(y)^r \sim \frac{Q(y)^r y^{\alpha}}{\Gamma(\alpha + 1)} \quad (y \downarrow 0). \quad (1139)$$

For  $0 < \beta < 1$ ,

$${}^C D_{0+}^{\beta} Q(y)^r \sim \frac{Q(y)^r y^{-\beta}}{\Gamma(1 - \beta)} \quad (y \downarrow 0). \quad (1140)$$

*Proof.* Put  $q = Q(y)$  and scale  $x = qt$  in [equation \(1113\)](#). The ratio of the two sides of [equation \(1139\)](#) is

$$r \int_0^1 t^{r-1} \left( 1 - \frac{F(qt)}{F(q)} \right)^{\alpha} dt. \quad (1141)$$

For each fixed  $t < 1$ , rapid variation [equation \(1098\)](#) makes the parenthesis tend to 1. The integrand is bounded by  $rt^{r-1}$ , so dominated convergence gives 1.

Likewise, [equation \(1132\)](#) yields the ratio

$$r \int_0^1 t^{r-1} \left( 1 - \frac{F(qt)}{F(q)} \right)^{-\beta} dt. \quad (1142)$$

For  $q \leq 1/2$ , convexity gives  $F(qt)/F(q) \leq t$ ; hence the integrand is bounded by  $rt^{r-1}(1-t)^{-\beta}$ , which is integrable because  $r > 0$  and  $\beta < 1$ . Dominated convergence again gives 1.  $\square$

**Corollary 6.2** (right endpoint). *As  $y \uparrow 1$ , the corresponding right-sided laws hold for  $(1 - Q(y))^r$  with  $1 - y$  in place of  $y$ .*

**Remark 6.3** (Slow convergence). The limit in [equation \(1140\)](#) is rigorous but numerically slow. For  $r = 1$ ,  $\beta = 0.4$ , the ratio in [equation \(1142\)](#) decreases only from approximately 1.1835 at  $y = 10^{-2}$  to 1.1127 at  $y = 10^{-7}$ . This is consistent with a first correction on the Lambert scale  $\rho(y)^{-1}$ , where

$$\rho(y) = \sqrt{\frac{2 \log(1/y)}{\log 2}}. \quad (1143)$$

A phase-resolved conjecture is stated in [conjecture 16.1](#).

## 7 A standard fractional bridge from Fabius to Rvachev

Because  $F$  is flat at 0, its left Riemann–Liouville and Caputo derivatives coincide for every positive noninteger order. Unlike the Fabius-clock operator, the following theorem concerns the ordinary Euclidean clock.

**Theorem 7.1** (all-order fractional  $F$ -up bridge). *Let  $n - 1 < \beta < n$  and  $0 < x < 1$ . Then*

$$D_{0+}^{\beta} F(x) = {}^C D_{0+}^{\beta} F(x) = 2^{\beta} \left( I_{-1+}^{n-\beta} \text{up}^{(n-1)} \right) (2x - 1). \quad (1144)$$

In particular, for  $0 < \beta < 1$ ,

$$D_{0+}^{\beta} F(x) = 2^{\beta} \left( I_{-1+}^{1-\beta} \text{up} \right) (2x - 1). \quad (1145)$$

*Proof.* Flatness removes every endpoint correction between the two fractional derivatives. Since  $F'(t) = 2 \operatorname{up}(2t - 1)$ ,

$$F^{(n)}(t) = 2^n \operatorname{up}^{(n-1)}(2t - 1). \quad (1146)$$

Insert this into  $I_{0+}^{n-\beta} F^{(n)}(x)$  and substitute  $s = 2t - 1$ . The factors are  $2^n$  from differentiation,  $2^{-1}$  from  $dt$ , and  $2^{-(n-\beta-1)}$  from the kernel, leaving exactly  $2^\beta$ .  $\square$

*Remark 7.2* (Nearby formal integral bridge). The displayed Riemann–Liouville/Caputo *derivative* theorem is not asserted to be formalized. What is now machine checked is the adjacent real positive-order integral identity

$$I_{-1+}^{\alpha+1} \operatorname{up}(x) = 2^\alpha I_{0+}^\alpha F((x+1)/2), \quad x \geq -1,$$

under the name `Fabius.fractionalVolterra_add_one_rvachevUp`. Passing from it to (1144) still requires the fractional-derivative and iterated derivative packaging stated in this report.

*Remark 7.3* (Interpolation endpoints). For  $0 < \beta < 1$ , letting  $\beta \downarrow 0$  in equation (1145) gives

$$F(x) = \int_{-1}^{2x-1} \operatorname{up}(t) dt,$$

whereas  $\beta \uparrow 1$  gives  $F'(x) = 2 \operatorname{up}(2x - 1)$ . Thus the identity is a literal continuum between the CDF and its density.

### 7.1 Fractional derivatives of $\operatorname{up}$ from the support boundary

On the left half of its support,  $\operatorname{up}(x) = F(x + 1)$ . Therefore, for  $-1 < x < 0$ ,

$$D_{-1+}^\beta \operatorname{up}(x) = 2^\beta \left( I_{-1+}^{n-\beta} \operatorname{up}^{(n-1)} \right) (2x + 1), \quad n - 1 < \beta < n. \quad (1147)$$

On the right half, reflection gives

$$D_{1-}^\beta \operatorname{up}(x) = 2^\beta \left( I_{-1+}^{n-\beta} \operatorname{up}^{(n-1)} \right) (1 - 2x), \quad 0 < x < 1, \quad (1148)$$

with the standard right-Caputo sign convention. These are boundary-anchored identities; they should not be confused with the translation-invariant Weyl operators in the next section.

### 7.2 Fractional primitives of $1 - F$

On  $[0, 1]$ ,  $\operatorname{up}(x) = 1 - F(x)$ . Hence

$$I_{0+}^\alpha \operatorname{up}(x) = \frac{x^\alpha}{\Gamma(\alpha + 1)} - I_{0+}^\alpha F(x), \quad (1149)$$

and the probabilistic formula equation (1111) gives

$$I_{0+}^\alpha \operatorname{up}(x) = \frac{x^\alpha - \mathbb{E}[(x - Y)_+^\alpha]}{\Gamma(\alpha + 1)}. \quad (1150)$$

At  $x = 1$ , symmetry yields

$$I_{0+}^\alpha F(1) = \frac{\mathcal{M}(\alpha)}{\Gamma(\alpha + 1)}, \quad \mathcal{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(y)^s dy, \quad (1151)$$

so endpoint fractional integration is exactly the entire inverse-moment function already developed in the corpus.

## 8 Whole-line Weyl derivatives and complex-order refinement

Boundary-anchored fractional derivatives break translation invariance. The compact support and entire Fourier product of up make Weyl calculus the natural whole-line alternative.

### 8.1 Fourier-multiplier definition

Following the standard Weyl multiplier conventions [5], choose principal branches and define, initially on Schwartz functions and then by tempered-distribution continuation,

$$\mathfrak{D}_+^\sigma f = \mathcal{F}^{-1}[(i\xi)^\sigma \widehat{f}(\xi)], \quad (1152)$$

$$\mathfrak{D}_-^\sigma f = \mathcal{F}^{-1}[(-i\xi)^\sigma \widehat{f}(\xi)]. \quad (1153)$$

For  $\Re\alpha > 0$ , the left Weyl integral is

$$(\mathfrak{I}_+^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_{-\infty}^x (x-t)^{\alpha-1} f(t) dt, \quad (1154)$$

whose multiplier is  $(i\xi)^{-\alpha}$  in the boundary-value sense.

### 8.2 The complex-order hierarchy

**Theorem 8.1** (Weyl–Rvachev refinement hierarchy). *Let*

$$W_\sigma = \mathfrak{D}_+^\sigma \text{ up}. \quad (1155)$$

*For every complex  $\sigma$  for which the terms are interpreted consistently as tempered distributions,*

$$W_\sigma(x) = 2^\sigma (W_{\sigma-1}(2x+1) - W_{\sigma-1}(2x-1)). \quad (1156)$$

*If  $n-1 < \beta < n$ , this becomes*

$$\mathfrak{I}_+^\beta \text{ up}(x) = 2^\beta \left[ (\mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)})(2x+1) - (\mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)})(2x-1) \right]. \quad (1157)$$

*At  $\beta = 1$  it reduces to Rvachev’s differential–difference formula (1090).*

*Proof.* Apply  $\mathfrak{D}_+^{\sigma-1}$  to [equation \(1090\)](#). Positive dilation obeys

$$\mathfrak{D}_+^\alpha [f(2x+a)] = 2^\alpha (\mathfrak{D}_+^\alpha f)(2x+a).$$

The prefactor 2 in [equation \(1090\)](#) therefore produces  $2^\sigma$ . Equivalently, multiply the Fourier refinement [equation \(1093\)](#) by  $(i\xi)^\sigma$  and invert. Finally,  $W_{\beta-1} = \mathfrak{I}_+^{n-\beta} \text{ up}^{(n-1)}$  by equality of Fourier multipliers.  $\square$

**Corollary 8.2** (right hierarchy and reflection). *Because up is even,*

$$\mathfrak{D}_-^\sigma \text{ up}(x) = \mathfrak{D}_+^\sigma \text{ up}(-x). \quad (1158)$$

*The corresponding right-sided refinement is obtained by reflecting [equation \(1156\)](#).*

### 8.3 Riesz symmetrization

For real  $0 < \beta < 2$ , define the Riesz fractional Laplacian independently by

$$(-\Delta)^{\beta/2} f(\xi) = |\xi|^\beta \widehat{f}(\xi). \quad (1159)$$

When  $\cos(\pi\beta/2) \neq 0$ ,

$$(-\Delta)^{\beta/2} f = \frac{\mathfrak{D}_+^\beta f + \mathfrak{D}_-^\beta f}{2 \cos(\pi\beta/2)}. \quad (1160)$$

At  $\beta = 1$  the quotient is understood by the Fourier definition (or via the Hilbert-transform derivative), not by separately evaluating the vanishing numerator and denominator.

## 9 Generalized Stieltjes transforms of complex order

The ordinary Cauchy transform is only the order-1 member of a much richer family. For real  $z > 1$  and  $\lambda \in \mathbb{C}$ , define

$$S_\lambda(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z-t)^\lambda} dt = \mathbb{E}(z - Z)^{-\lambda}, \quad (1161)$$

where the real logarithm fixes the power. Analytic continuation in  $z$  is available on the slit plane after choosing a branch.

### 9.1 Moment, derivative, and dyadic-order hierarchies

**Theorem 9.1** (complete generalized Stieltjes hierarchy). *For fixed  $z > 1$ ,  $S_\lambda(z)$  is entire in  $\lambda$ . For  $|z| > 1$ ,*

$$S_\lambda(z) = z^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda)_{2k}}{(2k)!} \frac{c_k}{z^{2k}}, \quad (1162)$$

locally normally in  $(\lambda, z)$ . Its  $z$  derivatives satisfy

$$\frac{d^n}{dz^n} S_\lambda(z) = (-1)^n (\lambda)_n S_{\lambda+n}(z). \quad (1163)$$

Moreover,

$$S_\lambda(z) = 2^{\lambda-1} \int_{-1}^1 S_\lambda(2z - v) dv, \quad (1164)$$

and, for  $\lambda \neq 1$ ,

$$S_\lambda(z) = \frac{2^{\lambda-1}}{1-\lambda} (S_{\lambda-1}(2z+1) - S_{\lambda-1}(2z-1)). \quad (1165)$$

At  $\lambda = 1$ , if

$$L(z) = \mathbb{E} \log(z - Z), \quad (1166)$$

then

$$S_1(z) = L(2z+1) - L(2z-1). \quad (1167)$$

*Proof.* For  $z > 1$ ,  $\log(z - Z)$  is bounded, so  $S_\lambda(z) = \mathbb{E} \exp[-\lambda \log(z - Z)]$  is entire. Expanding  $(1 - Z/z)^{-\lambda}$  and using symmetry removes odd moments, proving [equation \(1162\)](#). Differentiation under the integral gives [equation \(1163\)](#).

The random series satisfies

$$Z \stackrel{d}{=} \frac{V + Z'}{2}, \quad V \sim \text{Unif}[-1, 1], \quad (1168)$$

with  $Z'$  an independent copy. Conditioning on  $V$  gives [equation \(1164\)](#). Changing variables turns its integral into  $\int_{2z-1}^{2z+1} S_\lambda(w) dw$ . Since  $S'_{\lambda-1} = (1-\lambda)S_\lambda$ , this proves [equation \(1165\)](#). Taking the removable limit at  $\lambda = 1$  and using  $L' = S_1$  gives [equation \(1167\)](#).  $\square$

**Remark 9.2** (Interpolation between Appell polynomials and resolvents). For  $m \in \mathbb{N}_0$ ,

$$S_{-m}(z) = \mathbb{E}(z - Z)^m \quad (1169)$$

is a rational-coefficient Appell polynomial. Positive integer orders are ordinary higher resolvents. Thus [equation \(1165\)](#) is a single complex-order hierarchy connecting the polynomial and resolvent sides.

## 9.2 One-sided version

Define, for  $z > 1$ ,

$$R_\lambda(z) = \mathbb{E}(z - Y)^{-\lambda}. \quad (1170)$$

Since  $Y \stackrel{d}{=} (U + Y')/2$  with  $U$  uniform on  $[0, 1]$ ,

$$R_\lambda(z) = 2^\lambda \int_0^1 R_\lambda(2z - u) \, du, \quad (1171)$$

$$= \frac{2^\lambda}{1 - \lambda} (R_{\lambda-1}(2z) - R_{\lambda-1}(2z - 1)), \quad \lambda \neq 1. \quad (1172)$$

The centered and one-sided transforms are related by

$$S_\lambda(z) = 2^{-\lambda} R_\lambda\left(\frac{z+1}{2}\right). \quad (1173)$$

For  $\lambda > 0$ ,  $S_\lambda$  and  $R_\lambda$  are completely monotone on their real domains; all alternating derivatives are explicitly positive by [equation \(1163\)](#).

## 10 Complex-order resolvents involving $F$ and the inverse

### 10.1 Distribution and survival resolvents

For  $z > 1$  and  $\lambda \neq 1$ , Fubini or integration by parts gives

$$\int_0^1 \frac{F(x)}{(z-x)^\lambda} \, dx = \frac{R_{\lambda-1}(z) - (z-1)^{1-\lambda}}{1-\lambda}, \quad (1174)$$

$$\int_0^1 \frac{1-F(x)}{(z-x)^\lambda} \, dx = \frac{z^{1-\lambda} - R_{\lambda-1}(z)}{1-\lambda}. \quad (1175)$$

At  $\lambda = 1$ , the removable limits are

$$\int_0^1 \frac{F(x)}{z-x} \, dx = \mathbb{E} \log(z - Y) - \log(z - 1), \quad (1176)$$

$$\int_0^1 \frac{1-F(x)}{z-x} \, dx = \log z - \mathbb{E} \log(z - Y). \quad (1177)$$

These extend the integer resolvent formulas in the corpus to arbitrary complex order.

### 10.2 Inverse-composition resolvents

**Theorem 10.1** (complex inverse-resolvent transmutation). *Let  $H$  be absolutely continuous on  $[0, 1]$ ,  $z > 1$ , and  $\lambda \neq 1$ . Then*

$$T_{\lambda,H}(z) := \int_0^1 (z-y)^{-\lambda} H(Q(y)) \, dy \quad (1178)$$

$$= \frac{H(0)z^{1-\lambda} - H(1)(z-1)^{1-\lambda}}{1-\lambda} + \frac{1}{1-\lambda} \int_0^1 (z-F(x))^{1-\lambda} H'(x) \, dx. \quad (1179)$$

At  $\lambda = 1$ ,

$$T_{1,H}(z) = H(0) \log z - H(1) \log(z-1) + \int_0^1 H'(x) \log(z-F(x)) \, dx. \quad (1180)$$

Furthermore,

$$\frac{d^n}{dz^n} T_{\lambda,H}(z) = (-1)^n (\lambda)_n T_{\lambda+n,H}(z). \quad (1181)$$

*Proof.* Substitute  $y = F(x)$ :

$$T_{\lambda,H}(z) = \int_0^1 (z - F(x))^{-\lambda} H(x) F'(x) dx.$$

Since  $d(z - F(x))^{1-\lambda} = -(1-\lambda)(z - F(x))^{-\lambda} F'(x) dx$ , integration by parts proves [equation \(1179\)](#). Expanding at  $1 - \lambda = 0$  gives [equation \(1180\)](#); differentiation under the integral gives the hierarchy.  $\square$

For  $H(x) = x^p$  with  $p > 0$ ,

$$\int_0^1 (z - y)^{-\lambda} Q(y)^p dy = \frac{-(z - 1)^{1-\lambda} + p \int_0^1 x^{p-1} (z - F(x))^{1-\lambda} dx}{1 - \lambda}. \quad (1182)$$

This is the complex-order extension of the inverse logarithmic resolvent found in the latest corpus report.

### 10.3 Large- $z$ mixed-moment series

For  $|z| > 1$ ,

$$T_{\lambda,H}(z) = z^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda)_n}{n! z^n} \int_0^1 y^n H(Q(y)) dy. \quad (1183)$$

For  $H(x) = x^p$ , integration by parts after  $y = F(x)$  gives

$$\int_0^1 y^n Q(y)^p dy = \frac{1}{n+1} - \frac{p}{n+1} \int_0^1 x^{p-1} F(x)^{n+1} dx. \quad (1184)$$

Thus every coefficient is reduced to a beta-type Fabius integral. When  $p$  and  $n$  are nonnegative integers, finite dyadic formulas and rational moments can be inserted wherever the existing corpus machinery applies.

## 11 Riesz potentials and the fractional Laplacian of up

This section gives the cleanest new transform consequence. Compact support turns the nonlocal exterior fractional Laplacian into the generalized Stieltjes hierarchy exactly, not merely asymptotically.

### 11.1 Exact exterior fractional Laplacian

For  $0 < \beta < 2$ , use the standard one-dimensional normalization [8]:

$$C_{1,\beta} = \frac{\beta 2^{\beta-1} \Gamma((1+\beta)/2)}{\sqrt{\pi} \Gamma(1-\beta/2)}. \quad (1185)$$

Then

$$(-\Delta)^{\beta/2} f(x) = C_{1,\beta} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(t)}{|x - t|^{1+\beta}} dt. \quad (1186)$$

**Theorem 11.1** (exact exterior Stieltjes representation). *For  $0 < \beta < 2$  and  $|x| > 1$ ,*

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} S_{1+\beta}(|x|). \quad (1187)$$

Consequently,

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} |x|^{-1-\beta} \sum_{k=0}^{\infty} \frac{(1+\beta)_{2k}}{(2k)!} \frac{c_k}{|x|^{2k}}. \quad (1188)$$

The first terms are

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} |x|^{-1-\beta} \left[ 1 + \frac{(1+\beta)(2+\beta)}{18x^2} + \frac{19(1+\beta)(2+\beta)(3+\beta)(4+\beta)}{16200x^4} + O(x^{-6}) \right]. \quad (1189)$$

*Proof.* For  $|x| > 1$ ,  $\text{up}(x) = 0$  and the singularity is absent. Thus

$$(-\Delta)^{\beta/2} \text{up}(x) = -C_{1,\beta} \int_{-1}^1 \frac{\text{up}(t)}{|x-t|^{1+\beta}} dt.$$

For  $x > 1$  this is  $-C_{1,\beta} S_{1+\beta}(x)$ . Evenness gives the same formula with  $|x|$  for  $x < -1$ . Insert [equation \(1162\)](#) and the exact moments [equation \(1096\)](#).  $\square$

**Corollary 11.2** (sign and conservation). *The exterior fractional Laplacian is strictly negative. At the center,*

$$\begin{aligned} (-\Delta)^{\beta/2} \text{up}(0) &= C_{1,\beta} \int_{\mathbb{R}} \frac{1 - \text{up}(t)}{|t|^{1+\beta}} dt \\ &= 2C_{1,\beta} \left( \int_0^1 \frac{1 - \text{up}(t)}{t^{1+\beta}} dt + \frac{1}{\beta} \right) > 0. \end{aligned} \quad (1190)$$

Moreover,

$$\int_{\mathbb{R}} (-\Delta)^{\beta/2} \text{up}(x) dx = 0. \quad (1191)$$

Hence at least one zero occurs on each half-axis.

*Proof.* The exterior sign follows from [equation \(1187\)](#). Since  $\text{up}(0) = 1$ , the singular-integral definition at 0 gives the first line of [equation \(1190\)](#); splitting at  $|t| = 1$  gives the second. The function  $(-\Delta)^{\beta/2} \text{up}$  is integrable because it is smooth on compact sets and has the exterior decay [equation \(1188\)](#). Its integral is therefore its Fourier transform at 0, namely  $|0|^\beta \widehat{\text{up}}(0) = 0$ . Evenness and the opposite signs at 0 and outside the support force a zero on each half-axis.  $\square$

## 11.2 Riesz potentials

For  $0 < \alpha < 1$ , the Riesz potential with multiplier  $|\xi|^{-\alpha}$  is

$$(I_\alpha^R f)(x) = \kappa_\alpha \int_{\mathbb{R}} |x-t|^{\alpha-1} f(t) dt, \quad \kappa_\alpha = \frac{\Gamma((1-\alpha)/2)}{2^\alpha \sqrt{\pi} \Gamma(\alpha/2)}. \quad (1192)$$

**Theorem 11.3** (exterior Riesz-potential expansion). *For  $0 < \alpha < 1$  and  $|x| > 1$ ,*

$$I_\alpha^R \text{up}(x) = \kappa_\alpha S_{1-\alpha}(|x|) = \kappa_\alpha |x|^{\alpha-1} \sum_{k=0}^{\infty} \frac{(1-\alpha)_{2k}}{(2k)!} \frac{c_k}{|x|^{2k}}. \quad (1193)$$

*In particular, the leading far-field coefficient is exactly the unit mass of up; all corrections are rational polynomials in  $\alpha$ .*



### 11.3 A numerical one-node phenomenon

The accompanying experiment reconstructs  $\text{up}$  from the infinite sinc product, then evaluates the nonperiodic singular integral [equation \(1186\)](#). A symmetric second-difference removes the principal-value singularity and a Gauss–Jacobi rule integrates the remaining  $u^{1-\beta}$  weight. On a 601-point profile grid, exactly one positive sign change in  $(0, 1)$  is resolved for each tested order; Brent refinement gives the following approximations (the displayed seven decimal places are stable under a moderate change of quadrature order):

| $\beta$ | positive zero $a_\beta$ |
|---------|-------------------------|
| 0.25    | 0.6721595               |
| 0.50    | 0.6100983               |
| 0.75    | 0.5741282               |
| 1.00    | 0.5506118               |
| 1.25    | 0.5344913               |
| 1.50    | 0.5232645               |
| 1.75    | 0.5154602               |

**Conjecture 11.4** (one-node law). *For every  $0 < \beta < 2$ , the even function  $(-\Delta)^{\beta/2} \text{up}$  has exactly two simple zeros  $\pm a_\beta$ . It is positive on  $(-a_\beta, a_\beta)$  and negative outside. Moreover,  $a_\beta$  is strictly decreasing, with*

$$\lim_{\beta \downarrow 0} a_\beta = 1, \quad \lim_{\beta \uparrow 2} a_\beta = \frac{1}{2}. \quad (1194)$$

The endpoint  $1/2$  is natural: on  $(0, 1/2)$ ,  $-\text{up}'' = F'' = 4F'(2x) > 0$ , while the integer second derivative changes its Thue–Morse cell sign across  $x = 1/2$ . Proving uniqueness for noninteger order will likely require a variation-diminishing or total-positivity principle for the singular kernel combined with the special unimodality of  $\text{up}$ .

## 12 Fractional energies and autocorrelation transforms

The integer derivative norm law in the corpus samples a continuous spectral Mellin transform at even integers. Define

$$\mathcal{E}(s) = \frac{1}{2\pi} \int_{\mathbb{R}} |\xi|^s |\phi(\xi)|^2 d\xi, \quad \Re s > -1. \quad (1195)$$

The sinc product decays faster than every power, so this is holomorphic in the half-plane  $\Re s > -1$ .

**Proposition 12.1** (log-convex fractional energy). *For real  $s > -1$ ,  $\mathcal{E}(s) > 0$  and  $\log \mathcal{E}(s)$  is convex. At even integers,*

$$\mathcal{E}(2n) = \left\| \text{up}^{(n)} \right\|_2^2 = 2^{n(n+1)} \mathcal{E}(0). \quad (1196)$$

Consequently, for  $2n \leq s \leq 2n+2$ ,

$$\mathcal{E}(s) \leq \mathcal{E}(0) 2^{(n+1)(s-n)}. \quad (1197)$$

*Proof.* Positivity is immediate. Hölder’s inequality applied to  $|\xi|^s |\phi|^2 d\xi$  proves log-convexity. Parseval gives the first equality in [equation \(1196\)](#); the second is the existing derivative-equimeasurability law. Convex interpolation between the values at  $2n$  and  $2n+2$  gives [equation \(1197\)](#).  $\square$

For a shift  $a \in \mathbb{R}$ , define the fractional autocorrelation

$$\mathcal{C}_s(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\xi a} |\xi|^s |\phi(\xi)|^2 d\xi. \quad (1198)$$

Since  $\text{up}$  is even,

$$\mathcal{C}_s(a) = (-\Delta_a)^{s/2}(\text{up} * \text{up})(a), \quad \mathcal{C}_s(0) = \mathcal{E}(s). \quad (1199)$$

Thus the entire family of fractional Sobolev energies and shifted correlations is encoded by the same lacunary sinc product. For  $s = 2n$ , this reduces to integer derivatives; for noninteger  $s$ , it is a new continuous interpolation problem closely related to the shifted Stieltjes–Wigert spectral measure in the derivative-norm report.

## 13 Definite integrals and transform catalog

This section gathers a practical set of exact reductions. Most of the first group is already in the corpus; several composition identities at the end are new elementary consequences.

### 13.1 CDF–survival master identities

Let  $G$  be absolutely continuous on  $[0, 1]$ . Then

$$\int_0^1 G'(x)F(x) \, dx = G(1) - \mathbb{E}G(Y), \quad (1200)$$

$$\int_0^1 G'(x)(1 - F(x)) \, dx = \mathbb{E}G(Y) - G(0). \quad (1201)$$

Examples include

$$\int_0^1 x^{s-1}F(x) \, dx = \frac{1 - \mathcal{M}(s)}{s}, \quad (1202)$$

$$\int_0^1 x^{s-1}\text{up}(x) \, dx = \frac{\mathcal{M}(s)}{s}, \quad (1203)$$

$$\int_0^1 e^{tx}F(x) \, dx = \frac{e^t - P(t)}{t}, \quad (1204)$$

$$\int_0^1 e^{tx}\text{up}(x) \, dx = \frac{P(t) - 1}{t}. \quad (1205)$$

The apparent singularities at  $s = 0$  or  $t = 0$  are removable. Differentiating in the parameters generates logarithmic and polynomially weighted families.

### 13.2 Quantile transport and power duality

For every integrable  $H$ ,

$$\int_0^1 H(Q(y)) \, dy = \int_0^1 H(x)F'(x) \, dx = \mathbb{E}H(Y). \quad (1206)$$

For  $\Re p, \Re q > 0$ ,

$$p \int_0^1 x^{p-1}F(x)^q \, dx + q \int_0^1 y^{q-1}Q(y)^p \, dy = 1. \quad (1207)$$

The universal beta family is

$$\int_0^1 F(x)^a(1 - F(x))^b F'(x) \, dx = B(a+1, b+1), \quad \Re a, \Re b > -1. \quad (1208)$$

Using  $F' = 2\text{up}(2x - 1)$ ,

$$\int_0^1 F(x)^a \text{up}(x)^b \text{up}(2x - 1) \, dx = \frac{1}{2}B(a+1, b+1). \quad (1209)$$

This gives, for example,

$$\int_0^1 F(x)(1 - F(x))F'(x) dx = \frac{1}{6}. \quad (1210)$$

### 13.3 Full-line analytic kernels and Fourier transforms

If  $h(t) = \sum_{n \geq 0} a_n t^n$  is entire, then

$$\int_{-1}^1 h(t) \operatorname{up}(t) dt = \sum_{k=0}^{\infty} a_{2k} c_k. \quad (1211)$$

The Fourier, bilateral Laplace, Bessel, and generalized Stieltjes transforms are therefore

$$\int_{-1}^1 e^{i\xi t} \operatorname{up}(t) dt = \prod_{j \geq 1} \operatorname{sinc}(\xi/2^j), \quad (1212)$$

$$\int_{-1}^1 e^{st} \operatorname{up}(t) dt = \prod_{j \geq 1} \frac{\sinh(s/2^j)}{s/2^j}, \quad (1213)$$

$$\int_{-1}^1 J_0(at) \operatorname{up}(t) dt = \sum_{k=0}^{\infty} \frac{(-1)^k (a^2/4)^k}{(k!)^2} c_k, \quad (1214)$$

$$\int_{-1}^1 (z - t)^{-\lambda} \operatorname{up}(t) dt = S_{\lambda}(z). \quad (1215)$$

### 13.4 Product and order-statistic integrals

Let  $m, n \geq 1$ , let  $Y_1, Y_2, \dots$  be independent copies of  $Y$ ,  $A_m = \max(Y_1, \dots, Y_m)$ , and  $B_n = \min(Y_{m+1}, \dots, Y_{m+n})$ . Then

$$F(x)^m (1 - F(x))^n = \mathbb{P}(A_m \leq x < B_n), \quad (1216)$$

and hence, for real  $p > -1$ ,

$$\int_0^1 x^p F(x)^m (1 - F(x))^n dx = \frac{1}{p+1} \mathbb{E} [(B_n^{p+1} - A_m^{p+1})_+]. \quad (1217)$$

This is an exact probabilistic closed form even when no reduction to rational moments is available. The Lambert endpoint theory controls its large- $m$  or large- $n$  regimes.

### 13.5 Compositions with the inverse density

Equation 1088 yields a useful family that appears not to have been isolated in the audited integral reports. For  $p > -1$ ,

$$\int_0^1 (F'(Q(y)))^p dy = \int_0^1 F'(x)^{p+1} dx = 2^p \int_{-1}^1 \operatorname{up}(t)^{p+1} dt, \quad (1218)$$

$$\int_0^1 \operatorname{up}(2Q(y) - 1)^p dy = \int_{-1}^1 \operatorname{up}(t)^{p+1} dt, \quad (1219)$$

$$\int_0^1 Q'(y)^{-p} dy = 2^p \|\operatorname{up}\|_{p+1}^{p+1}. \quad (1220)$$

The proof is the single substitution  $y = F(x)$  followed by  $t = 2x - 1$ . These formulas tie inverse-derivative integrals directly to the complete  $L^p$  spectrum of the compact density.

## 14 Indefinite integral catalog

To avoid ambiguity, an “indefinite integral” below means the normalized primitive with zero initial data; arbitrary constants or lower-degree polynomials can be restored afterward.

### 14.1 Fabius and reflected Fabius functions

For  $0 \leq x \leq 1$ ,

$$\int_0^x F(t) dt = F(x/2), \quad (1221)$$

$$J^n F(x) = 2^{\binom{n}{2}} F(x/2^n), \quad (1222)$$

$$\int_0^x (1 - F(t)) dt = x - F(x/2), \quad (1223)$$

$$J^n(1 - F)(x) = \frac{x^n}{n!} - 2^{\binom{n}{2}} F(x/2^n). \quad (1224)$$

Because  $F(1 - x) = 1 - F(x)$ , these are also primitives of the reflected Fabius function. Polynomial and arbitrary complex-power weights are supplied by [equation \(1103\)](#).

### 14.2 Rvachev folds

On  $[-1, 0]$ ,  $\text{up}(x) = F(x + 1)$ , while on  $[0, 1]$ ,  $\text{up}(x) = 1 - F(x)$ . Thus a normalized first primitive based at  $-1$  is

$$\int_{-1}^x \text{up}(t) dt = \begin{cases} F((x + 1)/2), & -1 \leq x \leq 0, \\ 1 - F((1 - x)/2), & 0 \leq x \leq 1, \\ 1, & x \geq 1. \end{cases} \quad (1225)$$

Higher integer primitives follow either by repeated use of [equation \(1222\)](#) on each fold or from the exact truncated moment

$$I_{-1+}^n \text{up}(x) = \frac{1}{(n - 1)!} \mathbb{E}[(x - Z)_+^{n-1}]. \quad (1226)$$

For  $x \geq 1$  this becomes an Appell polynomial in  $x$  with rational coefficients.

### 14.3 Inverse Fabius function

The normalized first primitive is [equation \(1105\)](#). More generally, for  $\alpha > 0$  and  $p > 0$ ,

$$I_{0+}^\alpha Q^p(y) = \frac{p}{\Gamma(\alpha + 1)} \int_0^{Q(y)} x^{p-1} (y - F(x))^\alpha dx, \quad (1227)$$

while the Fabius-clock primitive is the elementary gamma law [equation \(1126\)](#). These two answers are complementary: the first retains Euclidean probability time and has a Fabius kernel; the second uses the inverse clock and becomes algebraic.

## 15 Numerical experiments and reproducibility

The accompanying script `numerical_experiments.py` reconstructs  $\text{up}$  by FFT inversion of a 48-factor sinc product on a  $2^{18}$  grid, builds monotone interpolants for  $F$  and  $Q$ , and compares formulas obtained from mathematically independent representations. The fractional-Laplacian profiles are then evaluated by the nonperiodic singular integral, not by a periodic Fourier multiplier, eliminating algebraic image contamination. Exact centered moments are generated by rational recurrence rather than by numerical quadrature. No random sampling is used.

### 15.1 Cross-check table

| identity                            | direct side     | transformed side | absolute error |
|-------------------------------------|-----------------|------------------|----------------|
| Generalized Stieltjes moment series | 0.466934130794  | 0.466934130794   | 1.665e-16      |
| Generalized Stieltjes hierarchy     | 0.466934130794  | 0.466934130794   | 1.110e-16      |
| Exterior Riesz moment series        | -0.189818171242 | -0.189818171242  | 7.272e-15      |
| Caputo inverse transmutation        | 0.556844544942  | 0.556845008522   | 4.636e-07      |
| Fractional $F$ -up bridge           | 0.569952059655  | 0.56995167835    | 3.813e-07      |
| Complex inverse resolvent           | 0.407510481292  | 0.407510481292   | 3.940e-13      |

The  $10^{-7}$ -scale errors in the two Caputo rows are the discretization errors of an independent L1 finite-difference approximation. The transform and moment-series identities agree near machine precision.

### 15.2 Endpoint Caputo convergence

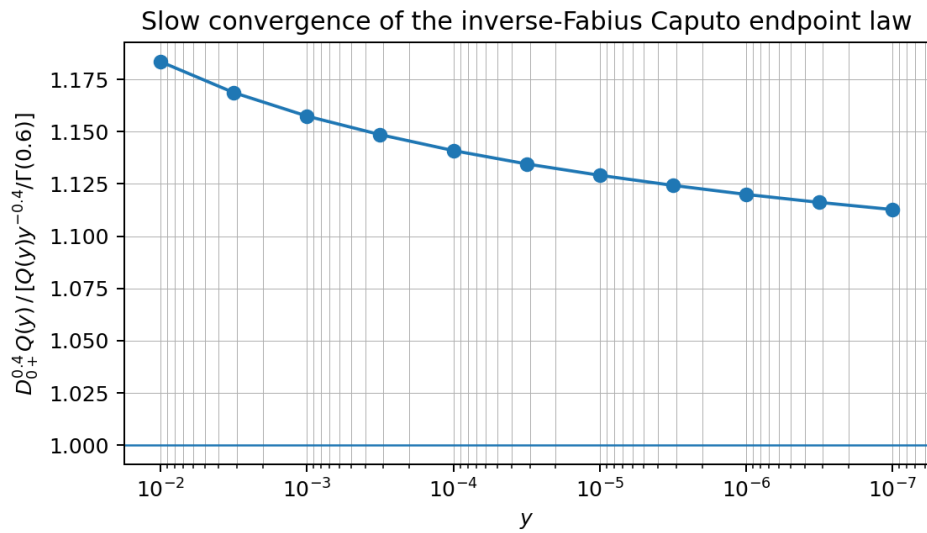


Figure 9: The ratio in [equation \(1140\)](#) for  $r = 1$  and  $\beta = 0.4$ . The theorem predicts convergence to 1; the slow drift exposes the scale of the first Lambert-type correction.

### 15.3 Fractional-Laplacian profiles

The script records all parameters and displayed numbers in `numerical_results.json`. Running

```
python numerical_experiments.py
```

regenerates both figures and the LaTeX table.

## 16 Conjectures and frontier directions

### 16.1 Phase-resolved Caputo endpoint correction

The proof of [theorem 6.1](#) discards the narrow layer  $t \approx 1$  where  $F(qt)/F(q)$  is non-negligible. The width of that layer is controlled by the same saddle parameter that appears in the inverse Lambert- $W$  expansion.

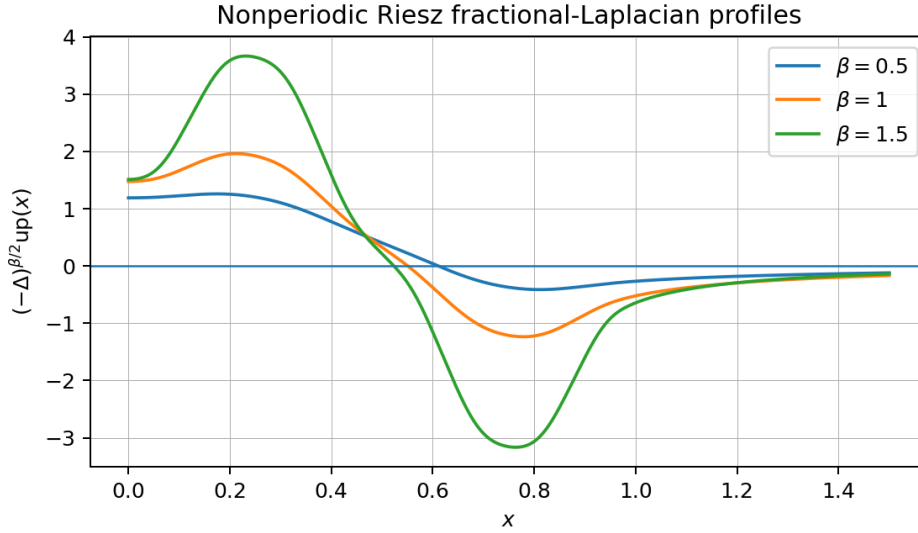


Figure 10: Nonperiodic singular-integral profiles of  $(-\Delta)^{\beta/2} \text{up}$  for three representative orders. Each profile has one positive zero in the plotted half-axis, motivating Conjecture 11.4.

**Conjecture 16.1** (Caputo endpoint phase). *For fixed  $r > 0$  and  $0 < \beta < 1$ , there exists a bounded one-periodic function  $A_{r,\beta}$  of the repository's endpoint phase  $\vartheta(y)$  such that*

$$\frac{{}^C D_{0+}^{\beta} Q(y)^r}{Q(y)^r y^{-\beta} / \Gamma(1-\beta)} = 1 + \frac{A_{r,\beta}(\vartheta(y))}{\rho(y)} + O(\rho(y)^{-2}), \quad (1228)$$

where  $\rho(y)$  is defined by equation (1143). The periodic term should be nonconstant but extremely small, as in the endpoint density asymptotics.

A route to proof is a boundary-layer rescaling of equation (1142), using the all-orders ratio asymptotic for  $F(qt)/F(q)$  uniformly when  $1-t = O(\rho(y)^{-1})$ .

## 16.2 The one-node law and total positivity

Conjecture 11.4 is stronger than the sign information proved in appendix 11. Three plausible approaches are:

1. represent the interior fractional Laplacian as a signed Stieltjes convolution and prove a variation-diminishing property;
2. use the infinite convolution construction of  $\text{up}$  and preservation of single crossing under uniform averaging;
3. establish strict total positivity for the appropriate fractional kernel after subtracting the center value.

A certified interval-arithmetic computation over a dense mesh in  $\beta$  would be a useful intermediate theorem, but not a substitute for uniqueness.

## 16.3 Fractional spectral interpolation

At the even integers,

$$\log_2 \frac{\mathcal{E}(2n)}{\mathcal{E}(0)} = n(n+1). \quad (1229)$$

The explicit spectral measure from the derivative-norm report is a nonclassical representative of a shifted Stieltjes–Wigert moment problem. Determining the full Mellin transform  $\mathcal{E}(s)$  would interpolate those quadratic samples and could reveal a small 2-periodic correction to the quadratic envelope

$$\frac{s^2}{4} + \frac{s}{2}. \quad (1230)$$

The exact integer samples alone do not force such periodicity; this is an open spectral-analysis direction rather than a formal conjecture at present.

## 16.4 General base and $q$ -deformation

Replace  $2^{-j}$  by normalized  $b^{-j}$  weights. The clock conjugacy, inverse transmutation, generalized Stieltjes hierarchy, and exterior Riesz expansion survive with minimal change. The order-lowering factors become  $b$ -dependent, while the rational moment coefficients become  $q = 1/b$  objects. The resulting family should provide a direct fractional-calculus bridge between the base- $b$  random series and Gaussian  $q$ -binomial arithmetic.

## 16.5 Arithmetic of fractional constants

For rational  $\beta$  and algebraic  $z > 1$ , the values

$$S_\beta(z), \quad {}^C D_{0+}^\beta Q(y)^r, \quad \mathcal{E}(\beta), \quad a_\beta \quad (1231)$$

are natural new constants. The Stieltjes series expresses the first through rational moments and gamma-polynomial coefficients. Padé approximation, Mahler-type functional equations, and denominator estimates from the  $q$ -binomial formulas may offer arithmetic information, although no irrationality claim is currently justified.

# 17 Lean formalization roadmap

The latest repository merge already formalizes the normalized integer Volterra operator and the exact Fabius primitive ladder. The new results can be layered on top in increasing order of analytic difficulty.

1. **Composition conjugacy.** Define  $\mathcal{C}_F$  and  $\mathcal{C}_Q$  on suitable function spaces; prove they are inverse order isomorphisms and transport semigroups.
2. **Fabius-clock integrals.** Formalize [equation \(1120\)](#) first for positive real orders and continuous functions using the substitution theorem. The power law then reduces to Mathlib’s beta integral.
3. **First-order Caputo transmutation.** The proof of [equation \(1130\)](#) requires only absolute continuity, inverse differentiation on  $(0, 1)$ , and a monotone change of variables. This is likely the most accessible genuinely new theorem.
4. **Endpoint asymptotics.** Package rapid variation and convexity into a dominated-convergence lemma for kernels  $t^{r-1}(1 - F(qt)/F(q))^{-\beta}$ .
5. **Generalized Stieltjes hierarchy — real-order stage discharged.** `StieltjesGeneralizedOrder.lean` formalizes the bounded variable, kernel integrability, digit integral, and order-lowering identity for every real  $\alpha > 1$  and real  $z > 1$ . Integer powers now also satisfy direct named recurrences throughout the complex slit domains. Complex order and a complex logarithmic branch remain later stages.
6. **Exterior Riesz theorem.** Avoid principal values entirely by restricting first to  $|x| > 1$ ; compact support reduces the proof to an ordinary integral and the Stieltjes identity.

7. **Weyl and full Riesz operators.** These require Fourier multiplier and tempered-distribution infrastructure and are a later target.

The numerical script can be converted into certified rational interval checks for selected transform identities and the conjectural zero locations, but the formal statements should remain independent of floating-point evidence.

## 18 Conclusion

The ordinary integral theory of the Fabius–Rvachev system is already unusually rigid: differentiation is conjugate to dyadic dilation, integer integration runs backward along the same orbit, and inverse substitution converts many areas into moments. The present investigation shows that this rigidity continues in several genuinely fractional directions.

The Fabius and inverse clocks give exact conjugate fractional calculi with semigroups, gamma power laws, and Mittag–Leffler eigenfunctions. In ordinary time, the inverse-composition transmutation removes the singular derivative of  $Q$  and replaces it by a Fabius-kernel integral; this leads to rigorous fractional endpoint laws. Standard fractional derivatives of  $F$  are exact fractional primitives of up, while whole-line Weyl derivatives satisfy a complex-order refinement hierarchy. On the transform side, one entire function  $S_\lambda(z)$  unifies Appell polynomials, Cauchy resolvents, logarithmic potentials, Riesz tails, and fractional-Laplacian exterior fields.

The strongest next targets are now sharply posed: prove the one-node law, resolve the first phase-dependent correction to the inverse Caputo endpoint asymptotic, determine the continuous fractional-energy interpolation of the integer derivative norms, and formalize the first-order transmutation and Stieltjes hierarchy in Lean. None requires inventing a new ad hoc numerical object; each is anchored in an exact identity proved above.

## A Formula index by operator

| Operator or transform               | Principal exact formula                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| Integer derivative of signed Fabius | $D^n \mathcal{F}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x)$                                     |
| Integer primitive                   | $J^n \mathcal{F}(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n)$                                       |
| Weighted Volterra primitive         | Newton series <a href="#">equation (1103)</a>                                                    |
| Inverse first primitive             | $\int_0^y Q(v) dv = yQ(y) - F(Q(y)/2)$                                                           |
| Standard RL primitive of $F$        | $I_{0+}^\alpha F(x) = \mathbb{E}[(x - Y)_+^\alpha] / \Gamma(\alpha + 1)$                         |
| Fabius-clock integral               | kernel <a href="#">equation (1120)</a>                                                           |
| Fabius-clock power law              | <a href="#">equations (1122) and (1123)</a>                                                      |
| Caputo derivative of $H \circ Q$    | transmutation <a href="#">equation (1130)</a>                                                    |
| Caputo derivative of $Q^r$          | <a href="#">equation (1132)</a> and endpoint law <a href="#">equation (1140)</a>                 |
| Standard derivative of $F$          | $D_{0+}^\beta F = 2^\beta I_{-1+}^{1-\beta} \text{up}(2x - 1)$ for $0 < \beta < 1$               |
| Weyl derivative of up               | complex-order hierarchy <a href="#">equation (1156)</a>                                          |
| Generalized Stieltjes transform     | moment series <a href="#">equation (1162)</a> and order lowering <a href="#">equation (1165)</a> |
| Complex $F$ resolvent               | <a href="#">equations (1174) and (1175)</a>                                                      |
| Complex inverse resolvent           | <a href="#">equation (1179)</a>                                                                  |



| Operator or transform         | Principal exact formula                                    |
|-------------------------------|------------------------------------------------------------|
| Exterior fractional Laplacian | $(-\Delta)^{\beta/2} \text{up} = -C_{1,\beta} S_{1+\beta}$ |
| Exterior Riesz potential      | $I_\alpha^R \text{up} = \kappa_\alpha S_{1-\alpha}$        |
| Fractional energy             | $\mathcal{E}(s) = (2\pi)^{-1} \int  \xi ^s  \phi ^2$       |

## B Exact low-order moments

For convenience, the first one-sided moments are

$$m_0 = 1, \quad m_1 = \frac{1}{2}, \quad m_2 = \frac{5}{18}, \quad m_3 = \frac{1}{6}, \quad m_4 = \frac{143}{1350}, \quad m_5 = \frac{19}{270}. \quad (1232)$$

The centered moments in [equation \(1096\)](#) satisfy

$$c_k = \frac{2m_{2k+1}}{2k+1}. \quad (1233)$$

Substitution in [equation \(1162\)](#) produces exact rational-coefficient large- $z$  expansions for every fixed complex order  $\lambda$ .

## C Computational files

The archive accompanying this report contains:

- `fabius_fractional_integrals_report.tex` – the present source;
- `fabius_fractional_integrals_report.pdf` – the rendered article;
- `numerical_experiments.py` – commented reproducibility code;
- `numerical_results.json` and `numerical_results.txt` – machine- and human-readable output;
- `numerical_results.tex` – the generated table;
- `caputo_endpoint_ratio.png` and `riesz_fractional_profiles.png` – generated figures;
- `README.md` and `SHA256SUMS.txt`.

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- [11] ProveIt Fourier-decay report family, sharp decay, zero multiplicities, and Thue–Morse lobe structure of the infinite sinc product, included under the documentation root at commit 3adc011108796355f1a874a5036f28098408d370.

## Part X

# Integral, Transform, and Fractional-Order Frontiers for the Fabius–Rvachev System

### Abstract

This report develops a broad integral calculus for the Fabius function  $F$ , its inverse  $Q = F^{-1}$ , and Rvachev’s compactly supported function up, after a recursive audit of the current and archived LaTeX corpus under the ProvelT documentation root. Existing dyadic derivative formulas, exact moment arithmetic, q-binomial and Thue–Morse structure, sinc products, Lambert– $W$  endpoint expansions, Bell and Bernoulli polynomial systems, repeated integration, weighted primitives, inverse-area identities, and the known obstruction to naive fractional self-similarity are treated as baseline results rather than reannounced as new.

The report first synthesizes ordinary and weighted antiderivatives, definite integrals, entire Mellin moments, negative and complex powers, Laplace/Fourier/Mellin/Stieltjes transforms, inverse-function dualities, order-statistic identities, and all integer orders of differentiation and integration. It then advances beyond the audited corpus in four directions. First, the inverse Fabius endpoint is placed in a fractional calculus: for  $T = \log(1/y)$  and  $\eta = yQ'(y)/Q(y) \sim \sqrt{\log 2/(2T)}$ ,

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta + o(\eta)),$$

while the Riemann–Liouville/Caputo derivative satisfies the dual digamma correction and the sharp strong threshold  $D_{0+}^{\sigma_p}(Q^\beta) \in L^p \iff \sigma_p \leq 1$  near the endpoint. Second, exact exterior moment series are obtained for the Riesz potential and fractional Laplacian of up; all coefficients are generated rationally from Bernoulli cumulants through complete Bell polynomials. Third, the Cauchy transform is shown to obey a logarithmic smoothing fixed point and an all-order resolvent hierarchy. Fourth, the log-periodic obstruction to fractional dyadic scaling is converted into an exact Floquet multiplier cocycle that restores the continuous dilation orbit on the global signed Fabius extension.

A deterministic, fully commented Python program reconstructs up from its infinite sinc product and checks the new nonlocal series, transform identities, and fractional endpoint corrections. “New” means not found in the recursively audited repository corpus at the time of writing; no claim of worldwide priority is made without a separate literature proof.

## 1 Corpus audit, scope, and novelty boundary

### 1.1 What was reviewed

The repository was traversed recursively through the documentation root, including the main monographs, glossary, Lean walkthrough, non-elementarity article, papers, canonical research-frontier volume, current frontier drafts, auxiliary generated TeX fragments, and frozen archive. The archive map was used to identify historical duplicates and their canonical successors; living sources and current drafts were checked directly at theorem and section level, with integration-specific full-text searches described in Appendix C.

The audit is anchored to the path-complete inventory at commit 32d6d36c51d803289e6d6a0dc0c37753766eba47, which records exactly sixty-seven LaTeX paths. A second traversal of the live main branch found five later frontier packages adding eleven visible TeX paths, including the self-sampling report’s six generated table fragments. Thus the current ledger contains seventy-eight paths: five core synthesis/reference sources, thirty-two frozen historical sources, thirty-four active frontier-report or auxiliary sources, and seven paper transcriptions [1, 2, 3].

The following table states the boundary used in this report.

| Imported baseline                                                                                                                                    | How it is used here                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Probability interpretation of $F$ , compact support and refinement equations for up, and the infinite sinc product                                   | Normalization and the common probabilistic language for all integrals.                                                                                                                                                                       |
| Exact dyadic values, rational moment recurrences, Bernoulli and $q$ -binomial structure                                                              | Inputs to the moment, Mellin, Newton-series, and transform calculations; these arithmetic results are not reproved in full.                                                                                                                  |
| Lambert- $W$ endpoint and inverse asymptotics, periodic phase corrections                                                                            | Imported as the asymptotic input for the endpoint fractional calculus and as comparison data for the Floquet phase.                                                                                                                          |
| Dyadic self-sampling quadrature, finite-prefix inverse acceleration, Rvachev-Appell, Newton/exponent-sequence, and inverse-moment polynomial systems | Explicitly excluded from novelty claims; only consequences needed for integration and transform formulas are recalled.                                                                                                                       |
| Canonical integration and repeated-integration notebooks, inverse-function reports, and weighted primitive identities                                | Treated as prior art. Ordinary inverse-pair, moment, and transform formulas below form a self-contained synthesis; the strongest corpus-relative claims are restricted to the fractional endpoint, nonlocal, resolvent, and Floquet results. |

### Strongest corpus-relative frontier results

After comparison with all seventy-eight paths, the following proved package was not located explicitly in the audited corpus: the Riemann-Liouville inverse-pair duality of [theorem 7.1](#); the digamma/trigamma endpoint expansions and sharp critical strong- $L^p$  law for  $Q^\beta$  in [theorems 9.2](#) and [9.4](#); the exact exterior fractional-Laplacian and Riesz-potential series and the fractional Sobolev-energy identity in [theorems 10.1, 10.2](#) and [10.4](#); the logarithmic Cauchy-transform fixed point and all-order resolvent hierarchy of [theorem 12.1](#); and the Floquet multiplier/cocycle that repairs fractional dyadic scaling in [theorem 11.1](#). The sinc-product Gini identity of [theorem 13.2](#) is another compact deduction not found in the comparison searches. The universal inverse-pair identity, entire moment/Newton calculus, negative-moment recurrence, beta transforms, and related ordinary formulas are presented as a useful self-contained synthesis and auxiliary deductions, without a priority claim. Since that audit, the proper full-interval and variable-endpoint absolutely-continuous inverse-pair identities have also become proved Lean theorems in the more general compact-interval forms recorded below; only the improper-singular extensions remain frontier claims.

## 1.2 Conventions

All powers of positive real variables use the real logarithm:  $x^s = \exp(s \log x)$ . The totalized Fabius distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1).$$

Its inverse  $Q : (0, 1) \rightarrow (0, 1)$  is the ordinary increasing inverse; endpoint values are understood by continuity. Angular frequency is used for Fourier transforms:  $\hat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) \, dx$ .

## 2 Probability model and transform baseline

### 2.1 The dyadic uniform sum

Let  $U_1, U_2, \dots$  be independent random variables uniformly distributed on  $[0, 1]$ , and put

$$Y = \sum_{k=1}^{\infty} \frac{U_k}{2^k}, \quad Z = 2Y - 1 = \sum_{k=1}^{\infty} \frac{2U_k - 1}{2^k}. \quad (1234)$$

Then  $Y \in [0, 1]$ ,  $Z \in [-1, 1]$ , and the laws are symmetric:

$$Y \stackrel{d}{=} 1 - Y, \quad Z \stackrel{d}{=} -Z. \quad (1235)$$

The Fabius function is the distribution function of  $Y$ ,

$$F(x) = \mathbb{P}(Y \leq x), \quad (1236)$$

and up is the density of  $Z$ . Thus

$$F'(x) = 2 \text{up}(2x - 1), \quad \text{up}(x) = \begin{cases} F(x + 1), & -1 \leq x \leq 0, \\ F(1 - x) = 1 - F(x), & 0 \leq x \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (1237)$$

In particular  $F(x) + F(1 - x) = 1$  on  $[0, 1]$ , and the quantile obeys

$$Q(1 - v) = 1 - Q(v). \quad (1238)$$

The self-decomposition

$$Y \stackrel{d}{=} \frac{U + Y'}{2} \quad (1239)$$

holds with  $U \sim \text{Unif}[0, 1]$  independent of a copy  $Y'$  of  $Y$ . It will generate both the positive- and negative-moment recurrences.

### 2.2 Moment-generating and Fourier products

Define

$$P(t) = \mathbb{E}e^{tY}, \quad \mathcal{U}(t) = \mathbb{E}e^{tZ}. \quad (1240)$$

Independence in [equation \(1234\)](#) gives the locally uniformly convergent products

$$P(t) = \prod_{k=1}^{\infty} \frac{e^{t/2^k} - 1}{t/2^k}, \quad (1241)$$

$$\mathcal{U}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}, \quad (1242)$$

$$P(t) = e^{t/2} \mathcal{U}(t/2). \quad (1243)$$

The characteristic function of the up law is therefore

$$\phi(t) = \widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \frac{\sin(t/2^k)}{t/2^k}. \quad (1244)$$

The scaling equation

$$P(2t) = \frac{e^t - 1}{t} P(t) \quad (1245)$$

is the exponential generating-function form of the moment recurrence below.

### 2.3 Rational moments, cumulants, and Bell polynomials

Write

$$m_n = \mathbb{E}Y^n = \int_0^1 Q(v)^n dv, \quad \mu_n = \mathbb{E}Z^n = \int_{-1}^1 x^n \text{up}(x) dx. \quad (1246)$$

Taking the  $n$ -th moment in [equation \(1239\)](#) gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (1247)$$

Thus every  $m_n$  is rational. The first values are

$$1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \dots \quad (1248)$$

The logarithm of [equation \(1242\)](#) and  $\log(\sinh z/z) = \sum_{r \geq 1} 2^{2r-1} B_{2r} z^{2r} / (r(2r)!)$  give the cumulants

$$\begin{aligned} \kappa_1(Y) &= \frac{1}{2}, \\ \kappa_{2r}(Y) &= \frac{B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0 \quad (r \geq 1), \end{aligned} \quad (1249)$$

$$\kappa_{2r}(Z) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Z) = 0. \quad (1250)$$

Consequently  $m_n$  and  $\mu_n$  are complete exponential Bell polynomials in these cumulants. A useful bridge between the one-sided and centered moments is

$$\mu_{2j} = 2 \int_0^1 x^{2j} \text{up}(x) dx = \frac{2m_{2j+1}}{2j+1}, \quad \mu_{2j+1} = 0. \quad (1251)$$

For example

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (1252)$$

## 3 A universal integration calculus for $F$ and $Q$

The classical area identity for a function and its inverse is only the first member of a much larger family. The following theorem is the main algebraic engine of the report.

### Exact Lean boundary for the inverse-pair identity

The generic theorem `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_of_lt_iff_lt` in `InversePairIntegral.lean` proves the complete proper-compact absolutely-continuous identity on arbitrary ordered compact source and level intervals. The clocks need only preserve their respective intervals and obey  $C(x) < u \iff x < Q(u)$ . There is no measurability assumption on either clock: the order adjunction forces monotonicity on both closed intervals, and the Lean proof constructs measurable clamped `IccExtend` versions internally. The proper compact variable-upper theorem is `Fabius.intervalIntegral_deriv_mul_comp_add_comp_mul_deriv_to_of_lt_iff_lt`.

The exact unit-interval Fabius wrapper is `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv`, which is precisely [equation \(1254\)](#). With both absolutely-continuous factors equal to the identity, it yields the named area-sum corollary `Fabius.intervalIntegral_fabiusReal_add_fabiusInv_eq_one`. The exact variable-upper Fabius

wrapper is `Fabius.intervalIntegral_deriv_mul_fabiusReal_add_fabiusInv_mul_deriv_to`, which is precisely [equation \(1253\)](#); its identity-function specialization is `Fabius.intervalIntegral_fabiusInv_to_eq_mul_sub_intervalIntegral_fabiusReal`. The bundled theorem below remains only partially formalized because its improper extensions through singular endpoints are not supplied by these proper-compact declarations.

**Theorem 3.1** (two-function inverse-pair identity). *Let  $A, B$  be absolutely continuous on  $[0, 1]$ . For every  $z \in [0, 1]$ ,*

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'(x)B(F(x)) \, dx + \int_0^{F(z)} A(Q(v))B'(v) \, dv. \quad (1253)$$

At  $z = 1$ ,

$$A(1)B(1) - A(0)B(0) = \int_0^1 A'(x)B(F(x)) \, dx + \int_0^1 A(Q(v))B'(v) \, dv. \quad (1254)$$

*The formulas remain valid under the usual improper-integral hypotheses whenever an endpoint is singular.*

*Proof.* Differentiate the product  $A(x)B(F(x))$ , integrate from 0 to  $z$ , and in the term containing  $F'(x)$  use the increasing substitution  $v = F(x)$ , so that  $x = Q(v)$ .  $\square$

**Remark 3.2.** No special functional equation of the Fabius function is needed in [theorem 3.1](#); it is a quantile identity. What is special here is that the moments and transforms of this particular quantile can be calculated by dyadic products and rational recurrences.

### 3.1 Power duality and logarithmic derivatives

Taking  $A(x) = x^p$ ,  $B(v) = v^q$  gives the following Mellin-type duality.

**Corollary 3.3** (power duality). *Initially for  $\Re p, \Re q > 0$ ,*

$$p \int_0^z x^{p-1} F(x)^q \, dx + q \int_0^{F(z)} Q(v)^p v^{q-1} \, dv = z^p F(z)^q. \quad (1255)$$

*In particular,*

$$p \int_0^1 x^{p-1} F(x)^q \, dx + q \int_0^1 v^{q-1} Q(v)^p \, dv = 1. \quad (1256)$$

*Both sides admit meromorphic or entire continuation in parameters whenever the endpoint flatness cancels the apparent singularities.*

For all logarithmic derivatives at once, define

$$J_{r,s}(p, q) = \int_0^1 x^{p-1} (\log x)^r F(x)^q (\log F(x))^s \, dx, \quad (1257)$$

$$K_{r,s}(p, q) = \int_0^1 v^{q-1} (\log v)^s Q(v)^p (\log Q(v))^r \, dv. \quad (1258)$$

Differentiating [equation \(1256\)](#) gives, for  $r + s > 0$ ,

$$pJ_{r,s} + rJ_{r-1,s} + qK_{r,s} + sK_{r,s-1} = 0, \quad (1259)$$

with  $pJ_{0,0} + qK_{0,0} = 1$ . Equation [equation \(1259\)](#) is a compact general formula for every mixed derivative of the two Mellin kernels.

### 3.2 A bivariate Laplace–Fourier identity

The exponential choice  $A(x) = e^{sx}$ ,  $B(v) = e^{tv}$  produces an especially simple transform relation.

**Theorem 3.4** (bivariate inverse-pair transform). *For all  $s, t \in \mathbb{C}$  and  $z \in [0, 1]$ ,*

$$s \int_0^z e^{sx+tF(x)} dx + t \int_0^{F(z)} e^{tv+sQ(v)} dv = e^{sz+tF(z)} - 1. \quad (1260)$$

Hence

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1. \quad (1261)$$

The identities are entire in  $(s, t)$ ; the cases  $s = 0$  or  $t = 0$  are understood by continuity.

Every mixed power integral involving  $x$  and  $F(x)$ , together with its quantile dual, is obtained by coefficient extraction from [equation \(1261\)](#). Purely imaginary  $s, t$  give a bivariate Fourier identity.

### 3.3 Weighted antiderivatives

Taking  $B(v) = v$  and a primitive  $A$  of a prescribed weight yields the useful master formula

$$\int_0^x A'(t)F(t) dt = A(x)F(x) - A(0)F(0) - \int_0^{F(x)} A(Q(v)) dv. \quad (1262)$$

For monomial weights this becomes the promised indefinite integral.

**Corollary 3.5** (monomial–Fabius antiderivative). *For  $p \neq -1$ , initially where the integrals converge and then by continuation,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1}F(x) - \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (1263)$$

The value at  $p = -1$  is removable:

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log Q(v) dv. \quad (1264)$$

On  $[0, 1]$ , where  $\text{up} = 1 - F$ ,

$$\int_0^x t^p \text{up}(t) dt = \frac{x^{p+1}(1 - F(x)) + \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (1265)$$

More generally, from [equation \(1255\)](#),

$$\int_0^x t^{p-1} F(t)^q dt = \frac{x^p F(x)^q}{p} - \frac{q}{p} \int_0^{F(x)} Q(v)^p v^{q-1} dv. \quad (1266)$$

Thus products of  $F$  and the positive half of  $\text{up}$  reduce to beta-weighted quantile integrals after expanding  $(1 - F)^n$ .

## 4 The entire complex moment function

**Definition 4.1.** For  $s \in \mathbb{C}$ , set

$$\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv. \quad (1267)$$

For nonnegative integers,  $\mathfrak{M}(n) = m_n$ .



### 4.1 Entire continuation

The endpoint flatness of  $F$  is much stronger than what is needed for ordinary positive moments: for every  $N > 0$ ,

$$F(x) = O_N(x^N) \quad (x \downarrow 0). \quad (1268)$$

**Theorem 4.2** (entire quantile Mellin function). *The function  $\mathfrak{M}$  in [equation \(1267\)](#) is entire. For every  $r \geq 0$ ,*

$$\mathfrak{M}^{(r)}(s) = \int_0^1 Q(v)^s (\log Q(v))^r dv = \mathbb{E}[Y^s (\log Y)^r]. \quad (1269)$$

*In particular all negative moments and all logarithmic moments exist.*

*Proof.* Let  $K \subset \mathbb{C}$  be compact and choose  $A > 0$  with  $\Re s \geq -A$  on  $K$ . For every fixed  $r$ , the integrand is bounded in modulus by  $Y^{-A} |\log Y|^r$ . Integration by parts gives

$$\mathbb{E}Y^{-A} = 1 + A \int_0^1 x^{-A-1} F(x) dx < \infty,$$

using [equation \(1268\)](#) with an exponent larger than  $A$ ; inserting powers of  $|\log x|$  does not alter convergence. Dominated differentiation on compact sets proves holomorphy and [equation \(1269\)](#).  $\square$

A first transform triangle follows immediately:

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad (1270)$$

$$\int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}, \quad (1271)$$

$$\int_0^1 Q(v)^{s-1} dv = \mathfrak{M}(s-1). \quad (1272)$$

The first quotient has a removable singularity at  $s = 0$  and is entire; the second is the meromorphic continuation of the classical Mellin transform of  $\text{up}|_{[0,1]}$  and has exactly one pole, at  $s = 0$ , with residue 1. Evenness gives

$$\int_{-1}^1 |x|^{s-1} \text{up}(x) dx = \frac{2\mathfrak{M}(s)}{s}. \quad (1273)$$

### 4.2 Superpolynomial moment decay

**Lemma 4.3.** *For every  $N > 0$ ,*

$$m_k = O_N(k^{-N}) \quad (k \rightarrow \infty). \quad (1274)$$

*Proof.* For  $k > 0$ , the tail formula and symmetry give

$$m_k = k \int_0^1 x^{k-1} (1 - F(x)) dx = k \int_0^1 x^{k-1} F(1-x) dx.$$

By [equation \(1268\)](#),  $F(1-x) \leq C_N(1-x)^N$ , hence

$$m_k \leq C_N k B(k, N+1) = O(k^{-N}).$$

$\square$

### 4.3 Two globally convergent Newton series

**Theorem 4.4** (Newton and central-even series). *Both of the following series converge locally uniformly for all  $s \in \mathbb{C}$ :*

$$\mathfrak{M}(s) = \sum_{k=0}^{\infty} (-1)^k \binom{s}{k} m_k, \quad (1275)$$

$$\mathfrak{M}(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j} = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{2m_{2j+1}}{2j+1}. \quad (1276)$$

*They therefore represent the entire function  $\mathfrak{M}$  on the whole plane.*

*Proof.* For  $0 < Y < 1$ , expand  $Y^s = (1 - (1 - Y))^s$ . Taking expectations and using  $1 - Y \stackrel{d}{=} Y$  gives equation (1275). On every compact set in  $s$ , the binomial coefficients have at most polynomial growth in  $k$ ; lemma 4.3 therefore justifies absolute, locally uniform convergence and interchange of sum and expectation.

For equation (1276), write  $Y = (1 + Z)/2$ , expand  $(1 + Z)^s$ , and use the evenness of the law of  $Z$ . The relation equation (1251) and lemma 4.3 again give locally uniform convergence.  $\square$

Several useful specializations are immediate. For an integer  $r \geq 1$ ,

$$\mathfrak{M}(-r) = \sum_{k=0}^{\infty} \binom{r+k-1}{k} m_k. \quad (1277)$$

Every term is rational and positive. In particular,

$$\mathfrak{M}(-1) = \sum_{k=0}^{\infty} m_k. \quad (1278)$$

Differentiation of equation (1275) at the origin yields logarithmic moments in terms of signed Stirling numbers of the first kind  $s(k, r)$ :

$$\mathfrak{M}^{(r)}(0) = r! \sum_{k=r}^{\infty} \frac{(-1)^k s(k, r)}{k!} m_k. \quad (1279)$$

For  $r = 1$ ,

$$-\mathbb{E} \log Y = \sum_{k=1}^{\infty} \frac{m_k}{k} = \int_0^1 \frac{F(x)}{x} dx. \quad (1280)$$

Numerically,

$$-\mathbb{E} \log Y = 0.758240000404408656 \dots \quad (1281)$$

### 4.4 A mixed reflected Mellin function

The inverse symmetry equation (1238) suggests the two-variable transform

$$\mathfrak{M}_2(a, b) = \int_0^1 Q(v)^a Q(1-v)^b dv = \mathbb{E}[Y^a (1-Y)^b]. \quad (1282)$$

**Proposition 4.5.** *The function  $\mathfrak{M}_2$  is entire on  $\mathbb{C}^2$ , symmetric in  $a, b$ , and has the locally uniformly convergent expansion*

$$\mathfrak{M}_2(a, b) = \sum_{k=0}^{\infty} (-1)^k \binom{b}{k} \mathfrak{M}(a+k). \quad (1283)$$

It also has the central expansion

$$\mathfrak{M}_2(a, b) = 2^{-a-b} \sum_{j,k \geq 0} (-1)^k \binom{a}{j} \binom{b}{k} \mu_{j+k}. \quad (1284)$$

If either parameter is a nonnegative integer, the corresponding binomial expansion terminates.

*Proof.* Endpoint flatness at both 0 and 1 gives local domination for arbitrary negative real parts, proving entire dependence on two parameters. Expand  $(1 - Y)^b$  to obtain [equation \(1283\)](#); super-polynomial decay of shifted moments gives local uniformity. The central formula follows from  $Y = (1 + Z)/2$ ,  $1 - Y = (1 - Z)/2$ .  $\square$

For example,

$$\int_0^1 Q(v)Q(1-v) dv = \frac{2}{9}, \quad \int_0^1 Q(v)^2 Q(1-v)^2 dv = \frac{34}{675}, \quad (1285)$$

and

$$\int_0^1 \frac{dv}{Q(v)(1-Q(v))} = 2\mathfrak{M}(-1). \quad (1286)$$

## 5 Negative moments: an exact recurrence and a new asymptotic constant

The Newton series [equation \(1277\)](#) is effective for moderate negative orders. A second construction is much more stable for large orders and reveals a new normalization law.

### 5.1 The recurrence

For  $s > 0$ , define

$$A(s) = \mathfrak{M}(-s) = \mathbb{E}Y^{-s}, \quad D(s) = \mathbb{E}(1 + Y)^{-s}. \quad (1287)$$

Conditioning in [equation \(1239\)](#) and integrating the uniform variable gives the following exact relation.

**Theorem 5.1** (negative-moment recurrence). *For every real  $s > 0$ ,*

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)). \quad (1288)$$

Equivalently, for  $s > 1$ ,

$$A(s) = \frac{2^s}{s-1} (A(s-1) - D(s-1)). \quad (1289)$$

Moreover  $D(s)$  has the rapidly convergent centered series

$$D(s) = \left(\frac{2}{3}\right)^s \sum_{j=0}^{\infty} \frac{(s)_{2j}}{(2j)! 3^{2j}} \mu_{2j}, \quad (1290)$$

where  $(s)_n = s(s+1) \cdots (s+n-1)$ .

*Proof.* By [equation \(1239\)](#), for  $s > 0$ ,

$$A(s+1) = 2^{s+1} \mathbb{E} \int_0^1 (U + Y')^{-s-1} dU.$$

For  $y > 0$ ,

$$\int_0^1 (u+y)^{-s-1} du = \frac{y^{-s} - (1+y)^{-s}}{s},$$

which gives [equation \(1288\)](#). For [equation \(1290\)](#), write  $1+Y = (3+Z)/2$ , expand  $(1+Z/3)^{-s}$ , and use the vanishing of odd moments. The factor  $3^{-2j}$  makes the series geometrically stable in addition to the superpolynomial decay of  $\mu_{2j}$ .  $\square$

The first values generated by this recurrence are

$$\begin{aligned} A(1) &= 2.31488614400581905 \dots, \\ A(2) &= 6.55899183845822858 \dots, \\ A(3) &= 24.3890791927221220 \dots, \\ A(4) &= 128.368907270620532 \dots, \\ A(5) &= 1025.15473735960333 \dots \end{aligned} \tag{1291}$$

## 5.2 A normalized limit and a periodic phase profile

Define

$$H(s) = \frac{\Gamma(s)A(s)}{2^{s(s+1)/2}}, \quad \rho(s) = \frac{D(s)}{A(s)}. \tag{1292}$$

Then [equation \(1288\)](#) becomes

$$H(s+1) = H(s)(1 - \rho(s)). \tag{1293}$$

Both factors are positive. Since  $\mathbb{P}(Y \leq 1/2) = 1/2$ ,

$$A(s) \geq 2^{s-1}, \quad D(s) \leq 1, \quad 0 < \rho(s) \leq 2^{1-s}. \tag{1294}$$

This elementary estimate is enough to prove convergence.

**Theorem 5.2** (periodic limiting profile). *For  $0 < \theta \leq 1$ , the product*

$$C(\theta) = H(\theta) \prod_{j=0}^{\infty} (1 - \rho(\theta + j)) \tag{1295}$$

*converges to a positive number. It extends by  $C(\theta+1) = C(\theta)$  to a positive, continuous, one-periodic function. As  $n \rightarrow \infty$ ,*

$$A(n+\theta) = C(\theta) \frac{2^{(n+\theta)(n+\theta+1)/2}}{\Gamma(n+\theta)} (1 + O_{\theta}(2^{-n})). \tag{1296}$$

*For integer  $r \rightarrow \infty$ ,*

$$\boxed{\mathfrak{M}(-r) = C_* \frac{2^{r(r+1)/2}}{\Gamma(r)} (1 + O(2^{-r}))}, \quad C_* = C(1) > 0. \tag{1297}$$

*Proof.* Iteration of [equation \(1293\)](#) gives

$$H(n+\theta) = H(\theta) \prod_{j=0}^{n-1} (1 - \rho(\theta + j)).$$

The sum of the tail bounds in [equation \(1294\)](#) is geometric. Hence the infinite product converges to a positive limit, and its omitted tail differs from 1 by  $O_{\theta}(2^{-n})$ . This proves [equation \(1296\)](#). Shifting the initial point by one and using [equation \(1293\)](#) proves exact periodicity. Uniform convergence on compact subintervals gives continuity; exact periodicity then removes the apparent endpoint at  $\theta = 0$ . The integer formula is the case  $\theta = 1$ .  $\square$

For integers the normalized sequence

$$C_r = \frac{(r-1)! \mathfrak{M}(-r)}{2^{r(r+1)/2}} \quad (1298)$$

is strictly decreasing and satisfies

$$C_{r+1} = C_r \left( 1 - \frac{D(r)}{A(r)} \right), \quad C_r \downarrow C_*. \quad (1299)$$

The product representation is

$$C_* = \frac{\mathfrak{M}(-1)}{2} \prod_{j=1}^{\infty} \left( 1 - \frac{D(j)}{A(j)} \right). \quad (1300)$$

The computation described in [appendix 15](#) gives

$$C_* = 0.750721980407870959 \dots \quad (1301)$$

The rapid stabilization is visible in

| $r$   | 1          | 2          | 4          | 6          | 8          | 12           |
|-------|------------|------------|------------|------------|------------|--------------|
| $C_r$ | 1.15744307 | 0.81987398 | 0.75216157 | 0.75072879 | 0.75072199 | 0.7507219804 |

### 5.3 Logarithmic asymptotics and entire-function growth

Stirling's expansion in [equation \(1297\)](#) gives a complete algebraic asymptotic series:

$$\begin{aligned} \log \mathfrak{M}(-r) &= \frac{\log 2}{2} r^2 - r \log r + \left( 1 + \frac{\log 2}{2} \right) r + \frac{1}{2} \log r + \log \frac{C_*}{\sqrt{2\pi}} \\ &\quad - \frac{1}{12r} + \frac{1}{360r^3} - \frac{1}{1260r^5} + \dots, \end{aligned} \quad (1302)$$

where

$$\log \frac{C_*}{\sqrt{2\pi}} = -1.20565842815389633 \dots \quad (1303)$$

This refines the growth one would obtain by a direct saddle-point transfer from the Lambert- $W$  endpoint expansion.

There is also a clean entire-function consequence. If  $M_{\mathfrak{M}}(R) = \max_{|s|=R} |\mathfrak{M}(s)|$ , then

$$|\mathfrak{M}(s)| \leq \mathbb{E} Y^{\Re s} \leq \mathbb{E} Y^{-R} = \mathfrak{M}(-R), \quad |s| = R, \quad (1304)$$

while equality occurs at  $s = -R$ . Thus

$$M_{\mathfrak{M}}(R) = \mathfrak{M}(-R). \quad (1305)$$

**Corollary 5.3.** *The entire function  $\mathfrak{M}$  has order 2 and type  $(\log 2)/2$ .*

### 5.4 Singular Fabius integrals

Putting  $s = p + 1$  in [equation \(1270\)](#) shows that

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p+1)}{p+1} \quad (1306)$$

is an entire function of  $p$ , despite the apparent pole at  $p = -1$ . Therefore all negative powers are integrable against  $F$ . For integers  $k \geq 2$ ,

$$\int_0^1 \frac{F(x)}{x^k} dx = \frac{\mathfrak{M}(1-k) - 1}{k-1}. \quad (1307)$$

The first examples are

$$\begin{aligned} \int_0^1 \frac{F(x)}{x} dx &= 0.758240000404408656\dots, \\ \int_0^1 \frac{F(x)}{x^2} dx &= 1.31488614400581905\dots, \\ \int_0^1 \frac{F(x)}{x^3} dx &= 2.77949591922911429\dots, \\ \int_0^1 \frac{F(x)}{x^4} dx &= 7.79635973090737400\dots \end{aligned} \quad (1308)$$

By [equation \(1297\)](#),

$$\int_0^1 \frac{F(x)}{x^k} dx \sim C_* \frac{2^{k(k-1)/2}}{\Gamma(k)}. \quad (1309)$$

This is an asymptotic law for a natural family of increasingly singular antiderivatives.

## 6 Definite weighted integrals

### 6.1 Polynomial and analytic weights

Let  $g$  be integrable and let  $G(x) = \int_0^x g(t) dt$ . Taking  $A = G$ ,  $B(v) = v$  in [theorem 3.1](#) gives

$$\int_0^1 g(x)F(x) dx = G(1) - \mathbb{E}G(Y), \quad (1310)$$

$$\int_0^1 g(x)(1 - F(x)) dx = \mathbb{E}G(Y). \quad (1311)$$

For a polynomial  $g$ , every quantity on the right is a rational combination of the moments  $m_n$ . For an entire  $g$ , the same formulas give convergent series whenever the Taylor expansion can be integrated termwise.

The monomial cases are

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p+1)}{p+1}, \quad (1312)$$

$$\int_0^1 x^p \text{up}(x) dx = \frac{\mathfrak{M}(p+1)}{p+1} \quad (\text{up} = 1 - F \text{ on } [0, 1]). \quad (1313)$$

For  $p \in \mathbb{N}$  these are rational. For example,

| $p$                            | 0   | 1     | 2    | 3         |
|--------------------------------|-----|-------|------|-----------|
| $\int_0^1 x^p F(x) dx$         | 1/2 | 13/36 | 5/18 | 1207/5400 |
| $\int_0^1 x^p \text{up}(x) dx$ | 1/2 | 5/36  | 1/18 | 143/5400  |

For symmetric weights  $g(x) = g(1-x)$ , symmetry alone gives

$$\int_0^1 g(x)F(x) dx = \frac{1}{2} \int_0^1 g(x) dx. \quad (1314)$$

## 6.2 Bernoulli-polynomial weights

Let

$$\beta_n = \mathbb{E}B_n(Y). \quad (1315)$$

Their exponential generating function follows from [equation \(1241\)](#):

$$\sum_{n=0}^{\infty} \beta_n \frac{t^n}{n!} = \frac{tP(t)}{e^t - 1}. \quad (1316)$$

Symmetry implies  $\beta_{2j+1} = 0$ . Since  $B'_{n+1} = (n+1)B_n$ , [equation \(1310\)](#) gives, for  $n \geq 1$ ,

$$\int_0^1 B_n(x)F(x) dx = -\frac{\beta_{n+1} - B_{n+1}}{n+1}. \quad (1317)$$

The first nontrivial expected Bernoulli values are

$$\beta_2 = -\frac{1}{18}, \quad \beta_4 = \frac{23}{1350}, \quad \beta_6 = -\frac{1543}{119070}, \quad \beta_8 = \frac{1196113}{65063250}. \quad (1318)$$

This supplies a direct integral bridge between the Bernoulli-polynomial structure and the Fabius distribution.

## 6.3 Beta weights

For  $\Re a, \Re b > 0$ , define

$$I_F(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1}F(x) dx. \quad (1319)$$

**Theorem 6.1** (beta-weighted Fabius integrals). *For  $\Re a, \Re b > 0$ ,*

$$I_F(a, b) = \mathbb{E}B_Y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} \mathfrak{M}(b+n), \quad (1320)$$

$$\int_0^1 x^{a-1}(1-x)^{b-1}(1-F(x)) dx = \mathbb{E}B_Y(a, b) = \sum_{n=0}^{\infty} \frac{(1-b)_n}{n!(a+n)} \mathfrak{M}(a+n). \quad (1321)$$

Here  $B_y$  is the incomplete beta function. If  $a$  in [equation \(1320\)](#), or  $b$  in [equation \(1321\)](#), is a positive integer, the corresponding series terminates. If both parameters are positive integers, the integrals are rational.

*Proof.* Use  $F(x) = \mathbb{E}\mathbf{1}_{\{Y \leq x\}}$  and Fubini:

$$I_F(a, b) = \mathbb{E} \int_Y^1 x^{a-1}(1-x)^{b-1} dx = \mathbb{E}B_{1-Y}(b, a) = \mathbb{E}B_Y(b, a),$$

where the last equality uses  $1-Y \stackrel{d}{=} Y$ . The hypergeometric expansion

$$B_y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} y^{b+n}$$

gives [equation \(1320\)](#). The endpoint flatness and [lemma 4.3](#) justify convergence beyond the ordinary boundary restrictions of the pointwise hypergeometric series. The second identity is analogous.  $\square$

For example, symmetry or direct termination gives

$$\int_0^1 x(1-x)F(x) dx = \frac{1}{12}. \quad (1322)$$

Differentiating [equation \(1320\)](#) in  $a$  and  $b$  produces all beta-logarithmic integrals with powers of  $\log x$  and  $\log(1-x)$ .

## 6.4 A dyadic half-interval identity

The lower-half differential equation  $F'(x) = 2F(2x)$  for  $0 \leq x \leq 1/2$  leads to exact quantile truncations.

**Proposition 6.2.** *For every complex  $p$ , with removable values interpreted by continuation,*

$$\int_0^{1/2} Q(v)^p dv = 2^{-p} \frac{1 - \Re(p+1)}{p+1}, \quad (1323)$$

$$\int_0^{1/2} x^p F(x) dx = \frac{2^{-p-1}}{p+1} \left[ \frac{1}{2} - \frac{1 - \Re(p+2)}{p+2} \right]. \quad (1324)$$

In particular,

$$\int_0^{1/2} F(x) dx = \frac{5}{72}. \quad (1325)$$

*Proof.* Change variables  $v = F(x)$  in the first integral and use the lower-half equation:

$$\int_0^{1/2} Q(v)^p dv = \int_0^{1/2} x^p F'(x) dx = 2^{-p} \int_0^1 y^p F(y) dy.$$

This is [equation \(1323\)](#). Substitute it into the incomplete formula [equation \(1263\)](#) at  $x = 1/2$  to obtain [equation \(1324\)](#).  $\square$

## 7 Fractional and integer antiderivatives

Repeated integration has a particularly transparent probabilistic meaning. The same calculus extends without change from integer order to the Riemann–Liouville operator. For  $\Re \alpha > 0$ , write

$$(I_0^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (1326)$$

### 7.1 Fractional inverse-pair duality

**Theorem 7.1** (fractional power duality). *Let  $\Re \alpha > 0$ . Initially for  $\Re q > 0$ ,*

$$(I_0^\alpha F^q)(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x-Q(v))^{\alpha} v^{q-1} dv. \quad (1327)$$

*Initially for  $\Re p > 0$ ,*

$$(I_0^\alpha Q^p)(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v-F(x))^{\alpha} x^{p-1} dx. \quad (1328)$$

*Both formulas continue in  $p$  or  $q$  wherever the endpoint terms admit continuation.*

*Proof.* Since  $d(x-t)^\alpha = -\alpha(x-t)^{\alpha-1} dt$ , integration by parts gives

$$I_0^\alpha F^q(x) = \frac{1}{\Gamma(\alpha+1)} \int_0^x (x-t)^\alpha d(F(t)^q).$$

The boundary terms vanish when  $\Re q > 0$ . Using  $d(F^q) = qF^{q-1} dF$  and substituting  $v = F(t)$  proves [equation \(1327\)](#). The proof of [equation \(1328\)](#) is identical after interchanging the inverse pair.  $\square$

The case  $q = 1$  has a direct truncated-moment interpretation.



**Corollary 7.2.** For  $\Re\alpha > 0$ ,

$$(I_0^\alpha F)(x) = \frac{\mathbb{E}[(x - Y)_+^\alpha]}{\Gamma(\alpha + 1)}. \quad (1329)$$

In particular,

$$(I_0^\alpha F)(1) = \frac{\mathfrak{M}(\alpha)}{\Gamma(\alpha + 1)}. \quad (1330)$$

*Proof.* In [equation \(1327\)](#), the integral over  $v$  is the expectation of  $(x - Q(V))_+^\alpha$  for a uniform  $V$ . Since  $Q(V) \stackrel{d}{=} Y$ , this is [equation \(1329\)](#). At  $x = 1$ , use  $1 - Y \stackrel{d}{=} Y$ .  $\square$

Formula [equation \(1330\)](#) says that the entire complex moment function is exactly the endpoint value of the fractional antiderivative, multiplied by  $\Gamma(\alpha + 1)$ . Thus the Newton series in [equation \(1275\)](#) is at the same time a global series for fractional repeated integration.

## 7.2 Repeated primitives of Rvachev's up function

For  $\Re\alpha > 0$ , define the left-anchored fractional integral

$$(I_{-1}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \int_{-1}^x (x - t)^{\alpha-1} \text{up}(t) dt. \quad (1331)$$

Because  $\text{up}$  is the density of  $Z$ ,

$$(I_{-1}^\alpha \text{up})(x) = \frac{\mathbb{E}[(x - Z)_+^{\alpha-1}]}{\Gamma(\alpha)}. \quad (1332)$$

At the right endpoint,

$$(I_{-1}^\alpha \text{up})(1) = \frac{2^{\alpha-1} \mathfrak{M}(\alpha - 1)}{\Gamma(\alpha)}. \quad (1333)$$

For an integer  $n \geq 1$ , [equation \(1332\)](#) becomes

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}[(x - Z)_+^{n-1}]. \quad (1334)$$

## 7.3 Appell polynomial tails

For  $x \geq 1$ , the positive-part signs in [equations \(1329\)](#) and [\(1334\)](#) disappear. Hence

$$(I_0^n F)(x) = \frac{1}{n!} \mathbb{E}(x - Y)^n, \quad x \geq 1, \quad (1335)$$

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x - Z)^{n-1}, \quad x \geq 1. \quad (1336)$$

The polynomial sequences

$$P_n(x) = \mathbb{E}(x - Y)^n, \quad R_n(x) = \mathbb{E}(x - Z)^n \quad (1337)$$

are Appell sequences:

$$P'_n(x) = nP_{n-1}(x), \quad R'_n(x) = nR_{n-1}(x). \quad (1338)$$

Their exponential generating functions are

$$\sum_{n \geq 0} P_n(x) \frac{t^n}{n!} = e^{tx} P(-t), \quad (1339)$$

$$\sum_{n \geq 0} R_n(x) \frac{t^n}{n!} = e^{tx} \mathcal{U}(-t). \quad (1340)$$

These polynomial tails and their inverse-moment projection role are part of the existing Rvachev–Appell research in the repository; here they arise as the exterior part of the fractional-truncated-moment formula.

#### 7.4 Ordinary $n$ -fold antiderivatives as incomplete moments

For an integer  $n \geq 1$ , the anchored  $n$ -fold antiderivative of  $F$  is

$$I_0^n F(x) = \frac{1}{n!} \int_0^{F(x)} (x - Q(v))^n dv. \quad (1341)$$

The same expression can be expanded finitely:

$$I_0^n F(x) = \frac{1}{n!} \sum_{j=0}^n (-1)^j \binom{n}{j} x^{n-j} \int_0^{F(x)} Q(v)^j dv. \quad (1342)$$

Thus every repeated primitive is determined by incomplete quantile moments. At a dyadic  $x$ , these can in turn be reduced by the exact dyadic machinery already developed in the repository.

For inverse powers, [equation \(1328\)](#) gives the complementary expression

$$I_0^n (Q^p)(v) = \frac{p}{n!} \int_0^{Q(v)} x^{p-1} (v - F(x))^n dx. \quad (1343)$$

The special case  $p = 1$  is a fractional extension of the area identity:

$$I_0^n Q(v) = \frac{1}{n!} \int_0^{Q(v)} (v - F(x))^n dx. \quad (1344)$$

For  $n = 1$ , this is

$$\int_0^v Q(w) dw = \int_0^{Q(v)} (v - F(x)) dx, \quad (1345)$$

the usual area formula in a form adapted to the Fabius quantile.

## 8 Derivatives of arbitrary order

### 8.1 Derivatives of the Fabius function

On the lower dyadic branch, repeated differentiation of  $F'(x) = 2F(2x)$  gives

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 < x < 2^{-n}. \quad (1346)$$

The full interval is obtained by the known dyadic/Thue–Morse reflection formulas. We do not duplicate those formulas here; the role of [equation \(1346\)](#) is to show why every inverse derivative becomes extremely singular near the endpoints.

### 8.2 Inverse derivatives and Bell polynomials

Let  $f_j = F^{(j)}(x)$ . Define rational differential expressions recursively by

$$\mathcal{R}_1(x) = \frac{1}{F'(x)}, \quad \mathcal{R}_{n+1}(x) = \frac{1}{F'(x)} \frac{d}{dx} \mathcal{R}_n(x). \quad (1347)$$

**Proposition 8.1** (all inverse derivatives). *For  $0 < v < 1$ ,*

$$Q^{(n)}(v) = \mathcal{R}_n(Q(v)). \quad (1348)$$

*Equivalently, for  $n \geq 2$ ,*

$$Q^{(n)}(v) = -\frac{1}{F'(Q(v))} \sum_{k=2}^n F^{(k)}(Q(v)) B_{n,k}(Q'(v), \dots, Q^{(n-k+1)}(v)), \quad (1349)$$

*where  $B_{n,k}$  is the partial exponential Bell polynomial.*

*Proof.* The operator relation follows by applying  $d/dv = (F'(x))^{-1} d/dx$  at  $x = Q(v)$ . The Bell recurrence follows by differentiating  $F(Q(v)) = v$   $n$  times and isolating the term containing  $Q^{(n)}$ .  $\square$

The first expressions are

$$\begin{aligned} Q' &= \frac{1}{f_1}, \\ Q'' &= -\frac{f_2}{f_1^3}, \\ Q''' &= \frac{3f_2^2 - f_1 f_3}{f_1^5}, \\ Q'''' &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{f_1^7}, \end{aligned} \quad (1350)$$

with every  $f_j$  evaluated at  $x = Q(v)$ . Combining these formulas with the Lambert– $W$  inverse endpoint expansion from the repository gives asymptotics of any fixed derivative order.

### 8.3 Derivatives of integral transforms

Since  $\mathfrak{M}$  is entire,

$$\mathfrak{M}^{(n)}(s) = \int_0^1 Q(v)^s (\log Q(v))^n dv. \quad (1351)$$

The bilateral up transform satisfies, away from its complex zeros,

$$\mathcal{U}^{(n)}(t) = \mathcal{U}(t) B_n(L_1(t), \dots, L_n(t)), \quad (1352)$$

where  $B_n$  is the complete exponential Bell polynomial and

$$L_j(t) = \frac{d^j}{dt^j} \log \mathcal{U}(t) = \sum_{k=1}^{\infty} 2^{-kj} \ell^{(j)}(t/2^k), \quad \ell(z) = \log \frac{\sinh z}{z}. \quad (1353)$$

At  $t = 0$ , these are precisely the cumulants in [equation \(1250\)](#); the continuation of [equation \(1352\)](#) through zeros is obtained by cancelling the apparent logarithmic singularities.

## 9 Endpoint fractional calculus of the inverse Fabius function

The inverse function is the most singular object in the Fabius–Rvachev system. Ordinary derivatives of  $Q$  blow up at both endpoints, while  $Q$  itself tends to zero more slowly than every positive power. This combination is exactly the regime in which fractional differentiation has a sharp critical exponent. The results in this section are not contained in the audited corpus: the repository develops the Lambert– $W$

inversion and integer inverse derivatives, but not the fractional endpoint law or its strong critical  $L^p$  inclusion.

Put

$$T = \log \frac{1}{y}, \quad \ell = \log 2, \quad \eta(y) = \frac{yQ'(y)}{Q(y)}. \quad (1354)$$

The dimensionless quantity  $\eta$  is the logarithmic elasticity of the quantile. It is the correct small parameter: powers of  $1/T$  alone miss the square-root scale and the repository's periodic Lambert phase.

**Proposition 9.1** (quantile scale and slow variation). *As  $y \downarrow 0$ ,*

$$Q(y) = \frac{\sqrt{T}}{e\sqrt{\ell}} \exp(-\sqrt{2\ell T}) \left(1 + O\left(\frac{\log T}{\sqrt{T}}\right)\right), \quad (1355)$$

$$\eta(y) = \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right). \quad (1356)$$

For each fixed  $c > 0$ ,

$$\frac{Q(cy)}{Q(y)} \rightarrow 1. \quad (1357)$$

Thus  $Q$ , and every positive power  $Q^\beta$ , is slowly varying at the origin in the multiplicative sense.

*Proof.* Equation (1355) is the leading term of the repository's corrected Lambert- $W$  inverse expansion. The full expansion is differentiable term by term; its logarithmic derivative gives (1356). Alternatively, the leading term contributes  $\sqrt{\ell/(2T)} - 1/(2T)$  to  $yQ'/Q$ , while differentiating the smooth periodic correction contributes  $O((\log T)/T)$ . Replacing  $y$  by  $cy$  changes  $T$  by the bounded number  $-\log c$ ; the difference of the square roots in (1355) therefore tends to zero. This proves (1357).  $\square$

For  $a > 0$ , let

$$H_a = \psi(a+1) + \gamma, \quad (1358)$$

where  $\psi$  is the digamma function and  $\gamma$  is Euler's constant. Thus  $H_n = 1 + 1/2 + \cdots + 1/n$  when  $n$  is a positive integer.

**Theorem 9.2** (digamma correction for fractional quantile primitives). *Let  $a > 0$  and  $\beta > 0$ . Then*

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta(y) + o(\eta(y))), \quad y \downarrow 0. \quad (1359)$$

Using (1356),

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} \left[1 - \beta H_a \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right)\right]. \quad (1360)$$

For  $a = n \in \mathbb{N}$ , this gives the first nontrivial endpoint correction to every  $n$ -fold antiderivative of every positive quantile power.

*Proof.* After  $v = yu$ ,

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a)} \int_0^1 (1-u)^{a-1} \left(\frac{Q(yu)}{Q(y)}\right)^\beta du. \quad (1361)$$

The differentiated endpoint expansion gives, uniformly on compact logarithmic  $u$ -ranges,

$$\log \frac{Q(yu)}{Q(y)} = \eta(y) \log u + o(\eta(y))(1 + |\log u|). \quad (1362)$$

Potter bounds for slowly varying functions dominate the remaining small- $u$  tail by an integrable power. We may therefore expand the exponential in (1362) under the beta integral. If  $U_a$  has density  $a(1-u)^{a-1}$  on  $(0, 1)$ , then

$$\mathbb{E} \log U_a = \psi(1) - \psi(a+1) = -H_a. \quad (1363)$$

The constant term in (1361) is  $1/\Gamma(a+1)$ ; the linear term is  $-\beta H_a \eta(y)$ . This proves (1359), and (1356) gives (1360).  $\square$

*Remark 9.3* (second endpoint correction). Let  $\delta(y) = y\eta'(y)$ , and put

$$V_a = H_a^2 + \psi_1(1) - \psi_1(a+1), \quad (1364)$$

where  $\psi_1$  is the trigamma function. Expanding the logarithmic endpoint jet one order further gives the formal, and under the repository's all-orders smooth expansion rigorous, refinement

$$\frac{\Gamma(a+1)I_{0+}^a(Q^\beta)(y)}{y^a Q(y)^\beta} = 1 - \beta H_a \eta + \frac{V_a}{2}(\beta^2 \eta^2 + \beta \delta) + o(\eta^2 + |\delta|). \quad (1365)$$

The appearance of digamma and trigamma values is forced by the first two logarithmic moments of a Beta(1,  $a$ ) variable. At integer  $a$ , these reduce to harmonic and second-order harmonic numbers, linking the fractional expansion directly to the Bernoulli/polygamma organization already present in the repository.

For  $0 < \sigma < 1$ , define

$$K_\sigma = \psi(1) - \psi(1-\sigma) = -\gamma - \psi(1-\sigma) > 0. \quad (1366)$$

**Theorem 9.4** (fractional inverse derivative and the sharp critical exponent). *Let  $0 < \sigma < 1$  and  $\beta > 0$ . Since  $Q(0) = 0$ , the left Caputo and Riemann–Liouville derivatives of  $Q^\beta$  coincide. As  $y \downarrow 0$ ,*

$$D_{0+}^\sigma(Q^\beta)(y) = \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} (1 + \beta K_\sigma \eta(y) + o(\eta(y))) \quad (1367)$$

$$= \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} \left[ 1 + \beta K_\sigma \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right) \right]. \quad (1368)$$

Consequently, for every  $0 < p < \infty$ ,

$$D_{0+}^\sigma(Q^\beta) \in L^p(0, \frac{1}{2}) \iff \sigma p \leq 1. \quad (1369)$$

By reflection, the same criterion holds for the right derivative  $D_{1-}^\sigma(1-Q)^\beta$  on  $(1/2, 1)$ . In particular, the critical exponent  $p = 1/\sigma$  is included in the strong  $L^p$  space, not merely in weak  $L^p$ .

*Proof.* Apply theorem 9.2 with  $a = 1 - \sigma$  and differentiate. Equivalently, compare with the exact model  $D_{0+}^\sigma y^\rho = \Gamma(1+\rho)y^{\rho-\sigma}/\Gamma(1+\rho-\sigma)$  and let the local logarithmic exponent  $\rho$  tend to  $\beta\eta(y)$ . The linear coefficient is  $\psi(1) - \psi(1-\sigma) = K_\sigma$ , which proves (1367). The endpoint expansion gives (1368).

It remains to decide integrability. The leading asymptotic reduces the question to

$$\int_0^\varepsilon y^{-\sigma p} Q(y)^{\beta p} dy. \quad (1370)$$

Set  $y = e^{-T}$ . By (1355), the logarithm of the transformed integrand, including  $dy = e^{-T} dT$ , is

$$(\sigma p - 1)T - \beta p \sqrt{2\ell T} + O(\log T). \quad (1371)$$

It tends linearly to  $-\infty$  if  $\sigma p < 1$ , linearly to  $+\infty$  if  $\sigma p > 1$ , and at  $\sigma p = 1$  retains the integrable stretched-exponential factor  $\exp(-\beta p \sqrt{2\ell T} + O(\log T))$ . This proves (1369). Reflection follows from  $1 - Q(1-y) = Q(y)$ .  $\square$

**Corollary 9.5** (integer endpoint as a limit). *The limiting case  $\sigma \uparrow 1$  predicts and is consistent with*

$$Q' \in L^1(0, 1), \quad Q' \notin L^p(0, 1) \quad (p > 1), \quad (1372)$$

*which is the known sharp first-inverse-derivative threshold. Thus (1369) is a genuine continuous-order interpolation of the integer result.*

## 10 Nonlocal fractional calculus of Rvachev's up-function

Riemann–Liouville operators are one-sided and adapted to antiderivatives. The Riesz potential and fractional Laplacian probe a different aspect: they are translation invariant and nonlocal, so a compactly supported function acquires explicit algebraic tails. For up, those tails are governed by rational moments and therefore by the same Bell–Bernoulli arithmetic as the local theory.

For  $0 < s < 1$ , use the one-dimensional normalization

$$(-\Delta)^s f(x) = c_{1,s} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(t)}{|x - t|^{1+2s}} dt, \quad c_{1,s} = \frac{4^s \Gamma(\frac{1}{2} + s)}{\sqrt{\pi} |\Gamma(-s)|}. \quad (1373)$$

For  $0 < a < 1$ , define the Riesz potential

$$(\mathcal{I}_a f)(x) = d_{1,a} \int_{\mathbb{R}} |x - t|^{a-1} f(t) dt, \quad d_{1,a} = \frac{\Gamma(\frac{1-a}{2})}{2^a \sqrt{\pi} \Gamma(\frac{a}{2})}. \quad (1374)$$

With the Fourier convention of this report, their multipliers are  $|\xi|^{2s}$  and  $|\xi|^{-a}$ , respectively.

**Theorem 10.1** (exact exterior series for the fractional Laplacian). *Let  $0 < s < 1$  and  $|x| > 1$ . Then*

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \sum_{j=0}^{\infty} \frac{(1+2s)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}}. \quad (1375)$$

*The series is absolutely and locally uniformly convergent on  $|x| > 1$ . Its first terms are*

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \left[ 1 + \frac{(1+2s)(1+s)}{9|x|^2} + \frac{19(1+2s)_4}{16200|x|^4} + \cdots \right]. \quad (1376)$$

*Proof.* Outside the support,  $\text{up}(x) = 0$ , so (1373) becomes

$$(-\Delta)^s \text{up}(x) = -c_{1,s} \int_{-1}^1 \frac{\text{up}(t)}{|x - t|^{1+2s}} dt. \quad (1377)$$

For  $x > 1$ , expand  $(x - t)^{-1-2s} = x^{-1-2s} \sum_{n \geq 0} (1+2s)_n (t/x)^n / n!$ . The expansion is uniformly absolutely convergent on every region  $x \geq 1 + \varepsilon$ . Evenness annihilates all odd moments and gives (1375). The case  $x < -1$  is the same by symmetry. Substituting  $\mu_2 = 1/9$  and  $\mu_4 = 19/675$  gives (1376).  $\square$

**Theorem 10.2** (exact exterior Riesz-potential series). *Let  $0 < a < 1$  and  $|x| > 1$ . Then*

$$(\mathcal{I}_a \text{up})(x) = d_{1,a} |x|^{a-1} \sum_{j=0}^{\infty} \frac{(1-a)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}}. \quad (1378)$$

*Again the series is absolutely and locally uniformly convergent. Thus every exterior coefficient is an explicit gamma factor times a rational Fabius–Rvachev moment.*

*Proof.* For  $x > 1$ , expand  $(x-t)^{a-1} = x^{a-1}(1-t/x)^{-(1-a)}$  in (1374); odd moments vanish. Reflection gives  $x < -1$ .  $\square$

**Proposition 10.3** (Bell–Bernoulli generation of every coefficient). *The even cumulants of the centered up law are*

$$\kappa_{2r} = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0. \quad (1379)$$

Consequently

$$\mu_{2j} = B_{2j}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2j}), \quad (1380)$$

where  $B_n$  is the complete exponential Bell polynomial. Equations (1375) and (1378) therefore give closed Bernoulli–Bell series for both nonlocal operators.

*Proof.* Take the logarithm of

$$\mathcal{U}(z) = \prod_{k \geq 1} \frac{\sinh(z/2^k)}{z/2^k}$$

and use

$$\log\left(\frac{\sinh z}{z}\right) = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} z^{2r}.$$

The geometric factor is  $\sum_{k \geq 1} 2^{-2rk} = 1/(2^{2r} - 1)$ , which gives (1379); the moment–cumulant relation is (1380).  $\square$

**Theorem 10.4** (fractional Sobolev energy as an infinite-sinc integral). *For every  $s \geq 0$ ,*

$$\mathcal{E}_s(\text{up}) := \int_{\mathbb{R}} \text{up}(x) (-\Delta)^s \text{up}(x) dx = \frac{1}{2\pi} \int_{\mathbb{R}} |\xi|^{2s} \prod_{k=1}^{\infty} \text{sinc}^2(\xi/2^k) d\xi. \quad (1381)$$

*The integral is finite for every  $s$ . For  $0 < s < 1$ , it also equals*

$$\mathcal{E}_s(\text{up}) = \frac{c_{1,s}}{2} \iint_{\mathbb{R}^2} \frac{|\text{up}(x) - \text{up}(y)|^2}{|x - y|^{1+2s}} dx dy. \quad (1382)$$

*Thus the full fractional Sobolev scale of up is encoded by one weighted family of infinite sinc-square integrals.*

*Proof.* Plancherel’s theorem and the Fourier multiplier  $|\xi|^{2s}$  give (1381). The up-function is  $C^\infty$  with compact support, so its Fourier transform decays faster than every power and all energies are finite. The standard symmetrization of the singular integral gives (1382).  $\square$

## 11 Fractional self-differential scaling and a dyadic Floquet corrector

The global signed self-differential extension  $\mathcal{F}$  satisfies  $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ . Integer differentiation therefore stays on a single dyadic dilation orbit. The repository’s Laplace analysis proves that standard fractional differentiation does not: a nonconstant log-periodic phase is the exact obstruction. Here we retain that obstruction and turn it into a covariant correction.

Let

$$L(s) = \int_0^\infty e^{-sx} \mathcal{F}(x) dx, \quad s > 0. \quad (1383)$$

The dilation equation  $L(s/2) = sL(s)$  is equivalent to

$$L(2^t) = 2^{-t(t+1)/2} \Pi(t), \quad \Pi(t+1) = \Pi(t), \quad (1384)$$

where  $\Pi$  is positive, nonconstant, and has minimal period one. Define the standard Laplace fractional derivative by  $\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s) = s^\alpha L(s)$  and the continuous dilation orbit by

$$(T_\alpha \mathcal{F})(x) = 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x). \quad (1385)$$

The known defect identity is

$$\frac{\mathcal{L}[T_\alpha \mathcal{F}](s)}{\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s)} = m_\alpha(s), \quad m_\alpha(s) = \frac{\Pi(\log_2 s - \alpha)}{\Pi(\log_2 s)}. \quad (1386)$$

In particular, the ratio is identically one exactly for integer  $\alpha$ .

**Theorem 11.1** (dyadic Floquet corrector and scale cocycle). *Let  $\mathcal{C}_\alpha$  be the Laplace multiplier with symbol  $m_\alpha$ :*

$$\mathcal{L}[\mathcal{C}_\alpha f](s) = m_\alpha(s) \mathcal{L}[f](s). \quad (1387)$$

Then, on the cyclic vector  $\mathcal{F}$ ,

$$T_\alpha \mathcal{F} = \mathcal{C}_\alpha \mathbf{D}^\alpha \mathcal{F} \quad (\alpha \in \mathbb{R}). \quad (1388)$$

The symbols satisfy the exact scale-cocycle law

$$m_{\alpha+\beta}(s) = m_\alpha(s/2^\beta) m_\beta(s). \quad (1389)$$

Moreover  $m_n \equiv 1$  for  $n \in \mathbb{Z}$ , so the corrector disappears at every integer order.

*Proof.* Equation (1388) is (1386) rewritten as an operator identity. For the cocycle,

$$m_\alpha(s/2^\beta) m_\beta(s) = \frac{\Pi(\log_2 s - \beta - \alpha)}{\Pi(\log_2 s - \beta)} \frac{\Pi(\log_2 s - \beta)}{\Pi(\log_2 s)} = m_{\alpha+\beta}(s).$$

Period one gives  $m_n = 1$ . □

**Remark 11.2.** The corrected family is not an ordinary translation-invariant semigroup; the argument shift  $s \mapsto s/2^\beta$  in (1389) is essential. It is better viewed as a Floquet cocycle over logarithmic frequency. This identifies a precise candidate for a fractional calculus adapted to the Fabius dyadic orbit, rather than forcing that orbit into the standard Riemann–Liouville semigroup.

## 12 Laplace, Fourier, Mellin, and resolvent transforms

### 12.1 One-sided Laplace transforms

Integration by parts against the distribution measure  $dF$  gives, for all  $t \in \mathbb{C}$  with removable values at  $t = 0$ ,

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad (1390)$$

$$\int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}. \quad (1391)$$

Consequently

$$\int_0^1 x^n e^{tx} F(x) dx = \frac{d^n}{dt^n} \left( \frac{e^t - P(t)}{t} \right), \quad (1392)$$

$$\int_0^1 x^n e^{tx} \text{up}(x) dx = \frac{d^n}{dt^n} \left( \frac{P(t) - 1}{t} \right). \quad (1393)$$

Purely imaginary  $t$  yields the corresponding sine and cosine transforms. On the full line,

$$\int_{-1}^1 e^{tx} \text{up}(x) dx = \mathcal{U}(t), \quad \int_{-1}^1 e^{i\omega x} \text{up}(x) dx = \phi(\omega). \quad (1394)$$



## 12.2 Stieltjes transforms and all resolvent powers

For  $z \in \mathbb{C} \setminus [0, 1]$ , define

$$G_Y(z) = \mathbb{E} \frac{1}{z - Y} = \int_0^1 \frac{dv}{z - Q(v)}. \quad (1395)$$

For  $|z| > 1$ ,

$$G_Y(z) = \sum_{n=0}^{\infty} \frac{m_n}{z^{n+1}}. \quad (1396)$$

Similarly, for  $z \notin [-1, 1]$ ,

$$G_Z(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (1397)$$

Applying [equation \(1310\)](#) to a resolvent primitive gives all higher powers. For every integer  $n \geq 1$ ,

$$\int_0^1 \frac{F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[ \frac{1}{(z - 1)^n} - \mathbb{E} \frac{1}{(z - Y)^n} \right], \quad (1398)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[ \mathbb{E} \frac{1}{(z - Y)^n} - \frac{1}{z^n} \right]. \quad (1399)$$

For  $n = 1$ ,

$$\int_0^1 \frac{F(x)}{(z - x)^2} dx = \frac{1}{z - 1} - G_Y(z), \quad (1400)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^2} dx = G_Y(z) - \frac{1}{z}. \quad (1401)$$

Using the logarithmic primitive, with a branch analytic off  $[0, 1]$ ,

$$\int_0^1 \frac{F(x)}{z - x} dx = \mathbb{E} \log(z - Y) - \log(z - 1), \quad (1402)$$

$$\int_0^1 \frac{1 - F(x)}{z - x} dx = \log z - \mathbb{E} \log(z - Y). \quad (1403)$$

These formulas furnish analytic continuations and boundary-value formulas for Hilbert transforms by approaching the cut from above and below.

## 12.3 A transform fixed point and the full resolvent hierarchy

The affine random fixed point for the centered up law produces more than a moment series. Let

$$R_n(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z - t)^n} dt, \quad z \in \mathbb{C} \setminus [-1, 1], \quad n \geq 1. \quad (1404)$$

Thus  $R_1 = G_Z$ . If  $X$  has density  $\text{up}$ , then  $X \stackrel{d}{=} (X' + U)/2$ , where  $U$  is uniform on  $[-1, 1]$  and independent of  $X' \stackrel{d}{=} X$ .

**Theorem 12.1** (logarithmic fixed point and higher resolvents). *With the logarithm branch normalized at infinity,*

$$R_1(z) = \mathbb{E} \log \left( \frac{2z - X + 1}{2z - X - 1} \right). \quad (1405)$$

It follows that

$$R_1'(z) = 2(R_1(2z + 1) - R_1(2z - 1)), \quad (1406)$$

and, for every  $n \geq 1$ ,

$$R_{n+1}(z) = \frac{2^n}{n} (R_n(2z - 1) - R_n(2z + 1)). \quad (1407)$$

For the inverse-Fabius resolvents

$$S_n(z) = \int_0^1 \frac{dv}{(z - Q(v))^n}, \quad z \in \mathbb{C} \setminus [0, 1], \quad (1408)$$

we have

$$S_n(z) = 2^n R_n(2z - 1), \quad (1409)$$

$$S_{n+1}(z) = \frac{2^{n+1}}{n} (S_n(2z - 1) - S_n(2z)). \quad (1410)$$

*Proof.* Condition on  $X'$  and integrate the uniform variable:

$$\mathbb{E} \left[ \frac{1}{z - (X' + U)/2} \middle| X' \right] = \int_{-1}^1 \frac{du}{2z - X' - u} = \log \frac{2z - X' + 1}{2z - X' - 1}.$$

This proves (1405). Differentiation gives (1406). Since  $R_n(z) = (-1)^{n-1} R_1^{(n-1)}(z)/(n-1)!$ , differentiating (1406) repeatedly gives (1407). Finally,  $Q(V)$  with  $V \sim \text{Unif}[0, 1]$  has the same law as  $Y = (X + 1)/2$ , which gives (1409); substitution into (1407) gives (1410).  $\square$

## 12.4 Entire-kernel calculus

If  $h(x) = \sum_{n \geq 0} a_n x^n$  is entire and the series is dominated on the compact support, then

$$\int_{-1}^1 h(x) \text{up}(x) dx = \sum_{n=0}^{\infty} a_n \mu_n, \quad (1411)$$

$$\int_0^1 h(x) dF(x) = \sum_{n=0}^{\infty} a_n m_n. \quad (1412)$$

For instance,

$$\int_{-1}^1 J_0(tx) \text{up}(x) dx = \sum_{n=0}^{\infty} \frac{(-1)^n (t^2/4)^n}{(n!)^2} \mu_{2n}. \quad (1413)$$

Together with equation (1310), this gives convergent moment series for Bessel, Airy, hypergeometric, and polylogarithmic kernels without requiring a separate quadrature of the flat endpoint function.

*Post-snapshot status.* The principle (1411) is now formal: `EntireKernelCalculus.lean` proves `Fabius.integral_tsum_eq_tsum_upMoment` — for any coefficient sequence with  $\sum_n |a_n| < \infty$ ,  $\int (\sum_n a_n x^n) d\mu_{\text{up}} = \sum_n a_n \mu_n$ ; since the support lies in  $(-1, 1)$ , the coefficients alone dominate, so every kernel entire on a neighbourhood of the closed support (Bessel, Airy, hypergeometric, polylogarithmic) integrates term by term through this single statement. The exponential instance is exported: `Fabius.mgf_id_rvachevMeasure_eq_tsum` gives the exact moment-generating series  $M(t) = \sum_{n \geq 0} t^n \mu_n / n!$  for every real  $t$ , joining the cumulant tail laws of the geometric-tail dictionary to the same moment data.

## 13 Products, order statistics, and correlation integrals

Integrals containing products of  $F$  and  $\text{up}$  are generally not rational, but they have exact probabilistic meanings that lead to useful transform formulas and asymptotics.

### 13.1 Products of distribution and survival powers

Let  $Y_1, \dots, Y_{m+n}$  be independent copies of  $Y$ , and define

$$A_m = \max(Y_1, \dots, Y_m), \quad B_n = \min(Y_{m+1}, \dots, Y_{m+n}). \quad (1414)$$

The two blocks are independent and

$$\mathbb{P}(A_m \leq x < B_n) = F(x)^m (1 - F(x))^n. \quad (1415)$$

**Theorem 13.1** (weighted order-statistic gap). *For integers  $m, n \geq 1$  and  $\Re p > -1$ ,*

$$\int_0^1 x^p F(x)^m (1 - F(x))^n dx = \frac{1}{p+1} \mathbb{E} \left[ (B_n^{p+1} - A_m^{p+1}) \mathbf{1}_{\{A_m < B_n\}} \right]. \quad (1416)$$

In particular,

$$\int_0^1 F(x)^m (1 - F(x))^n dx = \mathbb{E}(B_n - A_m)_+. \quad (1417)$$

*Proof.* Insert the indicator of the event in [equation \(1415\)](#), integrate, and apply Fubini. For each realization, the allowed interval is  $[A_m, B_n)$  if  $A_m < B_n$ , and empty otherwise. Its  $x^p$ -integral is the right side of [equation \(1416\)](#).  $\square$

Since  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$ , this theorem covers all integer products  $F^m \text{up}^n$  on the positive half of the support. Two useful special cases are

$$\int_0^1 F(x)^m dx = \mathbb{E}(1 - A_m) = \mathbb{E} \min(Y_1, \dots, Y_m), \quad (1418)$$

and

$$\int_0^1 F(x)^2 dx = \frac{1}{2} - \int_0^1 F(x)(1 - F(x)) dx. \quad (1419)$$

When the density factor is present, substitution gives a closed beta value for arbitrary complex exponents in their convergence domain:

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a+1, b+1). \quad (1420)$$

Using  $F'(x) = 2 \text{up}(2x-1)$ ,

$$2 \int_0^1 F(x)^a (1 - F(x))^b \text{up}(2x-1) dx = B(a+1, b+1). \quad (1421)$$

### 13.2 Quantiles and ordinary order statistics

If  $Y_{r:n}$  is the  $r$ -th order statistic of  $n$  independent Fabius variables, then  $Y_{r:n} = Q(V_{r:n})$  in distribution, where  $V_{r:n}$  has beta density. Consequently

$$\mathbb{E} Y_{r:n}^p = \frac{1}{B(r, n-r+1)} \int_0^1 v^{r-1} (1-v)^{n-r} Q(v)^p dv. \quad (1422)$$

Symmetry gives

$$\mathbb{E} Y_{r:n}^p = \mathbb{E}(1 - Y_{n+1-r:n})^p, \quad \mathbb{E} Y_{r:n} + \mathbb{E} Y_{n+1-r:n} = 1. \quad (1423)$$

Expanding the beta polynomial in [equation \(1422\)](#) reduces every integer  $n, r$  to a finite combination of the mixed quantile integrals appearing in [equation \(1256\)](#).

### 13.3 The Gini integral and the sinc product

For two independent copies  $Y, Y'$ , the standard distribution identity is

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y - Y'|. \quad (1424)$$

The difference is  $(Z - Z')/2$ , and its characteristic function is  $\phi(t/2)^2$ . The absolute-moment formula

$$\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty \frac{1 - \Re \varphi_X(t)}{t^2} dt \quad (1425)$$

therefore yields an integral containing only the infinite sinc product.

**Theorem 13.2** (Gini–sinc identity). *With  $\phi$  as in [equation \(1244\)](#),*

$$\boxed{\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2\pi} \int_0^\infty \frac{1 - \phi(t)^2}{t^2} dt.} \quad (1426)$$

Numerically,

$$\int_0^1 F(x)(1 - F(x)) dx = 0.0955632320160066 \dots, \quad (1427)$$

so

$$\int_0^1 F(x)^2 dx = 0.4044367679839934 \dots \quad (1428)$$

This is a direct link between a nonlinear real-space integral of  $F$  and the square of the Rvachev infinite product. It does not appear to simplify to the rational moment constants.

### 13.4 Autocorrelation and $L^2$ energy

Define

$$\mathcal{C}(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x + a) dx. \quad (1429)$$

Since  $\text{up}$  is even, the Wiener–Khinchin formula gives

$$\mathcal{C}(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} \phi(t)^2 dt. \quad (1430)$$

At  $a = 0$ , Parseval's identity gives

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = \frac{1}{\pi} \int_0^\infty \phi(t)^2 dt = 0.808873535967971 \dots \quad (1431)$$

More generally, derivatives of  $\mathcal{C}$  at the origin are convergent spectral moments whenever the corresponding weighted sinc-product integral converges; the fast decay results developed elsewhere in the repository provide quantitative truncation bounds.

## 14 Additional integral-transform identities

### 14.1 A transform relation for inverse resolvents

The general identity [theorem 3.1](#) can be applied in the  $v$ -variable as well. Taking  $A(x) = x$  and  $B(v) = \log(z/(z - v))$  gives

$$\int_0^1 \frac{Q(v)}{z - v} dv + \int_0^1 \log \frac{z}{z - F(x)} dx = \log \frac{z}{z - 1}. \quad (1432)$$

With  $B(v) = (z - v)^{-1}$ ,

$$\int_0^1 \frac{Q(v)}{(z - v)^2} dv + \int_0^1 \frac{dx}{z - F(x)} = \frac{1}{z - 1}. \quad (1433)$$

These are nonlinear resolvent dualities: a transform of the quantile is paired with a composition transform of the distribution function.

## 14.2 Frullani representations for logarithmic constants

For  $0 < Y \leq 1$ ,

$$-\log Y = \int_0^\infty \frac{e^{-tY} - e^{-t}}{t} dt. \quad (1434)$$

Averaging and using  $P(-t)$  gives

$$\int_0^1 \frac{F(x)}{x} dx = \int_0^\infty \frac{P(-t) - e^{-t}}{t} dt, \quad (1435)$$

where

$$P(-t) = \prod_{k=1}^\infty \frac{1 - e^{-t/2^k}}{t/2^k}. \quad (1436)$$

For  $a > 0$ , the negative moment itself has the Mellin–Laplace representation

$$\mathfrak{M}(-a) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} P(-t) dt. \quad (1437)$$

This is an independent numerical route to [equation \(1277\)](#) and connects the negative-moment constant  $C_*$  to the dyadic product on the positive Laplace axis.

## 14.3 A smoothing difference equation for $\mathfrak{M}$

Let

$$N(s) = \mathbb{E}(1 + Y)^s = \sum_{k=0}^\infty \binom{s}{k} m_k. \quad (1438)$$

The series is entire because of [lemma 4.3](#). Integrating the uniform variable in [equation \(1239\)](#) with an arbitrary complex exponent gives the entire difference relation

$$(s + 1)2^s \mathfrak{M}(s) = N(s + 1) - \mathfrak{M}(s + 1), \quad (1439)$$

with the value at  $s = -1$  interpreted by continuity. At nonnegative integer  $s$ , coefficient comparison recovers [equation \(1247\)](#); at negative  $s$ , it is equivalent to [equation \(1288\)](#). Thus the positive rational moments and the superexponentially growing negative moments are two sides of one entire difference equation.

## 15 Numerical experiments and reproducibility

The accompanying program `fractional_frontier_experiments.py` is fully commented and deterministic. It reconstructs up by FFT inversion of the sinc product on a period-four grid with  $2^{18}$  points; the support  $[-1, 1]$  prevents periodic overlap. The Fabius function is then obtained from  $F(x) = \text{up}(x - 1)$  on  $[0, 1]$ , and a monotone inverse gives  $Q$ . No Monte Carlo sampling or network access is used.

The program independently checks four parts of the new analysis:

1. the exterior series (1375) and (1378) against direct quadrature of their defining kernels;
2. the logarithmic Cauchy fixed point (1405), differential equation (1406), and hierarchy (1407);
3. the normalized endpoint ratio in (1359) and its first digamma correction;
4. the normalized fractional derivative in (1367). The derivative is obtained by a symmetric five-point differentiation of  $I^{1-\sigma}Q$ , avoiding the severe conditioning loss in direct division by the flat density  $F'$ .

| $s$  | $ x $ | direct $(-\Delta)^s$ up | moment series     | absolute error |
|------|-------|-------------------------|-------------------|----------------|
| 0.25 | 1.20  | $-1.80996700e-01$       | $-1.80996689e-01$ | $1.07e-08$     |
| 0.25 | 1.50  | $-1.20466958e-01$       | $-1.20466958e-01$ | $1.12e-12$     |
| 0.25 | 2.00  | $-7.45381034e-02$       | $-7.45381034e-02$ | $5.55e-17$     |
| 0.50 | 1.20  | $-2.95515474e-01$       | $-2.95515385e-01$ | $8.94e-08$     |
| 0.50 | 1.50  | $-1.67488207e-01$       | $-1.67488207e-01$ | $8.18e-12$     |
| 0.50 | 2.00  | $-8.70075045e-02$       | $-8.70075045e-02$ | $8.33e-17$     |
| 0.75 | 1.20  | $-2.93003324e-01$       | $-2.93002949e-01$ | $3.75e-07$     |
| 0.75 | 1.50  | $-1.39364026e-01$       | $-1.39364026e-01$ | $3.02e-11$     |
| 0.75 | 2.00  | $-6.03002358e-02$       | $-6.03002358e-02$ | $2.15e-16$     |

Table 25: Exterior fractional-Laplacian series checked against direct FFT-grid quadrature.

| $\alpha$ | $ x $ | direct $\mathcal{I}_\alpha$ up | moment series    | absolute error |
|----------|-------|--------------------------------|------------------|----------------|
| 0.25     | 1.20  | $1.38089169e-01$               | $1.38089169e-01$ | $4.12e-10$     |
| 0.25     | 1.50  | $1.14103403e-01$               | $1.14103403e-01$ | $5.18e-14$     |
| 0.25     | 2.00  | $9.04704471e-02$               | $9.04704471e-02$ | $4.16e-17$     |
| 0.50     | 1.20  | $3.76439711e-01$               | $3.76439711e-01$ | $3.17e-10$     |
| 0.50     | 1.50  | $3.32338157e-01$               | $3.32338157e-01$ | $4.24e-14$     |
| 0.50     | 2.00  | $2.85179706e-01$               | $2.85179706e-01$ | $1.11e-16$     |
| 0.75     | 1.20  | $1.03262489e+00$               | $1.03262489e+00$ | $1.72e-10$     |
| 0.75     | 1.50  | $9.71451305e-01$               | $9.71451305e-01$ | $2.50e-14$     |
| 0.75     | 2.00  | $9.00632222e-01$               | $9.00632222e-01$ | $1.11e-16$     |

Table 26: Exterior Riesz-potential series checked against direct quadrature.

| $y$       | $Q(y)$           | $\eta(y)$     | normalized integral | first correction | corrected error |
|-----------|------------------|---------------|---------------------|------------------|-----------------|
| $1.0e-02$ | $1.55942014e-01$ | $2.18739e-01$ | 0.87083647          | 0.83696799       | $3.39e-02$      |
| $3.0e-03$ | $1.21418852e-01$ | $1.97703e-01$ | 0.88052201          | 0.85264653       | $2.79e-02$      |
| $1.0e-03$ | $9.85392757e-02$ | $1.83167e-01$ | 0.88730767          | 0.86348039       | $2.38e-02$      |
| $3.0e-04$ | $7.96341581e-02$ | $1.71106e-01$ | 0.89338014          | 0.87246987       | $2.09e-02$      |
| $1.0e-04$ | $6.63361227e-02$ | $1.61734e-01$ | 0.89815279          | 0.87945530       | $1.87e-02$      |

Table 27: Fractional quantile primitive with  $\alpha = 0.65$  and  $\beta = 1.0$ . The normalized integral is  $\Gamma(\alpha + 1)I^\alpha(Q^\beta)/(y^\alpha Q(y)^\beta)$ .

At  $|x| = 1.2$ , where the exterior moment series converges most slowly among the displayed points, eighteen exact moments already give errors between  $10^{-8}$  and  $4 \times 10^{-7}$  for the fractional Laplacian; at  $|x| = 2$ , the discrepancy is at floating-point roundoff. The Cauchy-transform identities have residuals below  $10^{-16}$ . In the endpoint tables, the first digamma correction reduces the primitive error from about  $10^{-1}$  to about  $2 \times 10^{-2}$ , and the fractional-derivative error from about  $1.4 \times 10^{-1}$  to about  $1.5 \times 10^{-2}$  at the smallest resolved  $y$ . The trend is consistent with the predicted  $O((\log T)/T)$

| $y$       | $Q(y)$           | $\eta(y)$     | normalized derivative | first correction | corrected error |
|-----------|------------------|---------------|-----------------------|------------------|-----------------|
| $1.0e-02$ | $1.55942014e-01$ | $2.18739e-01$ | 1.18349375            | 1.21073362       | $2.72e-02$      |
| $3.0e-03$ | $1.21418852e-01$ | $1.97703e-01$ | 1.16810670            | 1.19046769       | $2.24e-02$      |
| $1.0e-03$ | $9.85392757e-02$ | $1.83167e-01$ | 1.15744377            | 1.17646395       | $1.90e-02$      |
| $3.0e-04$ | $7.96341581e-02$ | $1.71106e-01$ | 1.14819379            | 1.16484423       | $1.67e-02$      |
| $1.0e-04$ | $6.63361227e-02$ | $1.61734e-01$ | 1.14094770            | 1.15581494       | $1.49e-02$      |

Table 28: Caputo/Riemann–Liouville quantile derivative with  $\sigma = 0.4$  and  $\beta = 1.0$ . The normalized derivative is  $\Gamma(1 - \sigma)y^\sigma D^\sigma(Q^\beta)/Q(y)^\beta$ .

| transform identity             | absolute residual |
|--------------------------------|-------------------|
| logarithmic Cauchy fixed point | $0.000e+00$       |
| Cauchy differential equation   | $7.850e-17$       |
| resolvent hierarchy, $m = 1$   | $7.850e-17$       |
| resolvent hierarchy, $m = 2$   | $2.776e-17$       |
| resolvent hierarchy, $m = 3$   | $3.925e-17$       |

Table 29: Residuals for the transform fixed-point and higher-resolvent identities at  $z = 1.35 + 0.60i$ .

remainder; FFT resolution, rather than the asymptotic formula, becomes the limiting factor farther into the flat endpoint.

The script writes a human-readable ledger `numerical_results.txt`, a JSON file `numerical_results.json`, and the LaTeX tables included above. Running

```
python fractional_frontier_experiments.py
```

regenerates all three files.

## 16 Conjectures and frontier directions

### 16.1 Critical fractional Sobolev regularity of the inverse

**Conjecture 16.1** (sharp fractional Sobolev boundary). *For  $0 < s < 1$ ,  $1 \leq p < \infty$ , and  $\beta > 0$ ,*

$$Q^\beta \in W^{s,p}(0,1) \iff sp \leq 1. \quad (1440)$$

*At the critical line  $sp = 1$ , membership should be strong rather than merely weak, with the stretched-exponential inverse endpoint factor providing the additional integrability. The one-sided derivative theorem [theorem 9.4](#) proves the corresponding statement in the Riemann–Liouville endpoint scale. A proof of (1440) should combine a fractional Hardy equivalence with a two-sided localization and the repository’s all-orders inverse expansion.*

### 16.2 Function-space realization of the Floquet corrector

The symbol  $m_\alpha(s)$  in (1386) is bounded above and below on the positive axis for each fixed real  $\alpha$ , because  $\Pi$  is positive and periodic. It is therefore natural to expect a bounded causal operator on weighted Laplace–Hardy or Mellin–Sobolev spaces.

**Conjecture 16.2** (dyadic fractional calculus). *There is a natural Banach scale  $\mathcal{X}_\rho$  containing the global signed Fabius extension such that:*

1. *every  $\mathcal{C}_\alpha$  extends boundedly and invertibly to  $\mathcal{X}_\rho$ ;*
2. *the scale-cocycle identity (1389) becomes an operator cocycle under dyadic dilation;*

3.  $C_\alpha \mathbf{D}^\alpha$  is the unique continuous-order extension of the integer derivative orbit satisfying the cocycle and a mild spectral boundedness condition.

The Mellin variable  $\log_2 s$  turns the problem into periodic Floquet theory, while the nonzero Fourier modes of  $\log \Pi$  should control norm distortion.

### 16.3 Arithmetic nonlocal tails

For rational  $s$  or  $a$ , the series in [theorems 10.1](#) and [10.2](#) has rational moment coefficients multiplied by explicit gamma constants. The arithmetic nature of special values such as  $(-\Delta)^{1/2} \text{up}(2)$  and  $(\mathcal{I}_{1/2} \text{up})(2)$  is unknown.

**Conjecture 16.3** (noncollapse of exterior nonlocal values). *Except for identities forced by gamma reflection and the rational moment recurrence, the special values of the exterior fractional-Laplacian and Riesz-potential series at rational  $|x| > 1$  do not reduce to finite combinations of algebraic numbers and powers of  $\pi$ . A first tractable target is linear independence of several values over the field generated by the relevant gamma constants.*

### 16.4 Near-rigidity of the negative-moment phase

**Conjecture 16.4** (phase rigidity). *The continuous one-periodic limiting profile  $C(\theta)$  in [equation \(1295\)](#) is constant:*

$$C(\theta) \equiv C_*. \quad (1441)$$

A weaker and perhaps more realistic alternative is that  $C$  is nonconstant but its first Fourier harmonics are suppressed by the same spectral mechanism responsible for the extremely small periodic oscillations in the endpoint asymptotics studied elsewhere in the repository. The exact recurrence reduces this question to the Fourier analysis of the periodic product [equation \(1295\)](#). A rigorous high-precision test should compute  $A(\theta)$  independently by [equation \(1437\)](#), rather than by the same moment data used in the recurrence.

### 16.5 Completely regular growth of the entire Mellin function

The maximum modulus is known exactly through [equation \(1305\)](#), but directional cancellation remains open.

**Conjecture 16.5** (indicator function). *The entire function  $\mathfrak{M}$  has completely regular growth of order 2 with indicator*

$$h(\vartheta) = \frac{\log 2}{2} \max\{0, -\cos \vartheta\}^2. \quad (1442)$$

The upper bound follows immediately from  $|\mathfrak{M}(re^{i\vartheta})| \leq \mathfrak{M}(r \max(0, -\cos \vartheta))$ . The difficult part is a matching lower bound away from the negative real axis, where oscillation in  $e^{i\Im(s) \log Y}$  may create zeros and exceptional directions.

### 16.6 Arithmetic nature of the new constants

The quantities

$$C_*, \quad \mathfrak{M}(-r), \quad -\mathbb{E} \log Y, \quad \int_0^1 F(1-F), \quad \int_{\mathbb{R}} \text{up}^2 \quad (1443)$$

are defined by positive rational series or infinite sinc products, but no reduction to standard constants is visible.



**Conjecture 16.6.** *The constants  $C_*$ ,  $\mathfrak{M}(-r)$  for every positive integer  $r$ , and  $-\mathbb{E} \log Y$  are irrational; at least one of the stronger properties of transcendence or algebraic independence should hold among them.*

This is deliberately stated as a research target, not as numerical evidence for a specific transcendence theorem. Possible tools include Mahler-type functional equations for  $P(t)$ , arithmetic estimates for the rational Newton partial sums, and Padé approximants controlled by the moment recurrence.

### 16.7 Complete monotonicity of normalized errors

The sequence  $C_r$  is decreasing by [equation \(1299\)](#). Numerics suggest much more.

**Conjecture 16.7.** *The error sequence  $C_r - C_*$  is completely monotone:*

$$(-1)^k \Delta^k (C_r - C_*) \geq 0 \quad (r \geq 1, k \geq 0). \quad (1444)$$

A proof might represent  $C_r - C_*$  as a Hausdorff moment sequence derived from the positive tail product  $\prod_{j \geq r} (1 - \rho(j))$ .

### 16.8 Order-statistic endpoint regimes

When the sample size  $n$  grows, the minimum  $Y_{1:n}$  probes precisely the Lambert- $W$  endpoint regime. Since

$$\mathbb{P}(Y_{1:n} > x) = (1 - F(x))^n,$$

its moments are

$$\mathbb{E} Y_{1:n}^p = p \int_0^1 x^{p-1} (1 - F(x))^n dx. \quad (1445)$$

A promising direction is a uniform saddle-point expansion in  $p$  and  $n$ , with the periodic endpoint phase transported into order-statistic oscillations. The corresponding maximum follows by symmetry. This would connect the present integral calculus directly to the repository's inverse-Lambert and half-integer spectral programs.

### 16.9 General base and $q$ -deformations

Replace the dyadic weights by  $b^{-k}$ , normalized so the support remains  $[0, 1]$ . The distributional smoothing, inverse-pair calculus, and fractional identities survive unchanged, while the product  $P(t)$ , Bernoulli cumulants, and negative-moment normalization acquire base  $b$ . The expected leading law is

$$\mathbb{E} Y_b^{-r} \asymp C_b \frac{b^{r(r+1)/2}}{\Gamma(r)}. \quad (1446)$$

Determining  $C_b$ , its phase profile, and its relationship with  $q = 1/b$  binomial coefficients is a natural extension.

### 16.10 Formalization roadmap

The most accessible Lean targets are:

1. package the remaining improper-singular scope of [theorem 3.1](#); measure-level quantile transport, the measurable-free Galois/explicit-primitive/pointwise- $C^1$  layer-cake core, the arbitrary- $L^1$  weight and real-valued absolutely-continuous-primitive wrapper, and the complete and variable-upper proper-compact two-function absolutely-continuous identities are already formal;
2. the power identity [equation \(1256\)](#) and its polynomial corollaries;

3. the positive and negative moment recurrences;
4. positivity, monotonicity, and convergence of  $C_r$ ;
5. the order-statistic gap identity;
6. finite truncations and certified tail bounds for the Newton and sinc-product series.

The entire-function and fractional-integral portions require more analytic infrastructure, but can be decomposed into dominated-convergence, holomorphic-parameter integration, and beta-integral lemmas.

## 17 Summary of principal formulas

For convenient reference, the central formulas are collected here.

### Distribution and transforms

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

$$P(t) = \prod_{k \geq 1} \frac{e^{t/2^k} - 1}{t/2^k}, \quad \phi(t) = \prod_{k \geq 1} \frac{\sin(t/2^k)}{t/2^k},$$

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad \int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}.$$

### Inverse-pair integration

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'B \circ F + \int_0^{F(z)} A \circ Q B',$$

$$p \int_0^1 x^{p-1} F(x)^q dx + q \int_0^1 v^{q-1} Q(v)^p dv = 1,$$

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1.$$

### Entire moment calculus

$$\mathfrak{M}(s) = \int_0^1 Q(v)^s dv,$$

$$\mathfrak{M}(s) = \sum_{k \geq 0} (-1)^k \binom{s}{k} m_k = 2^{-s} \sum_{j \geq 0} \binom{s}{2j} \mu_{2j},$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad \int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}.$$

### Negative moments and fractional primitives

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)),$$

$$\mathfrak{M}(-r) \sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)},$$

$$I_0^\alpha F^q(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv,$$

$$I_0^\alpha Q^p(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx.$$

### New fractional endpoint laws

For  $T = \log(1/y)$ ,  $\eta = yQ'/Q \sim \sqrt{\log 2/(2T)}$ ,

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta + o(\eta)),$$

$$D_{0+}^\sigma(Q^\beta)(y) = \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} (1 + \beta K_\sigma \eta + o(\eta)),$$

$$D_{0+}^\sigma(Q^\beta) \in L^p(0, \tfrac{1}{2}) \iff \sigma p \leq 1.$$

### New nonlocal and resolvent formulas

For  $|x| > 1$ ,

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \sum_{j \geq 0} \frac{(1+2s)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}},$$

$$(\mathcal{I}_a \text{up})(x) = d_{1,a} |x|^{a-1} \sum_{j \geq 0} \frac{(1-a)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}},$$

$$R_{n+1}(z) = \frac{2^n}{n} (R_n(2z-1) - R_n(2z+1)).$$

The fractional self-differential orbit is restored exactly by the log-periodic multiplier  $m_\alpha(s) = \Pi(\log_2 s - \alpha)/\Pi(\log_2 s)$ , which obeys a dyadic scale-cocycle law.

## 18 Conclusion

The probability/quantile viewpoint converts the integral theory of the Fabius–Rvachev system into a coherent calculus rather than a list of isolated evaluations. The inverse-pair identity organizes indefinite weighted integrals; the entire moment function organizes fractional and negative powers; Riemann–Liouville operators organize repeated antiderivatives; and the infinite sinc product organizes correlation and fractional Sobolev energies. The inverse endpoint has a particularly clean continuous-order law: digamma coefficients interpolate the harmonic-number corrections, and the line  $\sigma p = 1$  is the exact strong integrability boundary. Nonlocal differentiation creates algebraic tails whose complete coefficient sequence remains rational before the universal gamma normalization. The negative-moment recurrence is also especially useful: it extracts an additional constant and an analytic periodic phase directly from the dyadic smoothing law, with exponentially fast normalization.

The next decisive steps are to determine whether the phase profile is exactly constant, to connect its Fourier modes with the spectral corrections already present in endpoint asymptotics, and to formalize the remaining improper-singular inverse-pair scope and the moment recurrences. These directions are sufficiently concrete to support both high-precision experiments and Lean proofs.

## A Exact low-order data

The exact positive moments used by the script are

| $n$ | $m_n = \mathbb{E}Y^n$ | $\mu_n = \mathbb{E}Z^n$ when shown |
|-----|-----------------------|------------------------------------|
| 0   | 1                     | 1                                  |
| 1   | 1/2                   | 0                                  |
| 2   | 5/18                  | 1/9                                |
| 3   | 1/6                   | 0                                  |
| 4   | 143/1350              | 19/675                             |
| 5   | 19/270                | 0                                  |
| 6   | 1153/23814            | 583/59535                          |
| 7   | 583/17010             | 0                                  |
| 8   | 1616353/65063250      | 132809/32531625                    |
| 9   | 132809/7229250        | 0                                  |
| 10  | 134926369/9762090030  | 46840699/24405225075               |

The centered entries can be generated more economically from  $\mu_{2j} = 2m_{2j+1}/(2j+1)$  than by expanding  $(2Y-1)^{2j}$ , which avoids severe cancellation at high order.

## B Repository audit categories

The recursive review used the following categories.

| Category                           | Role in the audit                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core expositions                   | Definitions, probability law, smoothness, dyadic derivatives, symmetry, Fourier product, moment arithmetic.                                                                 |
| Glossary and Lean walk-through     | Normalization checks and correspondence with formalized objects.                                                                                                            |
| Canonical research-frontier volume | Primary non-duplication comparison for asymptotics, Lambert- $W$ , Bell/Bernoulli, $q$ -binomial, interpolation, and extrapolation themes.                                  |
| Current drafts                     | Checks against dyadic self-sampling, inverse-function expansions, Rvachev-Appell systems, half-integer spectral effects, Fourier decay, and certification algorithms.       |
| Papers                             | Historical and arithmetic results, exact dyadic values, Fourier coefficients, and endpoint asymptotics.                                                                     |
| Frozen archive                     | Historical variants and absorbed integration material, checked through the archive map and canonical successors to prevent counting old snapshots as independent frontiers. |

## C Recursive TeX audit ledger

The following ledger records the authoritative sixty-seven-path pinned inventory, the eleven-path live-main supplement, the resulting seventy-eight-path current count, and the integration-specific exclusion searches used to avoid duplication.

```
Recursive TeX audit ledger
Repository: VladimirReshetnikov/ProveIt
Root: Analysis/FabiusFunction/docs
Audit date: 2026-08-27

Scope and counting convention
-----
The audit has two layers.

(1) A path-complete pinned baseline at commit
```

32d6d36c51d803289e6d6a0dc0c37753766eba47  
contains exactly 67 distinct \*.tex paths. Its authoritative inventory is  
Appendix A of Exponent\_Sequence\_Newton\_Rvachev\_Frontiers.tex.

(2) The live main-branch tree was then traversed for later additions. Five  
later draft packages add 11 visible \*.tex paths: seven files in the dyadic  
self-sampling package and one file each in the inverse, Newton/exponent,  
dyadic-inverse/Barnes, and q-Richardson/sinc packages. Thus the current  
tree inspected for this report contains 78 visible \*.tex paths:

```

5 core synthesis/reference sources
32 frozen historical sources
34 active frontier-report or auxiliary TeX sources
7 paper transcriptions
--
78 total

```

Generated table fragments were read together with their parent reports and  
underlying data. They were counted as TeX paths but not as independent  
mathematical claims. Frozen revisions were read for distinctive claims and  
provenance, but repeated theorems were counted once when setting the novelty  
boundary.

#### A. Authoritative 67-path pinned baseline

```

1. FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6. archive/early-notes/Fabius_Asymptotic/Fabius_Asymptotic.tex
7. archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the signed Thue-Morse sequence.tex
8. archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9. archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
10. archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
11. archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
12. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
13. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
14. archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
15. archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
16. archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
17. archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
18. archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
19. archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
20. archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
21. archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
22. archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
23. archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/
    Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
24. archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
25. archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
26. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
27. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
28. archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
29. archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
30. archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
31. archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
32. archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
33. archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex
34. archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
35. archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
36. archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
37. archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
38. semi-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex
39. semi-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.
    tex
40. semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex
41. semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex
42. semi-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex
43. semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex
44. semi-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.
    tex
45. semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
46. semi-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex
47. semi-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex
48. semi-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.
    tex
49. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
50. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
51. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/
    Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
52. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/
    Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
53. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/
    Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
54. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/
    asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
55. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
56. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
57. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
58. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/
    Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
59. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
60. semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
61. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63. papers/arXiv-1502.06738v2/Lacunary_products.tex
64. papers/arXiv-1702.05442v1/09-Function.tex

```

```

65. papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66. papers/arXiv-2211.13570v1/document.tex
67. papers/ubiq15/ubiq15-corrected.tex

B. Live-main supplement after the pinned baseline (11)
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68. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
69. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/appell_polynomials.tex
70. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/harmonic_tail_table.tex
71. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/quadrature_table.tex
72. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/quadrature_table_display.tex
73. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/spectral_check.tex
74. semi-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/spectral_check_display.tex
75. semi-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex
76. semi-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex
77. semi-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/fabius_frontier_dyadic_inverse_barnes.tex
78. semi-formalized-research-frontiers/drafts/fabius_frontier_results/q_richardson_sinc_frontier.tex

Integration-specific comparison protocol
-----
The primary exposition, canonical frontier synthesis, all active report sources,
auxiliary TeX fragments, paper transcriptions, and frozen integration/inverse
notebooks were compared at theorem and section level. Targeted literal and
close-variant searches included:

    antiderivative; indefinite integral; repeated primitive; weighted primitive;
    inverse-pair area identity; quantile integral; order statistic; beta transform;
    Mellin transform; Stieltjes transform; Cauchy transform; resolvent hierarchy;
    negative moment; complex moment; Newton series; central moment series;
    Riemann--Liouville; Caputo; fractional integral; fractional derivative;
    fractional Laplacian; Riesz potential; Sobolev energy; Gagliardo seminorm;
    digamma; trigamma; critical  $L_p$ ; Floquet multiplier; dilation cocycle;
    Gini mean difference; autocorrelation; sinc-square integral.

Prior-art boundary
-----
The audit treats as prior input: the probability/refinement definitions; exact
dyadic arithmetic;  $q$ -binomial and Thue--Morse identities; rational moment and
Bernoulli/Bell systems; the infinite sinc product and its spectral/Fourier-decay
analysis; repeated integration and Rvachev--Appell systems; Lambert- $W$  endpoint
and inverse asymptotics; Newton/exponent-sequence and  $q$ -Richardson machinery;
self-sampling quadrature; and the elementary weighted/inverse integration
identities already absorbed into the canonical frontier volume.

The strongest corpus-relative frontier claims of the accompanying report are
therefore restricted to the explicitly proved fractional endpoint corrections,
sharp critical  $L_p$  law, nonlocal exterior series and fractional energy identity,
Cauchy/logarithmic transform fixed point and resolvent hierarchy, and the
Floquet multiplier/cocycle that repairs fractional dyadic scaling. This is a
repository-relative nonduplication statement, not a claim of worldwide
historical priority.

```

## D Reproducibility instructions

From the directory containing this report, run

```
python fractional_frontier_experiments.py
```

to regenerate `new_numerical_results.tex`, `numerical_results.txt`, and `numerical_results.json`. Rebuild the PDF with

```

pdflatex -interaction=nonstopmode -halt-on-error \
    Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error \
    Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error \
    Integration_and_Transform_Frontiers.tex

```

The experiment uses NumPy, SciPy, and mpmath. All exact moments are represented by `Fraction`. `Fraction`; the FFT and quadratures are used only as independent floating-point checks of proved identities.

## References

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## Part XI

# Fractional Quantile Transmutation, Complex-Order Resolvents, and Nonlocal Rvachev Calculus

## Abstract

Let  $F$  be the Fabius distribution function on  $[0, 1]$ , let  $G = F^{-1}$ , and let  $\text{up}$  be Rvachev's compactly supported even density. This report develops an operator-level calculus for integrals and fractional derivatives involving all three functions. Its main result is a quantile–Volterra transmutation: for arbitrary complex integration order  $\beta$  with  $\Re\beta > 0$ , a Riemann–Liouville integral of  $\phi \circ G$  is converted into an ordinary integral with kernel  $(u - F(x))^\beta$ . The formula extends entirely in a power parameter, thereby covering positive, negative, and complex powers of  $G$ , as well as logarithmic powers obtained by parameter differentiation. A parallel transmutation is proved for Caputo derivatives of every noninteger order; the compact differential operator  $\mathcal{L}_F = (F')^{-1} \frac{d}{dx}$  packages all higher inverse-function Bell polynomials.

A two-parameter generalized Stieltjes transform  $\mathcal{S}_\lambda(z) = \int_0^1 (z - x)^{-\lambda} dF(x)$  is then developed. It satisfies a dyadic differential equation, an order-lowering refinement identity, a moment expansion, an exact fractional order-shift calculus in  $z$ , and a boundary-jump formula that recovers a right-sided fractional integral of  $F'$ . For Rvachev's function, Weyl and Riesz fractional derivatives obey an exact order-lowering dilation equation. Unlike ordinary derivatives, noninteger derivatives destroy compact support; outside  $[-1, 1]$  their algebraic tails have convergent expansions whose coefficients are the exact rational even moments of  $\text{up}$ .

Finally, if  $D = X_2 - X_1$  for independent Fabius variables, the normalized positive-part transform  $\mathcal{K}_\beta(a) = \mathbb{E}(D - a)_+^\beta / \Gamma(\beta + 1)$  extends to an entire function of the order  $\beta$  at every interior threshold. Positive orders are fractional stop-loss integrals, order 0 is a tail probability, order  $-1$  is the autocorrelation density, and negative integers are its derivatives. The whole hierarchy satisfies a triangular dyadic refinement equation and a fractional Fourier-energy identity. Numerical experiments reconstructing  $\text{up}$  from its infinite sinc product verify the new identities to between roughly  $10^{-9}$  and  $10^{-13}$ . The report distinguishes proved results, conditional endpoint refinements, and conjectural research directions, and records its novelty only relative to the audited repository snapshot rather than claiming exhaustive global priority.



# 1 Scope, corpus audit, and novelty boundary

## 1.1 Repository snapshot

The requested corpus was the recursive tree `Analysis/FabiusFunction/docs` in Vladimir Reshetnikov’s `ProveIt` repository [11]. The audit was pinned to commit `22f3939f162b8fe9481b4affcd6ea12e7f45e8fd`; pinning matters because the branch moved during the audit. GitHub’s indexed code search returned 84 `.tex` sources under this tree. They fall into four broad classes:

1. the principal human-readable exposition and mathematical glossary;
2. proved-result notes tied closely to the Lean development;
3. non-formalized frontier reports and successive drafts;
4. imported papers, translations, and OCR-corrected literature used as background sources.

All paths were recursively inventoried and searched for the terms relevant to integrals, transforms, inverse functions, Mellin and Fourier analysis, negative powers, and fractional calculus. Earlier versions and imported copies were de-duplicated by path, title, and mathematical content. The most important overlap checks were made against the two newest broad syntheses:

- `semi-formalized-research-frontiers/drafts/Fabius_Integral_and_Transform_Frontiers/fabius_integrals_and_transforms_frontier_report.tex`;
- `semi-formalized-research-frontiers/drafts/fabius_integral_frontiers_bundle/fabius_integral_frontiers.tex`.

These already subsume much of the material in the monomial-antiderivative, dyadic primitive-ladder, derivative-norm, inverse-beta, and Mellin-frontier reports. Consequently, this report does not spend pages reproving the standard polynomial primitive ladder or reproducing tables already present there.

## 1.2 What the repository already contained

The following topics were treated as part of the prior-art boundary for this snapshot:

- ordinary and iterated primitives of  $F$ ,  $1 - F$ , and  $\log$ , including monomial, polynomial, complex-power, and several negative-power weights;
- dyadic reduction of primitives and exact moment recurrences;
- causal fractional integrals of  $\log$ , Mellin transforms, and moment-tail expansions;
- inverse-Fabius integer layer-cake formulas, beta/Jacobi transforms, and inverse moments;
- the ordinary Cauchy–Stieltjes transform and its basic dyadic equation;
- derivative norms, Fourier–Parseval formulas, and basic autocorrelation identities;
- endpoint expansions expressed through lower Lambert  $W$ , including log-periodic corrections;
- the warning that a naive single-dilation fractional analogue of the integer derivative equation is obstructed by the dyadic log-periodic phase.

The present work uses those facts as infrastructure. In particular, it does not claim novelty for the probabilistic representation, the sinc product, the moment recurrences, or the endpoint Lambert structure.

### 1.3 Results that are new relative to the audited snapshot

The novelty comparison found no existing occurrence of the following operator systems in the repository:

1. the arbitrary-order quantile–Volterra transmutation (1471), including its entire continuation in a power parameter and the resulting logarithmic identities;
2. the all-orders Caputo transmutation through  $\mathcal{L}_F = (F')^{-1}D_x$ ;
3. the generalized Stieltjes family  $\mathcal{S}_\lambda(z)$  as a coupled complex-order/dyadic system, including fractional order shifts in the spectral variable and fractional boundary inversion;
4. the exact Weyl order-lowering dilation law and the rational-moment algebraic tails of Weyl and Riesz derivatives of up;
5. the entire order hierarchy  $\mathcal{K}_\beta(a)$  for Fabius differences, with negative orders equal to autocorrelation derivatives and with a triangular refinement equation valid by analytic continuation.

*Warning 1.1.* The word “new” below always means “not found in the audited repository at the pinned commit.” The external literature on fractional transforms is large, and no absolute priority claim is made. The formulas are nevertheless derived from first principles here and are suitable for independent checking or formalization.

## 2 Common probabilistic and analytic model

### 2.1 The three functions

We use the standard probabilistic realization of the Fabius distribution [5, 1]. Let  $U_1, U_2, \dots$  be independent random variables uniformly distributed on  $[0, 1]$ , and define

$$X := \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad Y := 2X - 1. \quad (1447)$$

Then  $X \in [0, 1]$ ,  $Y \in [-1, 1]$ , and

$$F(x) := \mathbb{P}\{X \leq x\}, \quad G(u) := F^{-1}(u), \quad \mathbb{P}\{Y \in dy\} = \text{up}(y) dy. \quad (1448)$$

The normalizations used throughout are

$$F'(x) = 2 \text{up}(2x - 1), \quad 0 < x < 1, \quad (1449)$$

$$F(1 - x) = 1 - F(x), \quad G(1 - u) = 1 - G(u), \quad (1450)$$

$$\text{up}(-y) = \text{up}(y), \quad \int_{-1}^1 \text{up}(y) dy = 1. \quad (1451)$$

In particular,

$$F(x) = \int_{-1}^{2x-1} \text{up}(y) dy, \quad \text{up}(y) = \begin{cases} F(1+y), & -1 \leq y \leq 0, \\ F(1-y), & 0 \leq y \leq 1, \\ 0, & |y| \geq 1. \end{cases} \quad (1452)$$

The random series has the fixed-point forms

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad Y \stackrel{d}{=} \frac{V + Y'}{2}, \quad (1453)$$

where  $U \sim \text{Unif}[0, 1]$ ,  $V \sim \text{Unif}[-1, 1]$ , and the primed variables are independent copies. This random-series and infinite-product view goes back to Jessen–Wintner and its later Fabius/Rvachev formulations [9, 5, 1]. With Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi x} f(x) dx$ ,

$$\widehat{\text{up}}(\xi) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{\xi}{2^j}\right), \quad \text{sinc } z := \frac{\sin z}{z}. \quad (1454)$$

## 2.2 Quantile and density dictionaries

For every integrable test function  $\psi$ ,

$$\int_0^1 \psi(G(u)) du = \int_0^1 \psi(x) dF(x) = \int_{-1}^1 \psi\left(\frac{1+y}{2}\right) \text{up}(y) dy. \quad (1455)$$

This simple identity is the common source of the inverse-function, Stieltjes, and autocorrelation formulas below. It gives, for example,

$$d_n := \mathbb{E}X^n = \int_0^1 G(u)^n du, \quad c_n := \mathbb{E}Y^{2n} = \int_{-1}^1 y^{2n} \text{up}(y) dy. \quad (1456)$$

The fixed-point equations yield exact rational recurrences; related moment recurrences are developed in [1, 8]:

$$d_0 = 1, \quad (2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{d_k}{n - k + 1}, \quad (1457)$$

$$c_0 = 1, \quad c_n = \frac{1}{(2n+1)(4^n - 1)} \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \quad (1458)$$

Thus

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}, \dots \quad (1459)$$

The value  $c_2 = 19/675$  is important in the nonlocal tail expansions; replacing it by  $1/75$  would correspond to a different and incorrect normalization.

## 2.3 Fractional-calculus conventions

We follow the standard Riemann–Liouville, Caputo, Weyl, and Riesz normalizations of [4, 12, 10]. For  $a < u$ ,  $\Re\beta > 0$ , and locally integrable  $h$ , the left Riemann–Liouville integral is

$$(I_{a+}^{\beta} h)(u) := \frac{1}{\Gamma(\beta)} \int_a^u (u-v)^{\beta-1} h(v) dv. \quad (1460)$$

For  $m-1 < \alpha < m$ ,  $m \in \mathbb{N}$ , the Caputo derivative is

$$({}^C D_{a+}^{\alpha} h)(u) := \frac{1}{\Gamma(m-\alpha)} \int_a^u (u-v)^{m-\alpha-1} h^{(m)}(v) dv. \quad (1461)$$

For  $0 < \alpha < 1$ , this reduces to the familiar integral of  $h'$ . Right-sided operators use the standard sign  $(-1)^m$ .

On the whole line, let  $k_{\sigma}(x) = x_{+}^{\sigma-1}/\Gamma(\sigma)$  denote the standard Gamma-normalized causal homogeneous distribution, continued entirely in  $\sigma$ . The left Weyl operator is  $D_{+}^{\alpha} f := k_{-\alpha} * f$ . Wherever the multiplier is an ordinary function, and elsewhere in the corresponding boundary-value distributional sense,

$$\widehat{D_{+}^{\alpha} f}(\xi) = (i\xi)^{\alpha} \widehat{f}(\xi), \quad (1462)$$

with the branch obtained from the causal kernel. This normalization removes zero-frequency polynomial ambiguities and makes the order family entire. The Riesz fractional Laplacian is

$$(-\Delta)^{\alpha/2} f(\xi) = |\xi|^\alpha \widehat{f}(\xi). \quad (1463)$$

Different fractional derivatives answer different boundary-value questions; no formula below silently identifies Caputo, Riemann–Liouville, Weyl, and Riesz operators.

### 3 An entire-order primitive ladder

Before turning to the genuinely new inverse-function formulas, it is useful to package ordinary derivatives and antiderivatives in a normalization that continues through every integer order.

#### 3.1 Fabius distribution ladder

For  $x \in (0, 1)$  and initially  $\Re\beta > -1$ , define

$$\mathfrak{F}_\beta(x) := \frac{1}{\Gamma(\beta+1)} \int_0^x (x-t)^\beta F'(t) dt. \quad (1464)$$

For  $\beta > 0$ , this is  $I_{0+}^\beta F(x)$ ; at  $\beta = 0$  it is  $F(x)$ .

**Proposition 3.1** (Entire order continuation). *For every fixed  $x \in (0, 1)$ ,  $\beta \mapsto \mathfrak{F}_\beta(x)$  extends to an entire function. It obeys*

$$\frac{\partial}{\partial x} \mathfrak{F}_\beta(x) = \mathfrak{F}_{\beta-1}(x) \quad (1465)$$

for all  $\beta \in \mathbb{C}$ , and at negative integers

$$\mathfrak{F}_{-m}(x) = F^{(m)}(x), \quad m \geq 1. \quad (1466)$$

*Proof.* Write  $s = x - t$ . Because  $F'$  and all of its derivatives vanish at 0, repeated integration by parts gives, for every  $N \geq 0$  and  $\Re\beta > -N - 1$ ,

$$\mathfrak{F}_\beta(x) = \frac{1}{\Gamma(\beta+N+1)} \int_0^x s^{\beta+N} F^{(N+1)}(x-s) ds. \quad (1467)$$

The formulas agree on overlaps and cover the whole  $\beta$ -plane. The reciprocal Gamma factor cancels the Mellin poles at the nonpositive integers. Differentiation proves (1465); substituting  $\beta = -m$  in (1467) with  $N = m$  gives  $F^{(m)}(x) - F^{(m)}(0) = F^{(m)}(x)$ .  $\square$

#### 3.2 Rvachev ladder and exterior moment series

For  $x > -1$  and initially  $\Re\sigma > 0$ , put

$$\mathfrak{U}_\sigma(x) := \frac{1}{\Gamma(\sigma)} \int_{-1}^{\min(x,1)} (x-t)^{\sigma-1} \text{up}(t) dt. \quad (1468)$$

For  $-1 < x < 1$ , flatness at  $-1$  again gives an entire continuation in  $\sigma$ , with

$$\partial_x \mathfrak{U}_\sigma = \mathfrak{U}_{\sigma-1}, \quad \mathfrak{U}_0 = \text{up}, \quad \mathfrak{U}_{-m} = \text{up}^{(m)}. \quad (1469)$$

For  $x > 1$ , the binomial series converges absolutely and yields

$$\mathfrak{U}_\sigma(x) = \frac{x^{\sigma-1}}{\Gamma(\sigma)} \sum_{j=0}^{\infty} \frac{(1-\sigma)_{2j}}{(2j)!} \frac{c_j}{x^{2j}}. \quad (1470)$$

The Weyl tail formula of [Appendix 7](#) is precisely the specialization  $\sigma = -\alpha$  of this exterior ladder. Thus ordinary antiderivatives, ordinary derivatives, and nonlocal fractional tails are not separate phenomena; they are different slices of one Gamma-normalized order family.

## 4 Quantile–Volterra transmutation

### 4.1 The master theorem

The basic obstacle in integrating  $G = F^{-1}$  is the derivative  $G'(u) = 1/F'(G(u))$ , which becomes very large at both endpoints. A layer-cake rearrangement removes this derivative completely.

**Theorem 4.1** (Quantile–Volterra transmutation). *Let  $0 \leq a < u < 1$ , put  $q_a = G(a)$  and  $q = G(u)$ , let  $\Re\beta > 0$ , and let  $\phi$  be absolutely continuous on  $[q_a, q]$ . Then*

$$I_{a+}^{\beta}[\phi(G) - \phi(q_a)](u) = \frac{1}{\Gamma(\beta + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta} dx. \quad (1471)$$

For  $a = 0$  and  $\phi(0) = 0$ ,

$$I_{0+}^{\beta}(\phi \circ G)(u) = \frac{1}{\Gamma(\beta + 1)} \int_0^{G(u)} \phi'(x)(u - F(x))^{\beta} dx. \quad (1472)$$

*Proof.* For  $v \in [a, u]$ ,

$$\phi(G(v)) - \phi(q_a) = \int_{q_a}^{G(v)} \phi'(x) dx \quad (1473)$$

$$= \int_{q_a}^q \phi'(x) \mathbf{1}_{\{F(x) \leq v\}} dx. \quad (1474)$$

Fubini's theorem is immediate for nonnegative  $\phi'$  and follows for an arbitrary absolutely continuous  $\phi$  by applying the same argument to its positive and negative variations. Hence

$$I_{a+}^{\beta}[\phi(G) - \phi(q_a)](u) = \frac{1}{\Gamma(\beta)} \int_{q_a}^q \phi'(x) \int_{F(x)}^u (u - v)^{\beta-1} dv dx \quad (1475)$$

$$= \frac{1}{\Gamma(\beta + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta} dx. \quad (1476)$$

For complex  $\beta$ , the kernel is positive real before exponentiation, so there is no branch ambiguity.  $\square$

---

The theorem trades an integral over probability level  $v$  containing the inverse  $G(v)$  for an integral over physical position  $x$  containing only the forward Fabius function. The singular inverse derivative never appears. For  $\beta = n \in \mathbb{N}$ , it gives every  $n$ -fold antiderivative at once; for noninteger  $\beta$ , the same formula is a genuine fractional primitive.

---

### 4.2 Powers, negative powers, and an entire parameter

Taking  $\phi(x) = x^p$  gives, initially for  $\Re p > 0$ ,

$$I_{0+}^{\beta} G^p(u) = \frac{p}{\Gamma(\beta + 1)} \int_0^q x^{p-1} (u - F(x))^{\beta} dx, \quad q = G(u). \quad (1477)$$

The endpoint flatness of  $F$  permits much more.

**Theorem 4.2** (Entire continuation in the power). *Fix  $u \in (0, 1)$  and  $\Re\beta > 0$ . The function  $p \mapsto I_{0+}^{\beta} G^p(u)$ , initially represented by (1477), extends to an entire function of  $p \in \mathbb{C}$ :*

$$I_{0+}^{\beta} G^p(u) = \frac{1}{\Gamma(\beta + 1)} \left[ u^{\beta} q^p + p \int_0^q x^{p-1} \left( (u - F(x))^{\beta} - u^{\beta} \right) dx \right]. \quad (1478)$$

Here  $q^p = \exp(p \log q)$  with the real logarithm.

*Proof.* Add and subtract  $u^\beta$  inside (1477); for  $\Re p > 0$ ,  $p \int_0^q x^{p-1} u^\beta dx = u^\beta q^p$ . Near  $x = 0$ ,

$$(u - F(x))^\beta - u^\beta = -\beta u^{\beta-1} F(x) + \mathcal{O}(F(x)^2). \quad (1479)$$

Since  $F(x)$  is flat at zero, the regularized Mellin integral converges locally uniformly for every  $p \in \mathbb{C}$ . It is therefore entire, as is  $q^p$ .  $\square$

This formula gives convergent representations for every negative and complex power. It also explains why negative inverse moments are much better behaved here than for an algebraic endpoint distribution: the superpolynomial flatness of  $F$  defeats every pole  $x^{p-1}$ .

### 4.3 Logarithmic powers and exponential generating functions

Differentiate (1478) at  $p = 0$ .

**Corollary 4.3** (Fractional logarithmic moments). *For every integer  $m \geq 1$ ,*

$$I_{0+}^\beta [(\log G)^m](u) = \frac{1}{\Gamma(\beta + 1)} \left[ u^\beta (\log q)^m \right. \quad (1480)$$

$$\left. + m \int_0^q \frac{(\log x)^{m-1}}{x} \left( (u - F(x))^\beta - u^\beta \right) dx \right]. \quad (1481)$$

Likewise, with  $\phi(x) = e^{sx}$ ,

$$I_{0+}^\beta (e^{sG} - 1)(u) = \frac{s}{\Gamma(\beta + 1)} \int_0^q e^{sx} (u - F(x))^\beta dx. \quad (1482)$$

Expanding either side in  $s$  simultaneously generates all positive integer powers. Differentiating with respect to  $s$  at nonzero values generates exponentially tilted inverse moments.

### 4.4 Definite integrals and fractional order statistics

At  $u = 1$ , (1472) becomes

$$I_{0+}^\beta (\phi \circ G)(1) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 \phi'(x) (1 - F(x))^\beta dx. \quad (1483)$$

For  $\phi(x) = x$  and symmetry,

$$I_{0+}^\beta G(1) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 F(x)^\beta dx. \quad (1484)$$

When  $\beta = n \in \mathbb{N}$ , if  $M_n = \max(X_1, \dots, X_n)$ , then  $\mathbb{P}(M_n \leq x) = F(x)^n$  and

$$n! I_{0+}^n G(1) = \int_0^1 F(x)^n dx = \mathbb{E}(1 - M_n). \quad (1485)$$

Thus the integration order continuously interpolates the expected gap above a maximum of an integer number of samples. Formula (1484) provides a natural “fractional sample-size” extension without inventing a noninteger collection of random variables. For  $n = 1$ , it gives  $\int_0^1 G(u) du = 1/2$ .

More generally,

$$I_{0+}^\beta G^p(1) = \frac{p}{\Gamma(\beta + 1)} \int_0^1 x^{p-1} (1 - F(x))^\beta dx, \quad (1486)$$

with entire continuation in  $p$  supplied by (1478). This is a two-parameter bridge between inverse moments and beta-like powers of the forward distribution.

#### 4.5 Log-convexity and Turan inequalities in the order

Suppose  $\phi' \geq 0$ . For real  $\beta > -1$ , define the direct-kernel order transform

$$T_\beta(u) := \int_{q_a}^q \phi'(x)(u - F(x))^\beta dx. \quad (1487)$$

For  $\beta > 0$  this equals  $\Gamma(\beta + 1)I_{a+}^\beta[\phi(G) - \phi(q_a)](u)$  by [Theorem 4.1](#); the direct integral also makes sense on the larger interval  $\beta > -1$ .

**Proposition 4.4** (Order log-convexity). *For real  $\beta > -1$  in the interval of convergence,  $T_\beta(u)$  is log-convex. In particular, whenever all three terms exist,*

$$T_\beta(u)^2 \leq T_{\beta-h}(u)T_{\beta+h}(u). \quad (1488)$$

If  $M = \phi(q) - \phi(q_a) > 0$ , then

$$\left(\frac{T_\beta(u)}{M}\right)^{1/\beta} \text{ is nondecreasing for } \beta > 0. \quad (1489)$$

*Proof.* Normalize  $\phi'(x) dx$  to a probability measure and apply Holder's inequality to the positive random variable  $u - F(x)$ . The second assertion is the monotonicity of power means.  $\square$

The Turan inequality is potentially useful for obtaining rigorous bounds on fractional primitives from adjacent integer orders already known exactly or recursively.

#### 4.6 Semigroup identity after transmutation

The Riemann–Liouville semigroup  $I^\gamma I^\beta = I^{\beta+\gamma}$  becomes the nontrivial forward-function identity

$$\begin{aligned} I_{a+}^\gamma \left[ \frac{1}{\Gamma(\beta + 1)} \int_{q_a}^{G(v)} \phi'(x)(v - F(x))^\beta dx \right] (u) \\ = \frac{1}{\Gamma(\beta + \gamma + 1)} \int_{q_a}^q \phi'(x)(u - F(x))^{\beta+\gamma} dx. \end{aligned} \quad (1490)$$

This is immediate from the theorem but useful computationally: two nested fractional integrations in the quantile variable collapse to one forward Fabius kernel. The generic theorem `Fabius.fractionalVolterra_add` does not by itself formalize (1490): this displayed identity also depends on the report-local quantile transmutation theorem, and no named Lean wrapper is exported.

### 5 Caputo derivatives of the inverse Fabius function

#### 5.1 Orders between zero and one

**Theorem 5.1** (Caputo quantile transmutation). *Let  $0 < \alpha < 1$ ,  $0 \leq a < u < 1$ , and let  $\phi$  be continuously differentiable on  $[G(a), G(u)]$ . Then*

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(1 - \alpha)} \int_{G(a)}^{G(u)} \phi'(x)(u - F(x))^{-\alpha} dx. \quad (1491)$$

Whenever the endpoint integral converges,  $a = 0$  is allowed.

*Proof.* By definition,

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(1-\alpha)} \int_a^u (u-v)^{-\alpha} \phi'(G(v)) G'(v) dv. \quad (1492)$$

Set  $v = F(x)$ . Since  $G'(F(x))F'(x) = 1$ , the inverse derivative cancels exactly, producing (1491).  $\square$

For powers,

$${}^C D_{0+}^\alpha G^p(u) = \frac{p}{\Gamma(1-\alpha)} \int_0^q x^{p-1} (u - F(x))^{-\alpha} dx, \quad \Re p > 0. \quad (1493)$$

For  $p = 1$ ,

$${}^C D_{0+}^\alpha G(u) = \frac{1}{\Gamma(1-\alpha)} \int_0^{G(u)} (u - F(x))^{-\alpha} dx. \quad (1494)$$

The singularity at  $x = G(u)$  is only of order  $(G(u) - x)^{-\alpha}$  and is therefore integrable.

With the standard right-Caputo sign convention and the symmetry of  $G$ ,

$${}^C D_{1-}^\alpha G(u) = -{}^C D_{0+}^\alpha G(1-u). \quad (1495)$$

## 5.2 Every noninteger Caputo order

The cancellation in Theorem 5.1 is the first member of a larger hierarchy. Define the first-order differential operator

$$\mathcal{L}_F := \frac{1}{F'(x)} \frac{d}{dx}. \quad (1496)$$

It is differentiation with respect to probability level, pulled back to the  $x$  coordinate.

**Lemma 5.2** (Inverse Bell polynomials in operator form). *For every integer  $m \geq 0$  and every smooth  $\phi$ ,*

$$\left. \frac{d^m}{du^m} \phi(G(u)) \right|_{u=F(x)} = \mathcal{L}_F^m \phi(x). \quad (1497)$$

*Proof.* The case  $m = 1$  is the inverse-function rule. If the formula holds at  $m$ , then differentiation in  $u$  at  $u = F(x)$  is exactly the operation  $(F'(x))^{-1} D_x = \mathcal{L}_F$ .  $\square$

The first three cases are

$$\mathcal{L}_F \phi = \frac{\phi'}{F'}, \quad (1498)$$

$$\mathcal{L}_F^2 \phi = \frac{\phi''}{(F')^2} - \frac{\phi' F''}{(F')^3}, \quad (1499)$$

$$\mathcal{L}_F^3 \phi = \frac{\phi'''}{(F')^3} - \frac{3\phi'' F'' + \phi' F'''}{(F')^4} + \frac{3\phi' (F'')^2}{(F')^5}. \quad (1500)$$

These are precisely the inverse-function Bell-polynomial expressions, but the operator notation scales to arbitrary order without combinatorial clutter.

**Theorem 5.3** (All-orders Caputo transmutation). *Let  $m-1 < \alpha < m$ ,  $m \in \mathbb{N}$ , and let  $0 < a < u < 1$ . If  $\phi$  is smooth on  $[G(a), G(u)]$ , then*

$${}^C D_{a+}^\alpha (\phi \circ G)(u) = \frac{1}{\Gamma(m-\alpha)} \int_{G(a)}^{G(u)} (u - F(x))^{m-\alpha-1} (\mathcal{L}_F^m \phi)(x) F'(x) dx. \quad (1501)$$

*The endpoint  $a = 0$  is valid whenever the displayed integral converges.*



*Proof.* Insert (1497) into the Caputo definition (1461) and substitute  $v = F(x)$ .  $\square$

For integer  $m$ , (1497) itself gives the complete formula for the  $m$ th derivative of any composite  $\phi(G(u))$ ; for an  $n$ -fold antiderivative, Theorem 4.1 with  $\beta = n$  gives the corresponding complete formula. Thus the two transmutations cover all integer and noninteger orders on both sides of order zero.

### 5.3 Abel inversion identity

If  $\phi(0) = 0$  and the relevant operators exist, the standard identity  $I_{0+}^\alpha {}^C D_{0+}^\alpha (\phi \circ G) = \phi \circ G$  becomes

$$I_{0+}^\alpha \left[ \frac{1}{\Gamma(1-\alpha)} \int_0^{G(v)} \phi'(x) (v - F(x))^{-\alpha} dx \right] (u) = \phi(G(u)). \quad (1502)$$

This is an Abel-type inversion formula whose data are expressed entirely through the forward function  $F$ . It suggests stable numerical schemes for  $G$  that avoid differentiating a numerically inverted CDF.

### 5.4 Endpoint asymptotics

Set

$$q = G(u), \quad \rho(u) := \frac{qF'(q)}{F(q)} = \frac{G(u)}{uG'(u)}. \quad (1503)$$

The sharp endpoint analysis in the audited repository [11], combined with standard regular-variation language [3], implies

$$\log \frac{1}{G(u)} \sim \sqrt{2 \log 2 \log \frac{1}{u}}, \quad \rho(u) \sim \sqrt{\frac{2 \log(1/u)}{\log 2}}. \quad (1504)$$

In particular,  $G$  is slowly varying at zero: for every fixed  $t > 0$ ,  $G(tu)/G(u) \rightarrow 1$  as  $u \downarrow 0$ .

**Proposition 5.4** (Leading fractional primitives). *For fixed  $\beta > 0$  and  $p > 0$ ,*

$$I_{0+}^\beta G^p(u) \sim \frac{u^\beta G(u)^p}{\Gamma(\beta+1)}, \quad u \downarrow 0. \quad (1505)$$

*Proof.* After  $v = ut$ ,

$$I_{0+}^\beta G^p(u) = \frac{u^\beta G(u)^p}{\Gamma(\beta)} \int_0^1 (1-t)^{\beta-1} \frac{G(ut)^p}{G(u)^p} dt. \quad (1506)$$

The uniform convergence theorem and Potter bounds for slowly varying functions give dominated convergence. The remaining integral is  $1/\beta$ .  $\square$

**Proposition 5.5** (Caputo endpoint laws). *For fixed  $0 < \alpha < 1$  and  $p > 0$ ,*

$${}^C D_{0+}^\alpha G^p(u) \sim \frac{u^{-\alpha} G(u)^p}{\Gamma(1-\alpha)}, \quad u \downarrow 0. \quad (1507)$$

For  $p = 1$ , writing  $s = 1 - u$ ,

$${}^C D_{0+}^\alpha G(1-s) \sim \Gamma(\alpha) \frac{s^{-\alpha} G(s)}{\rho(s)}, \quad s \downarrow 0. \quad (1508)$$

*Proof architecture.* The first assertion is the order-zero boundary case of the monotone-density theorem for fractional derivatives of slowly varying functions. It can also be derived from the cancellation-safe Marchaud form

$${}^C D_{0+}^\alpha f(u) = \frac{u^{-\alpha}}{\Gamma(1-\alpha)} \left[ f(u) + \alpha \int_0^1 \frac{f(u) - f(ut)}{(1-t)^{\alpha+1}} dt \right]. \quad (1509)$$

For the right endpoint, symmetry gives

$${}^C D_{0+}^\alpha G(1-s) = \frac{1}{\Gamma(1-\alpha)} \int_s^1 (r-s)^{-\alpha} G'(r) dr. \quad (1510)$$

Now  $G'(r) = r^{-1}G(r)/\rho(r)$  is regularly varying with index  $-1$ . Scaling  $r = sy$  gives  $\int_1^\infty (y-1)^{-\alpha} y^{-1} dy = \Gamma(1-\alpha)\Gamma(\alpha)$ .  $\square$

**Theorem 5.6** (Fractional integrability frontier). *For  $0 < \alpha < 1$  and  $0 < r < \infty$ ,*

$${}^C D_{0+}^\alpha G \in L^r(0,1) \iff r \leq \frac{1}{\alpha}. \quad (1511)$$

*The same threshold holds for  $G^p$  with any fixed  $p > 0$ .*

*Proof.* At either endpoint, Proposition 5.5 is a power  $u^{-\alpha}$  times a slowly varying factor. For  $\alpha r < 1$  the power is integrable; for  $\alpha r > 1$  no slowly varying factor can repair the power divergence. At the critical exponent  $r = 1/\alpha$ , use (1504): after  $L = \log(1/u)$ , the integral is bounded by an integral of  $\exp(-c\sqrt{L})$  times a power of  $L$ , which converges.  $\square$

The critical endpoint is unusually included. This is a direct consequence of the stretched-exponential-in- $\sqrt{\log}$  decay of  $G$ , and would fail for many more slowly decaying quantiles.

## 5.5 First correction and the log-periodic phase

The leading term uses only slow variation. A first correction requires a quantitative de Haan expansion. The repository's sharp endpoint theory contains a nonconstant dyadic periodic phase, so it is important not to hide that phase inside an unjustified constant.

*Conditional theorem 5.7* (First fractional correction). Assume that, in each of the weighted  $L^1$  spaces used below,

$$\frac{G(ut)}{G(u)} = 1 + \frac{\log t}{\rho(u)} + \mathcal{O}\left(\frac{(1+|\log t|)^2}{\rho(u)^2}\right) \quad (1512)$$

with an error uniform as  $u \downarrow 0$ . Then

$$\frac{\Gamma(\beta+1)I_{0+}^\beta G^p(u)}{u^\beta G(u)^p} = 1 - \frac{p[\psi(\beta+1) + \gamma]}{\rho(u)} + \mathcal{O}(\rho(u)^{-2}), \quad (1513)$$

where  $\psi$  is the digamma function and  $\gamma$  is Euler's constant. Moreover,

$$\frac{\Gamma(1-\alpha)u^{\alpha C} D_{0+}^\alpha G^p(u)}{G(u)^p} = 1 + \frac{p[-\psi(1-\alpha) - \gamma]}{\rho(u)} + \mathcal{O}(\rho(u)^{-2}). \quad (1514)$$

*Proof architecture.* For (1513), insert (1512) after  $v = ut$ . The coefficient is the beta expectation

$$\beta \int_0^1 (1-t)^{\beta-1} \log t dt = -\psi(\beta+1) - \gamma. \quad (1515)$$

For Caputo, use the Marchaud form. The coefficient is

$$\alpha \int_0^1 \frac{-\log t}{(1-t)^{\alpha+1}} dt = -\psi(1-\alpha) - \gamma. \quad (1516)$$

$\square$

**Conjecture 5.8** (Phase-refined de Haan expansion). *The Fabius inverse satisfies (1512) with a complete asymptotic expansion in  $\rho(u)^{-1}$  whose coefficients are polynomials in  $\log t$  and smooth periodic functions of the exact lower-Lambert dyadic phase. The periodic dependence first enters beyond the universal coefficients displayed in (1513) and (1514).*

This conjecture is deliberately compatible with the repository's proof that the endpoint phase is nonconstant; it does not replace the phase by a scalar.

## 6 A complex-order generalized Stieltjes system

### 6.1 Definition and elementary structure

The generalized Stieltjes class and its positivity theory provide the natural ambient transform language [13]. Work first on the principal slit plane

$$\Omega := \mathbb{C} \setminus (-\infty, 1]. \quad (1517)$$

For  $z \in \Omega$ , use the principal branch of  $\text{Log}(z - x)$  simultaneously for all  $x \in [0, 1]$  and define

$$\mathcal{S}_\lambda(z) := \int_0^1 (z - x)^{-\lambda} dF(x) = \int_0^1 (z - G(u))^{-\lambda} du. \quad (1518)$$

The upper and lower banks of the slit supply the boundary values used below; other branches live naturally on the universal cover of  $\mathbb{C} \setminus [0, 1]$ . For each fixed  $z \in \Omega$ ,  $\lambda \mapsto \mathcal{S}_\lambda(z)$  is entire, because the compact family  $\{\text{Log}(z - x) : 0 \leq x \leq 1\}$  is bounded. Parameter derivatives generate logarithmic kernels:

$$(-\partial_\lambda)^m \mathcal{S}_\lambda(z) = \int_0^1 (z - x)^{-\lambda} \text{Log}^m(z - x) dF(x). \quad (1519)$$

For  $z \in \Omega$  with  $|z| > 1$  the binomial expansion gives

$$\mathcal{S}_\lambda(z) = z^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda)_n}{n!} \frac{d_n}{z^n}. \quad (1520)$$

Every coefficient is rational and computable from (1457). At negative integer orders,  $\mathcal{S}_{-m}(z) = \mathbb{E}(z - X)^m$  is a polynomial; positive integer orders are the successive derivatives of the ordinary Stieltjes transform.

### 6.2 Dyadic order recurrence

**Theorem 6.1** (Dyadic generalized-Stieltjes equations). *For all  $z \in \Omega$  and all  $\lambda \in \mathbb{C}$ , by analytic continuation across the removable value  $\lambda = 1$ ,*

$$\mathcal{S}_\lambda(z) = \frac{2^\lambda}{1 - \lambda} [\mathcal{S}_{\lambda-1}(2z) - \mathcal{S}_{\lambda-1}(2z - 1)]. \quad (1521)$$

*It also satisfies the same-order differential-dyadic equation*

$$\partial_z \mathcal{S}_\lambda(z) = 2^{\lambda+1} [\mathcal{S}_\lambda(2z) - \mathcal{S}_\lambda(2z - 1)], \quad (1522)$$

*and the order-raising derivative identity*

$$\partial_z \mathcal{S}_\lambda(z) = -\lambda \mathcal{S}_{\lambda+1}(z). \quad (1523)$$

*Proof.* Using  $X \stackrel{d}{=} (U + X')/2$ ,

$$\mathcal{S}_\lambda(z) = 2^\lambda \mathbb{E} \int_0^1 (2z - X' - u)^{-\lambda} du \quad (1524)$$

$$= \frac{2^\lambda}{1-\lambda} \mathbb{E} \left[ (2z - X')^{1-\lambda} - (2z - 1 - X')^{1-\lambda} \right]. \quad (1525)$$

This is (1521). Differentiating it and using (1523) gives (1522); alternatively, differentiate under the defining integral and use the density refinement equation.  $\square$

At  $\lambda = 1$ , the removable limit is the logarithmic identity

$$\mathcal{S}_1(z) = 2\mathbb{E} [\text{Log}(2z - X) - \text{Log}(2z - 1 - X)]. \quad (1526)$$

For the centered variable  $Y$ , define

$$\mathcal{C}_\lambda(w) := \mathbb{E}(w - Y)^{-\lambda}. \quad (1527)$$

Then

$$\mathcal{S}_\lambda(z) = 2^\lambda \mathcal{C}_\lambda(2z - 1), \quad \mathcal{C}'_\lambda(w) = 2^\lambda [\mathcal{C}_\lambda(2w + 1) - \mathcal{C}_\lambda(2w - 1)]. \quad (1528)$$

The latter is a transform-level analogue of Rvachev's differential equation.

### 6.3 Laplace representation and exact fractional order shifts

For  $\Re z > 1$  and  $\Re \lambda > 0$ , the standard Laplace–Gamma representation [14] gives

$$\mathcal{S}_\lambda(z) = \frac{1}{\Gamma(\lambda)} \int_0^\infty t^{\lambda-1} e^{-zt} M_X(t) dt, \quad (1529)$$

$$M_X(t) = \mathbb{E} e^{tX} = e^{t/2} \widehat{\text{up}}(it/2). \quad (1530)$$

The sinc product becomes an infinite product of hyperbolic sinc factors on the imaginary axis.

Let  $A = -\partial_z$ , and define  $A^\nu$  on this Laplace class by multiplying the integrand by  $t^\nu$ . This is the natural Weyl fractional power of the positive generator  $-\partial_z$  on the right half-plane.

**Theorem 6.2** (Fractional spectral order shift). *Whenever  $\Re z > 1$ ,  $\Re \lambda > 0$ , and  $\Re(\lambda + \nu) > 0$ ,*

$$A^\nu \mathcal{S}_\lambda(z) = \frac{\Gamma(\lambda + \nu)}{\Gamma(\lambda)} \mathcal{S}_{\lambda+\nu}(z). \quad (1531)$$

*In particular, for  $n \in \mathbb{N}$ ,*

$$\frac{d^n}{dz^n} \mathcal{S}_\lambda(z) = (-1)^n (\lambda)_n \mathcal{S}_{\lambda+n}(z), \quad (1532)$$

*and the  $n$ -fold ordinary primitive is*

$$\int \cdots \int \mathcal{S}_\lambda(z) dz^n = (-1)^n \frac{\Gamma(\lambda - n)}{\Gamma(\lambda)} \mathcal{S}_{\lambda-n}(z) + P_{n-1}(z), \quad (1533)$$

*away from the exceptional integer parameters, with  $P_{n-1}$  an arbitrary polynomial of degree at most  $n - 1$ .*

*Proof.* Apply the multiplier  $t^\nu$  to (1529) and compare with the same representation at order  $\lambda + \nu$ .  $\square$

For  $\nu > 0$ , the inverse power  $A^{-\nu}$  is a fractional antiderivative with boundary condition at  $+\infty$ :

$$(A^{-\nu} f)(z) = \frac{1}{\Gamma(\nu)} \int_z^\infty (w - z)^{\nu-1} f(w) dw. \quad (1534)$$

Thus (1531) gives an exact continuum of spectral derivatives and antiderivatives, not merely an interpolation of integer formulas.

## 6.4 Boundary jump as a fractional integral of the density

**Theorem 6.3** (Fractional Plemelj–Abel jump). *Let  $0 < \lambda < 1$  and  $0 < x < 1$ . The nontangential boundary values satisfy*

$$\mathcal{S}_\lambda(x + i0) - \mathcal{S}_\lambda(x - i0) = -2i \sin(\pi\lambda) \int_x^1 (t - x)^{-\lambda} F'(t) dt. \quad (1535)$$

*Equivalently,*

$$I_{1-}^{1-\lambda} F'(x) = -\frac{\Gamma(\lambda)}{2\pi i} [\mathcal{S}_\lambda(x + i0) - \mathcal{S}_\lambda(x - i0)]. \quad (1536)$$

*Proof.* For  $t < x$ ,  $x - t$  lies on the positive real axis and has no jump. For  $t > x$ ,

$$(x - t + i0)^{-\lambda} - (x - t - i0)^{-\lambda} = -2i \sin(\pi\lambda) (t - x)^{-\lambda}. \quad (1537)$$

Integrate against  $F'(t) dt$  and use  $\Gamma(\lambda)\Gamma(1 - \lambda) = \pi/\sin(\pi\lambda)$ .  $\square$

At  $\lambda = 1$ , the usual Stieltjes inversion recovers the density itself. For  $0 < \lambda < 1$ , the jump recovers a right-sided fractional integral of the density; applying a right Riemann–Liouville derivative then reconstructs  $F'$ . This gives a continuum of inversion formulas connecting complex boundary data to fractional real-variable smoothing.

## 6.5 Endpoint values and inverse moments

Because the Fabius density is flat at both endpoints, all inverse moments exist. The endpoint value

$$\mathcal{S}_\lambda(1) = \mathbb{E}(1 - X)^{-\lambda} = \mathbb{E}X^{-\lambda} = \int_0^1 G(u)^{-\lambda} du \quad (1538)$$

therefore extends entire in  $\lambda$ . Formula (1521), interpreted at endpoint limits, offers a dyadic recursion for these inverse moments. Combining it with (1520) is a promising route to new  $q$ -series and  $q$ -binomial representations at dyadic spectral points.

# 7 Weyl and Riesz fractional calculus of Rvachev's function

## 7.1 Exact order-lowering dilation law

The ordinary Rvachev equation [1, 6] is

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (1539)$$

A naive attempt to replace the derivative by a fractional derivative while keeping up on the right is false. The correct equation lowers the order on the right-hand side.

**Theorem 7.1** (Weyl order-lowering equation). *For every  $\alpha \in \mathbb{C}$ , as tempered distributions and with a consistent branch of  $(i\xi)^\alpha$ ,*

$$D_+^\alpha \text{up}(x) = 2^\alpha [(D_+^{\alpha-1} \text{up})(2x + 1) - (D_+^{\alpha-1} \text{up})(2x - 1)]. \quad (1540)$$

*At  $\alpha = 1$  this is (1539); at integer  $n$  it gives all higher derivative equations.*

*Proof.* The Fourier transform of the right-hand side is

$$2^\alpha i \sin(\xi/2) (i\xi/2)^{\alpha-1} \widehat{\text{up}}(\xi/2) = 2i \sin(\xi/2) (i\xi)^{\alpha-1} \widehat{\text{up}}(\xi/2) \quad (1541)$$

$$= (i\xi)^\alpha \widehat{\text{up}}(\xi), \quad (1542)$$

where the last equality uses  $\widehat{\text{up}}(\xi) = \text{sinc}(\xi/2) \widehat{\text{up}}(\xi/2)$ .  $\square$

The theorem is the structurally correct fractional generalization: the self-similarity survives, but the dilation intertwines adjacent derivative orders. No contradiction with the repository's log-periodic obstruction arises, because the right side is not a scalar multiple of the original up.

## 7.2 One-sided Weyl tails

For  $0 < \alpha < 1$ , the left Marchaud–Weyl derivative is

$$D_+^\alpha f(x) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty \frac{f(x) - f(x-t)}{t^{1+\alpha}} dt. \quad (1543)$$

If  $x > 1$ , then  $\text{up}(x) = 0$  and only the translated support contributes.

**Theorem 7.2** (Convergent Weyl tail series). *For  $0 < \alpha < 1$  and  $x > 1$ ,*

$$D_+^\alpha \text{up}(x) = \frac{x^{-1-\alpha}}{\Gamma(-\alpha)} \sum_{j=0}^{\infty} \frac{(\alpha+1)_{2j}}{(2j)!} \frac{c_j}{x^{2j}}. \quad (1544)$$

*The series converges absolutely for every  $x > 1$ . In particular,*

$$D_+^\alpha \text{up}(x) = \frac{x^{-1-\alpha}}{\Gamma(-\alpha)} \left[ 1 + \frac{(\alpha+1)(\alpha+2)}{18x^2} + \frac{19(\alpha+1)_4}{16200x^4} + \dots \right]. \quad (1545)$$

*Proof.* For  $x > 1$ , (1543) becomes

$$D_+^\alpha \text{up}(x) = \frac{1}{\Gamma(-\alpha)} \int_{-1}^1 (x-y)^{-1-\alpha} \text{up}(y) dy. \quad (1546)$$

Expand  $(1-y/x)^{-1-\alpha}$ . Odd moments vanish and the even moments are  $c_j$ . Absolute convergence follows from  $|y/x| < 1$ .  $\square$

Since  $\Gamma(-\alpha) < 0$ , the left Weyl tail is negative to the right of the support. The right Weyl derivative has the reflected tail to the left. Ordinary integer derivatives retain compact support, but every genuinely noninteger Weyl derivative develops an algebraic tail. This is a sharp, quantified form of nonlocality.

## 7.3 Riesz fractional-Laplacian tails

For  $0 < \alpha < 2$ , the one-dimensional fractional Laplacian can be normalized as

$$(-\Delta)^{\alpha/2} f(x) = C_\alpha \text{pv} \int_{\mathbb{R}} \frac{f(x) - f(y)}{|x-y|^{1+\alpha}} dy, \quad C_\alpha = \frac{\Gamma(1+\alpha) \sin(\pi\alpha/2)}{\pi}. \quad (1547)$$

**Theorem 7.3** (Riesz tail series). *For  $0 < \alpha < 2$  and  $|x| > 1$ ,*

$$(-\Delta)^{\alpha/2} \text{up}(x) = -C_\alpha |x|^{-1-\alpha} \sum_{j=0}^{\infty} \frac{(\alpha+1)_{2j}}{(2j)!} \frac{c_j}{|x|^{2j}}. \quad (1548)$$

*Hence*

$$(-\Delta)^{\alpha/2} \text{up}(x) \sim -C_\alpha |x|^{-1-\alpha}. \quad (1549)$$

*Proof.* Outside the support,  $\text{up}(x) = 0$ , so the principal value is unnecessary and (1547) is minus a positive potential integral. Expand the kernel exactly as in Theorem 7.2.  $\square$

The coefficients in both nonlocal tails are rational and obey the explicit recurrence (1458). Thus a global fractional operation converts the compactly supported atomic function into an algebraically decaying function whose complete exterior expansion is arithmetically controlled by the same moment sequence that appears in the Fabius theory.

## 8 An entire-order hierarchy from Fabius differences

### 8.1 Autocorrelation and the difference density

Let  $X_1, X_2$  be independent copies of  $X$  and put

$$D := X_2 - X_1 \in [-1, 1]. \quad (1550)$$

Its density  $h$  is even. If

$$R(s) := \int_{\mathbb{R}} \text{up}(y) \text{up}(y+s) dy, \quad (1551)$$

then scaling  $D = (Y_2 - Y_1)/2$  gives

$$h(a) = 2R(2a). \quad (1552)$$

Equivalently,

$$h(a) = \int_0^1 F'(x)F'(x+a) dx, \quad (1553)$$

where  $F'$  is extended by zero outside  $[0, 1]$ . Thus  $h$  is  $C^\infty$  and flat at  $\pm 1$ .

### 8.2 Gamma-normalized positive-part order

For  $a \in (-1, 1)$  and initially  $\Re\beta > -1$ , define

$$\mathcal{K}_\beta(a) := \frac{\mathbb{E}(D-a)_+^\beta}{\Gamma(\beta+1)} = \frac{1}{\Gamma(\beta+1)} \int_a^1 (d-a)^\beta h(d) dd. \quad (1554)$$

**Warning 8.1.** The Gamma factor is part of the definition. For example,  $\mathbb{E}(D_+)^2 = 1/36$ , but  $\mathcal{K}_2(0) = \mathbb{E}(D_+)^2/\Gamma(3) = 1/72$ . This normalization is what makes the order ladder coefficient-free.

**Theorem 8.2** (Entire stop-loss order). *For every fixed  $a \in (-1, 1)$ ,  $\beta \mapsto \mathcal{K}_\beta(a)$  extends to an entire function. For every  $N \geq 0$  and  $\Re\beta > -N-1$ ,*

$$\mathcal{K}_\beta(a) = \frac{(-1)^N}{\Gamma(\beta+N+1)} \int_0^{1-a} t^{\beta+N} h^{(N)}(a+t) dt. \quad (1555)$$

*It satisfies the entire-order differential ladder*

$$-\partial_a \mathcal{K}_\beta(a) = \mathcal{K}_{\beta-1}(a), \quad (1556)$$

*and at negative integers*

$$\mathcal{K}_{-m}(a) = (-1)^{m-1} h^{(m-1)}(a), \quad m \geq 1. \quad (1557)$$

*Proof.* Set  $t = d - a$  in (1554). Repeated integration by parts moves derivatives onto  $h(a+t)$ . Every upper boundary term vanishes because  $h$  is flat at 1; every lower term vanishes in the initial half-plane and is then encoded by the zeros of  $1/\Gamma(\beta+1)$ . The resulting formulas (1555) agree on overlaps and give an entire continuation. Differentiating in  $a$  proves (1556); taking  $N = m$  in (1555) gives (1557).  $\square$

The order axis now has a concrete interpretation:

| $\beta$     | $\mathcal{K}_\beta(a)$                         |
|-------------|------------------------------------------------|
| $\beta > 0$ | fractional positive-part or stop-loss integral |
| 0           | $\mathbb{P}(D > a)$                            |
| -1          | $h(a)$                                         |
| -2          | $-h'(a)$                                       |
| -m          | $(-1)^{m-1} h^{(m-1)}(a).$                     |

(1558)

Thus a single entire function of the order joins tail probabilities, autocorrelations, derivatives, and fractional antiderivatives.

### 8.3 Triangular dyadic refinement

From (1453),

$$D \stackrel{d}{=} \frac{V + D'}{2}, \quad (1559)$$

where  $V = U_2 - U_1$  has triangular density  $\tau(v) = 1 - |v|$  on  $[-1, 1]$  and  $D'$  is an independent copy.

**Theorem 8.3** (Entire-order triangular refinement). *For every  $a \in \mathbb{R}$  and initially  $\Re\beta > -1$ ,*

$$\mathcal{K}_\beta(a) = 2^{-\beta} \int_{-1}^1 (1 - |v|) \mathcal{K}_\beta(2a - v) dv. \quad (1560)$$

For fixed interior  $a$ , the identity extends to every  $\beta \in \mathbb{C}$  by Theorem 8.2.

*Proof.* Condition on  $V = v$  in (1559):

$$(D - a)_+^\beta = 2^{-\beta} (D' - (2a - v))_+^\beta. \quad (1561)$$

Integrate against the triangular density and divide by  $\Gamma(\beta + 1)$ .  $\square$

At  $\beta = -1$ , (1560) is the density refinement equation; at lower negative integers it gives all derivative refinements. At positive orders it supplies a stable recursive equation for fractional stop-loss integrals.

### 8.4 A double inverse-Fabius integral

Using  $X_j = G(U_j)$  with independent uniforms,

$$\mathcal{K}_\beta(a) = \frac{1}{\Gamma(\beta + 1)} \int_0^1 \int_0^1 (G(v) - G(u) - a)_+^\beta du dv. \quad (1562)$$

This is a definite integral involving two inverse Fabius functions, while (1552) expresses the same hierarchy through Rvachev autocorrelation and (1553) through the forward density. It is therefore a literal three-function bridge.

At  $a = 0$  and  $\beta = 1$ , the Gini mean-difference identity gives the mixed forward-function integral

$$\mathcal{K}_1(0) = \frac{1}{2} \mathbb{E}|X_2 - X_1| = \int_0^1 F(x)(1 - F(x)) dx. \quad (1563)$$

Numerically this constant is 0.0955632320160...

### 8.5 Fractional Fourier energy

The characteristic function of  $D$  is

$$\varphi_D(t) = |\varphi_X(t)|^2 = \widehat{\text{up}}(t/2)^2. \quad (1564)$$

The standard fractional-moment identity [12, 10] now gives a transform formula.

**Theorem 8.4** (Fractional sinc-product energy). *For  $0 < \beta < 2$ ,*

$$\mathcal{K}_\beta(0) = \frac{\sin(\pi\beta/2)}{\pi} \int_0^\infty \frac{1 - \widehat{\text{up}}(t/2)^2}{t^{1+\beta}} dt. \quad (1565)$$

In particular,

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{\pi} \int_0^\infty \frac{1 - \widehat{\text{up}}(t/2)^2}{t^2} dt. \quad (1566)$$

*Proof.* For  $0 < \beta < 2$ ,

$$|x|^\beta = \frac{2\Gamma(\beta + 1) \sin(\pi\beta/2)}{\pi} \int_0^\infty \frac{1 - \cos(tx)}{t^{1+\beta}} dt. \quad (1567)$$

Take expectation with  $x = D$ , use symmetry  $\mathbb{E}(D_+)^{\beta} = \frac{1}{2} \mathbb{E}|D|^{\beta}$ , and divide by  $\Gamma(\beta + 1)$ .  $\square$



## 8.6 Exact even orders

For  $n \geq 0$ , independence and  $D = (Y_2 - Y_1)/2$  give

$$\mathcal{K}_{2n}(0) = \frac{1}{2(2n)!4^n} \sum_{k=0}^n \binom{2n}{2k} c_k c_{n-k}. \quad (1568)$$

The first values are

$$\mathcal{K}_2(0) = \frac{1}{72}, \quad (1569)$$

$$\mathcal{K}_4(0) = \frac{11}{64800}, \quad (1570)$$

$$\mathcal{K}_6(0) = \frac{211}{171460800}, \quad (1571)$$

$$\mathcal{K}_8(0) = \frac{31657}{5246700480000}, \quad (1572)$$

$$\mathcal{K}_{10}(0) = \frac{7647271}{354246723008640000}. \quad (1573)$$

These rational constants are new mixed integral evaluations; by (1562), each is a double integral of an even power of a difference of inverse Fabius values over the upper triangle  $v > u$ .

## 9 Numerical experiments

### 9.1 Reconstruction method

The accompanying script `numerical_experiments.py` reconstructs up directly from (1454). On a periodic interval of length 4 with  $N = 2^{17}$  Fourier modes it evaluates

$$\prod_{j=1}^{35} \text{sinc}(\xi/2^j) \quad (1574)$$

and applies an inverse FFT. The interval is wider than the exact support, so there is no wrap-around overlap. The CDF is obtained by cumulative quadrature and the inverse  $G$  by monotonicity-preserving PCHIP interpolation. Independent exact recurrences generate  $d_n$  and  $c_n$  as rational numbers.

The principal sanity checks, including the exact dyadic value  $F(1/4) = 5/72$  [7, 2, 11], are

$$\int_{-1}^1 \text{up} = 1, \quad \max \text{up} = 1, \quad \int_{-1}^1 y^2 \text{up}(y) dy = \frac{1}{9}, \quad F\left(\frac{1}{4}\right) = \frac{5}{72}. \quad (1575)$$

At full resolution, the computed value of  $F(1/4)$  differs from  $5/72$  by about  $1.6 \times 10^{-10}$ .

### 9.2 Independent checks

For the quantile theorem, the left side is integrated directly in  $u$  and the right side directly in  $x$ ; for negative powers, the regularized entire continuation is compared with the direct inverse integral. Caputo values are checked against a cancellation-safe Marchaud formula that uses only values of  $G$ , not the transmuted  $x$  integral. The generalized Stieltjes equations are checked by density quadrature, and the moment series uses exact rational  $d_n$ . Difference refinements use high-accuracy weighted quadrature, while the Fourier-energy check uses the sinc product independently.

Table 32: Independent checks of the quantile–Volterra and Caputo transmutations.

| case                              | direct form   | transmuted form | relative error         |
|-----------------------------------|---------------|-----------------|------------------------|
| $u = 0.2, \beta = 0.6, p = 1.0$   | 0.1226638614  | 0.1226638614    | $3.02 \times 10^{-13}$ |
| $u = 0.5, \beta = 1.4, p = 2.3$   | 0.0274018083  | 0.0274018083    | $2.1 \times 10^{-13}$  |
| $u = 0.8, \beta = 0.75, p = 0.4$  | 0.674404041   | 0.674404041     | $1.81 \times 10^{-15}$ |
| $u = 0.5, \beta = 0.8, p = -0.7$  | 1.280994219   | 1.280994219     | $2.86 \times 10^{-13}$ |
| $u = 0.8, \beta = 1.3, p = -1.2$  | 2.184447863   | 2.184447863     | $1.54 \times 10^{-13}$ |
| $u = 0.2, \beta = 0.6, p = -0.4$  | 0.7110303503  | 0.7110303503    | $1.31 \times 10^{-13}$ |
| $u = 0.6, \beta = 0.7, m = 1$     | -0.7015705216 | -0.7015705216   | $2.77 \times 10^{-14}$ |
| $u = 0.6, \beta = 0.7, m = 2$     | 0.7090525688  | 0.7090525688    | $4.17 \times 10^{-13}$ |
| $u = 0.6, \beta = 0.7, m = 3$     | -0.8030874837 | -0.8030874837   | $1.95 \times 10^{-13}$ |
| $u = 0.2, \alpha = 0.3, p = 1.0$  | 0.504741341   | 0.504741341     | $6.35 \times 10^{-12}$ |
| $u = 0.5, \alpha = 0.7, p = 1.0$  | 0.5952284038  | 0.5952284038    | $1.05 \times 10^{-12}$ |
| $u = 0.8, \alpha = 0.45, p = 1.0$ | 0.6895063772  | 0.6895063773    | $1.46 \times 10^{-10}$ |

Table 33: Generalized Stieltjes equations and moment-series convergence.

| $z$           | $\lambda$    | test                | truncation | absolute residual      |
|---------------|--------------|---------------------|------------|------------------------|
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | differential-dyadic | –          | $3.66 \times 10^{-16}$ |
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | order recurrence    | –          | $5 \times 10^{-16}$    |
| 2.2           | 1.4          | differential-dyadic | –          | $1.67 \times 10^{-16}$ |
| 2.2           | 1.4          | order recurrence    | –          | $3.33 \times 10^{-16}$ |
| $-0.7 + 0.5i$ | 0.8          | differential-dyadic | –          | $1.39 \times 10^{-16}$ |
| $-0.7 + 0.5i$ | 0.8          | order recurrence    | –          | $1.39 \times 10^{-15}$ |
| 2.2           | 1.4          | moment series       | 5          | 0.00154                |
| 2.2           | 1.4          | moment series       | 10         | $7.87 \times 10^{-6}$  |
| 2.2           | 1.4          | moment series       | 20         | $4.3 \times 10^{-10}$  |
| 2.2           | 1.4          | moment series       | 30         | $3.87 \times 10^{-14}$ |
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | moment series       | 5          | 0.00207                |
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | moment series       | 10         | $2.08 \times 10^{-5}$  |
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | moment series       | 20         | $6.92 \times 10^{-9}$  |
| $1.7 + 0.4i$  | $0.6 + 0.3i$ | moment series       | 30         | $4.62 \times 10^{-12}$ |

Table 34: Nonlocal tails and the entire-order difference hierarchy.

| test                 | parameters              | direct         | predicted      | relative error         |
|----------------------|-------------------------|----------------|----------------|------------------------|
| Weyl tail            | $x = 1.2, \alpha = 0.4$ | -0.2434690215  | -0.2434690025  | $7.8 \times 10^{-8}$   |
| Weyl tail            | $x = 1.5, \alpha = 0.8$ | -0.09670592448 | -0.09670592447 | $7.42 \times 10^{-11}$ |
| Weyl tail            | $x = 2.0, \alpha = 0.6$ | -0.09489107745 | -0.09489107745 | $2.63 \times 10^{-11}$ |
| $K_\beta$ refinement | $a = -0.2, \beta = 0.4$ | 0.5086799254   | 0.5086799254   | $1.79 \times 10^{-14}$ |
| $K_\beta$ refinement | $a = 0.0, \beta = 0.8$  | 0.1364947268   | 0.1364947268   | $9.35 \times 10^{-14}$ |
| $K_\beta$ refinement | $a = 0.25, \beta = 1.3$ | 0.00866136247  | 0.00866136247  | $4.45 \times 10^{-14}$ |
| $K_\beta$ refinement | $a = 0.5, \beta = 0.6$  | 0.002613518484 | 0.002613518484 | $8.63 \times 10^{-15}$ |
| fractional energy    | $\beta = 0.4$           | 0.2686797068   | 0.2686797079   | $3.99 \times 10^{-9}$  |
| fractional energy    | $\beta = 0.8$           | 0.1364947268   | 0.1364947274   | $4.11 \times 10^{-9}$  |
| fractional energy    | $\beta = 1.0$           | 0.09556323202  | 0.09556323241  | $4.11 \times 10^{-9}$  |
| fractional energy    | $\beta = 1.5$           | 0.03746967245  | 0.0374696726   | $4.11 \times 10^{-9}$  |

Table 35: Exact Gamma-normalized even values of the difference hierarchy.

| order | exact $K_{2n}(0)$                    | numerical value                 |
|-------|--------------------------------------|---------------------------------|
| 2     | $\frac{1}{72}$                       | 0.0138888888889                 |
| 4     | $\frac{64800}{211}$                  | 0.00016975308642                |
| 6     | $\frac{171460800}{31657}$            | $1.23060198016 \times 10^{-6}$  |
| 8     | $\frac{524670048000}{7647271}$       | $6.03369681967 \times 10^{-9}$  |
| 10    | $\frac{354246723008640000}{7647271}$ | $2.15874149378 \times 10^{-11}$ |

Table 36: Approach to the leading slowly-varying fractional-integral asymptotic for  $\beta = 0.7$ ,  $p = 1$ .

| $u$       | $\Gamma(\beta + 1)I^\beta G(u)/(u^\beta G(u))$ |
|-----------|------------------------------------------------|
| 0.01      | 0.8642340122                                   |
| 0.0001    | 0.8928590708                                   |
| $10^{-6}$ | 0.9076622860                                   |

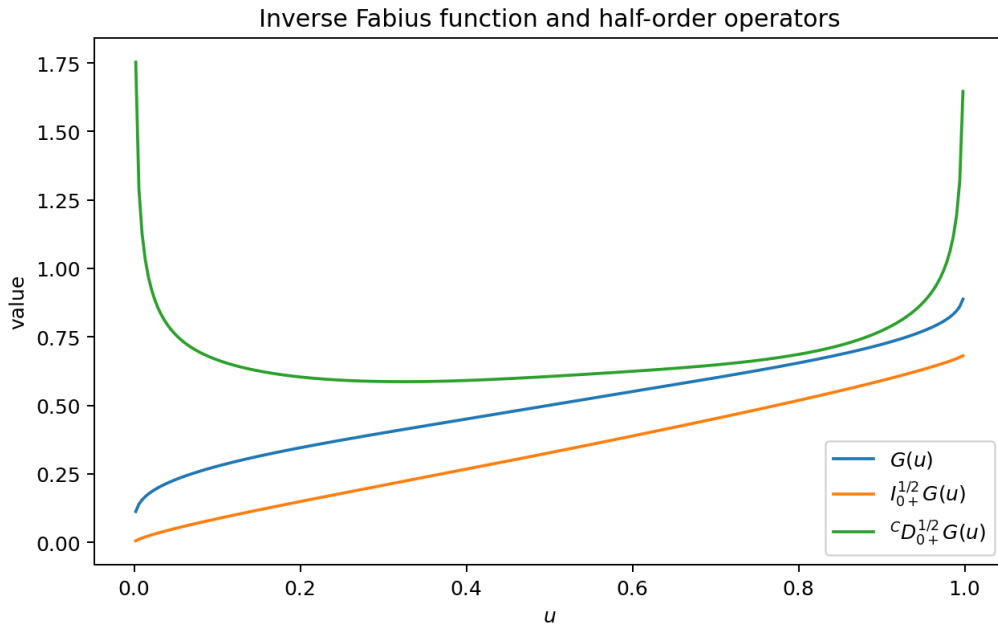


Figure 11: The inverse Fabius function together with its half-order Riemann–Liouville integral and half-order Caputo derivative. The Caputo operator amplifies endpoint structure, whereas the fractional integral smooths it.

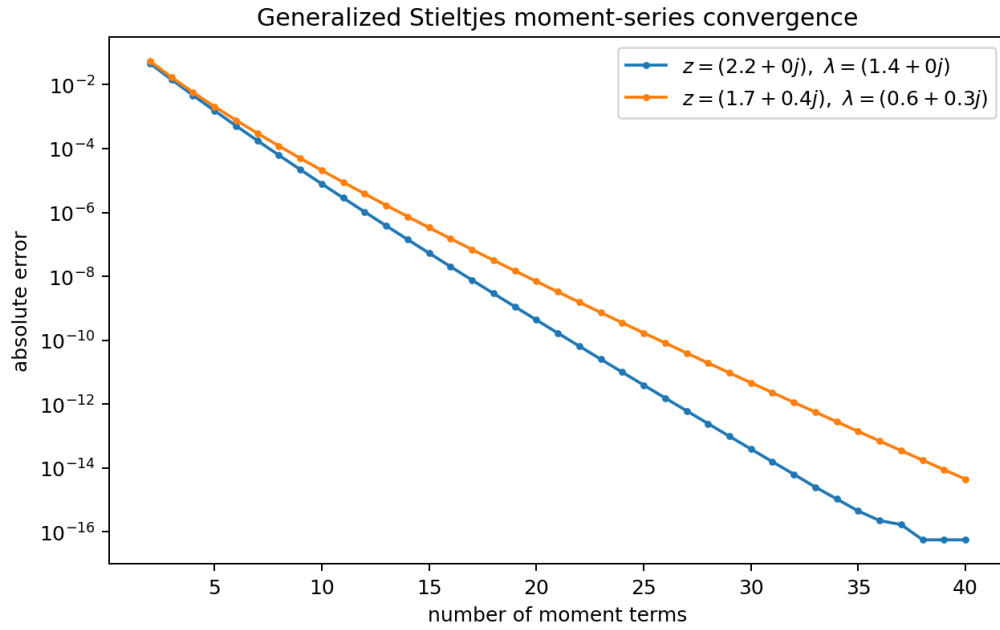


Figure 12: Convergence of the exact-moment expansion (1520) at two real/complex parameter pairs.

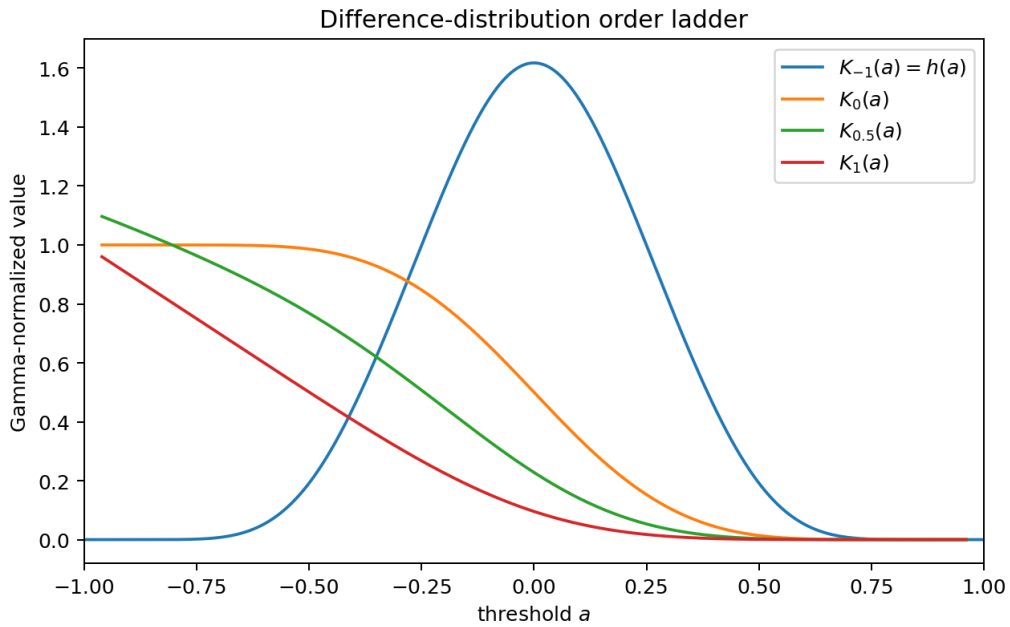


Figure 13: Several real-order slices of the entire difference hierarchy. The order  $-1$  slice is the difference density, order 0 is its tail, and positive orders are fractional positive-part integrals.

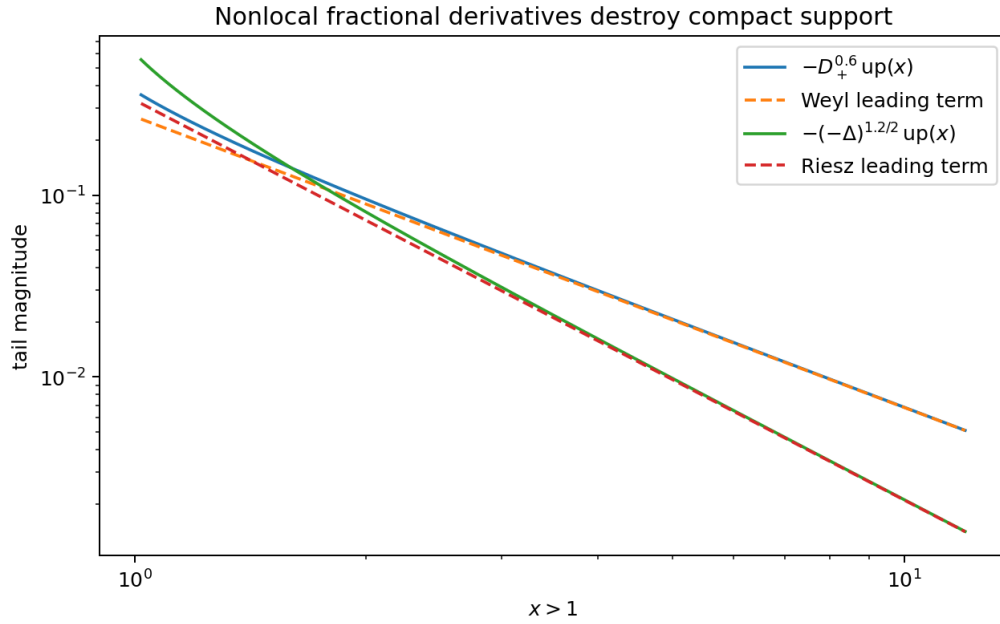


Figure 14: Weyl and Riesz fractional derivatives outside the support of  $up$ . Dashed curves are their leading  $x^{-1-\alpha}$  terms. The agreement illustrates how noninteger differentiation creates exact algebraic tails.

### 9.3 Boundary jump experiment

At  $x = 0.4$  and  $\lambda = 0.35$ , the predicted jump from (1535) was compared with  $\mathcal{S}_\lambda(x + i\varepsilon) - \mathcal{S}_\lambda(x - i\varepsilon)$ . The absolute discrepancies for  $\varepsilon = 10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}$  were, respectively,

$$5.89 \cdot 10^{-2}, \quad 1.80 \cdot 10^{-2}, \quad 6.01 \cdot 10^{-3}, \quad 1.81 \cdot 10^{-3}, \quad 6.02 \cdot 10^{-4}. \quad (1576)$$

The regular decrease is consistent with convergence to the fractional Plemelj boundary value. This is a deliberately harsher test than the off-cut functional equations, because it resolves a singular limit with a finite Fourier grid.

## 10 Further deductions and research directions

### 10.1 All-order endpoint expansions

Theorem 5.7 shows that once a weighted de Haan expansion for  $G(ut)/G(u)$  is known, fractional integration converts powers of  $\log t$  to digamma and polygamma constants. At higher orders one expects complete Bell polynomials in derivatives of the beta function. The missing ingredient is not fractional calculus but uniform control of the exact dyadic endpoint phase.

*Open problem 10.1* (Phase-aware Watson lemma). Prove a Watson–de Haan lemma for slowly varying functions whose second-order auxiliary function contains a smooth periodic term of a lower-Lambert phase, and apply it simultaneously to Riemann–Liouville and Caputo kernels.

A successful theorem would mechanically generate all correction terms in (1505) and (1507), with coefficients expressible through Bell polynomials and polygamma values. This is a natural meeting point of the repository’s Lambert- $W$ , Bell-polynomial, and fractional calculus strands.

## 10.2 A two-dimensional holonomic-like lattice

The pair of equations

$$\partial_z \mathcal{S}_\lambda = -\lambda \mathcal{S}_{\lambda+1}, \quad \mathcal{S}_\lambda(z) = \frac{2^\lambda}{1-\lambda} [\mathcal{S}_{\lambda-1}(2z) - \mathcal{S}_{\lambda-1}(2z-1)] \quad (1577)$$

forms a lattice in continuous order  $\lambda$  and dyadically transformed spectral position  $z$ . The fractional shift theorem (1531) fills the gaps between integer lattice levels.

*Open problem 10.2* (Special spectral points). Determine whether  $\mathcal{S}_\lambda(z)$  at dyadic  $z$  admits convergent  $q$ -binomial, continued-fraction, or Mahler-type representations that extend the known rational moment and dyadic-value formulas. Particularly natural points are  $z = 2^m$ ,  $z = 1 + 2^{-m}$ , and the centered points  $w = 2z - 1$  fixed or periodic under  $w \mapsto 2w \pm 1$ .

**Conjecture 10.3** (Dyadic resolvent rigidity). *For each real  $\lambda > 0$ , the differential–dyadic equation (1522), the asymptotic  $\mathcal{S}_\lambda(z) \sim z^{-\lambda}$  at infinity, and the generalized Stieltjes positivity conditions uniquely characterize  $\mathcal{S}_\lambda$ .*

This would be a transform-side uniqueness theorem parallel to the uniqueness of the original Rvachev function.

## 10.3 Zeros in the order plane

For fixed  $a \in (-1, 1)$ ,  $\mathcal{K}_\beta(a)$  is entire in  $\beta$ , and its values at negative integers are signed derivatives of  $h$ . The zeros in the order plane therefore interpolate the oscillatory geometry of every derivative of the autocorrelation density.

**Conjecture 10.4** (Symmetry-forced order zeros). *Evenness gives the exact lattice of forced zeros  $\mathcal{K}_{-2m}(0) = 0$  for  $m \geq 1$ . These zeros are eventually simple, and any additional sufficiently far-left real zeros interlace the forced lattice. A more flexible version asks whether the Jensen polynomials of a suitably shifted and rescaled order function become hyperbolic asymptotically.*

This is speculative, but it is numerically accessible because (1555) remains stable far to the left after choosing  $N$  large enough.

## 10.4 Spectral analysis of the triangular refinement operator

For each order define

$$(\mathcal{T}_\beta f)(a) := 2^{-\beta} \int_{-1}^1 (1 - |v|) f(2a - v) dv. \quad (1578)$$

Then  $\mathcal{K}_\beta$  is a fixed point of  $\mathcal{T}_\beta$ . The order enters both the scalar prefactor and the regularity class selected by the boundary conditions.

*Open problem 10.5* (Order-dependent refinement spectrum). Develop the spectrum of  $\mathcal{T}_\beta$  on weighted Sobolev, Mellin, or entire-function spaces. Determine whether analytic perturbation in  $\beta$  controls the zero set and growth of  $\mathcal{K}_\beta(a)$ .

The triangular kernel has Fourier transform  $\text{sinc}^2(\xi/2)$ , so this problem may connect the squared sinc product for the difference distribution to the original single sinc product for up.

## 10.5 Higher-dimensional differences and B-spline noise

For independent Fabius variables and coefficients  $\varepsilon_j \in \{\pm 1\}$ , consider

$$D_m := \sum_{j=1}^m \varepsilon_j X_j. \quad (1579)$$

The fixed-point noise becomes a compact B-spline distribution obtained from a signed sum of uniforms. The same construction gives an entire hierarchy

$$\mathcal{K}_{m,\beta}(a) := \frac{\mathbb{E}(D_m - a)_+^\beta}{\Gamma(\beta + 1)} \quad (1580)$$

and a refinement equation whose kernel is that B-spline. Even orders are finite polynomials in the rational moments  $d_n$  or  $c_n$ . This is a broad family of exact definite integrals involving several inverse Fabius functions.

## 10.6 Fractional operators and q-binomial arithmetic

The exact moment series in (1520), the exterior series (1544), and the exact even values (1568) are all built from rational moment sequences. Those sequences are already connected in the repository to  $q$ -Pochhammer symbols and Gaussian binomial coefficients at  $q = 2$  or  $q = 1/2$ .

*Open problem 10.6* (Arithmetic compression). Find direct  $q$ -hypergeometric forms for

$$\sum_{n \geq 0} \frac{(\lambda)_n}{n!} d_n z^{-n}, \quad \sum_{j \geq 0} \frac{(\alpha + 1)_{2j}}{(2j)!} c_j x^{-2j}, \quad (1581)$$

possibly after extracting the dominant lower-Lambert phase or applying a Borel transform. Determine denominator bounds for their Taylor coefficients as functions of the order parameters.

## 10.7 Formalization roadmap

The most direct Lean formalization order is:

1. prove the layer-cake form of Theorem 4.1 first for nonnegative  $\phi'$  and real  $\beta > 0$ , then extend by signed measures;
2. formalize the operator identity  $(D_u^m(\phi \circ G)) \circ F = \mathcal{L}_F^m \phi$  on compact subintervals of  $(0, 1)$ ;
3. derive the Caputo transmutation from change of variables;
4. define  $S_\lambda$  initially for real  $\lambda > 0$  and real  $z > 1$ , where all kernels are positive, then add complex analyticity;
5. formalize  $\mathcal{K}_\beta$  for real  $\beta > -1$ , its refinement equation, and the integer negative-order identities before attempting entire continuation;
6. use the already formalized sinc product and moment recurrences for the Weyl/Riesz exterior expansions.

The first, second, and fifth steps require mostly measure theory, ordinary calculus, and Gamma identities. They should be substantially easier than the sharp endpoint asymptotic development already present in the repository.

## 11 Conclusion

The integral theory of the Fabius and Rvachev functions becomes much more coherent when the integration order itself is treated as a variable. Three operator families organize most of the new structure:

1. the quantile–Volterra kernel  $(u - F(x))^\beta / \Gamma(\beta + 1)$ , which turns arbitrary-order primitives and Caputo derivatives of  $G$  into forward-function integrals;

2. the complex-order resolvent  $\mathcal{S}_\lambda(z)$ , whose spectral derivatives, dyadic refinements, moment expansions, and boundary jumps are all order shifts of one another;
3. the entire positive-part hierarchy  $\mathcal{K}_\beta(a)$ , which joins inverse-Fabius double integrals, Fabius density correlations, Rvachev autocorrelations, fractional energies, and all derivatives of the difference density.

The Weyl/Riesz analysis adds a complementary lesson: compact support is an integer-order phenomenon. Noninteger differentiation immediately produces algebraic tails, but the tails are not mysterious; every coefficient is an exact rational Rvachev moment. Together, these formulas connect the repository's existing themes – dyadic refinement, sinc products, rational moments, inverse asymptotics, Bell polynomials, Lambert  $W$ , and  $q$ -arithmetic – through a common continuous order parameter.

## A Proof details for analytic continuation

### A.1 Why the reciprocal Gamma factor is canonical

Let  $g \in C^\infty[0, L]$  be flat at  $L$ , and define initially for  $\Re \beta > -1$

$$M_g(\beta) := \frac{1}{\Gamma(\beta + 1)} \int_0^L t^\beta g(t) dt. \quad (1582)$$

Repeated integration by parts gives

$$M_g(\beta) = \frac{(-1)^N}{\Gamma(\beta + N + 1)} \int_0^L t^{\beta+N} g^{(N)}(t) dt. \quad (1583)$$

This is entire in  $\beta$ . At  $\beta = -m$ ,

$$M_g(-m) = (-1)^{m-1} g^{(m-1)}(0). \quad (1584)$$

Both [Theorem 8.2](#) and the ladders of [Appendix 3](#) are instances of this elementary lemma. The zeros of  $1/\Gamma(\beta + 1)$  are not an ornamental normalization: they cancel the poles of the local Mellin transform generated by the Taylor series of  $g$  at zero.

### A.2 Complex powers and branches

In the quantile formulas,  $u - F(x) \geq 0$  and  $x > 0$ , so  $(u - F(x))^\beta$  and  $x^p$  use real logarithms on the open integration interval. In the generalized Stieltjes transform,  $z - x$  moves in a compact subset of the principal cut plane when  $z \in \Omega$ ; a single branch of  $\text{Log}$  is fixed for all  $x \in [0, 1]$ . The upper and lower banks define the Plemelj values on  $(0, 1)$ . Statements on other cuts are branch-local and do not assert monodromy-free continuation around the support.

## B Moment recurrences and exact constants

From  $X = (U + X')/2$ ,

$$2^n d_n = \sum_{k=0}^n \binom{n}{k} \frac{d_k}{n - k + 1}. \quad (1585)$$

Separating the  $k = n$  term gives [\(1457\)](#). The first values are

$$d_0 = 1, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}, \quad d_5 = \frac{19}{270}, \quad d_6 = \frac{1153}{23814}. \quad (1586)$$



Similarly,  $Y = (V + Y')/2$  and  $\mathbb{E}V^{2m} = 1/(2m + 1)$  give

$$4^n c_n = \sum_{k=0}^n \binom{2n}{2k} \frac{c_k}{2(n-k) + 1}. \quad (1587)$$

After separating  $k = n$  and using

$$\frac{1}{2(n-k) + 1} \binom{2n}{2k} = \frac{1}{2n+1} \binom{2n+1}{2k}, \quad (1588)$$

one obtains (1458).

For the difference moments,

$$\mathbb{E}D^{2n} = \frac{1}{4^n} \sum_{k=0}^n \binom{2n}{2k} c_k c_{n-k}. \quad (1589)$$

Symmetry contributes the factor  $1/2$  for the positive half, and the Gamma normalization contributes  $(2n)!$ , giving (1568).

## C Reproducibility notes

The archive accompanying this report contains:

- `fabius_fractional_transform_frontiers.tex`, the source of this report;
- `fabius_fractional_transform_frontiers.pdf`, the rendered report;
- `numerical_experiments.py`, the fully commented experiment script;
- `numerical_results.txt` and `numerical_results.tex`, generated numerical output;
- four figures in both PDF and PNG forms;
- `README.md`, `CORPUS_AUDIT.md`, and `SHA256SUMS`.

The numerical script requires Python 3 with NumPy, SciPy, mpmath, and Matplotlib. Running

```
python numerical_experiments.py
pdflatex -interaction=nonstopmode -halt-on-error
    Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error
    Integration_and_Transform_Frontiers.tex
pdflatex -interaction=nonstopmode -halt-on-error
    Integration_and_Transform_Frontiers.tex
```

regenerates the tables, figures, and PDF. The `-quick` option uses a  $2^{15}$ -point grid for a fast smoke test; the reported values use  $2^{17}$ .

## D Corpus novelty matrix

| Topic                               | Prior repository coverage                                                    | Contribution of this report                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Polynomial and monomial primitives  | Extensive formulas for integer powers, negative powers, and dyadic reduction | Used only as baseline; not rederived in detail                                                                                 |
| Fractional primitives of $F$ and up | Causal fractional integrals and Mellin/moment-tail formulas                  | Gamma-normalized entire-order packaging and connection to nonlocal exterior tails                                              |
| Inverse Fabius integrals            | Integer layer-cake formulas, beta transforms, inverse moments                | Arbitrary complex-order quantile–Volterra theorem; entire power parameter; logarithmic and exponential families                |
| Fractional derivatives of $G$       | No Caputo transmutation found                                                | Exact orders $(0, 1)$ and arbitrary orders through $\mathcal{L}_F^m$ ; endpoint and $L^r$ theory                               |
| Stieltjes transforms                | Ordinary Cauchy/Stieltjes transform and a dyadic equation                    | Complex order, order recurrence, fractional spectral shifts, moment series, and fractional boundary inversion                  |
| Fractional derivatives of up        | Fourier norm spectrum and warning against naive scaling                      | Exact Weyl order-lowering dilation; convergent Weyl/Riesz algebraic tails                                                      |
| Autocorrelation and differences     | Basic correlation identities and selected norms                              | Entire positive-part order, negative-order derivatives, triangular refinement, inverse double integrals, and fractional energy |
| Endpoint asymptotics                | Sharp lower-Lambert and nonconstant periodic phase                           | Slow-variation consequences for fractional operators; phase-aware conditional correction and conjecture                        |

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## Part XII

# Integral and Transform Calculus for the Fabius–Rvachev–Quantile System

## Abstract

This report develops a unified calculus for indefinite and definite integrals, iterated and fractional antiderivatives, and Mellin, Laplace, Fourier, Cauchy–Stieltjes, and beta transforms involving the bounded Fabius function  $F$ , Rvachev’s compactly supported function  $\mathfrak{u}$ , and the inverse Fabius function  $G = F^{-1}$ . The repository corpus was audited at the pinned snapshot a6555a64671b; its recent monomial/fractional-calculus, inverse-asymptotic, Fourier-decay, and Bell–Bernoulli developments are treated as baseline rather than rediscovered.

The main new organizing principle is an adjacent-order-statistic spacing identity. It converts every weighted mixed integral  $\int h(x)F(x)^p(1 - F(x))^q dx$  into the expected increment of an antiderivative of  $h$  across two consecutive order statistics. This yields, in one stroke, explicit formulas for polynomial, Jacobi, exponential, trigonometric, logarithmic, Lambert- $W$ , Mellin, Fourier, and generalized resolvent weights, as well as a closed probabilistic form for every integer or fractional Volterra primitive. A complementary beta–quantile calculus gives incomplete antiderivatives in terms of truncated moments of  $G$ , a Pascal lattice, reflection identities, and strict checkerboard total positivity.

Further results include: a monotone beta-normalized balanced sequence converging to  $1/2$  faster than every inverse power; its logarithmic decay constant  $-1/(8 \log 2)$ , with a Lambert- $W$  saddle; a Cauchy–Stieltjes hierarchy for the up-law satisfying exact dyadic functional-differential equations; a CDF/quantile resolvent duality; endpoint derivative growth of order  $\exp((\log 2)n^2/2)$ ; and an exact complementary pair of sinc-product energy integrals for the previously opaque constant  $\int_0^1 F(x)^2 dx$ . A commented Python program independently checks the principal identities and is included with the source and PDF.

## 1 Scope, corpus audit, and status of the results

### 1.1 Audit boundary

The audit covered the `*.tex` sources under

[Analysis/FabiusFunction/docs](#)

including all subdirectories. The working snapshot was pinned to commit a6555a64671b, after the merge of the recent frontier-report stack. The pinned Git-tree enumeration used in the audit contained 87 TeX paths; its complete path manifest is included with this report. Byte-identical copies, source/PDF bundles, OCR transcriptions of external papers, and successive draft variants were grouped by content lineage; theorem-level comparison was performed against every content-distinct Fabius/Rvachev source and against the archive ledger.

The closest predecessors are the reports on monomial antiderivatives, complex-power Newton–Volterra expansions, fractional calculus, inverse-Fabius asymptotics, derivative norm spectra, and the consolidated frontier reports. Consequently, the following are treated here as *baseline*:

- the primitive ladder for  $x^\alpha F(x)$ , including the terminating integer-power case and convergent complex-power series;
- basic Volterra/Riemann–Liouville formulas for  $F$  and  $G$ ;
- endpoint asymptotics expressed through Lambert  $W_{-1}$ , periodic corrections, Bell polynomials, and Gamma–zeta operators;
- the infinite sinc product for  $\widehat{\mathfrak{u}}$ , its dyadic refinement, and the existing Fourier-decay analysis;

- exact dyadic values, rational moment recurrences, finite Thue–Morse splines, and the established  $q$ -binomial machinery.

A superscript <sup>N</sup> marks a result that appears new relative to the audited repository corpus; <sup>B</sup> marks a recovered or reorganized baseline fact; and <sup>C</sup> marks a conjecture. These labels are not claims of global priority in the mathematical literature.

## 1.2 Principal new contributions

The report's main contributions are:

1. the weighted adjacent-spacing theorem and its partial, iterated, and fractional antiderivative forms;
2. an incomplete beta-quantile antiderivative and a two-parameter beta-quantile lattice;
3. a bivariate resolvent for all mixed powers, with Möbius symmetry and strict checkerboard total positivity;
4. a beta-localization theorem, strict monotonicity of the balanced normalized sequence, superalgebraic convergence, and the exact leading logarithmic rate governed by a Lambert saddle;
5. a Cauchy–Stieltjes hierarchy for up, including boundary-value and endpoint nonanalyticity statements;
6. a generalized resolvent duality interchanging  $F$  and  $G$ ;
7. the sinc-energy identity and complementary Gini sum rule for  $\int_0^1 F^2$ ;
8. a certified rational finite-spline approximation scheme inherited from the Thue–Morse truncations, with an explicit coupling error bound.

## 2 The three functions and their probability models

### 2.1 The bounded Fabius distribution

Let  $(U_j)_{j \geq 1}$  be independent  $\text{Unif}[0, 1]$  variables and set

$$X = \sum_{j \geq 1} \frac{U_j}{2^j}. \quad (1590)$$

The bounded Fabius function is the distribution function

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1, \quad (1591)$$

continued by 0 to the left and 1 to the right whenever a genuine CDF is needed. It obeys

$$F(1-x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (1592)$$

and is strictly increasing on  $[0, 1]$ . Its inverse is

$$G(u) = F^{-1}(u), \quad 0 \leq u \leq 1, \quad (1593)$$

with  $G(1-u) = 1 - G(u)$ .

Write

$$d_s = \mathbb{E}X^s = \int_0^1 x^s F'(x) dx \quad (1594)$$

whenever the moment exists. For integer  $n \geq 1$ , the exact recurrence is

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (1595)$$

In particular,

$$d_0 = 1, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}.$$

## 2.2 Rvachev's compactly supported density

Put

$$Y = 2X - 1 = \sum_{j \geq 1} \frac{V_j}{2^j}, \quad V_j = 2U_j - 1 \sim \text{Unif}[-1, 1]. \quad (1596)$$

Then  $Y$  has density

$$\text{up}(y) = F(1 - |y|) \mathbf{1}_{[-1,1]}(y), \quad F'(x) = 2 \text{up}(2x - 1). \quad (1597)$$

Thus  $\text{up}$  is even and  $C^\infty$ , and

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)). \quad (1598)$$

We use the cycles-per-unit Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \text{sinc } t = \frac{\sin t}{t}. \quad (1599)$$

Then

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{j \geq 0} \text{sinc} \left( \frac{\pi \xi}{2^j} \right). \quad (1600)$$

The moment-generating functions are

$$M_X(z) = \mathbb{E} e^{zX} = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j}, \quad (1601)$$

$$M_Y(z) = \mathbb{E} e^{zY} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}. \quad (1602)$$

## 2.3 Bernoulli cumulants and Bell moments

The transform (1602) makes the Bell–Bernoulli structure transparent. Since

$$\log \frac{\sinh z}{z} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m},$$

one obtains

$$\log M_Y(z) = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!(2^{2m} - 1)} z^{2m}. \quad (1603)$$

Hence the nonzero cumulants are

$$\kappa_{2m}(Y) = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1}(Y) = 0, \quad (1604)$$

and the centered moments  $c_m = \mathbb{E} Y^{2m}$  are complete Bell polynomials in these cumulants. The first values are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}.$$

They also satisfy  $c_m = 2d_{2m+1}/(2m+1)$ .

### 3 Endpoint integrability: powers of a CDF versus powers of a quantile

The endpoint law established in the corpus has the logarithmic form

$$\log F(e^{-t}) = -\frac{t^2}{2 \log 2} + O(t \log t), \quad t \rightarrow +\infty, \quad (1605)$$

with much sharper Lambert-periodic expansions available in the baseline reports. This immediately creates a useful dichotomy.

**Proposition 3.1** (Integrability classification). *Let  $p, q, \alpha, \beta \in \mathbb{C}$ .*

- (i) *The mixed power  $F(x)^p(1 - F(x))^q$  is absolutely integrable on  $(0, 1)$  precisely when  $\Re p \geq 0$  and  $\Re q \geq 0$ .*
- (ii) *If  $\Re p > 0$ , then  $x^\alpha F(x)^p$  is integrable at 0 for every  $\alpha$ . If  $\Re p = 0$ , one needs  $\Re \alpha > -1$ ; if  $\Re p < 0$ , no algebraic weight repairs the divergence.*
- (iii) *Every fixed negative moment of  $X$  is finite:*

$$\mathbb{E}X^{-s} = \int_0^1 G(u)^{-s} du < \infty \quad (s > 0),$$

whereas  $\int_0^1 F(x)^{-s} dx = \infty$ .

- (iv) *For  $\Re \alpha > -1$ , the integral  $\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx$  converges when  $\Re q \geq 0$  and diverges when  $\Re q < 0$ .*

*Proof.* Put  $x = e^{-t}$ . By (1605), a negative power  $F(x)^{-s}$  contributes  $\exp(st^2/(2 \log 2) - t + O(t \log t)) dt$ , which diverges. A positive power has the opposite quadratic sign and dominates every algebraic singularity. For the quantile, inversion of (1605) gives  $G(e^{-L}) = \exp(-\sqrt{2 \log 2} L + O(\log L))$ . Thus  $G(u)^{-s} du$ , under  $u = e^{-L}$ , has exponent  $-L + O(\sqrt{L})$ , and is integrable. The remaining endpoint follows from symmetry, while  $\text{up}(1 - x) = F(x)$  on  $[0, 1]$ .  $\square$

This contrast explains why negative powers of  $F$  or  $\text{up}$  generally do not possess ordinary definite integrals, while inverse-Fabius negative moments are perfectly well behaved.

## 4 The adjacent-order-statistic spacing principle

### 4.1 The universal weighted identity

Let  $Z_1, \dots, Z_N$  be independent copies of  $X$ , and write  $Z_{1:N} \leq \dots \leq Z_{N:N}$  for their order statistics.

**Theorem 4.1** (Weighted adjacent-spacing theorem<sup>N</sup>). *Let  $p, q \in \mathbb{N}_{>0}$ ,  $N = p + q$ , and let  $h \in L^1(0, 1)$  have an absolutely continuous primitive  $H$ , so  $H' = h$  almost everywhere. Then*

$$\int_0^1 h(x) F(x)^p (1 - F(x))^q dx = \frac{\mathbb{E}[H(Z_{p+1:N}) - H(Z_{p:N})]}{\binom{N}{p}}. \quad (1606)$$

More generally, for  $0 \leq a < b \leq 1$ ,

$$\int_a^b h(x) F(x)^p (1 - F(x))^q dx = \mathbb{E} \left[ \mathbf{1}_{\{L < U\}} \int_a^b h(x) \mathbf{1}_{[L, U)}(x) dx \right], \quad (1607)$$

where  $L$  is the maximum of  $p$  independent copies of  $X$ , and  $U$  is the minimum of another  $q$  copies.

*Proof.* For every fixed  $x$ , independence gives

$$\mathbb{P}(L \leq x < U) = F(x)^p(1 - F(x))^q.$$

Fubini's theorem yields (1607). On  $\{L < U\}$ , all  $p$  observations in the first group precede all  $q$  observations in the second. Conditional on the unlabeled ordered sample, the labels are uniformly distributed among the  $\binom{N}{p}$  possibilities. Exactly one labeling has the first  $p$  observations in the lower group. On that labeling,  $L = Z_{p:N}$  and  $U = Z_{p+1:N}$ . Integrating  $h$  across the random gap gives (1606).  $\square$

The theorem is distribution-free: the Fabius law enters through the order statistics, not through the proof. Its value here is that the order statistics inherit the exact dyadic, inverse-quantile, and endpoint structures of  $F$ .

## 4.2 Moments, Jacobi weights, exponentials, and Lambert weights

Taking  $H(x) = x^{m+1}/(m+1)$  gives

$$\int_0^1 x^m F(x)^p(1 - F(x))^q dx = \frac{\mathbb{E}[Z_{p+1:N}^{m+1} - Z_{p:N}^{m+1}]}{(m+1)\binom{N}{p}}. \quad (1608)$$

For Jacobi weights, with the incomplete beta function  $B_x(a, b) = \int_0^x t^{a-1}(1-t)^{b-1} dt$ ,

$$\int_0^1 x^\alpha(1-x)^\beta F(x)^p(1 - F(x))^q dx = \frac{\mathbb{E}[B_{Z_{p+1:N}}(\alpha+1, \beta+1) - B_{Z_{p:N}}(\alpha+1, \beta+1)]}{\binom{N}{p}}. \quad (1609)$$

Differentiating in  $\alpha$  or  $\beta$  supplies all logarithmic Jacobi weights.

For  $z \neq 0$ ,

$$\int_0^1 e^{zx} F(x)^p(1 - F(x))^q dx = \frac{\mathbb{E}(e^{zZ_{p+1:N}} - e^{zZ_{p:N}})}{z\binom{N}{p}}, \quad (1610)$$

$$\int_0^1 e^{-2\pi i \xi x} F(x)^p(1 - F(x))^q dx = \frac{\mathbb{E}(e^{-2\pi i \xi Z_{p:N}} - e^{-2\pi i \xi Z_{p+1:N}})}{2\pi i \xi \binom{N}{p}}. \quad (1611)$$

The limits at  $z = 0$  or  $\xi = 0$  recover the mean spacing.

A Lambert- $W$  weight also has an elementary primitive:

$$\frac{d}{dx} \left( W(cx) + \frac{1}{2}W(cx)^2 \right) = \frac{W(cx)}{x}. \quad (1612)$$

Therefore, on any branch for which the path avoids the branch cut,

$$\int_0^1 \frac{W(cx)}{x} F(x)^p(1 - F(x))^q dx = \frac{\mathbb{E}[\mathcal{W}_c(Z_{p+1:N}) - \mathcal{W}_c(Z_{p:N})]}{\binom{N}{p}}, \quad (1613)$$

where  $\mathcal{W}_c(x) = W(cx) + \frac{1}{2}W(cx)^2$ . This is a direct integral appearance of Lambert  $W$ , distinct from its asymptotic role at the endpoint.



### 4.3 Partial, iterated, and fractional antiderivatives

For  $x \in [0, 1]$ , the partial primitive has the exact random-interval form

$$\int_0^x h(t)F(t)^p(1-F(t))^q dt = \mathbb{E}[\mathbf{1}_{\{L < x, L < U\}}(H(\min\{x, U\}) - H(L))]. \quad (1614)$$

This is already an indefinite integral: changing the lower endpoint adds only a constant.

The formula becomes especially sharp for the Riemann–Liouville operator

$$(I^\nu f)(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} f(t) dt, \quad \Re \nu > 0. \quad (1615)$$

**Theorem 4.2** (Fractional adjacent-spacing primitive<sup>N</sup>). *For  $p, q \in \mathbb{N}_{>0}$ ,  $N = p + q$ ,  $x \in [0, 1]$ , and  $\Re \nu > 0$ ,*

$$I^\nu[F^p(1-F)^q](x) = \frac{\mathbb{E}[(x - Z_{p:N})_+^\nu - (x - Z_{p+1:N})_+^\nu]}{\Gamma(\nu+1)\binom{N}{p}}. \quad (1616)$$

For integer  $n \geq 1$ , the right side is the normalized difference of two truncated  $n$ -th moments and is the unique  $n$ -fold primitive whose first  $n$  initial data vanish at 0.

*Proof.* Insert  $F(t)^p(1-F(t))^q = \mathbb{P}(L \leq t < U)$  into the Volterra integral and apply Fubini. For fixed  $L < U$ ,

$$\frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} \mathbf{1}_{[L,U)}(t) dt = \frac{(x-L)_+^\nu - (x-U)_+^\nu}{\Gamma(\nu+1)}.$$

The same random-label argument as in [theorem 4.1](#) supplies the factor  $\binom{N}{p}^{-1}$ . □

Endpoint cases are also useful. If  $M_p = Z_{p:p}$  and  $m_q = Z_{1:q}$ , then

$$I^\nu[F^p](x) = \frac{\mathbb{E}(x - M_p)_+^\nu}{\Gamma(\nu+1)}, \quad (1617)$$

$$I^\nu[(1-F)^q](x) = \frac{x^\nu - \mathbb{E}(x - m_q)_+^\nu}{\Gamma(\nu+1)}. \quad (1618)$$

For  $p = 1$ , (1617) collapses to the baseline Fabius ladder

$$\mathbb{E}(x - X)_+^n = n! 2^{\binom{n}{2}} F(x/2^n). \quad (1619)$$

Thus [theorem 4.2](#) is a genuine mixed-power extension of that ladder.

## 5 Incomplete beta-quantile antiderivatives

### 5.1 A closed incomplete formula

For  $r \in \mathbb{C}$ ,  $a, b > 0$ , and  $0 \leq y \leq 1$ , define

$$Q_r(a, b; y) = \int_0^y G(u)^r u^{a-1} (1-u)^{b-1} du, \quad Q_r(a, b) = Q_r(a, b; 1). \quad (1620)$$

**Theorem 5.1** (Incomplete beta-quantile primitive<sup>N</sup>). *Let  $m > -1$ ,  $p, q > 0$ ,  $a = F(x)$ . Whenever the displayed integrals converge,*

$$\int_0^x t^m F(t)^p (1-F(t))^q dt = \frac{x^{m+1} a^p (1-a)^q}{m+1}$$

$$-\frac{p}{m+1}Q_{m+1}(p, q+1; a) + \frac{q}{m+1}Q_{m+1}(p+1, q; a). \quad (1621)$$

Consequently,

$$\int_0^1 t^m F(t)^p (1-F(t))^q dt = \frac{qQ_{m+1}(p+1, q) - pQ_{m+1}(p, q+1)}{m+1}. \quad (1622)$$

*Proof.* Integrate by parts with  $t^m dt = d(t^{m+1}/(m+1))$ . The derivative of  $F^p(1-F)^q$  is

$$[pF^{p-1}(1-F)^q - qF^p(1-F)^{q-1}] F'.$$

Substitute  $u = F(t)$  in the two resulting terms. The endpoint term vanishes at  $x = 1$  when  $q > 0$ .  $\square$

Formula (1621) is a deterministic counterpart of the random-gap formula (1614). It is often better for symbolic work because all dependence on the inverse Fabius function is isolated in  $Q_r$ .

## 5.2 The beta-quantile lattice

The complete transforms obey exact algebraic identities.

**Proposition 5.2** (Pascal, reflection, and order-statistic laws<sup>N</sup>). *For admissible parameters,*

$$Q_r(a, b) = Q_r(a+1, b) + Q_r(a, b+1), \quad (1623)$$

$$Q_1(a, b) + Q_1(b, a) = B(a, b), \quad (1624)$$

$$Q_r(b, a) = \sum_{k=0}^r (-1)^k \binom{r}{k} Q_k(a, b) \quad (r \in \mathbb{N}). \quad (1625)$$

If  $a, b \in \mathbb{N}_{>0}$ , then

$$Q_r(a, b) = B(a, b) \mathbb{E} Z_{a:b-1}^r. \quad (1626)$$

*Proof.* Equation (1623) is the identity  $u^{a-1}(1-u)^{b-1} = u^a(1-u)^{b-1} + u^{a-1}(1-u)^b$ . For reflection, substitute  $u \mapsto 1-u$  and use  $G(1-u) = 1-G(u)$ . The same substitution followed by the binomial theorem gives (1625). Finally,  $F(Z_{a:b-1}) \sim \text{Beta}(a, b)$ , proving (1626).  $\square$

For  $q_s = Q_1(s, 1)$ , integration by parts gives

$$q_s = \int_0^1 u^{s-1} G(u) du = \frac{1-A_s}{s}, \quad A_s := \int_0^1 F(x)^s dx. \quad (1627)$$

When  $b \in \mathbb{N}$ ,

$$Q_1(a, b) = \sum_{j=0}^{b-1} (-1)^j \binom{b-1}{j} q_{a+j}. \quad (1628)$$

Thus the entire integer beta-quantile lattice is a finite binomial transform of the one-dimensional sequence  $(A_s)$ .

## 5.3 Unaligned products of $F$ and $G$

Compositions aligned with the inverse are immediate:

$$G(F(x)) = x, \quad F(G(u)) = u.$$

Products at the same external argument, such as  $F(x)^a G(x)^b$ , do not collapse. They nevertheless admit an exact two-dimensional layer-cake kernel. For  $a, b > 0$ ,

$$\int_0^1 F(x)^a G(x)^b dx = ab \int_0^1 \int_0^1 u^{a-1} v^{b-1} [1 - \max\{G(u), F(v)\}] du dv. \quad (1629)$$

Indeed,  $F(x)^a = a \int_0^1 u^{a-1} \mathbf{1}_{\{G(u) \leq x\}} du$ , while  $G(x)^b = b \int_0^1 v^{b-1} \mathbf{1}_{\{F(v) \leq x\}} dv$ . Formula (1629) is a promising starting point for genuinely nonlinear  $F$ - $G$  coupling constants.

## 6 The mixed-power lattice

Define

$$I_{p,q} = \int_0^1 F(x)^p (1 - F(x))^q dx, \quad p, q \geq 0. \quad (1630)$$

For integer arguments, the elementary partition  $1 = F + (1 - F)$  gives

$$I_{p,q} = I_{p+1,q} + I_{p,q+1}, \quad I_{p,q} = I_{q,p}. \quad (1631)$$

If  $\Delta A_p = A_{p+1} - A_p$ , then

$$I_{p,q} = (-1)^q \Delta^q A_p = \sum_{j=0}^q (-1)^j \binom{q}{j} A_{p+j}. \quad (1632)$$

For noninteger  $q$ , an Abel-regularized Newton series remains valid:

$$I_{p,q} = \lim_{r \uparrow 1} \sum_{j \geq 0} (-1)^j \binom{q}{j} r^j A_{p+j}, \quad \Re p, \Re q \geq 0. \quad (1633)$$

### 6.1 The bivariate resolvent

Let

$$R(z) = \int_0^1 \frac{dx}{1 - zF(x)} = \sum_{p \geq 0} A_p z^p, \quad |z| < 1. \quad (1634)$$

**Theorem 6.1** (Bivariate mixed-power generating function<sup>N</sup>). For  $|z|, |w| < 1$ , with removable points understood by analytic continuation,

$$\boxed{\sum_{p,q \geq 0} I_{p,q} z^p w^q = \frac{zR(z) + wR(w)}{z + w - zw}}. \quad (1635)$$

Moreover,

$$R(z) = \frac{1}{1-z} R\left(-\frac{z}{1-z}\right). \quad (1636)$$

*Proof.* Sum the geometric series under the integral:

$$\sum_{p,q \geq 0} I_{p,q} z^p w^q = \int_0^1 \frac{dx}{(1 - zF(x))(1 - w(1 - F(x)))}.$$

Partial fractions give coefficients  $z/(z + w - zw)$  and  $w/(z + w - zw)$ . Reflection  $x \mapsto 1 - x$  changes the second integral into  $R(w)$ . The same reflection applied directly to  $R$  gives (1636).  $\square$

Parameter differentiation supplies mixed logarithmic integrals:

$$\frac{\partial^{r+s}}{\partial p^r \partial q^s} I_{p,q} = \int_0^1 F(x)^p (1 - F(x))^q (\log F(x))^r (\log(1 - F(x)))^s dx. \quad (1637)$$

## 6.2 Strict checkerboard total positivity

The lattice has much stronger structure than Pascal recursion.

**Theorem 6.2** (Strict sign-regularity<sup>N</sup>). *Let  $p_1 < \dots < p_m$  and  $q_1 < \dots < q_m$  be nonnegative integers. Then*

$$(-1)^{\binom{m}{2}} \det[I_{p_i, q_j}]_{i,j=1}^m > 0. \quad (1638)$$

*Equivalently, reversing the order of the  $q$ -columns produces a strictly totally positive matrix.*

*Proof.* Under  $u = F(x)$ ,

$$I_{p,q} = \int_0^1 u^p (1-u)^q G'(u) \, du.$$

The continuous Cauchy–Binet (Andreief) identity gives

$$\det[I_{p_i, q_j}] = \frac{1}{m!} \int_{(0,1)^m} \det[u_k^{p_i}]_{i,k} \det[(1-u_k)^{q_j}]_{j,k} \prod_{k=1}^m G'(u_k) \, du_k.$$

On  $0 < u_1 < \dots < u_m < 1$ , the first generalized Vandermonde determinant is positive. The numbers  $1 - u_k$  occur in decreasing order, so the second has sign  $(-1)^{\binom{m}{2}}$ . Both are nonzero away from diagonals, and  $G' > 0$ , proving strictness.  $\square$

For  $m = 2$ , this yields the sharp reverse-TP inequality

$$I_{p,q} I_{p+1,q+1} < I_{p,q+1} I_{p+1,q}. \quad (1639)$$

For fixed  $q$ , the sequence  $p \mapsto I_{p,q}$  is a strict Hausdorff moment sequence and hence has positive Hankel minors; [theorem 6.2](#) adds the cross-parameter sign law.

## 7 Beta localization and asymptotics

### 7.1 Interior localization

Substitution  $u = F(x)$  gives, for a weight  $h$ ,

$$\int_0^1 h(x) F(x)^p (1-F(x))^q \, dx = B(p+1, q+1) \mathbb{E}K(B_{p,q}), \quad (1640)$$

where

$$K(u) = h(G(u))G'(u), \quad B_{p,q} \sim \text{Beta}(p+1, q+1). \quad (1641)$$

Let  $N = p+q+2$  and  $\mu = (p+1)/N$ . If  $p/N$  remains in a compact subinterval of  $(0, 1)$  and  $K \in C^4$  near  $\mu$ , then

$$\mathbb{E}K(B_{p,q}) = K(\mu) + \frac{\mu(1-\mu)}{2(N+1)} K''(\mu) + O(N^{-2}). \quad (1642)$$

In particular, when  $p+q = n$  and  $p/n \rightarrow \lambda \in (0, 1)$ ,

$$I_{p,q} \sim \sqrt{\frac{2\pi\lambda(1-\lambda)}{n}} \lambda^p (1-\lambda)^q G'(\lambda). \quad (1643)$$

This is the ordinary, algebraic interior regime.

## 7.2 The balanced sequence and central flatness

The balanced case is exceptional. Define

$$K_m = (2m+1) \binom{2m}{m} I_{m,m} = \mathbb{E}G'(B_m), \quad B_m \sim \text{Beta}(m+1, m+1). \quad (1644)$$

The center identity

$$F\left(\frac{1}{2} + h\right) = \frac{1}{2} + 2h - \text{sgn}(h)F(|h|) \quad (1645)$$

implies

$$F'\left(\frac{1}{2} + h\right) = 2[1 - F(2|h|)]. \quad (1646)$$

Because every derivative of  $F$  vanishes at 0, inverse differentiation gives

$$G'(1/2) = \frac{1}{2}, \quad G^{(r)}(1/2) = 0 \quad (r \geq 2). \quad (1647)$$

Thus  $G'$  is flat, but not locally constant, at its minimum.

**Theorem 7.1** (Monotone superalgebraic localization<sup>N</sup>). *The sequence  $(K_m)_{m \geq 0}$  is strictly decreasing and*

$$K_m \downarrow \frac{1}{2}. \quad (1648)$$

Moreover, for every  $A > 0$ ,

$$K_m - \frac{1}{2} = O_A(m^{-A}). \quad (1649)$$

*Proof.* The function  $G'$  is symmetric about  $1/2$ . On  $(1/2, 1)$ ,  $F'$  is strictly decreasing by (1646), so  $G'(u) = 1/F'(G(u))$  is strictly increasing. Hence  $G'$  increases with  $|u - 1/2|$ . The ratio of successive beta densities is a positive constant times  $u(1-u)$ , which strictly decreases with  $|u - 1/2|$ . A one-crossing argument therefore gives  $\mathbb{E}G'(B_{m+1}) < \mathbb{E}G'(B_m)$ . Concentration gives the limit  $G'(1/2) = 1/2$ .

For any  $A$ , flatness (1647) implies  $|G'(u) - 1/2| \leq C_A |u - 1/2|^{2A}$  in a central neighborhood. The corresponding beta moment is  $O(m^{-A})$ , while the complement has exponentially small beta probability. This proves (1649).  $\square$

## 7.3 A logarithmic law and Lambert saddle

Superalgebraic decay conceals a precise logarithmic scale.

**Theorem 7.2** (Balanced logarithmic asymptotic<sup>N</sup>). *Assume the endpoint logarithm (1605). Then*

$$\lim_{m \rightarrow \infty} \frac{\log(K_m - \frac{1}{2})}{(\log m)^2} = -\frac{1}{8 \log 2}. \quad (1650)$$

The saddle in the central displacement  $v = |u - 1/2| = e^{-t}$  is located at

$$t_m = \frac{1}{2}W(16m \log 2) + o(1). \quad (1651)$$

*Proof at logarithmic precision.* For  $u = 1/2 + v$ , write  $G(u) = 1/2 + h$ . Equations (1645)–(1646) imply  $v = 2h - F(h) \sim 2h$  and

$$G'(1/2 + v) - \frac{1}{2} = \frac{F(2h)}{2(1 - F(2h))}.$$

Consequently,

$$\log(G'(1/2 + v) - \frac{1}{2}) = -\frac{(\log(1/v))^2}{2 \log 2} + O(\log(1/v) \log \log(1/v)). \quad (1652)$$

The beta density contributes  $(1 - 4v^2)^m = \exp(-4mv^2 + O(mv^4))$ . At logarithmic precision the relevant exponent is therefore

$$\Psi_m(t) = 4me^{-2t} + \frac{t^2}{2 \log 2}.$$

Its stationary equation is  $8me^{-2t} = t/\log 2$ , or  $(2t)e^{2t} = 16m \log 2$ , giving (1651). Since  $t_m \sim \frac{1}{2} \log m$ ,

$$\Psi_m(t_m) = \frac{t_m^2 + t_m}{2 \log 2} = \frac{(\log m)^2}{8 \log 2} + o((\log m)^2).$$

Standard upper and lower Laplace bounds on windows around the saddle prove (1650).  $\square$

**Conjecture 7.3** (Full Lambert-periodic expansion<sup>C</sup>). *Let  $w_m = W(16m \log 2)$ . The quantity  $K_m - 1/2$  has a full expansion of the form*

$$K_m - \frac{1}{2} \sim \mathcal{A}(\vartheta_m) m^\delta w_m^\gamma \exp[-(w_m^2 + 2w_m)/(8 \log 2)] \left( 1 + \sum_{j \geq 1} \frac{P_j(w_m)}{w_m^j} \right), \quad (1653)$$

where  $\mathcal{A}$  is a nonconstant bounded periodic factor inherited from the endpoint phase,  $\vartheta_m$  is an affine function of  $\log_2 w_m$ , and the algebraic prefactor and polynomials  $P_j$  are determined by the all-orders endpoint expansion of  $F$ .

The exact periodic phase should be extracted from the corpus's Lambert-phase normal form rather than guessed from moderate values of  $m$ ; the convergence to the scale in (1650) is extremely slow.

## 8 Powers of Rvachev's up function

Because  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$  and  $\text{up}$  is even, powers of  $\text{up}$  reduce to inverse-beta moments.

**Theorem 8.1** (Power and weighted-power formulas<sup>N</sup>). *For  $\Re \alpha > -1$  and  $\Re q > 0$ ,*

$$\boxed{\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx = \frac{2q}{\alpha + 1} \int_0^1 G(u)^{\alpha+1} (1-u)^{q-1} du.} \quad (1654)$$

If  $q \in \mathbb{N}_{>0}$ , then

$$\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx = \frac{2}{\alpha + 1} \mathbb{E} Z_{1;q}^{\alpha+1}. \quad (1655)$$

In particular,

$$\int_{-1}^1 \text{up}(x)^q dx = 2A_q = 2\mathbb{E} Z_{1;q}. \quad (1656)$$

*Proof.* On  $[0, 1]$ , integrate  $x^\alpha (1 - F(x))^q$  by parts. The boundary terms vanish and substitution  $u = F(x)$  gives

$$\int_0^1 x^\alpha (1 - F(x))^q dx = \frac{q}{\alpha + 1} \int_0^1 G(u)^{\alpha+1} (1-u)^{q-1} du.$$

Double by evenness. For integer  $q$ , the beta variable  $\text{Beta}(1, q)$  is  $F(Z_{1;q})$ .  $\square$

The incomplete one-sided primitive is

$$\int_0^x t^\alpha \text{up}(t)^q dt = \frac{x^{\alpha+1}(1-F(x))^q}{\alpha+1} + \frac{q}{\alpha+1} Q_{\alpha+1}(1, q; F(x)). \quad (1657)$$

For  $q = 1$ ,

$$\int_{-1}^1 |x|^\alpha \text{up}(x) dx = \frac{2d_{\alpha+1}}{\alpha+1}. \quad (1658)$$

### 8.1 Coupling to the derivative of $F$

The derivative removes all distribution-specific complexity:

$$\int_0^1 F(x)^p (1-F(x))^q F'(x) dx = B(p+1, q+1), \quad (1659)$$

$$\int_0^1 F(x)^p (1-F(x))^q \text{up}(2x-1) dx = \frac{1}{2} B(p+1, q+1), \quad (1660)$$

$$\int_0^1 x^r F(x)^p (1-F(x))^q F'(x) dx = Q_r(p+1, q+1). \quad (1661)$$

These identities are exact for complex parameters throughout their domains of absolute convergence.

## 9 A transform dictionary

### 9.1 Transforms of the inverse Fabius function

The quantile substitution gives the master identity

$$\int_0^1 \Psi(G(u)) du = \mathbb{E}\Psi(X). \quad (1662)$$

Therefore

$$\int_0^1 e^{zG(u)} du = M_X(z) = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j}, \quad (1663)$$

$$\int_0^1 e^{-2\pi i \xi G(u)} du = e^{-\pi i \xi} \Phi(\xi/2), \quad (1664)$$

$$\int_0^1 (z - G(u))^{-\nu} du = \mathbb{E}(z - X)^{-\nu}, \quad (1665)$$

$$\int_0^1 G(u)^r (1 - G(u))^s du = \mathbb{E}X^r (1 - X)^s. \quad (1666)$$

When  $s \in \mathbb{N}$ , the last expression is the finite rational-moment sum

$$\sum_{k=0}^s (-1)^k \binom{s}{k} d_{r+k}. \quad (1667)$$

### 9.2 Linear transforms of $F$

Integration by parts gives

$$\int_0^1 e^{-sx} F(x) dx = \frac{M_X(-s) - e^{-s}}{s}, \quad (1668)$$

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (1669)$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - d_s}{s}, \quad (1670)$$

$$\int_0^1 x^{s-1} (1 - F(x)) dx = \frac{d_s}{s}. \quad (1671)$$

The removable values at  $s = 0$  or  $\xi = 0$  are taken by continuity.

For  $G$ , one obtains the nonlinear Laplace dual

$$\int_0^1 e^{-su} G(u) du = \frac{\int_0^1 e^{-sF(x)} dx - e^{-s}}{s} = \frac{\sum_{n \geq 0} \frac{(-s)^n}{n!} A_n - e^{-s}}{s}. \quad (1672)$$

Thus the Laplace transform of the quantile is encoded by the complete mixed-power edge sequence  $(A_n)$ .

### 9.3 Spacing transforms

For  $p, q > 0$ , the spacing theorem yields an entire transform package. With  $N = p + q$ ,

$$\begin{aligned} \mathcal{M}_{p,q}(s) &:= \int_0^1 x^{s-1} F(x)^p (1 - F(x))^q dx \\ &= \frac{\mathbb{E}(Z_{p+1:N}^s - Z_{p:N}^s)}{s \binom{N}{p}}, \end{aligned} \quad (1673)$$

$$\begin{aligned} \mathcal{L}_{p,q}(s) &:= \int_0^1 e^{-sx} F(x)^p (1 - F(x))^q dx \\ &= \frac{\mathbb{E}(e^{-sZ_{p:N}} - e^{-sZ_{p+1:N}})}{s \binom{N}{p}}, \end{aligned} \quad (1674)$$

$$\begin{aligned} \mathcal{C}_{p,q}(z) &:= \int_0^1 \frac{F(x)^p (1 - F(x))^q}{z - x} dx \\ &= \frac{1}{\binom{N}{p}} \mathbb{E} \log \frac{z - Z_{p:N}}{z - Z_{p+1:N}}. \end{aligned} \quad (1675)$$

More generally, for  $\nu \neq 1$ ,

$$\int_0^1 \frac{F(x)^p (1 - F(x))^q}{(z - x)^\nu} dx = \frac{\mathbb{E}[(z - Z_{p+1:N})^{1-\nu} - (z - Z_{p:N})^{1-\nu}]}{(\nu - 1) \binom{N}{p}}. \quad (1676)$$

All branch choices are understood consistently on a simply connected domain avoiding  $[0, 1]$ .

## 10 The Cauchy–Stieltjes hierarchy of the up-law

For  $z \in \mathbb{C} \setminus [-1, 1]$  and  $\Re \nu > 0$ , define

$$S_\nu(z) = \int_{-1}^1 \frac{\text{up}(y)}{(z - y)^\nu} dy = \mathbb{E}(z - Y)^{-\nu}. \quad (1677)$$



### 10.1 Laplace–sinh representation

For  $\Re z > 1$ ,

$$S_\nu(z) = \frac{1}{\Gamma(\nu)} \int_0^\infty t^{\nu-1} e^{-zt} \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j} dt. \quad (1678)$$

Combining (1603) with Watson expansion yields a complete large- $z$  Bell–Bernoulli asymptotic; because the support is compact, it is in fact convergent for  $|z| > 1$ :

$$S_\nu(z) = z^{-\nu} \sum_{m \geq 0} \frac{(\nu)_{2m}}{(2m)!} \frac{c_m}{z^{2m}}. \quad (1679)$$

Every coefficient is rational.

### 10.2 Dyadic functional equations

**Theorem 10.1** (Dyadic Stieltjes hierarchy<sup>N</sup>). *For  $\nu \neq 1$ ,*

$$S_\nu(z) = \frac{2^{\nu-1}}{\nu-1} [S_{\nu-1}(2z-1) - S_{\nu-1}(2z+1)]. \quad (1680)$$

At  $\nu = 1$ ,

$$S_1(z) = \mathbb{E} \log \frac{2z+1-Y}{2z-1-Y}. \quad (1681)$$

Furthermore,

$$S_\nu^{(m)}(z) = (-1)^m (\nu)_m S_{\nu+m}(z), \quad (1682)$$

$$S_1'(z) = 2 [S_1(2z+1) - S_1(2z-1)], \quad (1683)$$

$$S_1(-z) = -S_1(z). \quad (1684)$$

*Proof.* The self-similarity  $Y = (V + Y')/2$ , with  $V \sim \text{Unif}[-1, 1]$  independent of  $Y'$ , gives

$$S_\nu(z) = 2^{\nu-1} \mathbb{E} \int_{-1}^1 (2z - Y' - v)^{-\nu} dv.$$

The elementary integral in  $v$  gives (1680); its logarithmic limit is (1681). Differentiation proves (1682), and the case  $\nu = 2$  of (1680) gives (1683). Symmetry of  $Y$  gives (1684).  $\square$

### 10.3 Boundary values and a Hilbert-transform refinement law

For  $-1 < x < 1$ , the upper boundary value satisfies

$$S_{1,+}(x) = \text{PV} \int_{-1}^1 \frac{\text{up}(t)}{x-t} dt - \pi i \text{up}(x). \quad (1685)$$

Taking imaginary and real parts of (1683) yields respectively the known dyadic refinement of up and a parallel functional equation for its Hilbert transform. Thus the Cauchy transform packages the density and its harmonic conjugate into a single dyadic object.

## 10.4 Endpoint derivative growth

At the right endpoint,

$$\frac{(-1)^n}{n!} S_1^{(n)}(1) = 2^{-n-1} \mathbb{E} X^{-n-1}. \quad (1686)$$

The endpoint law (1605), by a one-dimensional Laplace principle, implies

$$\log \mathbb{E} X^{-s} = \frac{\log 2}{2} s^2 + O(s \log s). \quad (1687)$$

Hence

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \log \left| \frac{S_1^{(n)}(1)}{n!} \right| = \frac{\log 2}{2}. \quad (1688)$$

The one-sided derivatives at 1 all exist, but the Taylor series there has radius zero. This is a transform-level signature of the flat endpoint of up.

## 11 Generalized resolvent duality for $F$ and $G$

For  $z \notin [0, 1]$ , define

$$C_{F,\nu}(z) = \int_0^1 \frac{F(x)}{(z-x)^\nu} dx, \quad (1689)$$

$$C_{G,\nu}(z) = \int_0^1 \frac{G(u)}{(z-u)^\nu} du. \quad (1690)$$

**Theorem 11.1** (CDF/quantile resolvent duality<sup>N</sup>). For  $\nu \neq 1$ ,

$$C_{F,\nu}(z) = \frac{(z-1)^{1-\nu} - \mathbb{E}(z-X)^{1-\nu}}{\nu-1}, \quad (1691)$$

$$C_{G,\nu}(z) = \frac{(z-1)^{1-\nu} - \int_0^1 (z-F(x))^{1-\nu} dx}{\nu-1}. \quad (1692)$$

At  $\nu = 1$ ,

$$C_F(z) = \mathbb{E} \log(z-X) - \log(z-1), \quad (1693)$$

$$C_G(z) = \int_0^1 \log(z-F(x)) dx - \log(z-1). \quad (1694)$$

Both satisfy the same reflection law

$$C_F(z) - C_F(1-z) = C_G(z) - C_G(1-z) = \log \frac{z}{z-1}. \quad (1695)$$

*Proof.* Use the primitive  $(z-x)^{1-\nu}/(\nu-1)$  and integrate by parts. In the quantile case,  $\int H(u) dG(u) = \int H(F(x)) dx$ . The logarithmic formulas are the limits as  $\nu \rightarrow 1$ . Reflection follows by substituting  $x \mapsto 1-x$  and using the symmetries of  $F$  and  $G$ .  $\square$

Let

$$T(z) = \mathbb{E}(z-X)^{-1} = 2S_1(2z-1). \quad (1696)$$

The affine self-similarity  $X = (U + X')/2$  yields

$$T(z) = 2\mathbb{E} \log \frac{2z-X}{2z-1-X}, \quad (1697)$$

$$T'(z) = 4[T(2z) - T(2z-1)]. \quad (1698)$$

These formulas form a nonlinear/linear pair: (1697) is a logarithmic potential identity, while (1698) is a linear dyadic functional-differential equation.

## 12 Sinc-product energies and the constant $\int_0^1 F^2$

Set

$$A_2 = \int_0^1 F(x)^2 dx. \quad (1699)$$

Numerically,

$$A_2 = 0.404436767983985281465621491145 \dots \quad (1700)$$

No familiar elementary or classical-special-function closed form is known in the audited corpus. There is, however, an exact sinc-product energy form.

**Theorem 12.1** (Sinc energy and complementary Gini sum rule<sup>N</sup>). *With  $\Phi$  as in (1600),*

$$A_2 = \int_0^\infty \Phi(\xi)^2 d\xi, \quad (1701)$$

$$\mathbb{E}|X - X'| = 1 - 2A_2 = \frac{1}{2\pi^2} \int_0^\infty \frac{1 - \Phi(\xi)^2}{\xi^2} d\xi, \quad (1702)$$

$$\boxed{\int_0^\infty \Phi(\xi)^2 d\xi + \frac{1}{4\pi^2} \int_0^\infty \frac{1 - \Phi(\xi)^2}{\xi^2} d\xi = \frac{1}{2}.} \quad (1703)$$

*Proof.* Since  $\text{up}(x) = 1 - F(x)$  on  $[0, 1]$ , symmetry and Parseval give

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = 2A_2 = \int_{\mathbb{R}} \Phi(\xi)^2 d\xi,$$

which is (1701). The standard cycles-normalized fractional-energy identity at exponent one is

$$\mathbb{E}|Z| = \frac{1}{\pi^2} \int_0^\infty \frac{1 - \Re \varphi_Z(\xi)}{\xi^2} d\xi.$$

For  $Z = X - X'$ ,  $|\varphi_X(\xi)|^2 = \Phi(\xi/2)^2$ , yielding (1702) after a change of variables. Finally,  $\mathbb{E}|X - X'| = 2 \int_0^1 F(1 - F) = 1 - 2A_2$ .  $\square$

The integrand in (1702) is regular at the origin:

$$\lim_{\xi \rightarrow 0} \frac{1 - \Phi(\xi)^2}{\xi^2} = \frac{4\pi^2}{9}, \quad (1704)$$

because  $\text{Var}(Y) = 1/9$ .

For every integer  $q \geq 2$ , Fourier convolution gives the higher-power representation

$$A_q = \frac{1}{2} \int_{\mathbb{R}^{q-1}} \Phi(\xi_1) \cdots \Phi(\xi_{q-1}) \Phi\left(-\sum_{j=1}^{q-1} \xi_j\right) d\xi_1 \cdots d\xi_{q-1}. \quad (1705)$$

For  $0 < \alpha < 2$ , the full fractional energy is

$$\mathbb{E}|X - X'|^\alpha = \frac{2^{1-\alpha} \Gamma(1 + \alpha) \sin(\pi\alpha/2)}{\pi^{1+\alpha}} \int_0^\infty \frac{1 - \Phi(\xi/2)^2}{\xi^{1+\alpha}} d\xi. \quad (1706)$$

**Conjecture 12.2** (Arithmetic opacity of the energy constant<sup>C</sup>). *The number  $A_2$  is transcendental. More conservatively, it does not belong to the field generated over  $\overline{\mathbb{Q}}$  by finitely many values of Gamma, zeta, polylogarithm, and elementary functions at algebraic arguments.*

The conjecture is deliberately separated from the exact formulas (1701) and (1703); those formulas may instead suggest a new class of “dyadic sinc periods.”

## 13 General derivatives and antiderivatives of arbitrary order

### 13.1 Finite Thue–Morse derivative formulas

Let  $s_2(j)$  be the binary digit sum and  $T_n = n(n+1)/2$ . Iterating (1598) gives the baseline finite formulas

$$\text{up}^{(n)}(x) = 2^{T_n} \sum_{j=0}^{2^n-1} (-1)^{n+s_2(j)} \text{up}(2^n x + 2j + 1 - 2^n), \quad (1707)$$

$$F^{(n)}(x) = 2^{T_n} \sum_{j=0}^{2^{n-1}-1} (-1)^{n-1+s_2(j)} \text{up}(2^n x + 2j + 1 - 2^n), \quad n \geq 1. \quad (1708)$$

For a mixed power  $W_{p,q}(x) = F(x)^p(1 - F(x))^q$ , Faà di Bruno gives the complete derivative formula

$$W_{p,q}^{(n)}(x) = \sum_{k=0}^n \frac{d^k}{du^k} [u^p(1-u)^q] \Big|_{u=F(x)} B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)), \quad (1709)$$

where  $B_{n,k}$  is a partial exponential Bell polynomial and

$$\frac{d^k}{du^k} [u^p(1-u)^q] = \sum_{r=0}^k \binom{k}{r} p^r (-1)^{k-r} q^{k-r} u^{p-r} (1-u)^{q-k+r}. \quad (1710)$$

Substituting (1708) makes (1709) a finite expression in Thue–Morse translates of up.

### 13.2 Integer and fractional primitives of up

The normalized  $n$ -fold primitive from  $-\infty$  is

$$U_n(x) = \frac{1}{(n-1)!} \int_{-\infty}^x (x-t)^{n-1} \text{up}(t) dt. \quad (1711)$$

For  $x \leq 1$ ,

$$U_n(x) = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), \quad (1712)$$

whereas for  $x \geq 1$ , compact support gives the moment polynomial

$$U_n(x) = \sum_{k=0}^{n-1} \frac{(-1)^k \mu_k}{k!(n-1-k)!} x^{n-1-k}, \quad \mu_k = \mathbb{E}Y^k. \quad (1713)$$

The odd  $\mu_k$  vanish, and the even ones are the Bell–Bernoulli moments in (1603).

### 13.3 Fractional primitives of $F$ and $G$

The baseline probabilistic form for  $F$  is

$$I^\nu F(x) = \frac{1}{\Gamma(\nu+1)} \mathbb{E}(x-X)_+^\nu. \quad (1714)$$

For the inverse,

$$I^\nu G(y) = \frac{1}{\Gamma(\nu+1)} \int_0^{G(y)} (y-F(t))^\nu dt. \quad (1715)$$

The integer derivatives of the inverse are compactly expressed by the inverse operator

$$G^{(n)}(u) = \left[ \left( \frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \right]_{x=G(u)}. \quad (1716)$$

Equivalently, one may expand (1716) using Bell polynomials in the ratios  $F^{(k)}/F'$ . Equations (1709), (1616), (1712), and (1716) together provide a uniform answer to the “ $n$ -th derivative/antiderivative” part of the problem.

## 14 Finite Thue–Morse splines and certified series approximations

Let

$$X_M = \sum_{j=1}^M \frac{U_j}{2^j}, \quad F_M(x) = \mathbb{P}(X_M \leq x). \quad (1717)$$

The generalized Irwin–Hall formula is

$$F_M(x) = \frac{2^{M(M+1)/2}}{M!} \sum_{k=0}^{2^M-1} (-1)^{s_2(k)} \left( x - \frac{k}{2^M} \right)_+^M. \quad (1718)$$

Thus  $F_M$  is a rational piecewise polynomial on the dyadic mesh. The tail coupling satisfies

$$0 \leq X - X_M \leq 2^{-M} \quad \text{almost surely.} \quad (1719)$$

**Theorem 14.1** (Certified rational approximation<sup>N</sup>). *Let  $p, q \in \mathbb{N}_{>0}$ ,  $N = p + q$ , and let  $H$  be Lipschitz on  $[0, 1]$  with constant  $L$ . Define*

$$J = \int_0^1 H'(x) F(x)^p (1 - F(x))^q dx, \quad J_M = \int_0^1 H'(x) F_M(x)^p (1 - F_M(x))^q dx.$$

Then

$$|J - J_M| \leq \frac{2L}{\binom{N}{p}} 2^{-M}. \quad (1720)$$

If  $H'$  is a polynomial with rational coefficients, then  $J_M \in \mathbb{Q}$  and can be evaluated exactly by integrating finitely many dyadic polynomials.

*Proof.* Couple every observation with its truncation using the same uniforms. By (1719), every corresponding order statistic differs by at most  $2^{-M}$ . Apply (1606) to  $F$  and  $F_M$ ; the difference of the two adjacent-gap increments of  $H$  is at most  $2L2^{-M}$ . Formula (1718) proves rationality.  $\square$

Consequently, every integer mixed moment has a rational sequence representation with a certified exponential error. This complements the more rapidly convergent but structurally specialized  $q$ -Richardson schemes in the corpus. For noninteger powers, one may combine (1718) with the Abel series (1633), or use the beta–quantile integral directly.

## 15 Numerical experiments

The archive contains `fabius_integral_transforms.py`. It constructs a uniform-grid fixed point for  $F$ , computes exact rational moments by (1595), evaluates the sinc product, and checks independent sides of the principal identities. The complete output is in `numerical_results.txt`. Selected results are shown below.

| Check                                         | Absolute residual           |
|-----------------------------------------------|-----------------------------|
| $F(1/4) = 5/72$ , grid $2^{17}$               | $1.29 \times 10^{-11}$      |
| Weighted spacing identity, $e^x F^2(1 - F)^3$ | $3.35 \times 10^{-12}$      |
| Fractional spacing primitive, order 1.65      | $1.26 \times 10^{-12}$      |
| Cauchy equation (1683)                        | below $2.3 \times 10^{-16}$ |
| Sinc/Gini sum rule (1703)                     | $8.1 \times 10^{-15}$       |

The two independent estimates

$$\int_0^1 F(x)^2 dx \approx 0.4044367679787557$$

from the grid and

$$\int_0^{64} \Phi(\xi)^2 d\xi \approx 0.4044367679839853$$

from oscillatory quadrature agree to the expected grid accuracy. The normalized balanced values begin

$$\begin{aligned} K_1 &\approx 0.5733793921, & K_{10} &\approx 0.5030089983, \\ K_{100} &\approx 0.5000147741, & K_{1000} &\approx 0.50000000757. \end{aligned}$$

consistent with strict decrease and superalgebraic localization. They are far from the ultimate logarithmic regime in (1650), as the Lambert saddle moves only logarithmically.

## 16 Conjectures and frontier directions

### 16.1 Analytic continuation and non-holonomicity

**Conjecture 16.1** (No local crossing of the Cauchy cut<sup>C</sup>). *The Cauchy transform  $S_1$  admits no analytic continuation through any nonempty open subinterval of  $[-1, 1]$ . Equivalently, the density up is not real-analytic on any nonempty open interval.*

A proof would imply that  $S_1$  is not D-finite over  $\mathbb{C}(z)$ , since a D-finite function has only finitely many singular points in the finite plane. The dyadic equation (1683), together with the nowhere-analytic structure of  $F$ , is the natural route.

### 16.2 The mixed-power resolvent as a new special function

The function  $R$  in (1634) is a Stieltjes-type transform of the symmetric random variable  $F(U)$ , where  $U \sim \text{Unif}[0, 1]$ . It obeys the involutive Möbius law (1636), but no closed form is known.

*Open problem 16.2.* Determine the maximal analytic continuation of  $R$ , its boundary values on  $[1, \infty)$ , and whether its Möbius functional equation plus positivity characterizes the Fabius pushforward measure.

*Open problem 16.3.* Develop Padé and continued-fraction approximants to  $R$  from the exact Hausdorff moments  $A_n$ , and compare their pole distributions with the  $q$ -Richardson accelerants already present in the corpus.

### 16.3 Balanced Lambert tomography

The full conjecture conjecture 7.3 offers a new way to probe the periodic endpoint correction: balanced beta localization samples the flat central germ of  $G'$ , which is itself a reparameterized copy of the endpoint germ of  $F$ . Numerically fitting  $K_m$  after removing the Lambert saddle may recover the endpoint phase without evaluating  $F$  at extremely small arguments.

*Open problem 16.4.* Derive the first two explicit amplitude and phase terms in (1653), and design a confluent  $q$ -Richardson scheme in the variable  $w_m = W(16m \log 2)$  that preserves the periodic phase rather than averaging it away.

## 16.4 Dyadic sinc periods

The constants

$$\mathcal{P}_q = \int_{\mathbb{R}^{q-1}} \Phi(\xi_1) \cdots \Phi(\xi_{q-1}) \Phi\left(-\sum \xi_j\right) d\xi = 2A_q \quad (1721)$$

form a natural period family attached to the dyadic sinc product.

*Open problem 16.5.* Find recurrences, functional equations, or arithmetic obstructions for  $\mathcal{P}_q$ . Determine whether the complementary energy identity (1703) is the first member of a hierarchy involving fractional Riesz energies and higher convolution powers.

## 16.5 Formalization roadmap

The most Lean-friendly new results are:

1. the random-interval identity and adjacent-spacing theorem, using finite exchangeability and order statistics;
2. the Riemann–Liouville spacing formula, by Tonelli and an elementary truncated-power integral;
3. beta-quantile Pascal/reflection identities;
4. the Cauchy hierarchy, from the affine fixed-point law and integration of a uniform variable;
5. the sinc-energy identity, from Parseval and the standard Fourier formula for  $|x|$ ;
6. the certified finite-spline bound, from the deterministic tail coupling.

The balanced logarithmic asymptotic requires the endpoint theorem plus a quantitative Laplace principle and should follow only after the elementary transform layer is formalized.

## 17 Compact formula compendium

For convenient reference, the central formulas are collected here.

| Object                        | Formula                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Weighted mixed integral       | $\int h F^p (1 - F)^q = \binom{p+q}{p}^{-1} \mathbb{E}[H(Z_{p+1}) - H(Z_p)].$                                        |
| Fractional mixed primitive    | $I^\nu[F^p(1 - F)^q](x) = \frac{\mathbb{E}[(x - Z_p)_+^\nu - (x - Z_{p+1})_+^\nu]}{\Gamma(\nu + 1) \binom{p+q}{p}}.$ |
| Incomplete monomial primitive | Equation (1621) in terms of truncated $Q_{m+1}$ .                                                                    |
| Beta-quantile reflection      | $Q_1(a, b) + Q_1(b, a) = B(a, b).$                                                                                   |
| Mixed-power recurrence        | $I_{p,q} = I_{p+1,q} + I_{p,q+1} = (-1)^q \Delta^q A_p.$                                                             |
| Bivariate generating function | $\sum I_{p,q} z^p w^q = [zR(z) + wR(w)]/(z + w - zw).$                                                               |
| Balanced normalization        | $K_m = (2m + 1) \binom{2m}{m} I_{m,m} \downarrow 1/2$ , with $\log(K_m - 1/2) \sim -(\log m)^2/(8 \log 2).$          |

| Object                    | Formula                                                                              |
|---------------------------|--------------------------------------------------------------------------------------|
| Power of up               | $\int_{-1}^1  x ^\alpha \text{up}(x)^q dx = \frac{2q}{\alpha+1} Q_{\alpha+1}(1, q).$ |
| Inverse Fourier transform | $\int_0^1 e^{-2\pi i \xi G(u)} du = e^{-\pi i \xi} \Phi(\xi/2).$                     |
| up-Stieltjes hierarchy    | $S_\nu(z) = \frac{2^{\nu-1}}{\nu-1} [S_{\nu-1}(2z-1) - S_{\nu-1}(2z+1)].$            |
| CDF resolvent             | $C_{F,\nu}(z) = [(z-1)^{1-\nu} - \mathbb{E}(z-X)^{1-\nu}]/(\nu-1).$                  |
| Sinc energy               | $A_2 = \int_0^\infty \Phi^2$ , and equation (1703).                                  |
| Finite spline             | $F_M(x) = 2^{M(M+1)/2} M!^{-1} \sum_{k < 2^M} (-1)^{s_2(k)} (x - k/2^M)_+^M.$        |

## 18 Conclusion

The integration theory of the Fabius–Rvachev system becomes substantially more coherent when it is organized around three dual viewpoints:

1. *random intervals*: mixed CDF powers are occupation probabilities of gaps between two independent sample blocks;
2. *beta quantiles*: the substitution  $u = F(x)$  turns mixed powers into beta localization against  $G'$ ;
3. *self-similar transforms*: the affine fixed-point laws of  $X$  and  $Y$  induce exact dyadic equations for Cauchy, Stieltjes, Laplace, and Fourier transforms.

These viewpoints cover indefinite integrals, all integer and fractional primitives, broad families of definite transforms, and nonlinear inverse-Fabius integrals without requiring a fictitious elementary closed form. They also expose new frontier phenomena: strict sign-regularity, balanced beyond-all-orders localization, endpoint derivative growth, and dyadic sinc periods. The resulting formulas are compatible with the repository’s q-binomial, Lambert, Bell, Bernoulli, Thue–Morse, and sinc-product structures, but are not duplications of those earlier developments.

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